

GRADE 6 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 7) • Forward 2 (Grade 8)
82 question sets • 2465 questions

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Write a ratio that describes the relationship between two quantities

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Reading counts from a table or graph	3.MD.B.3
BACKWARD 2	Parts and totals	2.OA.A.1
BACKWARD 3	Fraction of a whole	3.NF.A.1
FORWARD 1	Ratios and unit rates with fractional quantities	7.RP.A.1
FORWARD 2	The unit rate as the slope of a graph	8.EE.B.5

SET 1 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A fruit bowl has 5 pears and 8 plums.

Write the ratio of pears to plums.

Ratio: _____

A choir has 14 sopranos and 9 altos.

Write the ratio of altos to sopranos.

Ratio: _____

The table shows the pets owned by students in a class.
Write the ratio of dogs to fish.

Pet	Dogs	Cats	Fish
Students	11	6	4

Ratio: _____

A parking lot has 13 cars and 7 trucks. There are no other vehicles.

Write the ratio of trucks to all of the vehicles.

Ratio: _____

A bag has 20 marbles. 12 are green and the rest are yellow.

Write the ratio of yellow marbles to green marbles.

Ratio: _____

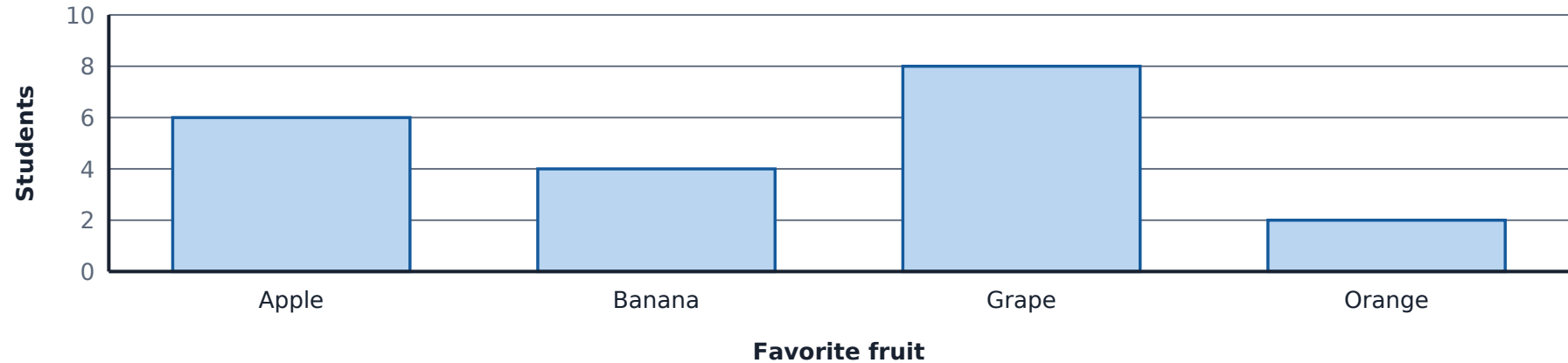
SET 1 • 6.RP.A.1

BACKWARD 1

Reading counts from a table or graph • Grade 3 • Measurement & Data •
3.MD.B.3

The graph shows the favorite fruits of a class.

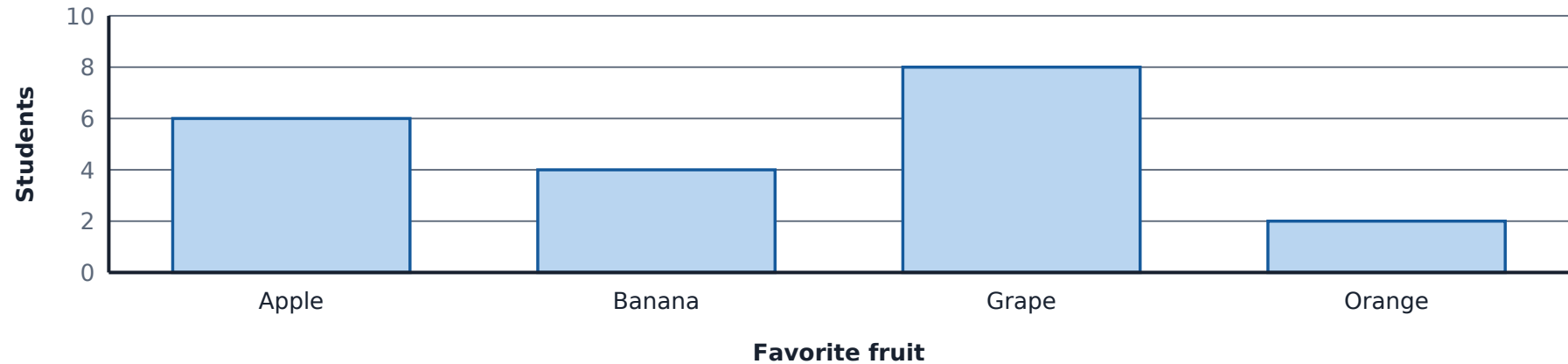
How many students chose grape?



Students: _____

The graph shows the favorite fruits of a class.

How many students chose banana?



Students: _____

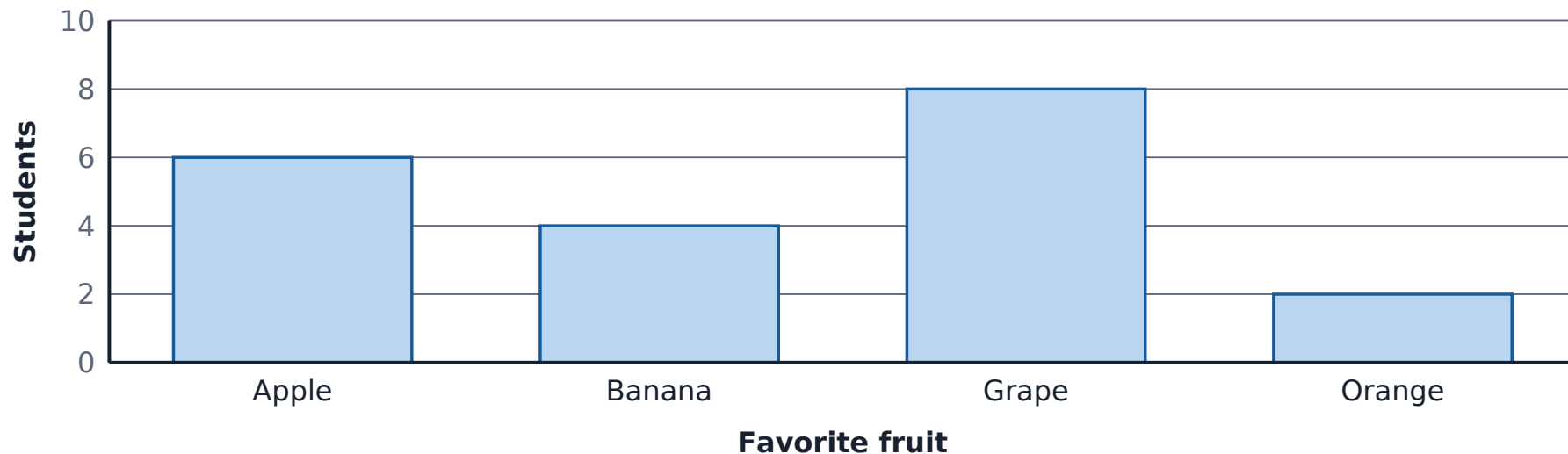
The table shows the pets owned by students.

How many students own cats?

Pet	Dogs	Cats	Fish
Students	11	6	4

Students: _____

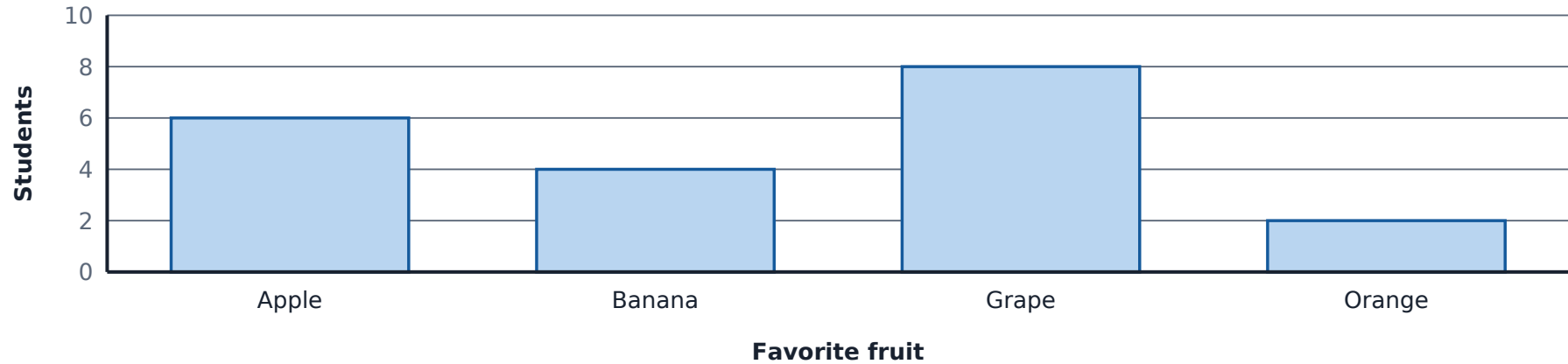
Which fruit was chosen by 2 students?



- A.** Grape **B.** Orange **C.** Banana **D.** Apple

True or false?

6 students chose apple.



A. True

B. False

SET 1 • 6.RP.A.1

BACKWARD 2

Parts and totals • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A team has 9 boys and 6 girls.

How many players are on the team?

Players: _____

A box holds 24 crayons. 15 are broken.

How many crayons are NOT broken?

Crayons: _____

True or false?

A shelf has 7 red books and 12 blue books. There are 19 books in all.

A. True

B. False

A garden has 30 plants. 18 are flowers and the rest are herbs.

How many herbs are there?

A. 12

B. 18

C. 22

D. 48

A bus has 17 adults and 26 children.

How many people are on the bus?

People: _____

SET 1 • 6.RP.A.1

BACKWARD 3

Fraction of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

The bar is divided into equal parts.

What fraction of the bar is shaded?



Fraction: _____

A pizza is cut into 6 equal slices. Ava eats 5 slices.
What fraction of the pizza does Ava eat?

Fraction: _____

True or false?

A bar is cut into 4 equal parts and 1 part is shaded. The shaded part is $\frac{1}{4}$ of the bar.

- A. True
- B. False

What fraction of the bar is shaded?



A. $\frac{2}{5}$

B. $\frac{3}{5}$

C. $\frac{2}{3}$

D. $\frac{5}{2}$

A field has 8 equal sections. Corn grows in 3 of the sections.

What fraction of the field has corn?

Fraction: _____

SET 1 • 6.RP.A.1

FORWARD 1

Ratios and unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A dressing uses $\frac{1}{2}$ cup of oil for every $\frac{3}{4}$ cup of vinegar.

How many cups of oil are used per cup of vinegar?

Cups of oil: _____

Mia paints $\frac{2}{3}$ of a fence in $\frac{1}{2}$ hour.

How much of the fence does she paint per hour?

Fence per hour: _____

A garden uses $\frac{3}{4}$ bag of soil for every $\frac{3}{8}$ bag of mulch.

How many bags of soil are used per bag of mulch?

A. $\frac{9}{32}$

B. $\frac{1}{2}$

C. $1\frac{1}{8}$

D. 2

A cyclist rides $2\frac{1}{2}$ miles in $\frac{1}{6}$ hour.

What is the speed in miles per hour?

Speed: _____

A pump moves $\frac{5}{8}$ gallon of water every $\frac{1}{4}$ minute.

How many gallons does it move per minute?

Gallons per minute: _____

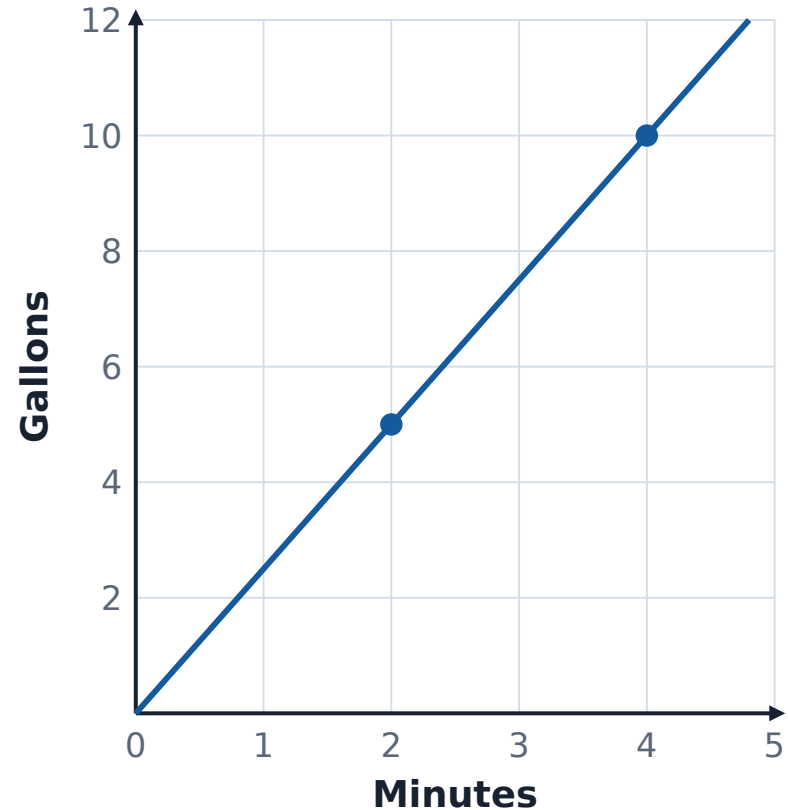
SET 1 • 6.RP.A.1

FORWARD 2

The unit rate as the slope of a graph • Grade 8 • Expressions & Equations •
8.EE.B.5

The graph shows a proportional relationship between minutes and gallons. What is the slope of the line?

Slope: _____



The graph of a proportional relationship passes through $(0, 0)$ and $(6, 9)$.

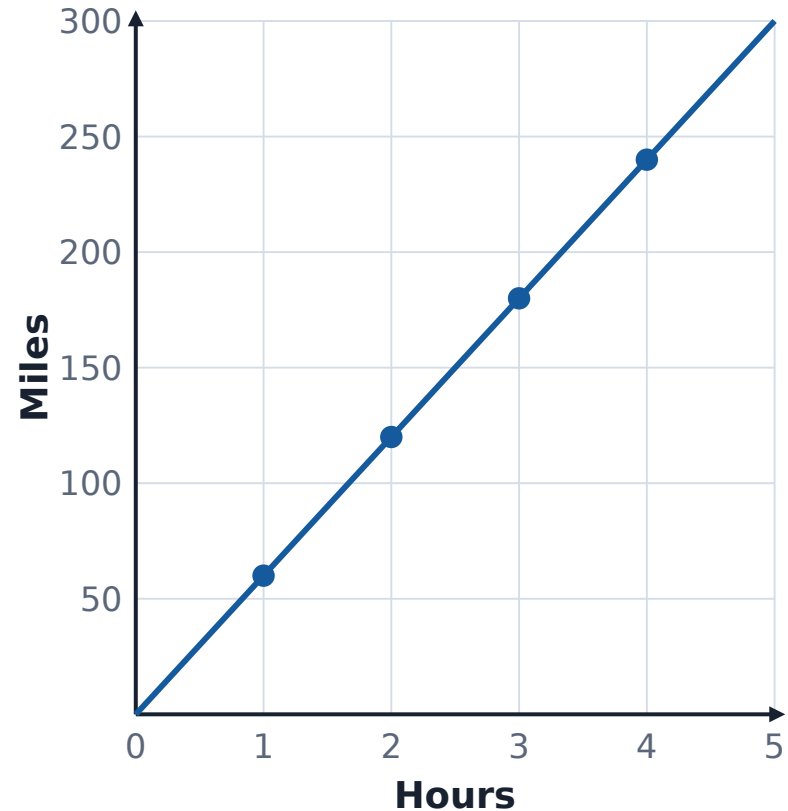
What is the slope?

Slope: _____

The graph shows the distance a car travels.

What is the slope of the line in miles per hour?

Slope: _____



A proportional relationship has a unit rate of 4 dollars per pound.

Which point is on its graph?

A. (3, 12)

B. (4, 1)

C. (3, 7)

D. (12, 3)

True or false?

For $y = 2.5x$, the unit rate is 2.5, and the slope of its graph is also 2.5.

- A. True
- B. False

Interpret ratio language ("for every," "to")

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Equal groups	3.OA.A.1
BACKWARD 2	Multiplicative comparison statements	4.OA.A.1
FORWARD 1	Constant of proportionality from a description	7.RP.A.2.b
FORWARD 2	Compare proportional relationships represented in different ways	8.EE.B.5

SET 2 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A recipe uses 5 cups of flour for every 2 cups of sugar.

Which statement describes the same ratio?

- A.** For every 7 cups of sugar, there are 5 cups of flour.
- B.** The ratio of sugar to flour is 2 : 5.
- C.** For every 2 cups of flour, there are 5 cups of sugar.
- D.** The ratio of flour to sugar is 2 : 5.

At an aquarium, the ratio of sharks to rays is 4 : 9.

Which statement is true?

- A.** For every 4 rays, there are 9 sharks.
- B.** For every 9 sharks, there are 4 rays.
- C.** For every 13 sharks, there are 4 rays.
- D.** For every 9 rays, there are 4 sharks.

The ratio of red beads to blue beads is 3 : 7.

Which statement is true?

- A.** For every 10 blue beads, there are 3 red beads.
- B.** For every 7 red beads, there are 3 blue beads.
- C.** For every 3 red beads, there are 7 blue beads.
- D.** For every 3 blue beads, there are 7 red beads.

At a camp, there are 2 counselors for every 9 campers.
What is the ratio of campers to counselors?

- A.** 2 : 9
- B.** 11 : 9
- C.** 9 : 2
- D.** 2 : 11

In a class vote, there were 4 votes for pizza for every 3 votes for tacos.

Which statement is true?

- A.** The ratio of pizza votes to taco votes is 3 : 4.
- B.** The ratio of pizza votes to all votes is 4 : 3.
- C.** There were 4 taco votes for every 3 pizza votes.
- D.** The ratio of taco votes to pizza votes is 3 : 4.

SET 2 • 6.RP.A.1

BACKWARD 1

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

4×6 can mean 4 groups with 6 in each group.

A. True

B. False

There are 3 bags with 5 apples in each bag.

Which expression shows the total?

A. 3×5

B. $5 - 3$

C. $3 + 5$

D. $5 \div 3$

There are 7 tables with 4 chairs at each table.
How many chairs are there?

Chairs: _____

True or false?

"2 wheels for every bike" means each bike has 2 wheels.

A. True

B. False

What does 6×2 mean?

- A.** 6 groups of 2
- B.** 62
- C.** 6 more than 2
- D.** 6 minus 2

SET 2 • 6.RP.A.1

BACKWARD 2

Multiplicative comparison statements • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

True or false?

24 is 3 times as many as 8.

A. True

B. False

Which equation shows "42 is 6 times as many as 7"?

A. $42 = 6 + 7$

B. $42 = 6 \times 7$

C. $42 - 7 = 6$

D. $7 = 6 \times 42$

True or false?

18 is 6 times as many as 3.

A. True

B. False

Fill in the blank.

45 is ____ times as many as 9.

Blank: _____

True or false?

20 is 4 times as many as 16.

A. True

B. False

SET 2 • 6.RP.A.1

FORWARD 1

Constant of proportionality from a description • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A candle burns 2 inches every 3 hours at a constant rate.

What is the constant of proportionality in inches per hour?

$k =$ _____

For every 5 laps, Jo swims 250 meters.

What is the constant of proportionality in meters per lap?

$k =$ _____

A machine fills 18 bottles for every 4 minutes.

What is the constant of proportionality in bottles per minute?

A. $\frac{2}{9}$

B. 4.5

C. 14

D. 72

For every 8 cups of water, a recipe uses 3 cups of rice.

What is the constant of proportionality in cups of rice per cup of water?

$k =$ _____

True or false?

For every 6 pounds of apples, the cost is \$9. The constant of proportionality is 1.5 dollars per pound.

- A.** True
- B.** False

SET 2 • 6.RP.A.1

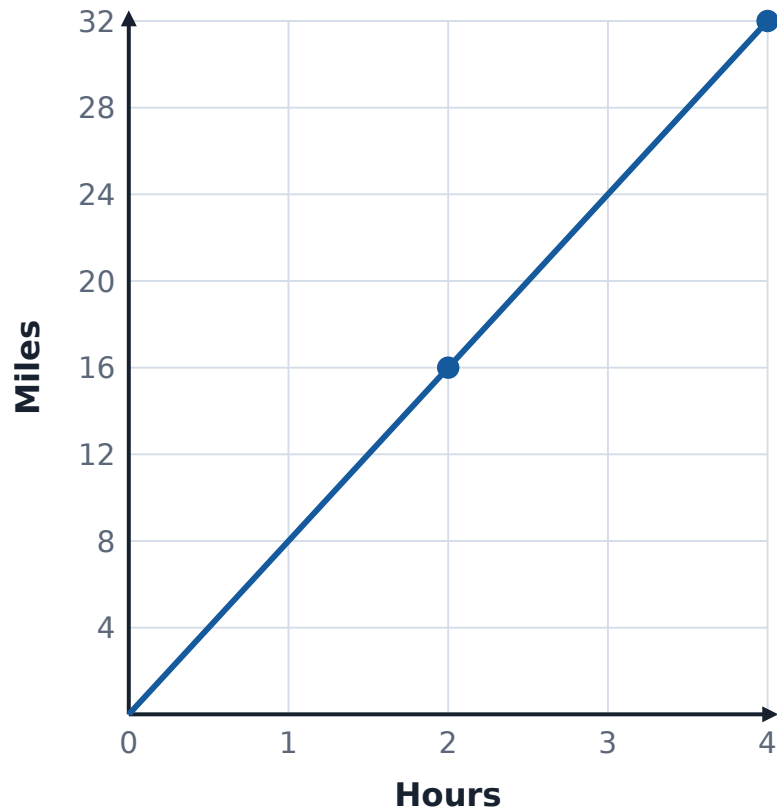
FORWARD 2

Compare proportional relationships represented in different ways • Grade 8 •
Expressions & Equations • 8.EE.B.5

Cyclist A's distance is shown in the graph. Cyclist B rides according to $d = 7.5t$, where d is miles and t is hours.

Which cyclist is faster?

- A.** Cyclist A
- B.** Cyclist B
- C.** They ride at the same speed.



Tank A fills according to $y = 12x$, where x is minutes and y is gallons. The graph of Tank B is a line through $(0, 0)$ and $(3, 45)$.

Which tank fills faster, and by how many gallons per minute?

Answer: _____

Store A: 2 pounds of apples cost \$3.50.

Store B: the cost is $c = 1.6p$ for p pounds.

Which store charges less per pound?

- A.** Store A
- B.** Store B
- C.** They charge the same.

Printer A prints 3 pages for every 2 seconds. Printer B prints according to $p = 1.4s$.

Which printer is faster?

- A.** Printer A
- B.** Printer B
- C.** They are the same.

Car A travels 150 miles in 3 hours at a constant speed.

Car B travels according to $d = 48t$.

How much faster is Car A, in miles per hour?

Miles per hour: _____

Find and describe a unit rate

MAIN

Grade 6 main question

6.RP.A.2

BACKWARD 1

Whole-number division

4.NBT.B.6

BACKWARD 2

Fraction as division

5.NF.B.3

BACKWARD 3

Meaning of "per" (the amount for each one)

3.OA.A.2

FORWARD 1

Unit rates with fractions

7.RP.A.1

FORWARD 2

Rate of change of a linear function

8.F.B.4

SET 3 • 6.RP.A.2

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A car travels 150 miles in 3 hours at a constant speed.
What is the unit rate in miles per hour?

Unit rate: _____

A 4-pound bag of rice costs \$12.

What is the price per pound?

Price per pound: _____

A punch recipe uses 5 cups of juice for every 8 cups of water.

How many cups of juice are used for each cup of water?

Cups of juice: _____

There are 2 teachers for every 30 students on a field trip.

How many students are there per teacher?

A. 15

B. 28

C. 30

D. 60

A printer prints 90 pages in 6 minutes.

How many pages does it print per minute?

Pages per minute: _____

SET 3 • 6.RP.A.2

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Find the quotient.

$$150 \div 3$$

Quotient: _____

Which is equal to $96 \div 4$?

- A.** 22
- B.** 23
- C.** 24
- D.** 26

True or false?

$$84 \div 6 = 14$$

A. True

B. False

Find the quotient.

$$212 \div 4$$

Quotient: _____

Find the quotient.

$$135 \div 5$$

Quotient: _____

SET 3 • 6.RP.A.2

BACKWARD 2

Fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $5 \div 8$ as a fraction.

Fraction: _____

True or false?

$\frac{2}{5}$ means the same as $2 \div 5$.

A. True

B. False

Four friends share 3 sandwiches equally.

How much sandwich does each friend get?

Each friend gets: _____

Which division expression is equal to $\frac{7}{8}$?

A. $8 \div 7$

B. $8 - 7$

C. $7 \div 8$

D. 7×8

A 5-yard ribbon is cut into 6 equal pieces.

How long is each piece?

Length: _____

SET 3 • 6.RP.A.2

BACKWARD 3

Meaning of "per" (the amount for each one) • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

12 apples are packed equally into 4 bags.

How many apples are in each bag?

Apples: _____

True or false?

"5 miles per hour" means 5 miles for each 1 hour.

A. True

B. False

Which phrase means the same as "\$2 per pound"?

- A.** \$2 for each pound
- B.** \$2 for all of the pounds
- C.** 2 pounds for each dollar
- D.** \$2 more than a pound

A farmer puts 24 eggs into cartons, 6 eggs per carton.
How many cartons does the farmer fill?

Cartons: _____

Jen reads 40 pages in 5 days. She reads the same number of pages each day.

How many pages does she read per day?

Pages: _____

SET 3 • 6.RP.A.2

FORWARD 1

Unit rates with fractions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.1

Kim walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

What is her speed in miles per hour?

Speed: _____

A painter paints $\frac{2}{3}$ of a wall in $\frac{1}{3}$ hour.

At this rate, how many walls can the painter paint per hour?

Walls per hour: _____

A snail crawls $\frac{5}{6}$ foot in $\frac{1}{3}$ minute.

How many feet does it crawl per minute?

Feet per minute: _____

A machine fills $\frac{3}{4}$ of a tank in $\frac{1}{2}$ hour.

How many tanks does it fill per hour?

A. $\frac{1}{4}$

B. $\frac{3}{8}$

C. $1\frac{1}{4}$

D. $1\frac{1}{2}$

Leo mows $\frac{7}{8}$ of an acre in $\frac{1}{2}$ hour.

How many acres does he mow per hour?

Acres per hour: _____

SET 3 • 6.RP.A.2

FORWARD 2

Rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What is the rate of change?

x	0	1	2	3
y	5	8	11	14

Rate of change: _____

A plumber charges a \$40 fee plus \$65 per hour.

Write a function for the cost C after h hours.

$C =$ _____

A line passes through $(2, 7)$ and $(6, 19)$.

What is the rate of change?

Rate of change: _____

A tank drains at a constant rate.

What is the rate of change in gallons per minute?

Minutes	0	5	10
Gallons	200	170	140

Rate of change: _____

A taxi ride costs \$3 plus \$2.50 per mile.

Which function gives the cost y for a ride of x miles?

A. $y = 3x + 2.5$

B. $y = 2.5x - 3$

C. $y = 2.5x + 3$

D. $y = 5.5x$

Find missing values in tables of equivalent ratios

MAIN Grade 6 main question

BACKWARD 1 Equivalent fractions

BACKWARD 2 Two related number patterns

BACKWARD 3 Multiples of a number

FORWARD 1 Constant of proportionality from a table

FORWARD 2 Rate of change and initial value from a table

6.RP.A.3.a

4.NF.A.1

5.OA.B.3

4.OA.B.4

7.RP.A.2.b

8.F.B.4

SET 4 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

The table shows equivalent ratios.

Find the missing value.

Cups of juice	2	4	6	10
Cups of water	3	6	9	?

Missing value: _____

Paint is mixed in the same ratio in every column.

Find the missing value.

Blue (cups)	3	6	?	15
Yellow (cups)	2	4	8	10

Missing value: _____

A swimmer keeps the same pace.

Find the missing value.

Laps	3	6	?	15
Minutes	8	16	24	40

Missing value: _____

The table shows equivalent ratios.

What is the missing value?

Tickets	4	8	12	?
Cost (\$)	10	20	30	45

A. 16

B. 18

C. 20

D. 35

A recipe uses 5 cups of oats for every 2 cups of raisins.
Find the missing values A and B.

Oats (cups)	5	10	A	25
Raisins (cups)	2	B	6	10

A = _____

B = _____

SET 4 • 6.RP.A.3.a

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{2}{3}$ is equivalent to $\frac{8}{12}$.

A. True

B. False

Find the missing number.

$$\frac{3}{4} = \frac{?}{20}$$

Missing number: _____

Which fraction is equivalent to $\frac{5}{6}$?

A. $\frac{5}{12}$

B. $\frac{10}{12}$

C. $\frac{6}{7}$

D. $\frac{10}{11}$

True or false?

$\frac{4}{5}$ is equivalent to $\frac{6}{7}$.

A. True

B. False

Find the missing number.

$$\frac{2}{5} = \frac{6}{?}$$

Missing number: _____

SET 4 • 6.RP.A.3.a

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A starts at 0 and adds 2.

Pattern B starts at 0 and adds 5.

What is the 5th number in Pattern B?

Number: _____

True or false?

Each number in Pattern B is 2 times the matching number in Pattern A.

Pattern A (add 4)	0	4	8	12
Pattern B (add 8)	0	8	16	24

A. True

B. False

Pattern A starts at 0 and adds 7.

Pattern B starts at 0 and adds 14.

What is the 4th number in Pattern A?

Number: _____

Pattern A: 0, 2, 4, 6, 8

Pattern B: 0, 6, 12, 18, 24

How does each number in Pattern B compare to the matching number in Pattern A?

A. 2 times as large

B. 3 times as large

C. 4 more

D. 6 times as large

Pattern A: 0, 1, 2, 3, ...

Pattern B: 0, 4, 8, 12, ...

What number in Pattern B matches 5 in Pattern A?

Number: _____

SET 4 • 6.RP.A.3.a

BACKWARD 3

Multiples of a number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four multiples of 3.

Multiples: _____

True or false?

40 is a multiple of 8.

A. True

B. False

Which number is a multiple of 5?

A. 32

B. 35

C. 48

D. 52

What is the 6th multiple of 4?

Multiple: _____

True or false?

27 is a multiple of 6.

A. True

B. False

SET 4 • 6.RP.A.3.a

FORWARD 1

Constant of proportionality from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality ($\frac{y}{x}$)?

x	2	5	8
y	7	17.5	28

k = _____

The table shows the cost of cheese.

What is the constant of proportionality in dollars per pound?

Pounds	3	6	9
Cost (\$)	7.50	15.00	22.50

$k =$ _____

The table shows a proportional relationship.
What is the constant of proportionality?

x	4	8	12
y	3	6	9

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. 3

D. 12

The table shows how far a toy car rolls.

What is the constant of proportionality in meters per second?

Seconds	2	3	5
Meters	1.2	1.8	3.0

$k =$ _____

True or false?

The constant of proportionality for the table is 6.

x	1	2	4
y	6	12	24

A. True

B. False

SET 4 • 6.RP.A.3.a

FORWARD 2

Rate of change and initial value from a table • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What are the rate of change and the initial value?

x	0	1	2	3
y	4	7	10	13

Rate of change: _____

Initial value: _____

The table shows the height of a plant.

What is the rate of change in cm per week?

Weeks	0	2	4
Height (cm)	6	9	12

Rate of change: _____

The table shows a linear function.

Which equation matches the table?

x	0	1	2	3
y	5	7	9	11

A. $y = x + 5$

B. $y = 5x + 2$

C. $y = 2x + 5$

D. $y = 2x$

The table shows a linear function.

Write an equation for the function.

x	1	2	3	4
y	9	13	17	21

$y =$ _____

True or false?

The initial value of the function in the table is 20.

x	0	1	2
y	20	17	14

A. True

B. False

Plot pairs of values from a ratio table on the coordinate plane

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Ordered pairs from two related patterns	5.OA.B.3
BACKWARD 2	Axes and ordered pairs	5.G.A.1
BACKWARD 3	Plotting points in the first quadrant	5.G.A.2
BACKWARD 4	Reading an axis marked by 2s, 3s, or 5s	3.MD.B.3
FORWARD 1	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a
FORWARD 2	Graph proportional relationships; interpret the slope	8.EE.B.5

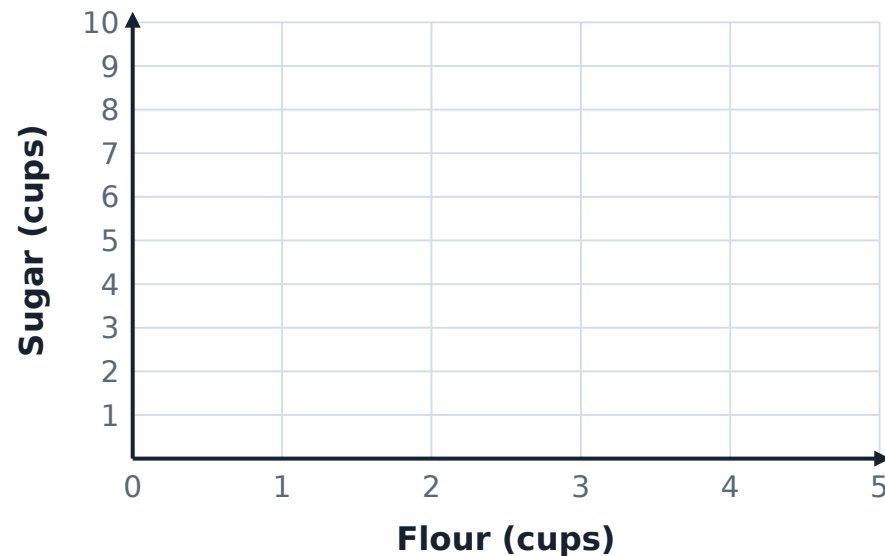
SET 5 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

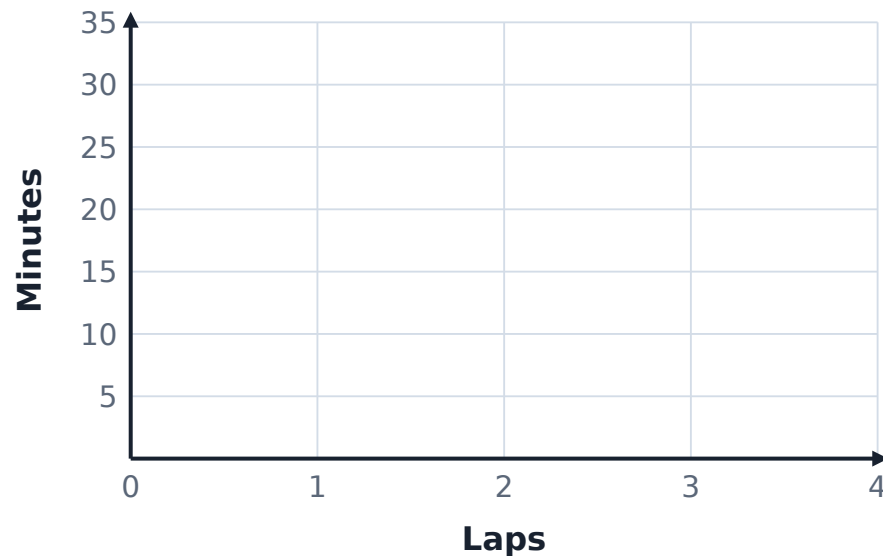
Plot the pairs of values from the table on the coordinate plane.

Flour (cups)	1	2	3	4
Sugar (cups)	2	4	6	8



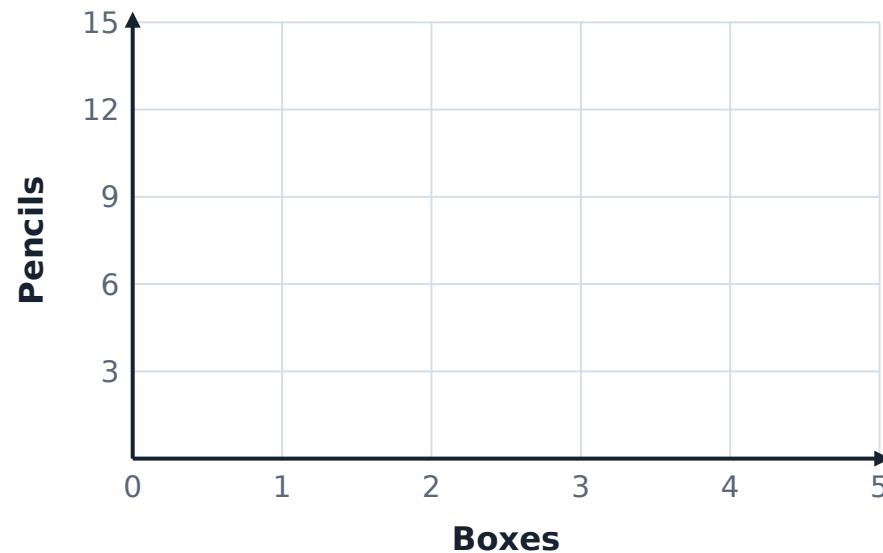
Plot the pairs of values from the table on the coordinate plane.

Laps	1	2	3
Minutes	10	20	30



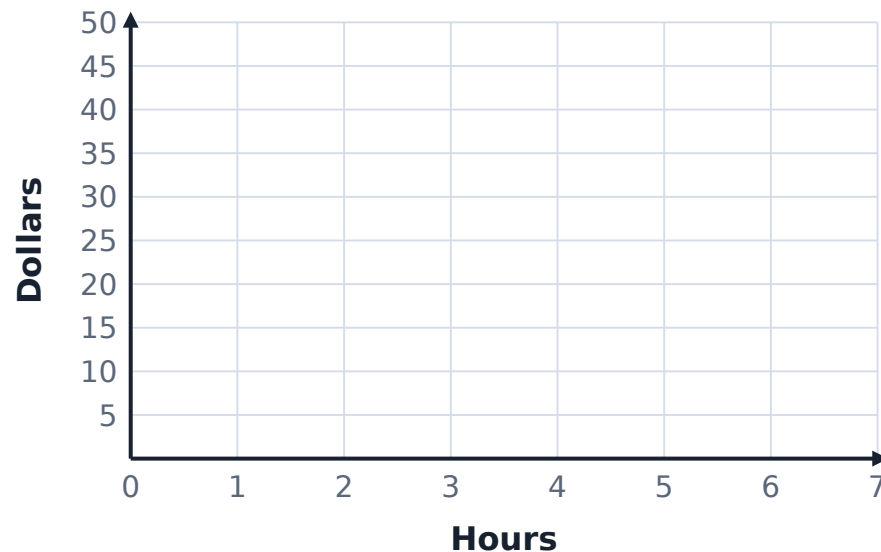
Plot the pairs of values from the table on the coordinate plane.

Boxes	1	2	3	4
Pencils	3	6	9	12



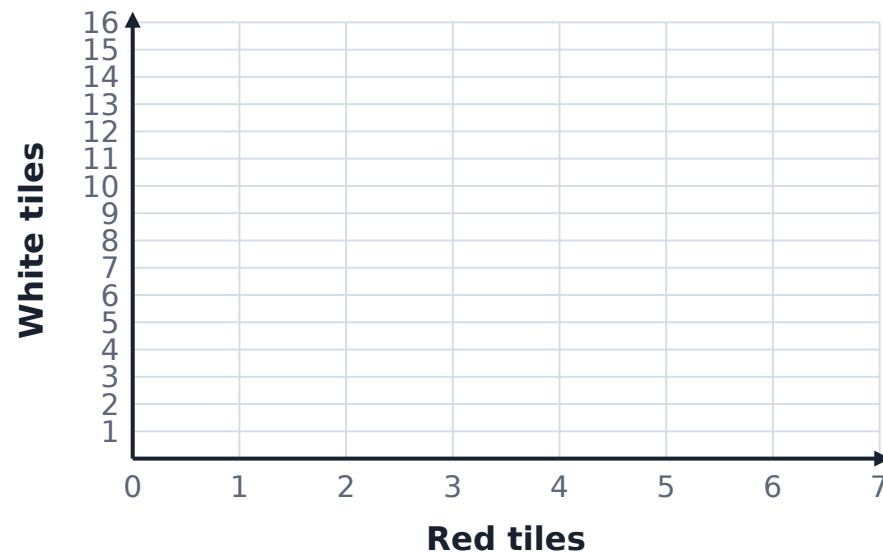
Plot the pairs of values from the table on the coordinate plane.

Hours	2	4	6
Dollars	15	30	45



Plot the pairs of values from the table on the coordinate plane.

Red tiles	2	4	6
White tiles	5	10	15



SET 5 • 6.RP.A.3.a

BACKWARD 1

Ordered pairs from two related patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 3, 6, 9

Write the ordered pair formed by the 3rd numbers of the patterns.

Ordered pair: _____

Which ordered pairs come from the table?

x	1	2	3
y	5	10	15

A. (1, 5), (2, 10), (3, 15)

B. (5, 1), (10, 2), (15, 3)

C. (1, 2), (5, 10), (15, 3)

D. (1, 10), (2, 5), (3, 15)

True or false?

The table gives the ordered pair (4, 12).

x	2	4	6
y	6	12	18

A. True

B. False

Write the ordered pair from the last column of the table.

x	1	2	3	4
y	7	14	21	28

Ordered pair: _____

True or false?

In an ordered pair, the first number comes from the top row of the table.

x	1	2
y	4	8

A. True

B. False

SET 5 • 6.RP.A.3.a

BACKWARD 2

Axes and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

To plot $(3, 7)$, start at $(0, 0)$, move 3 units right, then 7 units up.

A. True

B. False

In the ordered pair (6, 2), which number tells how far to move up?

A. 2

B. 4

C. 6

D. 8

What are the coordinates of the origin?

Origin: _____

True or false?

$(4, 9)$ and $(9, 4)$ name the same point.

A. True

B. False

Which axis is vertical?

A. The x-axis

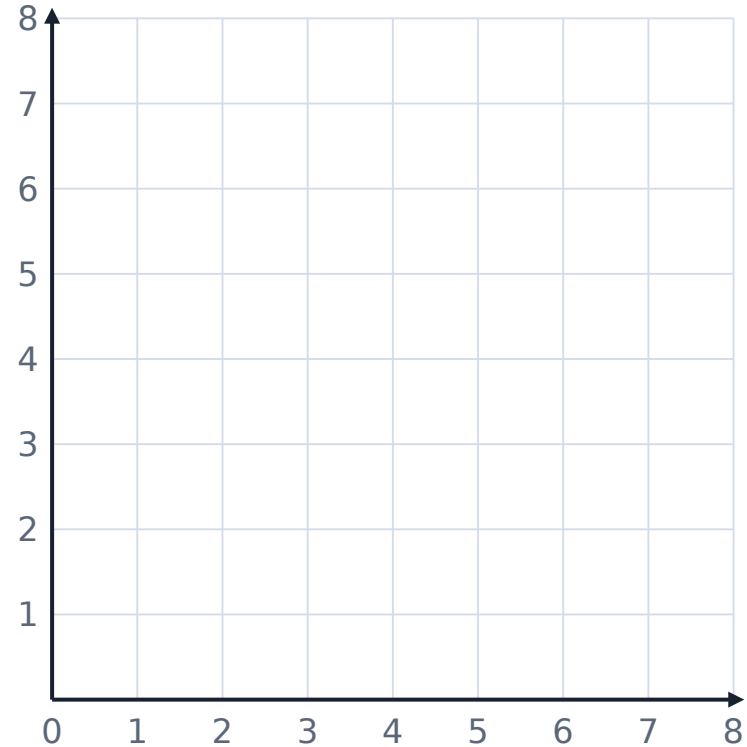
B. The y-axis

SET 5 • 6.RP.A.3.a

BACKWARD 3

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

Plot and label point A at (3, 5).

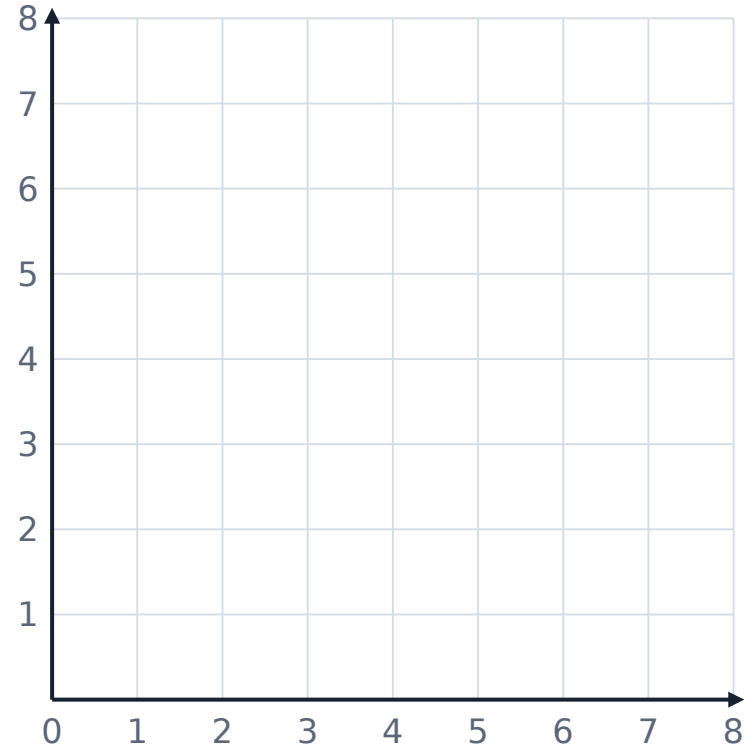


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q1/5

Plot and label point B at (6, 2).

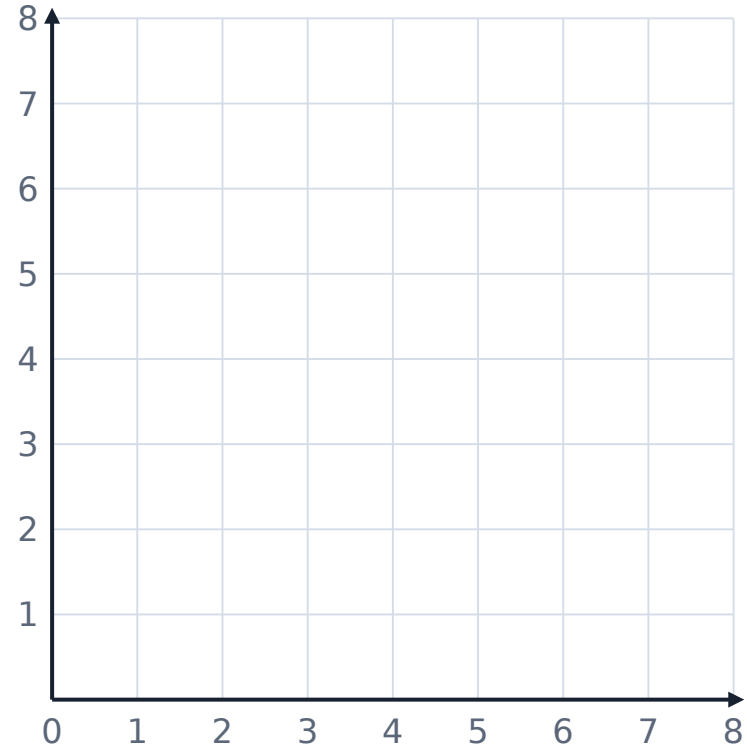


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

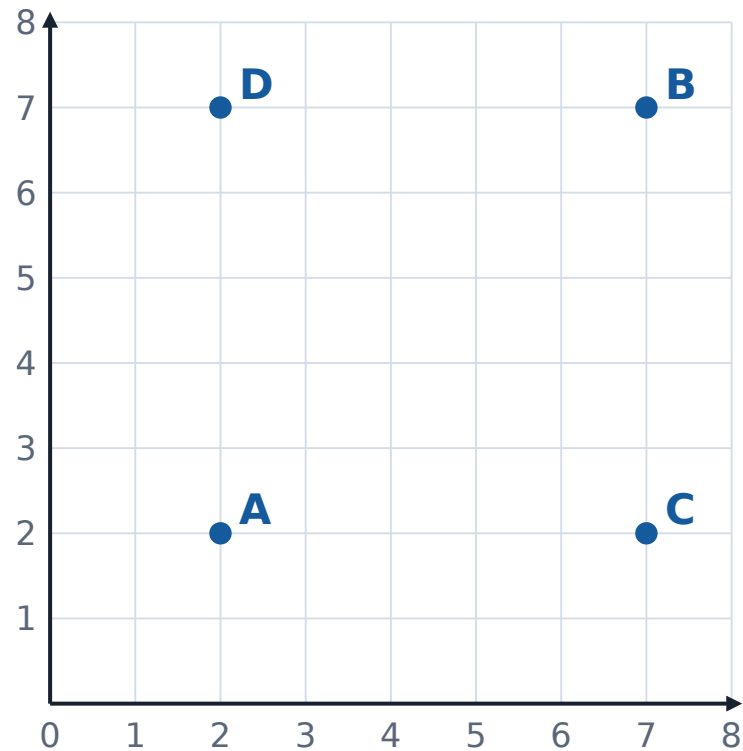
Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q2/5

Plot and label point C at (0, 4).

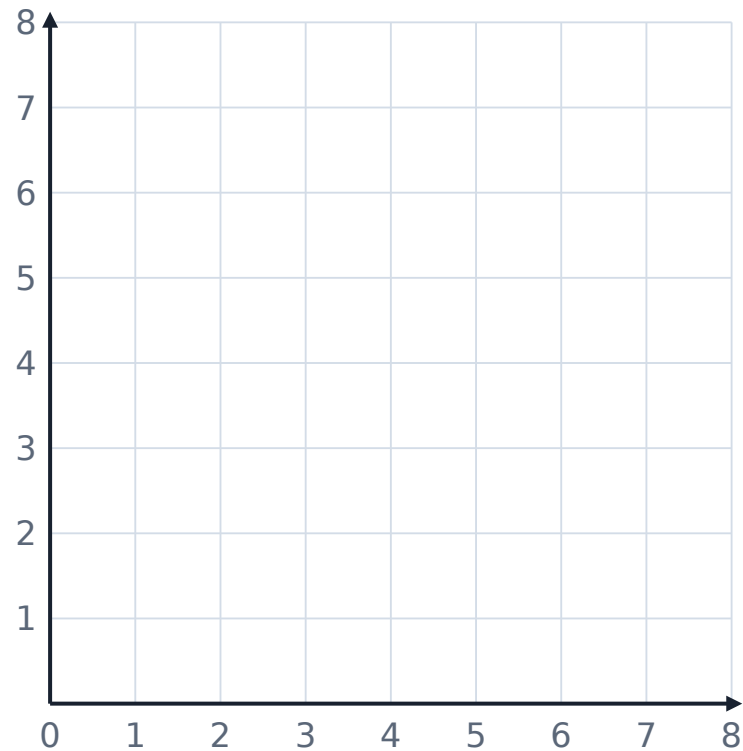


Which point is at (2, 7)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



Plot and label point D at (5, 5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 4

Reading an axis marked by 2s, 3s, or 5s • Grade 3 • Measurement & Data •
3.MD.B.3

On the y-axis, the grid lines are numbered 0, 5, 10, 15.
Where would 20 go?

- A.** At the 2nd grid line
- B.** At the 20th grid line
- C.** One grid line above 15
- D.** Halfway between 10 and 15

True or false?

On an axis numbered 0, 3, 6, 9, the value 12 is one grid line above 9.

A. True

B. False

On an axis numbered 0, 2, 4, 6, 8, where does the value 5 go?

Answer: _____

An axis is numbered 0, 5, 10, 15, 20.

How many units does each grid line stand for?

- A.** 1
- B.** 5
- C.** 10
- D.** 20

True or false?

On an axis numbered by 10s, the value 35 is halfway between 30 and 40.

- A.** True
- B.** False

SET 5 • 6.RP.A.3.a

FORWARD 1

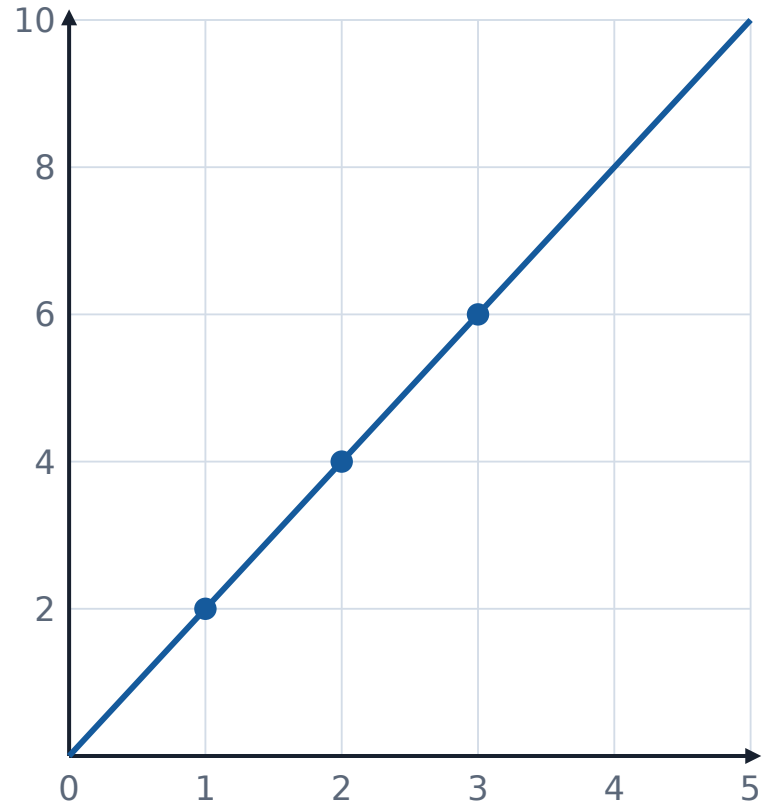
Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship.

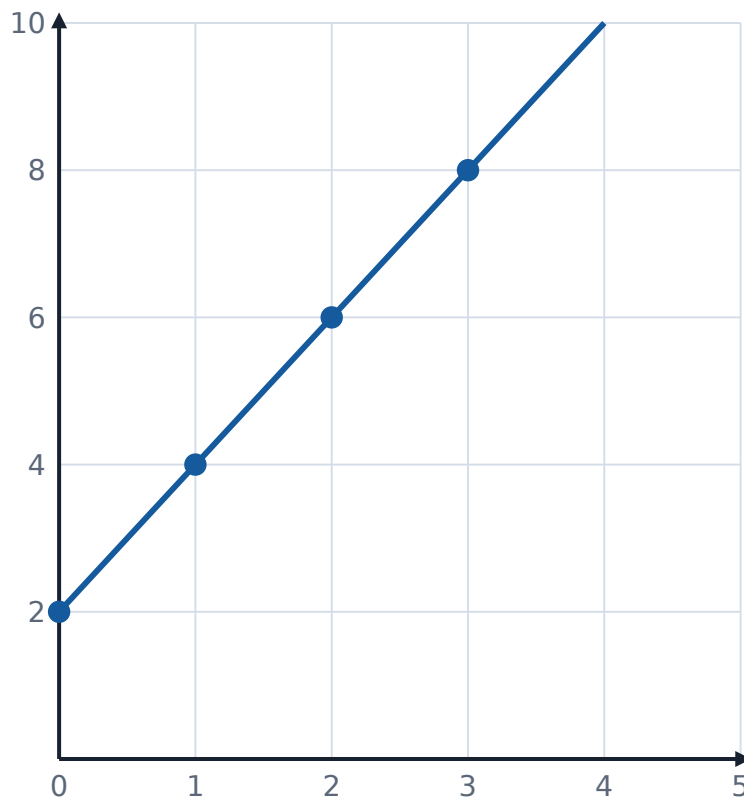
A. True

B. False



Which statement about the graph is true?

- A.** It is proportional because y increases by 2.
- B.** It is not proportional because the line does not pass through $(0, 0)$.
- C.** It is not proportional because the points are not on a line.
- D.** It is proportional because it is a straight line.

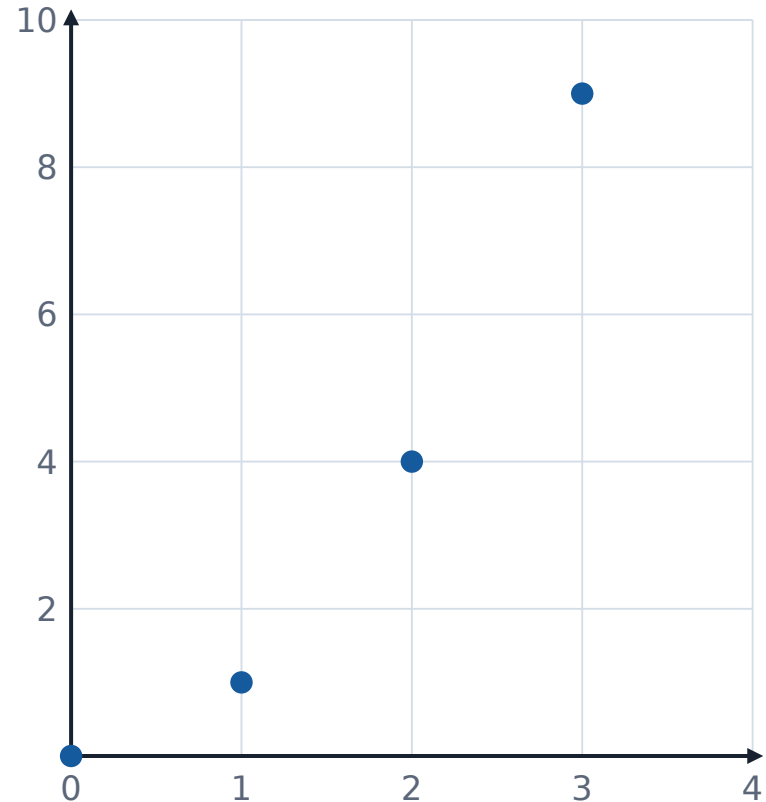


True or false?

The graph shows a proportional relationship.

A. True

B. False



The points $(2, 5)$, $(4, 10)$, and $(6, 15)$ are graphed.

Do they lie on a straight line through the origin? Write yes or no.

Answer: _____

Which set of points lies on a straight line through the origin?

A. $(0, 2), (1, 4), (2, 6)$

B. $(1, 3), (2, 4), (3, 5)$

C. $(2, 3), (4, 6), (6, 9)$

D. $(1, 1), (2, 4), (3, 9)$

SET 5 • 6.RP.A.3.a

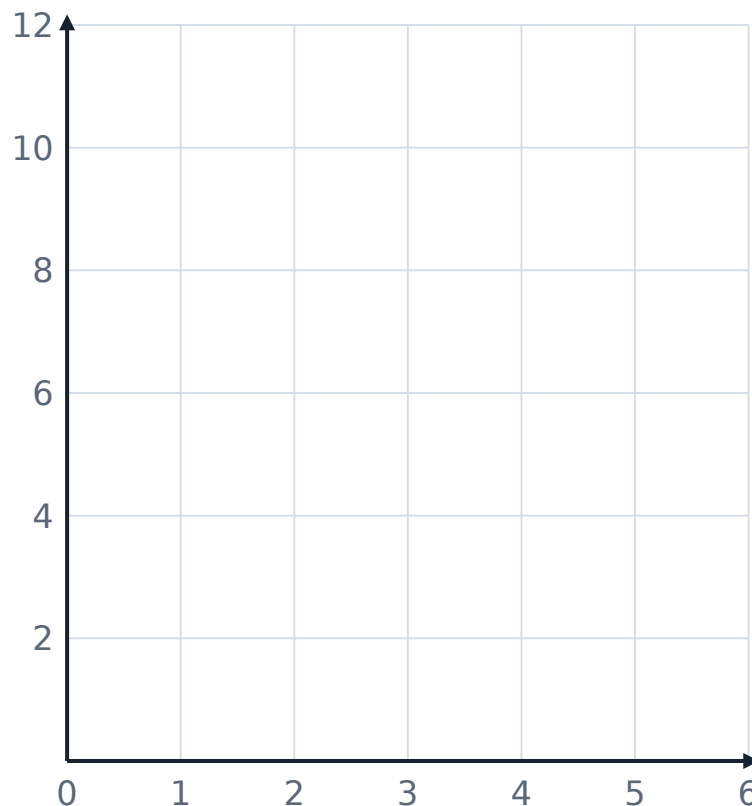
FORWARD 2

Graph proportional relationships; interpret the slope • Grade 8 • Expressions & Equations • 8.EE.B.5

Graph $y = 2x$ on the coordinate plane.

What is the slope of the line?

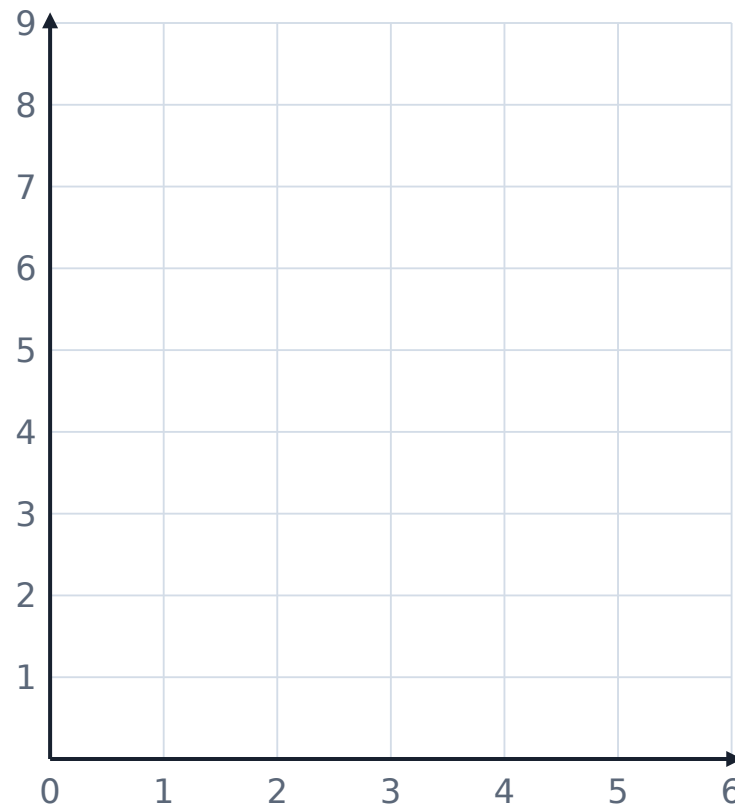
Slope: _____



Graph $y = 1.5x$ on the coordinate plane.

What is the slope of the line?

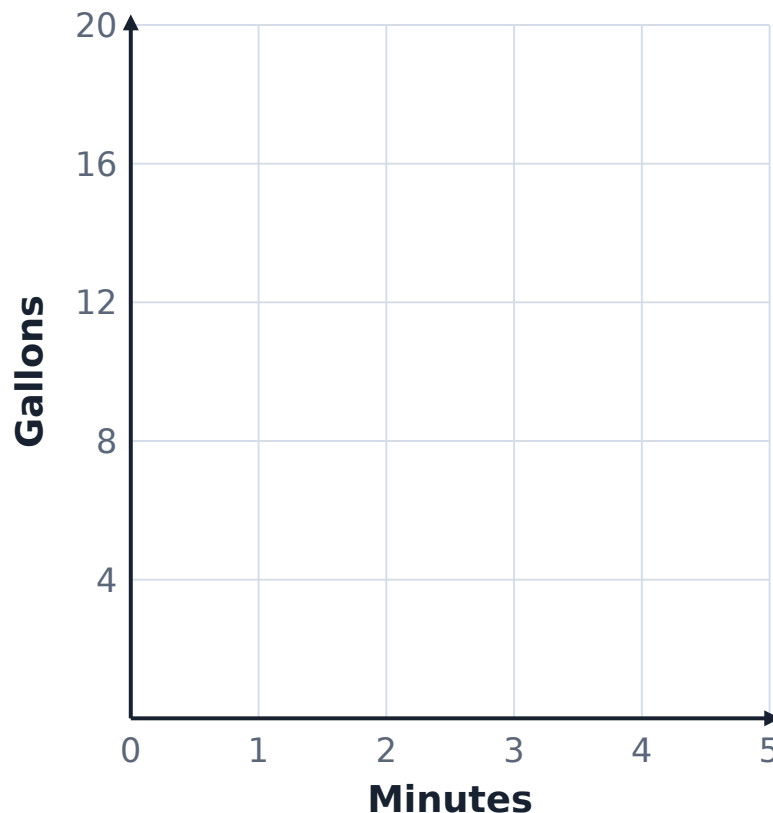
Slope: _____



Water flows at 4 gallons per minute. Graph the gallons y after x minutes.

What is the slope of the line?

Slope: _____



The graph of a proportional relationship passes through (5, 2).

What is the slope?

A. $\frac{2}{5}$

B. 2

C. $\frac{5}{2}$

D. 5

True or false?

The graph of $y = 3x$ passes through $(0, 0)$ and $(2, 6)$, and its slope is 3.

A. True

B. False

Use tables of equivalent ratios to compare ratios

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Comparing fractions	4.NF.A.2
BACKWARD 2	Common multiples	4.OA.B.4
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Compare unit rates involving fractions	7.RP.A.1
FORWARD 2	Compare proportional relationships represented in different ways	8.EE.B.5

SET 6 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Lemonade A uses 2 cups of mix for every 5 cups of water. Lemonade B uses 3 cups of mix for every 7 cups of water.

Use tables of equivalent ratios. Which lemonade has a stronger mix flavor?

A: Mix	2	4	6
A: Water	5	10	15

B: Mix	3	6	9
B: Water	7	14	21

A. Lemonade A

B. Lemonade B

C. They taste the same.

Class A has 4 boys for every 6 girls. Class B has 6 boys for every 9 girls.

Use tables of equivalent ratios. Which statement is true?

- A.** Class A has a greater ratio of boys to girls.
- B.** Class B has a greater ratio of boys to girls.
- C.** The ratios are equivalent.

Runner A runs 3 laps in 10 minutes. Runner B runs 4 laps in 12 minutes.

Use tables of equivalent ratios. Who runs faster?

- A.** Runner A
- B.** Runner B
- C.** They run at the same speed.

Paint A uses 1 part blue for every 3 parts white. Paint B uses 2 parts blue for every 5 parts white.

Use tables of equivalent ratios. Which paint is a darker blue?

- A.** Paint A
- B.** Paint B
- C.** They are the same color.

Store A sells 5 pens for \$3. Store B sells 8 pens for \$5.

Use tables of equivalent ratios. Which store gives more pens for the same amount of money?

- A.** Store A
- B.** Store B
- C.** They are the same.

SET 6 • 6.RP.A.3.a

BACKWARD 1

Comparing fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which fraction is greater?

A. $\frac{3}{7}$

B. $\frac{3}{5}$

True or false?

$$\frac{5}{8} > \frac{4}{8}$$

A. True

B. False

Which symbol makes the statement true?

$$\frac{2}{3} \text{ — } \frac{3}{4}$$

A. $<$

B. $>$

C. $=$

True or false?

$$\frac{1}{3} = \frac{2}{6}$$

A. True

B. False

Which fraction is greatest?

A. $\frac{1}{2}$

B. $\frac{7}{12}$

C. $\frac{2}{3}$

D. $\frac{5}{6}$

SET 6 • 6.RP.A.3.a

BACKWARD 2

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the multiples of 5 and the multiples of 7 up to 40.

Which number is on both lists?

Common multiple: _____

True or false?

30 is a multiple of both 5 and 6.

A. True

B. False

Which number is a multiple of both 3 and 4?

- A.** 9
- B.** 12
- C.** 14
- D.** 16

Name a number that is a multiple of both 6 and 9.

Number: _____

True or false?

20 is a multiple of both 4 and 8.

A. True

B. False

SET 6 • 6.RP.A.3.a

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{5} = \frac{?}{35}$$

Missing number: _____

True or false?

$$\frac{3}{7} = \frac{6}{14}$$

A. True

B. False

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{4}{9}$

B. $\frac{6}{9}$

C. $\frac{6}{8}$

D. $\frac{8}{10}$

Find the missing number.

$$\frac{1}{3} = \frac{5}{?}$$

Missing number: _____

True or false?

$$\frac{3}{10} = \frac{9}{20}$$

A. True

B. False

SET 6 • 6.RP.A.3.a

FORWARD 1

Compare unit rates involving fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ana walks $\frac{1}{2}$ mile in $\frac{1}{5}$ hour. Ben walks $\frac{3}{4}$ mile in $\frac{1}{3}$ hour.

Who walks faster?

- A.** Ana
- B.** Ben
- C.** They walk at the same speed.

Mix A uses $\frac{2}{3}$ cup of sugar per $\frac{1}{2}$ cup of water. Mix B uses $\frac{3}{4}$ cup of sugar per $\frac{2}{3}$ cup of water.

Which mix is sweeter?

- A.** Mix A
- B.** Mix B
- C.** They are the same.

Pump A moves $\frac{3}{8}$ gallon in $\frac{1}{4}$ minute. Pump B moves $\frac{1}{2}$ gallon in $\frac{1}{2}$ minute.

Which pump is faster, and what is its rate in gallons per minute?

Answer: _____

Snail A crawls $\frac{1}{6}$ foot in $\frac{1}{2}$ minute. Snail B crawls $\frac{1}{4}$ foot in $\frac{2}{3}$ minute.

Which snail is faster?

- A.** Snail A
- B.** Snail B
- C.** They are the same.

True or false?

$\frac{1}{2}$ mile in $\frac{1}{4}$ hour is a faster speed than $\frac{3}{4}$ mile in $\frac{1}{2}$ hour.

A. True

B. False

SET 6 • 6.RP.A.3.a

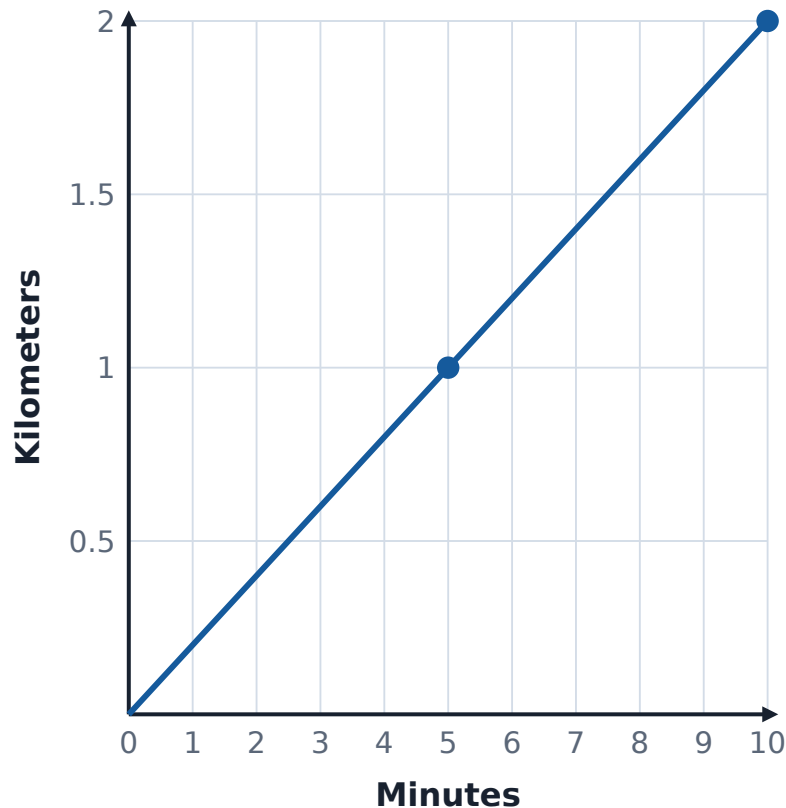
FORWARD 2

Compare proportional relationships represented in different ways • Grade 8 •
Expressions & Equations • 8.EE.B.5

Runner A's distance is shown in the graph. Runner B runs according to $d = 0.15t$, where d is kilometers and t is minutes.

Which runner is faster?

- A.** Runner A
- B.** Runner B
- C.** They are the same.



Machine A: $y = 45x$ bottles after x minutes. Machine B: the table shows 3 minutes $\rightarrow$ 120 bottles, 6 minutes $\rightarrow$ 240 bottles.

Which machine is faster?

- A.** Machine A
- B.** Machine B
- C.** They are the same.

Recipe A uses flour and sugar in the ratio $y = 1.5x$.

Recipe B's graph passes through $(0, 0)$ and $(4, 7)$.

Which recipe uses more flour y per cup of sugar x ?

Recipe: _____

Two proportional relationships: P is $y = 6x$. Q is a line through $(0, 0)$ and $(2, 13)$.

Which has the greater slope?

- A.** P
- B.** Q
- C.** They are the same.

Plan A: 20 minutes of calls cost \$3.00. Plan B:
 $c = 0.12m$, where m is minutes.

Which plan costs less per minute?

Plan: _____

Solve unit pricing problems

- MAIN** Grade 6 main question
- BACKWARD 1** Dividing money amounts
- BACKWARD 2** Multiplying money amounts
- BACKWARD 3** Finding the cost of one when each costs the same
- FORWARD 1** Equations for proportional relationships
- FORWARD 2** Build a linear function for a pricing situation

6.RP.A.3.b

5.NBT.B.7

5.NBT.B.7

3.OA.A.2

7.RP.A.2.c

8.F.B.4

SET 7 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

4 notebooks cost \$10.

At this rate, how much do 7 notebooks cost?

Cost: _____

3 pounds of apples cost \$5.40.

At this rate, how much do 5 pounds cost?

Cost: _____

6 cans of soup cost \$7.50.

At this rate, how much do 10 cans cost?

Cost: _____

5 movie tickets cost \$42.50.

At this rate, how much do 3 tickets cost?

- A.** \$8.50
- B.** \$25.50
- C.** \$28.33
- D.** \$127.50

8 yards of fabric cost \$36.

At this rate, how much do 3 yards cost?

Cost: _____

SET 7 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Find the quotient.

$$\$10.00 \div 4$$

Quotient: _____

Find the quotient.

$$\$5.40 \div 3$$

Quotient: _____

$$\$7.50 \div 6 = ?$$

- A.** \$0.80
- B.** \$1.20
- C.** \$1.25
- D.** \$12.50

Find the quotient.

$$\$36 \div 8$$

Quotient: _____

True or false?

$$\$42.50 \div 5 = \$8.50$$

A. True

B. False

SET 7 • 6.RP.A.3.b

BACKWARD 2

Multiplying money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Multiply.

$$\$2.50 \times 7$$

Product: _____

Multiply.

$$\$1.80 \times 5$$

Product: _____

True or false?

$$\$1.25 \times 10 = \$12.50$$

A. True

B. False

$$\$4.50 \times 3 = ?$$

- A.** \$1.50
- B.** \$7.50
- C.** \$12.50
- D.** \$13.50

Multiply.

$$\$8.50 \times 3$$

Product: _____

SET 7 • 6.RP.A.3.b

BACKWARD 3

Finding the cost of one when each costs the same • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

4 toys cost \$12 in all. Each toy costs the same.

How much does 1 toy cost?

Cost: _____

5 pencils cost 30 cents in all. Each pencil costs the same.

How much does 1 pencil cost?

Cost: _____

True or false?

6 stickers cost 18 cents in all. Each costs the same, so 1 sticker costs 3 cents.

A. True

B. False

8 cards cost \$16. Each card costs the same.

How much is 1 card?

- A.** \$2
- B.** \$8
- C.** \$24
- D.** \$128

3 books cost \$27 in all. Each book costs the same.

How much does 1 book cost?

Cost: _____

SET 7 • 6.RP.A.3.b

FORWARD 1

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Apples cost \$2.50 per pound.

Write an equation for the cost c of p pounds of apples.

Equation: _____

The table shows a proportional relationship.

Write an equation for the earnings e after h hours.

Hours (h)	2	3	5
Earnings (e)	36	54	90

Equation: _____

Pens cost \$0.75 each.

Which equation gives the cost c of n pens?

A. $c = \frac{n}{0.75}$

B. $c = n + 0.75$

C. $c = 0.75n$

D. $n = 0.75c$

5 pounds of rice cost \$8.

Write an equation for the cost c of p pounds of rice.

Equation: _____

True or false?

Grapes cost \$3.20 per pound. The equation $c = 3.2p$ gives the cost c of p pounds.

- A. True
- B. False

SET 7 • 6.RP.A.3.b

FORWARD 2

Build a linear function for a pricing situation • Grade 8 • Functions • 8.F.B.4

A store charges \$2.50 per pound of cherries plus a \$4 box fee.

Write a function for the total cost C of p pounds.

$C =$ _____

Tickets cost \$12 each plus a \$5 service fee per order.

Which function gives the cost C of t tickets?

A. $C = 5t + 12$

B. $C = 12t - 5$

C. $C = 12t + 5$

D. $C = 17t$

A delivery costs \$6 plus \$1.75 per mile.

What are the rate of change and the initial value?

Rate of change: _____

Initial value: _____

The table shows the cost of a gym membership.
Write a function for the cost C after m months.

Months (m)	0	1	2	3
Cost (C)	25	55	85	115

$C =$ _____

True or false?

A function for "\$3 per pound plus a \$2 bag fee" is $C = 3p + 2$.

- A. True
- B. False

Solve constant speed problems

MAIN	Grade 6 main question
BACKWARD 1	Whole-number division
BACKWARD 2	Multiplying by a one-digit number
BACKWARD 3	Hours and minutes
FORWARD 1	Speed as a unit rate with fractions
FORWARD 2	Compare speeds from graphs and equations

6.RP.A.3.b

4.NBT.B.6

4.NBT.B.5

4.MD.A.1

7.RP.A.1

8.EE.B.5

SET 8 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

A runner runs 12 kilometers in 60 minutes at a constant speed.

How far does the runner go in 25 minutes?

Distance: _____

A train travels 180 miles in 3 hours at a constant speed.
How far does it travel in 5 hours?

Distance: _____

A cyclist rides 45 miles in 3 hours at a constant speed.
How long does it take to ride 75 miles?

Time: _____

A snail moves 18 cm in 6 minutes at a constant speed.
How far does it move in 20 minutes?

- A.** 3 cm
- B.** 54 cm
- C.** 60 cm
- D.** 120 cm

A car travels 220 miles in 4 hours at a constant speed.
How long does it take to travel 385 miles?

Time: _____

SET 8 • 6.RP.A.3.b

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$180 \div 3$$

Quotient: _____

Divide.

$$220 \div 4$$

Quotient: _____

True or false?

$$385 \div 55 = 7$$

A. True

B. False

$$45 \div 3 = ?$$

- A.** 12
- B.** 14
- C.** 15
- D.** 135

Divide.

$$375 \div 5$$

Quotient: _____

SET 8 • 6.RP.A.3.b

BACKWARD 2

Multiplying by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$60 \times 5$$

Product: _____

Multiply.

$$15 \times 7$$

Product: _____

True or false?

$$55 \times 4 = 220$$

A. True

B. False

$$48 \times 6 = ?$$

- A.** 54
- B.** 248
- C.** 288
- D.** 2,448

Multiply.

$$125 \times 3$$

Product: _____

SET 8 • 6.RP.A.3.b

BACKWARD 3

Hours and minutes • Grade 4 • Measurement & Data • 4.MD.A.1

How many minutes are in 3 hours?

Minutes: _____

True or false?

There are 60 minutes in 1 hour.

A. True

B. False

Complete.

5 hours = ____ minutes

Minutes: _____

Which is longer?

- A.** 1 hour
- B.** 90 minutes
- C.** They are the same.

How many minutes are in $\frac{1}{2}$ hour?

Minutes: _____

SET 8 • 6.RP.A.3.b

FORWARD 1

Speed as a unit rate with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A hiker walks $\frac{3}{4}$ mile in $\frac{3}{8}$ hour.

What is the hiker's speed in miles per hour?

Speed: _____

A turtle crawls $\frac{1}{5}$ mile in $\frac{1}{2}$ hour.

What is its speed in miles per hour?

Speed: _____

A boat travels $2\frac{1}{2}$ miles in $\frac{1}{4}$ hour.

What is its speed in miles per hour?

A. $\frac{5}{8}$

B. $2\frac{3}{4}$

C. 5

D. 10

A skater glides $\frac{2}{3}$ kilometer in $\frac{1}{6}$ hour.

What is the skater's speed in kilometers per hour?

Speed: _____

True or false?

Going $\frac{1}{2}$ mile in $\frac{1}{10}$ hour is a speed of 5 miles per hour.

A. True

B. False

SET 8 • 6.RP.A.3.b

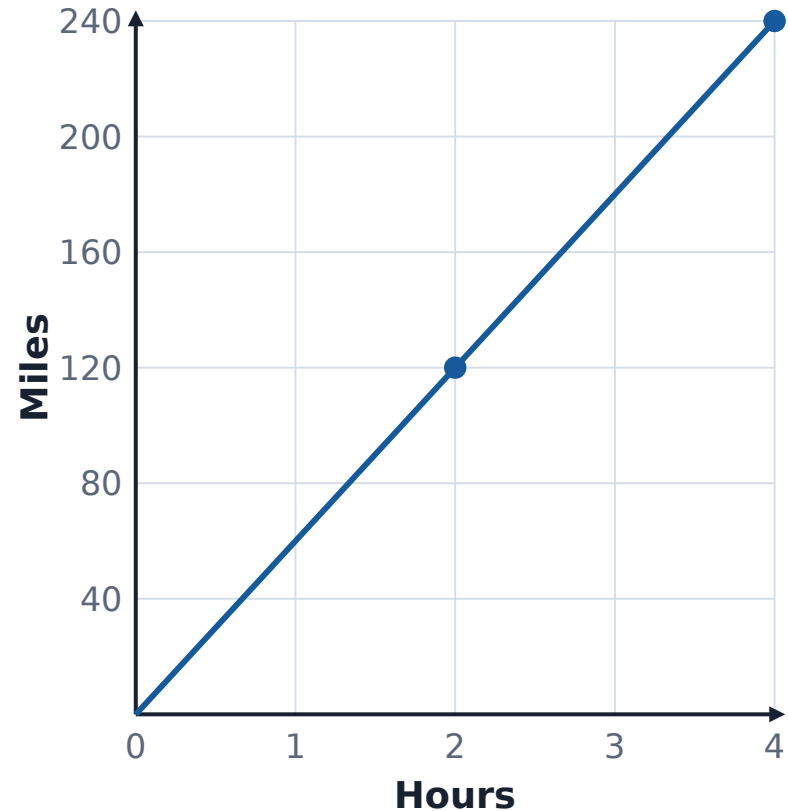
FORWARD 2

Compare speeds from graphs and equations • Grade 8 • Expressions & Equations • 8.EE.B.5

Car A's distance is shown in the graph. Car B travels according to $d = 55t$.

Which car is faster?

- A.** Car A
- B.** Car B
- C.** They are the same.



Train A travels according to $d = 80t$. Train B's graph is a line through $(0, 0)$ and $(3, 210)$.

Which train is faster, and by how many miles per hour?

Answer: _____

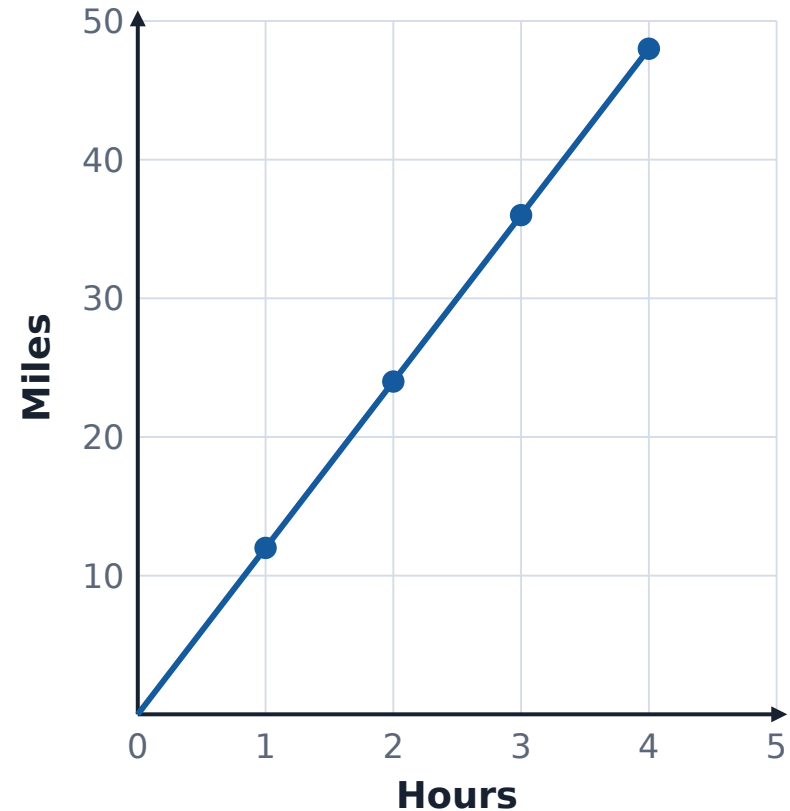
Walker A: $d = 3.5t$. Walker B: the graph passes through $(0, 0)$ and $(2, 8)$.

Who is faster?

- A.** Walker A
- B.** Walker B
- C.** They are the same.

What is the speed shown by the graph in miles per hour?

Speed: _____



True or false?

A line through $(0, 0)$ and $(5, 300)$ shows a greater speed than $d = 65t$.

A. True

B. False

Compare unit rates to find the better buy or the faster rate

MAIN	Grade 6 main question	6.RP.A.3.b
BACKWARD 1	Dividing money amounts	5.NBT.B.7
BACKWARD 2	Comparing decimals	5.NBT.A.3.b
BACKWARD 3	Finding the amount for one	3.OA.A.2
FORWARD 1	Compare deals using percents	7.RP.A.3
FORWARD 2	Compare two functions represented in different ways	8.F.A.2

SET 9 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

Which is the better buy?

A: 6 bottles of water for \$4.50

B: 10 bottles of water for \$7.00

A. A

B. B

C. Both are the same.

Which is the better buy?

A: a 12-ounce box of cereal for \$3.60

B: an 18-ounce box of cereal for \$4.95

A. A

B. B

C. Both are the same.

Which store has the lower price per pound?

A: 4 pounds of grapes for \$7.00

B: 3 pounds of grapes for \$5.40

A. A

B. B

C. Both are the same.

Which is the better buy?

A: 8 batteries for \$10.00

B: 12 batteries for \$14.40

A. A

B. B

C. Both are the same.

Which printer is faster?

A: 90 pages in 6 minutes

B: 140 pages in 10 minutes

A. A

B. B

C. Both are the same.

SET 9 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Find the quotient.

$$\$4.50 \div 6$$

Quotient: _____

Find the quotient.

$$\$3.60 \div 12$$

Quotient: _____

$$\$7.00 \div 10 = ?$$

- A.** \$0.07
- B.** \$0.70
- C.** \$7.00
- D.** \$70.00

Find the quotient.

$$\$14.40 \div 12$$

Quotient: _____

True or false?

$$\$5.40 \div 3 = \$1.80$$

A. True

B. False

SET 9 • 6.RP.A.3.b

BACKWARD 2

Comparing decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Which amount is less?

A. \$0.70

B. \$0.75

True or false?

$$0.8 > 0.75$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

0.275 0.3

Symbol: _____

Which number is greatest?

A. 1.075

B. 1.705

C. 1.75

D. 1.8

True or false?

$$1.25 < 1.2$$

A. True

B. False

SET 9 • 6.RP.A.3.b

BACKWARD 3

Finding the amount for one • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

3 bags hold 21 oranges in all, the same number in each.
How many oranges are in 1 bag?

Oranges: _____

A worker packs 32 boxes in 4 hours, the same number each hour.

How many boxes per hour?

Boxes: _____

True or false?

9 muffins cost \$18 in all, the same price each, so 1 muffin costs \$2.

A. True

B. False

40 pages are read in 8 days, the same number each day.

How many pages per day?

A. 5

B. 32

C. 48

D. 320

6 packs hold 54 cards in all, the same number in each.

How many cards are in 1 pack?

Cards: _____

SET 9 • 6.RP.A.3.b

FORWARD 1

Compare deals using percents • Grade 7 • Ratios & Proportional Relationships
• 7.RP.A.3

A \$60 jacket is on sale. Which deal saves more money?

- A.** \$10 off
- B.** 20% off
- C.** They save the same.

Which costs less after the discount?

A: a \$45 game at 30% off

B: a \$40 game at 20% off

A. A

B. B

C. They cost the same.

Store A sells a \$80 bike at 15% off. Store B sells the same bike for \$70.

Which store has the lower price, and by how much?

Answer: _____

Which is the better deal on a \$120 phone?

- A.** 25% off
- B.** \$25 off
- C.** They are the same.

True or false?

A \$50 item at 10% off costs the same as a \$55 item at 20% off.

- A. True
- B. False

SET 9 • 6.RP.A.3.b

FORWARD 2

Compare two functions represented in different ways • Grade 8 • Functions •
8.F.A.2

Plan A costs $y = 12x$ dollars for x months. Plan B is shown in the table.

Which plan has the greater rate of change?

Months (x)	2	4	6
Cost (y)	26	52	78

A. Plan A

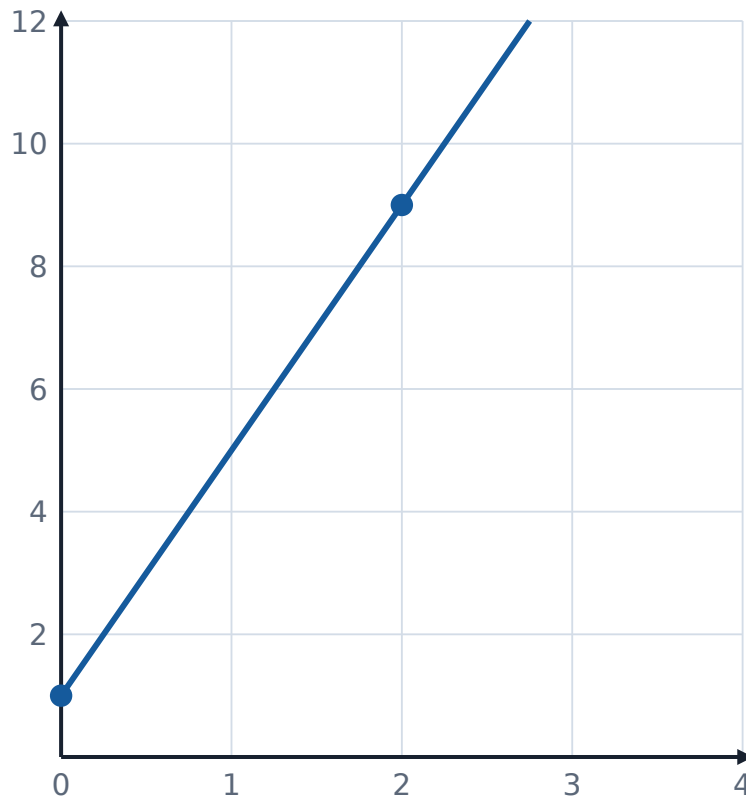
B. Plan B

C. They are the same.

Function A is $y = 3x + 4$. Function B is shown in the graph.

Which function has the greater rate of change?

- A.** Function A
- B.** Function B
- C.** They are the same.



Function A is shown in the table. Function B is $y = -3x + 2$.

Which function has the greater y-intercept?

x	0	1	2
y	5	3	1

- A.** Function A **B.** Function B **C.** They are the same.

Phone plan A costs \$30 plus \$0.05 per text. Phone plan B is a line through (0, 20) and (100, 35).

Which plan has the greater cost per text?

Plan: _____

True or false?

$y = 4x + 1$ has a greater rate of change than the function in the table.

x	0	1	2
y	0	5	10

A. True

B. False

Find a percent of a quantity as a rate per 100

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Hundredths as decimals	4.NF.C.6
BACKWARD 2	Tenths as hundredths	4.NF.C.5
BACKWARD 3	Fraction of a whole number	5.NF.B.4.a
BACKWARD 4	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Multi-step percent problems	7.RP.A.3
FORWARD 2	Build a linear function that uses a percent rate	8.F.B.4

SET 10 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 35% of 60?

Answer: _____

A jacket costs \$60. The sales tax is 8%.

How much is the tax?

Tax: _____

A class has 25 students. 40% of the students walk to school.

How many students walk to school?

Students: _____

A \$80 pair of shoes is 15% off.

How many dollars are taken off the price?

- A.** \$1.20
- B.** \$12
- C.** \$15
- D.** \$68

A school has 450 students. 12% of them play in the band.

How many students play in the band?

Students: _____

SET 10 • 6.RP.A.3.c

BACKWARD 1

Hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{45}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.07 = \frac{7}{100}$$

A. True

B. False

Which decimal is equal to $\frac{30}{100}$?

- A.** 0.03
- B.** 0.30
- C.** 3.0
- D.** 30.0

Write 0.58 as a fraction with a denominator of 100.

Fraction: _____

True or false?

$$\frac{8}{100} = 0.8$$

A. True

B. False

SET 10 • 6.RP.A.3.c

BACKWARD 2

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Find the missing number.

$$\frac{3}{10} = \frac{?}{100}$$

Missing number: _____

True or false?

$$\frac{6}{10} + \frac{3}{100} = \frac{63}{100}$$

A. True

B. False

Which fraction is equal to $\frac{7}{10}$?

A. $\frac{7}{100}$

B. $\frac{10}{70}$

C. $\frac{17}{100}$

D. $\frac{70}{100}$

Add.

$$\frac{2}{10} + \frac{45}{100}$$

Sum: _____

True or false?

$$\frac{9}{10} = \frac{90}{100}$$

A. True

B. False

SET 10 • 6.RP.A.3.c

BACKWARD 3

Fraction of a whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{35}{100} \times 60$$

Product: _____

Find $\frac{1}{4}$ of 60.

Answer: _____

True or false?

$$\frac{2}{5} \times 25 = 10$$

A. True

B. False

$$\frac{3}{4} \times 20 = ?$$

- A.** 5
- B.** 12
- C.** 15
- D.** 16

Find $\frac{7}{10}$ of 50.

Answer: _____

SET 10 • 6.RP.A.3.c

BACKWARD 4

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$0.35 \times 60$$

Product: _____

Multiply.

$$0.08 \times 60$$

Product: _____

True or false?

$$0.12 \times 450 = 54$$

A. True

B. False

$$0.4 \times 25 = ?$$

A. 0.1

B. 1

C. 10

D. 100

Multiply.

$$0.15 \times 80$$

Product: _____

SET 10 • 6.RP.A.3.c

FORWARD 1

Multi-step percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A meal costs \$40. You leave a 15% tip.

What is the total cost of the meal with the tip?

Total: _____

A pair of shoes costs \$80. They are on sale for 25% off.
What is the sale price?

Sale price: _____

A town's population grew from 2,000 to 2,300.
What is the percent increase?

Percent increase: _____

A \$50 game is marked up 20%. Then a 5% sales tax is added to the new price.

What is the final cost?

A. \$60.00

B. \$62.00

C. \$62.50

D. \$63.00

A salesperson earns a 6% commission. She sells \$3,500 in products.

How much commission does she earn?

Commission: _____

SET 10 • 6.RP.A.3.c

FORWARD 2

Build a linear function that uses a percent rate • Grade 8 • Functions • 8.F.B.4

A salesperson earns \$300 per week plus 5% of her sales S .

Write a function for her weekly earnings E .

$E =$ _____

A server earns \$40 per shift plus 18% of the food sales f .

Which function gives the earnings E ?

A. $E = 0.18f - 40$

B. $E = 0.18f + 40$

C. $E = 18f + 40$

D. $E = 40f + 0.18$

An agent earns \$500 per month plus 3% of her sales r .

What are the rate of change and the initial value of this function?

Rate of change: _____

Initial value: _____

A delivery driver earns \$60 per day plus 10% of the orders d delivered in dollars.

Write a function for the daily pay P .

$P =$ _____

True or false?

For $E = 0.04s + 250$, earnings increase by \$0.04 for each \$1 of sales s .

- A. True
- B. False

Find the whole, given a part and the percent

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Benchmark percents as fractions	4.NF.A.1
BACKWARD 2	Finding an unknown factor	3.OA.A.4
BACKWARD 3	Dividing a whole number by a unit fraction	5.NF.B.7.b
FORWARD 1	Percent problems: find the original amount	7.RP.A.3
FORWARD 2	Linear equations with decimal coefficients	8.EE.C.7.b

SET 11 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

12 is 25% of what number?

Answer: _____

A student answered 18 questions correctly. This was 90% of the questions on the test.

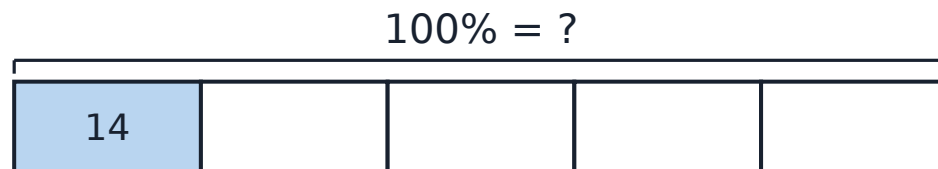
How many questions were on the test?

Questions: _____

Maria has saved \$45. This is 30% of the cost of a bike.
How much does the bike cost?

Cost: _____

The tape diagram shows that 20% of a number is 14.
What is the number?



A. 28

B. 34

C. 70

D. 280

In a survey, 36 students chose pizza. This was 40% of the students surveyed.

How many students were surveyed?

Students: _____

SET 11 • 6.RP.A.3.c

BACKWARD 1

Benchmark percents as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.A.1

Find the missing number.

$$\frac{25}{100} = \frac{1}{?}$$

Missing number: _____

Find the missing number.

$$\frac{50}{100} = \frac{?}{2}$$

Missing number: _____

True or false?

$$\frac{20}{100} = \frac{1}{5}$$

A. True

B. False

Which fraction is equivalent to $\frac{75}{100}$?

A. $\frac{1}{75}$

B. $\frac{7}{10}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

Find the missing number.

$$\frac{10}{100} = \frac{1}{?}$$

Missing number: _____

SET 11 • 6.RP.A.3.c

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$\square \times 4 = 36$$

$$\square = \underline{\hspace{10cm}}$$

What number goes in the box?

$$5 \times \square = 45$$

$$\square = \underline{\hspace{10cm}}$$

True or false?

If $6 \times \square = 42$, then $\square = 7$.

A. True

B. False

$$\square \times 8 = 56$$

What number goes in the box?

- A.** 6
- B.** 7
- C.** 8
- D.** 48

What number goes in the box?

$$9 \times \square = 63$$

$$\square = \underline{\hspace{10cm}}$$

SET 11 • 6.RP.A.3.c

BACKWARD 3

Dividing a whole number by a unit fraction • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b

Divide.

$$12 \div \frac{1}{4}$$

Quotient: _____

Divide.

$$5 \div \frac{1}{3}$$

Quotient: _____

True or false?

$$6 \div \frac{1}{2} = 3$$

A. True

B. False

How many fifths are in 3?

A. $\frac{3}{5}$

B. 5

C. 8

D. 15

Divide.

$$8 \div \frac{1}{5}$$

Quotient: _____

SET 11 • 6.RP.A.3.c

FORWARD 1

Percent problems: find the original amount • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

After a 20% discount, a jacket costs \$48.

What was the original price?

Original price: _____

A price increased by 10%. The new price is \$55.
What was the original price?

Original price: _____

The sales tax on an item was \$3.60. The tax rate is 6%.
What was the price before tax?

Price: _____

A town's population grew by 25% to 1,500 people.
What was the population before the increase?

- A.** 1,125
- B.** 1,200
- C.** 1,475
- D.** 1,875

A store marks up the price of a bike by 40%. The selling price is \$210.

How much did the store pay for the bike?

Cost: _____

SET 11 • 6.RP.A.3.c

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$x - 0.2x = 48$$

x = _____

Solve for x.

$$0.4x + 12 = 0.1x + 30$$

x = _____

Solve for x.

$$0.25(x + 40) = 30$$

x = _____

Which value of n solves the equation?

$$n + 0.15n = 92$$

- A.** 78.2
- B.** 80
- C.** 92.15
- D.** 105.8

Solve for y .

$$2.5y - 7 = 1.5y + 3$$

$y =$ _____

Convert measurement units using ratio reasoning

MAIN	Grade 6 main question	6.RP.A.3.d
BACKWARD 1	Measurement equivalents	4.MD.A.1
BACKWARD 2	Multiplying and dividing by powers of 10	5.NBT.A.2
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Rates with fractions and unit conversions	7.RP.A.1
FORWARD 2	Convert units for very large and very small quantities	8.EE.A.4

SET 12 • 6.RP.A.3.d

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.d

There are 3 feet in 1 yard.

How many feet are in 14 yards?

Feet: _____

There are 12 inches in 1 foot.

How many inches are in 4.5 feet?

Inches: _____

There are 16 ounces in 1 pound.

How many pounds are in 56 ounces?

Pounds: _____

A recipe needs 3 quarts of milk. There are 4 cups in 1 quart.

How many cups of milk are needed?

A. $\frac{3}{4}$ cup

B. $1\frac{1}{3}$ cups

C. 7 cups

D. 12 cups

1 inch = 2.54 centimeters.

A book is 10 inches tall. How many centimeters tall is it?

Centimeters: _____

SET 12 • 6.RP.A.3.d

BACKWARD 1

Measurement equivalents • Grade 4 • Measurement & Data • 4.MD.A.1

How many ounces are in 1 pound?

Ounces: _____

True or false?

There are 12 inches in 1 yard.

A. True

B. False

How many cups are in 1 quart?

- A.** 2
- B.** 4
- C.** 8
- D.** 16

How many grams are in 1 kilogram?

Grams: _____

True or false?

1 meter = 100 centimeters

A. True

B. False

SET 12 • 6.RP.A.3.d

BACKWARD 2

Multiplying and dividing by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$3.4 \times 1,000$$

Product: _____

Multiply.

$$2.54 \times 10$$

Product: _____

True or false?

$$0.6 \times 1,000 = 60$$

A. True

B. False

$$450 \div 100 = ?$$

- A.** 0.45
- B.** 4.5
- C.** 45
- D.** 45,000

Multiply.

$$7.25 \times 100$$

Product: _____

SET 12 • 6.RP.A.3.d

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$4.5 \times 12$$

Product: _____

Multiply.

$$3.5 \times 16$$

Product: _____

True or false?

$$1.5 \times 4 = 6$$

A. True

B. False

$$2.25 \times 4 = ?$$

A. 0.9

B. 8.1

C. 9

D. 90

Multiply.

$$6 \times 0.75$$

Product: _____

SET 12 • 6.RP.A.3.d

FORWARD 1

Rates with fractions and unit conversions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A faucet drips $\frac{1}{4}$ cup of water every $\frac{1}{3}$ minute.

How many cups does it drip per hour?

Cups per hour: _____

A snail travels $\frac{3}{8}$ foot every $\frac{1}{2}$ minute.

How many feet does it travel per hour?

Feet per hour: _____

A plant grows $\frac{1}{2}$ inch in $\frac{2}{3}$ week.

How many inches does it grow per week?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{3}{4}$

D. $1\frac{1}{3}$

A hose fills $\frac{3}{4}$ gallon in $\frac{1}{6}$ minute.

How many gallons does it fill per minute?

Gallons per minute: _____

A worker paints $\frac{2}{5}$ of a room in $\frac{1}{2}$ hour.

How many rooms can the worker paint in an 8-hour day?

Rooms: _____

SET 12 • 6.RP.A.3.d

FORWARD 2

Convert units for very large and very small quantities • Grade 8 • Expressions & Equations • 8.EE.A.4

$$1 \text{ km} = 10^6 \text{ mm.}$$

Write 3.2×10^6 millimeters in kilometers.

Kilometers: _____

Earth is about 1.5×10^8 km from the Sun. $1 \text{ km} = 10^3$ m.

How many meters is this? Write the answer in scientific notation.

Meters: _____

A sheet of paper is about 1×10^{-4} meter thick.

Which unit is the most appropriate for this measurement?

A. Miles

B. Meters

C. Millimeters

D. Kilometers

A virus is 1.2×10^{-7} meter long. 1 nanometer = 10^{-9} meter.

How many nanometers long is the virus?

Nanometers: _____

A glacier moves 3×10^{-1} meter per day. $1 \text{ m} = 10^2 \text{ cm}$.

How many centimeters does it move per day?

Centimeters: _____

Compute quotients of fractions and mixed numbers

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Multiplying fractions	5.NF.B.4.a
BACKWARD 2	Whole number $\div$ unit fraction	5.NF.B.7.b
BACKWARD 3	Unit fraction $\div$ whole number	5.NF.B.7.a
BACKWARD 4	Equivalent fractions	4.NF.A.1
BACKWARD 5	Mixed numbers as fractions	4.NF.B.3.b
FORWARD 1	Multiply and divide rational numbers	7.NS.A.2.c
FORWARD 2	Equations with fraction coefficients	8.EE.C.7.b

SET 13 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{5} \div \frac{2}{3}$$

Quotient: _____

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

Quotient: _____

Divide.

$$1\frac{1}{2} \div \frac{3}{4}$$

Quotient: _____

Which is equal to $\frac{4}{5} \div \frac{2}{3}$?

A. $\frac{2}{15}$

B. $\frac{8}{15}$

C. $\frac{5}{6}$

D. $1\frac{1}{5}$

Divide.

$$3\frac{1}{3} \div \frac{5}{6}$$

Quotient: _____

SET 13 • 6.NS.A.1

BACKWARD 1

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{5} \times \frac{2}{7}$$

Product: _____

True or false?

$$\frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$$

A. True

B. False

$$\frac{1}{2} \times \frac{5}{6} = ?$$

A. $\frac{1}{3}$

B. $\frac{5}{12}$

C. $\frac{5}{8}$

D. $\frac{6}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q3/5

Multiply.

$$\frac{4}{5} \times \frac{3}{2}$$

Product: _____

Multiply.

$$\frac{5}{6} \times 3$$

Product: _____

SET 13 • 6.NS.A.1

BACKWARD 2

Whole number $\div$ unit fraction • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$3 \div \frac{1}{4}$$

Quotient: _____

True or false?

$$4 \div \frac{1}{2} = 8$$

A. True

B. False

How many thirds are in 2?

A. $\frac{1}{6}$

B. $\frac{2}{3}$

C. 5

D. 6

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

Divide.

$$2 \div \frac{1}{8}$$

Quotient: _____

SET 13 • 6.NS.A.1

BACKWARD 3

Unit fraction $\div$ whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{4} \div 3$$

Quotient: _____

True or false?

$$\frac{1}{2} \div 3 = \frac{1}{6}$$

A. True

B. False

$$\frac{1}{5} \div 2 = ?$$

A. $\frac{1}{10}$

B. $\frac{2}{5}$

C. $\frac{5}{2}$

D. 10

Divide.

$$\frac{1}{6} \div 3$$

Quotient: _____

True or false?

$$\frac{1}{6} \div 2 = \frac{2}{6}$$

A. True

B. False

SET 13 • 6.NS.A.1

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{8}{12}$ is equivalent to $\frac{2}{3}$.

A. True

B. False

Find the missing number.

$$\frac{10}{15} = \frac{?}{3}$$

Missing number: _____

Which fraction is equivalent to $\frac{9}{12}$?

A. $\frac{2}{3}$

B. $\frac{3}{4}$

C. $\frac{12}{9}$

D. $\frac{9}{4}$

Find the missing number.

$$\frac{6}{8} = \frac{?}{4}$$

Missing number: _____

True or false?

$\frac{4}{6}$ is equivalent to $\frac{6}{8}$.

A. True

B. False

SET 13 • 6.NS.A.1

BACKWARD 5

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $1\frac{1}{2}$ as a fraction.

Fraction: _____

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

True or false?

$$2\frac{1}{4} = \frac{9}{4}$$

A. True

B. False

Which fraction is equal to $1\frac{2}{3}$?

A. $\frac{3}{3}$

B. $\frac{5}{3}$

C. $\frac{7}{3}$

D. $\frac{12}{3}$

Write $\frac{11}{4}$ as a mixed number.

Mixed number: _____

SET 13 • 6.NS.A.1

FORWARD 1

Multiply and divide rational numbers • Grade 7 • The Number System •
7.NS.A.2.c

Find the value.

$$\left(-\frac{2}{3}\right) \div \frac{4}{5}$$

Find the value.

$$-\frac{3}{4} \times \left(-\frac{8}{9}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q2/5

$$\left(-1\frac{1}{2}\right) \div \left(-\frac{3}{8}\right) = ?$$

A. -4

B. $-\frac{9}{16}$

C. $\frac{9}{16}$

D. 4

Find the value.

$$\frac{5}{6} \div \left(-\frac{5}{12}\right)$$

True or false?

$$\left(-\frac{1}{2}\right) \times \frac{4}{5} = -\frac{2}{5}$$

A. True

B. False

SET 13 • 6.NS.A.1

FORWARD 2

Equations with fraction coefficients • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{3}{4}(x - 2) = \frac{1}{2}x + 5$$

x = _____

Solve for x.

$$\frac{2}{3}x + 4 = \frac{1}{6}x + 7$$

x = _____

Solve for x.

$$\frac{x}{2} + \frac{x}{3} = 10$$

- A.** 5
- B.** 6
- C.** 12
- D.** 60

Solve for x.

$$\frac{1}{2}(4x + 6) = 3x - 5$$

x = _____

Solve for y.

$$\frac{5}{8}y - 1 = \frac{3}{8}y + 2$$

y = _____

Solve word problems involving division of fractions

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Division as "how many groups?"	3.OA.A.2
BACKWARD 2	Unit fractions build a fraction	3.NF.A.1
BACKWARD 3	Area with fractional side lengths	5.NF.B.4.b
BACKWARD 4	Word problems: dividing with unit fractions	5.NF.B.7.c
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Solve linear equations from contexts	8.EE.C.7.b

SET 14 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{2}{3}$ -cup scoops are in $\frac{5}{6}$ cup of flour?

Scoops: _____

A path is $\frac{3}{4}$ mile long. It is divided into sections that are each $\frac{1}{8}$ mile long.

How many sections are there?

Sections: _____

A rectangular patio has an area of $\frac{3}{5}$ square yard. Its length is $\frac{9}{10}$ yard.

What is its width?

Width: _____

A pitcher holds $\frac{9}{10}$ liter of juice. Each glass holds $\frac{3}{10}$ liter.

How many glasses can be filled?

A. $\frac{27}{100}$

B. $\frac{1}{3}$

C. 3

D. 6

Ana has $\frac{5}{6}$ pound of trail mix. She fills bags that each hold $\frac{5}{12}$ pound.

How many bags can she fill?

Bags: _____

SET 14 • 6.NS.A.1

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 4 are in 20?

Groups: _____

True or false?

$12 \div 3$ can mean "How many groups of 3 are in 12?"

- A. True
- B. False

Tom has 18 cookies. He puts 6 cookies in each bag.

Which equation finds the number of bags?

A. $18 \times 6 = \square$

B. $18 - 6 = \square$

C. $18 \div 6 = \square$

D. $6 \div 18 = \square$

24 students are split into teams of 8.

How many teams are there?

Teams: _____

How many 5s are in 35?

Answer: _____

SET 14 • 6.NS.A.1

BACKWARD 2

Unit fractions build a fraction • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make $\frac{3}{4}$?

Answer: _____

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

True or false?

$\frac{5}{6}$ is 5 parts of size $\frac{1}{6}$.

A. True

B. False

$\frac{3}{8}$ is made of how many parts of size $\frac{1}{8}$?

A. $\frac{1}{3}$

B. 3

C. 5

D. 8

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

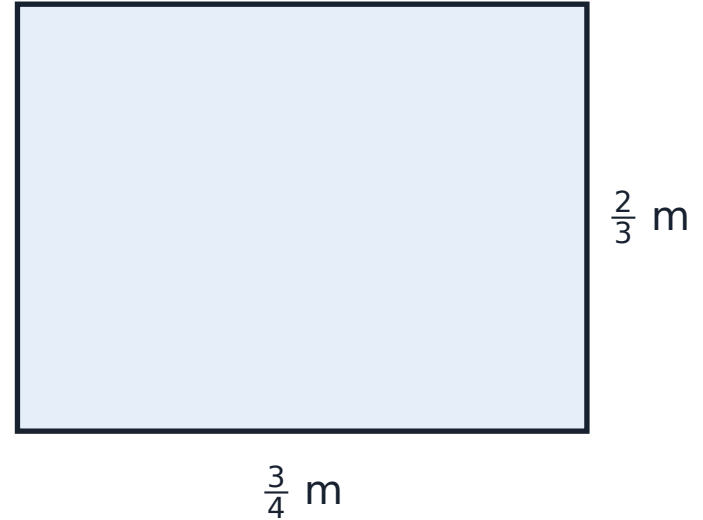
SET 14 • 6.NS.A.1

BACKWARD 3

Area with fractional side lengths • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

What is the area of the rectangle?

Area: _____



True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{3}$ foot has an area of $\frac{1}{6}$ square foot.

A. True

B. False

A rectangle is $\frac{2}{5}$ unit by $\frac{1}{2}$ unit.

What is its area?

A. $\frac{1}{5}$ square unit

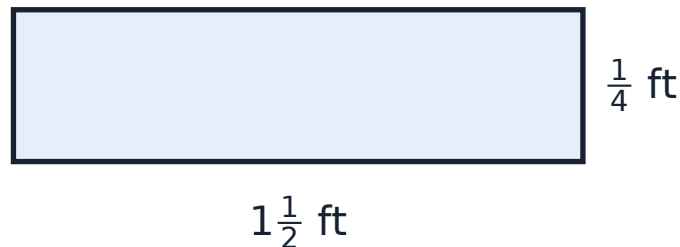
B. $\frac{2}{7}$ square unit

C. $\frac{3}{7}$ square unit

D. $\frac{9}{10}$ square unit

What is the area of the rectangle?

Area: _____



A garden is $\frac{5}{6}$ yard long and $\frac{3}{5}$ yard wide.

What is its area?

Area: _____

SET 14 • 6.NS.A.1

BACKWARD 4

Word problems: dividing with unit fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{4}$ -pound burgers can be made from 3 pounds of beef?

Burgers: _____

A 4-pound bag of rice is divided into $\frac{1}{2}$ -pound portions.

How many portions are there?

Portions: _____

Which expression finds how many $\frac{1}{5}$ -mile laps are in 3 miles?

A. $\frac{1}{5} \div 3$

B. $3 - \frac{1}{5}$

C. $3 \div \frac{1}{5}$

D. $3 \times \frac{1}{5}$

4 friends share $\frac{1}{3}$ pound of cheese equally.

How much cheese does each friend get?

Each friend: _____

True or false?

There are 10 half-hours in 5 hours.

A. True

B. False

SET 14 • 6.NS.A.1

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot each second.

How many seconds does it take to descend $5\frac{1}{4}$ feet?

Seconds: _____

The temperature was 4°F . It dropped $2\frac{1}{2}^{\circ}\text{F}$ each hour for 3 hours.

What was the final temperature?

Temperature: _____

A stock's value changed by $-\$1\frac{1}{4}$ each day for 4 days.

What was the total change?

- A.** $-\$5$
- B.** $-\$4\frac{1}{4}$
- C.** $-\$1$
- D.** $\$5$

A recipe needs $\frac{2}{3}$ cup of oil per batch. Jo has $3\frac{1}{3}$ cups of oil.

How many batches can Jo make?

Batches: _____

A hiker descends 1,250 feet in $\frac{5}{6}$ hour at a constant rate.

What is the change in elevation per hour?

Change per hour: _____

SET 14 • 6.NS.A.1

FORWARD 2

Solve linear equations from contexts • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{1}{2}(x + 6) = \frac{1}{3}(x + 12)$$

x = _____

Lee's age plus $\frac{1}{3}$ of his age is 24.

Write and solve an equation to find Lee's age.

Age: _____

Tank A has 40 gallons and drains $\frac{3}{4}$ gallon per minute.

Tank B has 10 gallons and fills at $1\frac{1}{4}$ gallons per minute.

After how many minutes will the tanks hold the same amount?

Minutes: _____

Solve for x.

$$\frac{2}{5}x - 3 = \frac{1}{5}x + 1$$

A. -10

B. 4

C. 10

D. 20

Solve for n.

$$\frac{3}{4}n + \frac{1}{2} = \frac{1}{4}n + 4$$

n = _____

Divide multi-digit numbers using the standard algorithm

MAIN	Grade 6 main question	6.NS.B.2
BACKWARD 1	Multiplying multi-digit numbers	5.NBT.B.5
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 3	Estimating quotients	5.NBT.B.6
BACKWARD 4	Multi-digit subtraction	4.NBT.B.4
FORWARD 1	Long division: fractions to decimals	7.NS.A.2.d
FORWARD 2	Repeating decimals and rational numbers	8.NS.A.1

SET 15 • 6.NS.B.2

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide.

$$4,508 \div 23$$

Quotient: _____

Use the standard algorithm to divide.

$$7,236 \div 36$$

Quotient: _____

Use the standard algorithm to divide.

$$9,975 \div 75$$

Quotient: _____

Use the standard algorithm to divide.

$$1,680 \div 48$$

A. 30

B. 35

C. 36

D. 350

Use the standard algorithm to divide.

$$25,494 \div 42$$

Quotient: _____

SET 15 • 6.NS.B.2

BACKWARD 1

Multiplying multi-digit numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Multiply.

$$46 \times 23$$

Product: _____

Multiply.

$$196 \times 23$$

Product: _____

True or false?

$$35 \times 48 = 1,680$$

A. True

B. False

$$133 \times 75 = ?$$

- A.** 1,596
- B.** 9,775
- C.** 9,875
- D.** 9,975

Multiply.

$$607 \times 42$$

Product: _____

SET 15 • 6.NS.B.2

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 4$$

Quotient: _____

Divide.

$$1,256 \div 8$$

Quotient: _____

True or false?

$$936 \div 3 = 312$$

A. True

B. False

$$714 \div 7 = ?$$

- A.** 12
- B.** 102
- C.** 120
- D.** 1,002

Divide.

$$2,345 \div 5$$

Quotient: _____

SET 15 • 6.NS.B.2

BACKWARD 3

Estimating quotients • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Which is the best estimate for $4,508 \div 23$?

- A.** 2
- B.** 20
- C.** 200
- D.** 2,000

Which is the best estimate for $7,236 \div 36$?

- A.** 2
- B.** 20
- C.** 200
- D.** 2,000

True or false?

9,975 $\div$ 75 is between 100 and 200.

A. True

B. False

Estimate $6,120 \div 29$ by rounding 29 to 30.

Estimate: _____

How many digits are in the quotient of $1,680 \div 48$?

A. 1

B. 2

C. 3

D. 4

SET 15 • 6.NS.B.2

BACKWARD 4

Multi-digit subtraction • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Subtract.

$$5,012 - 4,698$$

Difference: _____

Subtract.

$$3,000 - 1,847$$

Difference: _____

True or false?

$$7,236 - 7,200 = 36$$

A. True

B. False

$$9,975 - 9,000 = ?$$

- A.** 75
- B.** 975
- C.** 1,975
- D.** 9,075

Subtract.

$$6,203 - 2,875$$

Difference: _____

SET 15 • 6.NS.B.2

FORWARD 1

Long division: fractions to decimals • Grade 7 • The Number System •
7.NS.A.2.d

Use long division to write $\frac{5}{11}$ as a decimal.

Decimal: _____

Use long division to write $\frac{3}{8}$ as a decimal.

Decimal: _____

Which fraction has a repeating decimal?

A. $\frac{1}{5}$

B. $\frac{2}{3}$

C. $\frac{3}{4}$

D. $\frac{7}{8}$

True or false?

The decimal form of $\frac{7}{20}$ terminates.

A. True

B. False

Use long division to write $\frac{5}{6}$ as a decimal.

Decimal: _____

SET 15 • 6.NS.B.2

FORWARD 2

Repeating decimals and rational numbers • Grade 8 • The Number System •
8.NS.A.1

Write $0.272727\ldots$ as a fraction in simplest form.

Fraction: _____

Write 0.444... as a fraction.

Fraction: _____

Which number is irrational?

A. 0.333...

B. $\frac{5}{8}$

C. $\sqrt{16}$

D. $\sqrt{7}$

True or false?

Every repeating decimal is a rational number.

A. True

B. False

Write 1.1666... as a fraction.

Fraction: _____

Add multi-digit decimals

MAIN	Grade 6 main question
BACKWARD 1	Place value of decimal digits
BACKWARD 2	Equivalent decimals
BACKWARD 3	Lining up decimal points to add
BACKWARD 4	Multi-digit addition
FORWARD 1	Add rational numbers
FORWARD 2	Linear equations with decimal coefficients

6.NS.B.3

5.NBT.A.3.a

5.NBT.A.3.b

5.NBT.B.7

4.NBT.B.4

7.NS.A.1.d

8.EE.C.7.b

SET 16 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the sum.

$$315.4 + 28.75 + 6.309$$

Sum: _____

Find the sum.

$$47.08 + 9.6$$

Sum: _____

Find the sum.

$$128.5 + 3.47 + 60.009$$

Sum: _____

Find the sum.

$$6.75 + 12.9 + 0.48$$

- A.** 7.23
- B.** 19.23
- C.** 20.03
- D.** 20.13

Find the sum.

$$0.875 + 14.2 + 3.06$$

Sum: _____

SET 16 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.358, what place value does the digit 5 have?

- A.** Tenths
- B.** Hundredths
- C.** Ones
- D.** Thousandths

In 42.906, which digit is in the tenths place?

Digit: _____

True or false?

In 3.47, the digit 7 is in the hundredths place.

A. True

B. False

What is the value of the 8 in 0.584?

A. 0.008

B. 0.08

C. 0.8

D. 8

Write $5 + 0.3 + 0.07$ as a decimal.

Decimal: _____

SET 16 • 6.NS.B.3

BACKWARD 2

Equivalent decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

True or false?

2.4 and 2.40 have the same value.

A. True

B. False

True or false?

5.6 and 5.06 have the same value.

A. True

B. False

Which decimal is equal to 7.5?

- A.** 0.75
- B.** 7.005
- C.** 7.05
- D.** 7.50

Write 9.6 with a zero in the hundredths place.

Decimal: _____

True or false?

$$0.300 = 0.3$$

A. True

B. False

SET 16 • 6.NS.B.3

BACKWARD 3

Lining up decimal points to add • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $36.2 + 4.58$?

A.

$$\begin{array}{r} 36.2 \\ +4.58 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 36.20 \\ + 4.58 \\ \hline \end{array}$$

Which setup lines up the decimal points for $5.07 + 12.3$?

A.

$$\begin{array}{r} 5.07 \\ +12.30 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 5.07 \\ +12.3 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A. True
- B. False

$$\begin{array}{r} 8.4 \\ +2.75 \\ \hline \end{array}$$

Which setup lines up the decimal points for $128.5 + 3.47$?

A.

$$\begin{array}{r} 128.50 \\ + 3.47 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 128.5 \\ + 3.47 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 9.30 \\ + 0.45 \\ \hline \end{array}$$

SET 16 • 6.NS.B.3

BACKWARD 4

Multi-digit addition • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Add.

$$3,154 + 2,875$$

Sum: _____

Add.

$$4,605 + 2,798$$

Sum: _____

True or false?

$$1,975 + 3,489 = 5,464$$

A. True

B. False

$$6,758 + 1,265 = ?$$

A. 7,013

B. 7,923

C. 8,013

D. 8,023

Add.

$$12,480 + 9,635$$

Sum: _____

SET 16 • 6.NS.B.3

FORWARD 1

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-18.6 + 4.35 =$$

Find the value.

$$-24.5 + 3.27 =$$

Find the value.

$$-3.4 + (-9.15) =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q3/5

Find the value.

$$-6.25 + (-2.8) = ?$$

A. -9.05

B. -3.45

C. 3.45

D. 9.05

Find the value.

$$-15.75 + 20.3 =$$

SET 16 • 6.NS.B.3

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$-2.4x + 7.8 = 1.2x - 1.2$$

x = _____

Solve for x.

$$-1.5x + 6.4 = 0.5x - 1.6$$

x = _____

Solve for x.

$$0.8x + 7.2 = -1.2x + 1.2$$

x = _____

Solve for x.

$$-4.5x - 3.25 = -2.5x + 0.75$$

x = _____

Solve for x.

$$2.4(x + 1.5) = 1.4x + 6.6$$

x = _____

Subtract multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Place value of decimal digits

5.NBT.A.3.a

BACKWARD 2

Writing zeros to make equal decimal places

5.NBT.A.3.b

BACKWARD 3

Lining up decimal points to subtract

5.NBT.B.7

BACKWARD 4

Subtracting across zeros

4.NBT.B.4

FORWARD 1

Subtract rational numbers by adding the additive inverse

7.NS.A.1.c

FORWARD 2

Solve equations by collecting like terms

8.EE.C.7.b

SET 17 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the difference.

$$52.3 - 18.47$$

Difference: _____

Find the difference.

$$100 - 36.58$$

Difference: _____

Find the difference.

$$7.4 - 2.068$$

Difference: _____

Find the difference.

$$15.02 - 6.9$$

- A.** 8.12
- B.** 8.93
- C.** 9.12
- D.** 14.33

Find the difference.

$$243.6 - 87.95$$

Difference: _____

SET 17 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 2.068, what place value does the digit 6 have?

- A.** Tenths
- B.** Thousandths
- C.** Hundredths
- D.** Ones

In 87.95, which digit is in the hundredths place?

Digit: _____

True or false?

In 36.58, the digit 5 is in the tenths place.

A. True

B. False

What is the value of the 4 in 7.04?

- A.** 0.004
- B.** 0.04
- C.** 0.4
- D.** 4

Write $20 + 0.6 + 0.009$ as a decimal.

Decimal: _____

SET 17 • 6.NS.B.3

BACKWARD 2

Writing zeros to make equal decimal places • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.3.b

True or false?

$$7.4 = 7.400$$

A. True

B. False

Write 100 with two decimal places.

Decimal: _____

Which decimal is equal to 15.1?

- A.** 1.51
- B.** 15.001
- C.** 15.01
- D.** 15.10

Write 52.3 with a digit in the hundredths place, without changing its value.

Decimal: _____

True or false?

$$6.9 = 6.09$$

A. True

B. False

SET 17 • 6.NS.B.3

BACKWARD 3

Lining up decimal points to subtract • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Which setup lines up the decimal points for $15.6 - 3.42$?

A.

$$\begin{array}{r} 15.60 \\ - 3.42 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 15.6 \\ - 3.42 \\ \hline \end{array}$$

Which setup lines up the decimal points for $52.3 - 18.47$?

A.

$$\begin{array}{r} 52.3 \\ -18.47 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 52.30 \\ -18.47 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A. True
- B. False

$$\begin{array}{r} 7.4 \\ -2.068 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 100.00 \\ - 36.58 \\ \hline \end{array}$$

Which setup lines up the decimal points for $243.6 - 87.95$?

A.

$$\begin{array}{r} 243.60 \\ - 87.95 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 243.6 \\ - 87.95 \\ \hline \end{array}$$

SET 17 • 6.NS.B.3

BACKWARD 4

Subtracting across zeros • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Subtract.

$$5,000 - 2,847$$

Difference: _____

Subtract.

$$3,006 - 1,478$$

Difference: _____

True or false?

$$6,000 - 2,458 = 3,542$$

A. True

B. False

$$8,014 - 3,276 = ?$$

A. 4,738

B. 4,748

C. 4,838

D. 5,262

Subtract.

$$10,000 - 3,658$$

Difference: _____

SET 17 • 6.NS.B.3

FORWARD 1

Subtract rational numbers by adding the additive inverse • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$-4.2 - 3.75$$

Value: _____

Find the value.

$$2.5 - (-6.8)$$

Value: _____

Which expression is equal to $8.1 - 12.4$?

- A.** $12.4 - 8.1$
- B.** $-8.1 + 12.4$
- C.** $8.1 + (-12.4)$
- D.** $8.1 + 12.4$

Find the value.

$$-0.65 - (-1.2)$$

Value: _____

True or false?

$$-7.5 - 2.5 = -7.5 + (-2.5)$$

A. True

B. False

SET 17 • 6.NS.B.3

FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$5.2x - 3.75 = 2.2x + 8.25$$

x = _____

Solve for x.

$$0.9x + 12.5 = 0.4x + 10$$

x = _____

Solve for x.

$$6.3 - 1.2x = 0.8x - 1.7$$

A. -4

B. 0.4

C. 2.5

D. 4

Solve for x.

$$4.75 - x = 2x - 1.25$$

x = _____

Solve for x.

$$3.4x - 9.6 = 1.4x - 2.6$$

x = _____

Find and correct decimal alignment errors in addition and subtraction

MAIN	Grade 6 main question	6.NS.B.3
BACKWARD 1	Lining up decimal points	5.NBT.B.7
BACKWARD 2	Place value of decimal digits	5.NBT.A.3.a
BACKWARD 3	Estimating to check reasonableness	5.NBT.A.4
FORWARD 1	Find errors when adding and subtracting rational numbers	7.NS.A.1.d
FORWARD 2	Find errors in solving linear equations	8.EE.C.7.b

SET 18 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Jon put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 315.2 \\ 24.8 \\ +7.936 \\ \hline 11.336 \end{array}$$

Mistake: _____

Correct answer: _____

Ava put this example on the board.

What mistake was made?

What should the answer be?

$$\begin{array}{r} 56.4 \\ +3.25 \\ \hline 8.89 \end{array}$$

Mistake: _____

Correct answer: _____

Leo put this example on the board.

What mistake was made?

What should the answer be?

$$\begin{array}{r} 48.6 \\ -2.15 \\ \hline 2.71 \end{array}$$

Mistake: _____

Correct answer: _____

Mia put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 7.08 \\ +12.6 \\ \hline 8.34 \end{array}$$

Mistake: _____

Correct answer: _____

Ben put this example on the board.

What mistake was made?

What should the answer be?

$$\begin{array}{r} 125.6 \\ -3.48 \\ \hline 9.08 \end{array}$$

Mistake: _____

Correct answer: _____

SET 18 • 6.NS.B.3

BACKWARD 1

Lining up decimal points • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $56.4 + 3.25$?

A.

$$\begin{array}{r} 56.4 \\ + 3.25 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 56.40 \\ + 3.25 \\ \hline \end{array}$$

Which setup lines up the decimal points for $48.6 - 2.15$?

A.

$$\begin{array}{r} 48.60 \\ - 2.15 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 48.6 \\ - 2.15 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A. True
- B. False

$$\begin{array}{r} 7.08 \\ +12.6 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 125.60 \\ - 3.48 \\ \hline \end{array}$$

Which setup lines up the decimal points for $24.8 + 7.936$?

A.

$$\begin{array}{r} 24.8 \\ +7.936 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 24.800 \\ + 7.936 \\ \hline \end{array}$$

SET 18 • 6.NS.B.3

BACKWARD 2

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.936, what place value does the digit 3 have?

- A.** Hundredths
- B.** Tenths
- C.** Thousandths
- D.** Ones

In 315.2, which digit is in the ones place?

Digit: _____

True or false?

In 3.25, the digit 2 is in the tenths place.

A. True

B. False

What is the value of the 6 in 12.56?

- A.** 0.06
- B.** 0.6
- C.** 6
- D.** 60

In 48.6, which digit is in the tenths place?

Digit: _____

SET 18 • 6.NS.B.3

BACKWARD 3

Estimating to check reasonableness • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round each number to the nearest whole number. Then add to estimate.

$$315.2 + 24.8 + 7.936$$

Estimate: _____

Round 48.6 to the nearest whole number.

Rounded: _____

True or false?

7.08 + 12.6 is about 20.

A. True

B. False

Which is the best estimate for $125.6 - 3.48$?

- A.** 9
- B.** 90
- C.** 122
- D.** 130

Round 56.4 and 3.25 to the nearest whole numbers.
Then add.

Estimate: _____

SET 18 • 6.NS.B.3

FORWARD 1

Find errors when adding and subtracting rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Ray wrote: $-3.5 + 1.25 = -4.75$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

Tia wrote: $-6.2 - 2.4 = -3.8$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

Sam wrote: $4.5 - (-1.5) = 3$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

Kai wrote: $-7.25 + (-2.5) = -4.75$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

Lia wrote: $-0.8 + 2.3 = -1.5$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

SET 18 • 6.NS.B.3

FORWARD 2

Find errors in solving linear equations • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Kim solved $2.4x + 1.2 = 0.4x - 3.8$.

Step 1: $2x + 1.2 = -3.8$

Step 2: $2x = -2.6$

Step 3: $x = -1.3$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Jo solved $3.5x - 2 = 1.5x + 6$.

Step 1: $2x - 2 = 6$

Step 2: $2x = 4$

Step 3: $x = 2$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Ali solved $-1.2x + 5 = 0.8x - 3$.

Step 1: $-2x + 5 = -3$

Step 2: $-2x = -8$

Step 3: $x = -4$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Ned solved $0.5(x + 6) = 2x - 3$.

Step 1: $0.5x + 6 = 2x - 3$

Step 2: $9 = 1.5x$

Step 3: $x = 6$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Eve solved $4.2x - 1.5 = 2.2x + 2.5$.

Step 1: $2x - 1.5 = 2.5$

Step 2: $2x = 1$

Step 3: $x = 0.5$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Multiply multi-digit decimals

MAIN	Grade 6 main question
BACKWARD 1	Multiplying whole numbers
BACKWARD 2	Powers of 10 and the decimal point
BACKWARD 3	Multiplying tenths and hundredths
BACKWARD 4	Rounding decimals to estimate
FORWARD 1	Multiply rational numbers
FORWARD 2	Multiply numbers in scientific notation

6.NS.B.3

5.NBT.B.5

5.NBT.A.2

5.NBT.B.7

5.NBT.A.4

7.NS.A.2.c

8.EE.A.4

SET 19 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.6 \times 0.45$$

Product: _____

Multiply.

$$12.5 \times 3.8$$

Product: _____

Multiply.

$$4.25 \times 2.4$$

Product: _____

Multiply.

$$0.07 \times 0.6$$

A. 0.0042

B. 0.042

C. 0.42

D. 4.2

Multiply.

$$18.4 \times 0.35$$

Product: _____

SET 19 • 6.NS.B.3

BACKWARD 1

Multiplying whole numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Multiply.

$$36 \times 45$$

Product: _____

Multiply.

$$125 \times 38$$

Product: _____

True or false?

$$425 \times 24 = 10,200$$

A. True

B. False

$$184 \times 35 = ?$$

- A.** 1,472
- B.** 5,440
- C.** 6,340
- D.** 6,440

Multiply.

$$208 \times 15$$

Product: _____

SET 19 • 6.NS.B.3

BACKWARD 2

Powers of 10 and the decimal point • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$0.45 \times 10$$

Product: _____

Multiply.

$$3.6 \times 100$$

Product: _____

True or false?

$$24 \div 10 = 2.4$$

A. True

B. False

$$1,620 \div 1,000 = ?$$

- A.** 0.162
- B.** 1.62
- C.** 16.2
- D.** 162

Divide.

$$475 \div 10$$

Quotient: _____

SET 19 • 6.NS.B.3

BACKWARD 3

Multiplying tenths and hundredths • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$0.3 \times 0.2$$

Product: _____

Multiply.

$$0.5 \times 0.8$$

Product: _____

True or false?

$$0.4 \times 0.4 = 1.6$$

A. True

B. False

$$0.2 \times 0.03 = ?$$

A. 0.006

B. 0.06

C. 0.6

D. 6

Multiply.

$$1.2 \times 0.3$$

Product: _____

SET 19 • 6.NS.B.3

BACKWARD 4

Rounding decimals to estimate • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 3.62 to the nearest whole number.

Rounded: _____

Round 12.48 to the nearest tenth.

Rounded: _____

True or false?

4.25 rounded to the nearest whole number is 4.

A. True

B. False

Round 0.456 to the nearest hundredth.

A. 0.4

B. 0.45

C. 0.46

D. 0.5

Round 8.75 to the nearest whole number.

Rounded: _____

SET 19 • 6.NS.B.3

FORWARD 1

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the value.

$$(-3.6)(0.45)$$

Find the value.

$$(-2.5)(-4.2)$$

$$-0.8 \times 1.5 = ?$$

- A.** -1.2
- B.** -0.12
- C.** 0.12
- D.** 1.2

Find the value.

$$(-1.2)(-0.5)(-3)$$

True or false?

$$(-0.6)(0.7) = 0.42$$

A. True

B. False

SET 19 • 6.NS.B.3

FORWARD 2

Multiply numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Multiply. Write the answer in scientific notation.

$$(3.6 \times 10^4)(4.5 \times 10^{-2})$$

Product: _____

Multiply. Write the answer in scientific notation.

$$(2.5 \times 10^3)(4 \times 10^5)$$

Product: _____

$$(1.2 \times 10^6)(3 \times 10^{-4}) = ?$$

- A.** 4.2×10^2
- B.** 3.6×10^{10}
- C.** 3.6×10^2
- D.** 3.6×10^{-2}

A light-year is about 9.5×10^{12} km.

About how many kilometers are in 4.2 light-years? Write the answer in scientific notation.

Kilometers: _____

True or false?

$$(5 \times 10^3)(6 \times 10^2) = 3 \times 10^6$$

A. True

B. False

Divide multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Dividing by a two-digit number

5.NBT.B.6

BACKWARD 2

Using powers of 10 to shift the decimal point

5.NBT.A.2

BACKWARD 3

Dividing tenths and hundredths

5.NBT.B.7

FORWARD 1

Divide rational numbers

7.NS.A.2.b

FORWARD 2

Divide numbers in scientific notation

8.EE.A.4

SET 20 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$15.75 \div 0.35$$

Quotient: _____

Divide.

$$8.64 \div 2.4$$

Quotient: _____

Divide.

$$12.6 \div 0.45$$

Quotient: _____

Divide.

$$0.72 \div 0.08$$

A. 0.09

B. 0.9

C. 9

D. 90

Divide.

$$283.5 \div 13.5$$

Quotient: _____

SET 20 • 6.NS.B.3

BACKWARD 1

Dividing by a two-digit number • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$1,575 \div 35$$

Quotient: _____

Divide.

$$864 \div 24$$

Quotient: _____

True or false?

$$1,260 \div 45 = 28$$

A. True

B. False

$$672 \div 32 = ?$$

- A.** 12
- B.** 21
- C.** 22
- D.** 210

Divide.

$$2,835 \div 15$$

Quotient: _____

SET 20 • 6.NS.B.3

BACKWARD 2

Using powers of 10 to shift the decimal point • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.2

Which has the same quotient as $15.75 \div 0.35$?

A. $1,575 \div 35$

B. $157.5 \div 35$

C. $1,575 \div 3.5$

D. $15.75 \div 35$

True or false?

$8.64 \div 2.4$ has the same quotient as $86.4 \div 24$.

A. True

B. False

Multiply.

$$0.45 \times 100$$

Product: _____

Multiply.

$$12.6 \times 100$$

Product: _____

$$0.08 \times \underline{\quad} = 8$$

- A.** 0.1
- B.** 10
- C.** 100
- D.** 1,000

SET 20 • 6.NS.B.3

BACKWARD 3

Dividing tenths and hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

Divide.

$$2.4 \div 0.6$$

Quotient: _____

Divide.

$$3.5 \div 0.5$$

Quotient: _____

True or false?

$$1.2 \div 0.3 = 0.4$$

A. True

B. False

$$0.9 \div 0.3 = ?$$

A. 0.03

B. 0.3

C. 3

D. 30

Divide.

$$4.8 \div 4$$

Quotient: _____

SET 20 • 6.NS.B.3

FORWARD 1

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the value.

$$(-15.75) \div 0.35$$

Find the value.

$$-8.64 \div (-2.4)$$

True or false?

$$-(12 \div 4) = (-12) \div 4$$

A. True

B. False

$$7.2 \div (-0.9) = ?$$

A. -8

B. -0.8

C. 0.8

D. 8

Find the value.

$$-0.72 \div 0.08$$

SET 20 • 6.NS.B.3

FORWARD 2

Divide numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Divide. Write the answer in scientific notation.

$$(6.3 \times 10^8) \div (2.1 \times 10^3)$$

Quotient: _____

Divide. Write the answer in scientific notation.

$$(4.8 \times 10^{-2}) \div (1.6 \times 10^{-5})$$

Quotient: _____

$$(9 \times 10^6) \div (3 \times 10^9) = ?$$

A. 6×10^{-3}

B. 3×10^{15}

C. 3×10^3

D. 3×10^{-3}

Divide. Write the answer in scientific notation.

$$(7.5 \times 10^4) \div (2.5 \times 10^{-1})$$

Quotient: _____

True or false?

$$(8.4 \times 10^5) \div (4.2 \times 10^2) = 2 \times 10^3$$

A. True

B. False

Find the greatest common factor of two whole numbers

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Factors	4.OA.B.4
BACKWARD 2	Prime and composite numbers	4.OA.B.4
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Factor linear expressions using a common factor	7.EE.A.1
FORWARD 2	Solve equations using the distributive property	8.EE.C.7.b

SET 21 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

What is the greatest common factor of 18 and 45?

- A.** 3
- B.** 9
- C.** 15
- D.** 90

Find the greatest common factor of 28 and 42.

GCF: _____

Find the greatest common factor of 30 and 75.

GCF: _____

Find the greatest common factor of 16 and 40.

GCF: _____

SET 21 • 6.NS.B.4

BACKWARD 1

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 18.

Factors: _____

True or false?

7 is a factor of 42.

A. True

B. False

Which number is NOT a factor of 36?

- A.** 4
- B.** 8
- C.** 9
- D.** 12

True or false?

5 is a factor of 32.

A. True

B. False

List all the factors of 24.

Factors: _____

SET 21 • 6.NS.B.4

BACKWARD 2

Prime and composite numbers • Grade 4 • Operations & Algebraic Thinking •
4.OA.B.4

True or false?

13 is a prime number.

A. True

B. False

Which number is composite?

- A.** 2
- B.** 11
- C.** 17
- D.** 21

True or false?

1 is a prime number.

A. True

B. False

Is 29 prime or composite?

Answer: _____

Which number is prime?

A. 27

B. 31

C. 33

D. 39

SET 21 • 6.NS.B.4

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$42 \div 7$$

Quotient: _____

Divide.

$$36 \div 4$$

Quotient: _____

True or false?

$$45 \div 9 = 5$$

A. True

B. False

$$56 \div 8 = ?$$

A. 6

B. 7

C. 8

D. 9

Divide.

$$63 \div 9$$

Quotient: _____

SET 21 • 6.NS.B.4

FORWARD 1

Factor linear expressions using a common factor • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor using the greatest common factor.

$$18x + 24$$

Factored form: _____

Factor using the greatest common factor.

$$20a - 35$$

Factored form: _____

Which shows $16m + 40$ factored using the greatest common factor?

- A.** $8(2m + 5)$
- B.** $2(8m + 20)$
- C.** $8(2m + 40)$
- D.** $4(4m + 10)$

Factor using the greatest common factor.

$$27x + 45y$$

Factored form: _____

True or false?

$$14n - 21 = 7(2n - 3)$$

A. True

B. False

SET 21 • 6.NS.B.4

FORWARD 2

Solve equations using the distributive property • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$12x + 18 = 6(x + 5)$$

x = _____

Solve for x.

$$4(3x - 2) = 8(x + 1)$$

x = _____

Solve for x.

$$15x - 10 = 5(2x + 3)$$

A. -1

B. 1

C. 5

D. 25

Solve for x.

$$9(x + 2) = 6(x + 5)$$

x = _____

Solve for x.

$$10x + 25 = 5(x + 10)$$

x = _____

Find the least common multiple of two whole numbers

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Multiples	4.OA.B.4
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Add and subtract rational numbers with unlike denominators	7.NS.A.1.d
FORWARD 2	Equations with fractions: clearing denominators	8.EE.C.7.b

SET 22 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 8.

LCM: _____

Find the least common multiple of 4 and 10.

LCM: _____

What is the least common multiple of 9 and 12?

- A.** 3
- B.** 36
- C.** 72
- D.** 108

Find the least common multiple of 5 and 7.

LCM: _____

Find the least common multiple of 6 and 15.

LCM: _____

SET 22 • 6.NS.B.4

BACKWARD 1

Multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five multiples of 6.

Multiples: _____

True or false?

36 is a multiple of 8.

A. True

B. False

Which number is a multiple of both 4 and 6?

- A.** 8
- B.** 12
- C.** 18
- D.** 20

True or false?

45 is a multiple of 9.

A. True

B. False

What is the 4th multiple of 7?

Multiple: _____

SET 22 • 6.NS.B.4

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$6 \times 8$$

Product: _____

Multiply.

$$9 \times 4$$

Product: _____

True or false?

$$7 \times 5 = 35$$

A. True

B. False

$$3 \times 12 = ?$$

A. 15

B. 33

C. 36

D. 39

Multiply.

$$5 \times 6$$

Product: _____

SET 22 • 6.NS.B.4

FORWARD 1

Add and subtract rational numbers with unlike denominators • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{5}{6} + \frac{3}{8}$$

Find the value.

$$\frac{3}{4} - \left(-\frac{1}{6}\right)$$

$$-\frac{2}{3} - \frac{1}{4} = ?$$

A. $-\frac{11}{12}$

B. $-\frac{3}{7}$

C. $-\frac{1}{12}$

D. $\frac{5}{12}$

Find the value.

$$-1\frac{1}{2} + \frac{5}{6}$$

True or false?

$$\frac{3}{10} + \left(-\frac{2}{15}\right) = \frac{1}{6}$$

A. True

B. False

SET 22 • 6.NS.B.4

FORWARD 2

Equations with fractions: clearing denominators • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$\frac{x}{6} + \frac{x}{8} = 7$$

x = _____

Solve for x.

$$\frac{x}{4} - 1 = \frac{x}{6} + 2$$

x = _____

Solve for x.

$$\frac{2x}{3} + \frac{x}{2} = 14$$

A. 7

B. 12

C. $\frac{84}{5}$

D. 42

Solve for x.

$$\frac{x}{5} + 3 = \frac{x}{2}$$

x = _____

Solve for x.

$$\frac{(x+1)}{4} = \frac{(x-2)}{3}$$

x = _____

Use the distributive property to write a sum as the GCF times a sum

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Common factors	4.OA.B.4
BACKWARD 2	Distributive property with whole numbers	3.OA.B.5
BACKWARD 3	Parentheses and order of operations	5.OA.A.1
FORWARD 1	Factor and expand linear expressions	7.EE.A.1
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 23 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $45 + 20$ as a product of the GCF and a sum of two whole numbers.

Expression: _____

Which expression is equal to $42 + 56$ and uses the greatest common factor?

A. $2(21 + 28)$

B. $14(3 + 4)$

C. $14(3 + 8)$

D. $7(6 + 8)$

Use the greatest common factor to write $27 + 45$ as a product.

Expression: _____

True or false?

$$24 + 60 = 12(2 + 5)$$

A. True

B. False

Complete the equation.

$$30 + 75 = 15(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

SET 23 • 6.NS.B.4

BACKWARD 1

Common factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

5 is a factor of both 45 and 20.

A. True

B. False

List the factors of 12 and the factors of 18.

Which factors do they share?

Shared factors: _____

Which number is a factor of both 27 and 45?

- A.** 5
- B.** 6
- C.** 9
- D.** 15

List all the factors of 42.

Factors: _____

True or false?

6 is a factor of 56.

A. True

B. False

SET 23 • 6.NS.B.4

BACKWARD 2

Distributive property with whole numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$9 \times 6 = 9 \times 4 + 9 \times 2$$

A. True

B. False

True or false?

$$4 \times (9 + 2) = 4 \times 9 + 2$$

A. True

B. False

Which is equal to 6×12 ?

A. $6 \times 10 + 2$

B. $6 \times 10 \times 2$

C. $6 \times 10 + 6 \times 2$

D. $6 + 10 \times 2$

Fill in the blank.

$$5 \times 13 = 5 \times 10 + 5 \times \underline{\quad}$$

Blank: _____

Fill in the blank.

$$3 \times (4 + 5) = 3 \times 4 + 3 \times \underline{\hspace{1cm}}$$

Blank: _____

SET 23 • 6.NS.B.4

BACKWARD 3

Parentheses and order of operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$5 \times (9 + 4)$$

Value: _____

Evaluate.

$$14 \times (3 + 4)$$

Value: _____

True or false?

$$9 \times (3 + 5) = 32$$

A. True

B. False

$$12 \times (2 + 5) = ?$$

A. 17

B. 29

C. 60

D. 84

Evaluate.

$$(20 - 5) \times 3$$

Value: _____

SET 23 • 6.NS.B.4

FORWARD 1

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor.

$$12x - 18$$

Factored form: _____

Expand.

$$-3(2x - 5)$$

Expanded form: _____

Which expression is equivalent to $8y + 20$?

A. $4(2y + 20)$

B. $8(y + 20)$

C. $2(4y + 20)$

D. $4(2y + 5)$

Factor.

$$15a + 10b - 5$$

Factored form: _____

True or false?

$$\frac{1}{2}(6x - 4) = 3x - 2$$

A. True

B. False

SET 23 • 6.NS.B.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

How many solutions does the equation have?

$$4(x + 3) = 4x + 12$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$2(3x - 1) = 6x + 5$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$3(x + 2) = 2x + 9$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

What number goes in the box so that the equation has infinitely many solutions?

$$5(x - 2) = 5x - \square$$

$$\square = \underline{\hspace{10cm}}$$

Which equation has no solution?

A. $2(x + 4) = 2x + 5$

B. $2(x + 4) = x + 8$

C. $2(x + 4) = 2x + 8$

D. $2(x + 4) = 3x$

Use positive and negative numbers to represent quantities in context

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Whole numbers as distances from 0	2.MD.B.6
BACKWARD 2	Increase and decrease situations	2.OA.A.1
FORWARD 1	Adding integers as moving on a number line	7.NS.A.1.b
FORWARD 2	The sign of the rate of change in context	8.F.B.4

SET 24 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

A diver is 30 feet below sea level.

Write an integer to represent the diver's position.

Integer: _____

A company lost \$2,500 this month.

Write an integer to represent this change.

Integer: _____

A bank account shows a withdrawal of \$45.
Which number represents this change?

- A.** -45
- B.** -4.5
- C.** 0
- D.** 45

The temperature rose 12 degrees.

Write an integer to represent this change.

Integer: _____

A hawk flies 120 feet above sea level. A submarine is 120 feet below sea level.

Write an integer for each position.

Hawk: _____

Submarine: _____

SET 24 • 6.NS.C.5

BACKWARD 1

Whole numbers as distances from 0 • Grade 2 • Measurement & Data •
2.MD.B.6

What number does point A represent?



Number: _____

How many units from 0 is the number 6?

Units: _____

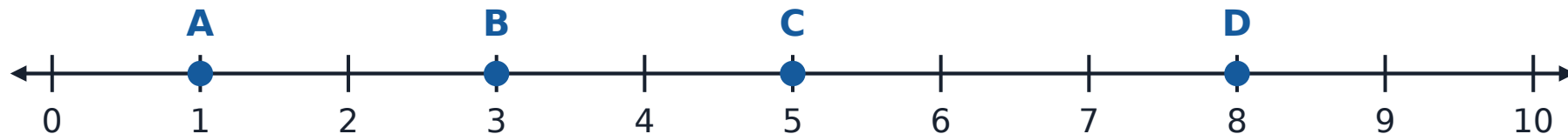
True or false?

On a number line, 9 is farther from 0 than 4 is.

A. True

B. False

Which point is 3 units from 0?



A. Point A

B. Point B

C. Point C

D. Point D

What number does point B represent?



Number: _____

SET 24 • 6.NS.C.5

BACKWARD 2

Increase and decrease situations • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

Kim had \$20. She spent \$8.

How much money does she have now?

Money: _____

A plant was 12 cm tall. It grew 5 cm.

How tall is the plant now?

Height: _____

Which situation describes a decrease?

- A.** A plant grows 4 inches.
- B.** Tom earns \$10.
- C.** A team gains 5 yards.
- D.** A balloon loses 6 feet of height.

True or false?

"The temperature fell 7 degrees" describes a decrease.

A. True

B. False

Which word describes an increase in money?

- A.** Deposit
- B.** Loss
- C.** Drop
- D.** Withdrawal

SET 24 • 6.NS.C.5

FORWARD 1

Adding integers as moving on a number line • Grade 7 • The Number System •
7.NS.A.1.b

The temperature is -4°F . It rises 9°F .

What is the new temperature?

Temperature: _____

A submarine is at -120 feet. It rises 45 feet.
What is its new position?

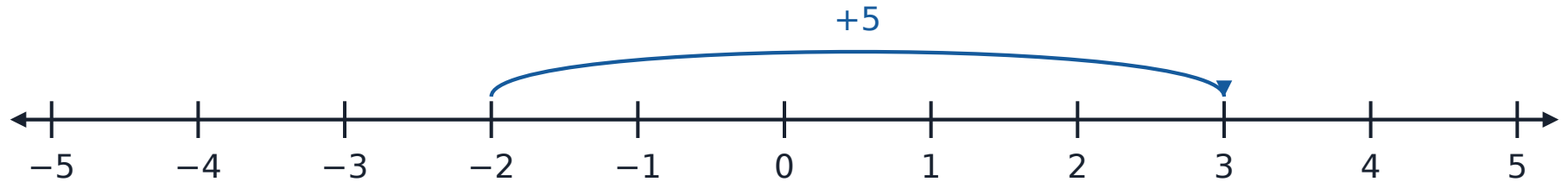
Position: _____

On a number line, where is $6 + (-4)$ compared to 6?

- A.** 4 units to the right of 6
- B.** 4 units to the left of 6
- C.** At 0
- D.** 6 units to the left of 4

The number line shows $-2 + 5$.

What is the sum?



Sum: _____

True or false?

A \$30 debt plus a \$30 deposit makes a balance of \$0.

A. True

B. False

SET 24 • 6.NS.C.5

FORWARD 2

The sign of the rate of change in context • Grade 8 • Functions • 8.F.B.4

A tank holds 60 gallons. It drains 3 gallons per minute.
Write a function for the amount A after t minutes.

$A =$ _____

An airplane at 12,000 feet descends 800 feet per minute.

What is the rate of change of its altitude?

- A.** –12,000 feet per minute **B.** –800 feet per minute
C. 800 feet per minute **D.** 12,000 feet per minute

The table shows a linear function.

What is the rate of change in degrees per hour?

Hours	0	2	4
Temperature (°F)	5	1	-3

Rate of change: _____

A diver starts at -10 feet and descends 4 feet per second.

Write a function for the diver's position d after t seconds.

$d =$ _____

True or false?

A line through $(0, 8)$ and $(4, 0)$ has a rate of change of -2 .

A. True

B. False

Explain the meaning of 0 in a real-world situation

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Zero as the starting point on a number line	2.MD.B.6
BACKWARD 2	A change of zero means no change	1.OA.D.8
FORWARD 1	Opposite quantities combine to make 0	7.NS.A.1.a
FORWARD 2	Interpret the initial value (the value when $x = 0$)	8.F.B.4

SET 25 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

On a Celsius thermometer, what does 0°C represent?

- A.** No temperature at all
- B.** The temperature at which water freezes
- C.** The hottest possible temperature
- D.** The temperature at which water boils

When numbers describe elevation, what does 0 represent?

- A.** Sea level
- B.** No elevation at all
- C.** The top of a mountain
- D.** The ocean floor

A bank account has a balance of \$0.

What does this mean?

- A.** The account has no money and owes nothing.
- B.** The account owes money.
- C.** The account has money in it.
- D.** The account earns interest.

A football team's change in yards on a play is 0.

What does 0 mean?

- A.** The team scored.
- B.** The team gained 10 yards.
- C.** The team neither gained nor lost yards.
- D.** The team lost all its yards.

A scale shows that your weight changed by 0 pounds since last month.

What does 0 mean in this situation?

Meaning: _____

SET 25 • 6.NS.C.5

BACKWARD 1

Zero as the starting point on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

True or false?

On a number line, 0 is the starting point for measuring distance.

- A. True
- B. False

What number is at point A?



A = _____

How far is 0 from itself on a number line?

- A.** 0 units
- B.** 1 unit
- C.** 2 units
- D.** 10 units

How far is 5 from 0 on a number line?

Units: _____

True or false?

A ruler starts measuring at 1.

A. True

B. False

SET 25 • 6.NS.C.5

BACKWARD 2

A change of zero means no change • Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

What number goes in the box?

$$8 + \square = 8$$

$$\square = \underline{\hspace{10cm}}$$

What number goes in the box?

$$15 - \square = 15$$

$$\square = \underline{\hspace{10cm}}$$

True or false?

A plant was 12 cm tall last week and is 12 cm tall now. It grew 0 cm.

A. True

B. False

A score went from 45 to 45.

What was the change?

- A.** 0
- B.** 1
- C.** 45
- D.** 90

What number goes in the box?

$$\square + 23 = 23$$

$$\square = \underline{\hspace{10cm}}$$

SET 25 • 6.NS.C.5

FORWARD 1

Opposite quantities combine to make 0 • Grade 7 • The Number System •
7.NS.A.1.a

Maya earns \$25 and then spends \$25.

What is the total change in her money?

Total change: _____

True or false?

$$-8 + 8 = 0$$

A. True

B. False

Which sum is equal to 0?

A. $-3.5 + 3.5$

B. $-3.5 + 0$

C. $-3.5 + (-3.5)$

D. $3.5 + 3.5$

A balloon rises 40 feet and then falls 40 feet.
What is its total change in height?

Total change: _____

What number added to 14.2 gives 0?

Number: _____

SET 25 • 6.NS.C.5

FORWARD 2

Interpret the initial value (the value when $x = 0$) • Grade 8 • Functions •
8.F.B.4

The water in a tank is $W = 60 - 3t$, where t is minutes.

What does 60 represent?

Meaning: _____

A hiker's elevation is $E = -40 + 15t$, where t is hours.

What does -40 represent?

- A.** The hiker starts 40 m below sea level.
- B.** The hiker ends 40 m below sea level.
- C.** The hiker climbs 40 m each hour.
- D.** The hike lasts 40 hours.

A phone's battery charge is $B = 100 - 8h$ after h hours.

What is B when $h = 0$, and what does it mean?

Value: _____

Meaning: _____

A savings account has $y = 25x + 150$ dollars after x weeks.

What does 150 represent?

- A.** The starting amount
- B.** The amount saved each week
- C.** The number of weeks
- D.** The amount after 25 weeks

True or false?

For the function $y = 4x$, the output is 0 when the input is 0.

A. True

B. False

Find opposites; the opposite of the opposite of a number

MAIN

Grade 6 main question

6.NS.C.6.a

BACKWARD 1

Distance from 0 on a number line

2.MD.B.6

BACKWARD 2

Positive and negative numbers in context

6.NS.C.5

FORWARD 1

Subtract by adding the opposite

7.NS.A.1.c

FORWARD 2

Solve equations by adding opposites to both sides

8.EE.C.7.b

SET 26 • 6.NS.C.6.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of -7 ?

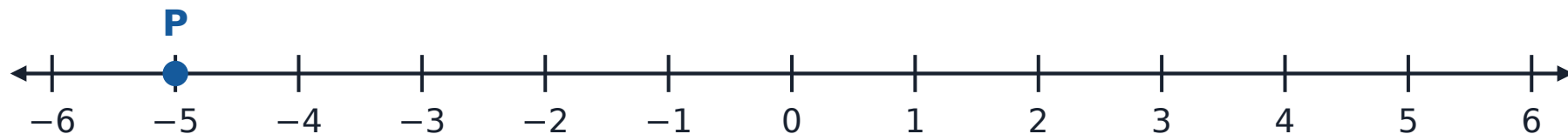
Opposite: _____

Simplify.

$$-(-12)$$

Value: _____

Which number is the opposite of the number at point P?



A. -5

B. 0

C. $\frac{1}{5}$

D. 5

True or false?

The opposite of 0 is 0.

A. True

B. False

What is the opposite of the opposite of 9?

Answer: _____

SET 26 • 6.NS.C.6.a

BACKWARD 1

Distance from 0 on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How far is point A from 0?



Units: _____

How many units from 0 is the number 8?

Units: _____

Which number is 2 units from 0 on this number line?



A. 0

B. 2

C. 4

D. 20

What number does point K represent?



Number: _____

True or false?

The distance from 0 to 7 is 7 units.

A. True

B. False

SET 26 • 6.NS.C.6.a

BACKWARD 2

Positive and negative numbers in context • Grade 6 • The Number System •
6.NS.C.5

True or false?

–5 can represent 5 degrees below zero.

A. True

B. False

Write an integer for a gain of 12 yards.

Integer: _____

Which integer represents 20 feet below sea level?

A. -20

B. -2

C. 0

D. 20

Write an integer for a loss of \$7.

Integer: _____

True or false?

A deposit of \$15 is written as -15 .

A. True

B. False

SET 26 • 6.NS.C.6.a

FORWARD 1

Subtract by adding the opposite • Grade 7 • The Number System • 7.NS.A.1.c

Rewrite $5 - 8$ as an addition expression.

Expression: _____

Find the value.

$$-3 - (-7)$$

Value: _____

True or false?

$$6 - (-2) = 6 + 2$$

A. True

B. False

Which expression is equal to $-4 - 9$?

A. $4 + 9$

B. $4 - (-9)$

C. $-4 + 9$

D. $-4 + (-9)$

Find the value.

$$2.5 - 6$$

Value: _____

SET 26 • 6.NS.C.6.a

FORWARD 2

Solve equations by adding opposites to both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$5x - 7 = 2x + 8$$

x = _____

Solve for x.

$$-3x + 4 = 3x - 8$$

x = _____

Solve for x.

$$6 - 2x = 4x - 12$$

A. -3

B. 1

C. 3

D. 9

Solve for x.

$$7x + 3 = -2x + 21$$

x = _____

Solve for x.

$$4(x - 1) = 2x + 10$$

x = _____

Use signs of coordinates to identify quadrants

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Axes, origin, and ordered pairs	5.G.A.1
BACKWARD 2	Negative numbers are on the opposite side of 0	6.NS.C.6.a
BACKWARD 3	Reading points in the first quadrant	5.G.A.2
FORWARD 1	Signs of products of rational numbers	7.NS.A.2.a
FORWARD 2	Transformations and the quadrant of the image	8.G.A.3

SET 27 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

In which quadrant is the point $(-4, 7)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

In which quadrant is the point $(3, -8)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

In which quadrant is the point $(-2.5, -6)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

A point has a negative x-coordinate and a positive y-coordinate.

In which quadrant is it?

Quadrant: _____

Which point is in Quadrant III?

A. $(-5, -1)$

B. $(5, 1)$

C. $(5, -1)$

D. $(-5, 1)$

SET 27 • 6.NS.C.6.b

BACKWARD 1

Axes, origin, and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

The origin is the point $(0, 0)$.

A. True

B. False

In the ordered pair (5, 2), which number tells how far to move along the x-axis?

A. 2

B. 3

C. 5

D. 7

What is the name of the point where the x-axis and the y-axis meet?

Name: _____

True or false?

In the ordered pair (3, 8), the y-coordinate is 3.

A. True

B. False

Which axis is horizontal?

A. The x-axis

B. The y-axis

SET 27 • 6.NS.C.6.b

BACKWARD 2

Negative numbers are on the opposite side of 0 • Grade 6 • The Number System • 6.NS.C.6.a

True or false?

On a horizontal number line, -4 is to the left of 0 .

- A. True
- B. False

On a vertical number line, where is -3 ?

- A.** Above 0
- B.** Above 3
- C.** Below 0
- D.** At 0

True or false?

On a vertical number line, 5 is below 0.

A. True

B. False

On a horizontal number line, is 2.5 to the left or the right of 0?

Answer: _____

Which number is to the left of 0 on a horizontal number line?

A. -6

B. 0

C. $\frac{1}{2}$

D. 6

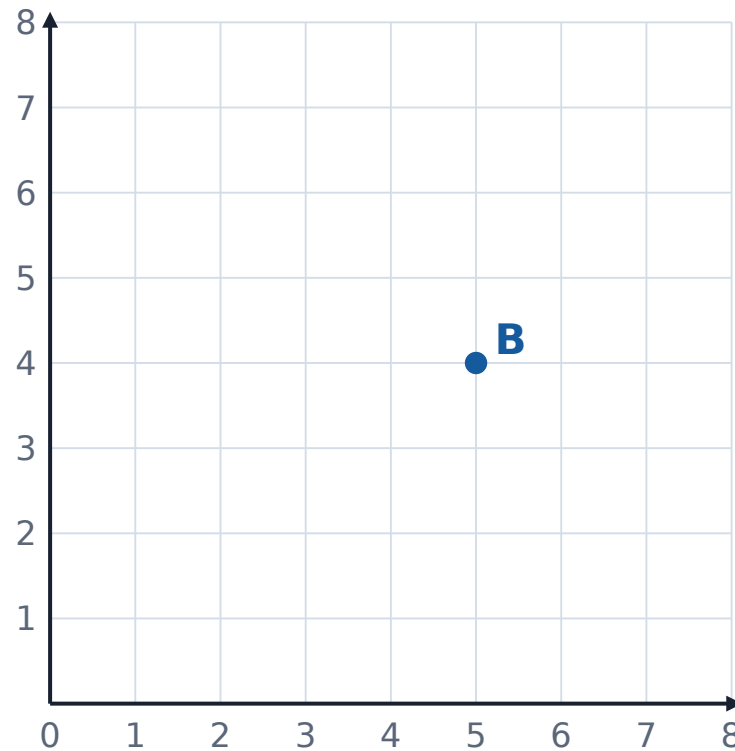
SET 27 • 6.NS.C.6.b

BACKWARD 3

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

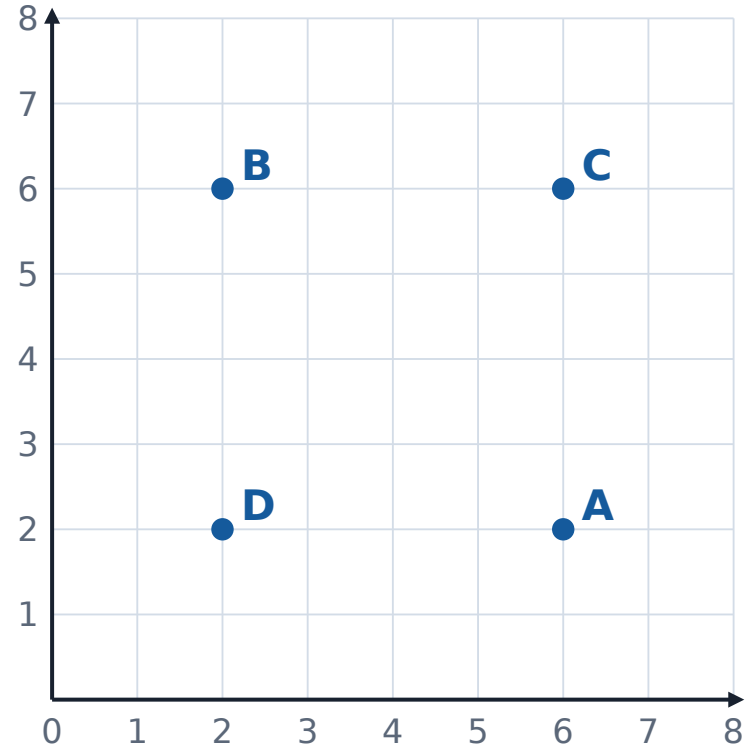
What are the coordinates of point B?

B = _____



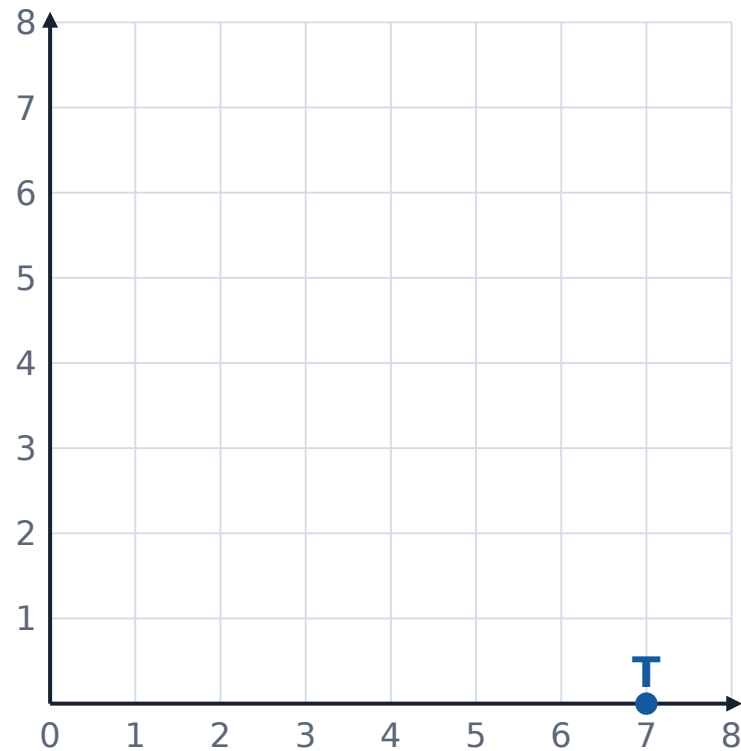
Which point is located at (2, 6)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



What are the coordinates of point T?

T = _____

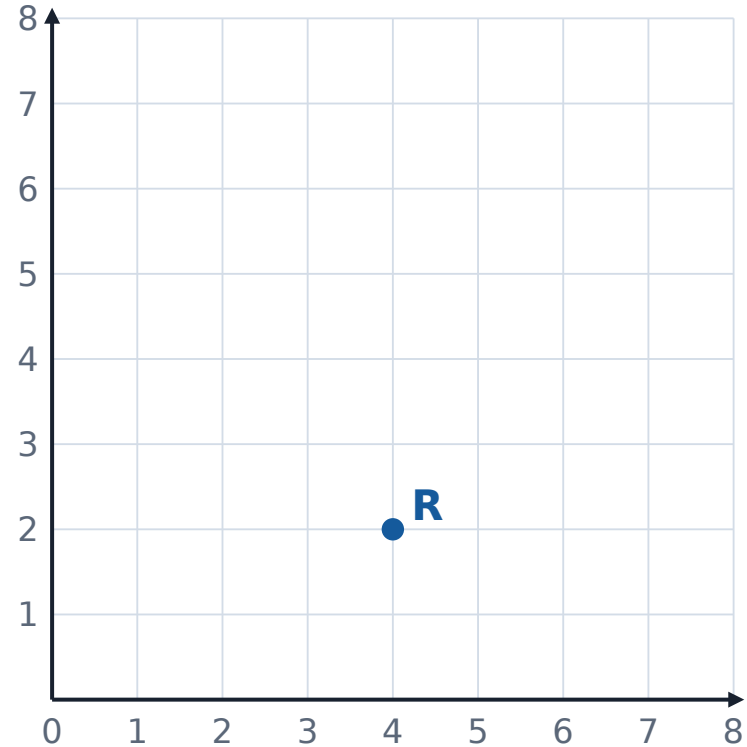


True or false?

Point R is located at (2, 4).

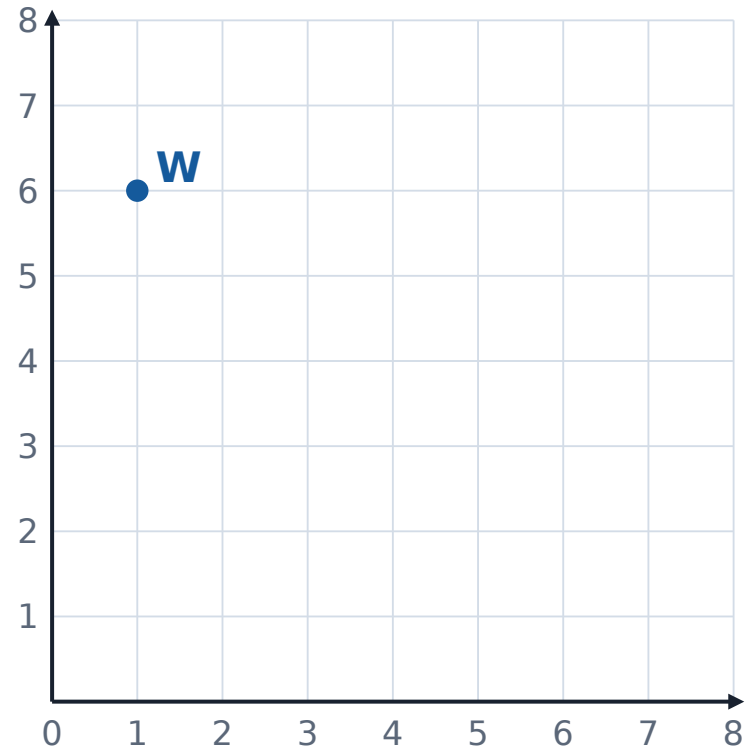
A. True

B. False



What are the coordinates of point W?

W = _____



SET 27 • 6.NS.C.6.b

FORWARD 1

Signs of products of rational numbers • Grade 7 • The Number System •
7.NS.A.2.a

The point (x, y) is in Quadrant II.

Is the product xy positive or negative?

Answer: _____

The point (x, y) is in Quadrant III.

Is the product xy positive or negative?

Answer: _____

For the point (x, y) , the product xy is positive and x is negative.

In which quadrant is the point?

A. Quadrant I

B. Quadrant II

C. Quadrant III

D. Quadrant IV

True or false?

If (x, y) is in Quadrant IV, then the product xy is negative.

A. True

B. False

Find the product of the coordinates of the point $(-3, 5)$.

Product: _____

SET 27 • 6.NS.C.6.b

FORWARD 2

Transformations and the quadrant of the image • Grade 8 • Geometry •
8.G.A.3

The point (2, 5) is rotated 180° about the origin.

What are the coordinates of the image, and in which quadrant is it?

Image: _____

Quadrant: _____

A point in Quadrant II is reflected across the y -axis.
In which quadrant is the image?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

The point $(-3, -4)$ is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

True or false?

Translating $(1, 2)$ 5 units to the left moves it into Quadrant II.

A. True

B. False

The point $(4, -1)$ is reflected across the x-axis.
In which quadrant is the image?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

Reflect points across one or both axes

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Opposite numbers	6.NS.C.6.a
BACKWARD 2	Lines of symmetry	4.G.A.3
BACKWARD 3	Reading coordinates in the first quadrant	5.G.A.2
FORWARD 1	Distance between a point and its reflection	7.NS.A.1.c
FORWARD 2	Reflections and rotations with coordinates	8.G.A.3

SET 28 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

The point $(3, -5)$ is reflected across the x -axis.
What are the coordinates of its image?

Image: _____

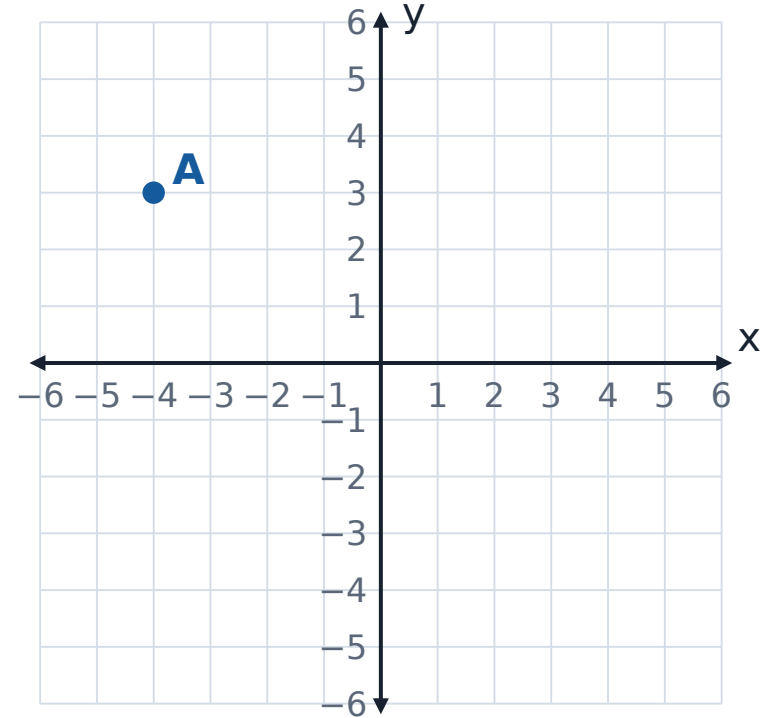
The point $(-2, 6)$ is reflected across the y -axis.
What are the coordinates of its image?

Image: _____

Point A is reflected across the y-axis.

What are the coordinates of its image?

Image: _____



Which point is the reflection of $(-7, -2)$ across the x-axis?

- A.** $(7, 2)$
- B.** $(-2, -7)$
- C.** $(-7, 2)$
- D.** $(7, -2)$

Point B(5, 1) is reflected across the y-axis and then across the x-axis.

What are the coordinates of the final image?

Image: _____

SET 28 • 6.NS.C.6.b

BACKWARD 1

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 6?

Opposite: _____

True or false?

The opposite of -9 is 9 .

A. True

B. False

Which pair of numbers are opposites?

- A.** 0 and 4
- B.** -4 and -4
- C.** 4 and $\frac{1}{4}$
- D.** -4 and 4

What is the opposite of -2.5 ?

Opposite: _____

True or false?

$$-(-3) = -3$$

A. True

B. False

SET 28 • 6.NS.C.6.b

BACKWARD 2

Lines of symmetry • Grade 4 • Geometry • 4.G.A.3

True or false?

A line of symmetry divides a figure into two matching halves.

A. True

B. False

How many lines of symmetry does a square have?

A. 0

B. 1

C. 2

D. 4

True or false?

The dashed line is a line of symmetry of the rectangle.

A. True

B. False



Which letter has a vertical line of symmetry?

A. F

B. A

C. J

D. R

How many lines of symmetry does a rectangle that is not a square have?

Lines: _____

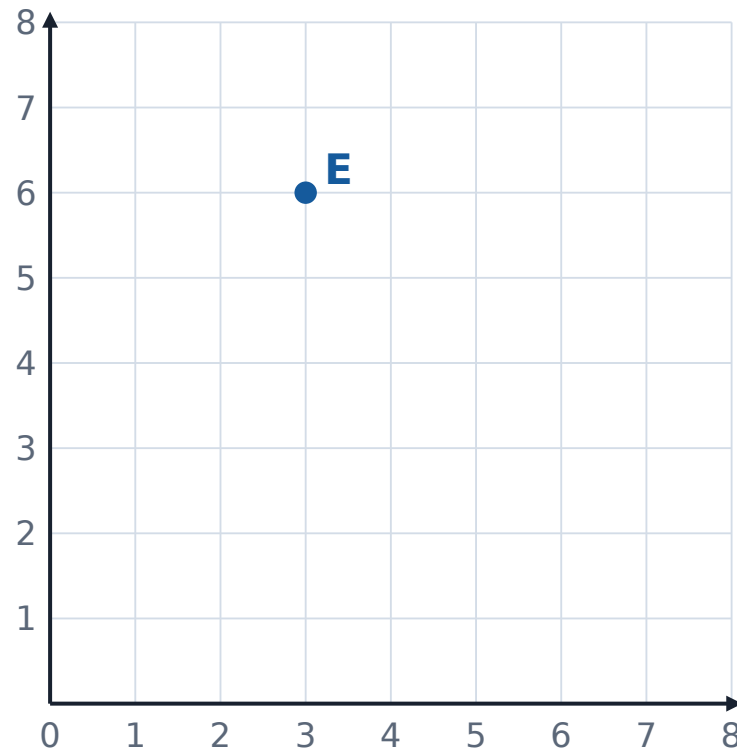
SET 28 • 6.NS.C.6.b

BACKWARD 3

Reading coordinates in the first quadrant • Grade 5 • Geometry • 5.G.A.2

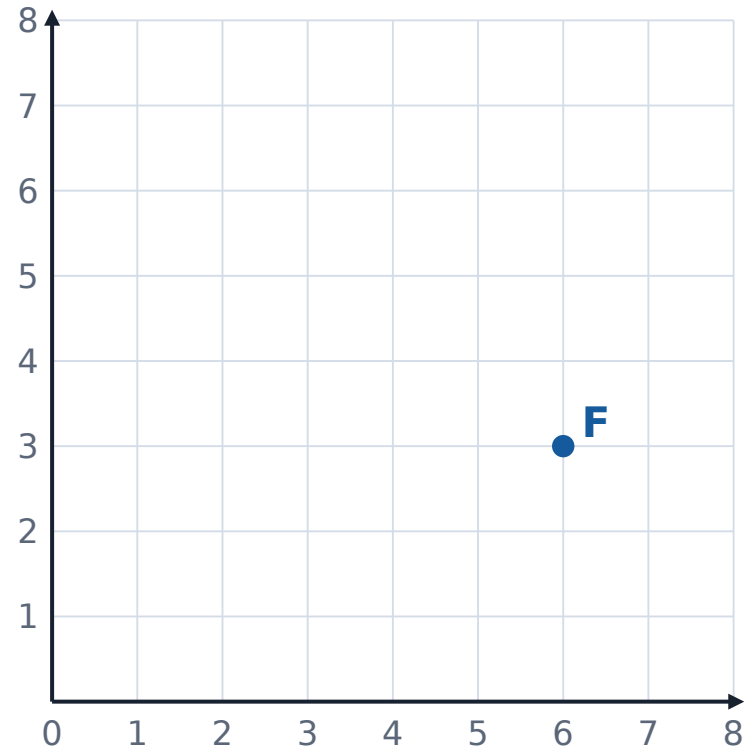
What are the coordinates of point E?

E = _____



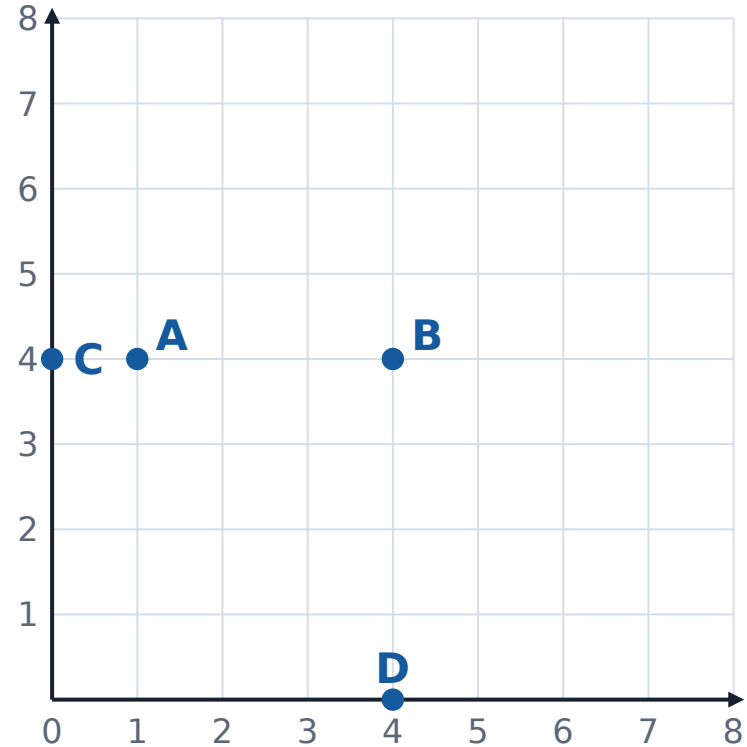
What are the coordinates of point F?

F = _____



Which point is located at (4, 0)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

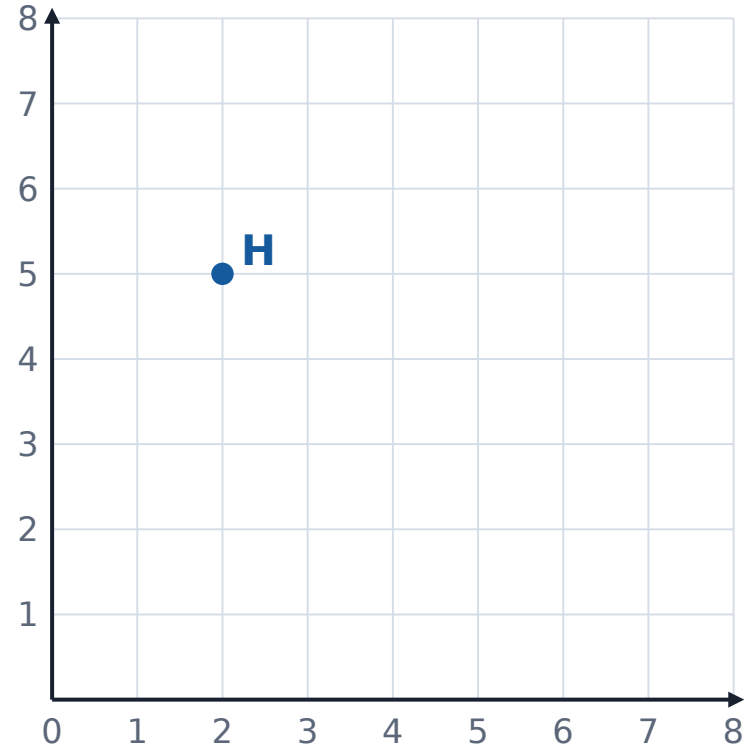


True or false?

Point H is located at (5, 2).

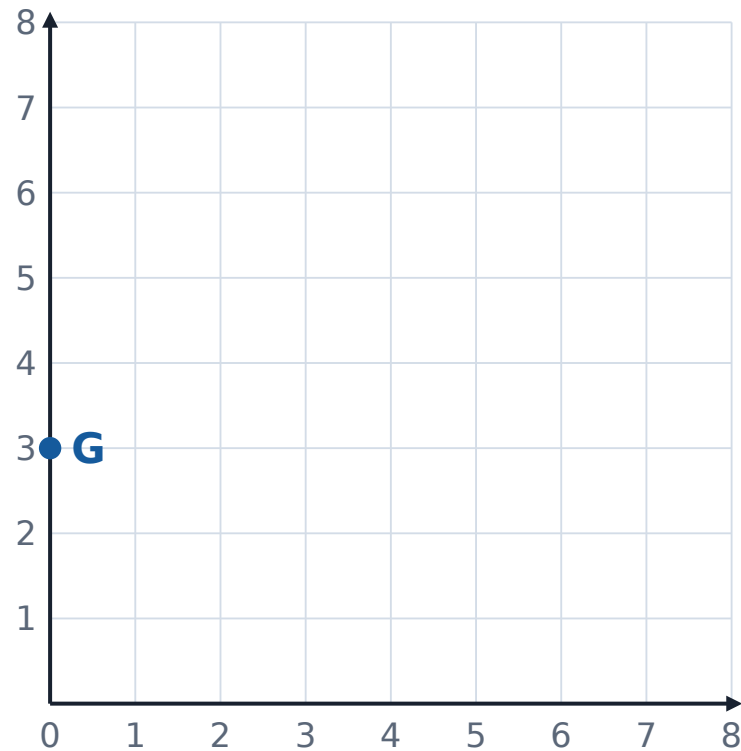
A. True

B. False



What are the coordinates of point G?

G = _____



SET 28 • 6.NS.C.6.b

FORWARD 1

Distance between a point and its reflection • Grade 7 • The Number System •
7.NS.A.1.c

The point $(3, -4.5)$ is reflected across the x-axis.
How far apart are the point and its image?

Distance: _____

The point $(-6.5, 2)$ is reflected across the y -axis.
How far apart are the point and its image?

Distance: _____

The point $(1, -2\frac{1}{2})$ is reflected across the x-axis.

How far apart are the point and its image?

A. -5

B. 0

C. $2\frac{1}{2}$

D. 5

A point in Quadrant I and its reflection across the y-axis are 11 units apart. The point's y-coordinate is 7.

What is the point's x-coordinate?

X = _____

True or false?

The point $(-4, 3)$ and its reflection across the x-axis are 6 units apart.

A. True

B. False

SET 28 • 6.NS.C.6.b

FORWARD 2

Reflections and rotations with coordinates • Grade 8 • Geometry • 8.G.A.3

A triangle has a vertex at $(2, -3)$. The triangle is reflected across the y -axis.

What are the coordinates of the image of that vertex?

Image: _____

Which transformation is described by $(x, y) \rightarrow (x, -y)$?

- A.** A rotation of 180°
- B.** A reflection across the x-axis
- C.** A translation
- D.** A reflection across the y-axis

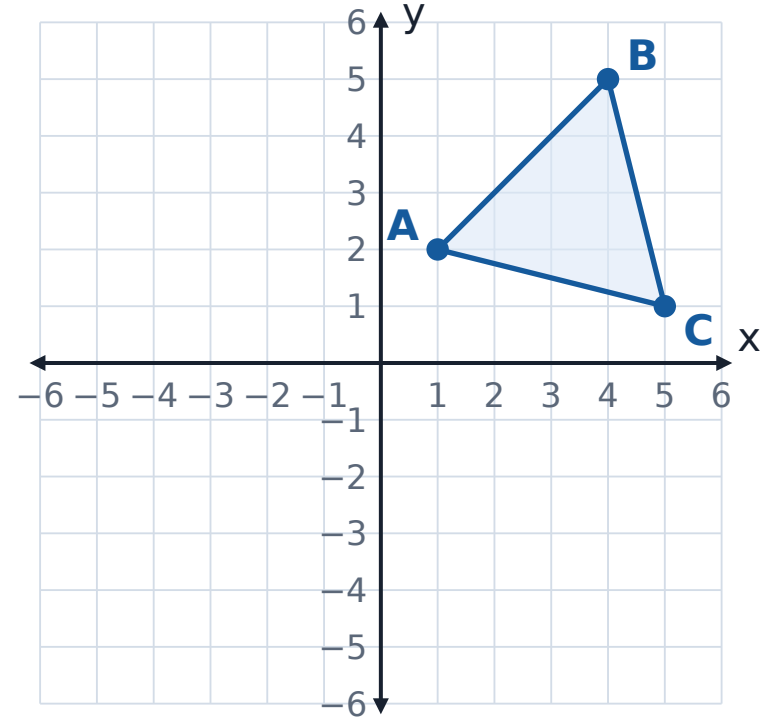
The point $(4, 1)$ is rotated 180° about the origin.
What are the coordinates of its image?

Image: _____

Triangle ABC is reflected
across the x-axis.

What are the coordinates of
 B' ?

$B' =$ _____



True or false?

Reflecting $(-3, 5)$ across the y -axis gives $(3, 5)$.

A. True

B. False

Name the rational number at a point on a number line

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Fractions on a number line	3.NF.A.2
BACKWARD 2	Decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Addition shown on a number line	7.NS.A.1.b
FORWARD 2	Approximate irrational numbers on a number line	8.NS.A.2

SET 29 • 6.NS.C.6.c

MAIN

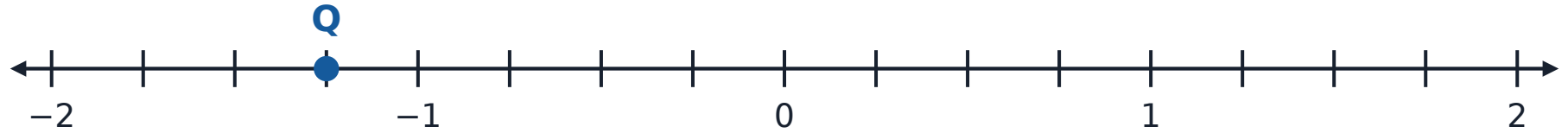
Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?



P = _____

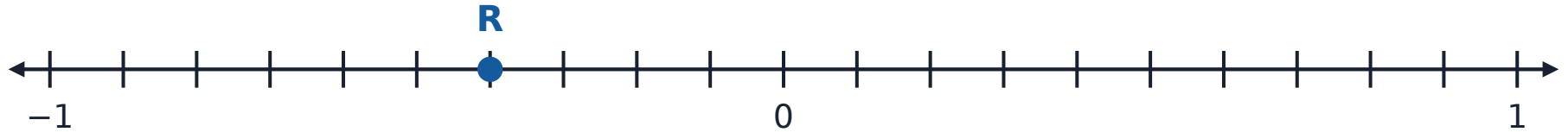
Write the number at point Q as a mixed number.



Q = _____

True or false?

Point R is located at -0.4 .



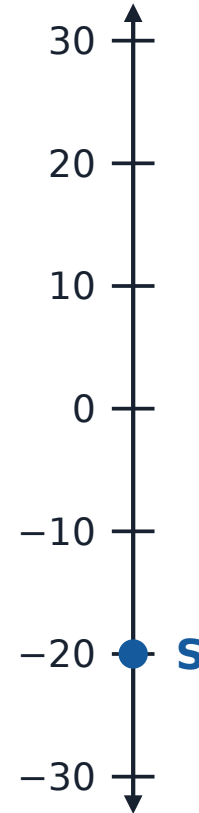
A. True

B. False

The number line shows
elevations in meters.

What elevation does point S
show?

S = _____



Write the number at point T as a fraction.



T = _____

SET 29 • 6.NS.C.6.c

BACKWARD 1

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction does point A represent?



A = _____

What fraction does point B represent?



B = _____

True or false?

Point C is located at $\frac{5}{6}$.



A. True

B. False

The space from 0 to 1 is split into equal parts.
How many equal parts are there?



A. 7

B. 8

C. 9

D. 10

Write the number at point D as a fraction.



D = _____

SET 29 • 6.NS.C.6.c

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal does point A represent?



A = _____

What decimal does point B represent?



B = _____

True or false?

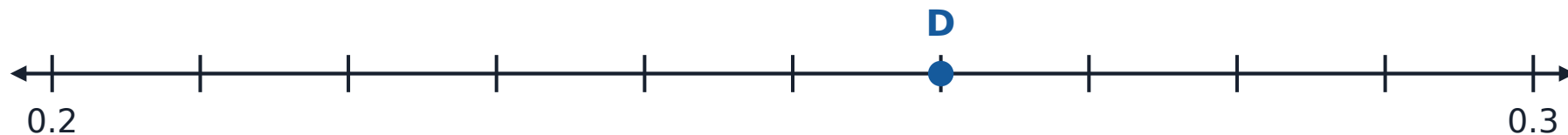
Point C is located at 0.3.



A. True

B. False

What decimal does point D represent?



A. 0.206

B. 0.26

C. 0.62

D. 2.6

What decimal does point E represent?



E = _____

SET 29 • 6.NS.C.6.c

BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point B (not shown) is the opposite of point A.

What number is at point B?



B = _____

True or false?

–2.5 and 2.5 are the same distance from 0.

A. True

B. False

What is the opposite of $1\frac{3}{4}$?

A. $-1\frac{3}{4}$

B. $-\frac{4}{7}$

C. $\frac{4}{7}$

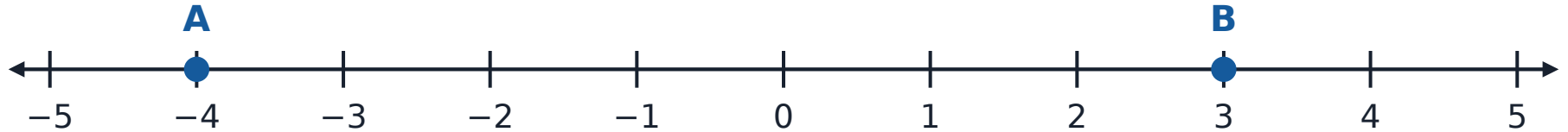
D. $1\frac{3}{4}$

What is the opposite of -0.8 ?

Opposite: _____

True or false?

Points A and B represent opposite numbers.



A. True

B. False

SET 29 • 6.NS.C.6.c

FORWARD 1

Addition shown on a number line • Grade 7 • The Number System • 7.NS.A.1.b

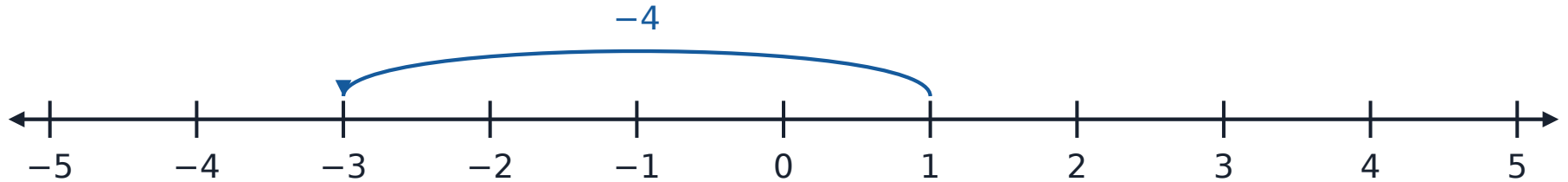
Start at -2.5 on a number line and move 4 units to the right.

Where do you end?

Number: _____

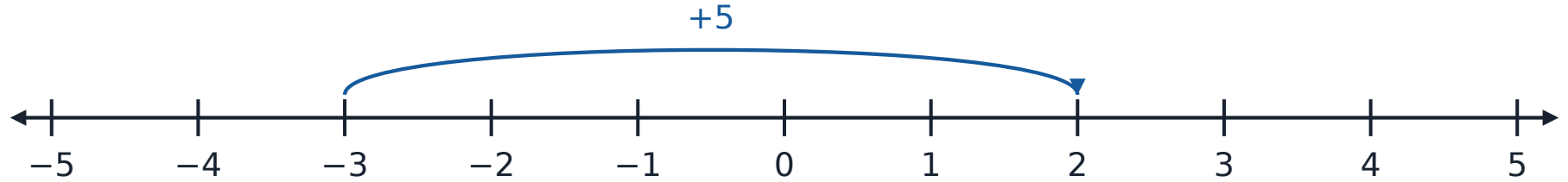
The number line shows $1 + (-4)$.

What is the sum?



Sum: _____

Which expression does the number line show?



A. $5 + 3$

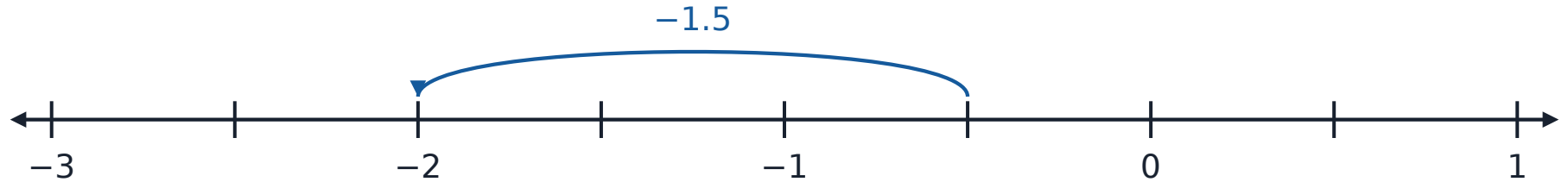
B. $-3 - 5$

C. $-3 + 5$

D. $2 + 5$

The number line shows $-0.5 + (-1.5)$.

What is the sum?



Sum: _____

True or false?

On a number line, $-3 + 7$ is 7 units to the right of -3 .

A. True

B. False

SET 29 • 6.NS.C.6.c

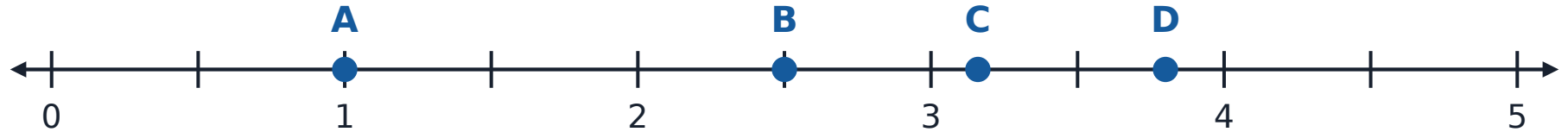
FORWARD 2

Approximate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

Between which two consecutive whole numbers is $\sqrt{20}$?

Between: _____

Which point best shows $\sqrt{10}$?



A. Point A

B. Point B

C. Point C

D. Point D

Estimate $\sqrt{50}$ to the nearest tenth.

Estimate: _____

Which list is ordered from least to greatest?

A. 2.5, $\sqrt{5}$, $\sqrt{8}$

B. $\sqrt{5}$, 2.5, $\sqrt{8}$

C. $\sqrt{8}$, $\sqrt{5}$, 2.5

D. $\sqrt{5}$, $\sqrt{8}$, 2.5

True or false?

π is between 3.1 and 3.2 on the number line.

A. True

B. False

Plot rational numbers on a horizontal or vertical number line

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Placing fractions on a number line	3.NF.A.2
BACKWARD 2	Placing decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Represent addition on a number line	7.NS.A.1.b
FORWARD 2	Locate irrational numbers approximately on a number line	8.NS.A.2

SET 30 • 6.NS.C.6.c

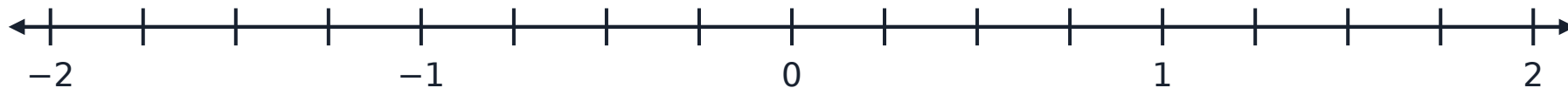
MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label point A at -2.5 on the number line.



Plot and label point B at $-\frac{3}{4}$ on the number line.

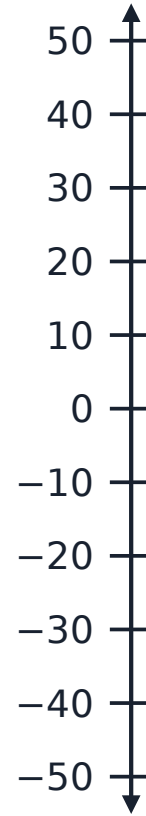


Plot and label -1.2 and its opposite on the number line.

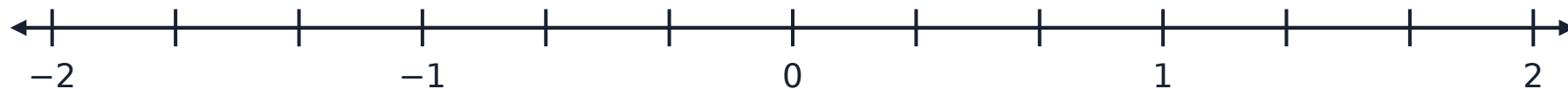


The number line shows
elevations in feet.

Plot and label point C at -35
feet.



Plot and label point D at $1\frac{2}{3}$ on the number line.



SET 30 • 6.NS.C.6.c

BACKWARD 1

Placing fractions on a number line • Grade 3 • Number & Operations–Fractions
• 3.NF.A.2

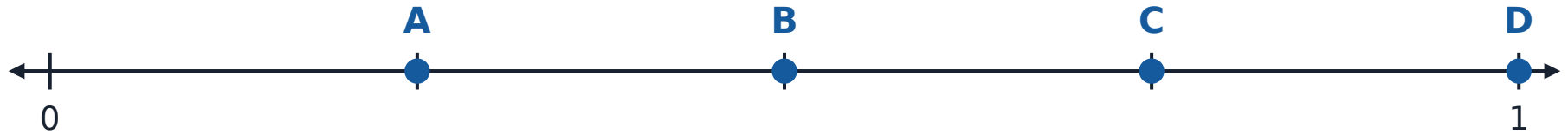
Plot and label $\frac{3}{4}$ on the number line.



Plot and label $\frac{2}{3}$ on the number line.



Which point is at $\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

Plot and label $\frac{5}{6}$ on the number line.



True or false?

To plot $\frac{3}{8}$, count 3 of the 8 equal parts from 0.



A. True

B. False

SET 30 • 6.NS.C.6.c

BACKWARD 2

Placing decimals on a number line • Grade 4 • Number & Operations–Fractions
• 4.NF.C.6

Plot and label 0.7 on the number line.



Plot and label 2.4 on the number line.



Which point is at 0.6?



A. Point A

B. Point B

C. Point C

D. Point D

Plot and label 0.35 on the number line.



True or false?

On a number line marked in tenths, 0.9 is one tick mark to the left of 1.

A. True

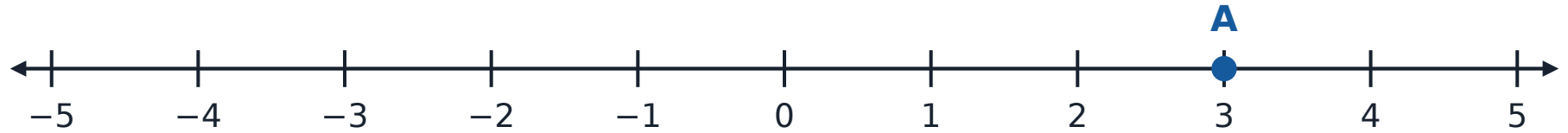
B. False

SET 30 • 6.NS.C.6.c

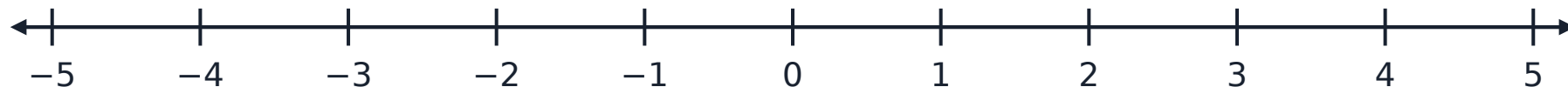
BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point A is at 3. Plot and label its opposite.



Plot and label 2 and -2 on the number line.



True or false?

A number and its opposite are the same distance from 0.

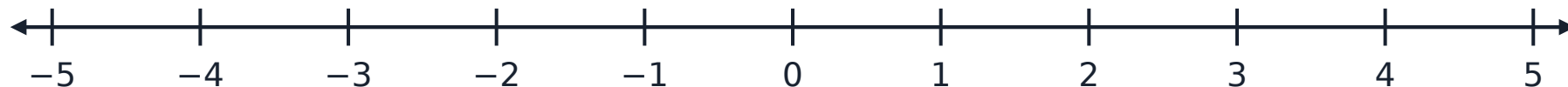
A. True

B. False

Point B is at -1.5 . Where is its opposite?

- A.** At -1.5
- B.** At 1.5
- C.** At 0
- D.** At 0.15

Plot and label the opposite of -4 .



SET 30 • 6.NS.C.6.c

FORWARD 1

Represent addition on a number line • Grade 7 • The Number System •
7.NS.A.1.b

Draw an arrow on the number line to show $-2 + 3.5$.

What is the sum?



Sum: _____

Draw arrows on the number line to show $1.5 + (-3)$.

What is the sum?

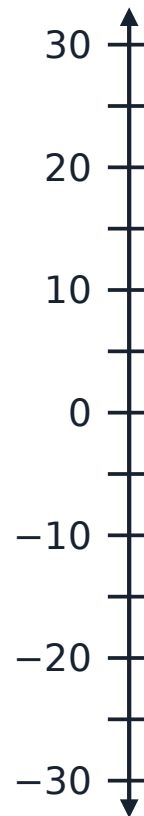


Sum: _____

Draw arrows on the vertical
number line to show
 $-10 + 25$.

What is the sum?

Sum: _____



An arrow on a number line starts at $-\frac{1}{2}$ and moves $\frac{3}{4}$ unit to the right.

Where does it end?

A. $-1\frac{1}{4}$

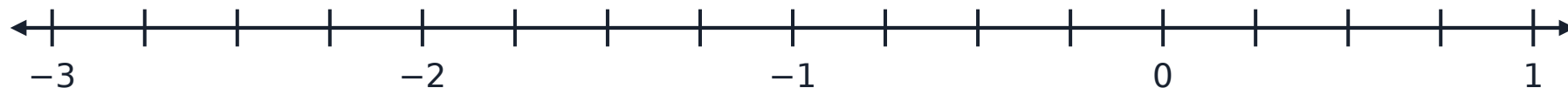
B. $-\frac{1}{4}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

Draw arrows on the number line to show $-1\frac{1}{4} + (-\frac{3}{4})$.

What is the sum?



Sum: _____

SET 30 • 6.NS.C.6.c

FORWARD 2

Locate irrational numbers approximately on a number line • Grade 8 • The Number System • 8.NS.A.2

Plot $\sqrt{10}$ at its approximate location on the number line.



Plot $\sqrt{2}$ at its approximate location on the number line.



Plot π at its approximate location on the number line.



Plot $-\sqrt{5}$ at its approximate location on the number line.



Plot $\sqrt{30}$ at its approximate location on the number line.



Name the coordinates of points in all four quadrants

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Ordered pairs and the axes	5.G.A.1
BACKWARD 2	Reading points in the first quadrant	5.G.A.2
BACKWARD 3	Quadrants and signs of coordinates	6.NS.C.6.b
FORWARD 1	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d
FORWARD 2	A function as a set of ordered pairs	8.F.A.1

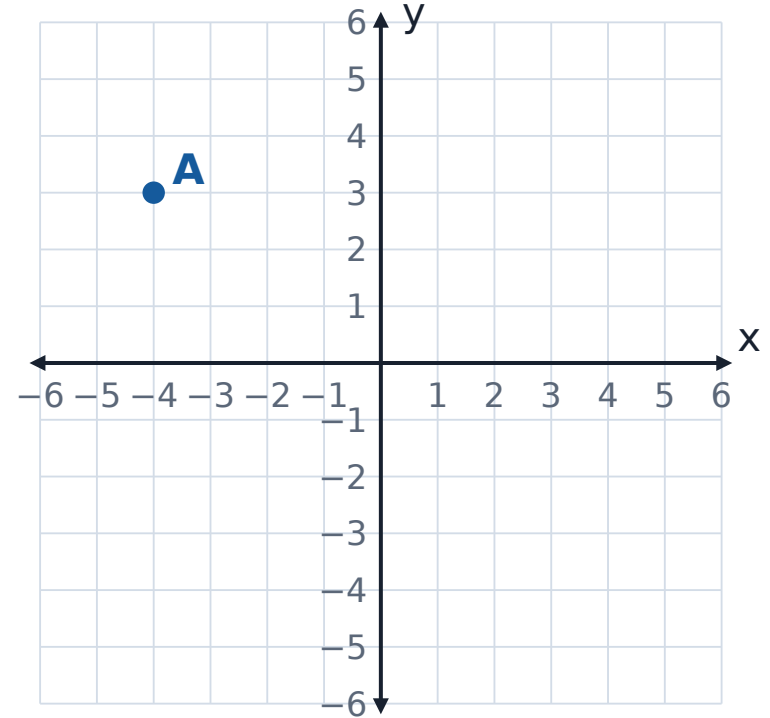
SET 31 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

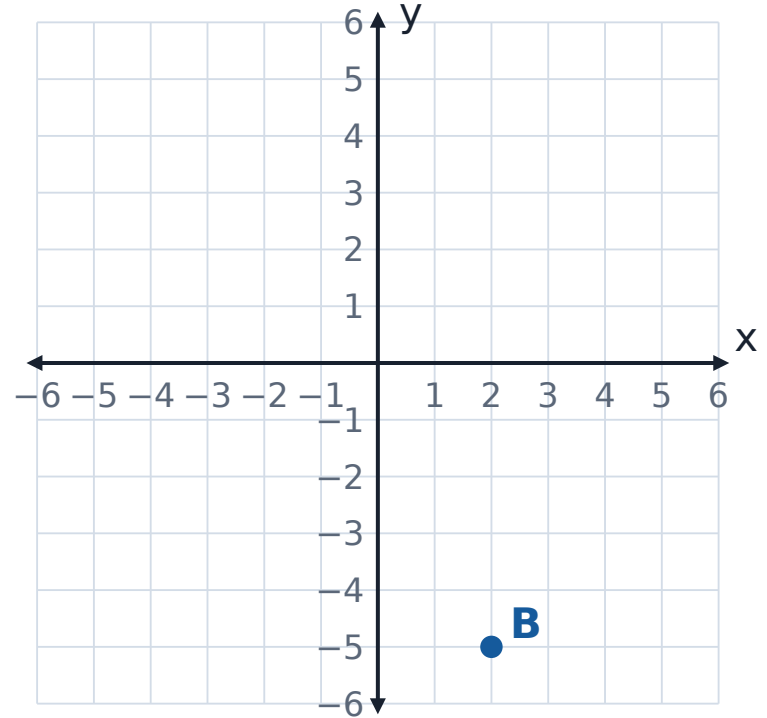
What are the coordinates of point A?

A = _____



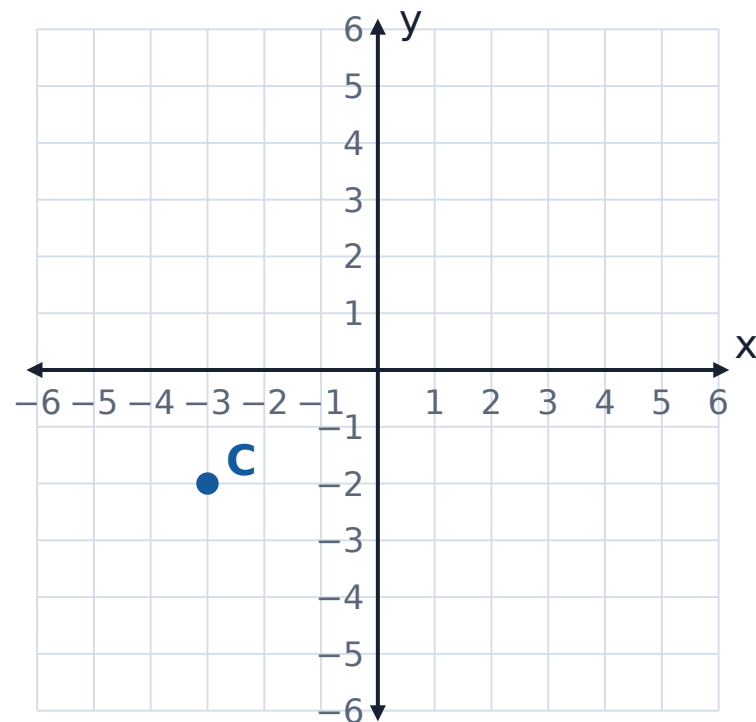
What are the coordinates of point B?

B = _____



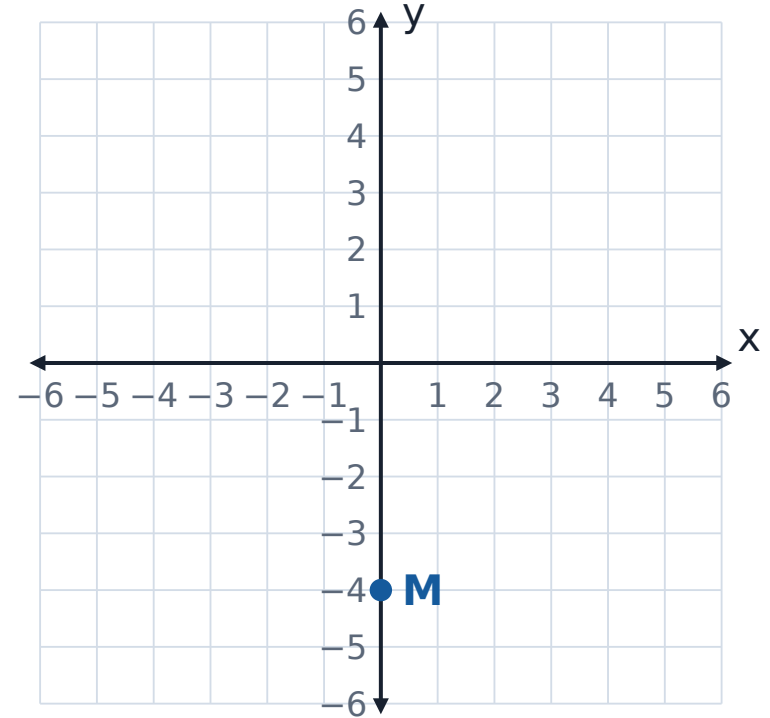
What are the coordinates of point C?

C = _____



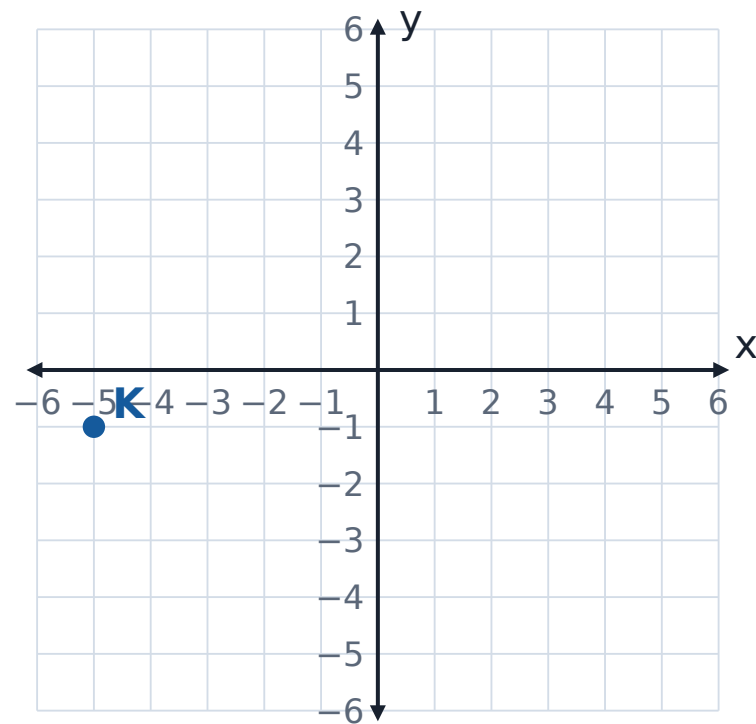
Which ordered pair names point M?

- A.** $(0, 4)$
- B.** $(4, 0)$
- C.** $(0, -4)$
- D.** $(-4, 0)$



What are the coordinates of point K?

K = _____



SET 31 • 6.NS.C.6.c

BACKWARD 1

Ordered pairs and the axes • Grade 5 • Geometry • 5.G.A.1

True or false?

The x-coordinate tells how far to move from the origin along the x-axis.

A. True

B. False

In the ordered pair $(7, 3)$, what is the y-coordinate?

- A.** 3
- B.** 4
- C.** 7
- D.** 10

The x-axis and the y-axis cross at a point.

What are its name and coordinates?

Name: _____

Coordinates: _____

True or false?

$(2, 5)$ and $(5, 2)$ name the same point.

A. True

B. False

In an ordered pair, which number is written first?

- A.** The smaller number
- B.** The x-coordinate
- C.** The y-coordinate
- D.** The larger number

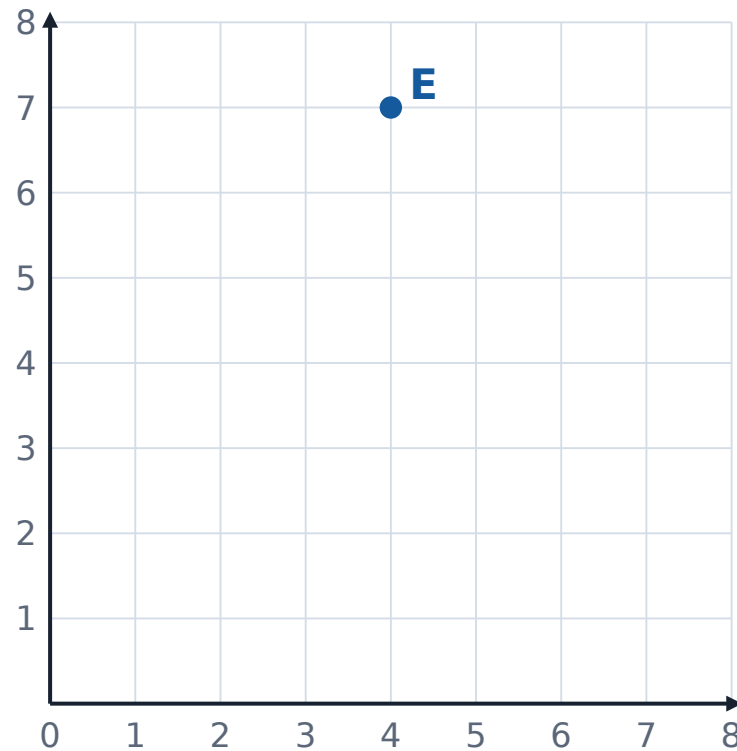
SET 31 • 6.NS.C.6.c

BACKWARD 2

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

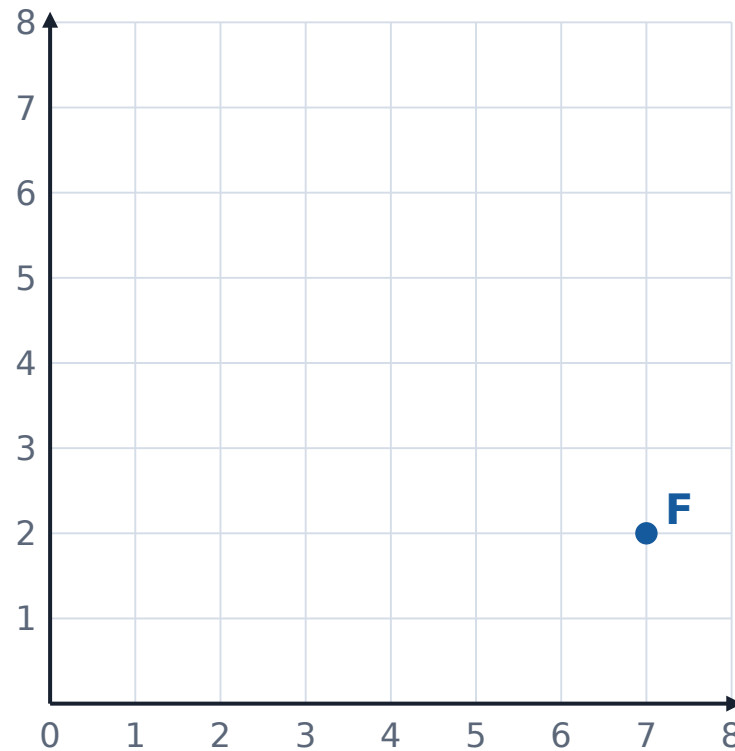
What are the coordinates of point E?

E = _____



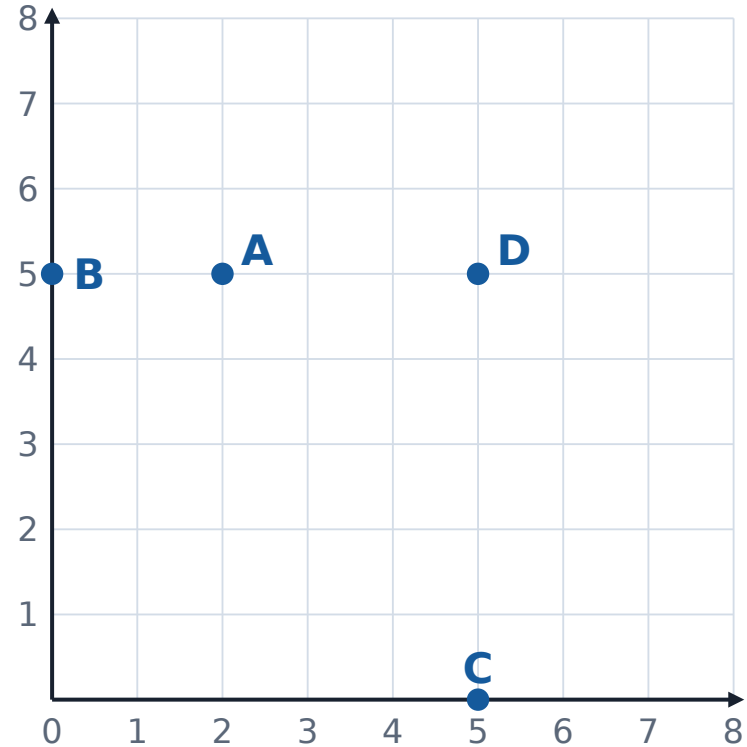
What are the coordinates of point F?

F = _____



Which point is located at (0, 5)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

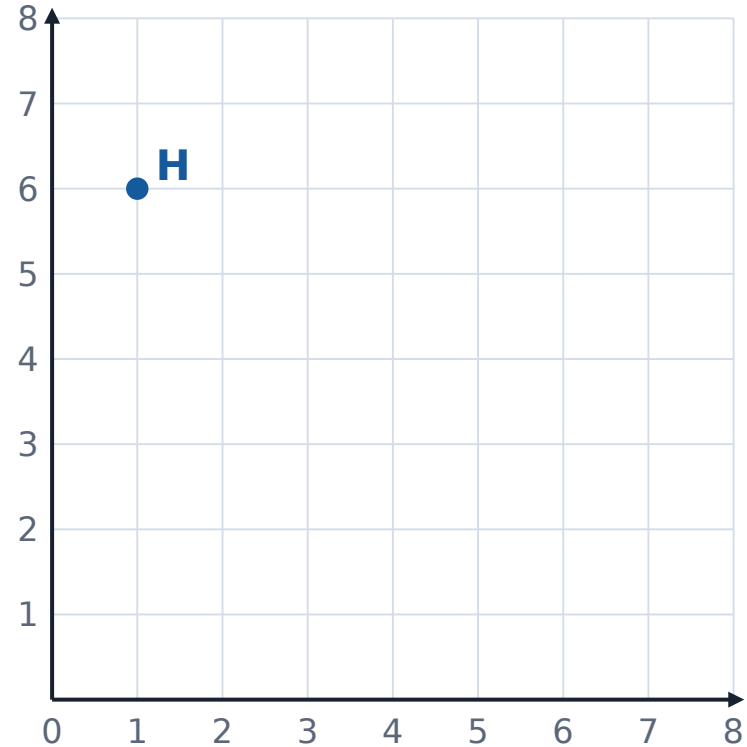


True or false?

Point H is located at (6, 1).

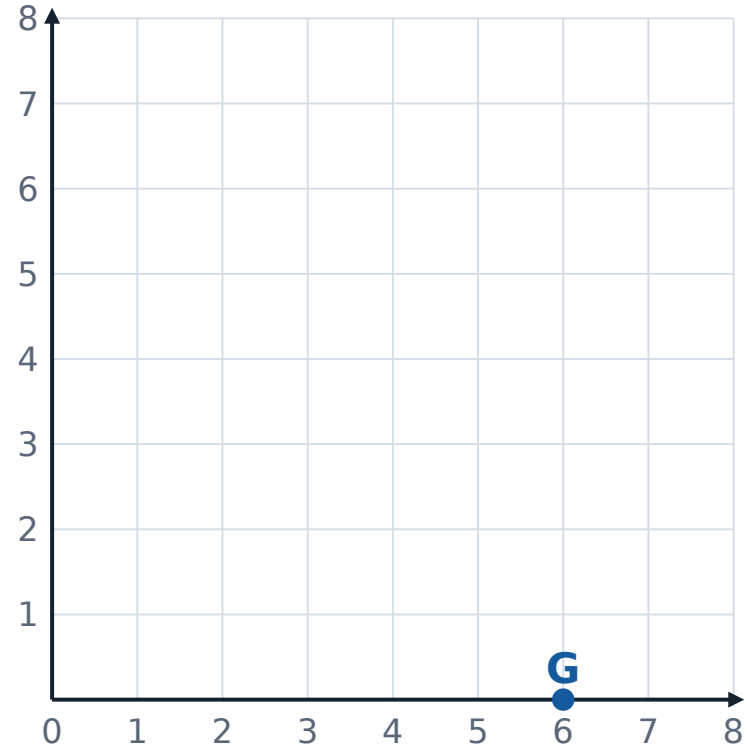
A. True

B. False



What are the coordinates of point G?

G = _____



SET 31 • 6.NS.C.6.c

BACKWARD 3

Quadrants and signs of coordinates • Grade 6 • The Number System •
6.NS.C.6.b

In which quadrant are both coordinates negative?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

In which quadrant is $(5, -2)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

True or false?

$(-3, 4)$ is in Quadrant II.

A. True

B. False

A point is in Quadrant IV.

Write "positive" or "negative" for each coordinate.

x-coordinate: _____

y-coordinate: _____

True or false?

The point $(0, -6)$ is in Quadrant III.

A. True

B. False

SET 31 • 6.NS.C.6.c

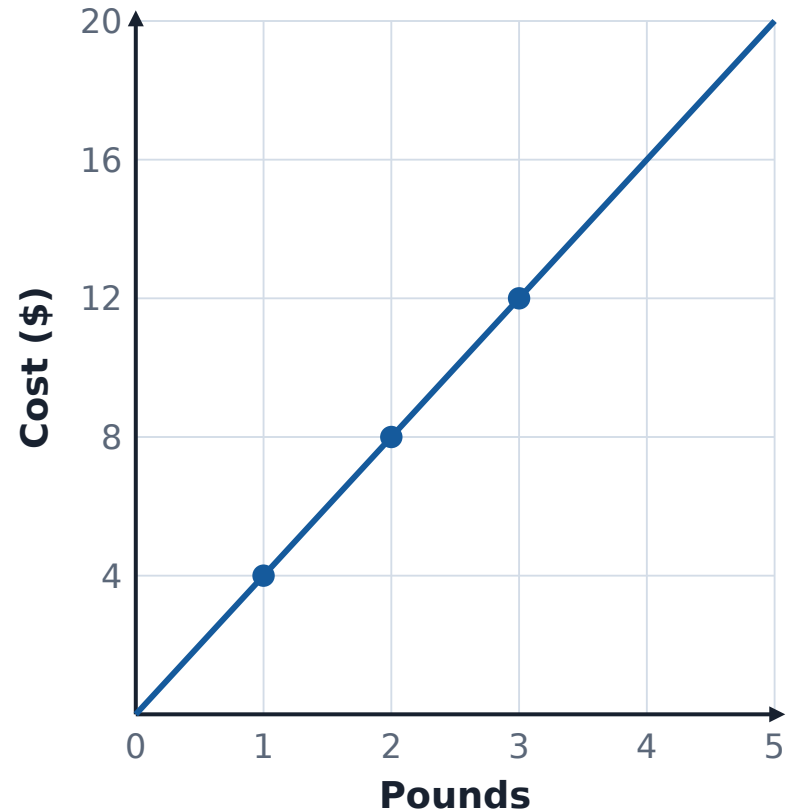
FORWARD 1

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of apples.

What does the point (2, 8) represent?

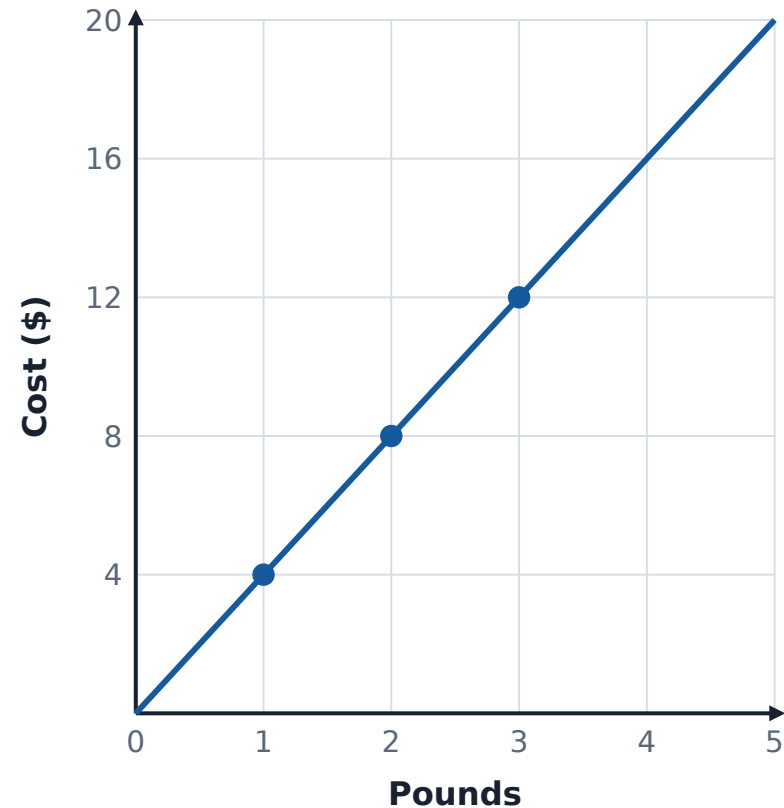
Meaning: _____



The graph shows the cost of apples.

What does the point (1, 4) represent?

- A.** 4 pounds cost \$1.
- B.** 1 pound costs \$4.
- C.** \$1 buys 4 pounds.
- D.** The cost starts at \$4.



True or false?

On the graph of a proportional relationship between pounds and cost, the point $(0, 0)$ means 0 pounds cost \$0.

- A.** True
- B.** False

The graph of a proportional relationship passes through $(1, r)$ and $(5, 30)$.

What is r ?

$r =$ _____

A graph shows the distance y in miles after x hours. It passes through $(3, 150)$.

What does this point mean?

- A.** In 3 hours, the distance is 150 miles.
- B.** The speed is 150 miles per hour.
- C.** In 150 hours, the distance is 3 miles.
- D.** The speed is 3 miles per hour.

SET 31 • 6.NS.C.6.c

FORWARD 2

A function as a set of ordered pairs • Grade 8 • Functions • 8.F.A.1

True or false?

The set of ordered pairs $(1, 2)$, $(2, 4)$, $(3, 6)$, $(4, 8)$ represents a function.

A. True

B. False

Which set of ordered pairs does NOT represent a function?

A. $(-1, 3), (0, 5), (1, 7)$

B. $(1, 4), (2, 4), (3, 4)$

C. $(0, 0), (1, 1), (2, 2)$

D. $(2, 5), (2, 7), (3, 9)$

Does the set $(4, 1), (5, 2), (4, 3)$ represent a function?
Write yes or no.

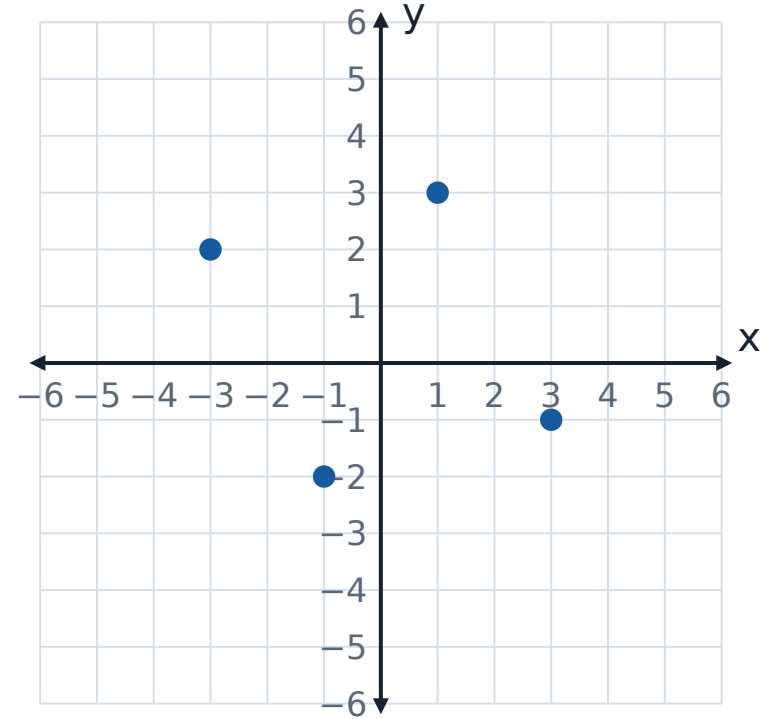
Answer: _____

True or false?

The graph represents a function.

A. True

B. False



A function assigns to each input ____.

- A.** at least two outputs
- B.** no output
- C.** the same output as the input
- D.** exactly one output

Plot points in all four quadrants

MAIN

Grade 6 main question

6.NS.C.6.c

BACKWARD 1

Which direction to move first

5.G.A.1

BACKWARD 2

Plotting points in the first quadrant

5.G.A.2

BACKWARD 3

Direction from the sign of a coordinate

6.NS.C.6.b

FORWARD 1

Graph pairs to decide whether a relationship is proportional

7.RP.A.2.a

FORWARD 2

Graph proportional relationships

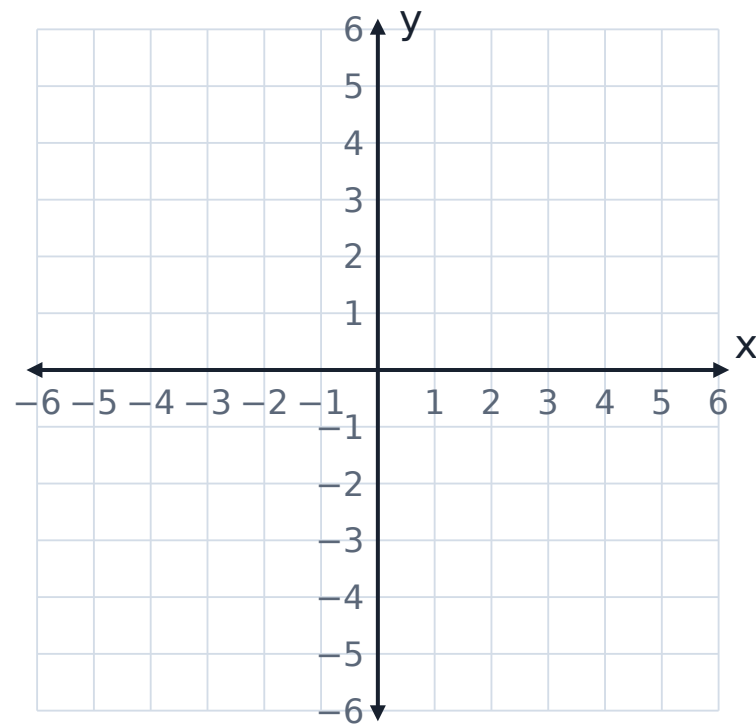
8.EE.B.5

SET 32 • 6.NS.C.6.c

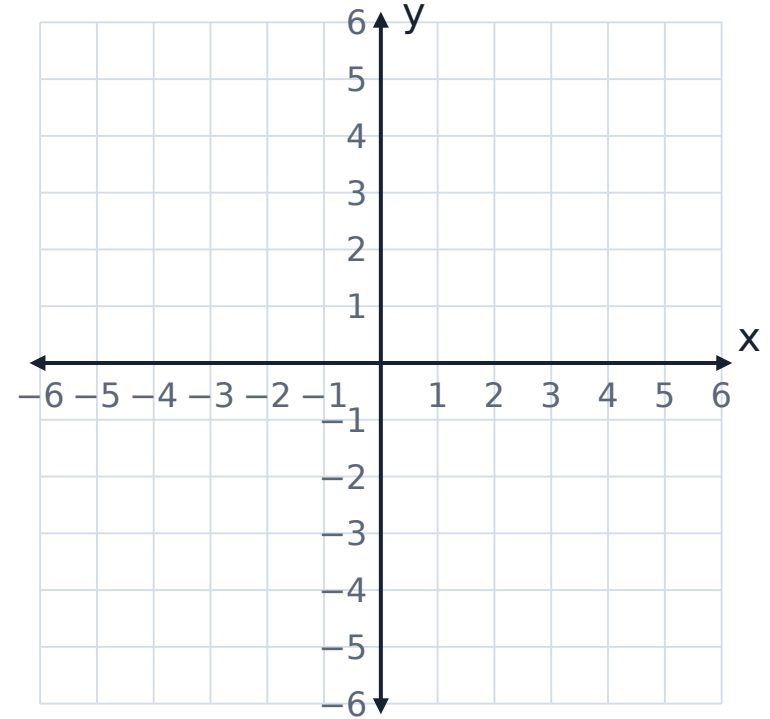
MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

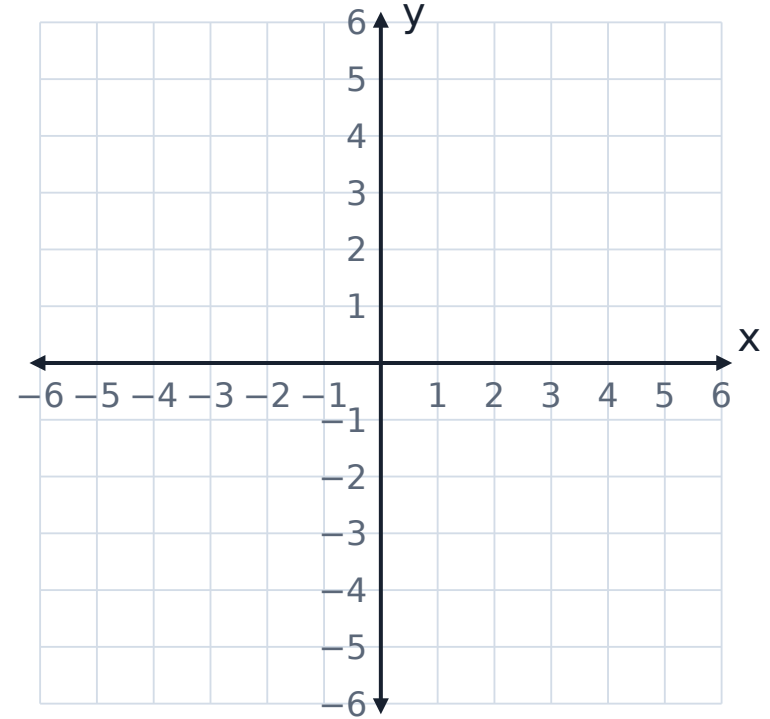
Plot and label point $P(-3, 4)$
and point $Q(2, -5)$.



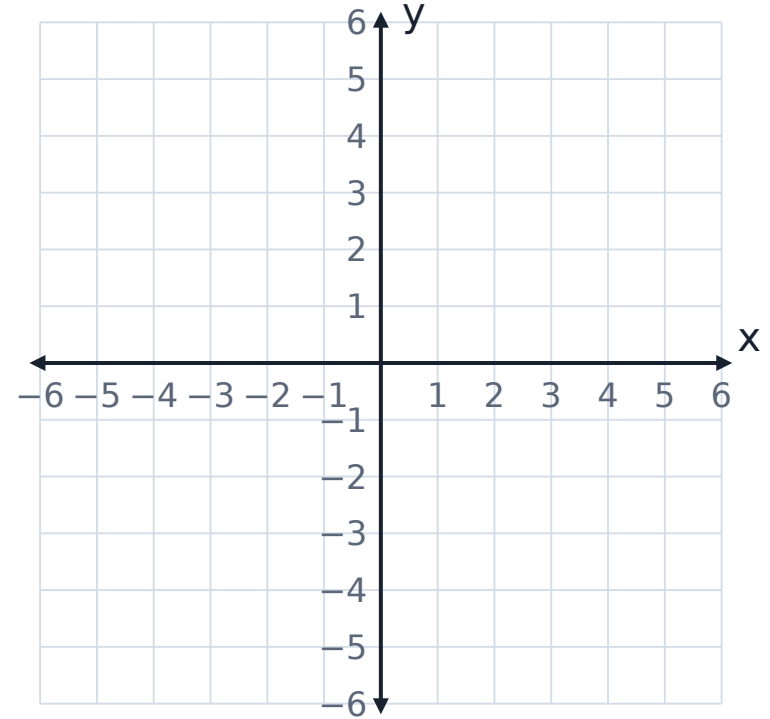
Plot and label point $R(-5, -2)$
and point $S(4, 1)$.



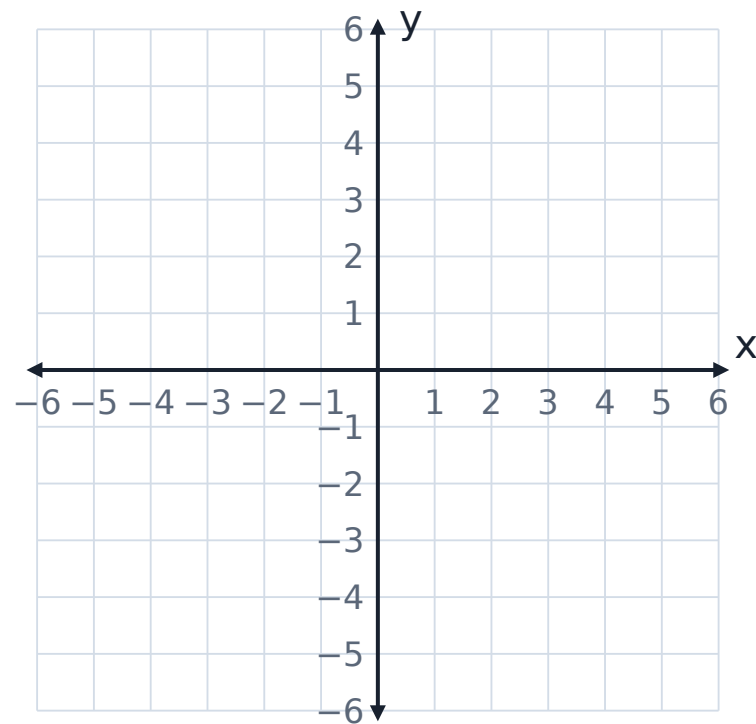
Plot and label point $T(0, -3)$
and point $U(-4, 0)$.



Plot and label point $V(-2.5, 3)$ and point $W(1.5, -2)$.



Plot and label point $X(-6, -6)$
and point $Y(5, -4)$.



SET 32 • 6.NS.C.6.c

BACKWARD 1

Which direction to move first • Grade 5 • Geometry • 5.G.A.1

True or false?

To plot $(6, 1)$, first move 6 units along the x-axis.

A. True

B. False

To plot (2, 7) from the origin, what do you do first?

- A.** Move 7 units up
- B.** Move 2 units right
- C.** Move 2 units up
- D.** Move 7 units right

True or false?

To plot $(0, 5)$, you do not move left or right.

A. True

B. False

To plot (4, 3), how many units up do you move?

Units: _____

Which ordered pair is 3 units right and 6 units up from the origin?

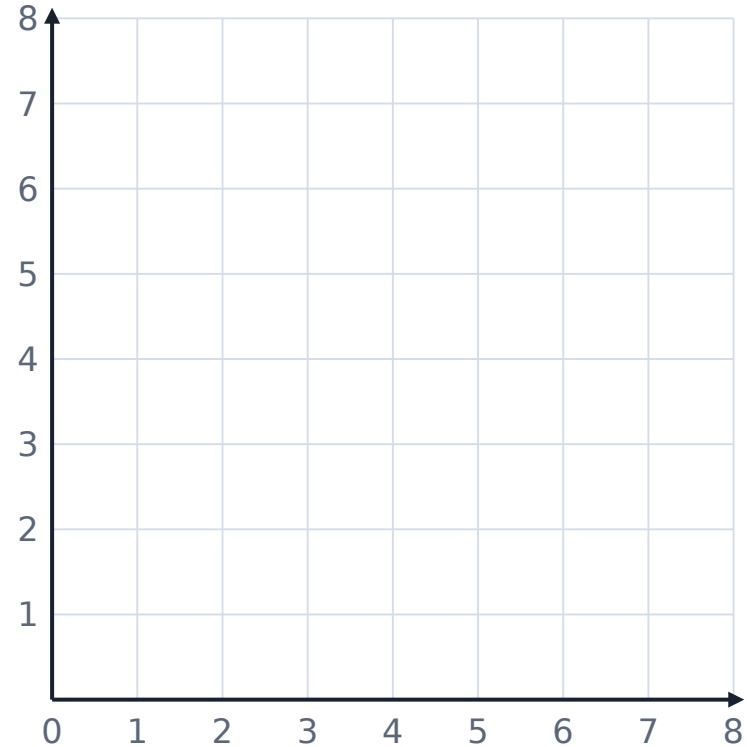
- A.** (3, 3)
- B.** (3, 6)
- C.** (6, 3)
- D.** (6, 6)

SET 32 • 6.NS.C.6.c

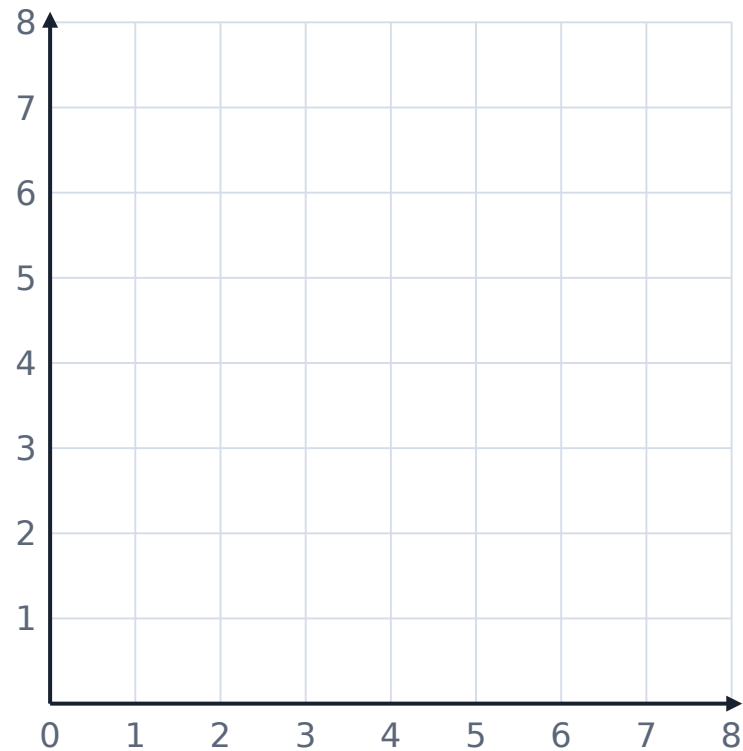
BACKWARD 2

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

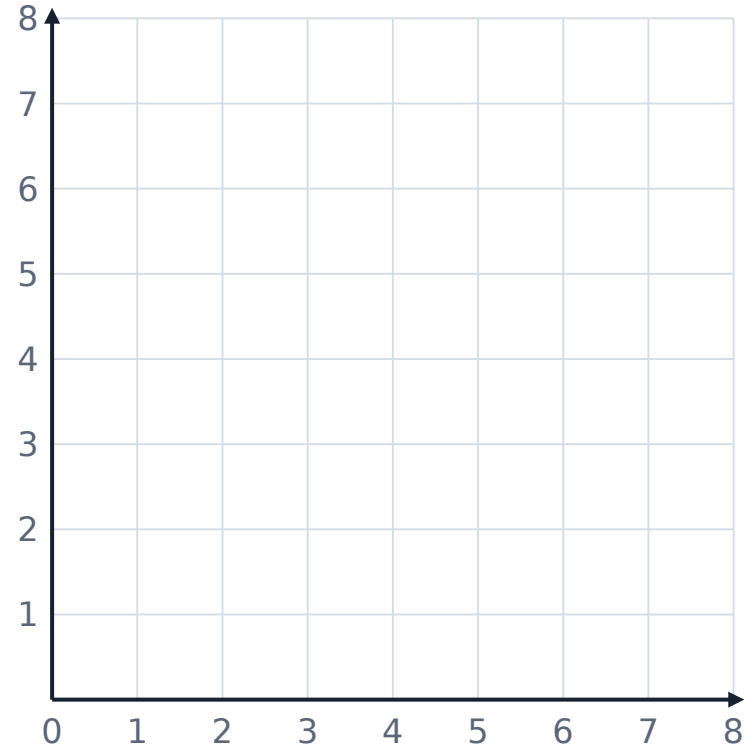
Plot and label point A at (2, 6).



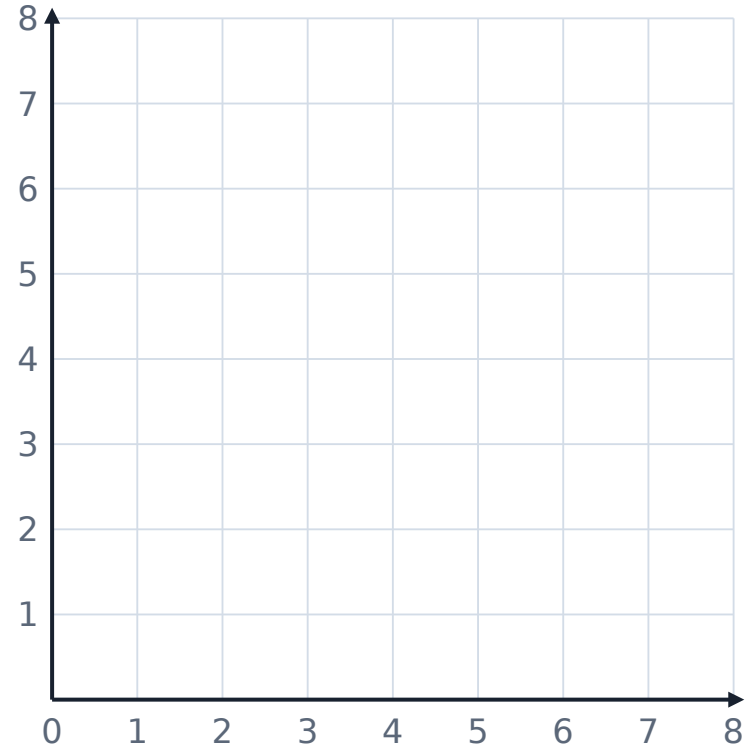
Plot and label point B at (5, 1).



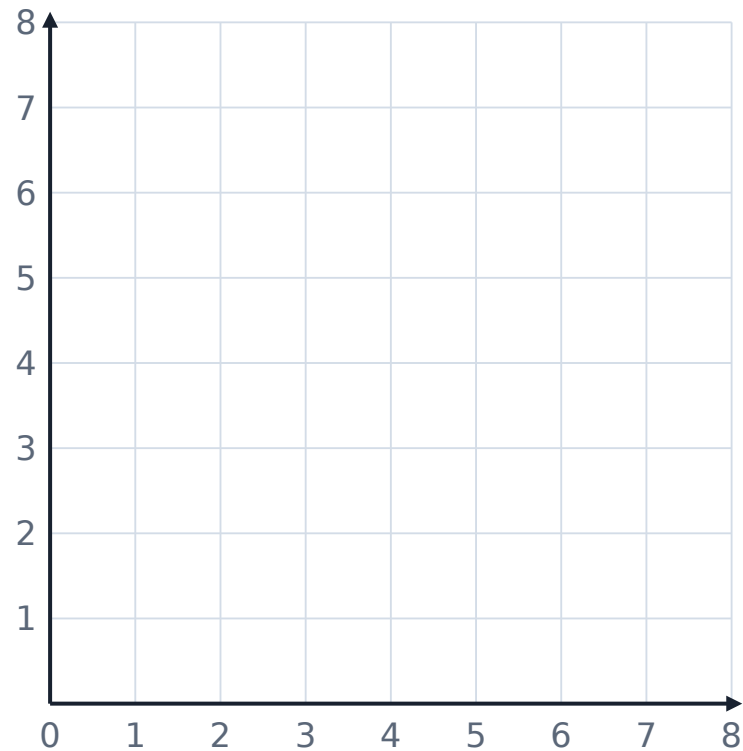
Plot and label point C at (0, 7).



Plot and label point D at (7, 7).



Plot and label point E at (5, 0).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 6.NS.C.6.c

BACKWARD 3

Direction from the sign of a coordinate • Grade 6 • The Number System •
6.NS.C.6.b

True or false?

To plot a point with a negative x-coordinate, you move left from the origin.

- A.** True
- B.** False

To plot $(-2, -5)$, which directions do you move?

- A.** Right 2, down 5
- B.** Right 2, up 5
- C.** Left 2, down 5
- D.** Left 5, down 2

True or false?

To plot $(3, -4)$, you move up 4 units.

A. True

B. False

To plot $(-6, 1)$, do you move left or right?

Answer: _____

Which point is left of the y -axis and above the x -axis?

- A.** $(-3, 2)$
- B.** $(3, -2)$
- C.** $(3, 2)$
- D.** $(-3, -2)$

SET 32 • 6.NS.C.6.c

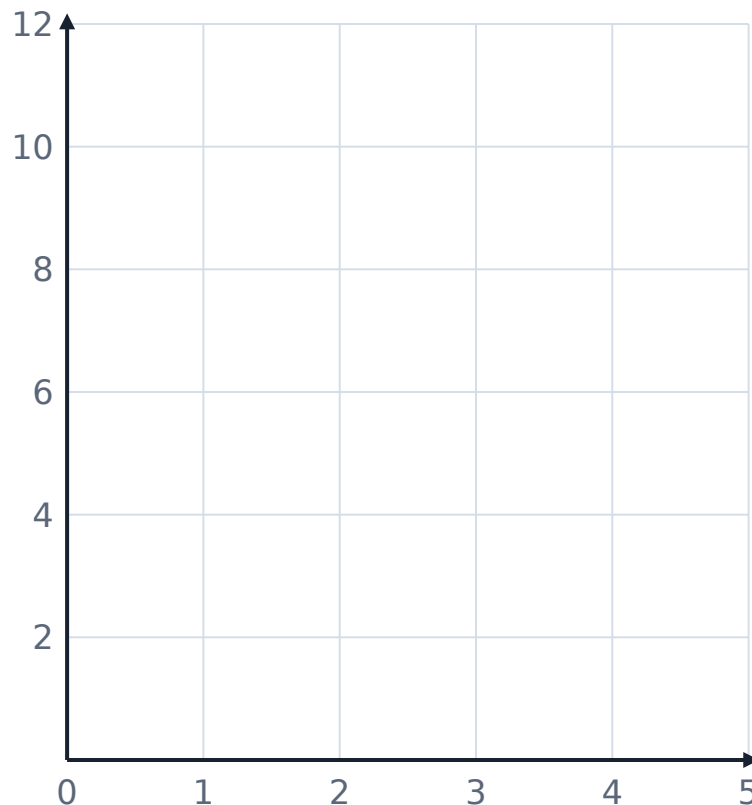
FORWARD 1

Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs $(1, 3)$, $(2, 6)$,
and $(3, 9)$.

Do they lie on a straight line
through the origin?

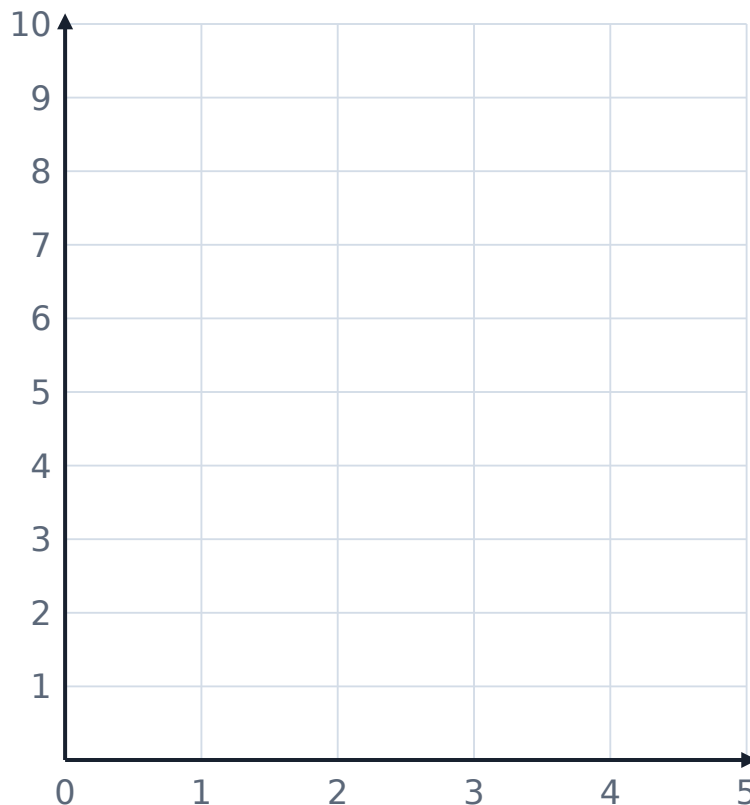
Answer: _____



Graph the pairs $(1, 3)$, $(2, 5)$,
and $(3, 7)$.

Is the relationship
proportional?

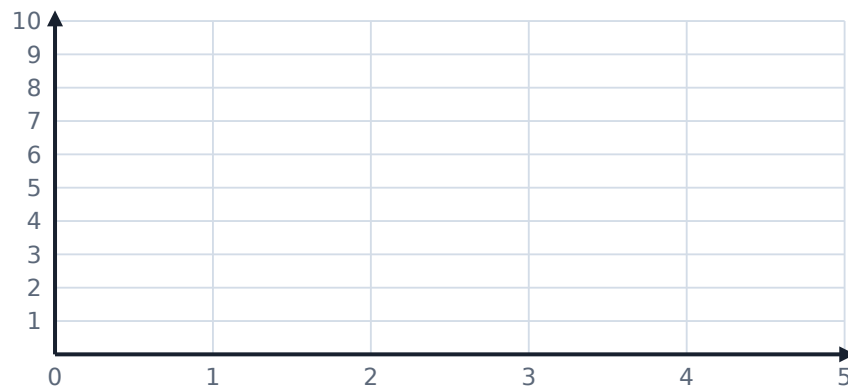
Answer: _____



Graph the pairs from the table.

Is the relationship proportional?

x	1	2	4
y	2.5	5	10



Answer: _____

True or false?

The points $(2, 1)$, $(4, 2)$, and $(6, 3)$ lie on a line through the origin, so the relationship is proportional.

- A. True
- B. False

Which set of pairs would graph as a proportional relationship?

A. (1, 1), (2, 4), (3, 9)

B. (1, 3), (2, 4), (3, 5)

C. (0, 2), (1, 4), (2, 6)

D. (2, 3), (4, 6), (6, 9)

SET 32 • 6.NS.C.6.c

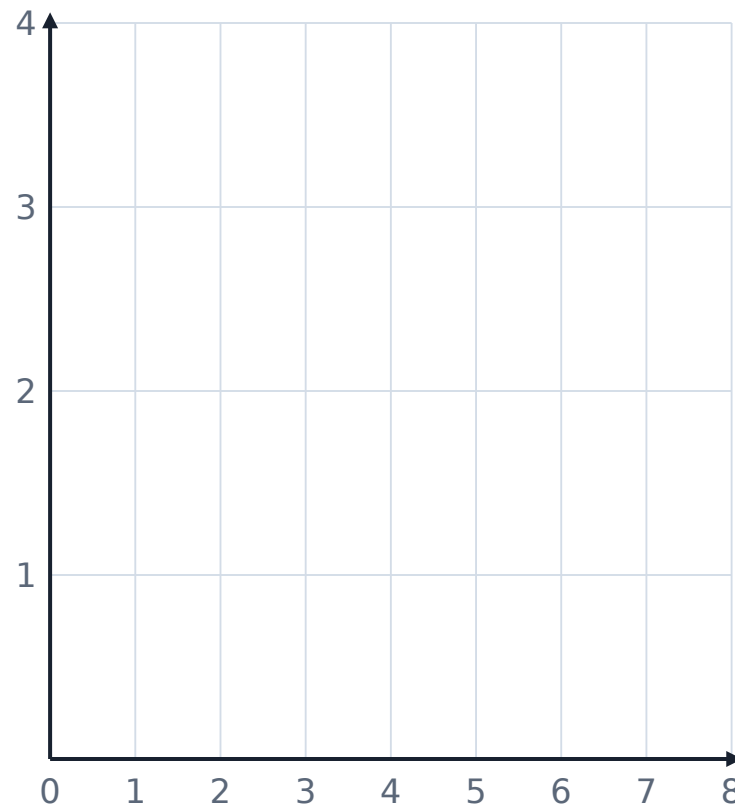
FORWARD 2

Graph proportional relationships • Grade 8 • Expressions & Equations •
8.EE.B.5

Graph $y = 0.5x$ on the coordinate plane.

What is the slope of the line?

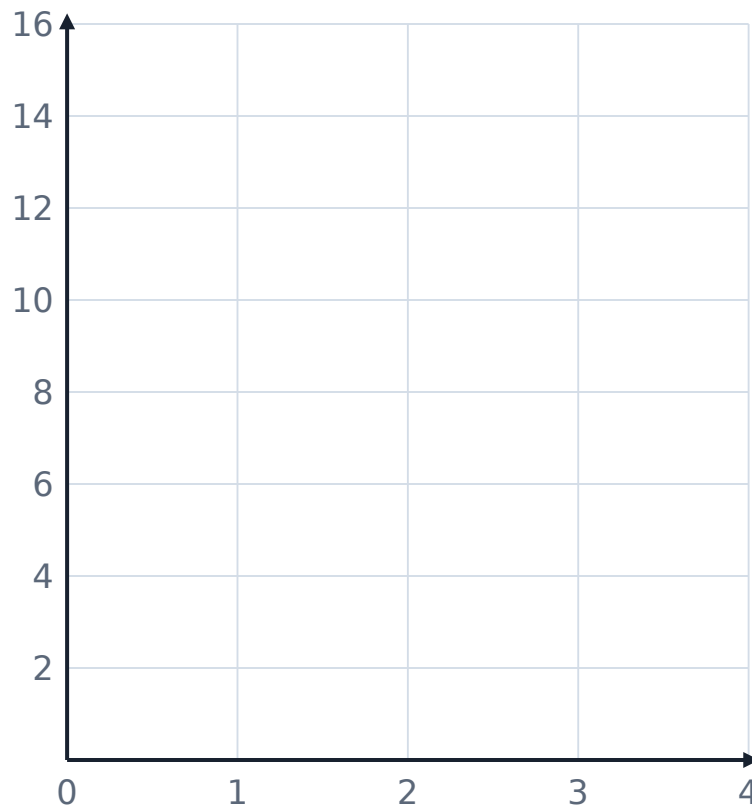
Slope: _____



Graph $y = 4x$ on the coordinate plane.

What is the slope of the line?

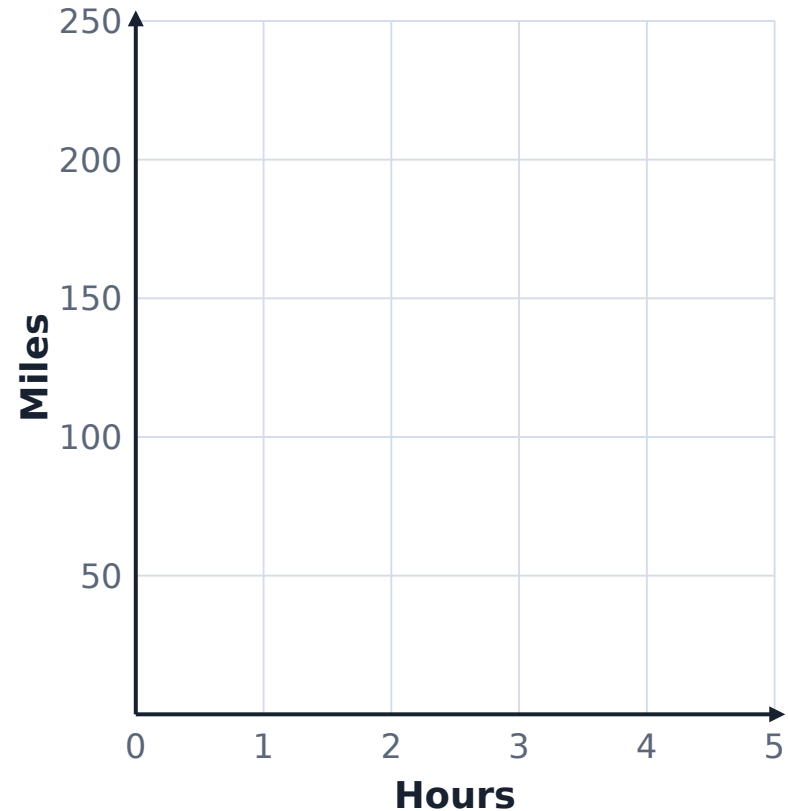
Slope: _____



A car travels 50 miles per hour. Graph the distance y after x hours.

What is the slope?

Slope: _____



Which point is on the graph of $y = 3x$?

A. (3, 1)

B. (12, 4)

C. (4, 12)

D. (4, 7)

True or false?

The graph of $y = \frac{2}{3}x$ passes through (3, 2).

A. True

B. False

Interpret inequalities as relative position on a number line

MAIN	Grade 6 main question
BACKWARD 1	Comparison symbols
BACKWARD 2	Position on a number line
BACKWARD 3	Negative numbers on a number line
FORWARD 1	Solve and graph inequalities
FORWARD 2	Compare and order irrational numbers

6.NS.C.7.a

2.NBT.A.4

2.MD.B.6

6.NS.C.6.c

7.EE.B.4.b

8.NS.A.2

SET 33 • 6.NS.C.7.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.a

$$-4 > -10$$

Which statement explains this inequality?

- A.** -4 is to the left of -10 on a number line.
- B.** 4 is less than 10 .
- C.** -4 is farther from 0 than -10 .
- D.** -4 is to the right of -10 on a number line.

True or false?

The number line shows that $-6 > -1$.



A. True

B. False

Write an inequality that compares -4 and -9 .

Inequality: _____

Which statement is true?

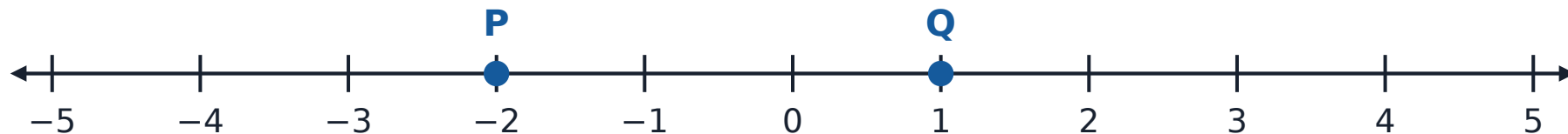
A. $-5 < -2$

B. $-2 < -5$

C. $-5 > 0$

D. $-5 > -2$

Write an inequality using the numbers at points P and Q.



Inequality: _____

SET 33 • 6.NS.C.7.a

BACKWARD 1

Comparison symbols • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

452 425

Symbol: _____

True or false?

$$67 < 76$$

A. True

B. False

Which statement is true?

A. $309 < 290$

B. $309 = 290$

C. $309 > 290$

Write $>$, $<$, or $=$ to compare.

138 183

Symbol: _____

True or false?

500 > 505

A. True

B. False

SET 33 • 6.NS.C.7.a

BACKWARD 2

Position on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

True or false?

On a number line, 8 is to the right of 3.

A. True

B. False

Which number is farthest to the left on a number line?

A. 2

B. 5

C. 7

D. 9

Which point is farther to the left, A or B?



Point: _____

True or false?

Numbers get greater as you move to the left on a number line.

A. True

B. False

Which number is to the right of 14 on a number line?

- A.** 9
- B.** 11
- C.** 14
- D.** 19

SET 33 • 6.NS.C.7.a

BACKWARD 3

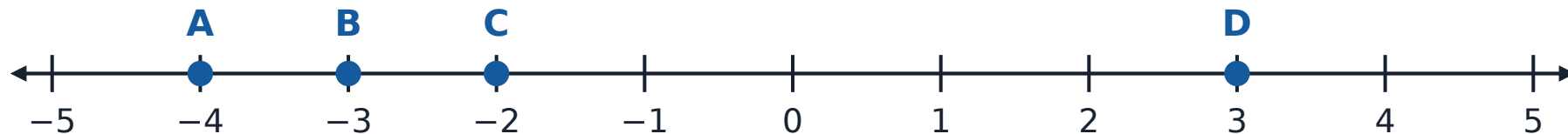
Negative numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?



A = _____

Which point is at -3 ?



A. Point A

B. Point B

C. Point C

D. Point D

True or false?

-4 is to the left of -1 on a number line.

A. True

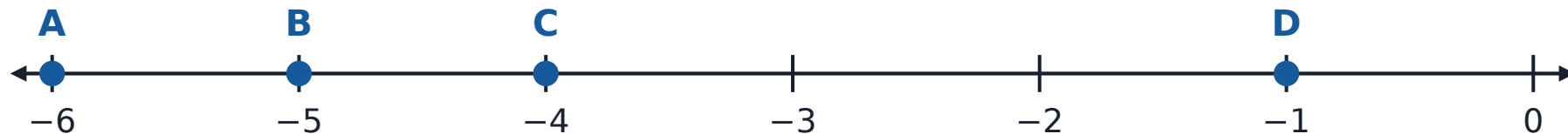
B. False

What number is at point B?



B = _____

Which point is at -5 ?



A. Point A

B. Point B

C. Point C

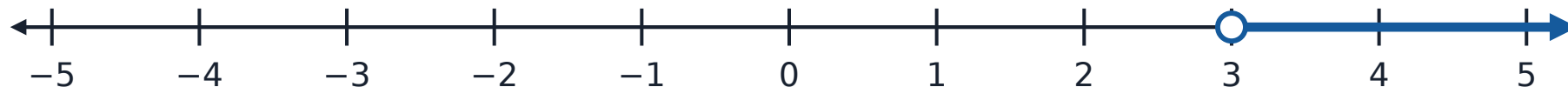
D. Point D

SET 33 • 6.NS.C.7.a

FORWARD 1

Solve and graph inequalities • Grade 7 • Expressions & Equations • 7.EE.B.4.b

Which inequality does the graph show?



A. $x > 3$

B. $x \leq 3$

C. $x < 3$

D. $x \geq 3$

Solve the inequality.

$$2x + 5 > 11$$

Solution: _____

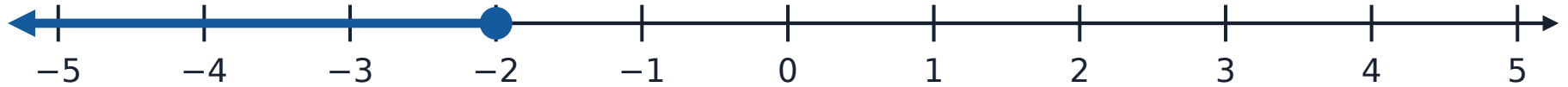
Solve the inequality.

$$-3x + 4 \leq 16$$

Solution: _____

True or false?

The graph shows the solutions of $4x - 1 \leq -9$.



A. True

B. False

Solve the inequality.

$$5x - 2 < 18$$

Solution: _____

SET 33 • 6.NS.C.7.a

FORWARD 2

Compare and order irrational numbers • Grade 8 • The Number System •
8.NS.A.2

Which is greater, $\sqrt{50}$ or 7.2?

Greater: _____

Which number is between 3 and 4?

A. $\sqrt{12}$

B. $\sqrt{8}$

C. $\sqrt{3}$

D. $\sqrt{17}$

True or false?

$$\sqrt{30} < 5.5$$

A. True

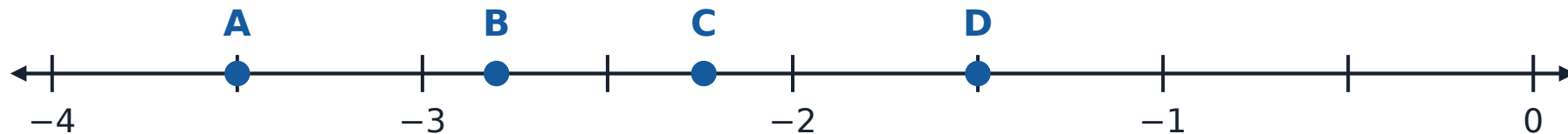
B. False

Order from least to greatest.

π , $\sqrt{8}$, 3.1

Order: _____

Which point best shows $-\sqrt{5}$?



A. Point A

B. Point B

C. Point C

D. Point D

Write, interpret, and explain order statements in real-world contexts

MAIN	Grade 6 main question	6.NS.C.7.b
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Inequality symbols	1.NBT.B.3
BACKWARD 3	Order as position on a number line	6.NS.C.7.a
FORWARD 1	Write and solve inequalities for real-world situations	7.EE.B.4.b
FORWARD 2	Compare quantities written with powers of 10	8.EE.A.3

SET 34 • 6.NS.C.7.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.b

A temperature of -2°F is warmer than a temperature of -11°F .

Write an inequality to show this.

Inequality: _____

Account A has a balance of $-\$25$. Account B has a balance of $-\$40$.

Which statement is true?

- A.** $-25 > -40$, so Account A has the greater balance.
- B.** $-40 > -25$, so Account B has the greater balance.
- C.** $-25 < -40$, so Account A owes more.
- D.** The balances are equal.

A fish swims at -12 feet. A turtle swims at -5 feet.

Write an inequality to compare the positions. Which animal is deeper?

Inequality: _____

Deeper animal: _____

True or false?

$-15 < -10$ means that -15°F is colder than -10°F .

A. True

B. False

The lowest point in a valley is -86 meters. A beach is at 0 meters.

Which statement is true?

- A.** $-86 < 0$, so the valley's lowest point is lower.
- B.** $-86 = 0$
- C.** $0 < -86$, so the beach is lower.
- D.** $-86 > 0$, so the valley's lowest point is higher.

SET 34 • 6.NS.C.7.b

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 12 degrees below zero.

Integer: _____

True or false?

A debt of \$30 can be written as -30 .

A. True

B. False

Which situation could be represented by -6 ?

- A.** A loss of 6 yards
- B.** A deposit of \$6
- C.** A gain of 6 yards
- D.** 6 feet above sea level

Write an integer for 50 feet below sea level.

Integer: _____

True or false?

A temperature of 4 degrees above zero is written as -4 .

A. True

B. False

SET 34 • 6.NS.C.7.b

BACKWARD 2

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

$$42 \text{ ___ } 24$$

A. $<$

B. $>$

C. $=$

True or false?

$$15 > 51$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

73 37

Symbol: _____

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

True or false?

$$89 < 98$$

A. True

B. False

SET 34 • 6.NS.C.7.b

BACKWARD 3

Order as position on a number line • Grade 6 • The Number System •
6.NS.C.7.a

True or false?

$-2 > -6$ because -2 is to the right of -6 on a number line.

A. True

B. False

Which number is greatest?

A. -9

B. -7

C. -4

D. -1

–5 is to the left of 3 on a number line.

Write an inequality to show this.

Inequality: _____

True or false?

-8 is to the right of -3 on a number line.

A. True

B. False

Which number is to the left of -4 on a number line?

A. -6

B. -2

C. 0

D. 4

SET 34 • 6.NS.C.7.b

FORWARD 1

Write and solve inequalities for real-world situations • Grade 7 • Expressions & Equations • 7.EE.B.4.b

A submarine at -40 m descends 12 m per minute. It must stay above -160 m.

Write and solve an inequality for the number of minutes t it can descend.

Solution: _____

You have \$25 and spend \$4 per day. You want to keep at least \$5.

Write and solve an inequality for the number of days d .

Solution: _____

A freezer is at 2°C . It cools 3°C per hour.

For which numbers of hours h is its temperature below -10°C ?

A. $h > 4$

B. $h > -4$

C. $h < -4$

D. $h < 4$

A diver at -8 ft descends 5 ft per minute.

Write and solve an inequality to find when the diver is deeper than -48 ft.

Solution: _____

True or false?

The solution of $-2x + 1 > 9$ is $x < -4$.

A. True

B. False

SET 34 • 6.NS.C.7.b

FORWARD 2

Compare quantities written with powers of 10 • Grade 8 • Expressions & Equations • 8.EE.A.3

A mountain is about 9×10^3 m tall. An ocean trench is about -1.1×10^4 m deep.

Which is farther from sea level?

Answer: _____

Which number is greater?

A. 3×10^{-2}

B. 8×10^{-3}

A city has about 2×10^6 people. A town has about 5×10^4 people.

How many times as many people live in the city?

Times as many: _____

Order from least to greatest.

$$4 \times 10^3, 9 \times 10^2, 1 \times 10^4$$

Order: _____

True or false?

$$-5 \times 10^3 < -2 \times 10^3$$

A. True

B. False

Find absolute value as distance from 0

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Distance on a number line	2.MD.B.6
BACKWARD 2	Opposite numbers	6.NS.C.6.a
FORWARD 1	Distance between rational numbers as $ p - q $	7.NS.A.1.c
FORWARD 2	Square roots and cube roots as solutions	8.EE.A.2

SET 35 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-18|$$

Value: _____

Which number has the greatest absolute value?

A. -12

B. -5

C. 8

D. 10

How far is -7.5 from 0 on a number line?

Distance: _____

True or false?

$$|-6| = |6|$$

A. True

B. False

Which two numbers have an absolute value of 11?

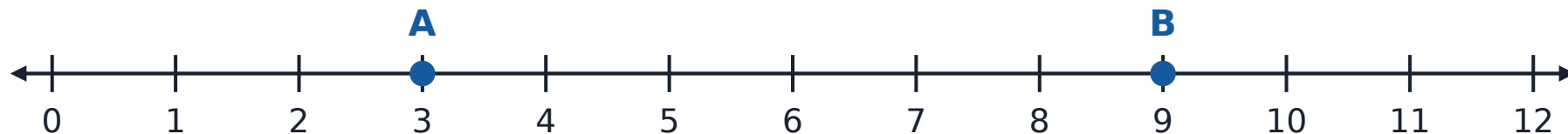
Numbers: _____

SET 35 • 6.NS.C.7.c

BACKWARD 1

Distance on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How many units apart are points A and B?



Units: _____

True or false?

The distance from 0 to 5 is the same as the distance from 5 to 10.

A. True

B. False

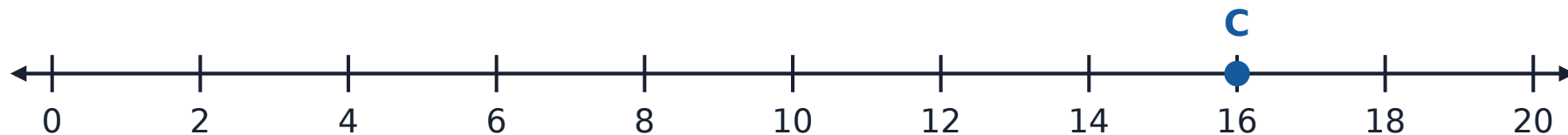
How many units from 0 is the number 11?

- A.** 0
- B.** 1
- C.** 10
- D.** 11

What is the distance from 4 to 10 on a number line?

Distance: _____

How many units is point C from 0?



Units: _____

SET 35 • 6.NS.C.7.c

BACKWARD 2

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 15?

Opposite: _____

True or false?

Opposite numbers are the same distance from 0.

A. True

B. False

Which number is the opposite of $-\frac{2}{3}$?

A. $-\frac{3}{2}$

B. $-\frac{2}{3}$

C. $\frac{2}{3}$

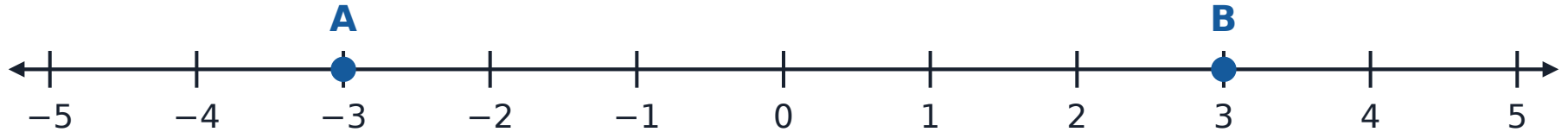
D. $\frac{3}{2}$

What is the opposite of 0?

Opposite: _____

True or false?

Points A and B represent opposite numbers.



A. True

B. False

SET 35 • 6.NS.C.7.c

FORWARD 1

Distance between rational numbers as $|p - q|$ • Grade 7 • The Number System
• 7.NS.A.1.c

Find the distance between -4 and 7 on a number line.

Distance: _____

Find the value.

$$|-3.5 - 2|$$

Value: _____

Which expression gives the distance between -9 and -2 ?

A. $-9 - 2$

B. $|-9 - (-2)|$

C. $|-9 + (-2)|$

D. $-9 + 2$

True or false?

The distance between -6 and 6 is 0 .

A. True

B. False

Find the distance between $-2\frac{1}{2}$ and $1\frac{1}{4}$ on a number line.

Distance: _____

SET 35 • 6.NS.C.7.c

FORWARD 2

Square roots and cube roots as solutions • Grade 8 • Expressions & Equations •
8.EE.A.2

Solve.

$$x^2 = 49$$

$x =$ _____

Solve.

$$x^3 = 64$$

$x =$ _____

What are the solutions of $x^2 = 81$?

- A.** $x = -9$ only
- B.** $x = 9$ only
- C.** $x = 40.5$
- D.** $x = 9$ or $x = -9$

Solve. Write the answer using a square root symbol.

$$x^2 = 20$$

$x =$ _____

True or false?

Both 6 and -6 are solutions of $x^2 = 36$.

A. True

B. False

Interpret absolute value as magnitude in a real-world situation

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Distance from 0	2.MD.B.6
BACKWARD 3	Opposites are the same distance from 0	6.NS.C.6.a
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare magnitudes with powers of 10	8.EE.A.3

SET 36 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

An account balance is $-\$45$.

What is the absolute value of the balance, and what does it describe?

Absolute value: _____

It describes: _____

A diver is at an elevation of -42 feet.

How far is the diver from sea level?

Distance: _____

A stock's value changed by $-\$8.50$ today.

What is the size of the change?

- A.** $-\$8.50$
- B.** $\$0$
- C.** $\$8.50$
- D.** $\$17.00$

The temperature is -13°F .

How many degrees below zero is it?

Degrees: _____

A submarine is at -275 m. A plane is at 275 m.

Which is farther from sea level? Explain using absolute value.

Answer: _____

SET 36 • 6.NS.C.7.c

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

True or false?

–30 can represent owing \$30.

A. True

B. False

Write an integer for 42 feet below sea level.

Integer: _____

Which situation is represented by $+25$?

- A.** A loss of 25 yards
- B.** 25 feet below sea level
- C.** A debt of \$25
- D.** A deposit of \$25

Write an integer for a debt of \$18.

Integer: _____

True or false?

A temperature of 0°F means there is no temperature.

A. True

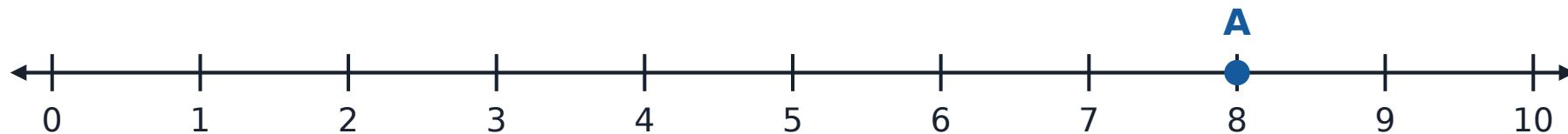
B. False

SET 36 • 6.NS.C.7.c

BACKWARD 2

Distance from 0 • Grade 2 • Measurement & Data • 2.MD.B.6

How many units from 0 is point A?



Units: _____

True or false?

The distance from 0 to 13 is 13 units.

A. True

B. False

Which number is 5 units from 0 on this number line?



A. 0

B. 5

C. 10

D. 50

How far is 20 from 0 on a number line?

Distance: _____

True or false?

The number that is 0 units from 0 is 1.

A. True

B. False

SET 36 • 6.NS.C.7.c

BACKWARD 3

Opposites are the same distance from 0 • Grade 6 • The Number System •
6.NS.C.6.a

True or false?

–42 and 42 are the same distance from 0.

A. True

B. False

What is the opposite of -275 ?

Opposite: _____

Which number is the same distance from 0 as -8.5 ?

A. -5.8

B. 0

C. 8.5

D. 85

True or false?

The opposite of 13 is -13 .

A. True

B. False

Name the number that is the same distance from 0 as -45 , on the other side of 0.

Number: _____

SET 36 • 6.NS.C.7.c

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A company lost \$2.5 million in one year and earned a profit of \$1.8 million the next year.

What is the total for the two years?

Total: _____

The temperature was -6°F . It dropped 11°F .
What is the new temperature?

Temperature: _____

A diver at -35.5 feet rises 12.25 feet.

What is the diver's new position?

- A.** -47.75 feet
- B.** -23.25 feet
- C.** -22.75 feet
- D.** 23.25 feet

A hiker goes from an elevation of 1,200 feet to an elevation of -80 feet.

What is the change in elevation?

Change: _____

True or false?

If the temperature drops from -4°C to -9°C , it gets 5°C colder.

A. True

B. False

SET 36 • 6.NS.C.7.c

FORWARD 2

Compare magnitudes with powers of 10 • Grade 8 • Expressions & Equations •
8.EE.A.3

One company's debt is 2×10^6 dollars. Another company's debt is 4×10^4 dollars.

How many times as large is the first debt?

Times as large: _____

The deepest ocean point is about -1×10^4 meters. A lake bottom is about -5×10^1 meters.

About how many times as deep is the ocean point?

A. 5

B. 20

C. 200

D. 2,000

Estimate 7,980,000 as a single digit times a power of 10.

Estimate: _____

True or false?

6×10^9 is 300 times as large as 2×10^7 .

A. True

B. False

Estimate 0.0000031 as a single digit times a power of 10.

Estimate: _____

Distinguish comparisons of absolute value from statements about order

MAIN	Grade 6 main question	6.NS.C.7.d
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Ordering negative numbers	6.NS.C.7.a
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare the size and order of irrational numbers	8.NS.A.2

SET 37 • 6.NS.C.7.d

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.d

True or false?

An account balance less than $-\$50$ means a debt greater than $\$50$.

- A. True
- B. False

Which account balance shows the greatest debt?

A. $-\$55$

B. $-\$38$

C. $-\$20$

D. $\$10$

True or false?

$-12 < -5$, so $|-12| < |-5|$.

A. True

B. False

Two temperatures are -15°F and -9°F .

Which temperature is colder? Which has the greater absolute value?

Colder: _____

Greater absolute value: _____

A submarine is at -300 feet. A diver is at -45 feet.

Which statement is true?

- A.** The diver is deeper because $-45 > -300$.
- B.** The submarine is deeper because $-300 > -45$.
- C.** The submarine is deeper because $|-300| > |-45|$.
- D.** They are at the same depth.

SET 37 • 6.NS.C.7.d

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-25|$$

Value: _____

Find the value.

$|14|$

Value: _____

True or false?

$$|-3| = -3$$

A. True

B. False

Which number has an absolute value of 7?

A. -7

B. $-\frac{1}{7}$

C. 0

D. $\frac{1}{7}$

Find the value.

$$|-2.5|$$

Value: _____

SET 37 • 6.NS.C.7.d

BACKWARD 2

Ordering negative numbers • Grade 6 • The Number System • 6.NS.C.7.a

Which number is least?

A. -20

B. -12

C. -5

D. 0

True or false?

$$-9 > -4$$

A. True

B. False

Order from least to greatest.

$-3, -8, 1$

Order: _____

True or false?

$$-1 > -100$$

A. True

B. False

Which number is greater?

A. -16

B. -6

SET 37 • 6.NS.C.7.d

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for a debt of \$40.

Integer: _____

What could -8 represent?

- A.** A deposit of \$8
- B.** A gain of 8 yards
- C.** 8 feet below sea level
- D.** 8 feet above sea level

True or false?

An account balance of $-\$10$ means you owe $\$10$.

- A. True
- B. False

Write an integer for 15 degrees below zero.

Integer: _____

True or false?

A 20-foot descent can be written as +20.

A. True

B. False

SET 37 • 6.NS.C.7.d

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Two accounts have balances of $-\$42.50$ and $-\$18.75$.
How much greater is the larger debt?

Difference: _____

A diver at -24.5 m rises 9.75 m and then descends 15 m.

What is the diver's final position?

Position: _____

The temperature at 6 a.m. was -8.5°F . At noon it was 3.5°F .

How much did the temperature rise?

A. -12°F

B. -5°F

C. 5°F

D. 12°F

Maria's balance is $-\$36$. She pays back $\frac{1}{4}$ of what she owes.

What is her new balance?

Balance: _____

True or false?

A change from -12 m to -30 m is a change of -18 m.

A. True

B. False

SET 37 • 6.NS.C.7.d

FORWARD 2

Compare the size and order of irrational numbers • Grade 8 • The Number System • 8.NS.A.2

Which number has the greater absolute value?

A. 4.3

B. $-\sqrt{20}$

True or false?

$$-\sqrt{10} < -3$$

A. True

B. False

Order from least to greatest.

$-\sqrt{5}$, -2.5 , -2

Order: _____

Which number is farther from 0 on a number line?

A. 3.1

B. $-\pi$

Between which two consecutive integers is $-\sqrt{30}$?

Between: _____

Find distances between points with the same first or second coordinate

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Naming points on the coordinate plane	6.NS.C.6.c
BACKWARD 3	Distance between whole numbers	2.MD.B.6
FORWARD 1	Distance between rational numbers	7.NS.A.1.c
FORWARD 2	Distance between points using the Pythagorean Theorem	8.G.B.8

SET 38 • 6.NS.C.8

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-3, 4)$ and $(5, 4)$.

Distance: _____

Find the distance between $(2, -6)$ and $(2, 5)$.

Distance: _____

What is the distance between $(-7, -2)$ and $(-1, -2)$?

A. -6

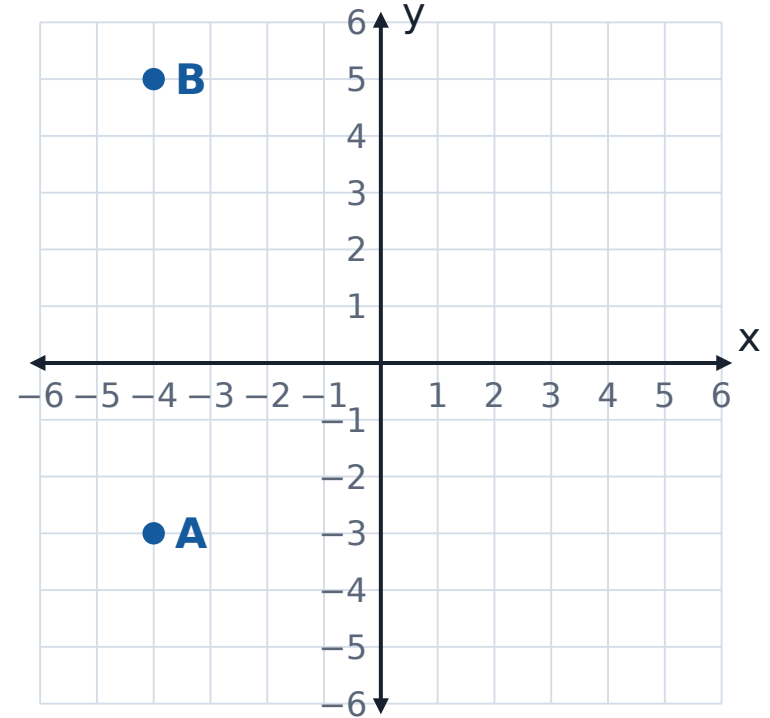
B. 3

C. 6

D. 8

What is the distance between points A and B?

Distance: _____



Find the distance between $(-2.5, 1)$ and $(3.5, 1)$.

Distance: _____

SET 38 • 6.NS.C.8

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-7|$$

Value: _____

True or false?

$$|-3| + |5| = 8$$

A. True

B. False

$$|-4| + |6| = ?$$

A. -10

B. -2

C. 2

D. 10

Find the value.

$$|-9| + |2|$$

Value: _____

True or false?

The distance from -5 to 0 is $|-5|$.

A. True

B. False

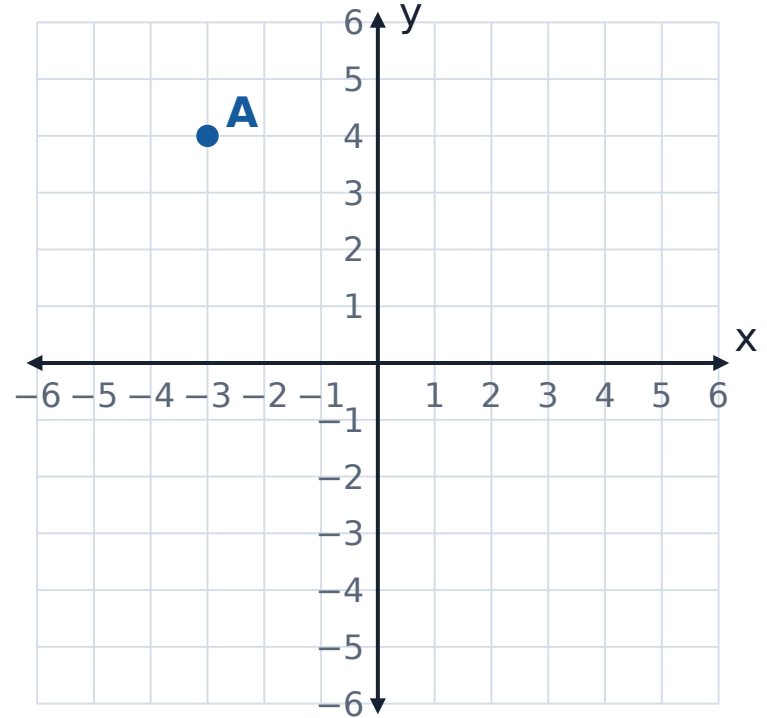
SET 38 • 6.NS.C.8

BACKWARD 2

Naming points on the coordinate plane • Grade 6 • The Number System •
6.NS.C.6.c

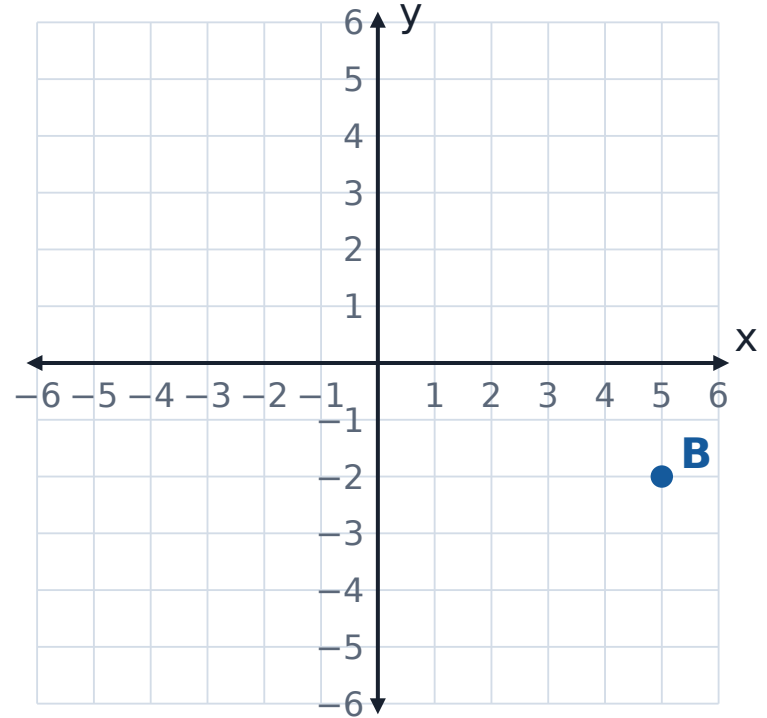
What are the coordinates of point A?

A = _____



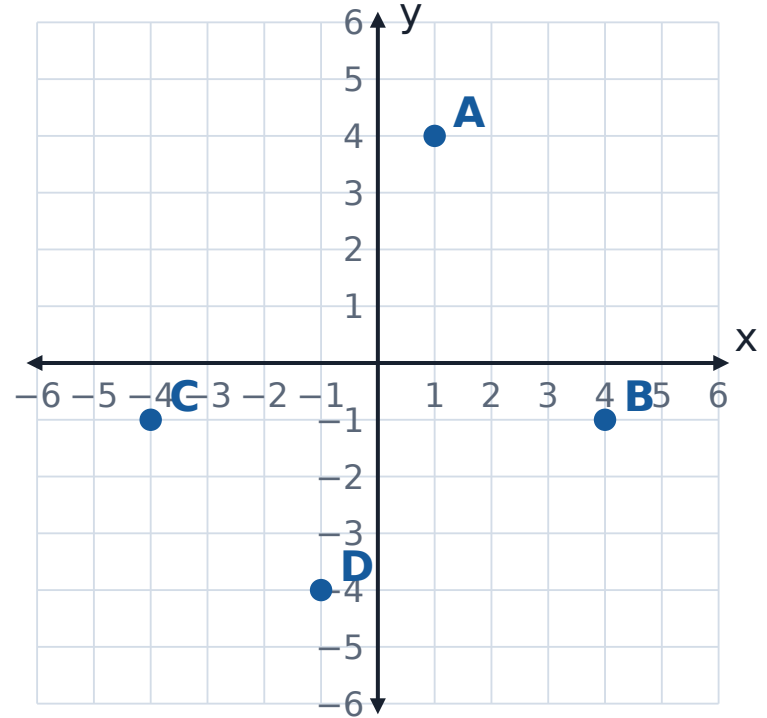
What are the coordinates of point B?

B = _____



Which point is at $(-1, -4)$?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

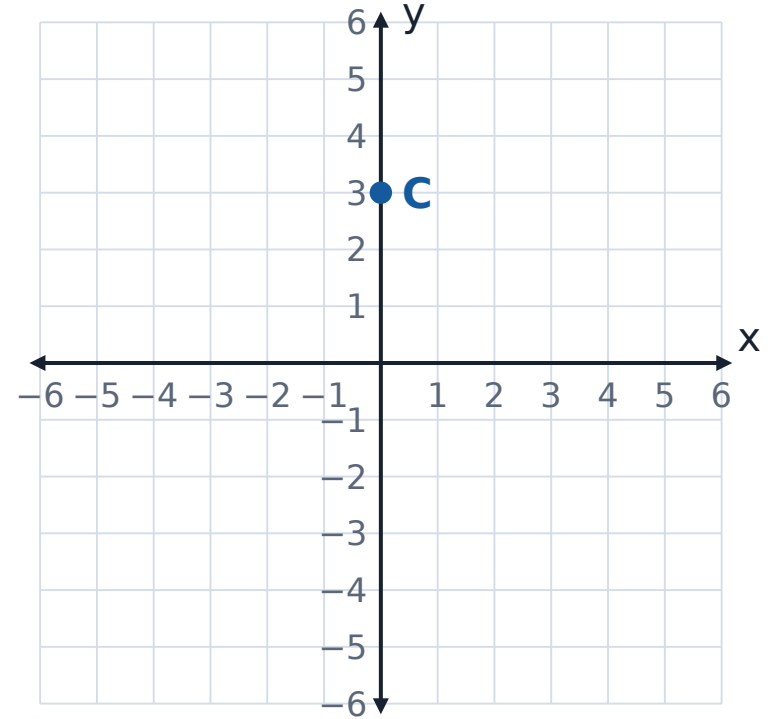


True or false?

Point C is located at $(3, 0)$.

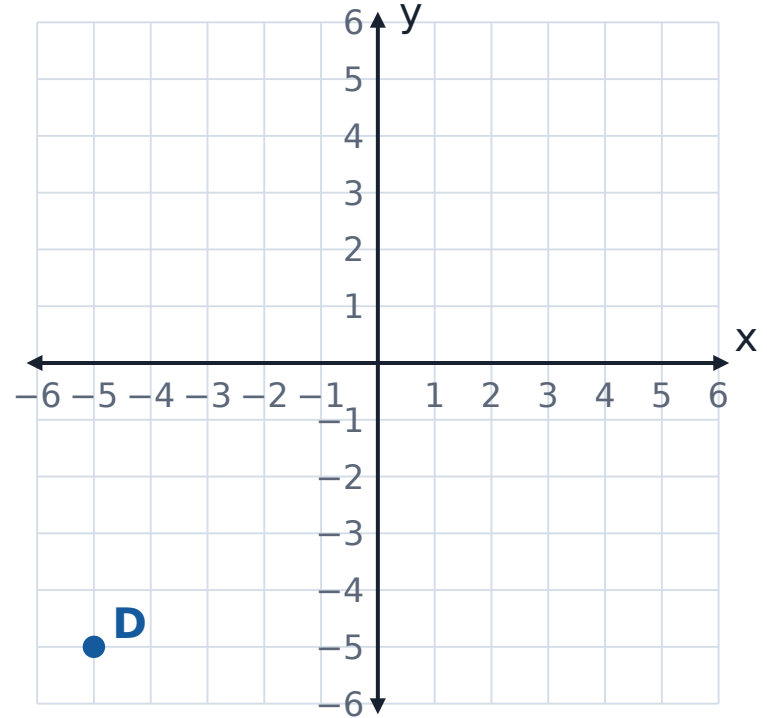
A. True

B. False



What are the coordinates of point D?

D = _____



SET 38 • 6.NS.C.8

BACKWARD 3

Distance between whole numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 3 to 11 on a number line?

Distance: _____

How many units apart are points A and B?



Units: _____

True or false?

The distance from 12 to 4 is 8 units.

A. True

B. False

What is the distance from 5 to 14?

- A.** 8
- B.** 9
- C.** 10
- D.** 19

What is the distance from 0 to 13 on a number line?

Distance: _____

SET 38 • 6.NS.C.8

FORWARD 1

Distance between rational numbers • Grade 7 • The Number System •
7.NS.A.1.c

Find the distance between -2.5 and 4.5 on a number line.

Distance: _____

Find the value.

$$|-8.4 - 3.6|$$

Value: _____

What is the distance between $-3\frac{1}{2}$ and $2\frac{1}{4}$ on a number line?

A. $-5\frac{3}{4}$

B. $1\frac{1}{4}$

C. $5\frac{3}{4}$

D. $6\frac{1}{4}$

True or false?

The distance between -7 and -2 is $|-7 - (-2)|$, which is 5.

A. True

B. False

Find the distance between -4 and 2.5 on a number line.

Distance: _____

SET 38 • 6.NS.C.8

FORWARD 2

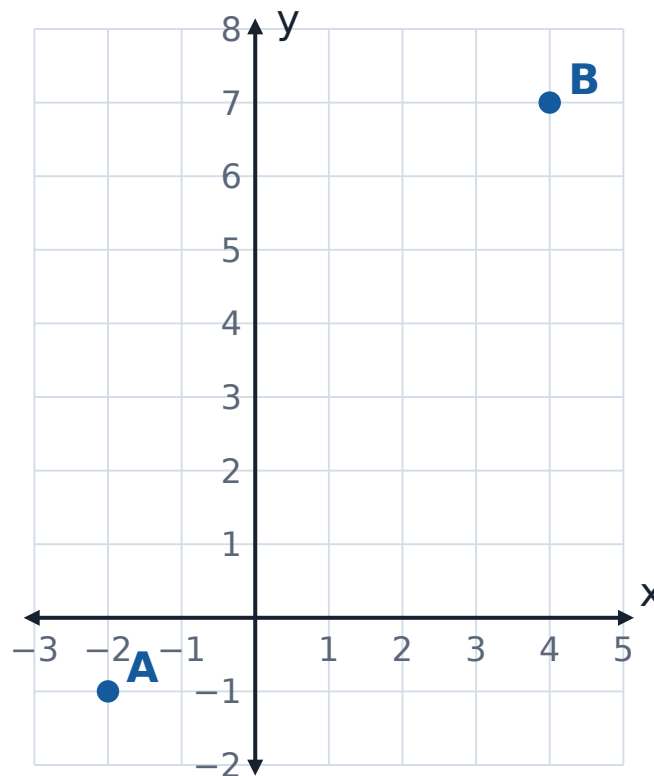
Distance between points using the Pythagorean Theorem • Grade 8 •
Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

Find the distance between points A and B.

Distance: _____



What is the distance between (0, 0) and (5, 12)?

- A.** $\sqrt{119}$
- B.** 17
- C.** 13
- D.** 7

Find the distance between $(-3, 5)$ and $(1, 2)$.

Distance: _____

True or false?

The distance between $(2, 1)$ and $(5, 5)$ is 5.

A. True

B. False

Solve real-world problems by graphing points in all four quadrants

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value as distance	6.NS.C.7.c
BACKWARD 2	Plotting points in all four quadrants	6.NS.C.6.c
BACKWARD 3	Perimeter of rectangles	3.MD.D.8
BACKWARD 4	Area of rectangles	3.MD.C.7.b
FORWARD 1	Use a map scale on a coordinate grid	7.G.A.1
FORWARD 2	Straight-line distances on a map	8.G.B.8

SET 39 • 6.NS.C.8

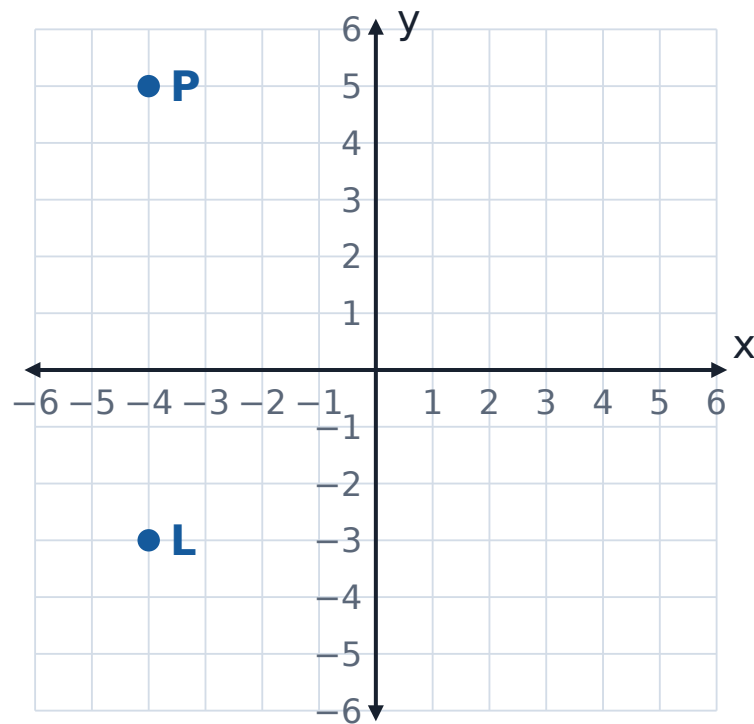
MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

On the map, each unit is 1 block.

How many blocks is it from the library L to the park P?

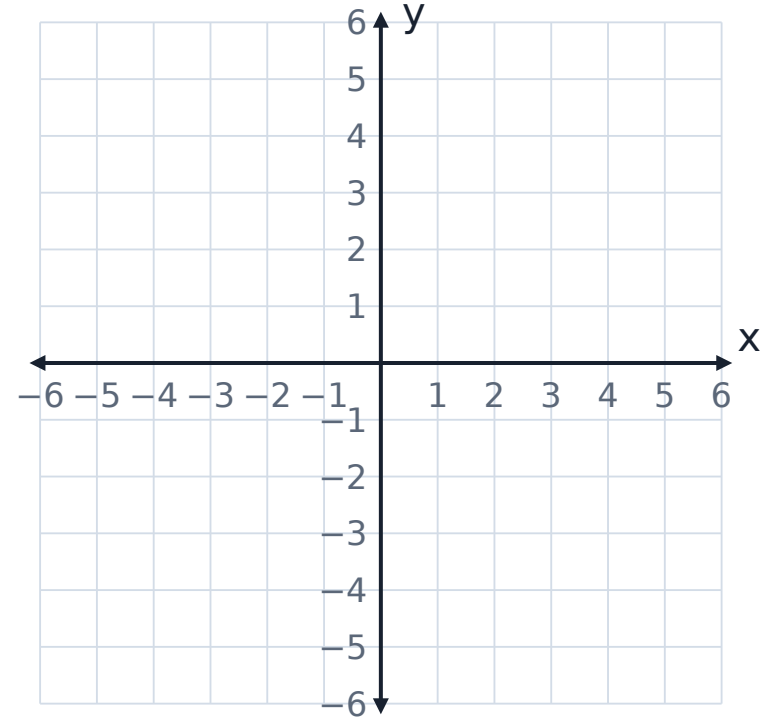
Blocks: _____



A park is a rectangle with corners at $(-2, 3)$, $(4, 3)$, $(4, -1)$, and $(-2, -1)$. Each unit is 1 meter.

Graph the park. What is its perimeter?

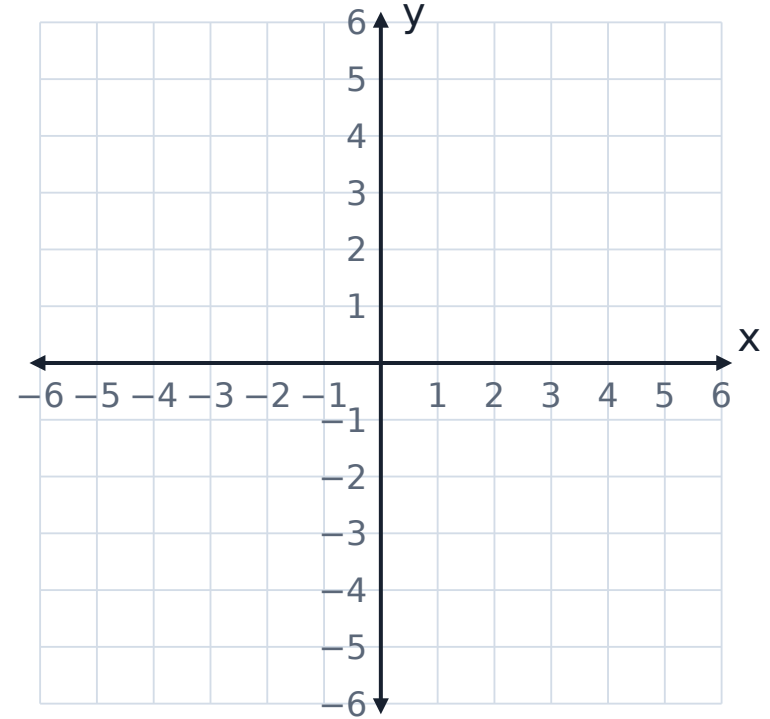
Perimeter: _____



The school is at $(-3, 2)$ and the pool is at $(-3, -4)$. Each unit is 1 block.

Graph both places. How many blocks apart are they?

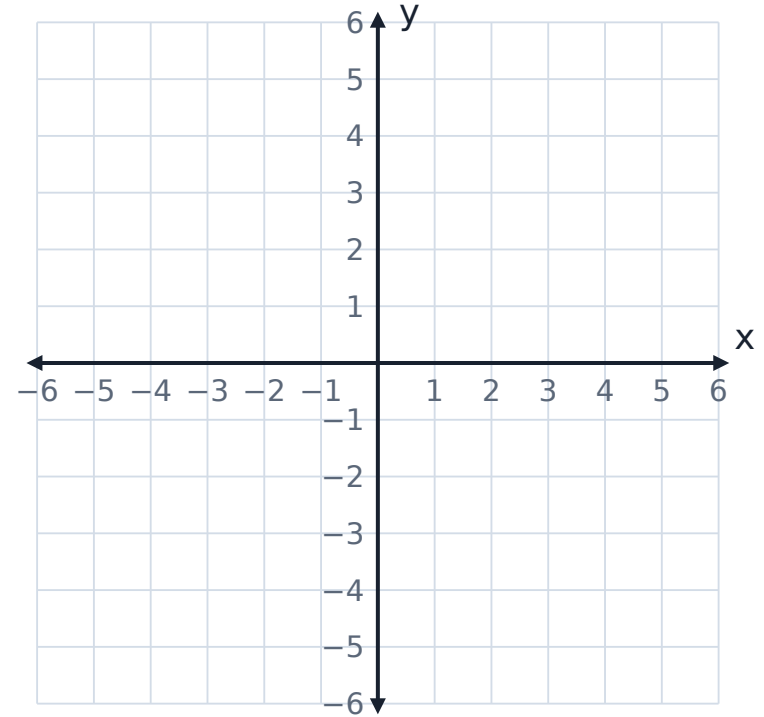
Blocks: _____



A garden is a rectangle with corners at $(-5, 1)$, $(1, 1)$, $(1, -3)$, and $(-5, -3)$. Each unit is 1 meter.

What is the area of the garden?

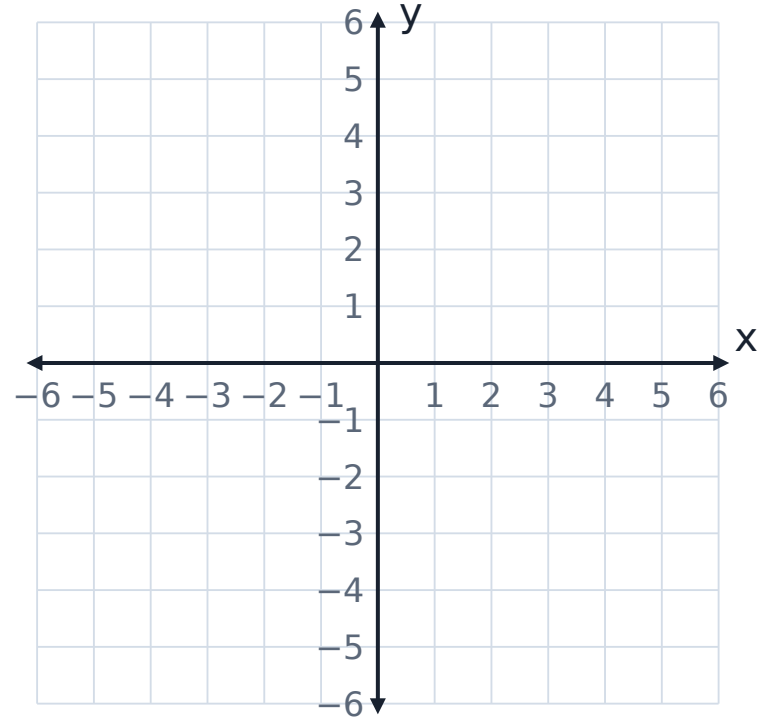
- | | |
|----------------------------|----------------------------|
| A. 10 square meters | B. 12 square meters |
| C. 20 square meters | D. 24 square meters |



A hiker starts at $(-5, 0)$, walks to $(3, 0)$, and then walks to $(3, 6)$. Each unit is 1 km.

Graph the path. How far does the hiker walk?

Distance: _____



SET 39 • 6.NS.C.8

BACKWARD 1

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-4| + |5|$$

Value: _____

True or false?

The distance from -3 to 0 is 3 .

A. True

B. False

How far is -6 from 0 ?

A. -6

B. 0

C. 6

D. 12

Find the value.

$$|-2| + |4|$$

Value: _____

True or false?

$$|-5| = -5$$

A. True

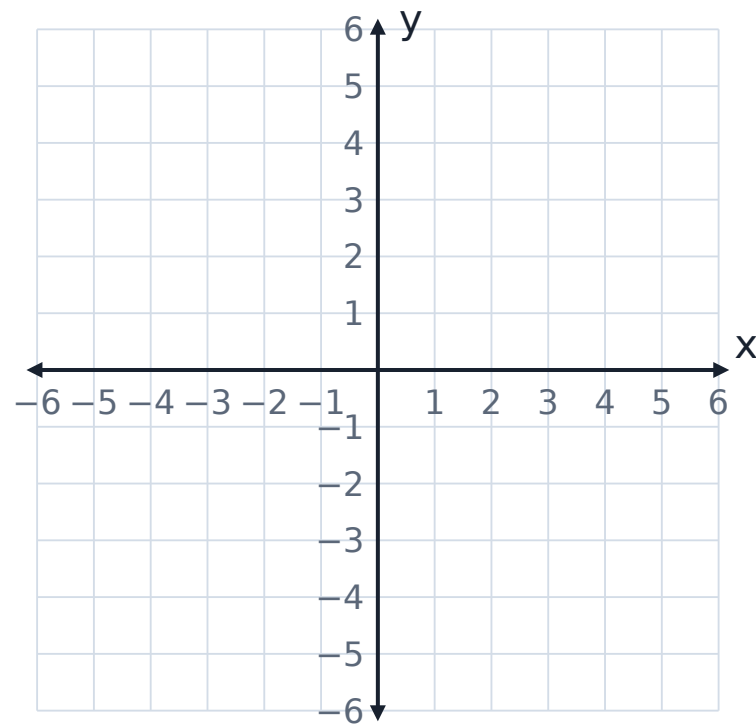
B. False

SET 39 • 6.NS.C.8

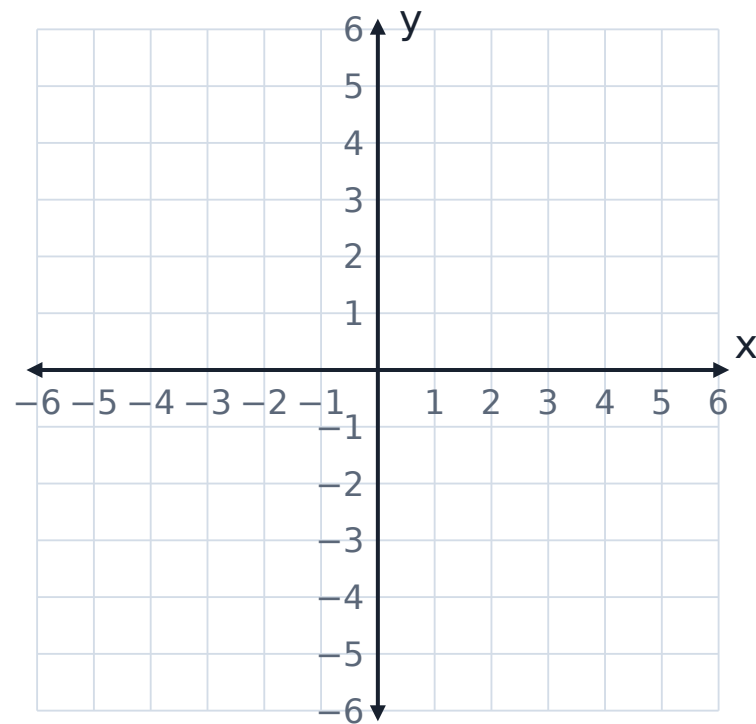
BACKWARD 2

Plotting points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

Plot and label point A($-2, 3$).



Plot and label point B(4, -1).



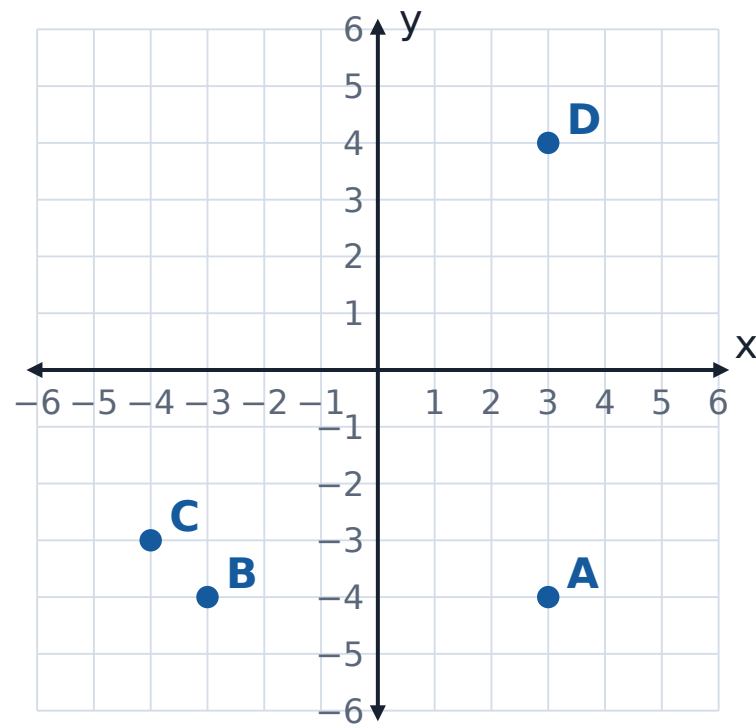
Which point is at $(-3, -4)$?

A. Point A

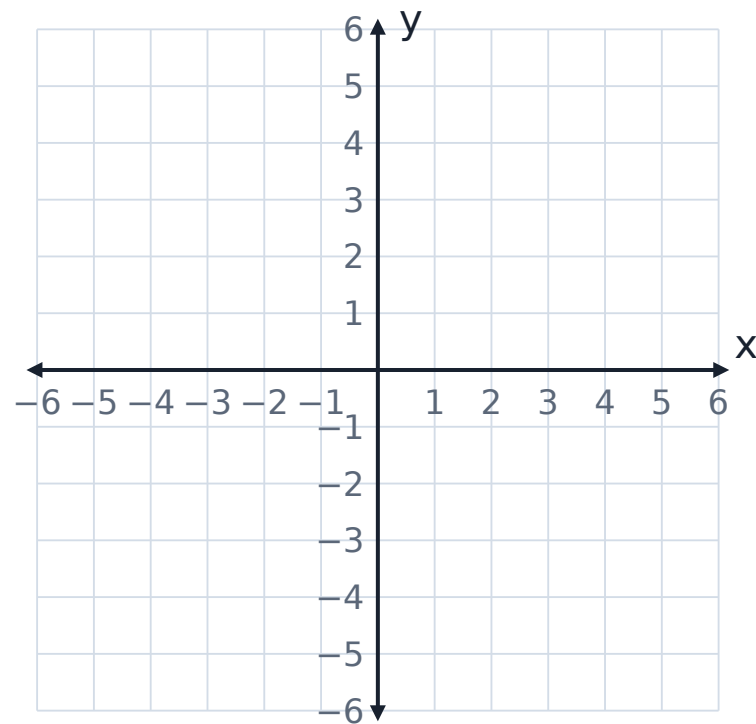
B. Point B

C. Point C

D. Point D



Plot and label point C(0, -5).

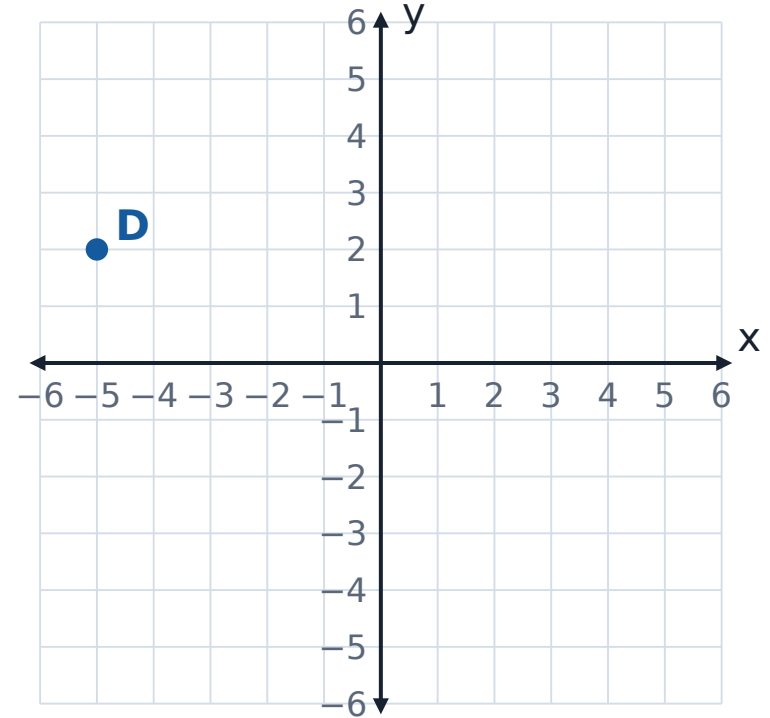


True or false?

Point D is located at $(-5, 2)$.

A. True

B. False



SET 39 • 6.NS.C.8

BACKWARD 3

Perimeter of rectangles • Grade 3 • Measurement & Data • 3.MD.D.8

What is the perimeter of the rectangle?

Perimeter: _____



A rectangle is 9 cm long and 3 cm wide.

What is its perimeter?

Perimeter: _____

True or false?

A square with 5-meter sides has a perimeter of 25 meters.

- A.** True
- B.** False

A rectangle is 8 inches long and 2 inches wide.

What is its perimeter?

- A.** 10 inches
- B.** 16 inches
- C.** 18 inches
- D.** 20 inches

A rectangle has a perimeter of 18 feet and a length of 5 feet.

What is its width?

Width: _____

SET 39 • 6.NS.C.8

BACKWARD 4

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 6 m long and 4 m wide.

What is its area?

Area: _____

True or false?

A rectangle that is 7 units by 3 units has an area of 21 square units.

- A. True
- B. False

A rectangle is 5 ft by 8 ft.

What is its area?

- A.** 13 square feet
- B.** 26 square feet
- C.** 40 square feet
- D.** 80 square feet

What is the area of the rectangle?

Area: _____



A square has 6-inch sides.

What is its area?

Area: _____

SET 39 • 6.NS.C.8

FORWARD 1

Use a map scale on a coordinate grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is $\frac{1}{2}$ mile. The school is at $(-3, 2)$ and the park is at $(5, 2)$.

What is the actual distance between them?

Distance: _____

On a trail map, each unit is 250 meters. The trail goes straight from $(0, -4)$ to $(0, 6)$.

How long is the actual trail?

Length: _____

On a map, each unit is 1.5 km. A park has corners at $(-2, 1)$, $(4, 1)$, $(4, -3)$, and $(-2, -3)$.

What is the actual perimeter of the park?

A. 13.3 km

B. 20 km

C. 30 km

D. 36 km

On a plan, each unit is 20 feet. A garden has corners at $(-1, -1)$, $(3, -1)$, $(3, 2)$, and $(-1, 2)$.

What are the actual length and width?

Length and width: _____

True or false?

On a map where each unit is 5 miles, the points $(-3, 0)$ and $(3, 0)$ are 30 miles apart.

A. True

B. False

SET 39 • 6.NS.C.8

FORWARD 2

Straight-line distances on a map • Grade 8 • Geometry • 8.G.B.8

On a map, the library is at $(-2, 1)$ and the pool is at $(4, 9)$. Each unit is 1 block.

What is the straight-line distance between them?

Distance: _____

A drone flies from $(0, 0)$ to $(9, 12)$. Each unit is 1 meter.
How far does it fly in a straight line?

Distance: _____

Home is at $(-3, -1)$ and school is at $(1, 2)$. Each unit is 1 km.

What is the straight-line distance?

A. 5 km

B. 25 km

C. 7 km

D. $\sqrt{7}$ km

Find the straight-line distance from $(-4, 6)$ to $(2, -2)$.

Distance: _____

True or false?

The straight-line distance from $(1, 1)$ to $(7, 9)$ is 10.

A. True

B. False

Evaluate numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Order of operations with parentheses	5.OA.A.1
BACKWARD 3	Repeated multiplication	3.OA.C.7
FORWARD 1	Products of signed numbers	7.NS.A.2.a
FORWARD 2	Evaluate expressions with integer exponents	8.EE.A.1

SET 40 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2^4 + 3^2$$

Value: _____

Evaluate.

$$3 \times 2^3$$

Value: _____

Evaluate.

$$(5 - 2)^3 \div 9$$

Value: _____

Evaluate.

$$5^2 - 4 \times 2$$

A. -3

B. 17

C. 21

D. 42

Evaluate.

$$\left(\frac{2}{3}\right)^2$$

Value: _____

SET 40 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write 10^3 as a whole number.

Number: _____

True or false?

$$10^4 = 10,000$$

A. True

B. False

Which is equal to 100,000?

A. 10^5

B. 5^{10}

C. 10^4

D. 10^6

Write 1,000,000 as a power of 10.

Power of 10: _____

True or false?

$$10^2 = 20$$

A. True

B. False

SET 40 • 6.EE.A.1

BACKWARD 2

Order of operations with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$(8 + 4) \div 3$$

Value: _____

Evaluate.

$$20 - (3 \times 4)$$

Value: _____

True or false?

$$2 \times (3 + 5) = 11$$

A. True

B. False

Evaluate.

$$18 \div (2 + 4)$$

A. 3

B. 6

C. 9

D. 13

Evaluate.

$$(9 - 3) \times 5$$

Value: _____

SET 40 • 6.EE.A.1

BACKWARD 3

Repeated multiplication • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$2 \times 2 \times 2$$

Product: _____

Multiply.

$$5 \times 5$$

Product: _____

True or false?

$$3 \times 3 \times 3 = 9$$

A. True

B. False

$$4 \times 4 \times 4 = ?$$

A. 12

B. 16

C. 44

D. 64

Multiply.

$$6 \times 6$$

Product: _____

SET 40 • 6.EE.A.1

FORWARD 1

Products of signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Evaluate.

$$(-3)^2$$

Value: _____

Evaluate.

$$(-2)^3$$

Value: _____

True or false?

$$-3^2 = (-3)^2$$

A. True

B. False

$$(-1)^5 = ?$$

A. -5

B. -1

C. 1

D. 5

Evaluate.

$$\left(-\frac{1}{2}\right)^2$$

Value: _____

SET 40 • 6.EE.A.1

FORWARD 2

Evaluate expressions with integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate.

$$2^{-3}$$

Value: _____

Evaluate.

$$5^{-2} \times 5^4$$

Value: _____

$$10^{-2} = ?$$

A. -100

B. -20

C. $\frac{1}{100}$

D. 0.1

Evaluate.

$$\frac{4^5}{4^3}$$

Value: _____

True or false?

$$7^0 = 1$$

A. True

B. False

Write numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Multiplying with repeated factors	3.OA.C.7
FORWARD 1	Powers of negative numbers	7.NS.A.2.a
FORWARD 2	Write expressions as a single power	8.EE.A.1

SET 41 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Write using an exponent.

$$7 \times 7 \times 7 \times 7$$

Expression: _____

Which expression is equal to $3 \times 3 \times 3 \times 3 \times 3$?

A. 5^3

B. 15

C. 3^5

D. 3×5

Write using an exponent.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Expression: _____

Write using an exponent.

$$0.4 \times 0.4$$

Expression: _____

Write using exponents.

$$2 \times 2 \times 2 \times 5 \times 5$$

Expression: _____

SET 41 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write $10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

True or false?

$$10 \times 10 \times 10 \times 10 = 10^4$$

A. True

B. False

Which is equal to 10^5 ?

A. $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$

B. $10 \times 10 \times 10 \times 10 \times 10$

C. 100,000,000

D. 10×5

How many factors of 10 are in 10^6 ?

Factors: _____

True or false?

$$10^2 = 10 \times 2$$

A. True

B. False

SET 41 • 6.EE.A.1

BACKWARD 2

Multiplying with repeated factors • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$2 \times 2 \times 2 \times 2$$

Product: _____

Multiply.

$$3 \times 3$$

Product: _____

True or false?

$$5 \times 5 \times 5 = 15$$

A. True

B. False

How many times is 7 used as a factor in $7 \times 7 \times 7$?

- A.** 1
- B.** 3
- C.** 7
- D.** 21

Multiply.

$$4 \times 4$$

Product: _____

SET 41 • 6.EE.A.1

FORWARD 1

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2.a

Write using an exponent.

$$(-3)(-3)(-3)(-3)$$

Expression: _____

Write $(-2)(-2)(-2)$ using an exponent. Is its value positive or negative?

Expression: _____

Sign: _____

Which expression is equal to $(-5)^2$?

A. -5×2

B. $(-5)(-5)$

C. -5×5

D. $-(5 \times 5)$

True or false?

$(-1)^6$ is positive.

A. True

B. False

Write using an exponent.

$$\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)$$

Expression: _____

SET 41 • 6.EE.A.1

FORWARD 2

Write expressions as a single power • Grade 8 • Expressions & Equations •
8.EE.A.1

Write as a single power.

$$4^3 \times 4^5$$

Single power: _____

Write as a single power.

$$(2^3)^4$$

Single power: _____

Which is equal to $\frac{6^7}{6^2}$?

A. 6^{14}

B. 6^5

C. 6^9

D. 1^5

Write as a single power.

$$3^4 \times 3^{-6}$$

Single power: _____

True or false?

$$5^2 \times 5^3 = 5^6$$

A. True

B. False

Write expressions in which letters stand for numbers

MAIN	Grade 6 main question	6.EE.A.2.a
BACKWARD 1	Words for operations	5.OA.A.2
BACKWARD 2	A symbol for an unknown number	3.OA.A.4
BACKWARD 3	Order matters in subtraction and division	3.OA.B.5
FORWARD 1	Write and solve equations of the form $px + q = r$	7.EE.B.4.a
FORWARD 2	Construct a function from a description	8.F.B.4

SET 42 • 6.EE.A.2.a

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.a

Write an expression for "subtract m from 12."

Expression: _____

Write an expression for "5 more than twice a number n ."

Expression: _____

Which expression means "the quotient of x and 4"?

A. $4x$

B. $\frac{4}{x}$

C. $x - 4$

D. $\frac{x}{4}$

Write an expression for "3 less than the product of 7 and a number m ."

Expression: _____

True or false?

"8 less than k" can be written as $8 - k$.

A. True

B. False

SET 42 • 6.EE.A.2.a

BACKWARD 1

Words for operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Which expression shows "add 6 and 9, then multiply by 4"?

A. $6 + 9 \times 4$

B. $4 \times (6 + 9)$

C. $(4 + 6) \times 9$

D. $4 \times 6 + 9$

Write an expression for "subtract 4 from 10, then divide by 2."

Expression: _____

True or false?

"The product of 6 and 9" means 6×9 .

A. True

B. False

Which word describes the answer to a subtraction problem?

- A.** Quotient
- B.** Difference
- C.** Product
- D.** Sum

Write an expression for "the sum of 12 and 5, multiplied by 3."

Expression: _____

SET 42 • 6.EE.A.2.a

BACKWARD 2

A symbol for an unknown number • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.4

What number goes in the box?

$$\square + 7 = 12$$

$$\square = \underline{\hspace{10em}}$$

True or false?

In $7 \times \square = 42$, the box stands for 6.

A. True

B. False

$$\square - 5 = 9$$

What number goes in the box?

- A.** 4
- B.** 9
- C.** 14
- D.** 45

What number goes in the box?

$$20 \div \square = 4$$

$$\square = \underline{\hspace{10em}}$$

What number goes in the box?

$$\square \times 3 = 27$$

$$\square = \underline{\hspace{10cm}}$$

SET 42 • 6.EE.A.2.a

BACKWARD 3

Order matters in subtraction and division • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$7 - 3 = 3 - 7$$

A. True

B. False

True or false?

$$6 \times 4 = 4 \times 6$$

A. True

B. False

True or false?

$$12 \div 3 = 3 \div 12$$

A. True

B. False

Which statement is true?

A. $10 \div 2 = 2 \div 10$

B. $5 + 9 = 9 + 5$

C. $8 - 2 = 2 - 8$

D. $9 - 5 = 5 - 9$

True or false?

$$3 + 11 = 11 + 3$$

A. True

B. False

SET 42 • 6.EE.A.2.a

FORWARD 1

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A rectangle has a perimeter of 46 inches. Its width is 8 inches.

Write and solve an equation to find its length.

Length: _____

Jo has \$40. She buys 3 shirts that cost the same amount and has \$4 left.

How much does each shirt cost?

Cost: _____

Which equation represents "5 more than 3 times a number is 26"?

A. $3(n + 5) = 26$

B. $5n + 3 = 26$

C. $3n + 5 = 26$

D. $3n - 5 = 26$

Twice a number minus 7 is 15.

Write and solve an equation to find the number.

Number: _____

A phone plan costs \$30 per month plus \$0.10 per text.
One month's bill is \$42.

How many texts were sent?

Texts: _____

SET 42 • 6.EE.A.2.a

FORWARD 2

Construct a function from a description • Grade 8 • Functions • 8.F.B.4

A pool has 500 gallons of water. It is filled at 25 gallons per minute.

Write a function for the gallons G after t minutes.

$G =$ _____

Tickets cost \$4 each plus a \$6 fee for the order.

Which function gives the total cost y for x tickets?

A. $y = 10x$

B. $y = 4x + 6$

C. $y = 6x + 4$

D. $y = 4x - 6$

The table shows a linear function.

Write an equation for the function.

x	0	1	2
y	7	10	13

$y =$ _____

A line passes through $(0, -2)$ and $(3, 4)$.

Write an equation for the line.

$y =$ _____

True or false?

The function $y = 15x + 20$ has an initial value of 15.

A. True

B. False

Identify parts of an expression: term, sum, product, factor, quotient, coefficient

MAIN	Grade 6 main question	6.EE.A.2.b
BACKWARD 1	Sum, difference, product, and quotient	5.OA.A.2
BACKWARD 2	Factors	4.OA.B.4
BACKWARD 3	Interpreting expressions without evaluating	5.OA.A.2
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Interpret the parts of a linear function	8.F.B.4

SET 43 • 6.EE.A.2.b

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.b

In the expression $4(9 + 3)$, how can the 4 be described?

- A.** A sum
- B.** A factor
- C.** A term of $9 + 3$
- D.** A quotient

How many terms are in the expression $3x + 5y - 2$?

Terms: _____

What is the coefficient of y in $4x + 9y$?

Coefficient: _____

True or false?

In $5(m + n)$, the quantity $(m + n)$ is a single factor and also a sum of two terms.

A. True

B. False

In the expression $7m + 3$, which part is the constant term?

- A.** 3
- B.** $7m$
- C.** m
- D.** 7

SET 43 • 6.EE.A.2.b

BACKWARD 1

Sum, difference, product, and quotient • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$8 + 7$ is a sum.

A. True

B. False

What is 6×9 called?

- A.** A quotient
- B.** A product
- C.** A sum
- D.** A difference

Is $24 \div 6$ a sum, difference, product, or quotient?

Answer: _____

True or false?

15 – 4 is called a product.

A. True

B. False

Which expression is a quotient?

A. 20×5

B. $20 + 5$

C. $20 - 5$

D. $20 \div 5$

SET 43 • 6.EE.A.2.b

BACKWARD 2

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

In $4 \times 7 = 28$, the numbers 4 and 7 are factors of 28.

A. True

B. False

Name the two factors in 5×9 .

Factors: _____

Which number is NOT a factor of 20?

- A.** 4
- B.** 5
- C.** 6
- D.** 10

True or false?

3 is a factor of 12.

A. True

B. False

Write one factor pair of 18.

Factor pair: _____

SET 43 • 6.EE.A.2.b

BACKWARD 3

Interpreting expressions without evaluating • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$4 \times (2,365 + 718)$ is four times as large as $2,365 + 718$.

A. True

B. False

How does $5 \times (26 + 4)$ compare to $26 + 4$?

- A.** It is the same.
- B.** It is 5 times as large.
- C.** It is 5 more.
- D.** It is 5 less.

True or false?

$(9 + 4) \times 2$ is 2 more than $9 + 4$.

A. True

B. False

Which expression is 4 times as large as $125 - 25$?

A. $(125 - 25) \div 4$

B. $4 \times (125 - 25)$

C. $125 - 25 \times 4$

D. $4 + (125 - 25)$

Without calculating, how many times as large is
 $10 \times (63 + 7)$ as $63 + 7$?

Times as large: _____

SET 43 • 6.EE.A.2.b

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$$4x + 3 - 2x + 5$$

Simplified: _____

Simplify.

$$0.5y + 1.25y$$

Simplified: _____

Which pair shows like terms?

A. $3a$ and $-7a$

B. x^2 and x

C. $3a$ and $3b$

D. $4x$ and 4

Simplify.

$$-3(x - 2) + 5x$$

Simplified: _____

True or false?

$$\frac{1}{2}x + \frac{1}{4}x = \frac{3}{4}x$$

A. True

B. False

SET 43 • 6.EE.A.2.b

FORWARD 2

Interpret the parts of a linear function • Grade 8 • Functions • 8.F.B.4

The cost to bowl is $C = 4.5n + 12$, where n is the number of games.

What does 12 represent?

Meaning: _____

In $y = -3x + 7$, what is the rate of change?

A. x

B. 3

C. -3

D. 7

The gas left in a car's tank is $G = 14 - 0.04m$, where m is miles driven.

What is the initial value?

Initial value: _____

A car's distance from home is $d = 60t + 25$, where t is hours.

What does 60 represent?

- A.** The speed in miles per hour
- B.** The starting distance
- C.** The total distance
- D.** The total time

True or false?

In $y = 8x$, the initial value is 0.

A. True

B. False

Evaluate expressions and formulas for given values

MAIN	Grade 6 main question	6.EE.A.2.c
BACKWARD 1	Whole-number exponents	6.EE.A.1
BACKWARD 2	Order of operations with grouping symbols	5.OA.A.1
BACKWARD 3	A letter stands for a number	3.OA.D.8
BACKWARD 4	Multiplying fractions	5.NF.B.4.a
FORWARD 1	Evaluate expressions with rational numbers	7.NS.A.3
FORWARD 2	Volume formulas for cylinders, cones, and spheres	8.G.C.9

SET 44 • 6.EE.A.2.c

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x^2 + 2$ when $x = 4$.

Value: _____

The volume of a cube is $V = s^3$.

Find V when $s = \frac{2}{3}$ foot.

$V =$ _____

The perimeter of a rectangle is $P = 2l + 2w$.

Find P when $l = 7.5$ cm and $w = 4$ cm.

$P =$ _____

Evaluate $2(a + b)$ when $a = 3.5$ and $b = 4$.

A. 7.5

B. 9.5

C. 11.5

D. 15

Evaluate $\frac{x}{4} + 5y$ when $x = 12$ and $y = 2$.

Value: _____

SET 44 • 6.EE.A.2.c

BACKWARD 1

Whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$4^2$$

Value: _____

Evaluate.

$$\left(\frac{1}{2}\right)^3$$

Value: _____

True or false?

$$3^2 = 6$$

A. True

B. False

$$2^5 = ?$$

A. 10

B. 16

C. 25

D. 32

Evaluate.

$$5^2 + 1$$

Value: _____

SET 44 • 6.EE.A.2.c

BACKWARD 2

Order of operations with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$3 \times (16 + 2)$$

Value: _____

Evaluate.

$$2 \times (6 + 4)$$

Value: _____

True or false?

$$(3 + 4) \times 2 = 11$$

A. True

B. False

Evaluate.

$$(12 \div 4) + 10$$

A. 1

B. 6

C. 13

D. 22

Evaluate.

$$[2 \times (5 + 3)] - 4$$

Value: _____

SET 44 • 6.EE.A.2.c

BACKWARD 3

A letter stands for a number • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

If $n = 6$, what is $n + 9$?

Value: _____

True or false?

If $a = 4$, then $a \times 5 = 20$.

A. True

B. False

If $k = 7$, what is $3 \times k$?

A. 10

B. 21

C. 37

D. 73

If $m = 20$, what is $m - 8$?

Value: _____

If $t = 3$, what is $4 \times t + 2$?

Value: _____

SET 44 • 6.EE.A.2.c

BACKWARD 4

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$$

Product: _____

Multiply.

$$6 \times \frac{1}{3}$$

Product: _____

True or false?

$$\frac{2}{3} \times \frac{2}{3} = \frac{4}{6}$$

A. True

B. False

$$\frac{3}{4} \times 8 = ?$$

A. $\frac{3}{32}$

B. 2

C. 6

D. 24

Multiply.

$$\frac{1}{2} \times 12$$

Product: _____

SET 44 • 6.EE.A.2.c

FORWARD 1

Evaluate expressions with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Use $C = \frac{5}{9}(F - 32)$ to find C when $F = -4$.

$C =$ _____

Evaluate $-2x + 5$ when $x = -3.5$.

Value: _____

Evaluate $a^2 - b$ when $a = -3$ and $b = 4$.

A. -13

B. -5

C. 5

D. 13

Evaluate $\frac{(x+y)}{2}$ when $x = -6.5$ and $y = 2.5$.

Value: _____

True or false?

When $x = -2$, the value of $x^3 + 1$ is -7 .

A. True

B. False

SET 44 • 6.EE.A.2.c

FORWARD 2

Volume formulas for cylinders, cones, and spheres • Grade 8 • Geometry •
8.G.C.9

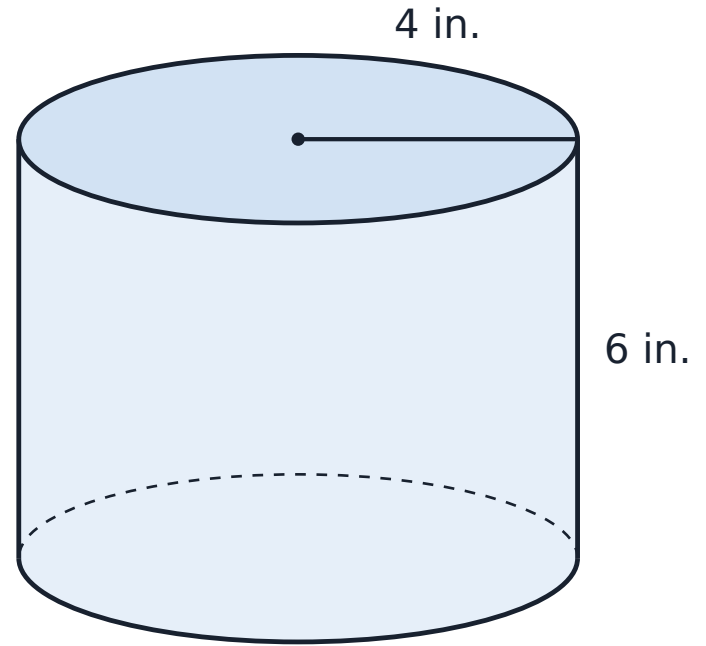
A cylinder has a radius of 3 cm and a height of 10 cm.
Find its volume. Use 3.14 for π .

Volume: _____

Find the volume of the cylinder.

Use 3.14 for π .

Volume: _____



A cone has a radius of 3 units and a height of 4 units.

What is its volume in terms of π ?

A. 36π

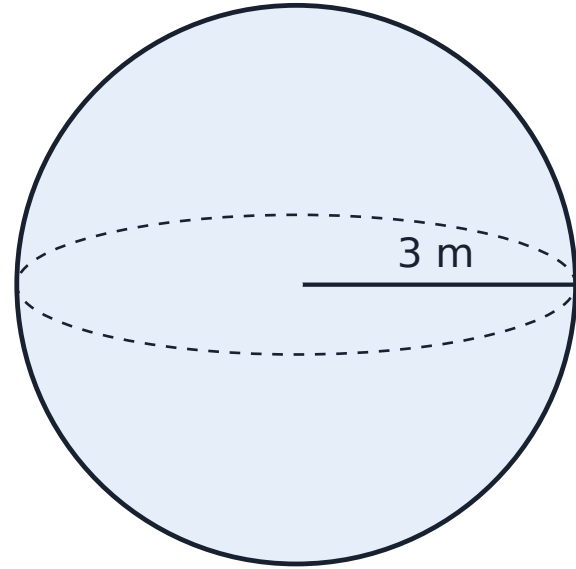
B. 48π

C. 4π

D. 12π

Find the volume of the sphere
in terms of π .

Volume: _____



True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 20π .

A. True

B. False

Use the distributive property to expand expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Distributive property with numbers	3.OA.B.5
BACKWARD 2	Equal groups	3.OA.A.1
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Expand linear expressions with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by expanding and collecting like terms	8.EE.C.7.b

SET 45 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(4 + x)$$

Expression: _____

Expand.

$$5(3a - 2)$$

Expression: _____

Which expression is equivalent to $7(2m + 5)$?

- A.** $9m + 35$
- B.** $2m + 35$
- C.** $14m + 5$
- D.** $14m + 35$

Expand.

$$2(x + 3y + 4)$$

Expression: _____

Expand.

$$4(0.5y + 3)$$

Expression: _____

SET 45 • 6.EE.A.3

BACKWARD 1

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$6 \times (4 + 3) = 6 \times 4 + 6 \times 3$$

A. True

B. False

Fill in the blank.

$$5 \times (2 + 7) = 5 \times 2 + 5 \times \underline{\quad}$$

Blank: _____

Which is equal to 3×14 ?

A. $30 + 4 + 3$

B. $3 \times 10 + 4$

C. $3 \times 10 + 3 \times 4$

D. $3 + 10 \times 4$

True or false?

$$7 \times 12 = 7 \times 10 + 2$$

A. True

B. False

Fill in the blank.

$$8 \times (5 + 1) = 8 \times 5 + \underline{\hspace{2cm}}$$

Blank: _____

SET 45 • 6.EE.A.3

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 2)$ means 3 groups of $(4 + 2)$.

A. True

B. False

Which means 4 groups of $(5 + 1)$?

A. $4 + (5 + 1)$

B. $4 \times (5 + 1)$

C. $4 \times 5 + 1$

D. $(4 + 5) \times 1$

How many groups of $(2 + 6)$ are in $5 \times (2 + 6)$?

Groups: _____

True or false?

$$2 \times (3 + 4) = (3 + 4) + (3 + 4)$$

A. True

B. False

Write 6×3 as a sum of equal groups.

Sum: _____

SET 45 • 6.EE.A.3

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$4 \times 0.5$$

Product: _____

Multiply.

$$3 \times 1.2$$

Product: _____

True or false?

$$6 \times 0.25 = 1.5$$

A. True

B. False

$$5 \times 0.4 = ?$$

A. 0.2

B. 2

C. 5.4

D. 20

Multiply.

$$8 \times 0.75$$

Product: _____

SET 45 • 6.EE.A.3

FORWARD 1

Expand linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$$\frac{1}{2}(8x - 6)$$

Expression: _____

Expand.

$$-4(2y - 3)$$

Expression: _____

Which is equivalent to $0.5(6a + 10)$?

- A.** $3a + 5$
- B.** $6a + 5$
- C.** $3.5a + 5$
- D.** $3a + 10$

Expand.

$$-\frac{2}{3}(9m + 3)$$

Expression: _____

True or false?

$$-2(x - 7) = -2x + 14$$

A. True

B. False

SET 45 • 6.EE.A.3

FORWARD 2

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$3(x + 2) = 5x - 4$$

x = _____

Solve for x.

$$2(4x - 1) - 3x = 18$$

x = _____

Solve for x.

$$-2(x - 5) = 3(x + 5)$$

A. -5

B. -1

C. 1

D. 5

Solve for x.

$$6(x - 1) = 4(x + 2)$$

x = _____

Solve for x.

$$5(2x + 1) = 3(3x + 4)$$

x = _____

Use the distributive property to factor expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Greatest common factor	6.NS.B.4
BACKWARD 2	Distributive property with numbers	3.OA.B.5
BACKWARD 3	Factors	4.OA.B.4
FORWARD 1	Factor linear expressions with rational coefficients	7.EE.A.1
FORWARD 2	Use factoring to decide how many solutions an equation has	8.EE.C.7.a

SET 46 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Factor using the greatest common factor.

$$24x + 16$$

Expression: _____

Factor using the greatest common factor.

$$18y + 30$$

Expression: _____

Which shows $15a + 25b$ factored using the greatest common factor?

- A.** $15(a + 10b)$
- B.** $5(3a + 5b)$
- C.** $3(5a + 25b)$
- D.** $5(3a + 25b)$

Factor using the greatest common factor.

$$36m + 27n$$

Expression: _____

Factor using the greatest common factor.

$$14x + 21$$

Expression: _____

SET 46 • 6.EE.A.3

BACKWARD 1

Greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 16.

GCF: _____

Find the greatest common factor of 18 and 30.

GCF: _____

True or false?

The greatest common factor of 15 and 25 is 5.

- A. True
- B. False

What is the greatest common factor of 36 and 27?

- A.** 3
- B.** 6
- C.** 9
- D.** 12

Find the greatest common factor of 14 and 21.

GCF: _____

SET 46 • 6.EE.A.3

BACKWARD 2

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 \times 3 + 8 \times 2 = 8 \times (3 + 2)$$

A. True

B. False

Fill in the blank.

$$6 \times 4 + 6 \times 5 = 6 \times (4 + \underline{\quad})$$

Blank: _____

Which is equal to $7 \times 2 + 7 \times 9$?

A. $14 + 9$

B. $7 \times 2 \times 9$

C. $7 \times (2 + 9)$

D. $7 + 2 + 9$

True or false?

$$5 \times 6 + 3 \times 6 = 6 \times (5 + 3)$$

A. True

B. False

Fill in the blank.

$$9 \times 7 + 9 \times 3 = \underline{\quad} \times (7 + 3)$$

Blank: _____

SET 46 • 6.EE.A.3

BACKWARD 3

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 16.

Factors: _____

True or false?

6 is a factor of 30.

A. True

B. False

Which number is a factor of 27?

- A.** 4
- B.** 6
- C.** 9
- D.** 12

List all the factors of 21.

Factors: _____

True or false?

8 is a factor of 36.

A. True

B. False

SET 46 • 6.EE.A.3

FORWARD 1

Factor linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor out 2.5.

$$2.5x + 10$$

Expression: _____

Factor out -4 .

$$-4x + 12$$

Expression: _____

Which is equivalent to $\frac{1}{2}x + 3$?

A. $\frac{1}{2}(x + \frac{3}{2})$

B. $\frac{1}{2}(x + 3)$

C. $\frac{1}{2}(x + 6)$

D. $2(x + 6)$

Factor out 0.6.

$$0.6y - 1.8$$

Expression: _____

True or false?

$$-5a - 15 = -5(a + 3)$$

A. True

B. False

SET 46 • 6.EE.A.3

FORWARD 2

Use factoring to decide how many solutions an equation has • Grade 8 •
Expressions & Equations • 8.EE.C.7.a

Factor the left side to decide. How many solutions does the equation have?

$$6x + 9 = 3(2x + 3)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$12x + 8 = 4(3x + 1)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$10x - 15 = 5(2x - 3)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$8x + 4 = 4(x + 1)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

What number goes in the box so that the equation has infinitely many solutions?

$$14x + 21 = 7(2x + \square)$$

$\square =$ _____

Combine like terms to write equivalent expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Repeated addition as multiplication	3.OA.A.1
BACKWARD 2	Changing the order and grouping of addends	3.OA.B.5
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by collecting like terms	8.EE.C.7.b

SET 47 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Write an equivalent expression with one term.

$$y + y + y + y$$

Expression: _____

Simplify.

$$5x + 3x$$

Expression: _____

Which expression is equivalent to $7a + 2 + 3a$?

A. $7a + 5a$

B. $10a + 5$

C. $10a + 2$

D. $12a$

Simplify.

$$4m + 6 + 2m + 1$$

Expression: _____

Simplify.

$$9n - 4n + 3$$

Expression: _____

SET 47 • 6.EE.A.3

BACKWARD 1

Repeated addition as multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$4 + 4 + 4 = 3 \times 4$$

A. True

B. False

Write $7 + 7 + 7 + 7$ as a multiplication expression.

Expression: _____

What does 4×9 mean?

- A.** 4 groups of 9
- B.** 9 minus 4
- C.** 4 more than 9
- D.** The number 49

True or false?

$$2 + 2 + 2 + 2 + 2 = 2 \times 2$$

A. True

B. False

Write $9 + 9$ as a multiplication expression.

Expression: _____

SET 47 • 6.EE.A.3

BACKWARD 2

Changing the order and grouping of addends • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$4 + 6 + 2 = 4 + 2 + 6$$

A. True

B. False

Fill in the blank.

$$7 + 5 + 3 = 7 + 3 + \underline{\quad}$$

Blank: _____

Which is equal to $(8 + 2) + 5$?

A. $8 + 2 \times 5$

B. $8 \times (2 + 5)$

C. $(8 - 2) + 5$

D. $8 + (2 + 5)$

True or false?

$$9 + 1 + 9 = 9 + 9 + 1$$

A. True

B. False

Fill in the blank.

$$6 + 3 + 4 = 6 + 4 + \underline{\quad}$$

Blank: _____

SET 47 • 6.EE.A.3

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$$2.25y + 0.75y$$

Expression: _____

Simplify.

$$\frac{1}{2}x + \frac{1}{4}x$$

Expression: _____

Simplify.

$$6x - 2 + 3x + 7$$

Expression: _____

Simplify.

$$-3a + 7a - 2$$

A. $4a - 2$

B. $4a + 2$

C. $2a$

D. $-10a - 2$

True or false?

$$2.5\text{m} - \text{m} = 1.5\text{m}$$

A. True

B. False

SET 47 • 6.EE.A.3

FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$3x + 5x - 4 = 20$$

x = _____

Solve for y .

$$7y - 2y + 3 = 2y + 18$$

$y =$ _____

Solve for a.

$$4a + 6 - a = 2a + 10$$

A. -4

B. $\frac{16}{5}$

C. 4

D. 16

Solve for n.

$$0.5n + 1.5n - 3 = n + 5$$

n = _____

Solve for k.

$$6k - 2k + 1 = 3k - 4$$

k = _____

Identify when two expressions are equivalent

MAIN	Grade 6 main question	6.EE.A.4
BACKWARD 1	Commutative and associative properties	3.OA.B.5
BACKWARD 2	Evaluate an expression for a given value	6.EE.A.2.c
BACKWARD 3	Equal groups	3.OA.A.1
FORWARD 1	Rewrite expressions to show how quantities are related	7.EE.A.2
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 48 • 6.EE.A.4

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.4

True or false?

$m + m + m + m$ and $4m$ are equivalent expressions.

A. True

B. False

Which expression is equivalent to $2(x + 4)$?

A. $x + 8$

B. $2x + 4$

C. $2x + 6$

D. $2x + 8$

True or false?

$3(x + 2)$ and $3x + 2$ are equivalent expressions.

A. True

B. False

Which expression is equivalent to $6a + 4a$?

A. $6a + 4$

B. $10a$

C. $24a$

D. $10a^2$

Which pair of expressions is equivalent?

A. $5(a + 3)$ and $5a + 15$

B. $5(a + 3)$ and $a + 15$

C. $5(a + 3)$ and $5a + 3$

D. $5(a + 3)$ and $8a$

SET 48 • 6.EE.A.4

BACKWARD 1

Commutative and associative properties • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 + 5 = 5 + 8$$

A. True

B. False

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

A. True

B. False

Which equation shows the commutative property of multiplication?

A. $6 \times 7 = 42$

B. $6 \times 7 = 7 \times 6$

C. $6 \times (7 + 1) = 6 \times 7 + 6$

D. $6 + 0 = 6$

True or false?

$$9 - 4 = 4 - 9$$

A. True

B. False

Fill in the blank.

$$3 \times 8 = 8 \times \underline{\quad}$$

Blank: _____

SET 48 • 6.EE.A.4

BACKWARD 2

Evaluate an expression for a given value • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4x + 2x$ when $x = 3$.

Value: _____

Evaluate $3(x + 2)$ when $x = 5$.

Value: _____

True or false?

When $x = 2$, the value of $3x + 2$ is 8.

A. True

B. False

Evaluate $2(x + 4)$ when $x = 1$.

- A.** 6
- B.** 9
- C.** 10
- D.** 12

Evaluate $5a + 15$ when $a = 2$.

Value: _____

SET 48 • 6.EE.A.4

BACKWARD 3

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$6 + 6 + 6 + 6 = 4 \times 6$$

A. True

B. False

Which expression means 3 groups of 8?

A. $3 + 8$

B. 38

C. $8 - 3$

D. 3×8

Write $5 + 5 + 5$ as a multiplication expression.

Expression: _____

True or false?

$$7 \times 2 = 7 + 7$$

A. True

B. False

How many groups of 4 are in $4 + 4 + 4 + 4 + 4$?

Groups: _____

SET 48 • 6.EE.A.4

FORWARD 1

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price c plus an 8% tax can be written as $c + 0.08c$.

Which expression is equivalent?

- A.** $0.08c$
- B.** $1.08c$
- C.** $1.8c$
- D.** $8.08c$

True or false?

$$a - 0.2a = 0.8a$$

A. True

B. False

Write an expression with one term that is equivalent to $n - 0.25n$.

Expression: _____

Which expression shows the price p after a 30% discount?

- A.** $0.3p$
- B.** $0.7p$
- C.** $1.3p$
- D.** $p - 30$

The perimeter of a square is $s + s + s + s$.

Write an equivalent expression with one term.

Expression: _____

SET 48 • 6.EE.A.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

True or false?

The equation $2(x + 3) = 2x + 6$ has infinitely many solutions.

A. True

B. False

How many solutions does the equation have?

$$5x - 3 = 5x + 1$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$3(2x + 4) = 6x + 12$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

Write a number in the box so the equation has no solution.

$$4x + 7 = 4x + \square$$

$$\square = \underline{\hspace{10cm}}$$

Which equation has exactly one solution?

A. $3x + 1 = 2x + 1$

B. $3x + 1 = 3x + 2$

C. $3x + 1 = 3x + 1$

D. $3(x + 1) = 3x + 3$

Use substitution to decide whether a number makes an equation or inequality true

MAIN	Grade 6 main question	6.EE.B.5
BACKWARD 1	Evaluate an expression	6.EE.A.2.c
BACKWARD 2	Comparing numbers	2.NBT.A.4
BACKWARD 3	Meaning of the equal sign	1.OA.D.7
FORWARD 1	Solve equations of the form $px + q = r$ and $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solutions of a system of two linear equations	8.EE.C.8.a

SET 49 • 6.EE.B.5

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.5

Which value from the set $\{2, 4, 6, 8\}$ makes $3x + 1 = 19$ true?

A. 2

B. 4

C. 6

D. 8

True or false?

$x = 5$ is a solution of $2x - 3 > 8$.

A. True

B. False

Which numbers from the set $\{1, 3, 5, 7\}$ make $x + 4 < 10$ true?

Numbers: _____

Is $y = 2.5$ a solution of $4y = 10$?

A. Yes

B. No

Which value from the set $\{12, 15, 18\}$ makes $n \div 3 = 6$ true?

$n =$ _____

SET 49 • 6.EE.B.5

BACKWARD 1

Evaluate an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x + 1$ when $x = 6$.

Value: _____

Evaluate $2x - 3$ when $x = 5$.

Value: _____

True or false?

When $y = 4$, the value of $5y$ is 20.

A. True

B. False

Evaluate $x + 4$ when $x = 7$.

- A.** 3
- B.** 11
- C.** 28
- D.** 47

Evaluate $n \div 3$ when $n = 15$.

Value: _____

SET 49 • 6.EE.B.5

BACKWARD 2

Comparing numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

19 21

Symbol: _____

True or false?

$$7 > 8$$

A. True

B. False

Which statement is true?

A. $9 > 6$

B. $10 < 5$

C. $11 < 10$

D. $6 > 9$

True or false?

$$24 = 24$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

105 150

Symbol: _____

SET 49 • 6.EE.B.5

BACKWARD 3

Meaning of the equal sign • Grade 1 • Operations & Algebraic Thinking •
1.OA.D.7

True or false?

$$4 + 5 = 9 + 0$$

A. True

B. False

True or false?

$$12 = 15 - 3$$

A. True

B. False

True or false?

$$3 + 6 = 5 + 3$$

A. True

B. False

Which equation is true?

A. $4 + 3 = 5 + 3$

B. $4 + 3 = 5 + 2$

C. $7 = 4 - 3$

D. $4 + 3 = 8$

True or false?

$$10 - 2 = 7$$

A. True

B. False

SET 49 • 6.EE.B.5

FORWARD 1

Solve equations of the form $px + q = r$ and $p(x + q) = r$ • Grade 7 •
Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 7 = 22$$

x = _____

Solve for x.

$$4(x - 2) = 20$$

x = _____

Solve for x.

$$0.5x + 6 = 10$$

A. 2

B. 8

C. 20

D. 32

Solve for x.

$$\frac{2}{3}x - 4 = 10$$

x = _____

A movie ticket costs \$12. Popcorn costs p dollars. Two tickets and one popcorn cost \$31.

What is p ?

$p =$ _____

SET 49 • 6.EE.B.5

FORWARD 2

Solutions of a system of two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.a

True or false?

$(2, 3)$ is a solution of both $y = x + 1$ and $y = 2x - 1$.

A. True

B. False

Which point is a solution of both $y = 2x$ and $y = x + 3$?

A. (0, 3)

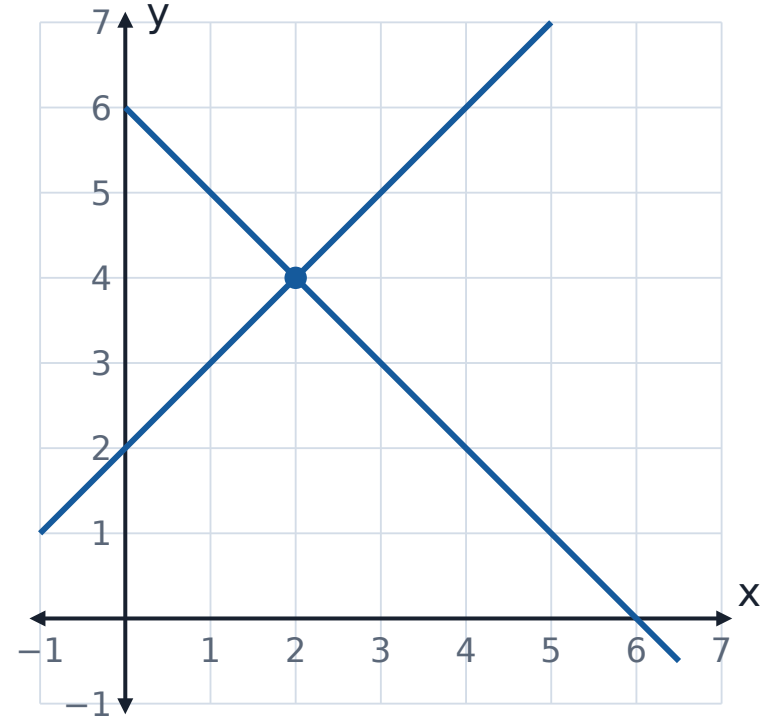
B. (2, 5)

C. (1, 2)

D. (3, 6)

What is the solution of the system shown by the graph?

Solution: _____



True or false?

(1, 4) is a solution of both $y = 3x + 1$ and $y = -x + 5$.

A. True

B. False

Is $(0, 2)$ a solution of both $y = 2x + 2$ and $y = x - 2$?

A. Yes

B. No

Use variables to write expressions for real-world problems

MAIN	Grade 6 main question	6.EE.B.6
BACKWARD 1	Word problems with an unknown number	3.OA.A.3
BACKWARD 2	Multiplicative comparison word problems	4.OA.A.2
BACKWARD 3	Number patterns that follow a rule	4.OA.C.5
FORWARD 1	Write and solve equations for word problems	7.EE.B.4.a
FORWARD 2	Model a linear relationship with a function	8.F.B.4

SET 50 • 6.EE.B.6

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost \$8 each.

Write an expression for the cost of t tickets.

Expression: _____

Kim is k years old. Sam is 3 years older than Kim.

Write an expression for Sam's age.

Expression: _____

A pizza is cut into s slices. 4 friends share the slices equally.

Which expression shows how many slices each friend gets?

A. $4s$

B. $s - 4$

C. $\frac{s}{4}$

D. $s + 4$

The length of a rectangle is 5 inches more than its width w .

Write an expression for the length.

Expression: _____

A gym charges \$20 to join plus \$15 per month.

Write an expression for the total cost after m months.

Expression: _____

SET 50 • 6.EE.B.6

BACKWARD 1

Word problems with an unknown number • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 bags with the same number of apples.
There are 42 apples in all.

How many apples are in each bag?

Apples: _____

There are 5 boxes with $\square$ pencils in each box. There are 35 pencils in all.

Which equation matches?

A. $35 \times 5 = \square$

B. $\square - 5 = 35$

C. $5 + \square = 35$

D. $5 \times \square = 35$

True or false?

If $4 \times \square = 28$, then $\square = 7$.

A. True

B. False

There are 8 rows with $\square$ chairs in each row. There are 56 chairs in all.

What number is $\square$?

$\square =$ _____

Jan split 45 stickers equally among 9 friends.
How many stickers did each friend get?

Stickers: _____

SET 50 • 6.EE.B.6

BACKWARD 2

Multiplicative comparison word problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A red ribbon is 7 inches long. A blue ribbon is 3 times as long.

How long is the blue ribbon?

Length: _____

Ben has 6 books. Ali has 4 times as many books as Ben.
Which equation finds b , the number of books Ali has?

A. $b = 4 \times 6$

B. $b = 4 + 6$

C. $b = 6 \div 4$

D. $b = 6 - 4$

True or false?

A cat weighs 5 pounds. A dog weighs 8 times as much. The dog weighs 40 pounds.

A. True

B. False

A bush is 4 feet tall. A tree is 5 times as tall.

How tall is the tree?

Height: _____

Sara ran 24 miles. That is 3 times as far as Tom ran.

How far did Tom run?

Miles: _____

SET 50 • 6.EE.B.6

BACKWARD 3

Number patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking
• 4.OA.C.5

The rule is "add 4." The pattern starts at 3.
Write the first five numbers.

Numbers: _____

The rule is "multiply by 2." The pattern starts at 1.
What is the 5th number?

- A.** 8
- B.** 10
- C.** 16
- D.** 32

True or false?

The rule "add 5," starting at 5, gives 5, 10, 15, 20.

A. True

B. False

What is the rule for this pattern?

7, 11, 15, 19, ...

Rule: _____

The rule is "add 6." The pattern starts at 2.

What is the 4th number?

Number: _____

SET 50 • 6.EE.B.6

FORWARD 1

Write and solve equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Sam has \$60. He spends \$8 each week.

Write and solve an equation to find when he will have \$20 left.

Weeks: _____

A rectangle has a perimeter of 50 cm and a length of 15 cm.

Which equation can be used to find its width w ?

A. $2(15 + w) = 50$

B. $2(15w) = 50$

C. $15 + 2w = 50$

D. $15 + w = 50$

A taxi charges \$3 plus \$2 per mile. The fare was \$17.

How many miles was the ride?

Miles: _____

You buy 4 notebooks and a \$6 backpack. You spend \$18 in all.

How much does each notebook cost?

Cost: _____

True or false?

"Five less than twice a number is 17" can be written as $2n - 5 = 17$.

A. True

B. False

SET 50 • 6.EE.B.6

FORWARD 2

Model a linear relationship with a function • Grade 8 • Functions • 8.F.B.4

A plant is 4 cm tall. It grows 1.5 cm each week.
Write a function for its height h after w weeks.

$h =$ _____

Which situation matches $y = 2x + 10$, where y is dollars in savings after x weeks?

- A.** Save \$12 each week.
- B.** Start with \$10 and spend \$2 each week.
- C.** Start with \$10 and save \$2 each week.
- D.** Start with \$2 and save \$10 each week.

The table shows savings over time.

Write a function for the savings y after x weeks.

Weeks (x)	0	1	2
Savings (y)	30	45	60

$y =$ _____

A rental truck costs \$50 plus \$0.25 per mile.

Write a function for the cost C of driving m miles.

$C =$ _____

True or false?

A line through $(0, 5)$ and $(2, 9)$ has an initial value of 5 and a rate of change of 2.

A. True

B. False

Solve one-step equations of the form

$$x + p = q$$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Subtraction as an unknown-addend problem	1.OA.B.4
BACKWARD 2	Adding and subtracting decimals	5.NBT.B.7
BACKWARD 3	Adding and subtracting fractions and mixed numbers	5.NF.A.1
FORWARD 1	Solve two-step equations	7.EE.B.4.a
FORWARD 2	Solve equations with variables on both sides	8.EE.C.7.b

SET 51 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 7.5 = 12$$

x = _____

Solve for n.

$$n - \frac{3}{4} = 5\frac{1}{2}$$

A. $4\frac{3}{4}$

B. $5\frac{1}{4}$

C. $6\frac{1}{4}$

D. $6\frac{3}{4}$

Solve for y.

$$y + 2\frac{1}{3} = 6$$

y = _____

Solve for m.

$$18.25 + m = 30$$

m = _____

Solve for k.

$$k - 4.8 = 9.2$$

$$k = \underline{\hspace{10em}}$$

SET 51 • 6.EE.B.7

BACKWARD 1

Subtraction as an unknown-addend problem • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4

True or false?

$12 - 7 = \square$ can be solved by thinking
 $7 + \square = 12$.

A. True

B. False

What number goes in the box?

$$7 + \square = 15$$

$$\square = \underline{\hspace{10em}}$$

To find $16 - 9$, think $9 + \square = 16$.

What number goes in the box?

- A.** 6
- B.** 7
- C.** 8
- D.** 25

What number goes in the box?

$$\square + 8 = 13$$

$$\square = \underline{\hspace{10em}}$$

True or false?

To find $11 - 5$, you can think $5 + 6 = 11$.

A. True

B. False

SET 51 • 6.EE.B.7

BACKWARD 2

Adding and subtracting decimals • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

Subtract.

$$12 - 7.5$$

Difference: _____

Subtract.

$$30 - 18.25$$

Difference: _____

True or false?

$$9.2 + 4.8 = 14$$

A. True

B. False

$$6.25 + 2.8 = ?$$

A. 6.53

B. 8.33

C. 9.05

D. 9.5

Add.

$$3.7 + 5.45$$

Sum: _____

SET 51 • 6.EE.B.7

BACKWARD 3

Adding and subtracting fractions and mixed numbers • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$6 - 2\frac{1}{3}$$

Difference: _____

Add.

$$5\frac{1}{2} + \frac{3}{4}$$

Sum: _____

True or false?

$$\frac{2}{3} + \frac{1}{6} = \frac{5}{6}$$

A. True

B. False

$$\frac{7}{8} - \frac{1}{4} = ?$$

A. $\frac{1}{2}$

B. $\frac{5}{8}$

C. $\frac{3}{4}$

D. $\frac{6}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q4/5

Add.

$$1\frac{2}{5} + \frac{1}{2}$$

Sum: _____

SET 51 • 6.EE.B.7

FORWARD 1

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 8 = 29$$

x = _____

Solve for x.

$$-2x + 5 = 17$$

x = _____

Solve for x.

$$\frac{x}{4} - 3 = 2$$

A. -4

B. $\frac{5}{4}$

C. 5

D. 20

Solve for x.

$$0.6x + 1.5 = 4.5$$

x = _____

Solve for x.

$$5(x + 2) = 45$$

x = _____

SET 51 • 6.EE.B.7

FORWARD 2

Solve equations with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$7x - 4 = 3x + 12$$

x = _____

Solve for x.

$$-5x + 9 = x - 3$$

x = _____

Solve for x.

$$2.5x + 4 = 1.5x + 9$$

A. -5

B. 5

C. 6.5

D. 13

Solve for x.

$$\frac{1}{2}x + 6 = 2x - 3$$

x = _____

Solve for x.

$$3(2x - 5) = 4x + 1$$

x = _____

Solve one-step equations of the form

$$px = q$$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Dividing by a fraction	6.NS.A.1
BACKWARD 3	Dividing decimals	5.NBT.B.7
FORWARD 1	Solve equations of the form $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solve multi-step linear equations	8.EE.C.7.b

SET 52 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$4x = 30$$

x = _____

Solve for y.

$$\frac{2}{3}y = 12$$

$$y = \underline{\hspace{10cm}}$$

Solve for m.

$$0.6m = 4.2$$

m = _____

Solve for n.

$$5n = 3.5$$

A. -1.5

B. 0.7

C. 1.43

D. 17.5

Solve for k.

$$\frac{3}{4}k = \frac{9}{10}$$

k = _____

SET 52 • 6.EE.B.7

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $48 \div 6$ by thinking $6 \times \square = 48$.

$\square =$ _____

True or false?

$63 \div 7 = 9$ because $7 \times 9 = 63$.

A. True

B. False

Which fact helps you find $45 \div 5$?

A. $9 - 5 = 4$

B. $45 \times 5 = 225$

C. $5 \times 9 = 45$

D. $5 + 40 = 45$

What number goes in the box?

$$\square \times 6 = 42$$

$$\square = \underline{\hspace{10cm}}$$

True or false?

$24 \div 4 = 5$ because $4 \times 5 = 24$.

A. True

B. False

SET 52 • 6.EE.B.7

BACKWARD 2

Dividing by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$12 \div \frac{2}{3}$$

Quotient: _____

Divide.

$$\frac{9}{10} \div \frac{3}{4}$$

Quotient: _____

True or false?

$$5 \div \frac{5}{8} = 8$$

A. True

B. False

$$\frac{9}{10} \div \frac{3}{5} = ?$$

A. $\frac{27}{50}$

B. $\frac{2}{3}$

C. $1\frac{1}{3}$

D. $1\frac{1}{2}$

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

SET 52 • 6.EE.B.7

BACKWARD 3

Dividing decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Divide.

$$4.2 \div 0.6$$

Quotient: _____

Divide.

$$30 \div 4$$

Quotient: _____

True or false?

$$3.5 \div 5 = 0.7$$

A. True

B. False

$$2.4 \div 0.3 = ?$$

A. 0.8

B. 7

C. 8

D. 80

Divide.

$$7.2 \div 9$$

Quotient: _____

SET 52 • 6.EE.B.7

FORWARD 1

Solve equations of the form $p(x + q) = r$ • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Solve for x.

$$3(x - 4) = 21$$

x = _____

Solve for x.

$$\frac{1}{2}(x + 6) = 9$$

x = _____

Solve for x.

$$3(x + 1.5) = 12$$

A. 2

B. 2.5

C. 4

D. 5.5

Solve for x.

$$-2(x - 3) = 14$$

x = _____

Solve for x.

$$0.5(x + 8) = 7$$

x = _____

SET 52 • 6.EE.B.7

FORWARD 2

Solve multi-step linear equations • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$3(2x - 1) = 4x + 7$$

x = _____

Solve for x.

$$\frac{2}{3}(x + 3) = x - 1$$

x = _____

Solve for x.

$$5x - 2(x + 1) = 10$$

A. 2

B. $\frac{8}{3}$

C. 4

D. 12

Solve for x.

$$0.4(10x - 5) = 2x + 4$$

x = _____

Solve for x.

$$6x = 2(x + 8)$$

x = _____

Write and solve one-step equations for real-world problems

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Word problems with an unknown factor	3.OA.A.3
BACKWARD 2	Word problems with an unknown addend	2.OA.A.1
BACKWARD 3	A letter stands for an unknown quantity	3.OA.D.8
FORWARD 1	Write and solve two-step equations for word problems	7.EE.B.4.a
FORWARD 2	Solve real-world problems with variables on both sides	8.EE.C.7.b

SET 53 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

A pack of 6 pens costs \$4.50.

Write and solve an equation to find the cost p of one pen.

Equation: _____

$p =$ _____

After spending \$12.75, Sam has \$30.25 left.

Write and solve an equation to find how much money m Sam started with.

Equation: _____

$m =$ _____

A board is cut into 5 equal pieces. Each piece is 1.8 meters long.

Write and solve an equation to find the length L of the board.

Equation: _____

$L =$ _____

Maya is 1.35 m tall. She is 0.4 m taller than her brother.
Which equation can be used to find her brother's height b ?

A. $0.4b = 1.35$

B. $1.35 + 0.4 = b$

C. $b - 0.4 = 1.35$

D. $b + 0.4 = 1.35$

A recipe uses $\frac{3}{4}$ cup of sugar per batch. Kim used 6 cups of sugar.

Write and solve an equation to find the number of batches b .

Equation: _____

$b =$ _____

SET 53 • 6.EE.B.7

BACKWARD 1

Word problems with an unknown factor • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

4 friends share some grapes equally. Each gets 9 grapes.

How many grapes were there?

Grapes: _____

There are 7 teams with $\square$ players each. There are 56 players in all.

Which equation matches?

A. $7 \times \square = 56$

B. $56 \times 7 = \square$

C. $\square - 7 = 56$

D. $7 + \square = 56$

True or false?

If $3 \times \square = 24$, then $\square = 8$.

A. True

B. False

A teacher puts 40 books into 5 equal stacks.
How many books are in each stack?

Books: _____

Each box holds 6 eggs. There are 54 eggs.
How many boxes are there?

Boxes: _____

SET 53 • 6.EE.B.7

BACKWARD 2

Word problems with an unknown addend • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Lin had some stickers. She got 15 more. Now she has 42.

How many did she have at first?

Stickers: _____

A plant grew 8 cm and is now 35 cm tall.

Which equation finds the starting height h ?

A. $8h = 35$

B. $h - 8 = 35$

C. $h + 35 = 8$

D. $h + 8 = 35$

True or false?

Tom spent \$17 and has \$26 left, so he started with \$43.

A. True

B. False

A bus had 30 people. Some got off. Now there are 18.
How many got off?

People: _____

Ada read 24 pages on Monday and some more on Tuesday. She read 61 pages in all.

How many pages did she read on Tuesday?

Pages: _____

SET 53 • 6.EE.B.7

BACKWARD 3

A letter stands for an unknown quantity • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

True or false?

In $5 \times n = 40$, the letter n stands for 8.

A. True

B. False

In the equation $p + 9 = 20$, what number is p ?

$p =$ _____

Which equation matches "a number plus 6 is 15"?

A. $n + 15 = 6$

B. $n - 6 = 15$

C. $6n = 15$

D. $n + 6 = 15$

In the equation $3 \times k = 27$, what number is k ?

$k =$ _____

True or false?

In $m - 4 = 10$, the letter m stands for 6.

A. True

B. False

SET 53 • 6.EE.B.7

FORWARD 1

Write and solve two-step equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a \$25 fee plus \$15 per month. Jen paid \$130.

Write and solve an equation to find the number of months m .

$m =$ _____

A plumber charges \$45 plus \$60 per hour. A bill is \$225.
How many hours did the plumber work?

Hours: _____

A class buys 12 tickets and pays a \$15 group fee. The total is \$159.

Which equation finds the cost t of one ticket?

A. $12t - 15 = 159$

B. $12t + 15 = 159$

C. $12(t + 15) = 159$

D. $15t + 12 = 159$

Mo saves \$18 per week and already has \$54. He wants \$180.

How many weeks w will it take?

$W =$ _____

True or false?

A \$6 pizza plus 3 toppings costs \$10.50. Each topping costs \$1.50.

- A. True
- B. False

SET 53 • 6.EE.B.7

FORWARD 2

Solve real-world problems with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges \$20 plus \$3 per shirt. Store B charges \$5 per shirt.

Write and solve an equation to find when the costs are equal.

Shirts: _____

Ana has \$120 and spends \$8 per week. Ben has \$40 and saves \$12 per week.

After how many weeks will they have the same amount?

Weeks: _____

Taxi A charges \$4 plus \$2.50 per mile. Taxi B charges \$1 plus \$3 per mile.

For how many miles do the rides cost the same?

A. 3

B. 5

C. 6

D. 10

A candle 30 cm tall burns 1.5 cm per hour. Another candle 20 cm tall burns 0.5 cm per hour.

After how many hours are they the same height?

Hours: _____

Plan A: \$15 per month plus \$0.10 per text. Plan B: \$0.25 per text.

For how many texts do the plans cost the same?

Texts: _____

Write an inequality $x > c$ or $x < c$ for a real-world condition

MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Inequality symbols	1.NBT.B.3
BACKWARD 2	Meaning of "at least," "at most," "more than," and "fewer than"	6.NS.C.7.b
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Write and solve inequalities for real-world problems	7.EE.B.4.b
FORWARD 2	Use two linear equations to decide when one quantity is greater...	8.EE.C.8.c

SET 54 • 6.EE.B.8

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Riders must be at least 48 inches tall.

Write an inequality for the height h of a rider.

Inequality: _____

A bag can hold at most 20 pounds.

Which inequality describes the weight w the bag can hold?

A. $w \geq 20$

B. $w < 20$

C. $w \leq 20$

D. $w > 20$

The temperature must stay below -3°C .

Write an inequality for the temperature t .

Inequality: _____

More than 25 people must sign up for a trip to happen.
Write an inequality for the number of people p .

Inequality: _____

Children 12 years old and younger get a discount.

Write an inequality for the age a of a child who gets the discount.

Inequality: _____

SET 54 • 6.EE.B.8

BACKWARD 1

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

58 85

A. $<$

B. $>$

C. $=$

True or false?

$$30 > 29$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

64 46

Symbol: _____

Which symbol means "is greater than"?

A. $<$

B. $>$

C. $=$

True or false?

$$17 < 12$$

A. True

B. False

SET 54 • 6.EE.B.8

BACKWARD 2

Meaning of "at least," "at most," "more than," and "fewer than" • Grade 6 • The Number System • 6.NS.C.7.b

True or false?

"At least 10" includes the number 10.

A. True

B. False

What does "at most 6" mean?

- A.** 6 or more
- B.** Less than 6
- C.** More than 6
- D.** 6 or less

True or false?

"Fewer than 8" includes the number 8.

A. True

B. False

Which number is NOT "at least 12"?

- A.** 11
- B.** 12
- C.** 13
- D.** 20

True or false?

9 is "more than 7."

A. True

B. False

SET 54 • 6.EE.B.8

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 3 degrees below zero.

Integer: _____

True or false?

-10°C is colder than -3°C .

A. True

B. False

Which number represents 15 feet below sea level?

- A.** -15
- B.** -1.5
- C.** 0
- D.** 15

Write an integer for a temperature of 7 degrees below zero.

Integer: _____

True or false?

"Below -3°C " includes -5°C .

A. True

B. False

SET 54 • 6.EE.B.8

FORWARD 1

Write and solve inequalities for real-world problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

You have \$50. Each ride costs \$6.

Write and solve an inequality to find how many rides r you can afford.

Solution: _____

A salesperson earns \$80 plus \$12 for each sale.

Write and solve an inequality for the number of sales s needed to earn more than \$200.

Solution: _____

A phone has 75% battery and uses 5% per hour. It must stay above 20%.

Which inequality gives the hours h ?

A. $75 - 5h > 20$

B. $75 + 5h > 20$

C. $75 - 5h < 20$

D. $5h - 75 > 20$

A van can carry at most 1,200 pounds. The driver weighs 180 pounds. Each box weighs 60 pounds.

Write and solve an inequality for the number of boxes b .

Solution: _____

True or false?

The solution of $3x + 2 \geq 17$ is $x \geq 5$.

A. True

B. False

FORWARD 2

Use two linear equations to decide when one quantity is greater • Grade 8 •
Expressions & Equations • 8.EE.C.8.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Plan A costs \$20 plus \$5 per month. Plan B costs \$35 plus \$2 per month.

After how many months does Plan A cost more than Plan B?

Months: _____

Tank A has 100 gallons and drains 6 gallons per minute.
Tank B has 40 gallons and fills 4 gallons per minute.
After how many minutes does Tank B hold more water?

Minutes: _____

Sara has \$50 and saves \$8 per week. Tom has \$90 and saves \$3 per week.

When will Sara have more money than Tom?

- A.** After 8 weeks
- B.** After 3 weeks
- C.** After 5 weeks
- D.** Never

Gym X charges \$60 plus \$4 per class. Gym Y charges \$12 per class.

For how many classes is Gym X cheaper?

Classes: _____

True or false?

$y = 3x + 2$ and $y = x + 10$ are equal when $x = 4$.
For $x > 4$, $3x + 2$ is greater.

A. True

B. False

Represent solutions of $x > c$ or $x < c$ on a number line

MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Locating numbers on a number line	6.NS.C.6.c
BACKWARD 2	Greater numbers are to the right	6.NS.C.7.a
BACKWARD 3	Meaning of inequality symbols	1.NBT.B.3
FORWARD 1	Solve and graph two-step inequalities	7.EE.B.4.b
FORWARD 2	Linear equations with one, no, or infinitely many solutions — near...	8.EE.C.7.a

SET 55 • 6.EE.B.8

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Graph the solutions of $x > 3$ on the number line.



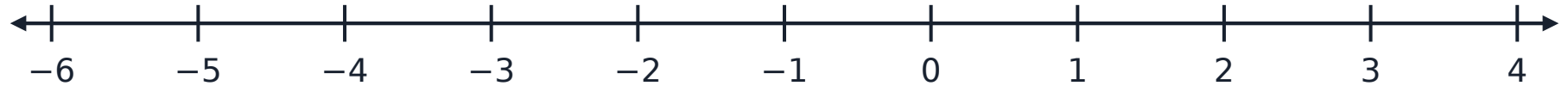
Graph the solutions of $x \leq -1$ on the number line.



Graph the solutions of $x < 2.5$ on the number line.



Graph the solutions of $x \geq -4$ on the number line.



Graph the solutions of $x > -1.5$ on the number line.



SET 55 • 6.EE.B.8

BACKWARD 1

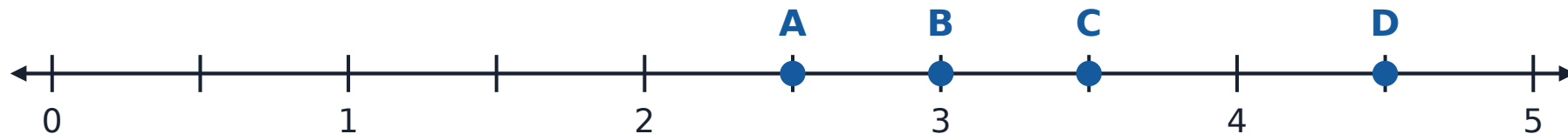
Locating numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?



A = _____

Which point is at 3.5?



A. Point A

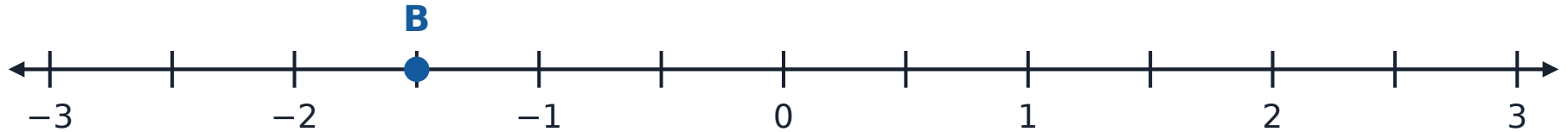
B. Point B

C. Point C

D. Point D

True or false?

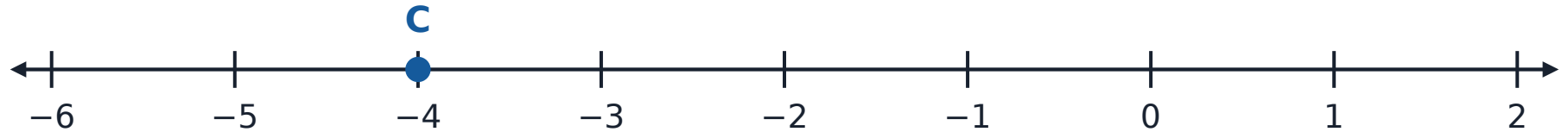
Point B is at -1.5 .



A. True

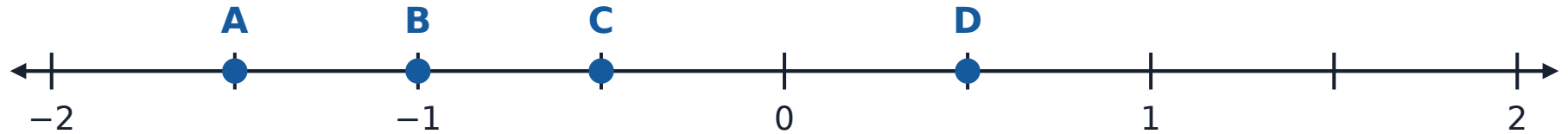
B. False

What number is at point C?



C = _____

Which point is at $-\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

SET 55 • 6.EE.B.8

BACKWARD 2

Greater numbers are to the right • Grade 6 • The Number System • 6.NS.C.7.a

True or false?

Every number to the right of 3 on a number line is greater than 3.

A. True

B. False

Which numbers are less than -1 ?

- A.** Numbers to the right of -1
- B.** Numbers to the left of -1
- C.** Only -2
- D.** Only 0

True or false?

-3 is greater than -1 .

A. True

B. False

Is 2.5 to the left or the right of 2 on a number line?

Answer: _____

Which number is greater than -4 ?

A. -6

B. -5

C. -4.5

D. -3

SET 55 • 6.EE.B.8

BACKWARD 3

Meaning of inequality symbols • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

What does the symbol $>$ mean?

- A.** Is equal to
- B.** Is greater than
- C.** Is less than

True or false?

The symbol $<$ means "is less than."

- A. True
- B. False

Which statement is true?

A. $8 < 5$

B. $8 = 5$

C. $8 > 5$

Write the symbol for "is less than."

Symbol: _____

True or false?

$$12 > 21$$

A. True

B. False

SET 55 • 6.EE.B.8

FORWARD 1

Solve and graph two-step inequalities • Grade 7 • Expressions & Equations •
7.EE.B.4.b

Solve $2x + 1 > 7$. Then graph the solutions.



Solution: _____

Solve $4x - 3 \leq 5$. Then graph the solutions.



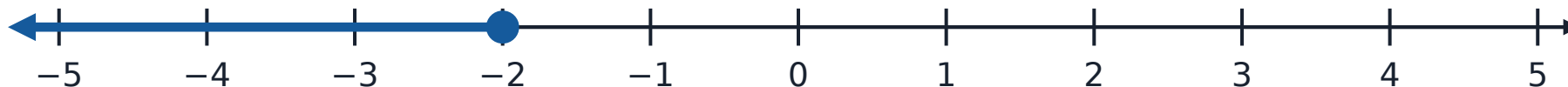
Solution: _____

Solve $-2x + 4 < 10$. Then graph the solutions.



Solution: _____

Which inequality does the graph show?



A. $x < -2$

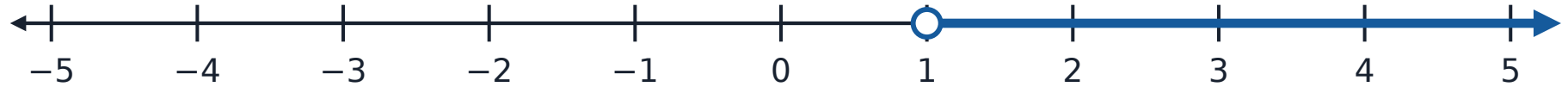
B. $x > -2$

C. $x \geq -2$

D. $x \leq -2$

True or false?

The graph shows the solutions of $3x + 2 > 5$.



A. True

B. False

FORWARD 2

Linear equations with one, no, or infinitely many solutions • Grade 8 •
Expressions & Equations • 8.EE.C.7.a

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

True or false?

$x + 3 = x + 3$ has infinitely many solutions.

A. True

B. False

How many solutions does the equation have?

$$2x + 5 = 2x - 1$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

Which equation has infinitely many solutions?

A. $4x - 1 = x$

B. $4x = 0$

C. $4(x - 1) = 4x - 4$

D. $4(x - 1) = 4x - 1$

How many solutions does the equation have?

$$3x - 7 = 3x - 7$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$6x - 2 = 4x + 8$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

Identify the independent and dependent variables

MAIN Grade 6 main question

6.EE.C.9

BACKWARD 1 Input-output rules

4.OA.C.5

BACKWARD 2 Two related number patterns

5.OA.B.3

FORWARD 1 Write an equation for a proportional relationship

7.RP.A.2.c

FORWARD 2 Functions: each input has exactly one output

8.F.A.1

SET 56 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

The equation $c = 2.5p$ gives the cost c of p pounds of grapes.

Which is the independent variable?

- A.** p (pounds)
- B.** c (cost)
- C.** 2.5
- D.** There is none.

Money earned m depends on the number of hours worked h .

Which is the dependent variable?

Dependent variable: _____

A plant's height depends on the number of days it grows.

Which is the independent variable?

- A.** The plant's height
- B.** The number of days
- C.** Neither
- D.** Both

In $y = 4x + 3$, which variable is the dependent variable?

Dependent variable: _____

True or false?

Ice cream sales s depend on the temperature t .
So t is the independent variable.

- A.** True
- B.** False

SET 56 • 6.EE.C.9

BACKWARD 1

Input-output rules • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "multiply by 3."

What is the output when the input is 7?

Output: _____

True or false?

The rule is "add 6." When the input is 4, the output is 10.

A. True

B. False

What is the rule?

In	2	4	6
Out	6	8	10

A. Multiply by 2

B. Multiply by 3

C. Add 2

D. Add 4

The rule is "subtract 2."

What is the output when the input is 15?

Output: _____

True or false?

In an input-output table, the output depends on the input.

A. True

B. False

SET 56 • 6.EE.C.9

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

True or false?

Pattern A: 0, 2, 4, 6

Pattern B: 0, 6, 12, 18

Each number in Pattern B is 3 times the matching number in Pattern A.

A. True

B. False

The rule is "add 5," starting at 0.

What number comes next? 0, 5, 10, ____

Number: _____

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

When Pattern A is 12, what is the matching number in Pattern B?

Number: _____

Pattern A starts at 0 and adds 1. Pattern B starts at 0 and adds 4.

Which ordered pairs come from the patterns?

A. (0, 4), (1, 5), (2, 6)

B. (0, 0), (1, 4), (2, 8)

C. (1, 4), (2, 5), (3, 6)

D. (0, 0), (4, 1), (8, 2)

True or false?

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Each number in Pattern B is 5 more than the matching number in Pattern A.

A. True

B. False

SET 56 • 6.EE.C.9

FORWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Tomatoes cost \$3.20 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

The table shows a proportional relationship.
Write an equation for the cost c of n ounces.

Ounces (n)	4	10
Cost (c)	\$1.20	\$3.00

Equation: _____

Which equation represents a proportional relationship?

A. $y = 4.5x$

B. $y = 4.5x + 1$

C. $y = x + 4.5$

D. $y = 4.5$

A car uses 3 gallons of gas to travel 84 miles.

Write an equation for the miles m traveled on g gallons.

Equation: _____

True or false?

$y = \frac{2}{3}x$ represents a proportional relationship.

A. True

B. False

SET 56 • 6.EE.C.9

FORWARD 2

Functions: each input has exactly one output • Grade 8 • Functions • 8.F.A.1

True or false?

A function assigns exactly one output to each input.

A. True

B. False

Which table does NOT represent a function?

A. x: 0, 1, 2 y: 0, 2, 4

B. x: 1, 1, 2 y: 3, 5, 7

C. x: 2, 3, 4 y: 5, 6, 7

D. x: 1, 2, 3 y: 4, 4, 4

In the function $C = 12t$, which variable is the input?

Input: _____

True or false?

The set of ordered pairs $(1, 2)$, $(2, 3)$, $(1, 4)$ represents a function.

- A. True
- B. False

A function gives the height of a ball after t seconds.
What is the input?

- A.** The speed
- B.** The ball
- C.** The height
- D.** The time t

Write an equation for the relationship shown in a table or graph

MAIN	Grade 6 main question	6.EE.C.9
BACKWARD 1	Finding a rule from a table	4.OA.C.5
BACKWARD 2	Reading points on a graph	5.G.A.2
BACKWARD 3	Ordered pairs from two patterns	5.OA.B.3
FORWARD 1	Equations for proportional relationships from tables and graphs	7.RP.A.2.c
FORWARD 2	Write a linear function from a table or graph	8.F.B.4

SET 57 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation that shows how y depends on x .

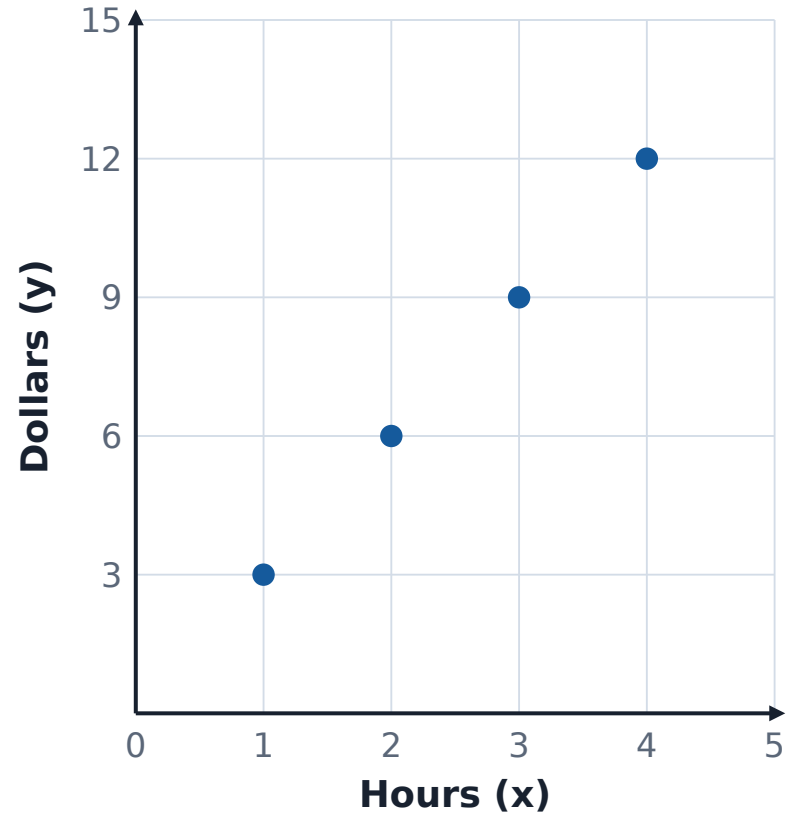
x	1	2	3	4
y	6	12	18	24

$y =$ _____

The graph shows y dollars earned for x hours worked.

Write an equation for the relationship.

$y =$ _____



The table shows a plant's height h after w weeks.
Write an equation for h in terms of w .

Weeks (w)	0	1	2	3
Height (h)	5	7	9	11

$h =$ _____

Which equation matches the table?

x	1	2	3
y	4	5	6

A. $y = 4x$

B. $y = x + 4$

C. $y = x + 3$

D. $y = 3x$

The table shows the cost c of n bags of popcorn.
Write an equation for c in terms of n .

Bags (n)	2	4	6
Cost (c)	7	14	21

$C =$ _____

SET 57 • 6.EE.C.9

BACKWARD 1

Finding a rule from a table • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

What is the rule?

In	1	2	3	4
Out	5	10	15	20

Rule: _____

What is the rule?

In	3	5	7
Out	7	9	11

Rule: _____

True or false?

The rule for the table is "multiply by 4."

In	1	2	3
Out	4	8	12

A. True

B. False

What is the rule?

In	3	6	9
Out	1	2	3

A. Multiply by 3

B. Subtract 2

C. Add 3

D. Divide by 3

The rule is "multiply by 6." What is the output when the input is 5?

Output: _____

SET 57 • 6.EE.C.9

BACKWARD 2

Reading points on a graph • Grade 5 • Geometry • 5.G.A.2

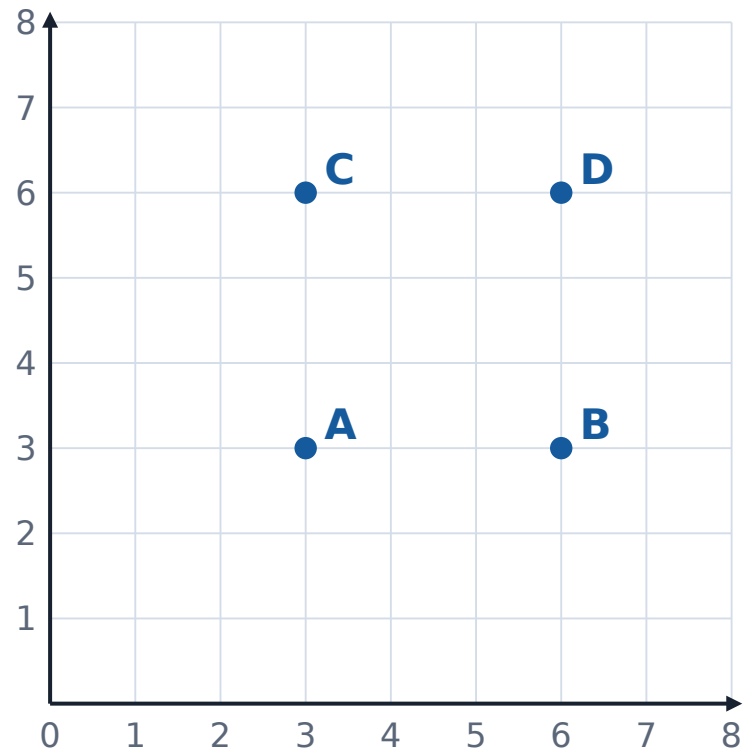
Which point is at (3, 6)?

A. Point A

B. Point B

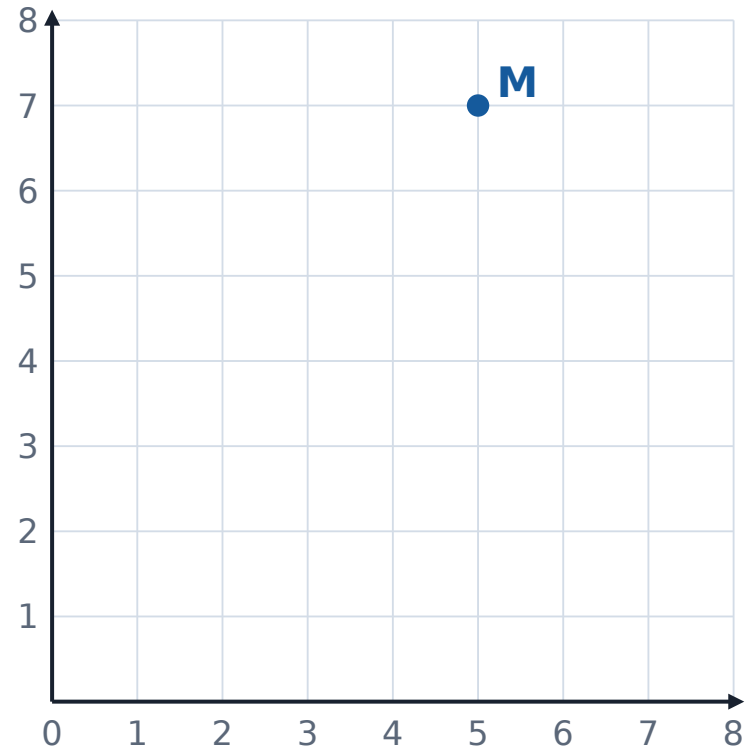
C. Point C

D. Point D



What are the coordinates of point M?

M = _____



True or false?

The point $(2, 8)$ is 2 units right and 8 units up from the origin.

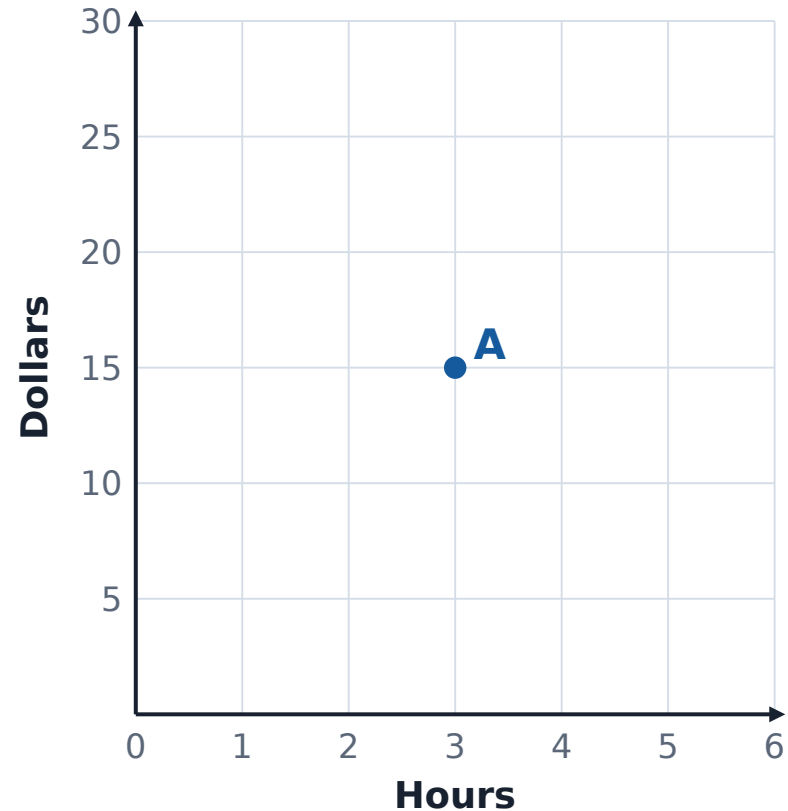
A. True

B. False

The graph shows dollars earned after working some hours.

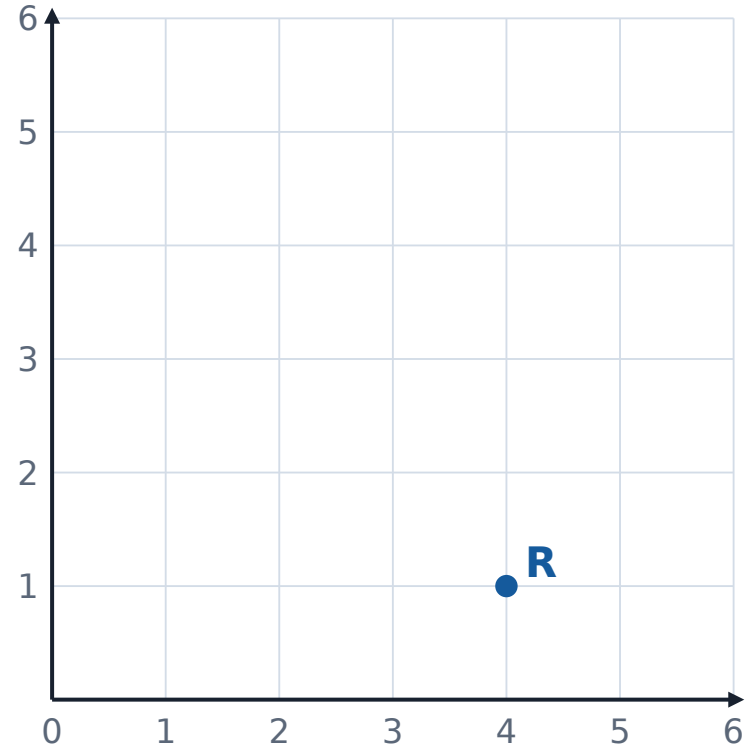
How many dollars were earned after 3 hours?

Dollars: _____



Which ordered pair names point R?

- A.** (4, 4)
- B.** (1, 4)
- C.** (1, 1)
- D.** (4, 1)



SET 57 • 6.EE.C.9

BACKWARD 3

Ordered pairs from two patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Write the ordered pair formed by the 4th numbers.

Ordered pair: _____

True or false?

The pairs (1, 3), (2, 6), (3, 9) show that each y-value is 3 times the x-value.

A. True

B. False

Pattern A adds 2. Pattern B adds 6. Both start at 0.
How does each B number compare to its A number?

- A.** 2 more
- B.** 3 times as large
- C.** 4 more
- D.** 6 times as large

Pattern A: 0, 2, 4

Pattern B: 0, 14, 28

What is the rule that changes each A number into its B number?

Rule: _____

True or false?

The pairs $(0, 0)$, $(1, 4)$, $(2, 8)$ come from Pattern A "add 4" and Pattern B "add 1."

- A. True
- B. False

SET 57 • 6.EE.C.9

FORWARD 1

Equations for proportional relationships from tables and graphs • Grade 7 •
Ratios & Proportional Relationships • 7.RP.A.2.c

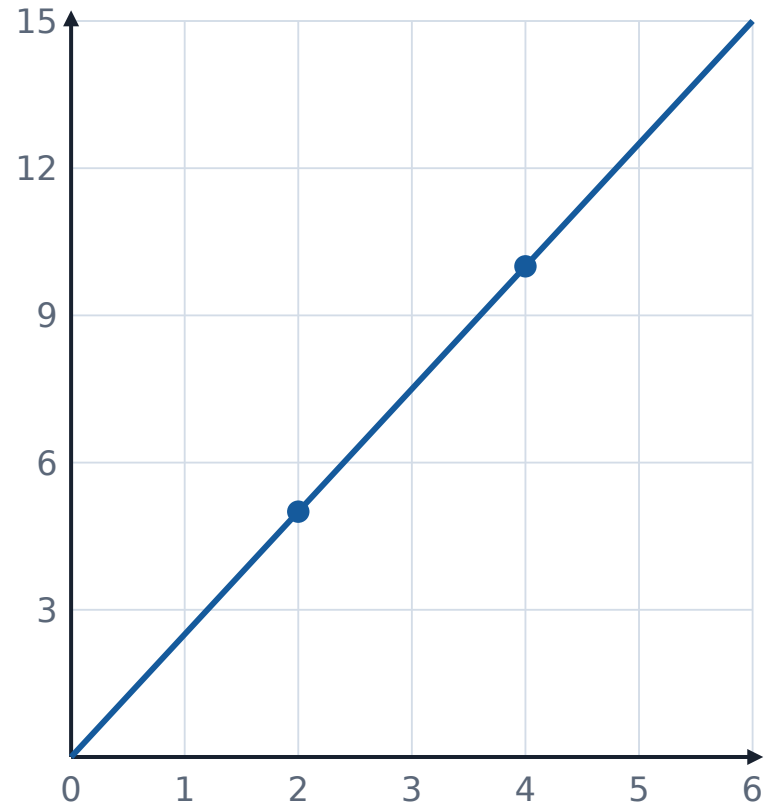
Write an equation for the proportional relationship in the table.

x	2	5	8
y	7	17.5	28

$y =$ _____

Write an equation for the proportional relationship in the graph.

$y =$ _____



The table shows a proportional relationship.
Which equation matches?

x	4	8	12
y	3	6	9

A. $y = 0.75x$

B. $y = 1.33x$

C. $y = 3x$

D. $y = x - 1$

Write an equation for the cost c of p pounds.

Pounds (p)	3	6	9
Cost (c)	7.50	15.00	22.50

$C =$ _____

True or false?

The table shows the equation $y = 6x$.

x	1	2	4
y	6	12	24

A. True

B. False

SET 57 • 6.EE.C.9

FORWARD 2

Write a linear function from a table or graph • Grade 8 • Functions • 8.F.B.4

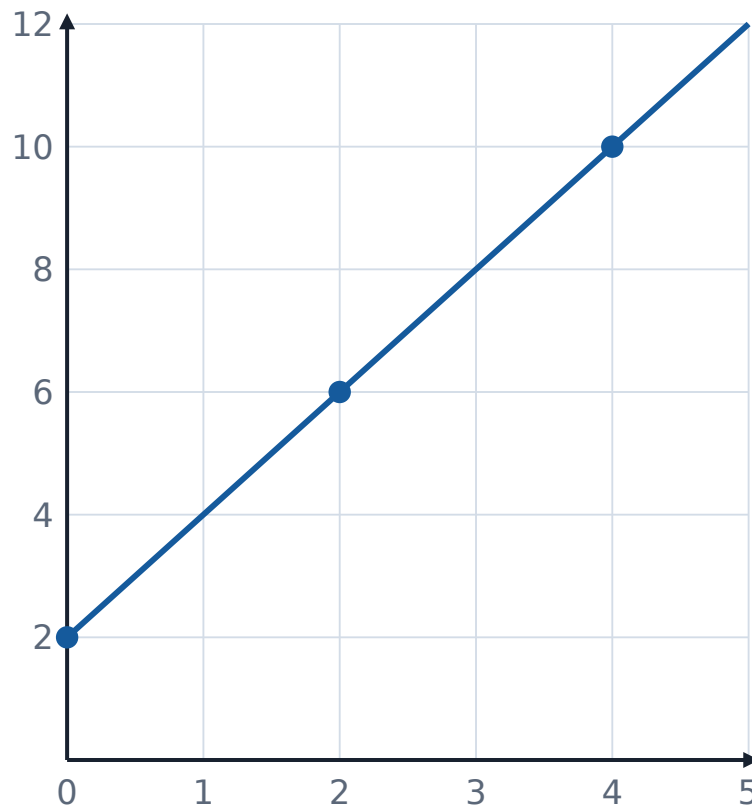
Write a function for the table.

x	0	1	2	3
y	4	7	10	13

$y =$ _____

Write a function for the line in the graph.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q2/5

Which function matches the table?

x	0	1	2
y	10	8	6

A. $y = -2x + 10$

B. $y = 10x - 2$

C. $y = -2x$

D. $y = 2x + 10$

Write a function for the table.

x	1	2	3	4
y	9	13	17	21

$y =$ _____

True or false?

The line through $(0, 1)$ and $(3, 7)$ has the equation $y = 2x + 1$.

A. True

B. False

Area of right triangles and other triangles

MAIN

Grade 6 main question

6.G.A.1

BACKWARD 1

Area of rectangles

3.MD.C.7.b

BACKWARD 2

Half of a product

5.NF.B.4.a

BACKWARD 3

Right angles and perpendicular segments

4.G.A.1

FORWARD 1

Area and volume of figures made of triangles

7.G.B.6

FORWARD 2

Use the Pythagorean Theorem in right triangles

8.G.B.7

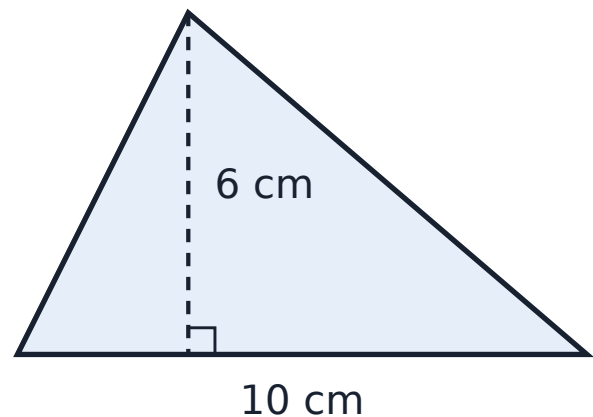
SET 58 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

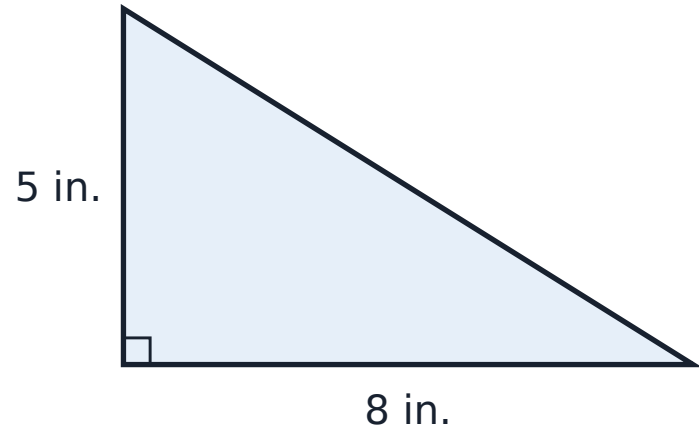
Find the area of the triangle.

Area: _____



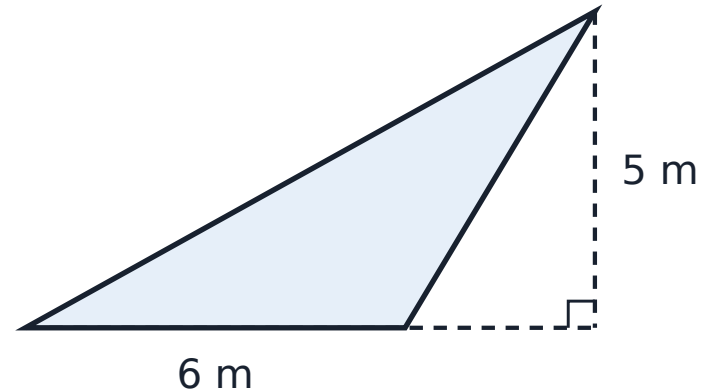
Find the area of the right triangle.

Area: _____



Find the area of the triangle.

Area: _____



A triangle has a base of 12 feet and a height of 7 feet.

What is its area?

- A.** 19 square feet
- B.** 38 square feet
- C.** 42 square feet
- D.** 84 square feet

A triangular sign has a base of 3.5 feet and a height of 2 feet.

What is its area?

Area: _____

SET 58 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



A rectangle is 9 feet long and 6 feet wide.

What is its area?

Area: _____

True or false?

A rectangle that is 5 cm long and 3 cm wide has an area of 15 square centimeters.

A. True

B. False

A rectangle is 12 m long and 4 m wide.

What is its area?

- A.** 16 square meters
- B.** 24 square meters
- C.** 32 square meters
- D.** 48 square meters

Find the area of the rectangle.

Area: _____



SET 58 • 6.G.A.1

BACKWARD 2

Half of a product • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times 10 \times 6$$

Product: _____

Multiply.

$$\frac{1}{2} \times 14$$

Product: _____

True or false?

$$\frac{1}{2} \times 8 \times 5 = 20$$

A. True

B. False

$$\frac{1}{2} \times 12 \times 7 = ?$$

A. 19

B. 21

C. 42

D. 84

Multiply.

$$\frac{1}{2} \times 9 \times 4$$

Product: _____

SET 58 • 6.G.A.1

BACKWARD 3

Right angles and perpendicular segments • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

A. True

B. False

Which angle measure is a right angle?

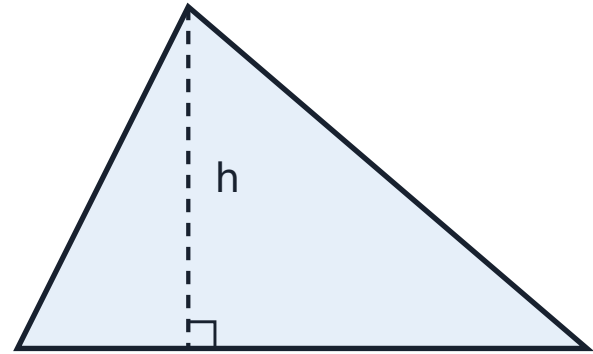
- A.** 45°
- B.** 90°
- C.** 100°
- D.** 180°

True or false?

The dashed segment is perpendicular to the base of the triangle.

A. True

B. False



How many degrees are in a right angle?

Degrees: _____

True or false?

A square has 4 right angles.

A. True

B. False

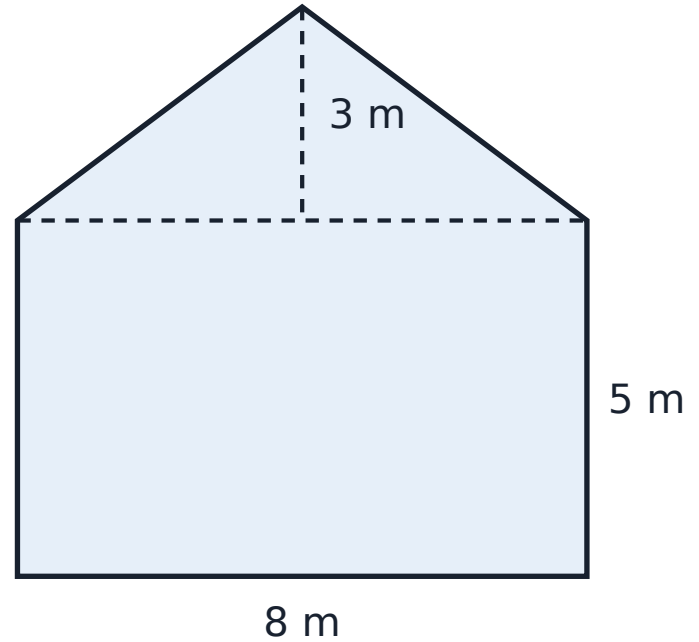
SET 58 • 6.G.A.1

FORWARD 1

Area and volume of figures made of triangles • Grade 7 • Geometry • 7.G.B.6

Find the area of the figure.

Area: _____



A kite is made of two triangles. Each triangle has a base of 10 inches and a height of 4 inches.

What is the area of the kite?

Area: _____

A triangle with a base of 4 cm and a height of 3 cm is cut out of a rectangle that is 12 cm by 6 cm.

What is the area of the part that is left?

- A.** 60 square cm
- B.** 66 square cm
- C.** 72 square cm
- D.** 78 square cm

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

True or false?

A figure made of a 6-by-4 rectangle and a triangle with a base of 3 and a height of 4 has an area of 30 square units.

- A.** True
- B.** False

SET 58 • 6.G.A.1

FORWARD 2

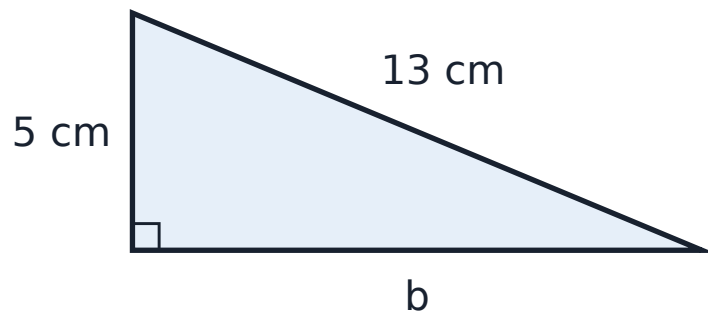
Use the Pythagorean Theorem in right triangles • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.
What is the length of the hypotenuse?

Hypotenuse: _____

Find the length b .

$b =$ _____



A 10-foot ladder leans against a wall. The top reaches 8 feet up the wall.

How far is the bottom of the ladder from the wall?

A. 2 feet

B. 6 feet

C. 12.8 feet

D. 18 feet

An isosceles triangle has a base of 16 inches and two sides of 10 inches.

What is its area?

Area: _____

A right triangle has legs of 9 m and 12 m.
What is the length of the hypotenuse?

Hypotenuse: _____

Area of parallelograms and trapezoids

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area by splitting into rectangles	3.MD.C.7.d
BACKWARD 3	Classifying quadrilaterals	5.G.B.4
BACKWARD 4	Height is perpendicular to the base	4.G.A.1
FORWARD 1	Area of figures made of quadrilaterals in real-world problems	7.G.B.6
FORWARD 2	Use the Pythagorean Theorem to find a missing height	8.G.B.7

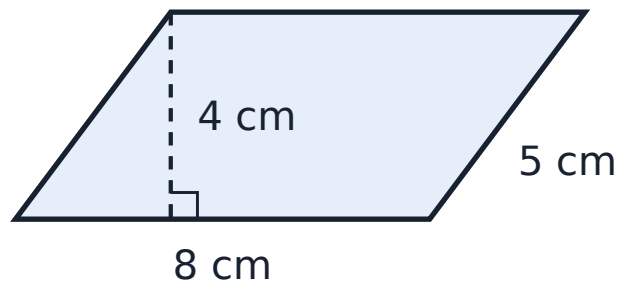
SET 59 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

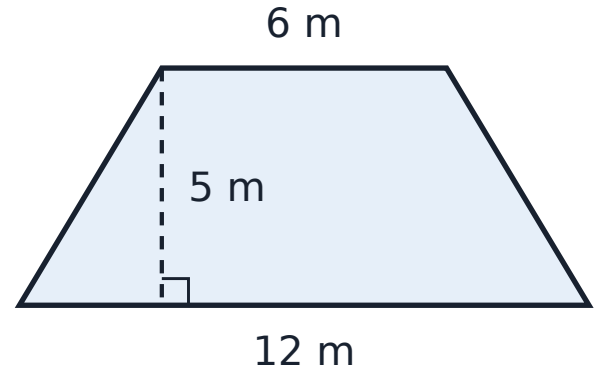
Find the area of the
parallelogram.

Area: _____



Find the area of the trapezoid.

Area: _____



A parallelogram has a base of 9 feet and a height of 5 feet.

What is its area?

A. 14 square feet

B. 22.5 square feet

C. 28 square feet

D. 45 square feet

A trapezoid has bases of 7 inches and 11 inches. Its height is 4 inches.

What is its area?

Area: _____

A parallelogram has a base of 12.5 meters and a height of 4 meters.

What is its area?

Area: _____

SET 59 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



A rectangle is 11 cm long and 5 cm wide.
What is its area?

Area: _____

True or false?

A square with 6-meter sides has an area of 24 square meters.

A. True

B. False

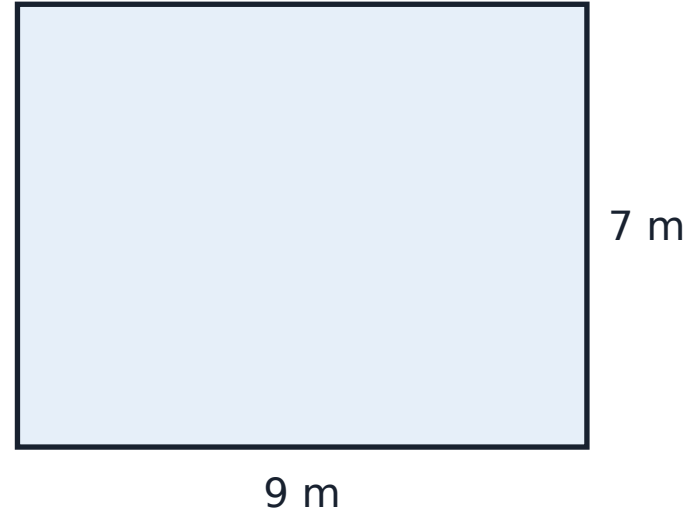
A rectangle is 15 inches long and 2 inches wide.

What is its area?

- A.** 17 square inches
- B.** 30 square inches
- C.** 34 square inches
- D.** 60 square inches

Find the area of the rectangle.

Area: _____



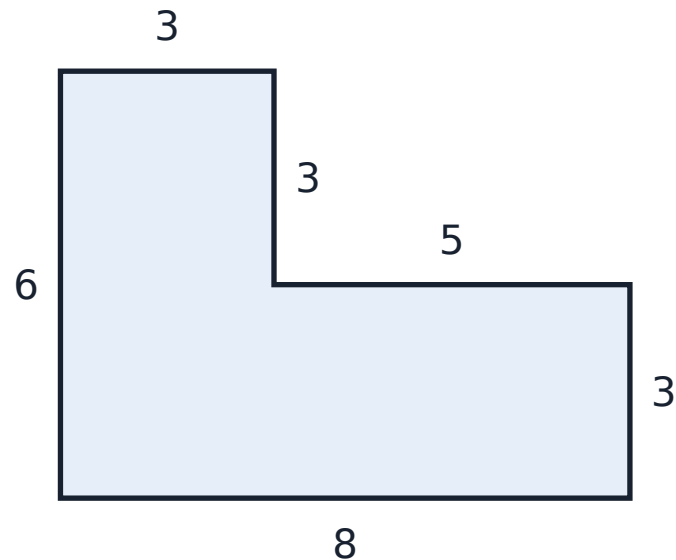
SET 59 • 6.G.A.1

BACKWARD 2

Area by splitting into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

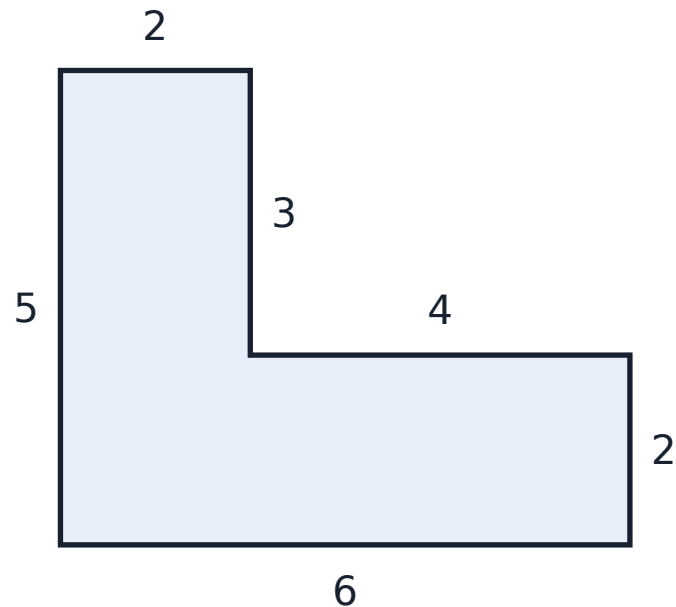
Find the area of the figure.

Area: _____



Find the area of the figure.

Area: _____



True or false?

The area of an L-shaped figure can be found by adding the areas of two rectangles.

A. True

B. False

A figure is made of a 4-by-5 rectangle and a 2-by-3 rectangle that do not overlap.

What is its area?

A. 14 square units

B. 20 square units

C. 26 square units

D. 120 square units

A figure is made of a 10-by-2 rectangle and a 3-by-3 square that do not overlap.

What is its area?

Area: _____

SET 59 • 6.G.A.1

BACKWARD 3

Classifying quadrilaterals • Grade 5 • Geometry • 5.G.B.4

True or false?

Every rectangle is a parallelogram.

A. True

B. False

True or false?

Every parallelogram is a rectangle.

A. True

B. False

Which quadrilateral always has four right angles?

- A.** Trapezoid
- B.** Rhombus
- C.** Parallelogram
- D.** Rectangle

How many pairs of parallel sides does a parallelogram have?

A. 0

B. 1

C. 2

D. 4

True or false?

A square is a rhombus.

A. True

B. False

SET 59 • 6.G.A.1

BACKWARD 4

Height is perpendicular to the base • Grade 4 • Geometry • 4.G.A.1

True or false?

The height of a parallelogram is perpendicular to its base.

A. True

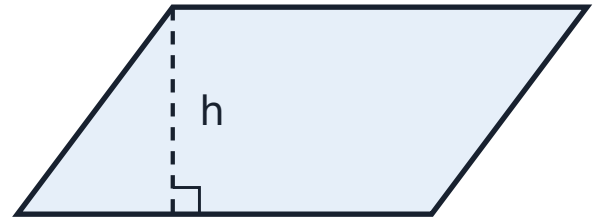
B. False

True or false?

The dashed segment shows the height of the parallelogram.

A. True

B. False



Which statement describes perpendicular lines?

- A.** They meet at right angles.
- B.** They never meet.
- C.** They meet at a 45° angle.
- D.** They are always the same length.

True or false?

The slanted side of a parallelogram is always its height.

A. True

B. False

What is the measure of the angle between a base and its height?

Angle: _____

SET 59 • 6.G.A.1

FORWARD 1

Area of figures made of quadrilaterals in real-world problems • Grade 7 •
Geometry • 7.G.B.6

A patio is made of a rectangle that is 10 ft by 6 ft and a trapezoid with bases of 6 ft and 4 ft and a height of 3 ft.
What is the area of the patio?

Area: _____

A wall is 12 ft by 8 ft. It has a window shaped like a parallelogram with a base of 3 ft and a height of 2 ft.

What is the area of the wall that is not window?

Area: _____

A sign is a trapezoid with bases of 1.5 m and 2.5 m and a height of 1.2 m.

What is its area?

- A.** 2.4 square meters
- B.** 3.6 square meters
- C.** 4.8 square meters
- D.** 5.2 square meters

A garden is made of two parallelograms that do not overlap. Each has a base of 7 m and a height of 4 m.

What is the total area?

Area: _____

True or false?

A trapezoid with bases of 6 and 10 and a height of 5 has the same area as a rectangle that is 8 by 5.

- A. True
- B. False

SET 59 • 6.G.A.1

FORWARD 2

Use the Pythagorean Theorem to find a missing height • Grade 8 • Geometry •
8.G.B.7

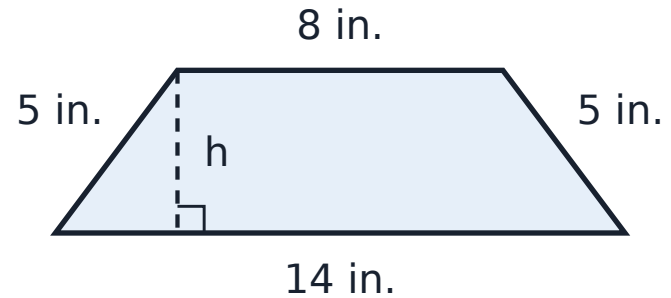
A parallelogram has a slanted side of 5 cm. The slanted side, the height, and a 3 cm piece of the base form a right triangle.

What is the height?

Height: _____

Find the height h of the trapezoid.

$h =$ _____



A ramp forms a right triangle. The ramp is 13 ft long and its base is 12 ft long.

How tall is the ramp?

A. 1 ft

B. 5 ft

C. 17.7 ft

D. 25 ft

A parallelogram has a slanted side of 10 m. The slanted side, the height, and a 6 m piece of the base form a right triangle.

What is the height?

Height: _____

True or false?

A trapezoid has a slanted side of 10 in. The slanted side, the height, and a 6 in. piece of the base form a right triangle. The height is 8 in.

A. True

B. False

Area of polygons by composing and decomposing

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectilinear figures	3.MD.C.7.d
BACKWARD 2	Missing side lengths of rectangles	4.MD.A.3
BACKWARD 3	Area of triangles	6.G.A.1
FORWARD 1	Area of composite figures in real-world problems	7.G.B.6
FORWARD 2	Volume of composite solids	8.G.C.9

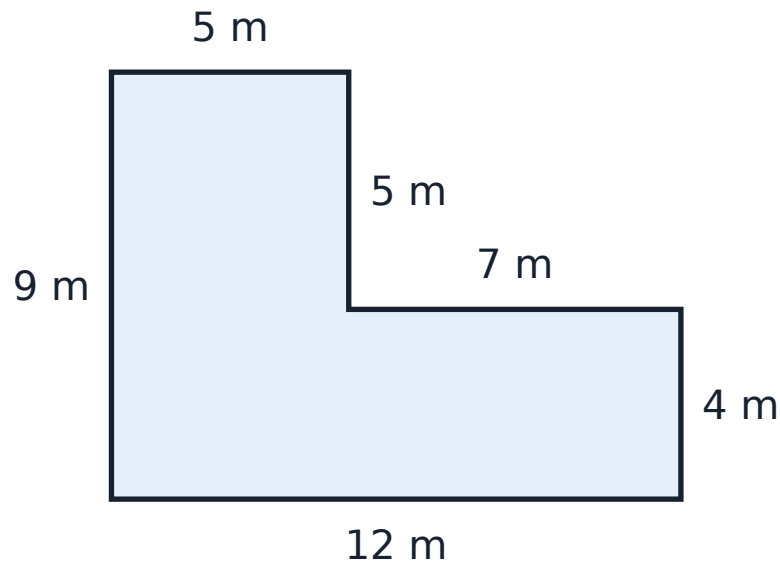
SET 60 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

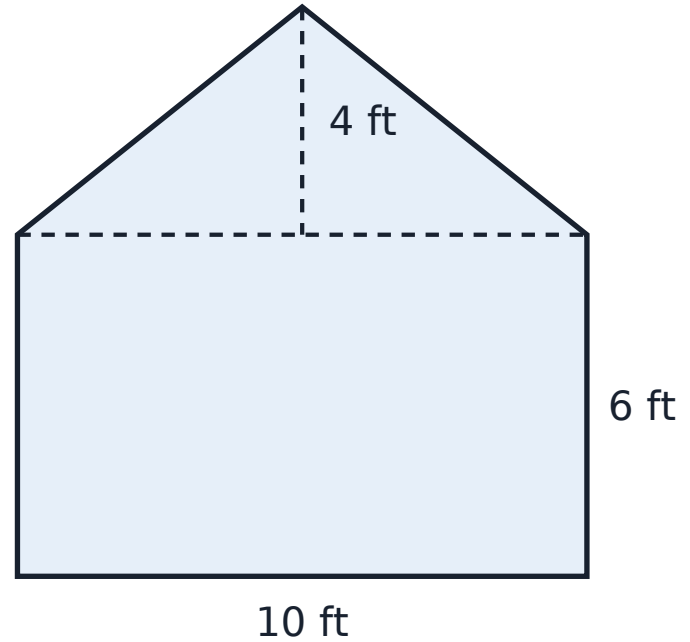
Find the area of the figure.

Area: _____



Find the area of the figure.

Area: _____



A rectangular yard is 20 m by 15 m. A triangular garden in the yard has a base of 6 m and a height of 5 m.

What is the area of the yard that is NOT garden?

Area: _____

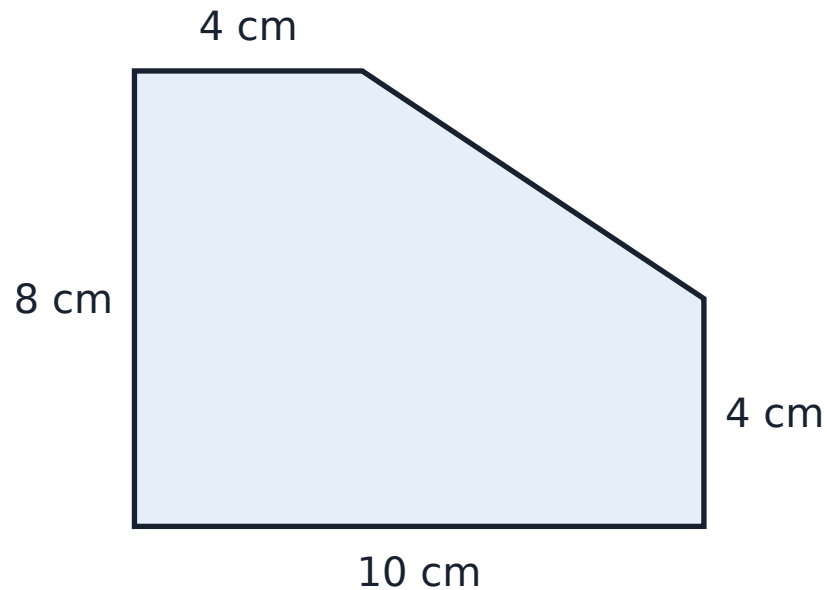
A polygon is split into a rectangle that is 8 by 3 and two triangles. Each triangle has a base of 2 and a height of 3.

What is the area of the polygon?

- A.** 24 square units
- B.** 27 square units
- C.** 30 square units
- D.** 36 square units

Find the area of the figure.

Area: _____



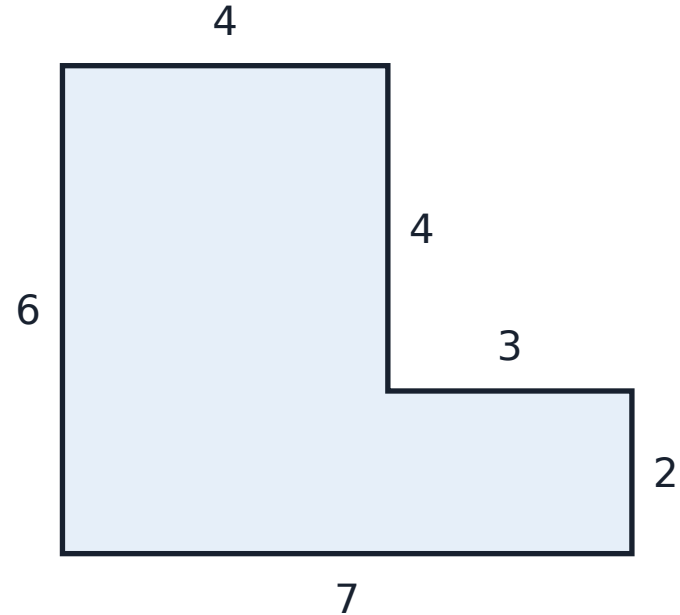
SET 60 • 6.G.A.1

BACKWARD 1

Area of rectilinear figures • Grade 3 • Measurement & Data • 3.MD.C.7.d

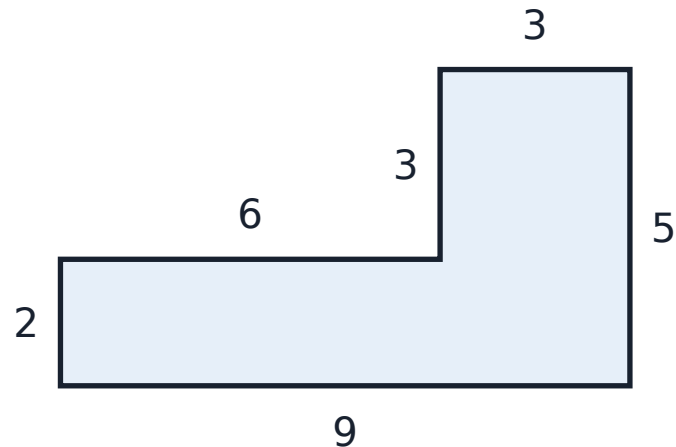
Find the area of the figure.

Area: _____



Find the area of the figure.

Area: _____



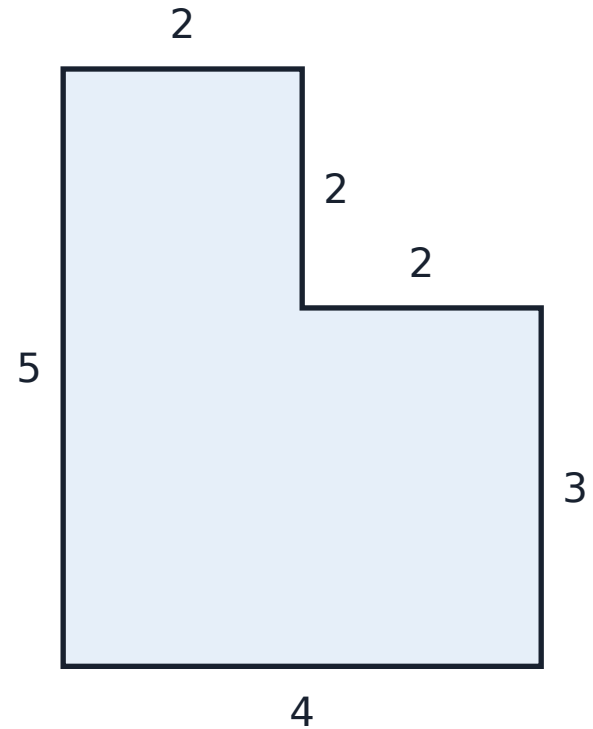
True or false?

A figure made of a 5-by-2 rectangle and a 3-by-4 rectangle has an area of 22 square units.

- A. True
- B. False

Which expression gives the area of the figure?

- A.** 4×5
- B.** $4 + 3 + 2 + 2$
- C.** $4 \times 3 \times 2$
- D.** $4 \times 3 + 2 \times 2$



A figure is made of two rectangles: 6 by 3 and 2 by 4.
What is its area?

Area: _____

SET 60 • 6.G.A.1

BACKWARD 2

Missing side lengths of rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 48 square feet and a width of 6 feet.

What is its length?

Length: _____

A rectangle has a perimeter of 30 cm and a length of 9 cm.

What is its width?

Width: _____

True or false?

A rectangle with an area of 36 square cm and a length of 9 cm has a width of 4 cm.

- A. True
- B. False

A rectangle has an area of 56 square inches and a width of 7 inches.

What is its length?

A. 8 inches

B. 9 inches

C. 49 inches

D. 63 inches

The rectangle has an area of 60 square meters.

What is the missing side length?



Length: _____

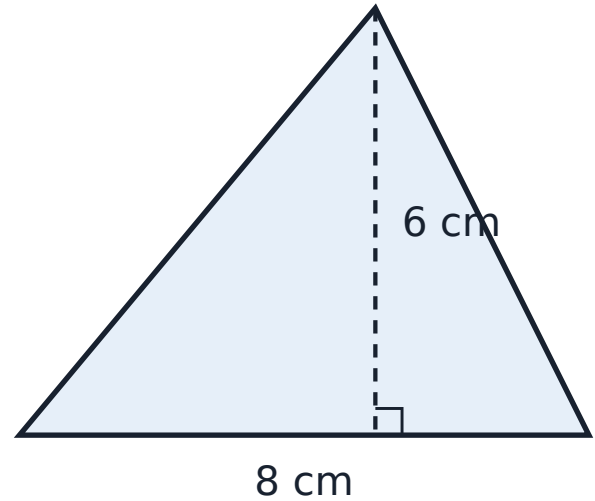
SET 60 • 6.G.A.1

BACKWARD 3

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



A triangle has a base of 10 m and a height of 3 m.

What is its area?

Area: _____

True or false?

A right triangle with legs of 4 in. and 6 in. has an area of 24 square inches.

A. True

B. False

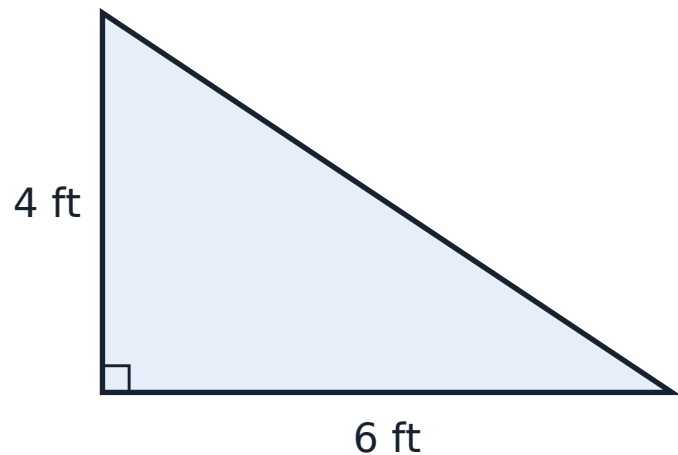
A triangle has a base of 5 ft and a height of 8 ft.

What is its area?

- A.** 13 square feet
- B.** 20 square feet
- C.** 26 square feet
- D.** 40 square feet

Find the area of the triangle.

Area: _____



SET 60 • 6.G.A.1

FORWARD 1

Area of composite figures in real-world problems • Grade 7 • Geometry •
7.G.B.6

A deck is made of a rectangle that is 12 ft by 10 ft and a triangle with a base of 10 ft and a height of 6 ft.

What is the area of the deck?

Area: _____

A trapezoid-shaped garden has bases of 8 m and 14 m and a height of 5 m. It has a square pond with 2 m sides.

What is the area of the garden without the pond?

Area: _____

A regular hexagon is split into 6 equal triangles. Each triangle has a base of 4 cm and a height of 3.5 cm.

What is the area of the hexagon?

- A.** 21 square cm
- B.** 24 square cm
- C.** 42 square cm
- D.** 84 square cm

An L-shaped prism has a base area of 30 square feet and a height of 4 feet.

What is its volume?

Volume: _____

True or false?

A figure made of a 5-by-5 square and a triangle with a base of 5 and a height of 2 has an area of 30 square units.

- A.** True
- B.** False

SET 60 • 6.G.A.1

FORWARD 2

Volume of composite solids • Grade 8 • Geometry • 8.G.C.9

A silo is a cylinder with a radius of 3 m and a height of 10 m, topped by a hemisphere with a radius of 3 m.

What is its volume in terms of π ?

Volume: _____

A cone has a radius of 3 in. and a height of 4 in. A hemisphere with a radius of 3 in. sits on top.

What is the total volume in terms of π ?

Volume: _____

A cone with a radius of 2 and a height of 3 is cut out of a cylinder with a radius of 2 and a height of 6.

What volume is left?

A. 20π

B. 28π

C. 24π

D. 4π

A cylinder and a cone each have a radius of 1 and a height of 2.

How much more volume does the cylinder have, in terms of π ?

Difference: _____

True or false?

A sphere with a radius of 3 has the same volume as a cylinder with a radius of 3 and a height of 4.

A. True

B. False

Find volume by packing a prism with unit-fraction cubes

MAIN	Grade 6 main question	6.G.A.2
BACKWARD 1	Volume by counting unit cubes	5.MD.C.4
BACKWARD 2	The volume of one small cube	5.NF.B.4.a
BACKWARD 3	Volume as the number of unit cubes	5.MD.C.3.b
FORWARD 1	Volume of prisms in real-world problems	7.G.B.6
FORWARD 2	Volume of cylinders, cones, and spheres	8.G.C.9

SET 61 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

How many cubes with $\frac{1}{2}$ -inch edges fill a box that is 2 in. by $1\frac{1}{2}$ in. by 1 in.?

Cubes: _____

A box is 3 in. by 2 in. by $1\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill the box?

Cubes: _____

True or false?

A prism is packed with 48 cubes that each have $\frac{1}{2}$ -unit edges. Its volume is 6 cubic units.

- A.** True
- B.** False

A prism is packed with 64 cubes that each have $\frac{1}{4}$ -inch edges.

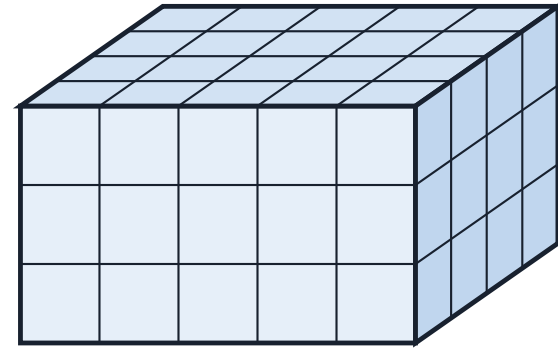
What is the volume of the prism?

- A.** $\frac{1}{4}$ cubic inch
- B.** 1 cubic inch
- C.** 4 cubic inches
- D.** 16 cubic inches

The prism is packed with
cubes that have $\frac{1}{2}$ -cm edges.

It is 5 cubes long, 4 cubes
wide, and 3 cubes tall.

What is its volume?



Volume: _____

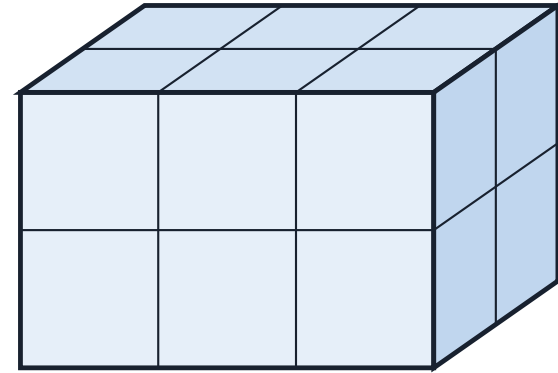
SET 61 • 6.G.A.2

BACKWARD 1

Volume by counting unit cubes • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic unit.

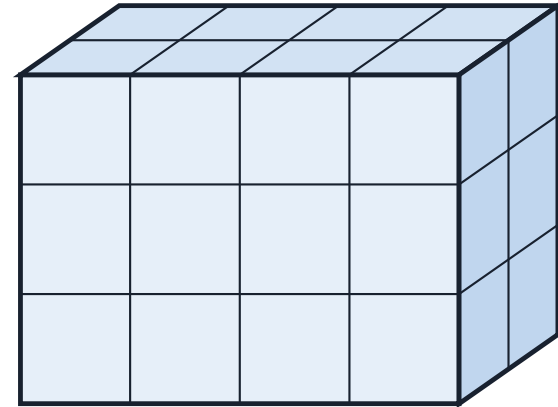
What is the volume of the prism?



Volume: _____

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

True or false?

A prism made of 2 layers with 6 unit cubes in each layer has a volume of 12 cubic units.

- A. True
- B. False

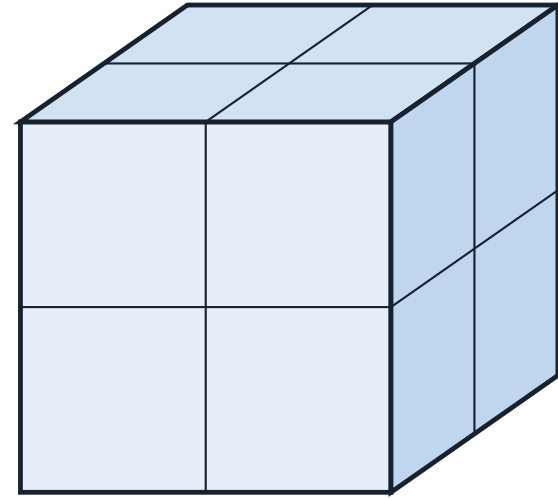
A box holds 3 layers with 5 unit cubes in each layer.

What is its volume?

- A.** 8 cubic units
- B.** 15 cubic units
- C.** 35 cubic units
- D.** 53 cubic units

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

SET 61 • 6.G.A.2

BACKWARD 2

The volume of one small cube • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Product: _____

Multiply.

$$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$$

Product: _____

True or false?

$$\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

A. True

B. False

Multiply.

$$24 \times \frac{1}{8}$$

A. $\frac{1}{3}$

B. 3

C. 32

D. 192

Multiply.

$$64 \times \frac{1}{64}$$

Product: _____

SET 61 • 6.G.A.2

BACKWARD 3

Volume as the number of unit cubes • Grade 5 • Measurement & Data •
5.MD.C.3.b

True or false?

A solid that can be packed with 20 unit cubes with no gaps has a volume of 20 cubic units.

- A.** True
- B.** False

A box is filled with 36 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 6 cubic units
- B.** 12 cubic units
- C.** 36 cubic units
- D.** 36 square units

A box is filled with 45 unit cubes with no gaps.
What is its volume?

Volume: _____

True or false?

Volume is measured in square units.

A. True

B. False

Which unit could be used to measure volume?

- A.** Square centimeters
- B.** Cubic centimeters
- C.** Centimeters
- D.** Kilograms

SET 61 • 6.G.A.2

FORWARD 1

Volume of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 5 in. and a height of 4 in. The prism is 8 in. long.

What is its volume?

Volume: _____

A prism has a base area of $12\frac{1}{2}$ square cm and a height of 4 cm.

What is its volume?

Volume: _____

A cube has edges of 2.5 cm.

What is its volume?

- A.** 6.25 cubic cm
- B.** 7.5 cubic cm
- C.** 15 cubic cm
- D.** 15.625 cubic cm

An aquarium is 30 in. long, 12 in. wide, and 16 in. tall. It is filled with water to a height of 12 in.

What is the volume of the water?

Volume: _____

True or false?

A triangular prism with a base area of 15 square units and a length of 6 units has a volume of 90 cubic units.

A. True

B. False

SET 61 • 6.G.A.2

FORWARD 2

Volume of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

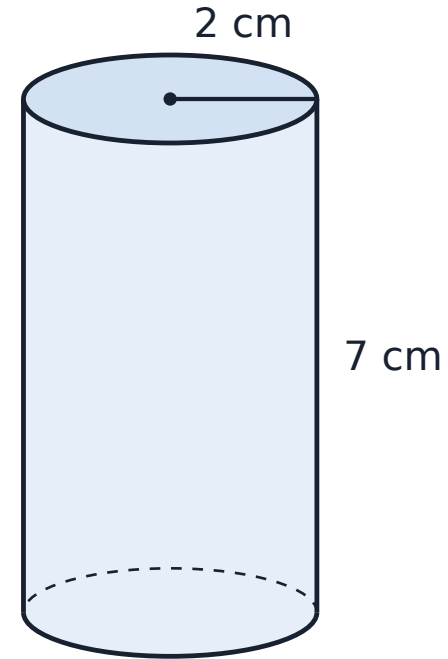
A cylinder has a radius of 5 and a height of 4.
What is its volume in terms of π ?

Volume: _____

Find the volume of the cylinder.

Use 3.14 for π .

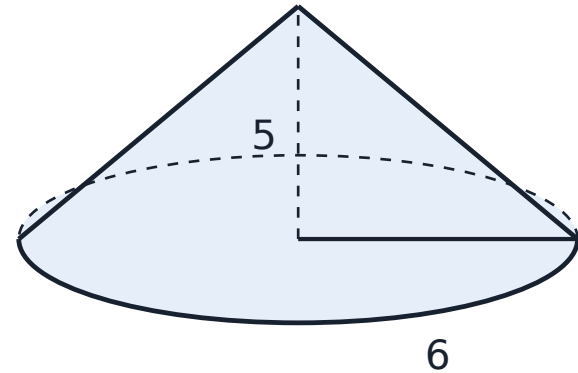
Volume: _____



A cone has a radius of 6 and a height of 5.

What is its volume?

- A.** 150π **B.** 180π
C. 30π **D.** 60π



A sphere has a radius of 6.

What is its volume in terms of π ?

Volume: _____

True or false?

A cylinder has a greater volume than a cone with the same radius and height.

A. True

B. False

Apply $V = l w h$ and $V = b h$ with fractional edge lengths

MAIN

Grade 6 main question

6.G.A.2

BACKWARD 1

Volume formula with whole numbers

5.MD.C.5.b

BACKWARD 2

Multiplying mixed numbers

5.NF.B.6

BACKWARD 3

Multiplying fractions

5.NF.B.4.a

FORWARD 1

Volume of triangular and rectangular prisms

7.G.B.6

FORWARD 2

Volume of cylinders using $V = Bh$

8.G.C.9

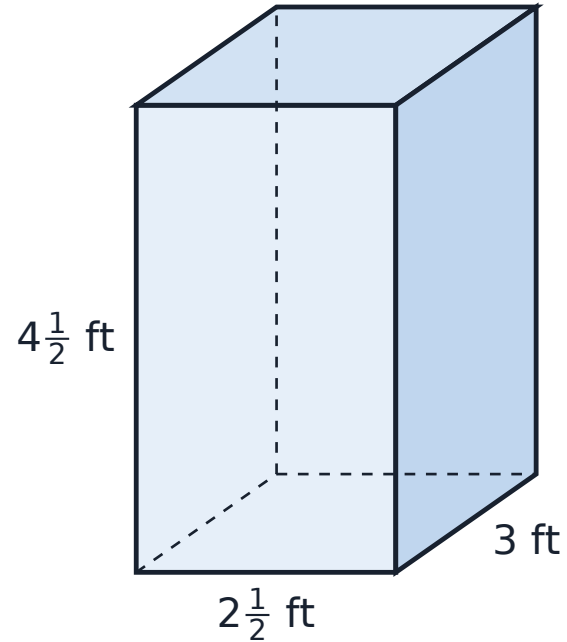
SET 62 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

Find the volume of the rectangular prism.

Volume: _____



A box is $1\frac{1}{2}$ ft long, 2 ft wide, and $\frac{3}{4}$ ft tall.

What is its volume?

- A.** $2\frac{1}{4}$ cubic feet
- B.** $2\frac{3}{4}$ cubic feet
- C.** 3 cubic feet
- D.** $4\frac{1}{4}$ cubic feet

Use $V = l \times w \times h$ to find the volume of a prism with $l = 5$ cm, $w = 2\frac{1}{2}$ cm, and $h = 1\frac{1}{5}$ cm.

$V =$ _____

A cube has edges of $1\frac{1}{2}$ feet.

What is its volume?

Volume: _____

A prism has a base area of $4\frac{1}{2}$ square inches and a height of $2\frac{2}{3}$ inches.

Use $V = b \times h$ to find its volume.

$V =$ _____

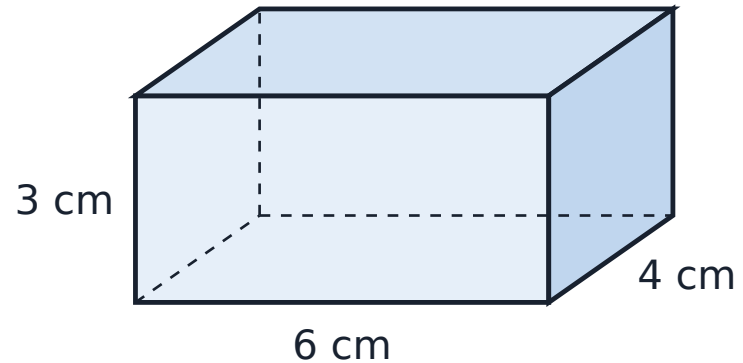
SET 62 • 6.G.A.2

BACKWARD 1

Volume formula with whole numbers • Grade 5 • Measurement & Data •
5.MD.C.5.b

Find the volume of the prism.

Volume: _____



A prism has a length of 10 m, a width of 2 m, and a height of 5 m.

What is its volume?

Volume: _____

True or false?

A cube with 3-inch edges has a volume of 27 cubic inches.

A. True

B. False

A box is 8 in. by 5 in. by 2 in.

What is its volume?

- A.** 15 cubic inches
- B.** 40 cubic inches
- C.** 80 cubic inches
- D.** 160 cubic inches

A prism has a base area of 20 square cm and a height of 6 cm.

What is its volume?

Volume: _____

SET 62 • 6.G.A.2

BACKWARD 2

Multiplying mixed numbers • Grade 5 • Number & Operations–Fractions •
5.NF.B.6

Multiply.

$$2\frac{1}{2} \times 3$$

Product: _____

Multiply.

$$1\frac{1}{2} \times \frac{3}{4}$$

Product: _____

True or false?

$$2\frac{1}{2} \times 2 = 5$$

A. True

B. False

$$3 \times 4\frac{1}{2} = ?$$

A. $7\frac{1}{2}$

B. $12\frac{1}{6}$

C. $12\frac{1}{2}$

D. $13\frac{1}{2}$

Multiply.

$$1\frac{1}{3} \times 1\frac{1}{2}$$

Product: _____

SET 62 • 6.G.A.2

BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{2}{3}$$

Product: _____

Multiply.

$$\frac{5}{2} \times \frac{9}{2}$$

Product: _____

True or false?

$$\frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$$

A. True

B. False

$$\frac{1}{2} \times \frac{4}{5} = ?$$

A. $\frac{2}{5}$

B. $\frac{4}{7}$

C. $\frac{5}{7}$

D. $\frac{8}{5}$

Multiply.

$$\frac{6}{5} \times 5$$

Product: _____

SET 62 • 6.G.A.2

FORWARD 1

Volume of triangular and rectangular prisms • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 9 cm long.

What is its volume?

Volume: _____

A tent is a triangular prism. Its triangle has a base of 2.4 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

A prism has a base area of 7.5 square feet and a height of 2.4 feet.

What is its volume?

- A.** 9.9 cubic feet
- B.** 15 cubic feet
- C.** 18 cubic feet
- D.** 180 cubic feet

A box is 4.5 cm by 2 cm by 3.2 cm.

What is its volume?

Volume: _____

True or false?

A triangular prism with a triangle area of 10 square inches and a length of 7 inches has a volume of 70 cubic inches.

A. True

B. False

SET 62 • 6.G.A.2

FORWARD 2

Volume of cylinders using $V = Bh$ • Grade 8 • Geometry • 8.G.C.9

A cylinder has a base area of 12π square cm and a height of 5 cm.

What is its volume in terms of π ?

Volume: _____

A can has a radius of 3 in. and a height of 8 in.
What is its volume in terms of π ?

Volume: _____

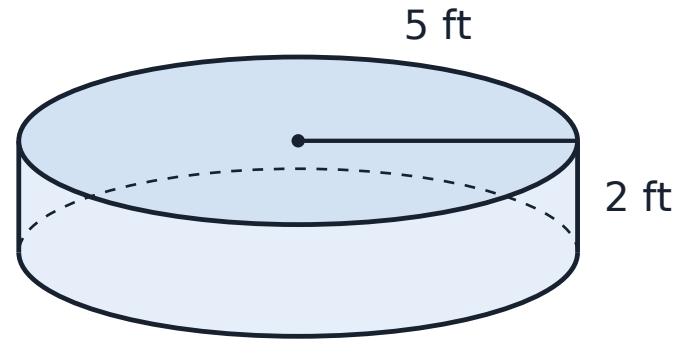
A cylinder has a radius of 1.5 m and a height of 4 m.

What is its volume?

- A.** 12π cubic meters
- B.** 9π cubic meters
- C.** 3π cubic meters
- D.** 6π cubic meters

Find the volume of the cylinder in terms of π .

Volume: _____



True or false?

Doubling the height of a cylinder doubles its volume.

A. True

B. False

Draw polygons in the coordinate plane from the coordinates of their vertices

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Plotting points in all four quadrants	6.NS.C.6.c
BACKWARD 2	Attributes of quadrilaterals	3.G.A.1
BACKWARD 3	Naming shapes by sides and angles	2.G.A.1
FORWARD 1	Draw shapes with given conditions	7.G.A.2
FORWARD 2	Draw images of polygons under transformations	8.G.A.3

SET 63 • 6.G.A.3

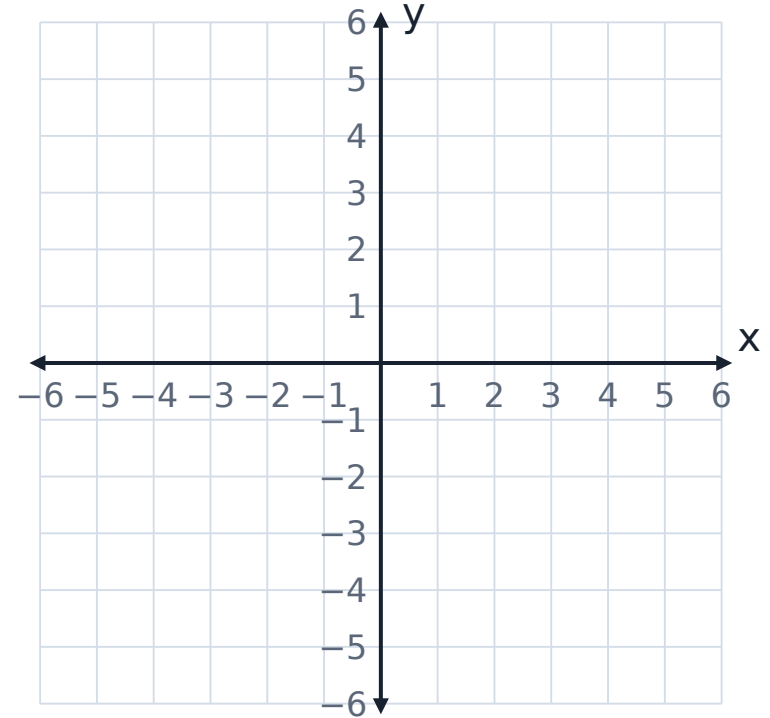
MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

Plot $A(-3, 2)$, $B(4, 2)$, $C(4, -3)$, and $D(-3, -3)$. Connect them in order.

What polygon did you draw?

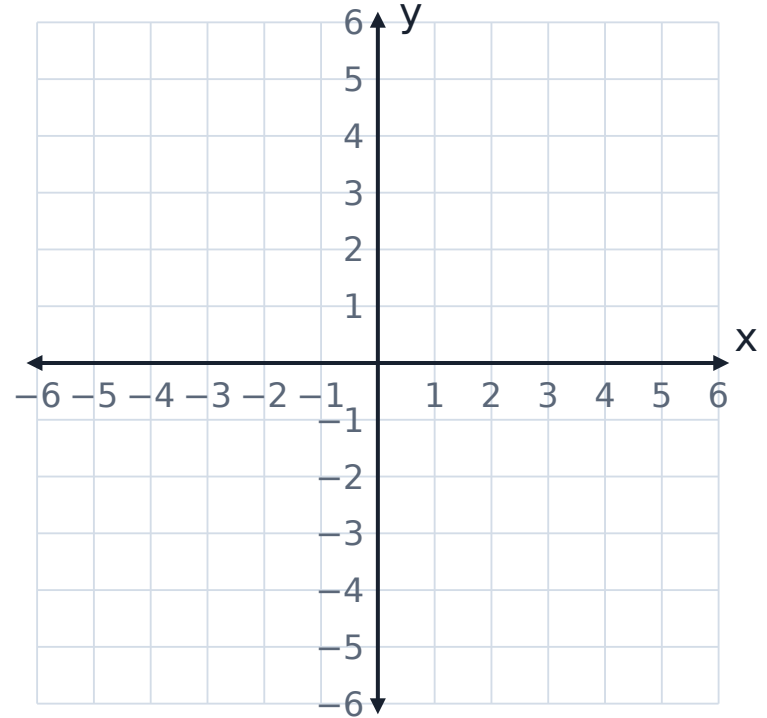
Polygon: _____



Plot $P(0, 4)$, $Q(3, -2)$, and $R(-3, -2)$. Connect them in order.

What polygon did you draw?

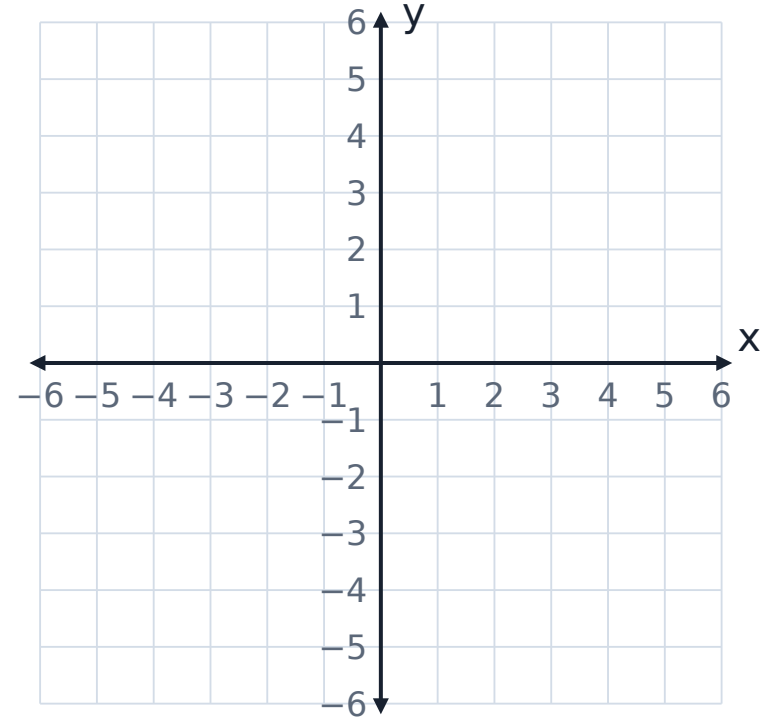
Polygon: _____



Plot $E(-2, 1)$, $F(2, 1)$, $G(4, -3)$, and $H(-4, -3)$. Connect them in order.

What polygon did you draw?

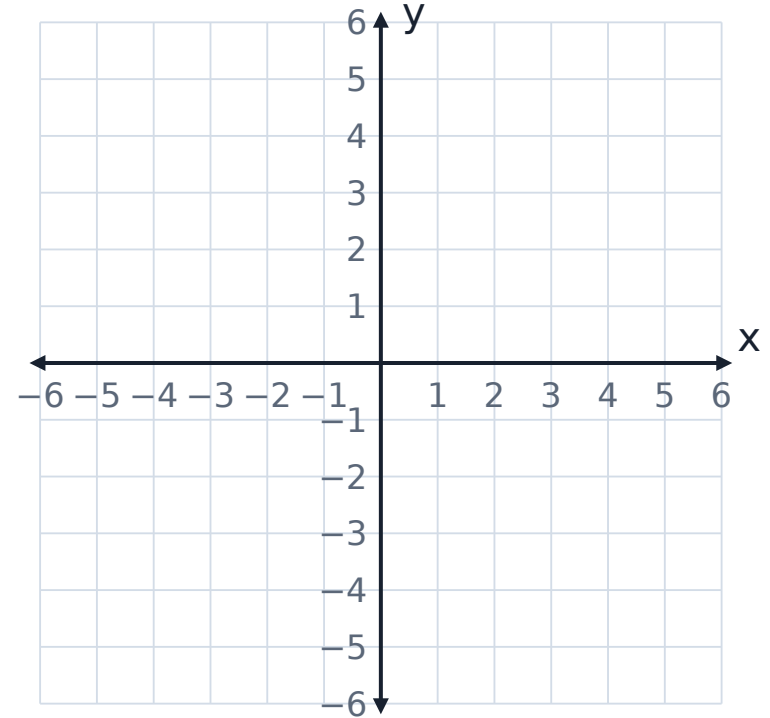
Polygon: _____



Plot $J(-1, 3)$, $K(3, 3)$, $L(3, -1)$,
and $M(-1, -1)$. Connect
them in order.

What polygon did you draw?

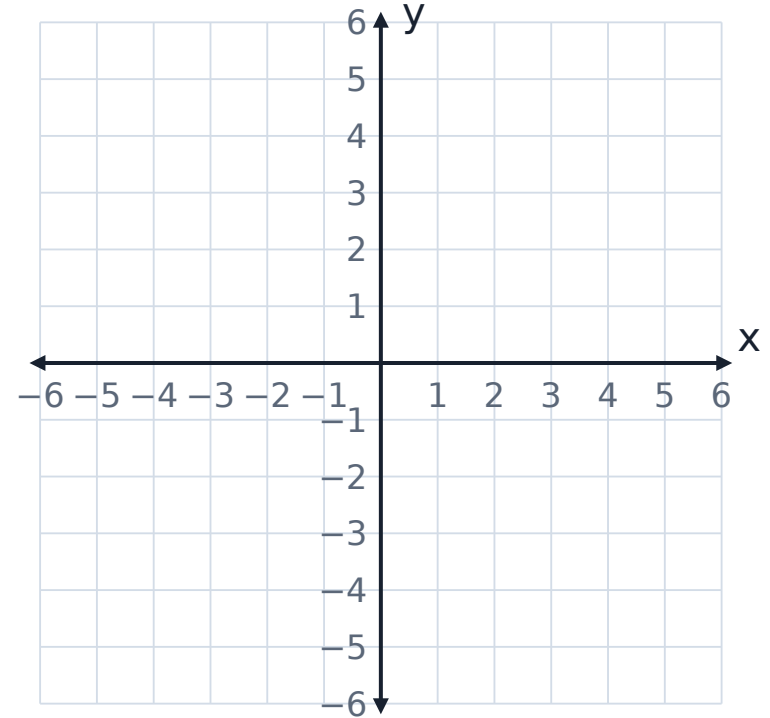
Polygon: _____



Plot $W(-5, -2)$, $X(-2, 3)$, $Y(4, 3)$, and $Z(1, -2)$. Connect them in order.

What polygon did you draw?

Polygon: _____

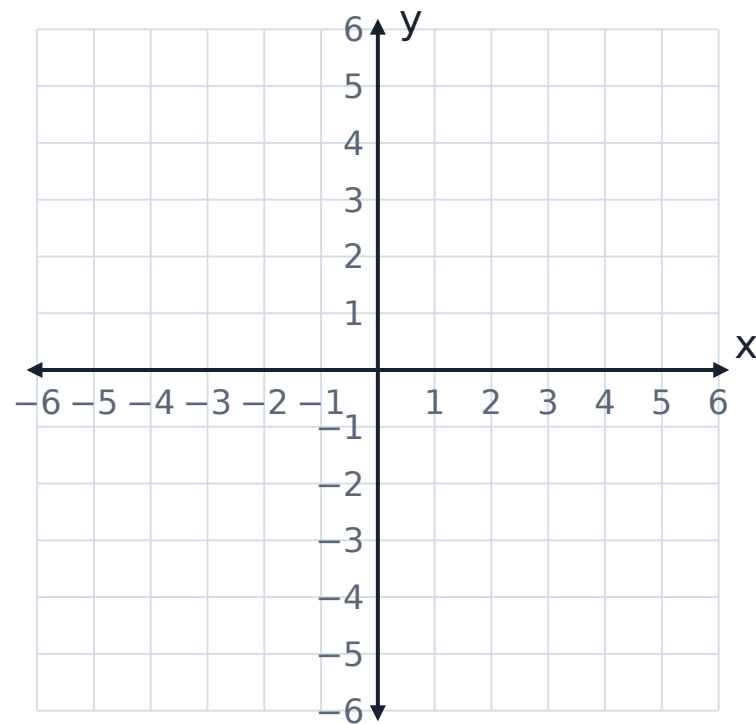


SET 63 • 6.G.A.3

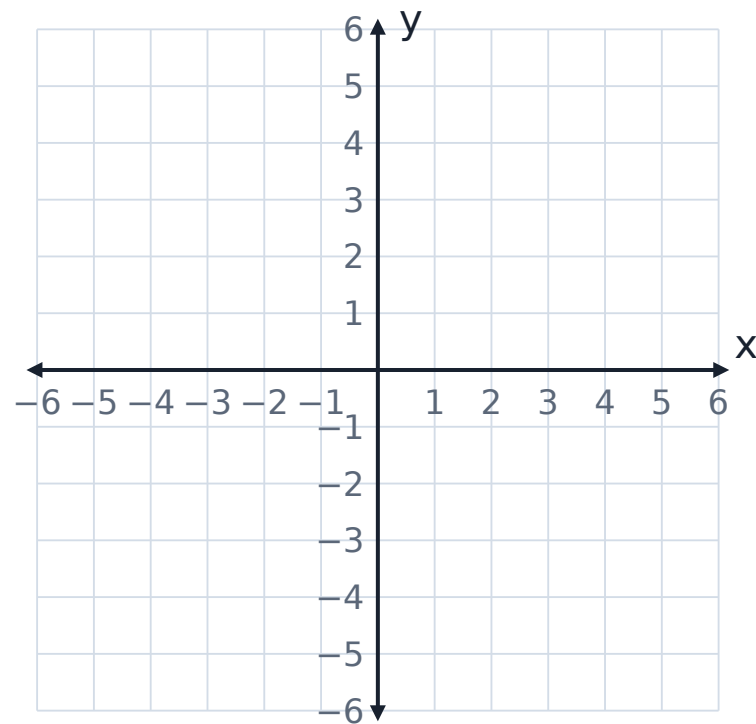
BACKWARD 1

Plotting points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

Plot and label point A($-3, 2$).



Plot and label point B(4, -3).



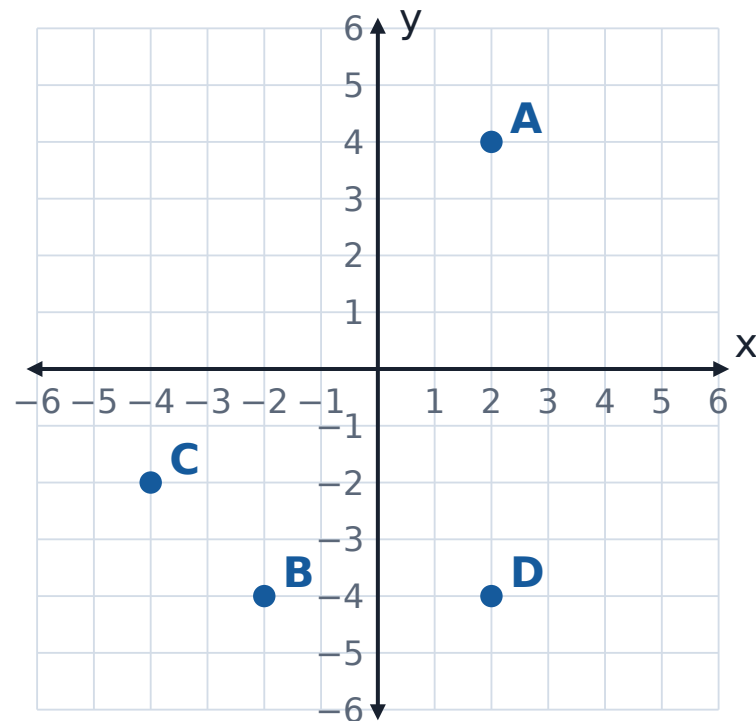
Which point is at $(-2, -4)$?

A. Point A

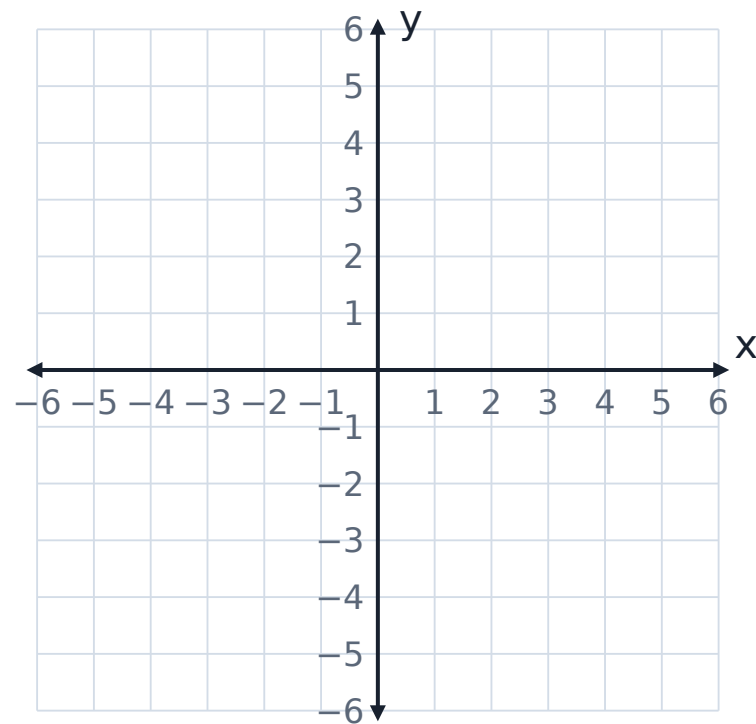
B. Point B

C. Point C

D. Point D



Plot and label point C(0, 4).

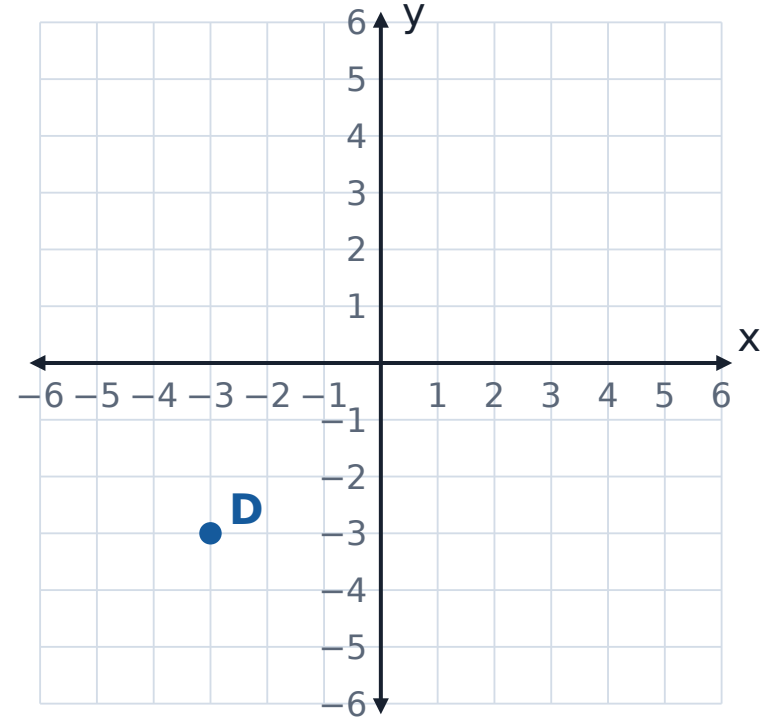


True or false?

Point D is at $(-3, -3)$.

A. True

B. False



SET 63 • 6.G.A.3

BACKWARD 2

Attributes of quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle has 4 right angles.

A. True

B. False

True or false?

Opposite sides of a rectangle have the same length.

A. True

B. False

Which shape always has 4 sides of equal length and 4 right angles?

- A.** Square
- B.** Rectangle
- C.** Trapezoid
- D.** Rhombus

True or false?

Every quadrilateral has 4 right angles.

A. True

B. False

How many sides does a quadrilateral have?

Sides: _____

SET 63 • 6.G.A.3

BACKWARD 3

Naming shapes by sides and angles • Grade 2 • Geometry • 2.G.A.1

A shape has 3 sides and 3 angles.

What is it called?

Shape: _____

How many sides does a pentagon have?

A. 4

B. 5

C. 6

D. 8

True or false?

A hexagon has 6 sides.

A. True

B. False

A shape has 4 sides and 4 angles.

What is it called?

Shape: _____

Which shape has the most sides?

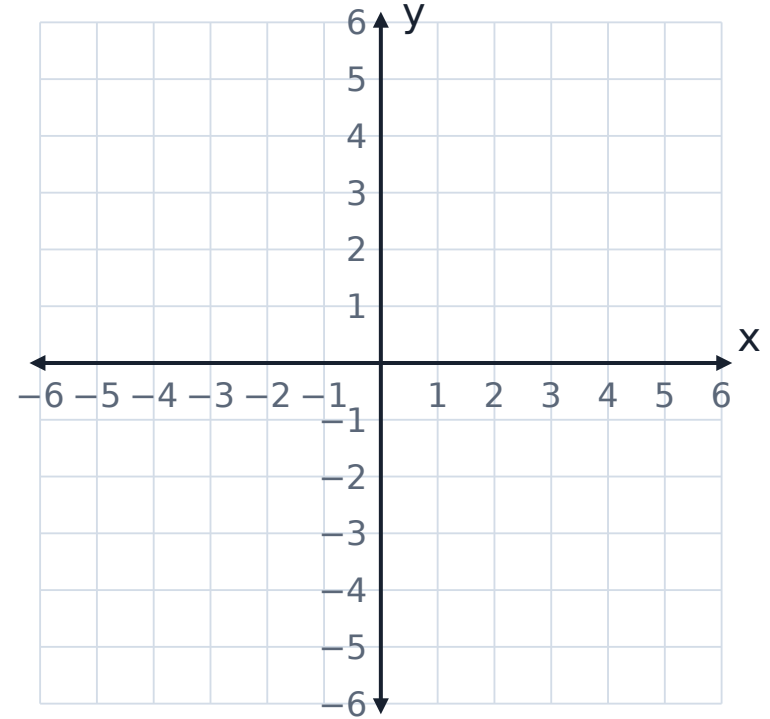
- A.** Pentagon
- B.** Quadrilateral
- C.** Triangle
- D.** Hexagon

SET 63 • 6.G.A.3

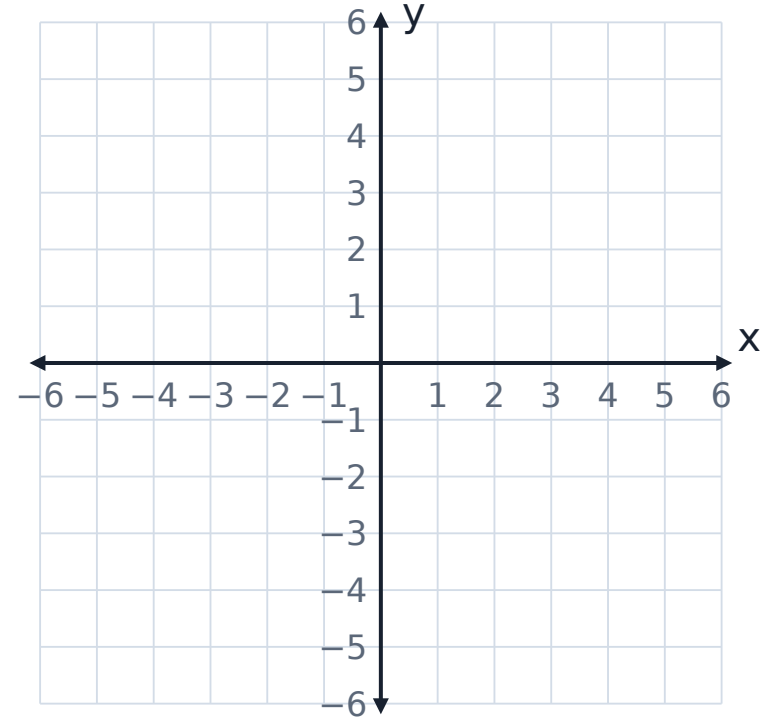
FORWARD 1

Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Draw a rectangle with an area of 12 square units. One vertex is at the origin.



Draw a right triangle with legs of 4 units and 3 units. One vertex is at $(-2, -2)$.



Can a triangle have side lengths of 2 cm, 3 cm, and 6 cm? Write yes or no.

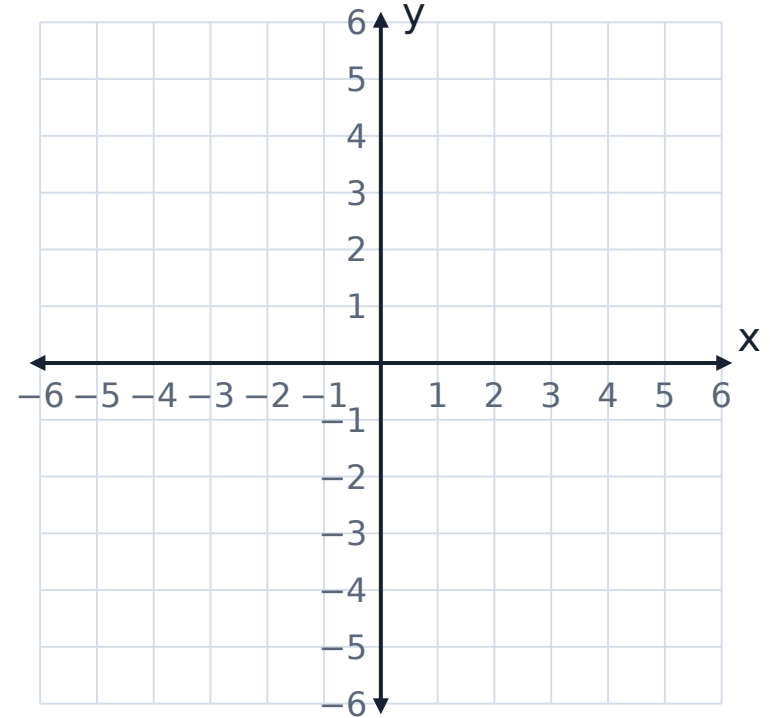
Answer: _____

A triangle has angles of 90° and 45° .

What is the third angle?

- A.** 45°
- B.** 55°
- C.** 90°
- D.** 135°

Draw a square with a side length of 5 units. One vertex is at $(-3, 1)$.



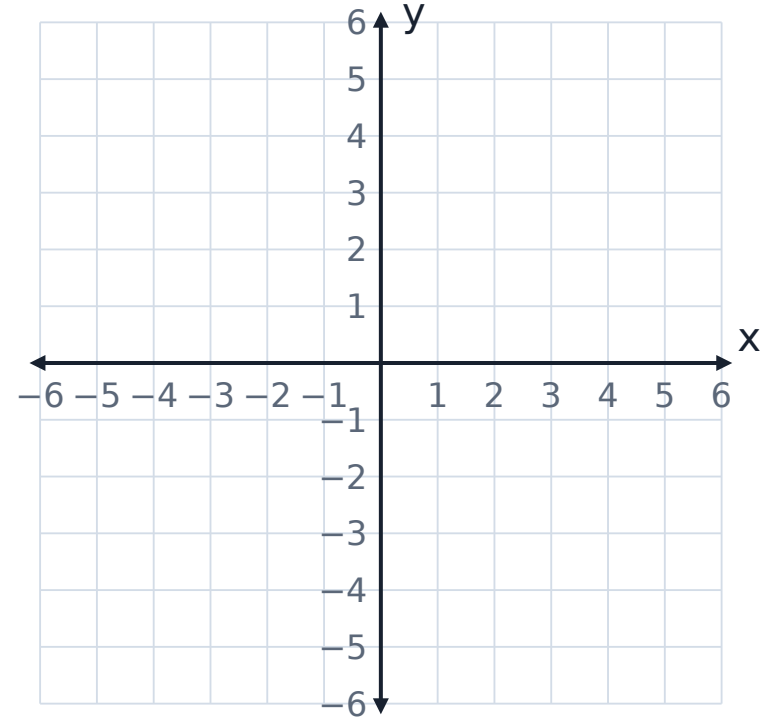
SET 63 • 6.G.A.3

FORWARD 2

Draw images of polygons under transformations • Grade 8 • Geometry •
8.G.A.3

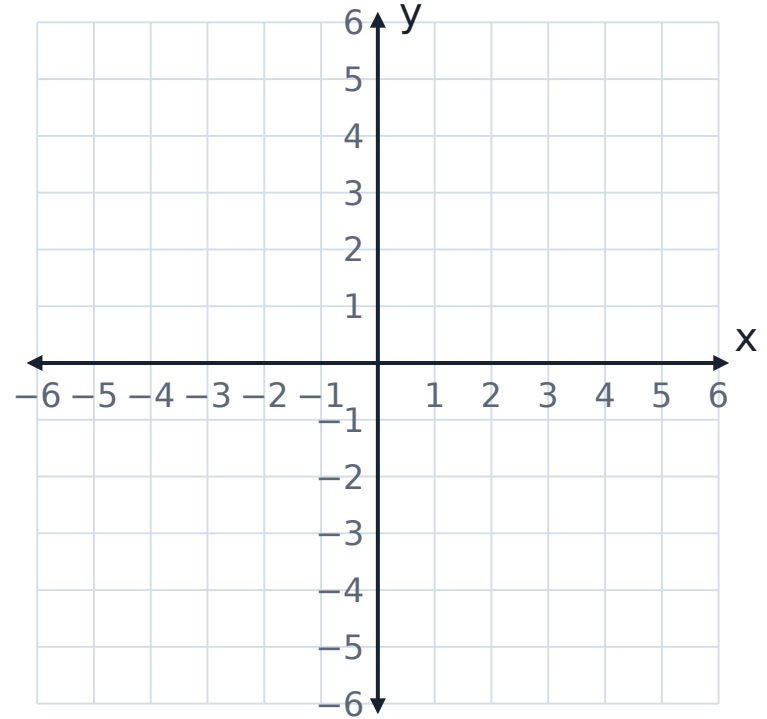
Triangle ABC has vertices
 $A(1, 1)$, $B(4, 1)$, and $C(1, 3)$.

Draw its image after a
translation 5 units left and 2
units down.



Rectangle PQRS has vertices $P(1, 2)$, $Q(4, 2)$, $R(4, 4)$, and $S(1, 4)$.

Draw its image after a reflection across the x-axis.



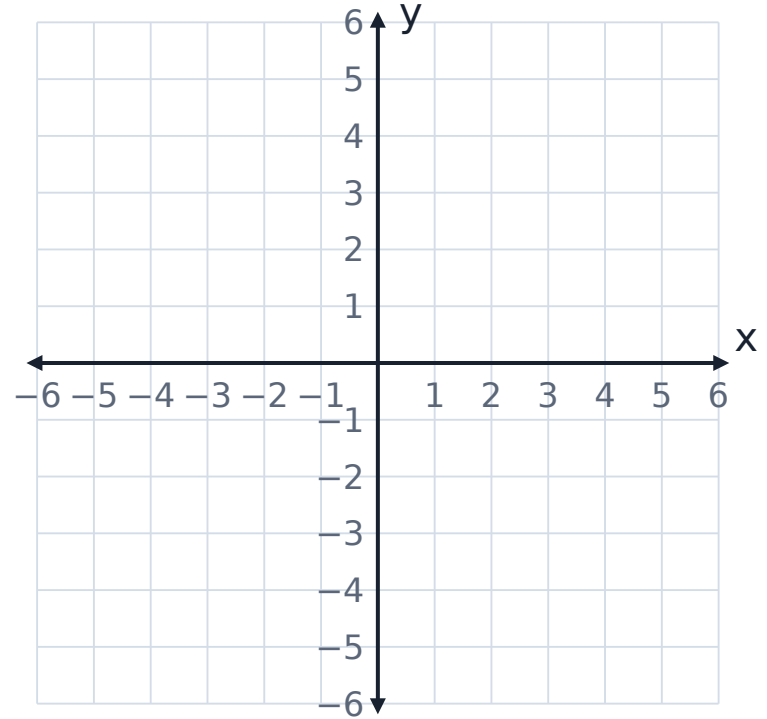
Triangle ABC has vertices A(2, 1), B(4, 3), and C(2, 3). It is rotated 180° about the origin.

What are the coordinates of B'?

B' = _____

Square EFGH has vertices $E(1, 1)$, $F(2, 1)$, $G(2, 2)$, and $H(1, 2)$.

Draw its image after a dilation by a scale factor of 3 centered at the origin.



True or false?

A rectangle with vertices $(1, 1)$, $(3, 1)$, $(3, 2)$, $(1, 2)$ is reflected across the y -axis. One image vertex is $(-3, 2)$.

A. True

B. False

Use coordinates to find side lengths and perimeters of polygons

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Distance between points with the same x or y	6.NS.C.8
BACKWARD 2	Absolute value	6.NS.C.7.c
BACKWARD 3	Perimeter of polygons	3.MD.D.8
FORWARD 1	Lengths between points with rational coordinates	7.NS.A.1.c
FORWARD 2	Find slanted side lengths with the Pythagorean Theorem	8.G.B.8

SET 64 • 6.G.A.3

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

A rectangle has vertices at $(-3, 2)$, $(4, 2)$, $(4, -5)$, and $(-3, -5)$.

What is its perimeter?

Perimeter: _____

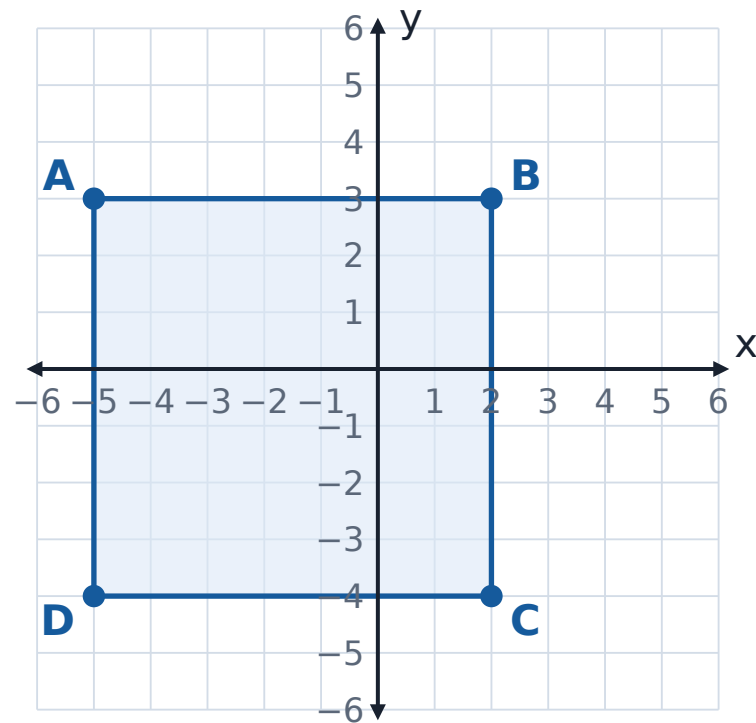
One side of a polygon goes from $(-2, -3)$ to $(-2, 5)$.

How long is this side?

Length: _____

What is the length of side AB?

AB = _____



A triangle has vertices at $(-4, 1)$, $(2, 1)$, and $(2, -7)$.

How long is the horizontal side?

- A.** 2 units
- B.** 6 units
- C.** 8 units
- D.** 10 units

A rectangle has vertices at $(-5, -2)$, $(1, -2)$, $(1, 3)$, and $(-5, 3)$.

What is its perimeter?

Perimeter: _____

SET 64 • 6.G.A.3

BACKWARD 1

Distance between points with the same x or y • Grade 6 • The Number System
• 6.NS.C.8

Find the distance between $(-3, 2)$ and $(4, 2)$.

Distance: _____

Find the distance between $(1, -4)$ and $(1, 6)$.

Distance: _____

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 10.

A. True

B. False

What is the distance between $(2, -3)$ and $(2, -8)$?

A. -11

B. -5

C. 5

D. 11

Find the distance between $(-6, -1)$ and $(-2, -1)$.

Distance: _____

SET 64 • 6.G.A.3

BACKWARD 2

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-5| + |2|$$

Value: _____

True or false?

$$|-7| = 7$$

A. True

B. False

$$|-3| + |4| = ?$$

A. -7

B. -1

C. 1

D. 7

Find the value.

$$|-6|$$

Value: _____

True or false?

$$|-2| + |-2| = 0$$

A. True

B. False

SET 64 • 6.G.A.3

BACKWARD 3

Perimeter of polygons • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of the rectangle.

Perimeter: _____



A square has sides of 9 inches.

What is its perimeter?

Perimeter: _____

True or false?

A rectangle that is 10 m by 4 m has a perimeter of 28 m.

A. True

B. False

A triangle has sides of 5 cm, 6 cm, and 7 cm.

What is its perimeter?

- A.** 11 cm
- B.** 18 cm
- C.** 30 cm
- D.** 210 cm

A rectangle is 12 ft long and 5 ft wide.

What is its perimeter?

Perimeter: _____

SET 64 • 6.G.A.3

FORWARD 1

Lengths between points with rational coordinates • Grade 7 • The Number System • 7.NS.A.1.c

A side of a polygon goes from $(-2.5, 3)$ to $(4, 3)$.

How long is the side?

Length: _____

A rectangle has vertices at $(-1.5, 2)$, $(3.5, 2)$, $(3.5, -1)$, and $(-1.5, -1)$.

What is its perimeter?

Perimeter: _____

How long is the segment from $(0, -3\frac{1}{2})$ to $(0, 1\frac{1}{4})$?

A. $-4\frac{3}{4}$

B. $2\frac{1}{4}$

C. $4\frac{3}{4}$

D. $5\frac{1}{4}$

A square has vertices at $(-2.5, -2.5)$, $(2.5, -2.5)$, $(2.5, 2.5)$, and $(-2.5, 2.5)$.

What is the length of one side?

Length: _____

True or false?

The segment from $(-4.2, 1)$ to $(1.8, 1)$ is 6 units long.

- A. True
- B. False

SET 64 • 6.G.A.3

FORWARD 2

Find slanted side lengths with the Pythagorean Theorem • Grade 8 • Geometry
• 8.G.B.8

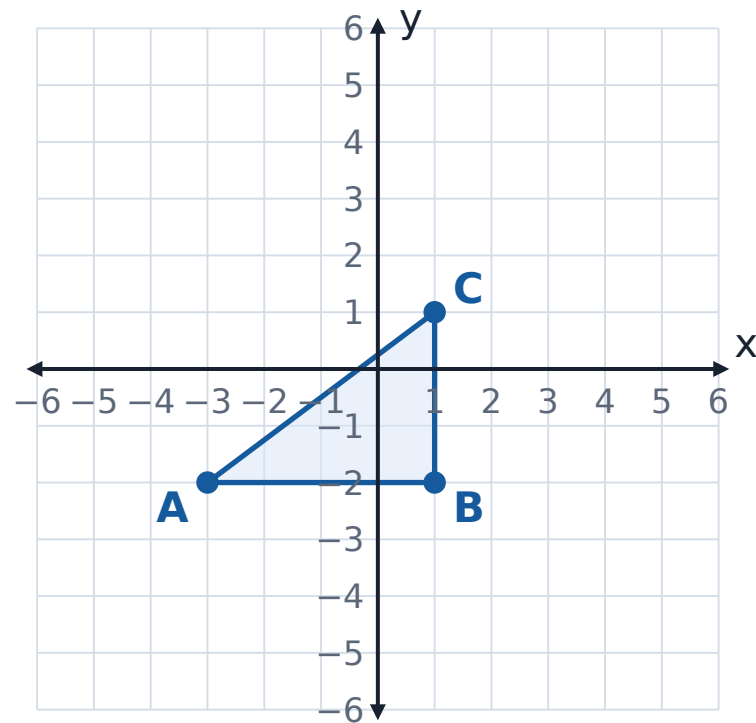
A triangle has vertices at $(0, 0)$, $(6, 0)$, and $(6, 8)$.

How long is the slanted side?

Length: _____

Find the length of side AC.

AC = _____



A parallelogram has a side from $(1, 1)$ to $(4, 5)$.

How long is that side?

- A.** 3
- B.** 5
- C.** 7
- D.** 25

Find the perimeter of the triangle with vertices $(0, 0)$, $(5, 0)$, and $(0, 12)$.

Perimeter: _____

True or false?

The side from $(-2, 3)$ to $(4, -5)$ is 10 units long.

A. True

B. False

Represent three-dimensional figures using nets

MAIN	Grade 6 main question	6.G.A.4
BACKWARD 1	Faces of three-dimensional shapes	2.G.A.1
BACKWARD 2	Shapes of faces	3.G.A.1
FORWARD 1	Cross sections of three-dimensional figures	7.G.A.3
FORWARD 2	Volume of a cylinder described by its net	8.G.C.9

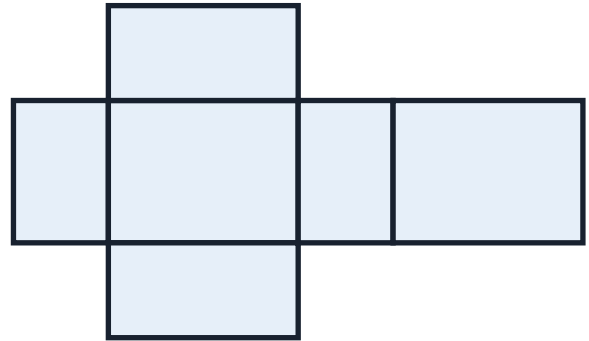
SET 65 • 6.G.A.4

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

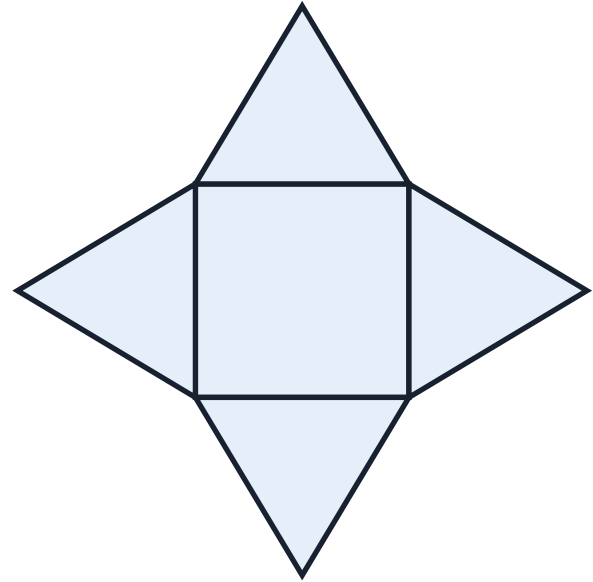
What three-dimensional figure does the net form?

- A.** Triangular prism
- B.** Rectangular prism
- C.** Square pyramid
- D.** Cube



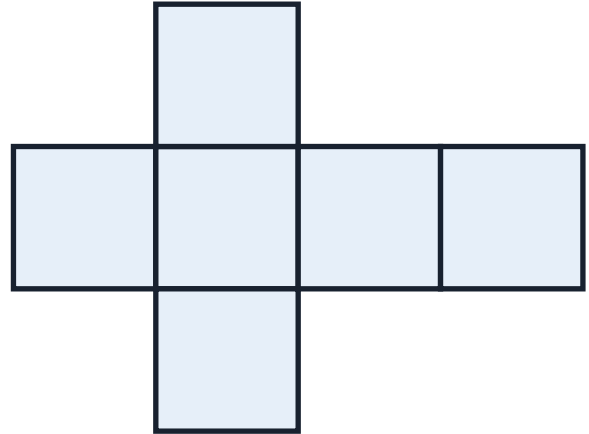
What three-dimensional figure does the net form?

- A.** Cube
- B.** Triangular pyramid
- C.** Square pyramid
- D.** Triangular prism



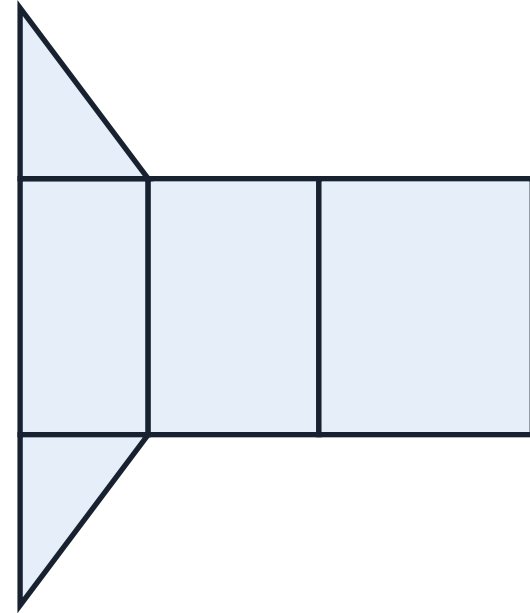
What three-dimensional figure does the net form?

- A.** Rectangular prism
- B.** Triangular prism
- C.** Cube
- D.** Square pyramid



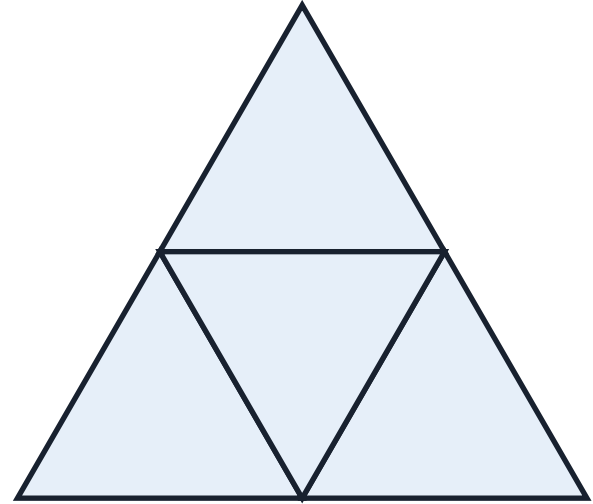
What three-dimensional figure does the net form?

- A.** Triangular prism
- B.** Triangular pyramid
- C.** Square pyramid
- D.** Rectangular prism



What three-dimensional figure does the net form?

- A.** Cube
- B.** Square pyramid
- C.** Triangular pyramid
- D.** Triangular prism



SET 65 • 6.G.A.4

BACKWARD 1

Faces of three-dimensional shapes • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

How many faces does a rectangular prism have?

- A.** 4
- B.** 6
- C.** 8
- D.** 12

True or false?

A square pyramid has 5 faces.

A. True

B. False

How many faces does a triangular prism have?

Faces: _____

How many triangular faces does a square pyramid have?

A. 1

B. 3

C. 4

D. 5

SET 65 • 6.G.A.4

BACKWARD 2

Shapes of faces • Grade 3 • Geometry • 3.G.A.1

What shape are the faces of a cube?

- A.** Triangles
- B.** Pentagons
- C.** Circles
- D.** Squares

True or false?

Every square is also a rectangle.

A. True

B. False

What shape is the base of a square pyramid?

- A.** Square
- B.** Circle
- C.** Triangle
- D.** Rectangle that is not a square

True or false?

The two ends of a triangular prism are triangles.

A. True

B. False

What shape are the side faces of a triangular prism?

Shape: _____

SET 65 • 6.G.A.4

FORWARD 1

Cross sections of three-dimensional figures • Grade 7 • Geometry • 7.G.A.3

A rectangular prism is sliced parallel to its base.
What shape is the cross section?

- A.** Rectangle
- B.** Circle
- C.** Pentagon
- D.** Triangle

A square pyramid is sliced parallel to its base.
What shape is the cross section?

- A.** Square
- B.** Rectangle that is not a square
- C.** Circle
- D.** Triangle

A cube is sliced through the middle, parallel to one face.
What shape is the cross section?

Shape: _____

A square pyramid is sliced vertically through its top vertex.

What shape is the cross section?

A. Pentagon

B. Square

C. Rectangle

D. Triangle

True or false?

A triangular prism sliced parallel to its triangular bases makes a triangle cross section.

- A.** True
- B.** False

SET 65 • 6.G.A.4

FORWARD 2

Volume of a cylinder described by its net • Grade 8 • Geometry • 8.G.C.9

A net of a cylinder has two circles with a radius of 3 cm and a rectangle that is 8 cm tall.

What is the volume of the cylinder in terms of π ?

Volume: _____

A net of a cylinder has two circles with a radius of 5 in. and a rectangle that is 4 in. tall.

What is the volume in terms of π ?

Volume: _____

A net of a cylinder has circles with a diameter of 6 m.
The rectangle is 10 m tall.

What is the volume?

- A.** 30π cubic meters
- B.** 90π cubic meters
- C.** 360π cubic meters
- D.** 60π cubic meters

A net of a cylinder has two circles with a radius of 2 ft and a rectangle that is 7 ft tall.

What is the volume? Use 3.14 for π .

Volume: _____

True or false?

In the net of a cylinder, the height of the rectangle is the height of the cylinder.

- A. True
- B. False

Use nets to find surface area

MAIN	Grade 6 main question	6.G.A.4
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area of triangles	6.G.A.1
BACKWARD 3	Adding the areas of parts	3.MD.C.7.d
FORWARD 1	Surface area of prisms in real-world problems	7.G.B.6
FORWARD 2	Pythagorean Theorem in three dimensions	8.G.B.7

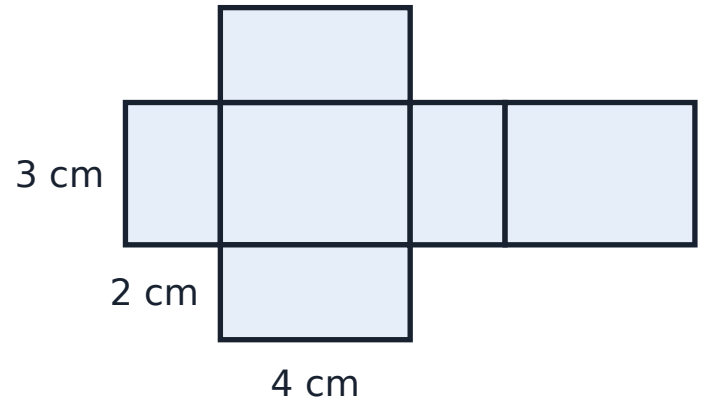
SET 66 • 6.G.A.4

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

The net folds into a rectangular prism.

What is the surface area of the prism?

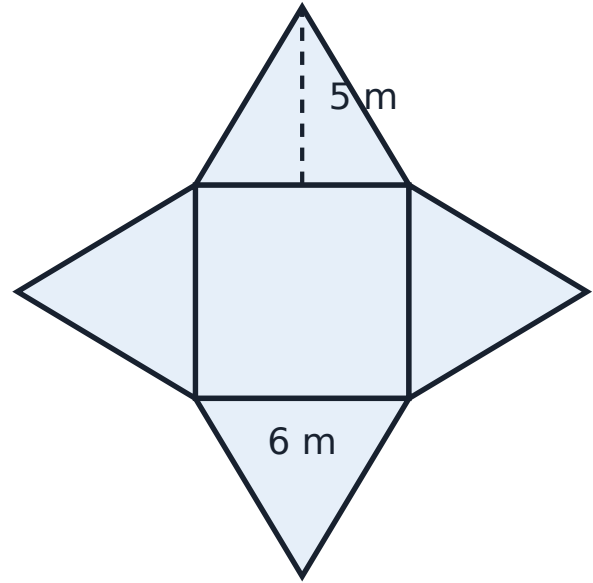


Surface area: _____

The net folds into a square pyramid.

What is the surface area of the pyramid?

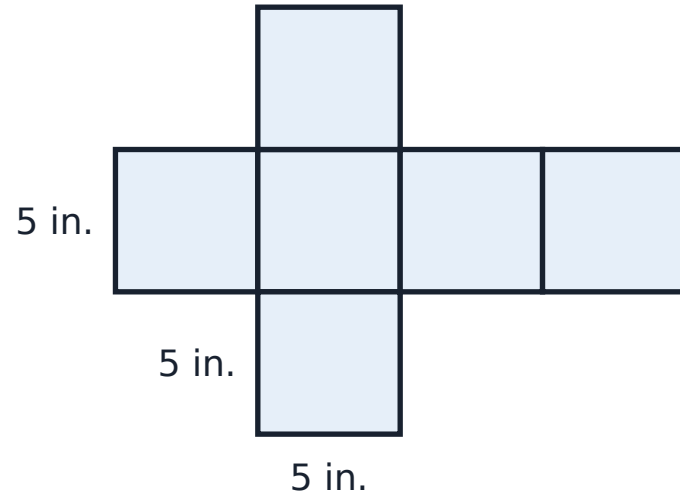
Surface area: _____



The net folds into a cube.

What is the surface area of the cube?

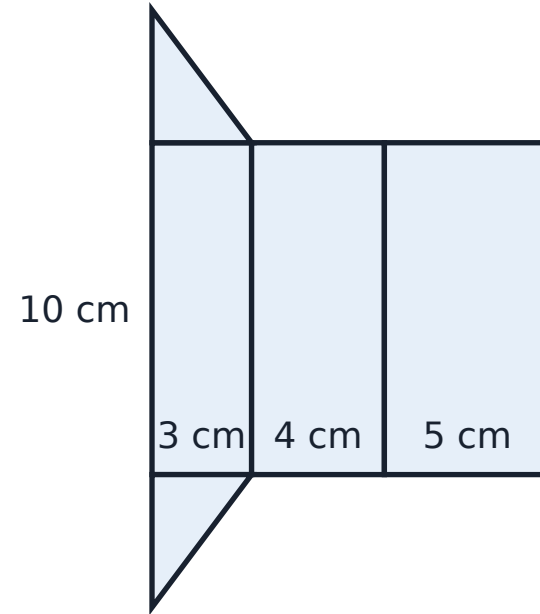
Surface area: _____



The net folds into a triangular prism.

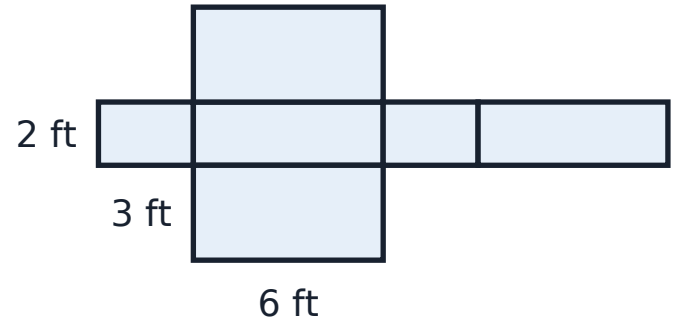
What is the surface area of the prism?

Surface area: _____



The net folds into a rectangular prism.

What is the surface area of the prism?



Surface area: _____

SET 66 • 6.G.A.4

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



A rectangle is 5 m by 7 m.

What is its area?

Area: _____

True or false?

A rectangle that is 3 units by 6 units has an area of 18 square units.

- A.** True
- B.** False

A rectangle is 6 m by 2 m.

What is its area?

- A.** 8 square meters
- B.** 12 square meters
- C.** 16 square meters
- D.** 24 square meters

A square face has 5-inch edges.

What is its area?

Area: _____

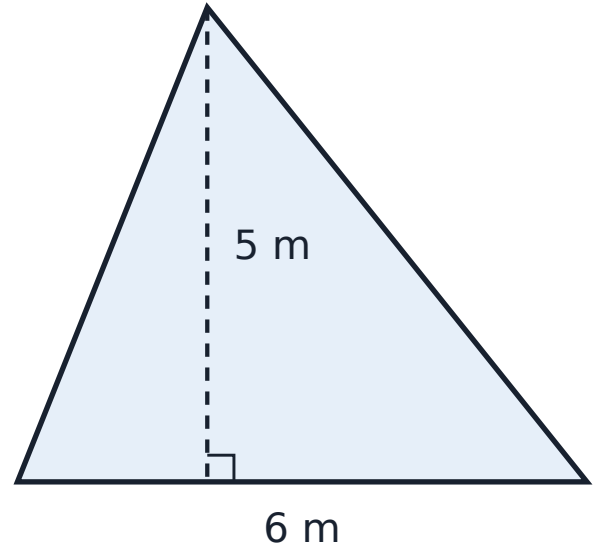
SET 66 • 6.G.A.4

BACKWARD 2

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



A triangle has a base of 8 cm and a height of 3 cm.

What is its area?

Area: _____

True or false?

A triangle with a base of 10 and a height of 4 has an area of 40 square units.

A. True

B. False

A right triangle has legs of 3 cm and 4 cm.

What is its area?

- A.** 6 square cm
- B.** 7 square cm
- C.** 10 square cm
- D.** 12 square cm

Four triangles each have a base of 6 m and a height of 5 m.

What is their total area?

Total area: _____

SET 66 • 6.G.A.4

BACKWARD 3

Adding the areas of parts • Grade 3 • Measurement & Data • 3.MD.C.7.d

A figure is made of a 4-by-3 rectangle and a 4-by-2 rectangle that do not overlap.

What is its total area?

Area: _____

True or false?

A figure made of two 5-by-5 squares has an area of 50 square units.

- A.** True
- B.** False

A figure is made of rectangles with areas of 12, 8, and 6 square units.

What is its total area?

A. 14 square units

B. 20 square units

C. 26 square units

D. 576 square units

Six squares each have an area of 9 square inches.
What is their total area?

Area: _____

A figure is made of rectangles that are 2 by 6, 2 by 6,
and 3 by 6.

What is its total area?

Area: _____

SET 66 • 6.G.A.4

FORWARD 1

Surface area of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A cube has edges of 2.5 cm.

What is its surface area?

Surface area: _____

A rectangular prism is 5 in. by 4 in. by 2.5 in.

What is its surface area?

- A.** 42.5 square inches
- B.** 50 square inches
- C.** 65 square inches
- D.** 85 square inches

A gift box is 8 in. by 6 in. by 4 in.

How much wrapping paper covers the box exactly?

Paper: _____

A box with no lid is 10 in. long, 4 in. wide, and 3 in. tall.
How much cardboard is needed to make its 5 faces?

Cardboard: _____

True or false?

The surface area of a prism is the sum of the areas of all of its faces.

A. True

B. False

SET 66 • 6.G.A.4

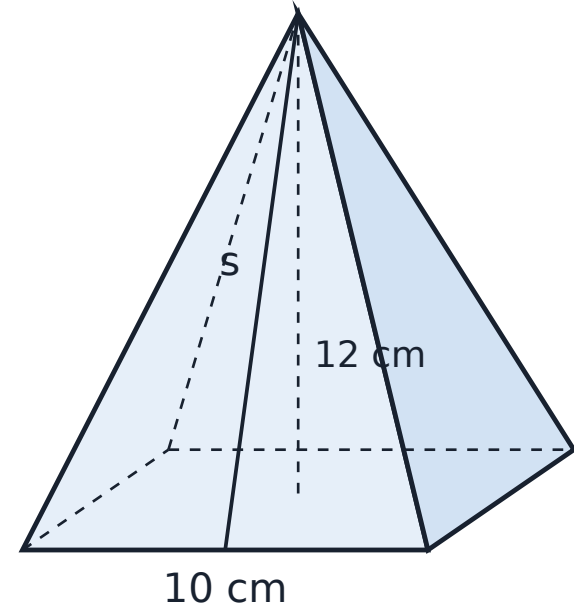
FORWARD 2

Pythagorean Theorem in three dimensions • Grade 8 • Geometry • 8.G.B.7

The square pyramid has a base edge of 10 cm and a height of 12 cm.

What is its slant height s ?

$s =$ _____



A box is 3 in. by 4 in. by 12 in.

How long is the longest diagonal from one corner to the opposite corner?

Length: _____

A square pyramid has a base edge of 16 cm and a slant height of 10 cm.

What is its height?

A. 6 cm

B. 8 cm

C. 12.8 cm

D. 18.9 cm

A cone has a radius of 5 in. and a slant height of 13 in.
What is its height?

Height: _____

True or false?

The bottom of a box is 6 in. by 8 in. The diagonal across the bottom is 10 in.

A. True

B. False

Recognize statistical questions

MAIN Grade 6 main question

BACKWARD 1 Reading a bar graph

BACKWARD 2 Reading a line plot

FORWARD 1 Random samples and representative samples

FORWARD 2 Statistical questions about two variables and scatter plots

6.SP.A.1

3.MD.B.3

4.MD.B.4

7.SP.A.1

8.SP.A.1

SET 67 • 6.SP.A.1

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.1

Which question is a statistical question?

- A.** How many hours did I sleep last night?
- B.** How many days are in a week?
- C.** What time does our school start?
- D.** How many hours of sleep do students in my class get each night?

True or false?

"How tall is the flagpole at our school?" is a statistical question.

A. True

B. False

Which question is NOT a statistical question?

- A.** How much do the dogs at the shelter weigh?
- B.** What is the population of our city right now?
- C.** How long do students spend on homework each day?
- D.** How many pets do the students in grade 6 have?

True or false?

"What are the ages of the teachers at our school?" is a statistical question.

A. True

B. False

Which question is a statistical question?

- A.** How many minutes do the players on our team practice each week?
- B.** How many minutes are in an hour?
- C.** How many minutes did Coach Lee practice today?
- D.** How many players are on a soccer field at once?

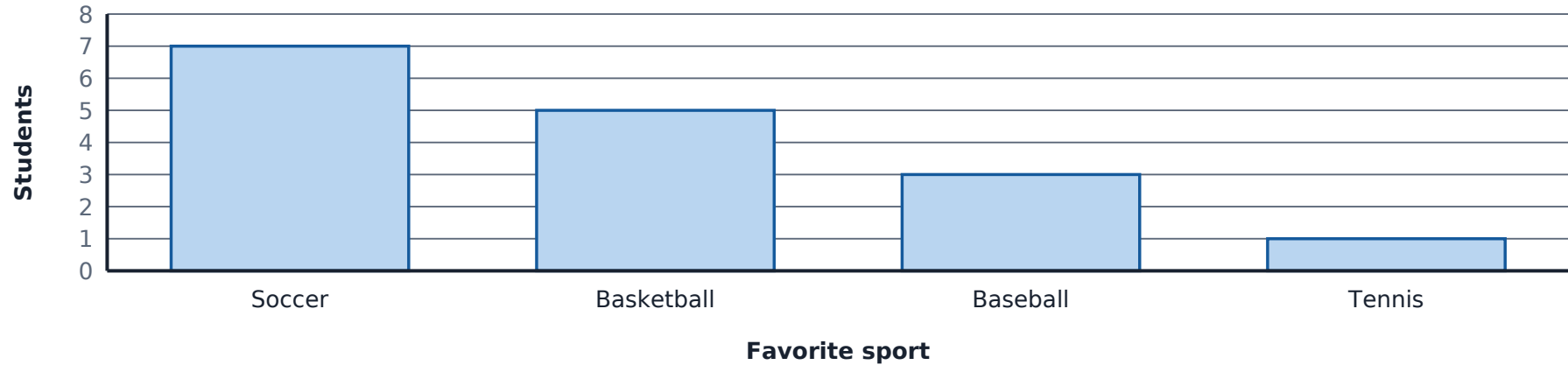
SET 67 • 6.SP.A.1

BACKWARD 1

Reading a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

The graph shows the favorite sports of a class.

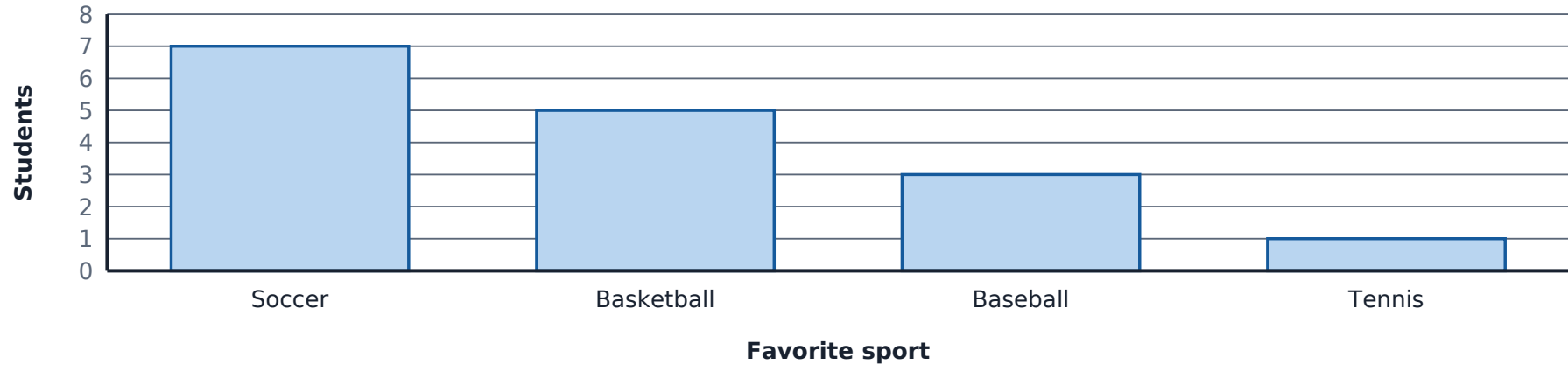
How many students chose basketball?



Students: _____

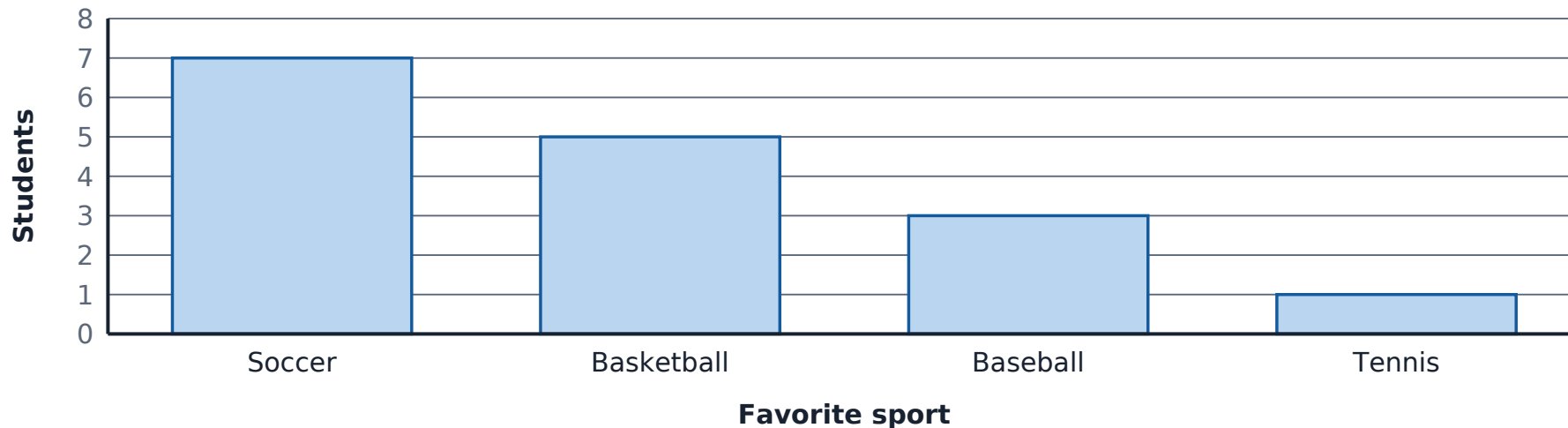
The graph shows the favorite sports of a class.

How many more students chose soccer than baseball?



Students: _____

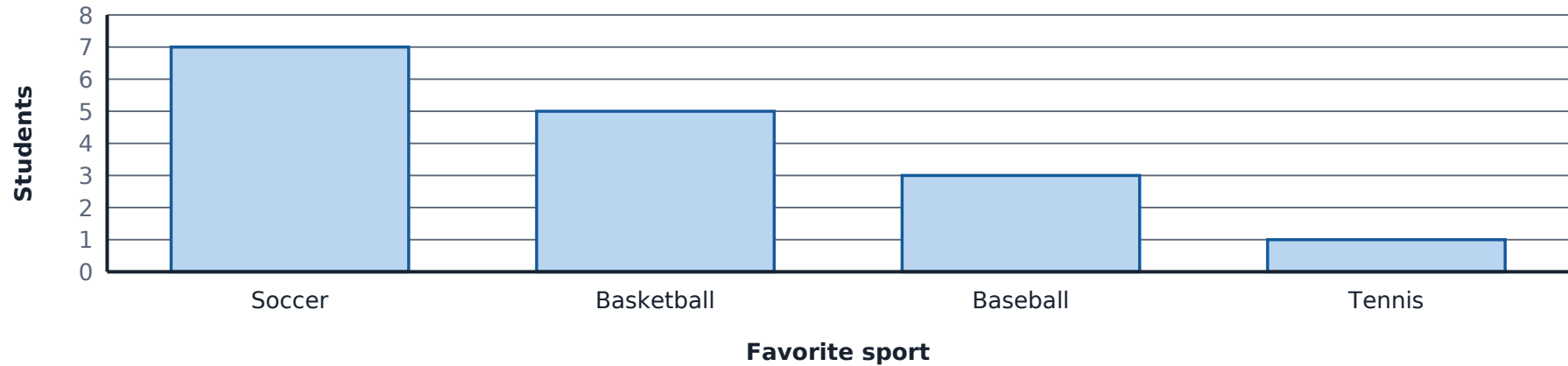
Which sport did 3 students choose?



- A.** Baseball **B.** Tennis **C.** Soccer **D.** Basketball

The graph shows the favorite sports of a class.

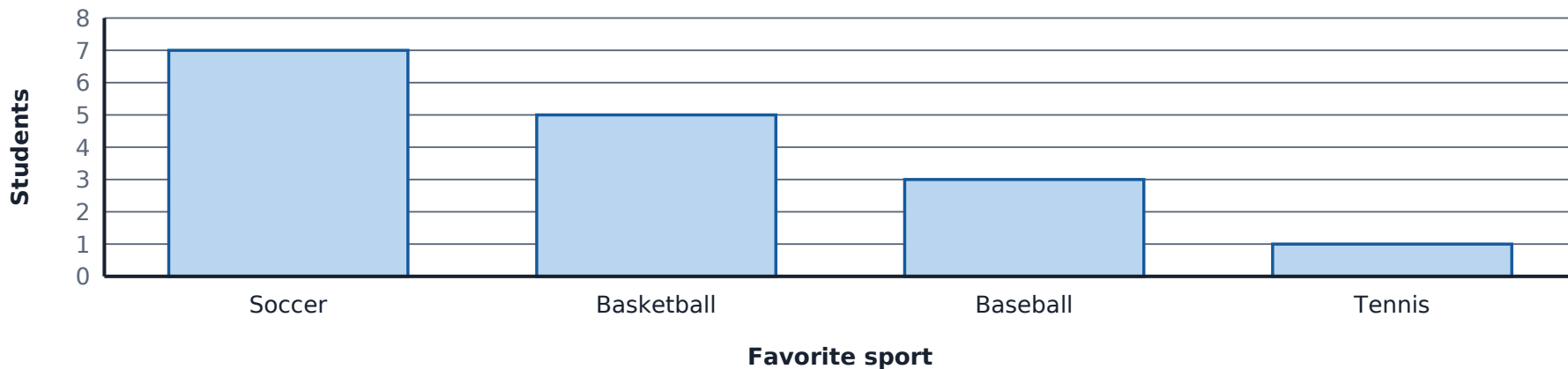
How many students answered in all?



Students: _____

True or false?

More students chose basketball than soccer.



A. True

B. False

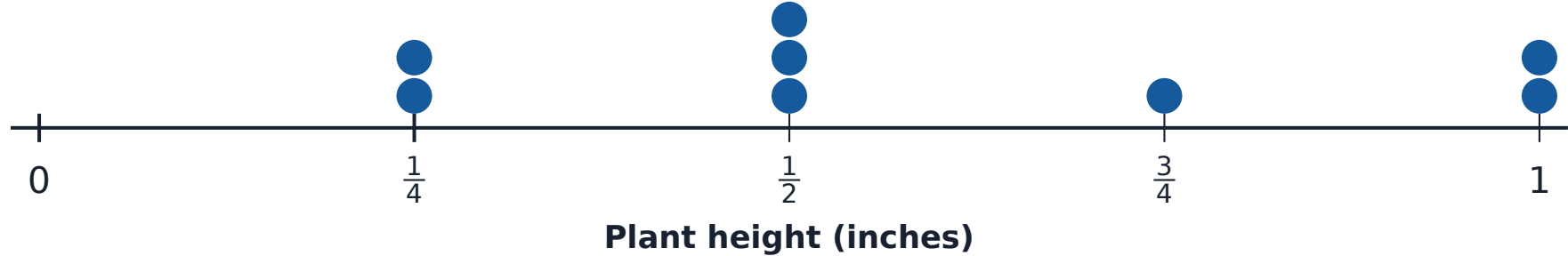
SET 67 • 6.SP.A.1

BACKWARD 2

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows the heights of plants.

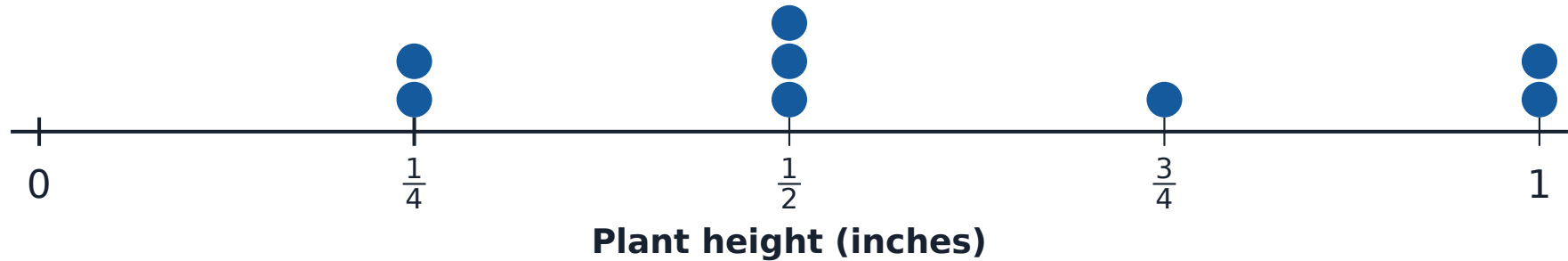
How many plants are $\frac{1}{2}$ inch tall?



Plants: _____

The line plot shows the heights of plants.

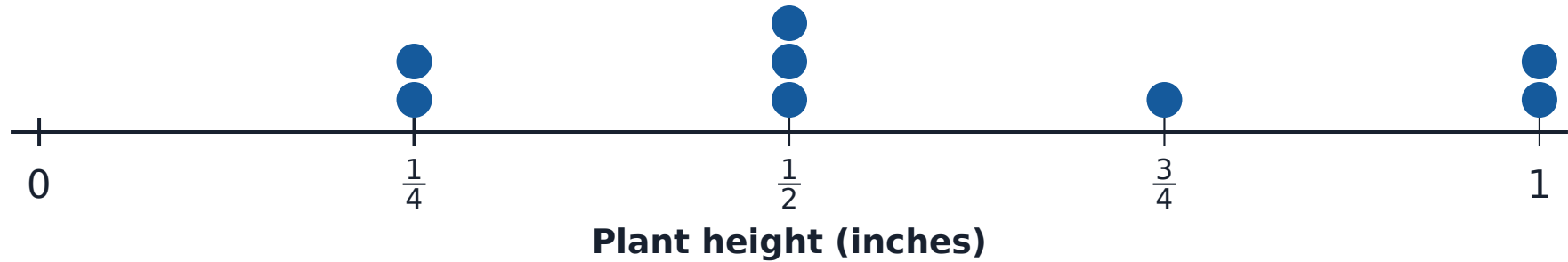
How many plants were measured?



Plants: _____

True or false?

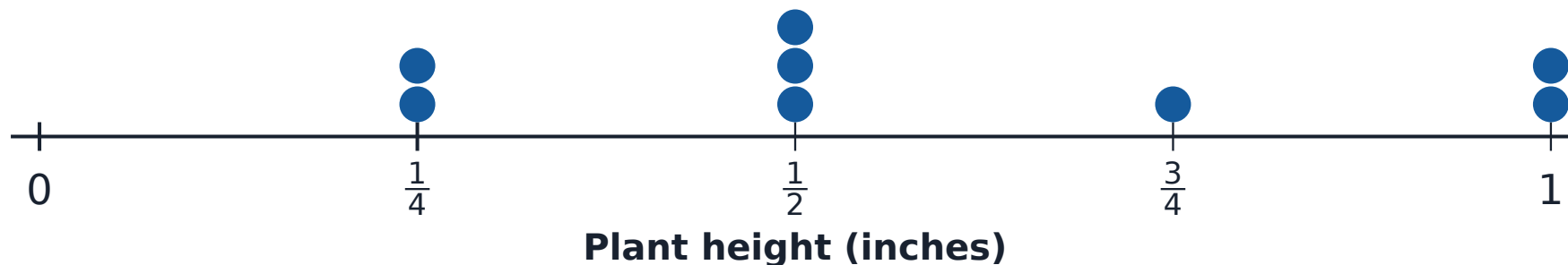
The most common plant height is $\frac{3}{4}$ inch.



A. True

B. False

Which height appears least often?



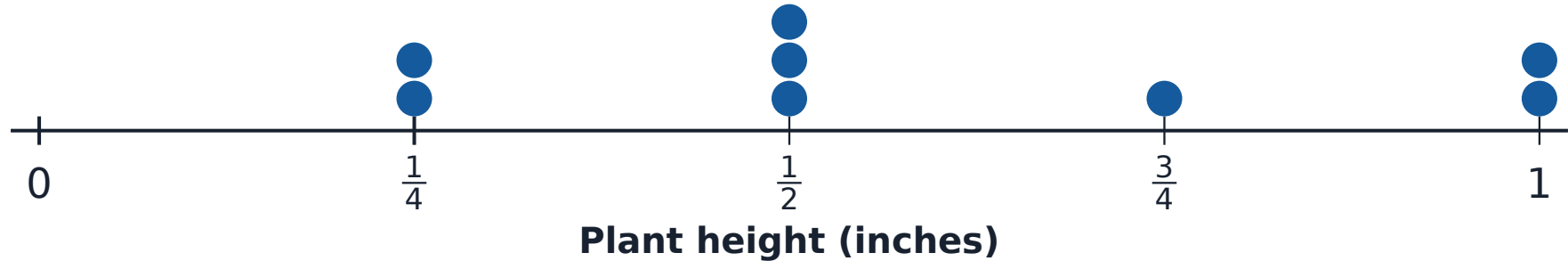
A. $\frac{1}{4}$ inch

B. $\frac{1}{2}$ inch

C. $\frac{3}{4}$ inch

D. 1 inch

What is the difference between the tallest and the shortest plant heights?



Difference: _____

SET 67 • 6.SP.A.1

FORWARD 1

Random samples and representative samples • Grade 7 • Statistics & Probability • 7.SP.A.1

A principal wants to know the favorite lunch of all 600 students.

Which sample is most likely to represent all students?

- A.** The students on the soccer team
- B.** 50 students chosen at random from the whole school
- C.** The 30 students in one class
- D.** The first 50 students in line on pizza day

True or false?

A random sample is more likely to represent a population than a sample of your friends.

A. True

B. False

A reporter asks only basketball players, "What is your favorite sport?"

Why is this sample biased?

Reason: _____

A random sample of 40 students out of 800 shows that 10 walk to school.

About how many of the 800 students walk to school?

A. 10

B. 40

C. 80

D. 200

True or false?

Conclusions about a population from a sample are valid only if the sample represents the population.

- A. True
- B. False

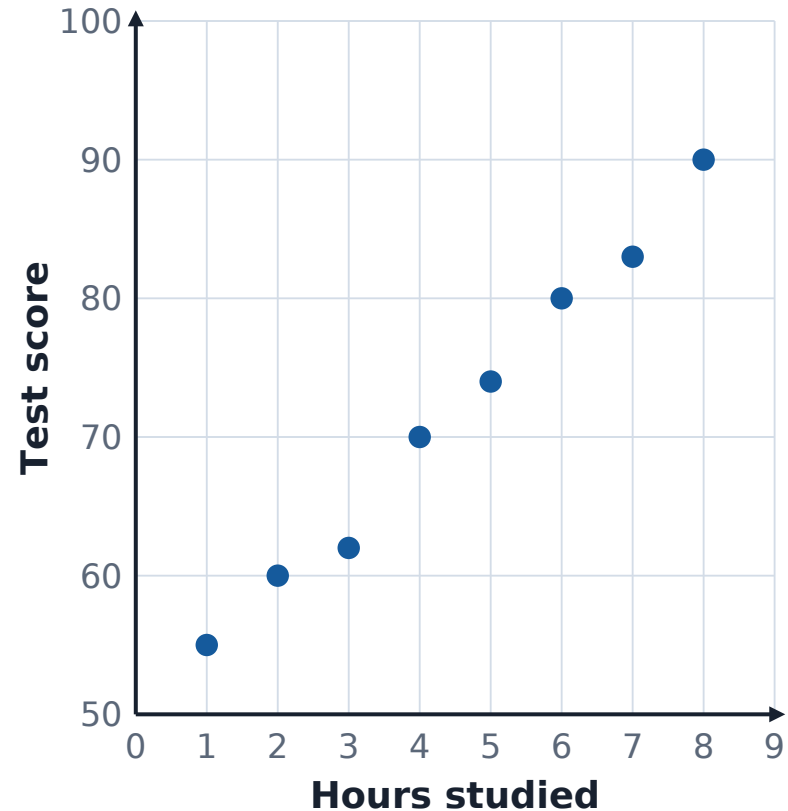
SET 67 • 6.SP.A.1

FORWARD 2

Statistical questions about two variables and scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

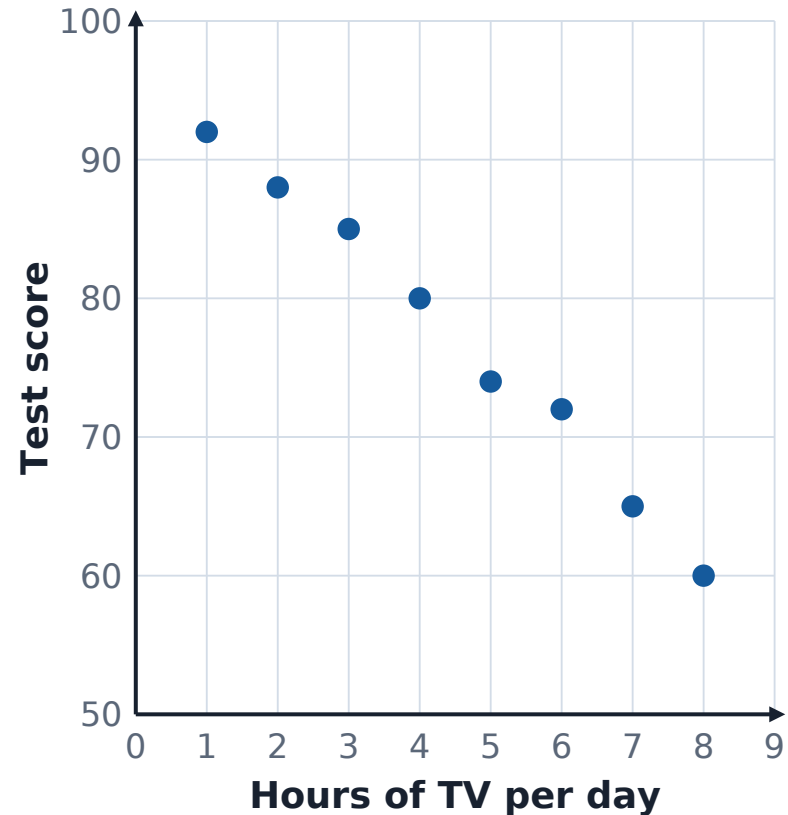
What type of association does the scatter plot show?

- A.** No association
- B.** Nonlinear
- C.** Positive linear
- D.** Negative linear



What type of association does the scatter plot show?

- A.** Negative linear
- B.** Positive linear
- C.** No association
- D.** Nonlinear



True or false?

A scatter plot shows data about two variables for each person or object.

- A.** True
- B.** False

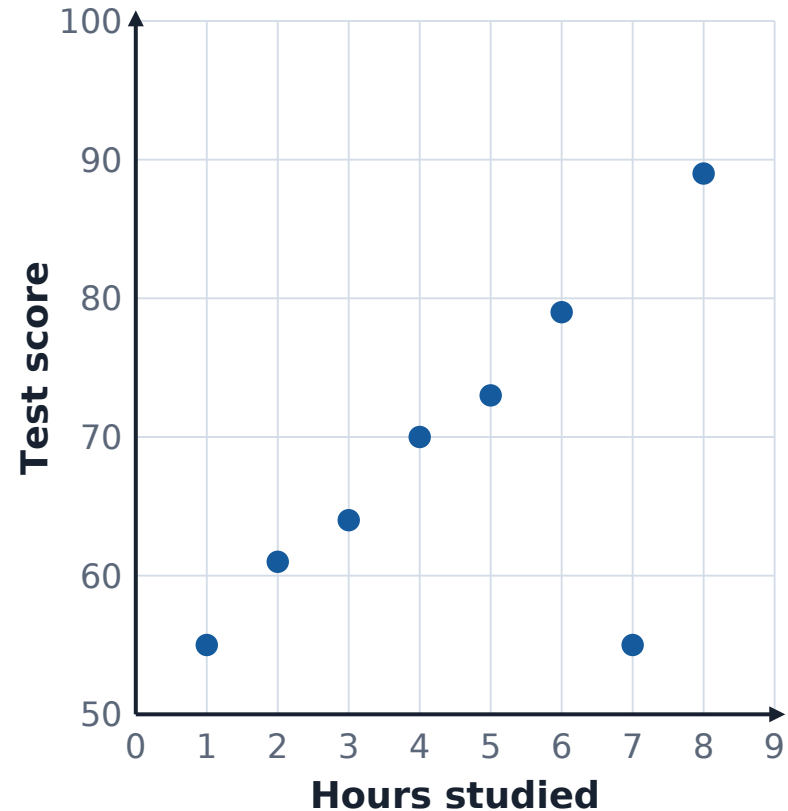
Which question would be best answered with a scatter plot?

- A.** How many students ride the bus?
- B.** What is the most common eye color?
- C.** Is there a relationship between height and arm span?
- D.** What is the median age in the class?

Which point is an outlier?

Write its coordinates.

Outlier: _____



Describe the overall shape of a distribution

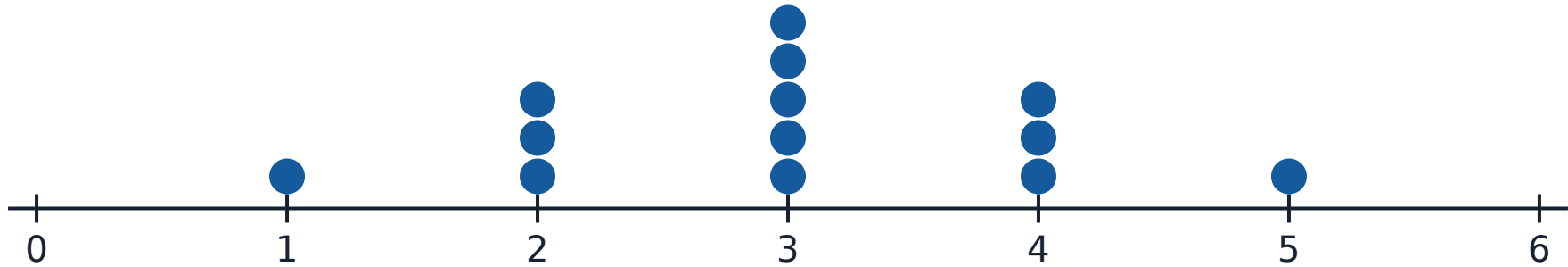
MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Most and least in a bar graph	3.MD.B.3
FORWARD 1	Compare two distributions visually	7.SP.B.3
FORWARD 2	Linear and nonlinear patterns in scatter plots	8.SP.A.1

SET 68 • 6.SP.A.2

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

Which word best describes the shape of the data?



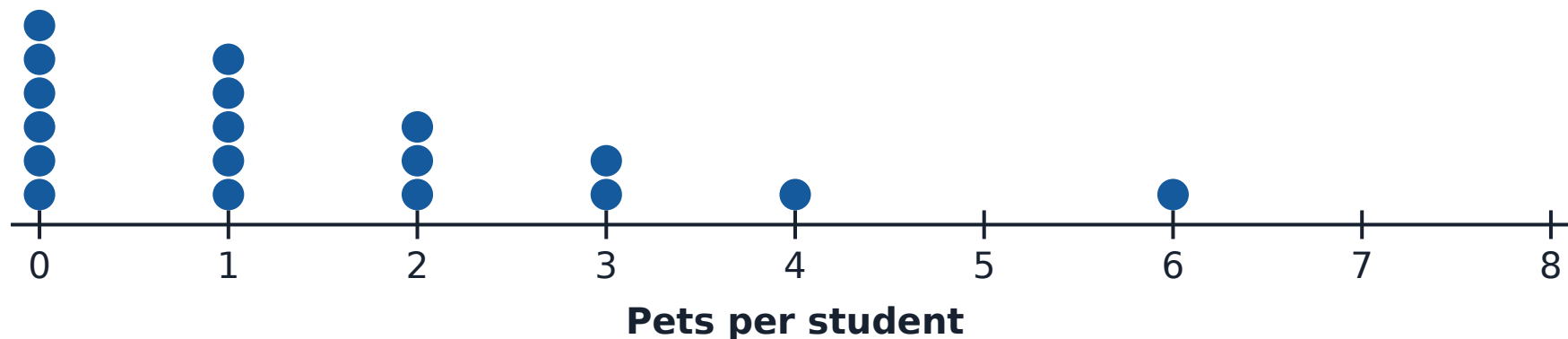
A. Uniform

B. Symmetric

C. Skewed left

D. Skewed right

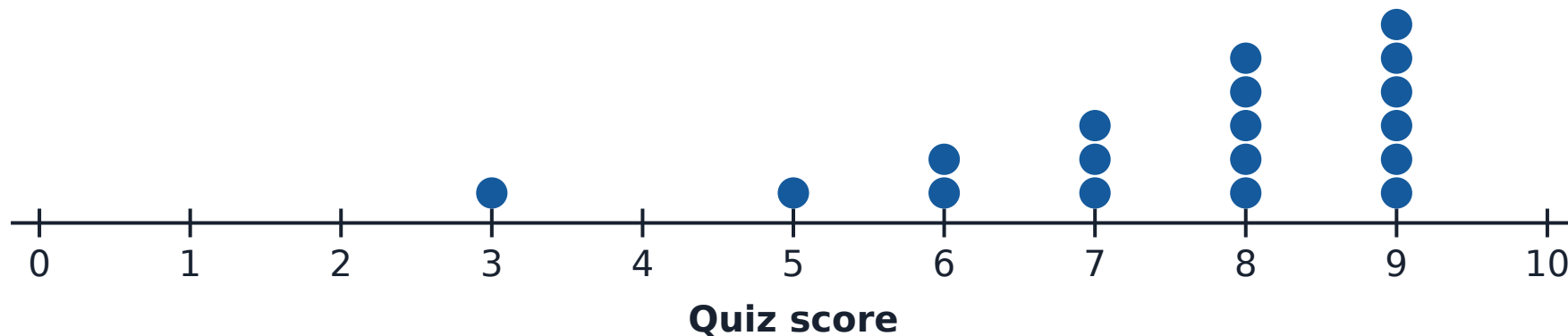
Which word best describes the shape of the data?



- A.** Skewed left
- C.** Skewed right

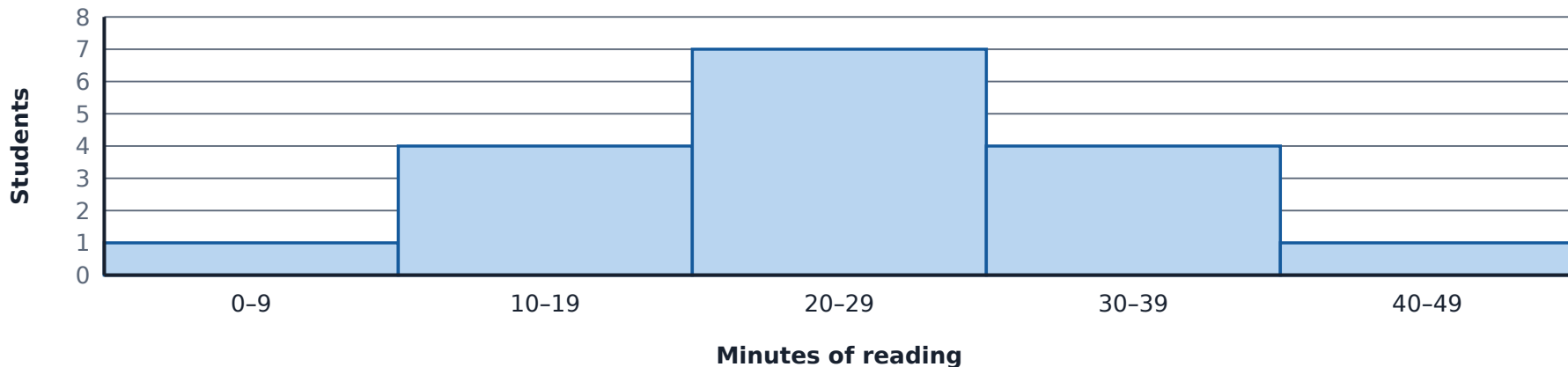
- B.** Symmetric
- D.** Uniform

Which word best describes the shape of the data?



- A.** Symmetric
- B.** Skewed left
- C.** Skewed right
- D.** Uniform

Which word best describes the shape of the data in the histogram?



A. Skewed right

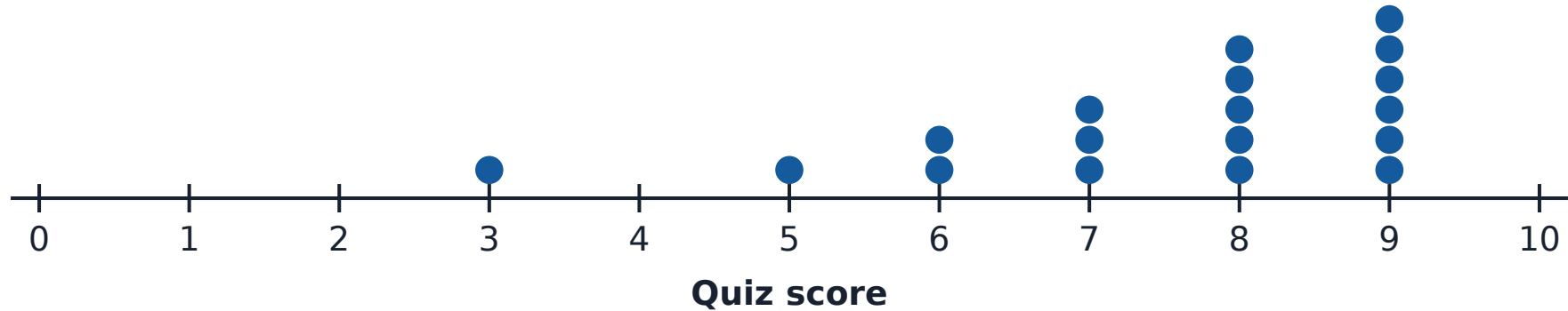
B. Skewed left

C. Uniform

D. Symmetric

True or false?

The data are skewed right.



A. True

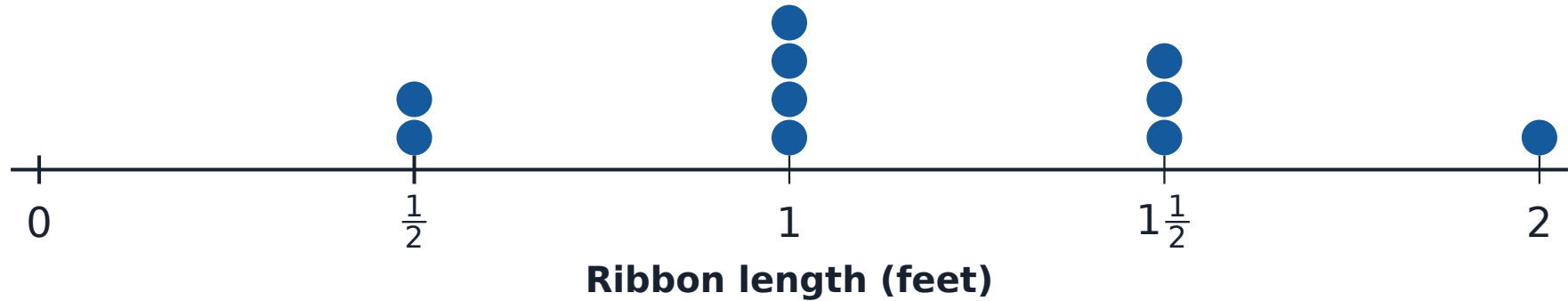
B. False

SET 68 • 6.SP.A.2

BACKWARD 1

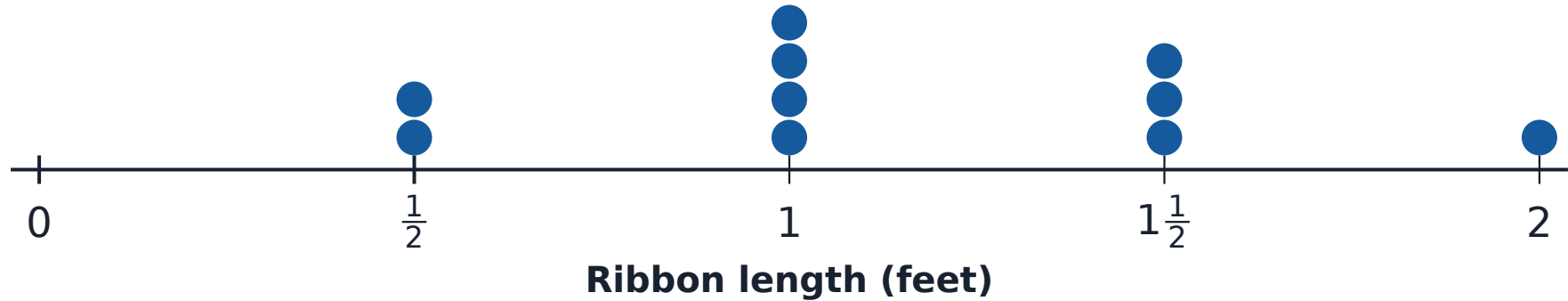
Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many ribbons are 1 foot long?



Ribbons: _____

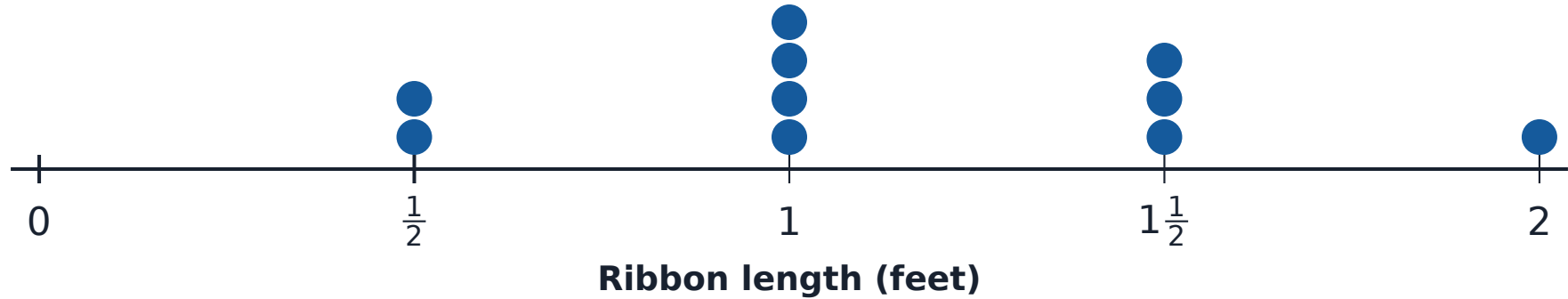
How many ribbons are there in all?



Ribbons: _____

True or false?

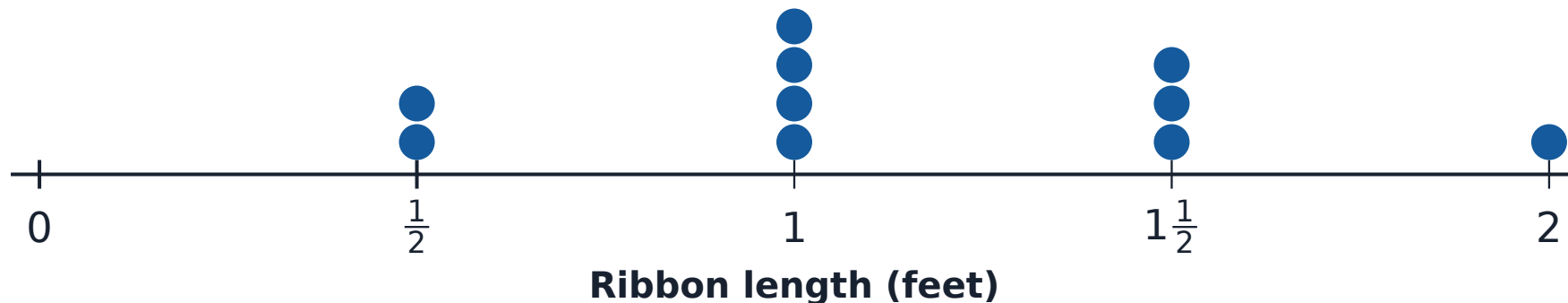
Two ribbons are $\frac{1}{2}$ foot long.



A. True

B. False

Which ribbon length is most common?



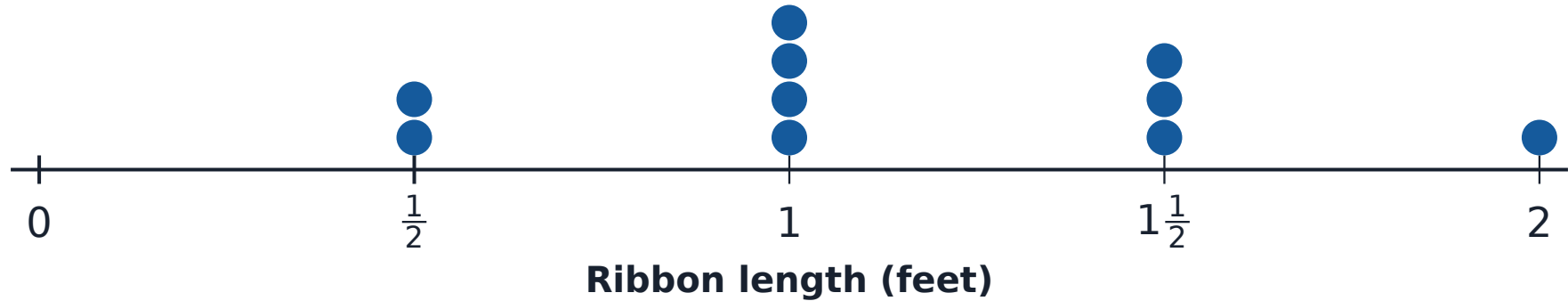
A. $\frac{1}{2}$ foot

B. 1 foot

C. $1\frac{1}{2}$ feet

D. 2 feet

How many ribbons are longer than 1 foot?



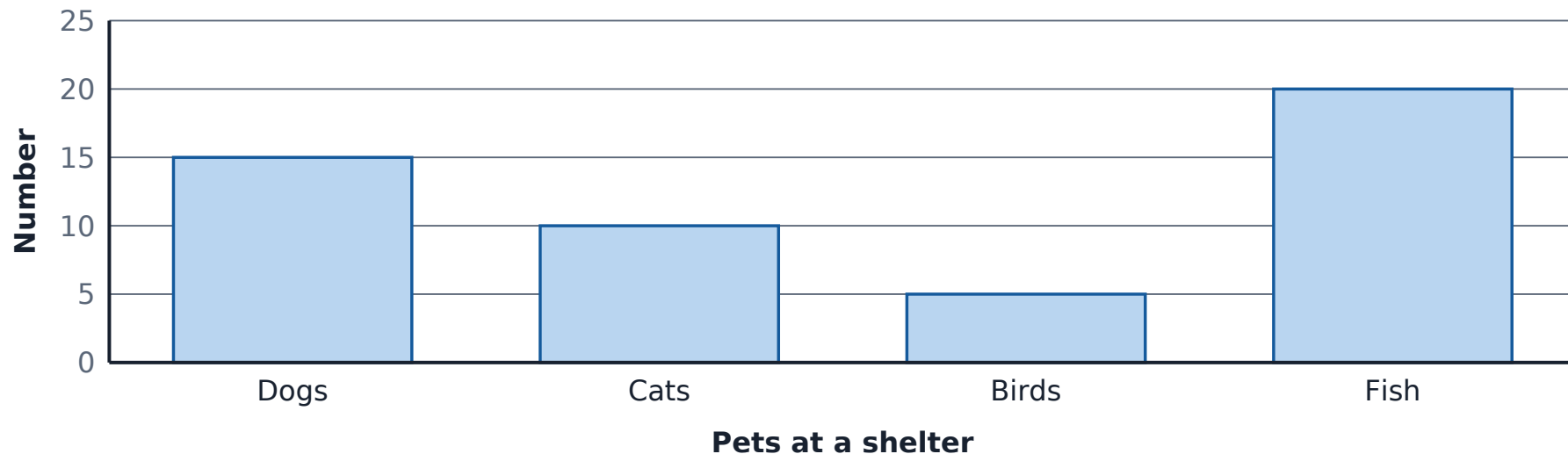
Ribbons: _____

SET 68 • 6.SP.A.2

BACKWARD 2

Most and least in a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

Which pet does the shelter have the most of?



A. Birds

B. Fish

C. Dogs

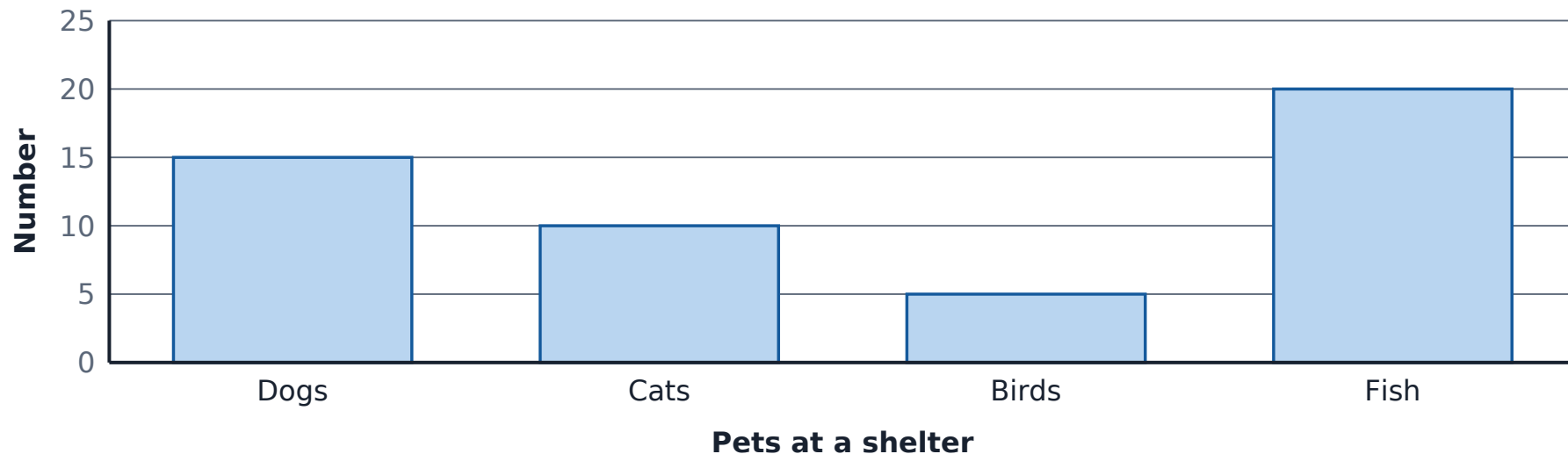
D. Cats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q1/5

Which pet does the shelter have the fewest of?



A. Dogs

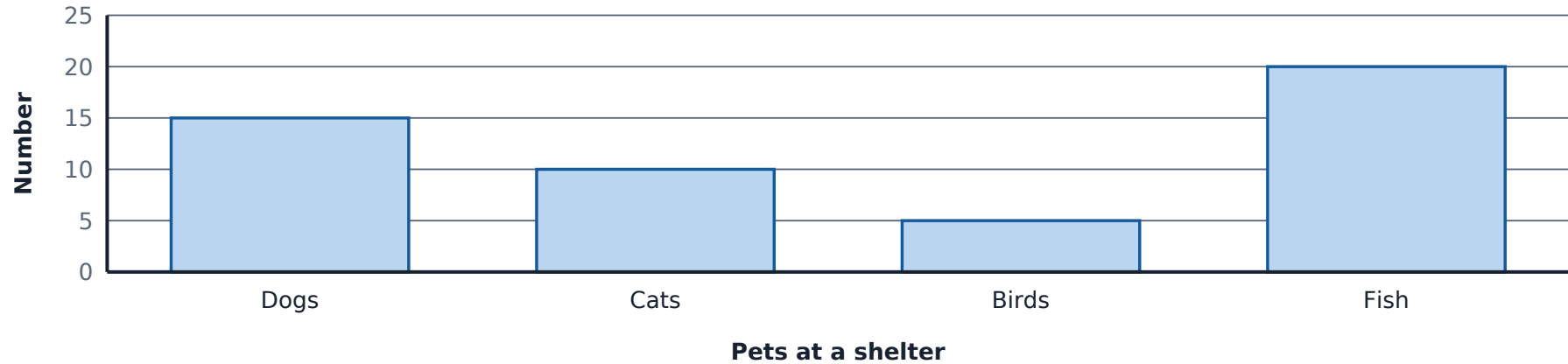
B. Birds

C. Cats

D. Fish

True or false?

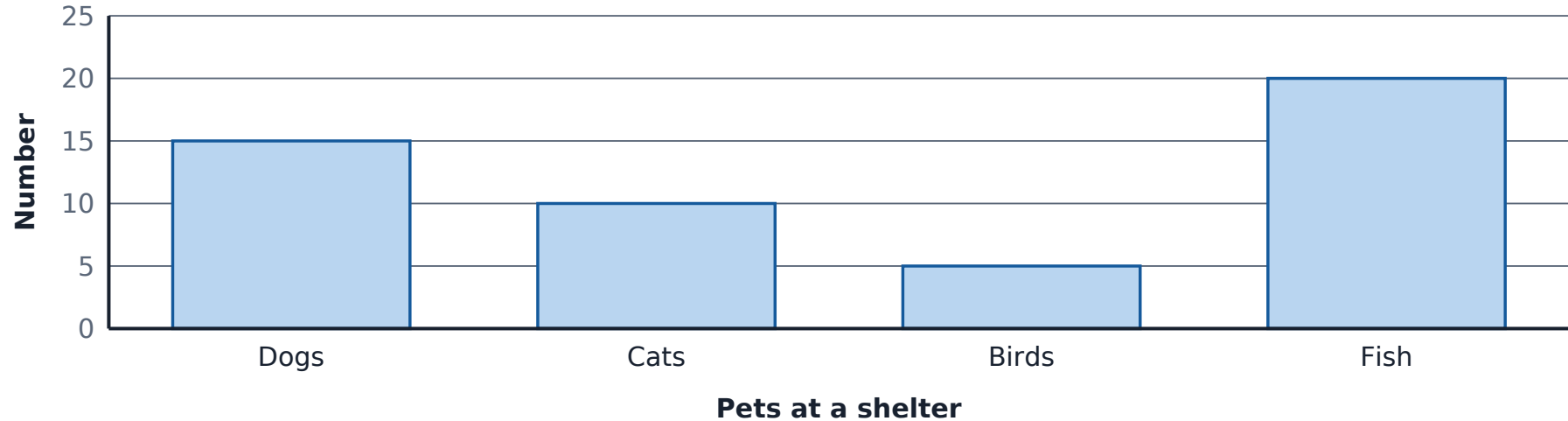
The shelter has more dogs than cats.



A. True

B. False

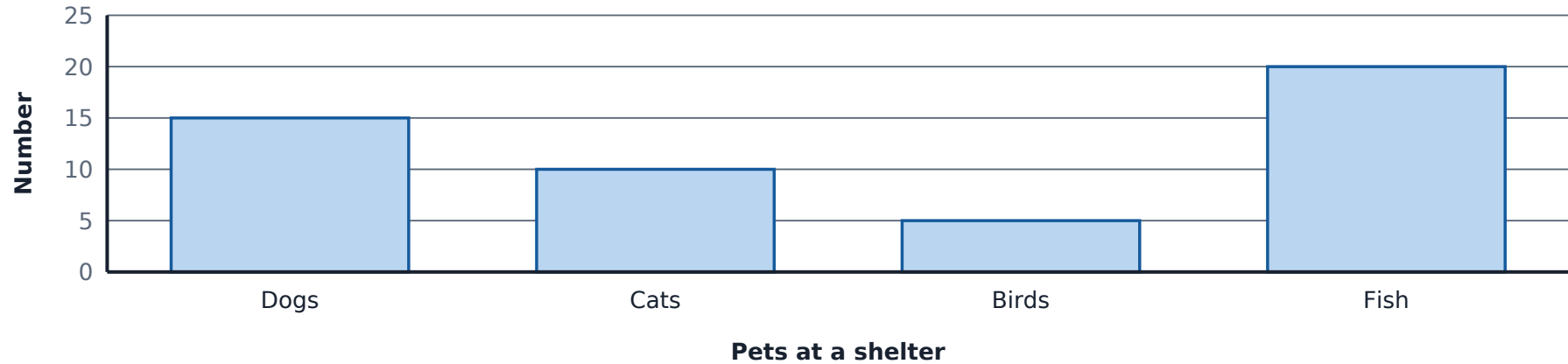
How many more fish than birds are at the shelter?



Number: _____

True or false?

The tallest bar shows the smallest number.



A. True

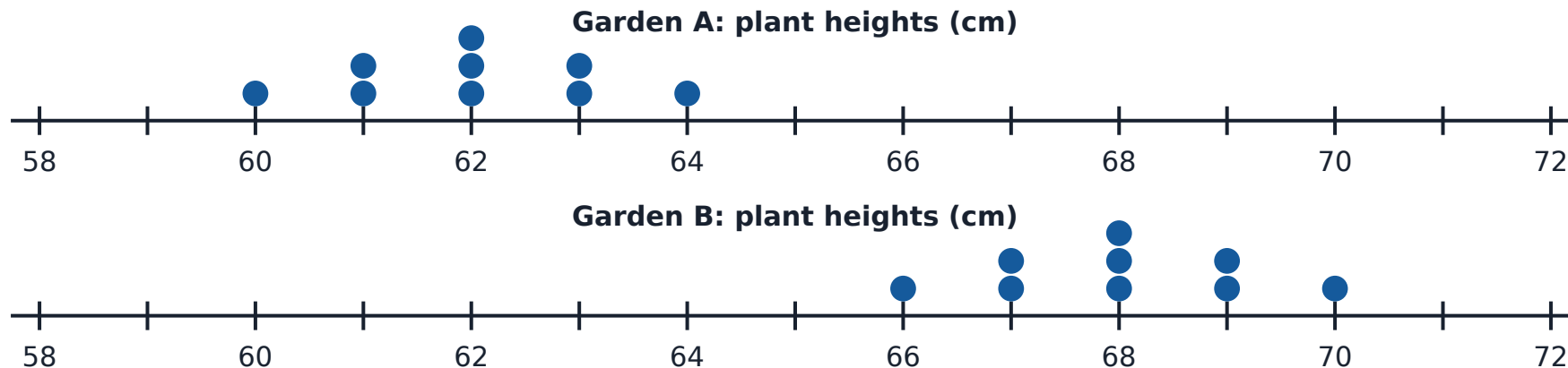
B. False

SET 68 • 6.SP.A.2

FORWARD 1

Compare two distributions visually • Grade 7 • Statistics & Probability •
7.SP.B.3

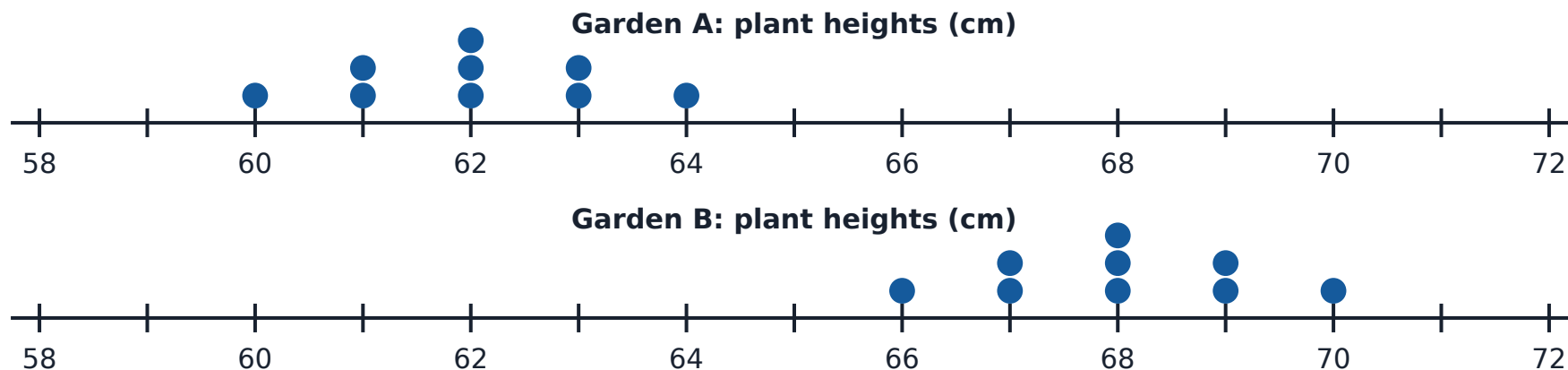
About how many centimeters greater is the center of Garden B's plant heights?



Centimeters: _____

True or false?

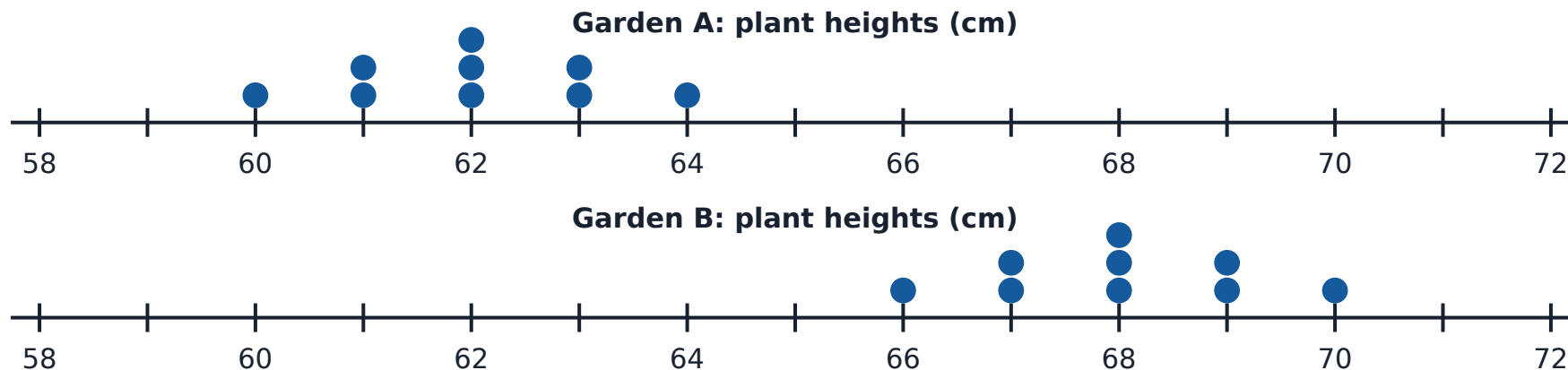
The two distributions have about the same shape and spread.



A. True

B. False

Which statement is true?



- A.** Garden B has more variability. **B.** Garden A's plants are typically taller.
- C.** The distributions do not overlap. **D.** The centers are equal.

Garden A's mean height is 62 cm and Garden B's is 68 cm. Each has a MAD of about 1 cm.

The difference in means is about how many times the MAD?

Times: _____

Two classes have mean scores of 78 and 84. Each has a MAD of 6.

The difference in means is how many times the MAD?

A. 0.5

B. 1

C. 2

D. 6

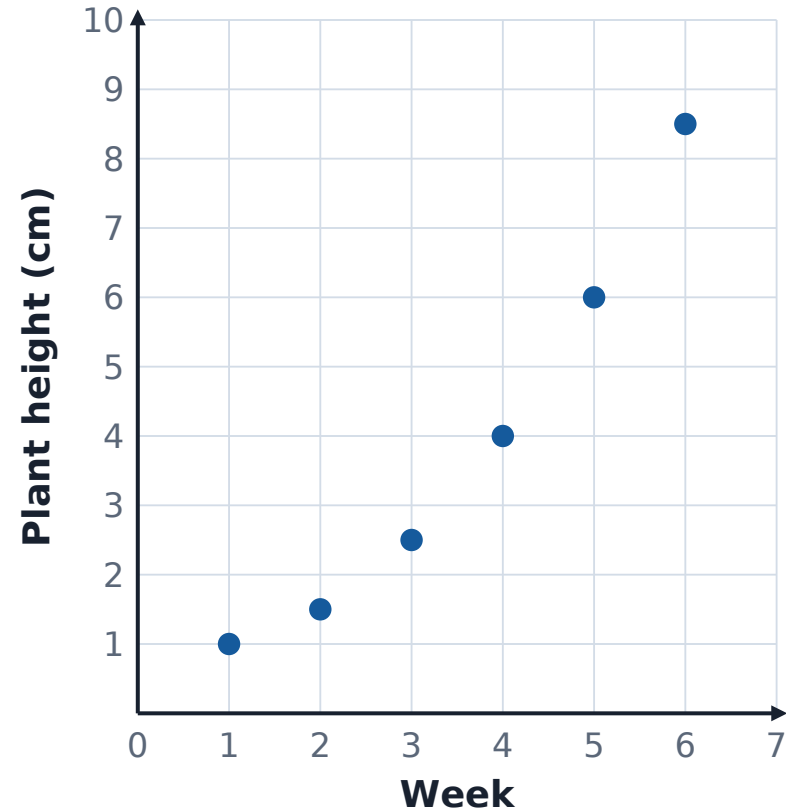
SET 68 • 6.SP.A.2

FORWARD 2

Linear and nonlinear patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

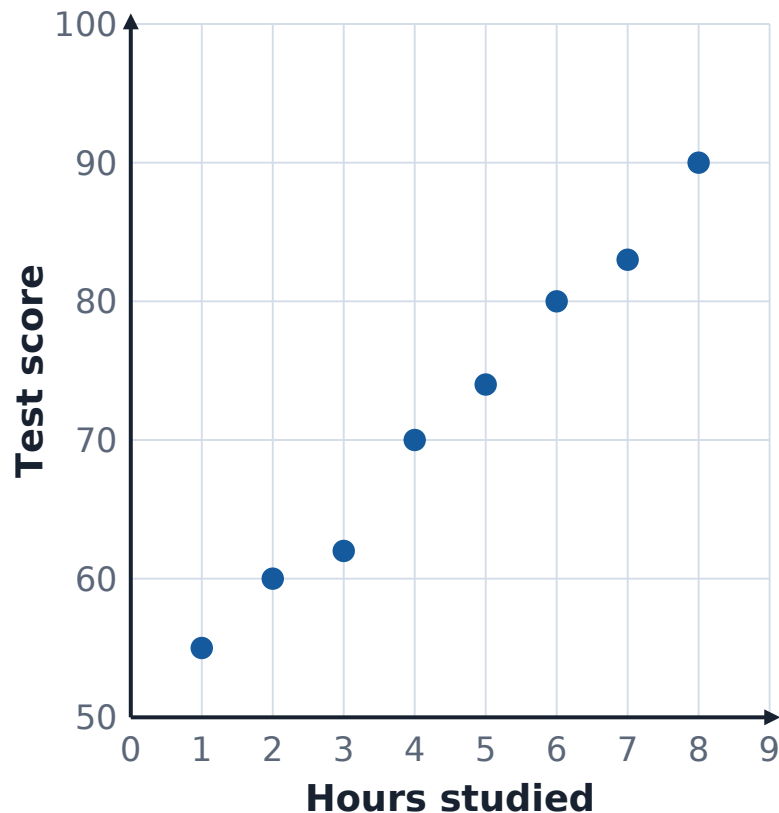
Which describes the pattern in the scatter plot?

- A.** Nonlinear
- B.** No association
- C.** Negative linear
- D.** Constant



Is the pattern in the scatter plot linear or nonlinear?

- A.** Linear
- B.** Nonlinear

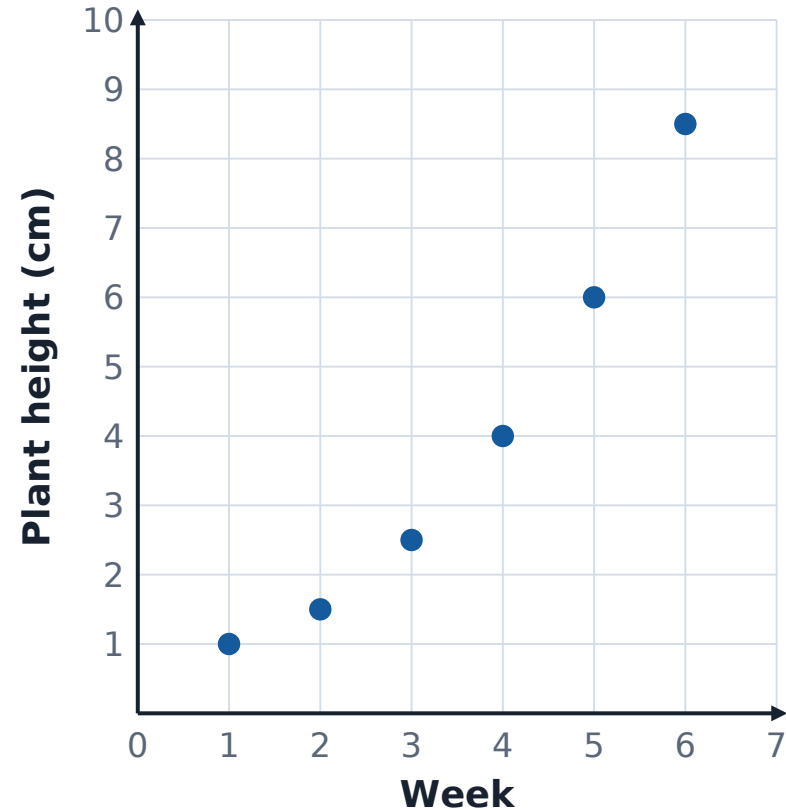


True or false?

The pattern in the scatter plot is linear.

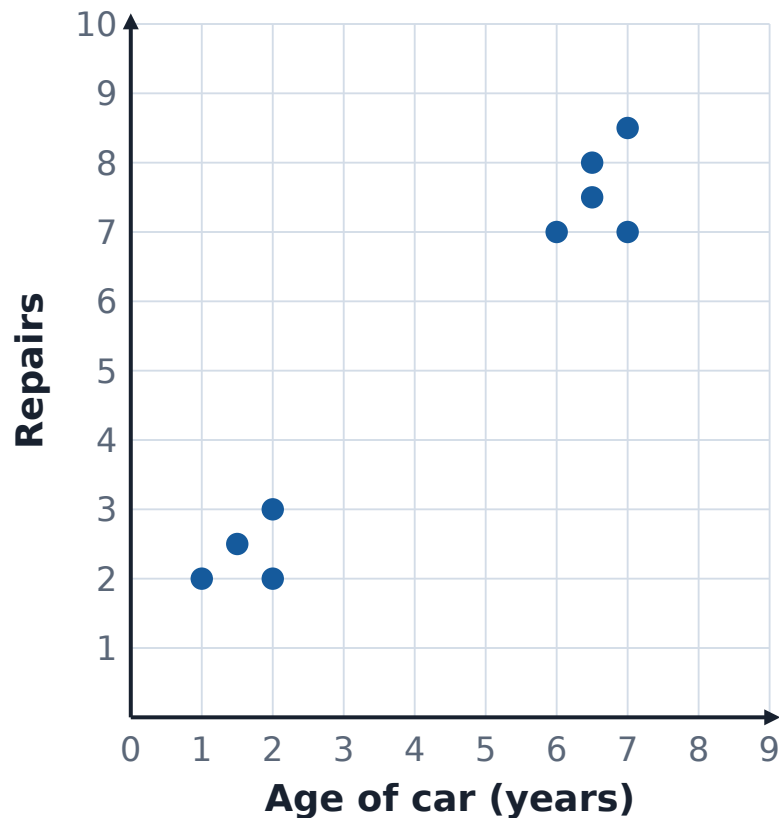
A. True

B. False



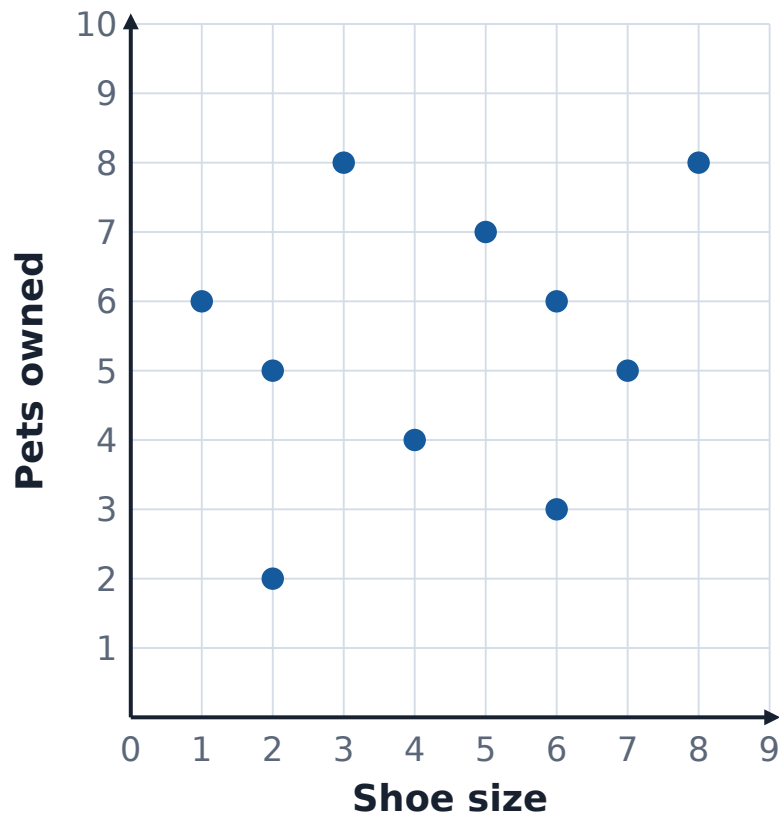
How many clusters does the scatter plot show?

Clusters: _____



Which describes the scatter plot?

- A.** Positive linear
- B.** Nonlinear
- C.** Negative linear
- D.** No association



Describe the center of a distribution

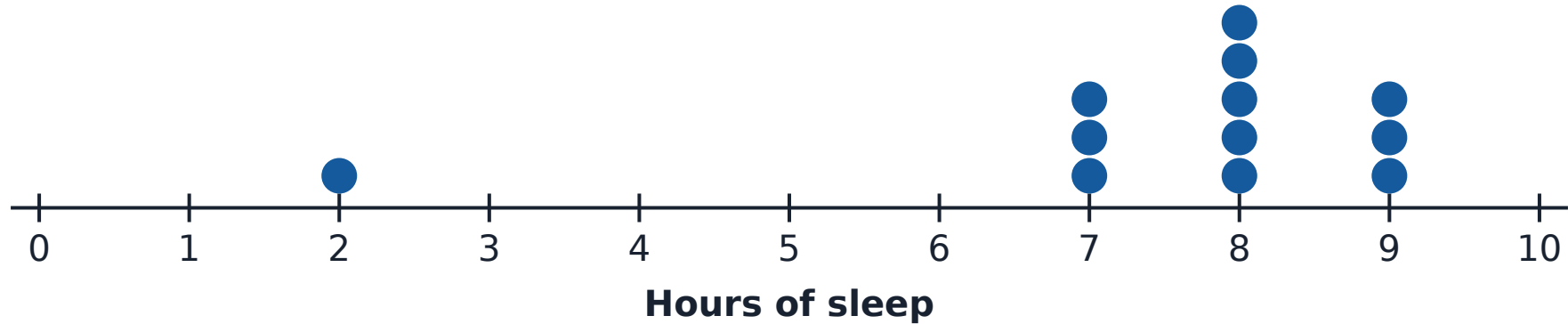
MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Ordering numbers	2.NBT.A.4
BACKWARD 3	Halfway between two numbers	2.MD.B.6
FORWARD 1	Compare the centers of two populations	7.SP.B.4
FORWARD 2	Use a line of best fit to make predictions — nearest related (no di...	8.SP.A.2

SET 69 • 6.SP.A.2

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

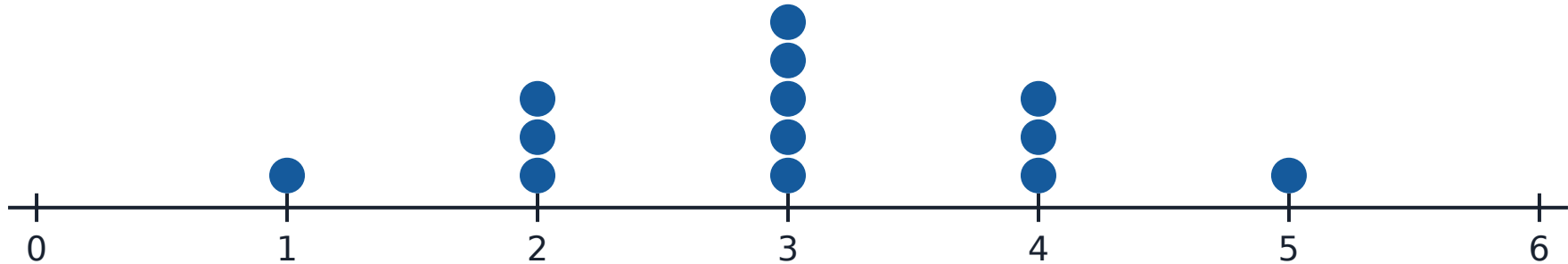
Around what value do most of the data cluster?



Value: _____

The data are symmetric.

What value is at the center of the distribution?



Value: _____

In which interval is the center of the data?



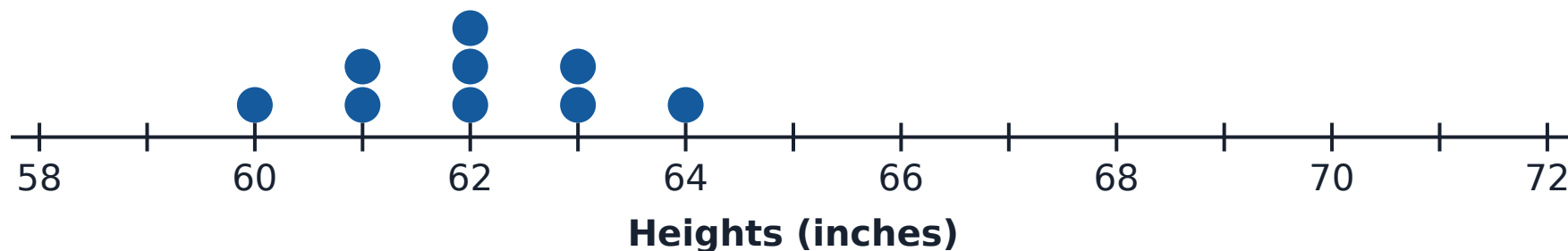
- A.** 0-9 **B.** 10-19 **C.** 20-29 **D.** 30-39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q3/5

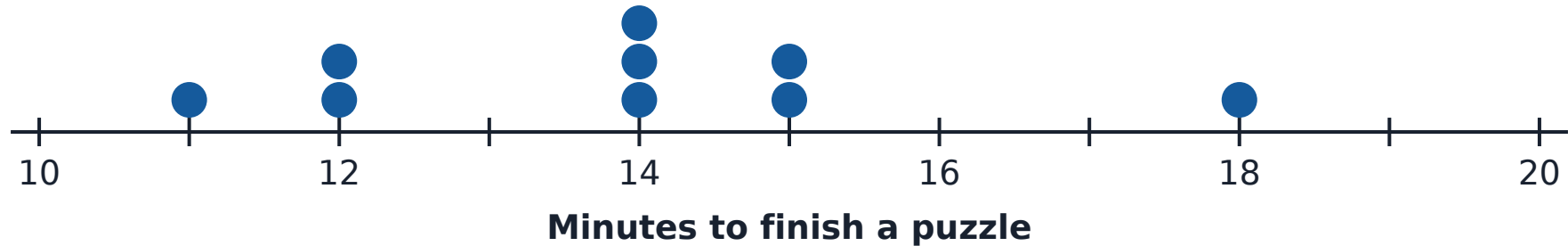
What value is at the center of the heights?



Value: _____

True or false?

The center of the distribution is about 12.



A. True

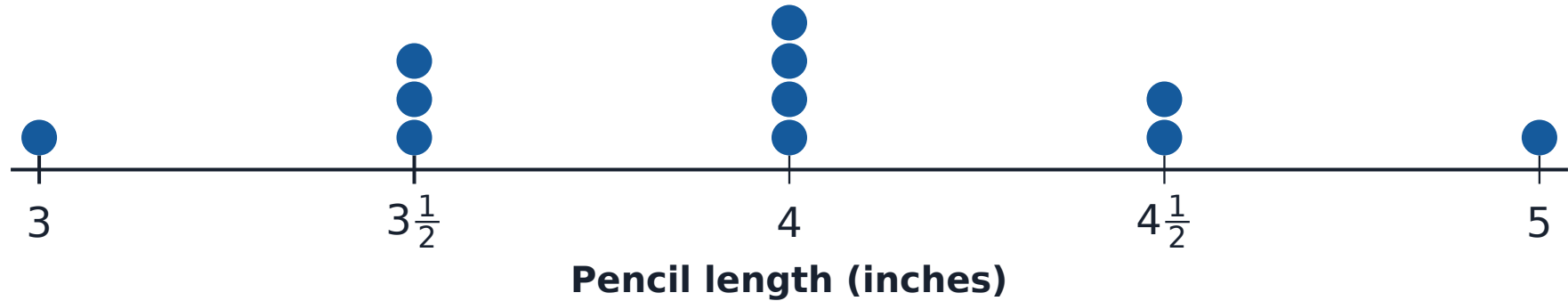
B. False

SET 69 • 6.SP.A.2

BACKWARD 1

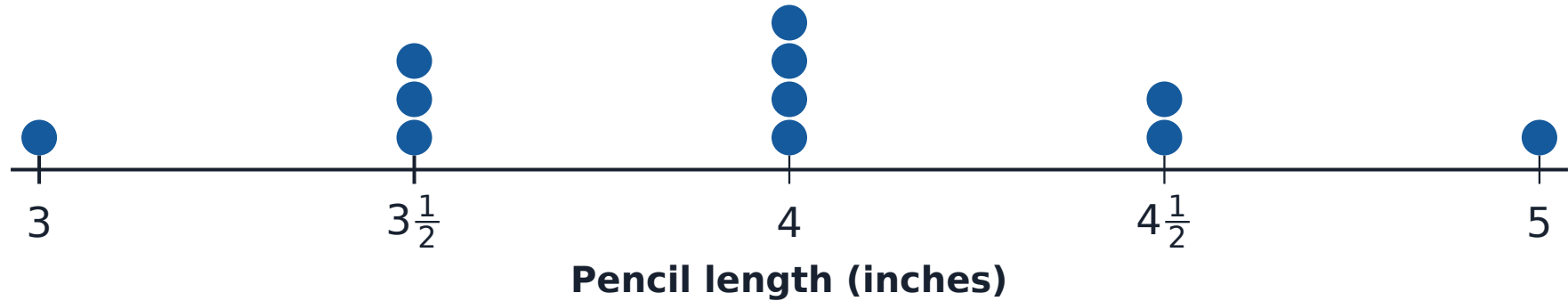
Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many pencils are 4 inches long?



Pencils: _____

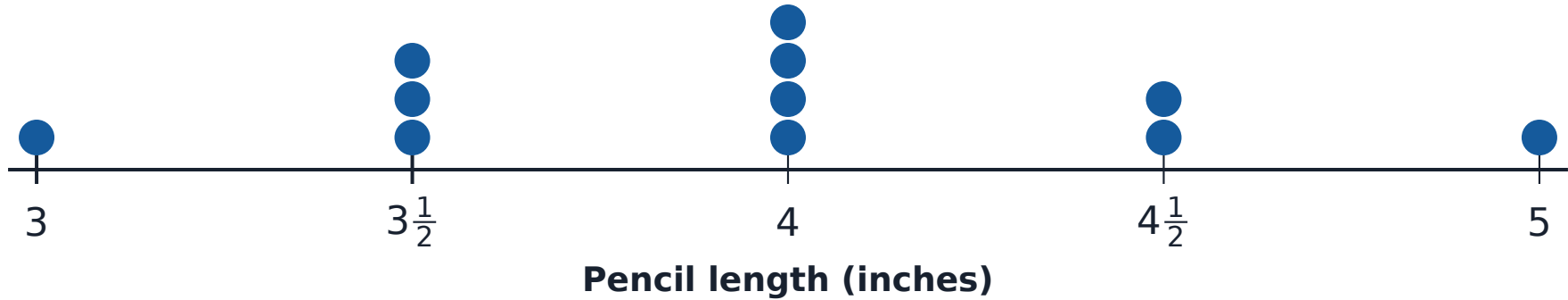
How many pencils were measured?



Pencils: _____

True or false?

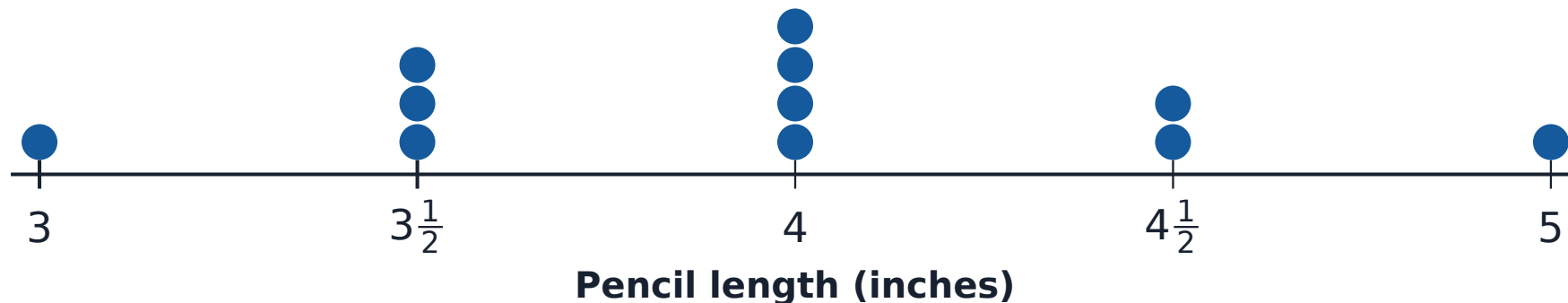
Three pencils are $3\frac{1}{2}$ inches long.



A. True

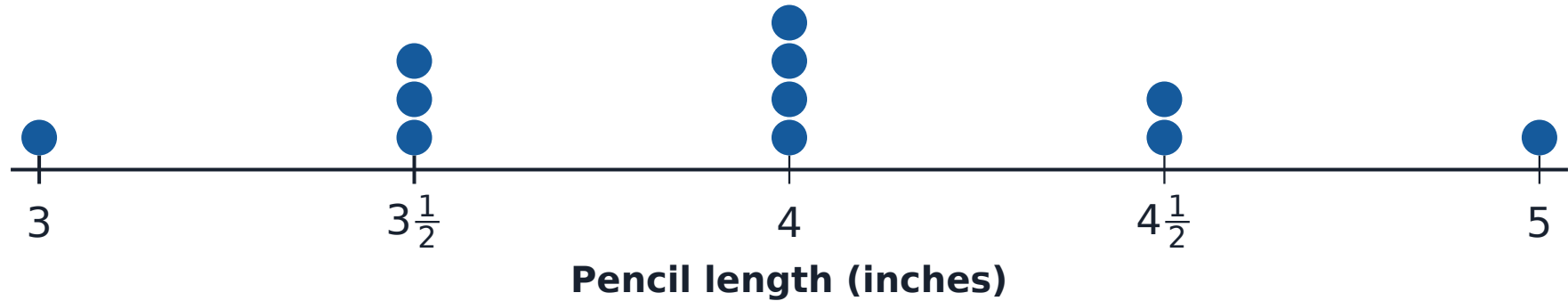
B. False

Which pencil length is most common?



- A.** 3 inches **B.** $3\frac{1}{2}$ inches **C.** 4 inches **D.** 5 inches

How many pencils are shorter than 4 inches?



Pencils: _____

SET 69 • 6.SP.A.2

BACKWARD 2

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

42, 24, 36, 18

Order: _____

Which number is greatest?

- A.** 53
- B.** 299
- C.** 305
- D.** 350

True or false?

These numbers are in order from least to greatest: 12, 21, 19, 30

- A. True
- B. False

Order from least to greatest.

105, 150, 115

Order: _____

Which number is least?

- A.** 89
- B.** 91
- C.** 98
- D.** 100

SET 69 • 6.SP.A.2

BACKWARD 3

Halfway between two numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What number is halfway between 10 and 20?

Number: _____

What number is halfway between points A and B?



Number: _____

True or false?

The number halfway between 60 and 64 is 62.

- A. True
- B. False

What number is halfway between 4 and 12?

- A.** 6
- B.** 8
- C.** 10
- D.** 16

What number is halfway between 0 and 18?

Number: _____

SET 69 • 6.SP.A.2

FORWARD 1

Compare the centers of two populations • Grade 7 • Statistics & Probability •
7.SP.B.4

Class A has a median score of 82. Class B has a median score of 75. The variability is similar.

Which statement is true?

- A.** The classes scored the same.
- B.** Class A typically scored higher.
- C.** Class B typically scored higher.
- D.** Class A's scores vary more.

Random samples: Town A has a mean age of 34 years.
Town B has a mean age of 42 years.

What is the difference in mean ages?

Difference: _____

In random samples, 7th graders sleep a median of 8 hours, and 6th graders sleep a median of 9 hours.

Which group typically sleeps more?

Group: _____

Sample A has a mean of 12 and Sample B has a mean of 15. The samples have similar variability.

Which conclusion is best?

- A.** The populations are the same.
- B.** Values in population A tend to be greater.
- C.** Values in population B tend to be greater.
- D.** No conclusion is possible.

True or false?

To compare the centers of two populations, you can compare the medians of random samples.

- A.** True
- B.** False

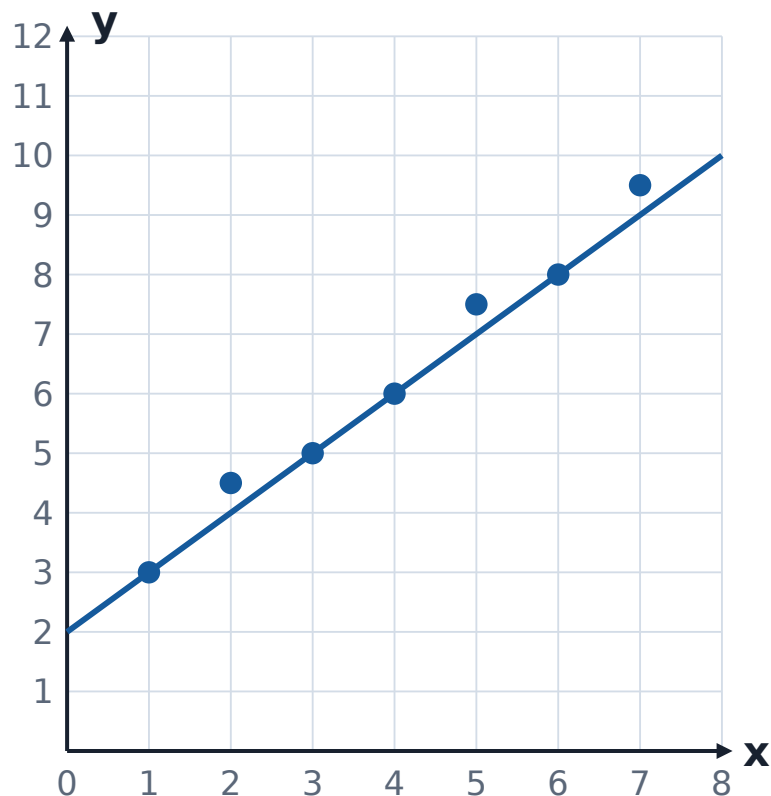
FORWARD 2

Use a line of best fit to make predictions • Grade 8 • Statistics & Probability •
8.SP.A.2

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

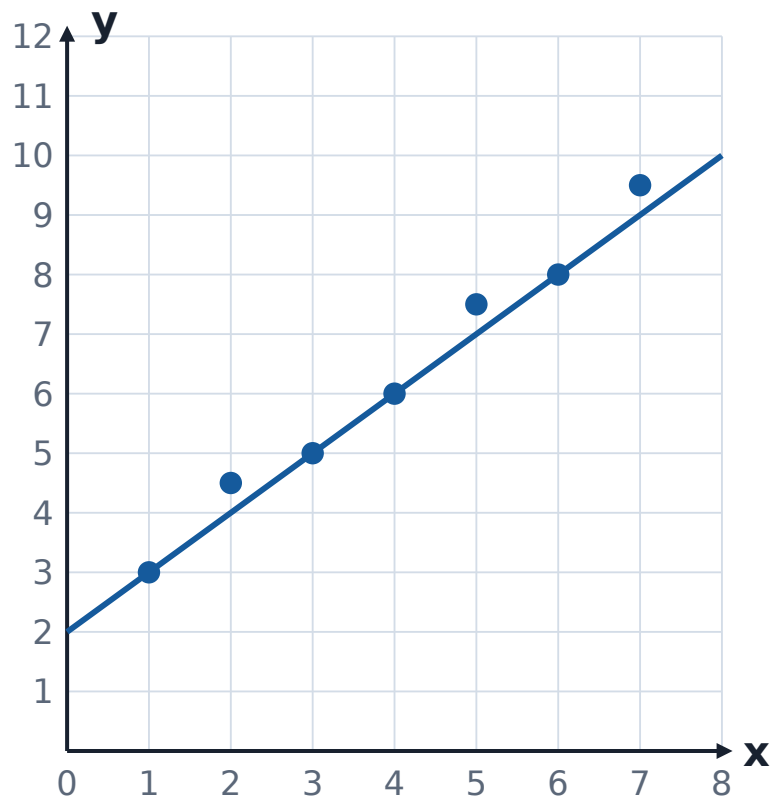
Use the line of best fit to
predict y when $x = 6$.

$y =$ _____



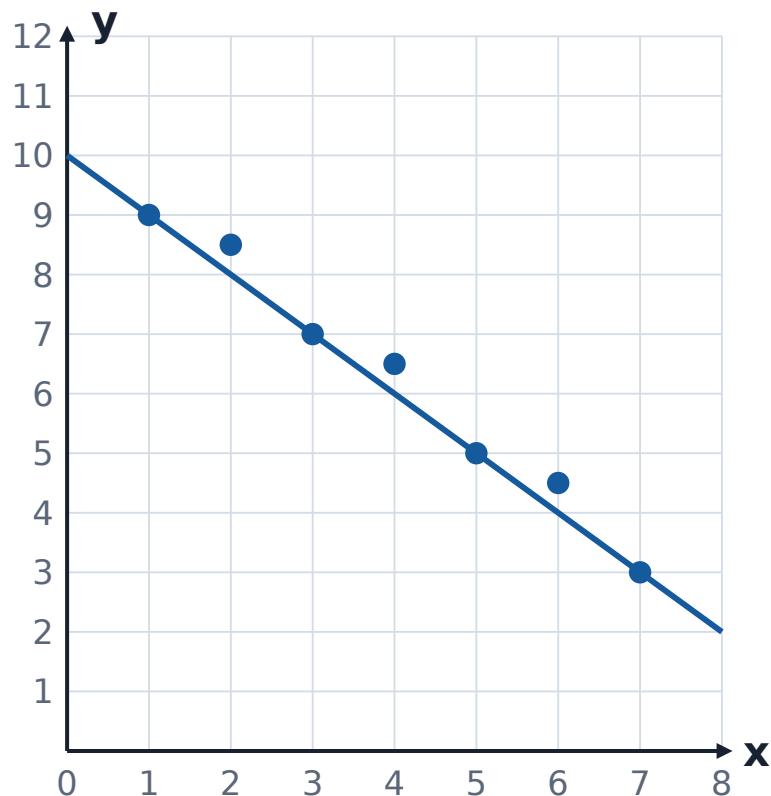
Use the line of best fit to
predict y when $x = 2$.

$y =$ _____



Use the line of best fit to
predict y when $x = 4$.

$y =$ _____



Which statement about a line of best fit is true?

- A.** It must pass through every point.
- B.** It shows the overall trend of the data.
- C.** It must pass through the origin.
- D.** It is always horizontal.

True or false?

A line of best fit usually has about the same number of points above it as below it.

- A.** True
- B.** False

Describe the spread of a distribution

MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Finding a difference	2.NBT.B.5
BACKWARD 2	Reading a line plot	4.MD.B.4
BACKWARD 3	Greatest and least numbers	2.NBT.A.4
FORWARD 1	Compare variability of two distributions	7.SP.B.3
FORWARD 2	Judge how closely data fit a line	8.SP.A.2

SET 70 • 6.SP.A.2

MAIN

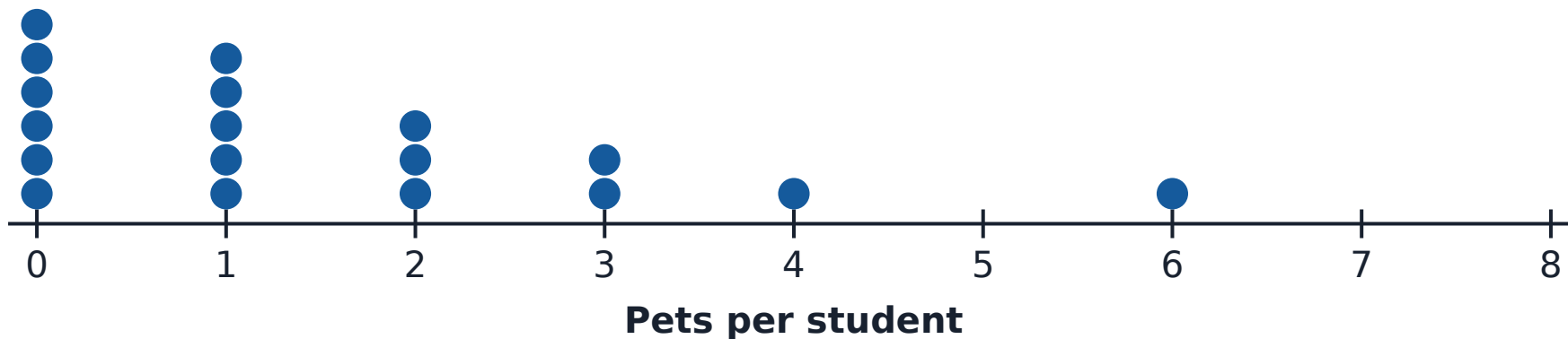
Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

What is the range of the data?



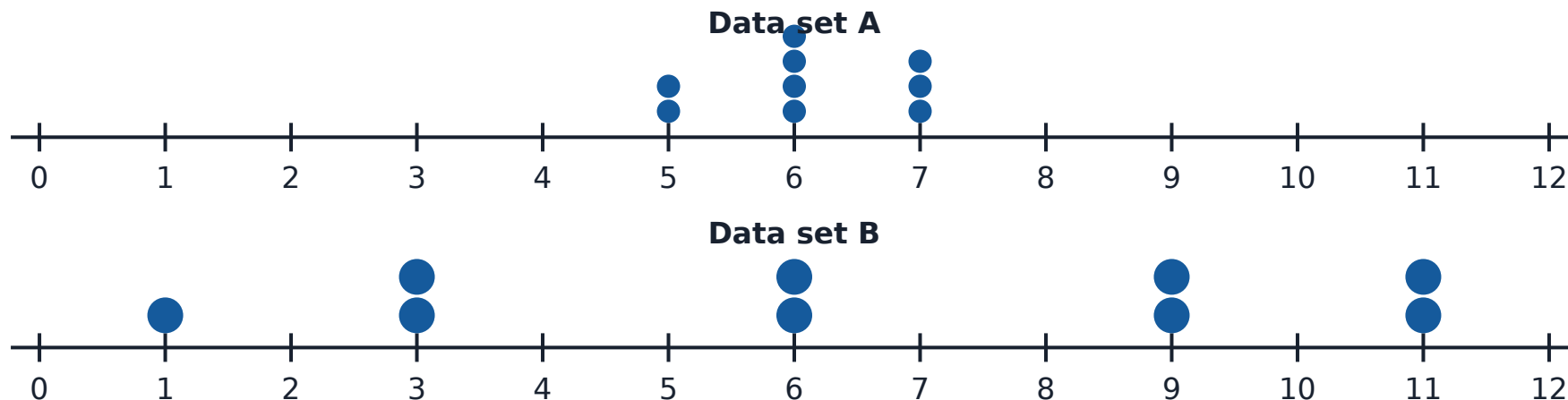
Range: _____

What is the range of the data?



Range: _____

Which data set has the greater spread?

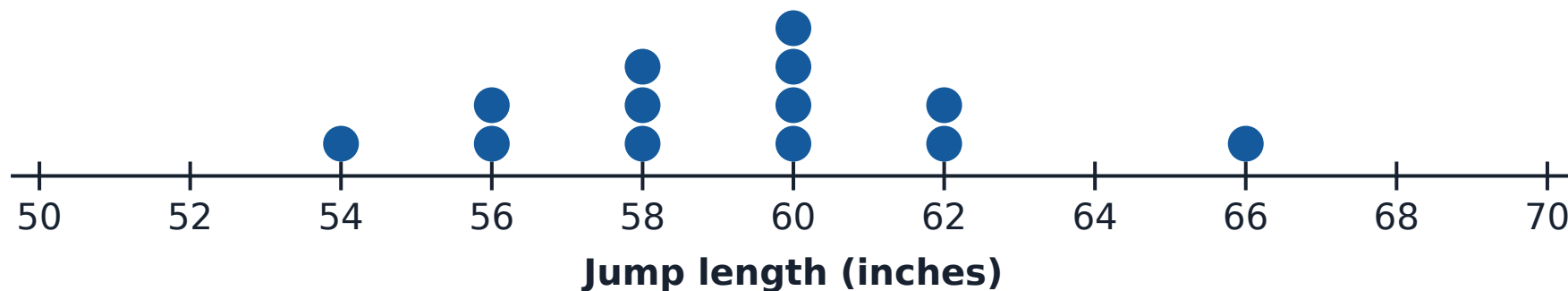


A. Data set A

B. Data set B

C. They have the same spread.

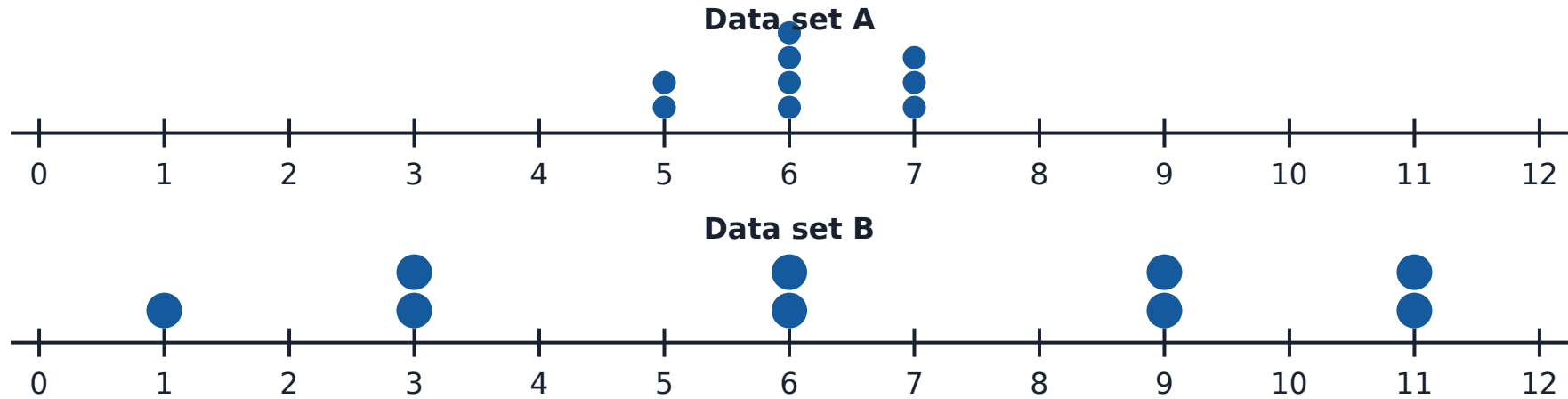
What is the range of the jump lengths?



Range: _____

True or false?

Data set A has a range of 2.



A. True

B. False

SET 70 • 6.SP.A.2

BACKWARD 1

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

The greatest value is 47. The least value is 18.

What is the difference?

Difference: _____

Subtract.

$$83 - 29$$

Difference: _____

True or false?

$$62 - 35 = 27$$

A. True

B. False

$$91 - 46 = ?$$

- A.** 45
- B.** 47
- C.** 55
- D.** 137

What is the difference between 66 and 54?

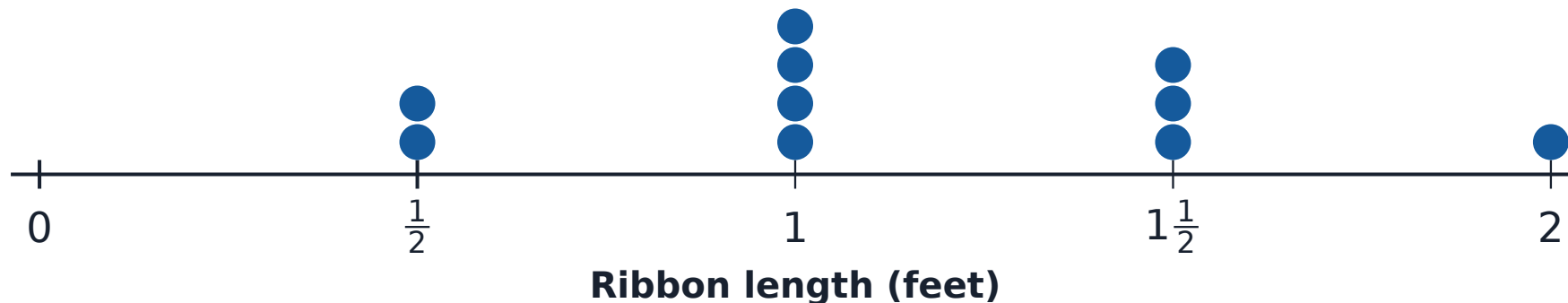
Difference: _____

SET 70 • 6.SP.A.2

BACKWARD 2

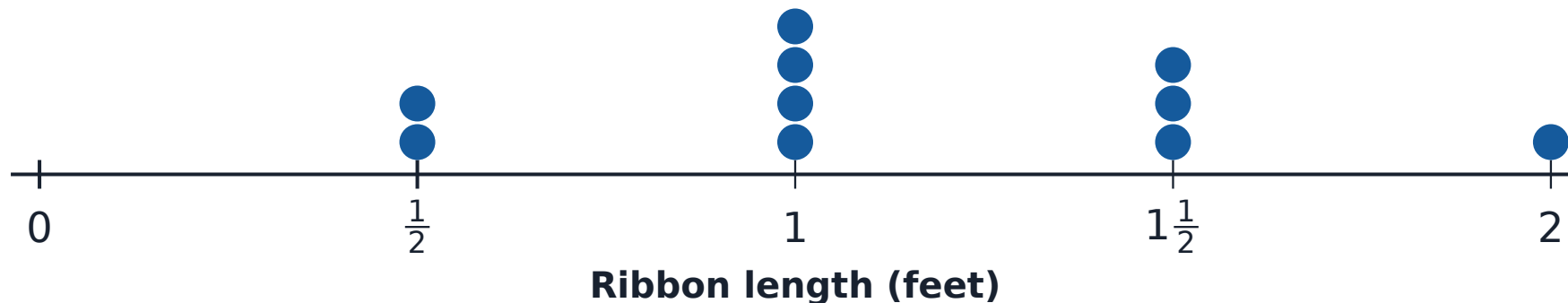
Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

What is the longest ribbon length shown?



Length: _____

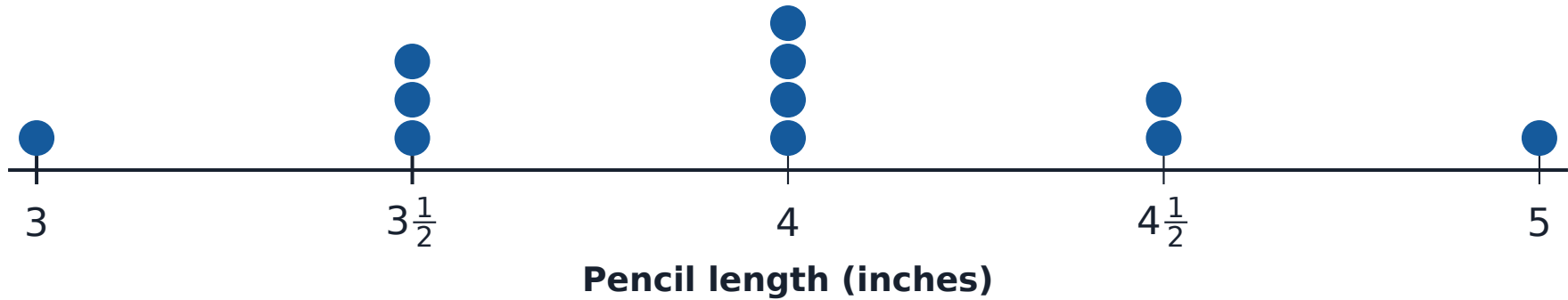
What is the shortest ribbon length shown?



Length: _____

True or false?

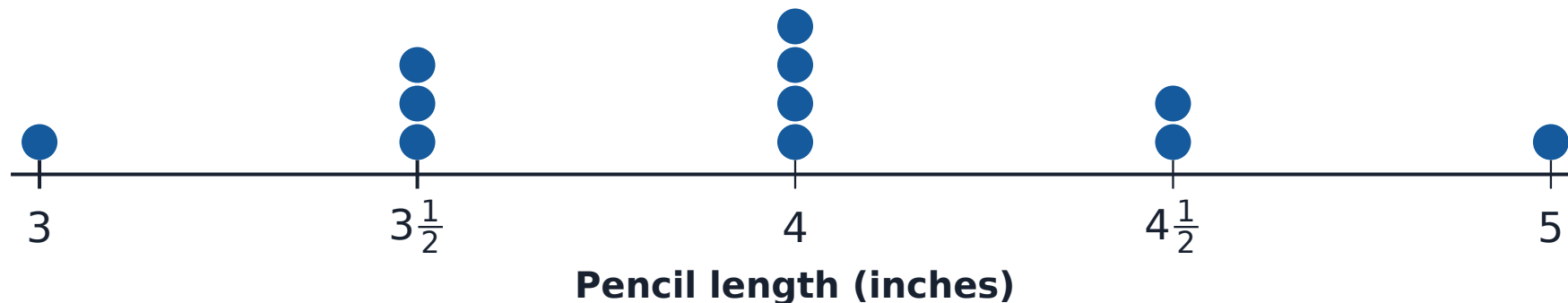
The longest pencil is 5 inches.



A. True

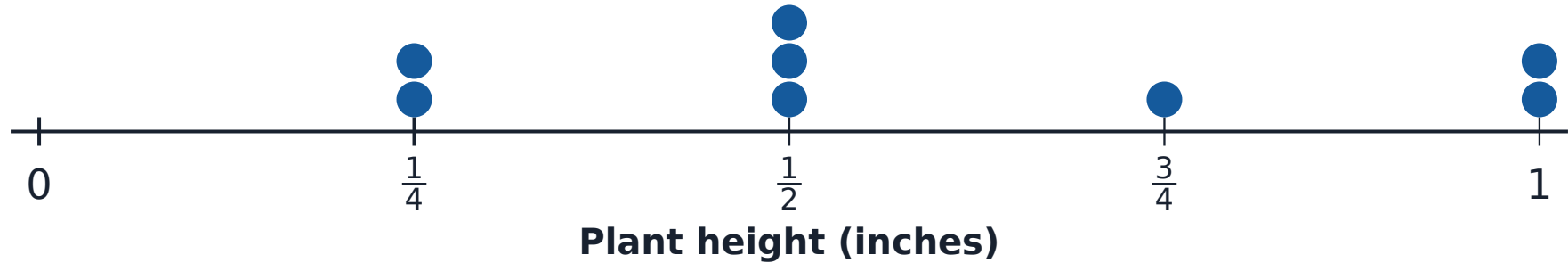
B. False

What is the shortest pencil length?



- A.** 3 inches **B.** $3\frac{1}{2}$ inches **C.** 4 inches **D.** 5 inches

How many plants are 1 inch tall?



Plants: _____

SET 70 • 6.SP.A.2

BACKWARD 3

Greatest and least numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

What is the greatest number?

18, 81, 58, 85

Greatest: _____

What is the least number?

204, 240, 402, 42

Least: _____

True or false?

The least number in the list 33, 30, 13, 31 is 13.

A. True

B. False

Which number is greatest?

A. 207

B. 270

C. 702

D. 720

What is the least number?

66, 56, 65, 60

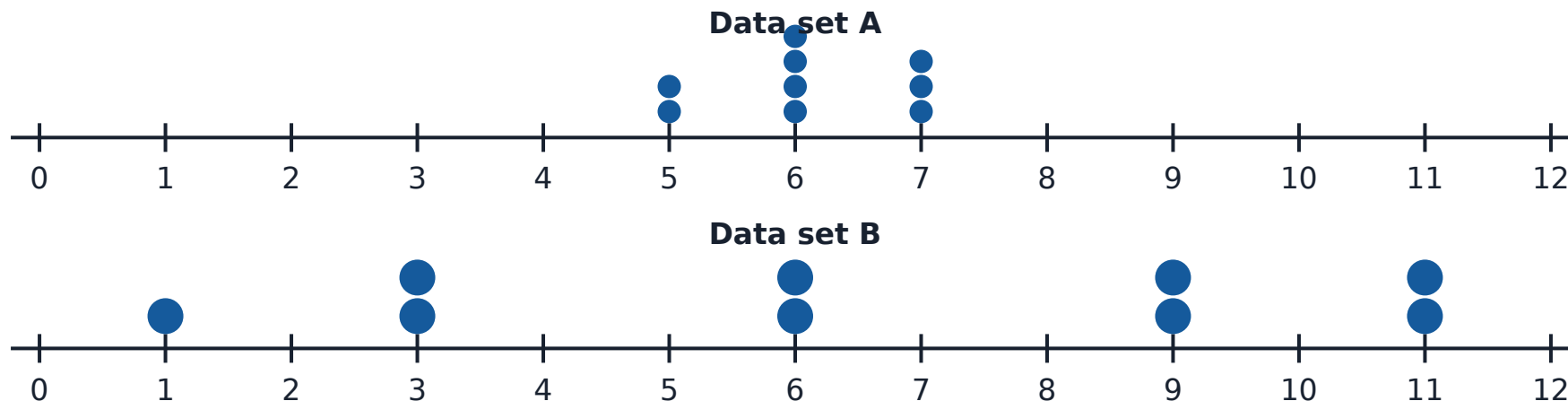
Least: _____

SET 70 • 6.SP.A.2

FORWARD 1

Compare variability of two distributions • Grade 7 • Statistics & Probability •
7.SP.B.3

Which data set has more variability?



A. Data set A

B. Data set B

C. They are the same.

Class A's mean is 60 and Class B's mean is 66. Each class has a MAD of 3.

The difference in means is how many times the MAD?

Times: _____

True or false?

Two data sets with means of 50 and 56 and MADs of 3 have means that differ by 2 MADs.

- A. True
- B. False

Data set A has a mean of 20 and a MAD of 2. Data set B has a mean of 21 and a MAD of 6.

Which statement is true?

- A.** The means differ by 3 MADs.
- B.** Data set A's values are more spread out.
- C.** Data set B's values are more spread out.
- D.** The data sets do not overlap.

Two data sets have means of 8 and 14. Each has a MAD of 1.5.

The difference in means is how many times the MAD?

Times: _____

SET 70 • 6.SP.A.2

FORWARD 2

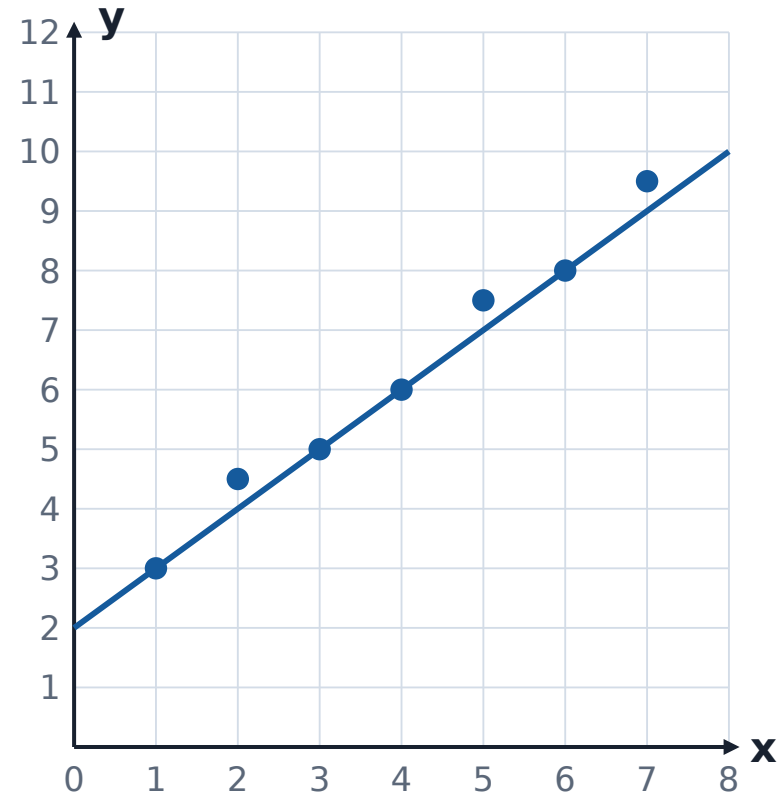
Judge how closely data fit a line • Grade 8 • Statistics & Probability • 8.SP.A.2

True or false?

The line is a good fit for the data.

A. True

B. False

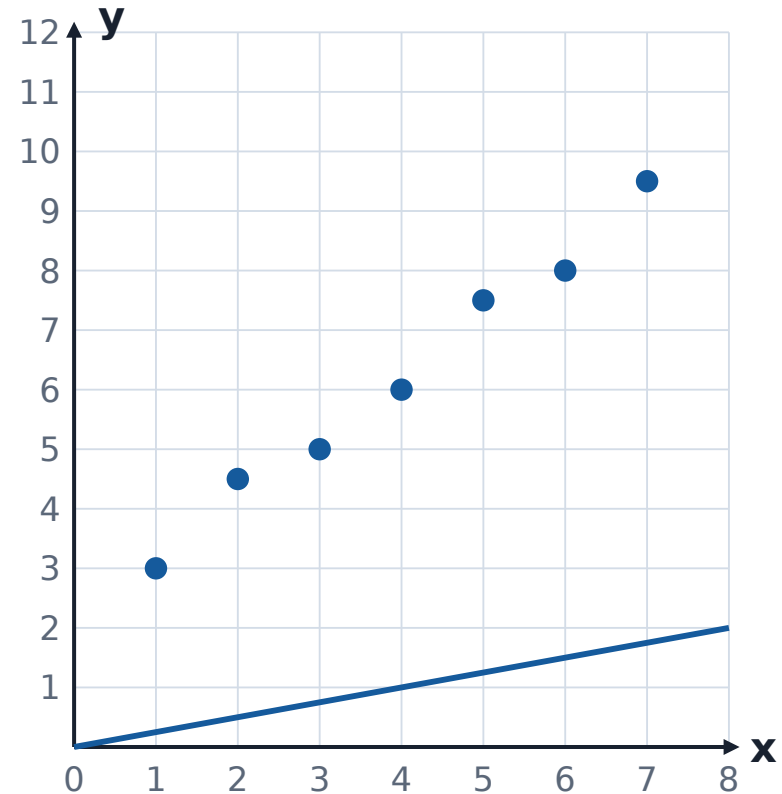


True or false?

The line is a good fit for the data.

A. True

B. False



Which is a sign of a good line of best fit?

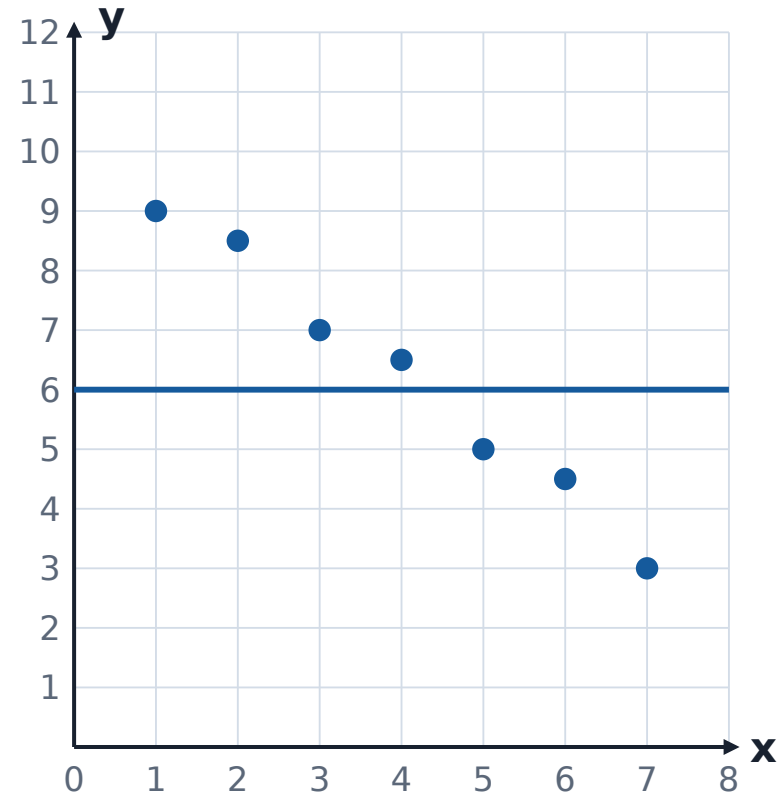
- A.** The line goes through only one point.
- B.** All points are below the line.
- C.** Points are close to the line on both sides.
- D.** All points are above the line.

True or false?

The line is a good fit for the data.

A. True

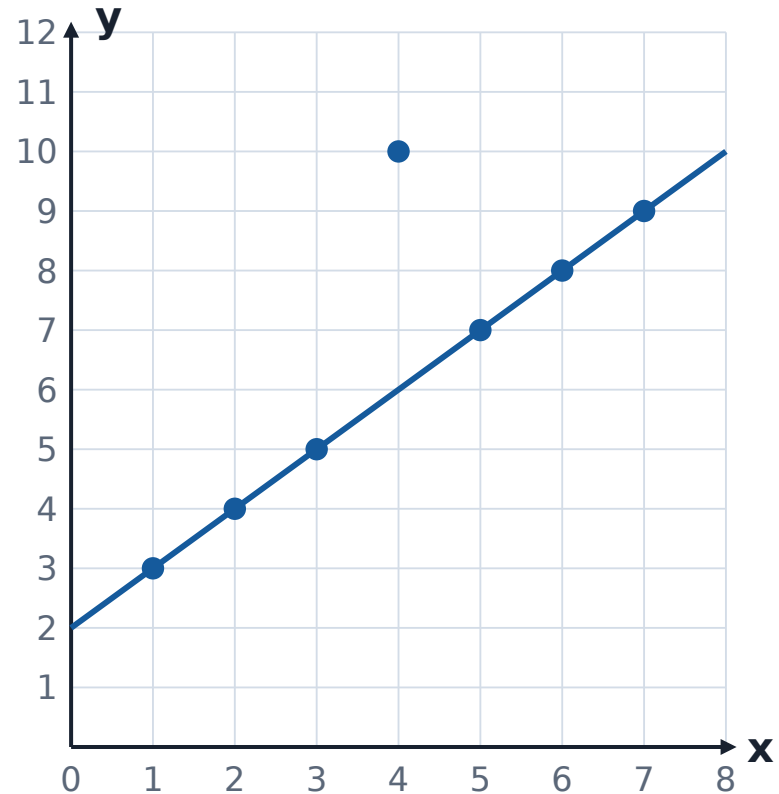
B. False



Which point is farthest from the line?

Write its coordinates.

Point: _____



A measure of center vs. a measure of variation

MAIN	Grade 6 main question	6.SP.A.3
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Sharing a total equally	5.MD.B.2
BACKWARD 3	Finding a difference	2.NBT.B.5
FORWARD 1	Compare populations using center and variability	7.SP.B.4
FORWARD 2	Interpret the slope and intercept of a linear model — nearest rela...	8.SP.A.3

SET 71 • 6.SP.A.3

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.3

Which measure describes the center of a data set?

- A.** Interquartile range
- B.** Mean absolute deviation
- C.** Median
- D.** Range

Which measure describes how much the values in a data set vary?

- A.** Mean
- B.** The middle value
- C.** Median
- D.** Interquartile range

True or false?

A measure of center summarizes all of the values in a data set with a single number.

- A.** True
- B.** False

True or false?

A measure of variation describes how the values in a data set vary with a single number.

- A.** True
- B.** False

Which measure is a measure of variation?

- A.** Median
- B.** Mean
- C.** Mean absolute deviation
- D.** The middle value

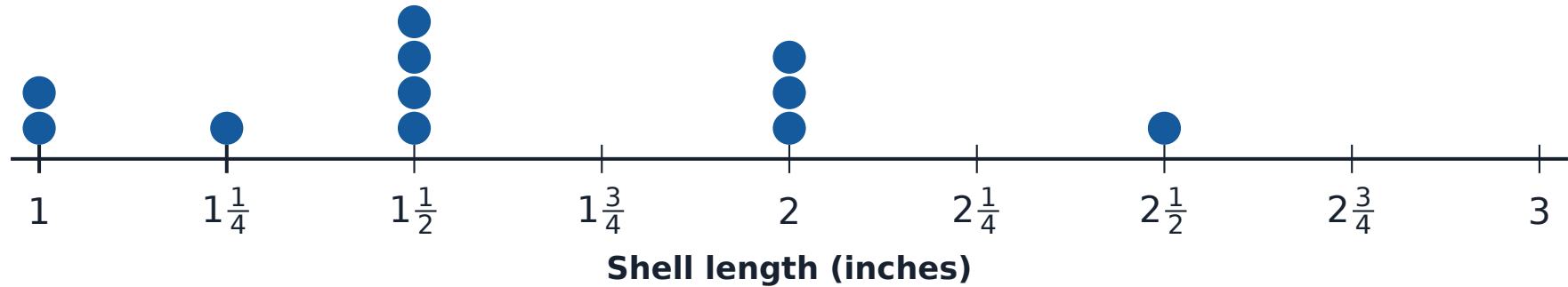
SET 71 • 6.SP.A.3

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows shell lengths.

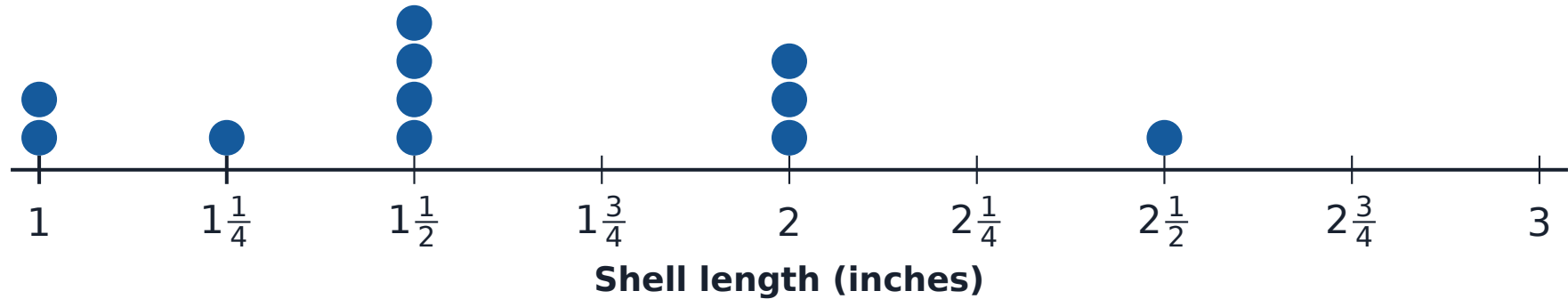
How many shells are $1\frac{1}{2}$ inches long?



Shells: _____

The line plot shows shell lengths.

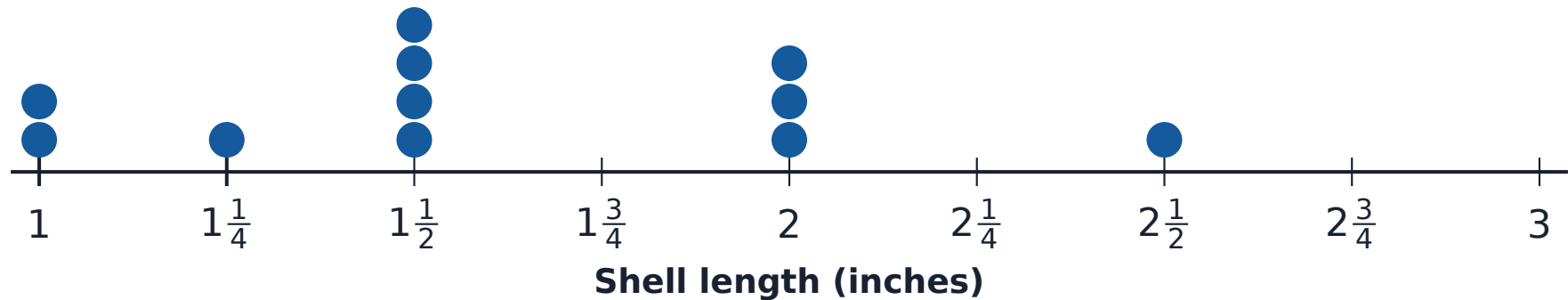
How many shells are there in all?



Shells: _____

True or false?

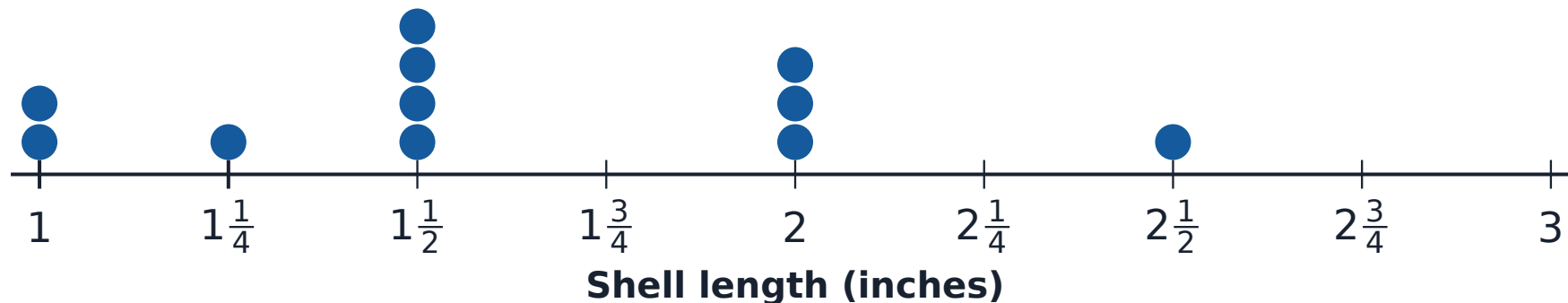
Three shells are 2 inches long.



A. True

B. False

Which shell length is most common?



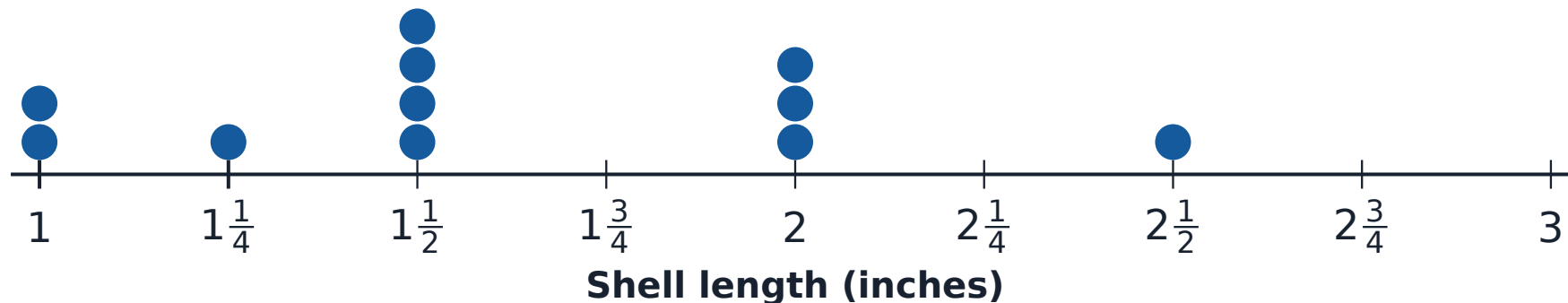
A. 1 inch

B. $1\frac{1}{2}$
inches

C. 2 inches

D. $2\frac{1}{2}$
inches

What is the total length of the 2-inch shells?



Total length: _____

SET 71 • 6.SP.A.3

BACKWARD 2

Sharing a total equally • Grade 5 • Measurement & Data • 5.MD.B.2

Three friends have 4, 6, and 8 stickers. They share all of the stickers equally.

How many stickers does each friend get?

Stickers: _____

Four jars hold 2, 3, 5, and 6 liters of water. The water is shared equally among the jars.

How much water is in each jar?

Liters: _____

True or false?

Cups holding 1, 2, and 3 liters are shared equally.
Each cup then holds 2 liters.

- A. True
- B. False

Five friends collected 3, 5, 6, 7, and 9 cans. They share the cans equally.

How many cans does each friend get?

A. 5

B. 6

C. 7

D. 30

Two bottles hold $\frac{1}{4}$ liter and $\frac{3}{4}$ liter. The juice is shared equally between the bottles.

How much juice is in each bottle?

Liters: _____

SET 71 • 6.SP.A.3

BACKWARD 3

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$76 - 38$$

Difference: _____

The highest score is 95 and the lowest is 58.

What is the difference?

Difference: _____

True or false?

$$100 - 47 = 53$$

A. True

B. False

$$84 - 29 = ?$$

- A.** 45
- B.** 55
- C.** 65
- D.** 113

Subtract.

$$62 - 17$$

Difference: _____

SET 71 • 6.SP.A.3

FORWARD 1

Compare populations using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.4

Class A has a median score of 82 and an IQR of 10.

Class B has a median score of 75 and an IQR of 10.

Which statement is true?

A. Class A typically scored higher.

B. Class A's scores vary more.

C. Class B typically scored higher.

D. The classes scored the same.

True or false?

Two samples have the same median. The sample with the greater IQR has more variability.

- A.** True
- B.** False

Which measure is best for comparing the variability of two samples?

- A.** IQR
- B.** Sample size
- C.** Median
- D.** Mean

Sample A: mean 40, MAD 2. Sample B: mean 40, MAD 7.
Which sample has values that vary more?

Sample: _____

Store A's daily sales have a median of \$520. Store B's have a median of \$610. Their IQRs are similar.

Which store typically has greater daily sales?

Store: _____

FORWARD 2

Interpret the slope and intercept of a linear model • Grade 8 • Statistics & Probability • 8.SP.A.3

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The model $y = 2.5x + 40$ predicts a test score y after x hours of study.

What does the slope 2.5 mean?

Meaning: _____

The model $y = 3x + 12$ gives a plant's height y (cm) after x weeks.

What does 12 represent?

- A.** The number of weeks
- B.** The height at week 0
- C.** The height after 12 weeks
- D.** The growth per week

True or false?

In the model $y = -0.5x + 30$, y decreases by 0.5 for each increase of 1 in x .

- A. True
- B. False

The model $y = 1.2x + 5$ gives the length y (cm) of a fish at x months old.

What does 5 represent?

Meaning: _____

In the model $y = 45x + 200$, what does 45 represent if y is dollars saved after x weeks?

- A.** Total savings
- B.** Starting savings
- C.** Number of weeks
- D.** Dollars saved per week

Display numerical data in a dot plot

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Line plots of whole-number measurements	2.MD.D.9
BACKWARD 2	Line plots with fractional measurements	4.MD.B.4
BACKWARD 3	Ordering numbers	2.NBT.A.4
FORWARD 1	Compare two dot plots	7.SP.B.3
FORWARD 2	Construct a scatter plot	8.SP.A.1

SET 72 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

Data: 2, 4, 4, 5, 7, 7, 7, 9

Make a dot plot of the data on the number line.



Number of pets: 0, 1, 1, 2, 2, 2, 3, 5

Make a dot plot of the data on the number line.



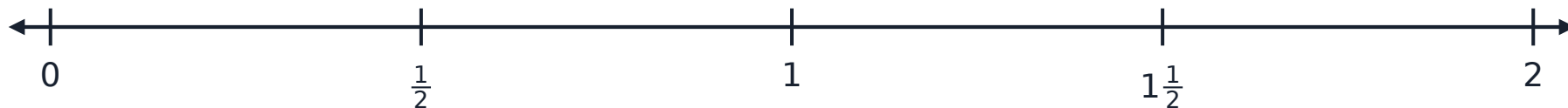
Quiz scores: 6, 7, 7, 8, 8, 8, 9, 10, 10

Make a dot plot of the data on the number line.



Ribbon lengths (feet): $\frac{1}{2}$, 1, 1, $1\frac{1}{2}$, 2, 1

Make a dot plot of the data on the number line.



Hours of sleep: 7, 8, 9, 8, 10, 7, 8

Make a dot plot of the data on the number line.



SET 72 • 6.SP.B.4

BACKWARD 1

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 3, 4, 4, 5, 4

In a line plot, how many marks are drawn at 4?

Marks: _____

Lengths (cm): 2, 3, 3, 5

Make a line plot of the lengths.



True or false?

In a line plot of 6, 6, 7, 9, there are two marks above 6.

- A.** True
- B.** False

Which number line is best for a line plot of 12, 14, 15, 15?

- A.** A number line from 0 to 100 by tens
- B.** A number line from 20 to 30
- C.** A number line from 10 to 16
- D.** A number line from 0 to 5

Lengths (feet): 8, 9, 9, 9, 11

Which value gets the most marks?

Value: _____

SET 72 • 6.SP.B.4

BACKWARD 2

Line plots with fractional measurements • Grade 4 • Measurement & Data •
4.MD.B.4

Lengths (inches): $\frac{1}{2}$, 1, 1, $1\frac{1}{2}$, 1, $\frac{1}{2}$

In a line plot, how many X's are drawn at 1?

X's: _____

True or false?

In a line plot of 2, $2\frac{1}{2}$, $2\frac{1}{2}$, 3, there are two X's above $2\frac{1}{2}$.

A. True

B. False

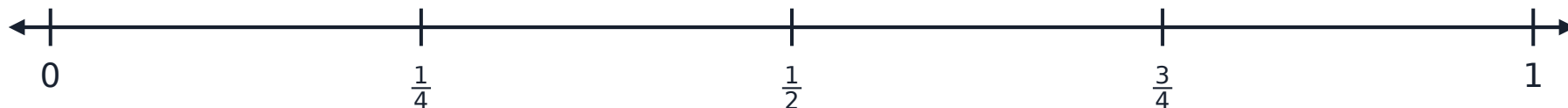
Which number line would you use for these lengths?

$$\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{1}{2}$$

- A.** A number line from 0 to 10 marked in ones
- B.** A number line from 0 to 100 marked in tens
- C.** A number line from 1 to 2 marked in halves
- D.** A number line from 0 to 1 marked in fourths

Lengths (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$

Make a line plot of the lengths.



True or false?

A line plot shows each data value as a mark above a number line.

- A.** True
- B.** False

SET 72 • 6.SP.B.4

BACKWARD 3

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

7, 2, 9, 4, 4

Order: _____

Which list is in order from least to greatest?

A. 6, 7, 8, 10

B. 10, 8, 7, 6

C. 7, 6, 8, 10

D. 6, 8, 7, 10

True or false?

These numbers are in order from least to greatest: 1, 2, 5, 3

- A. True
- B. False

What are the least and greatest values?

8, 10, 7, 9, 7

Least: _____

Greatest: _____

Which number is least?

A. 0

B. 1

C. 2

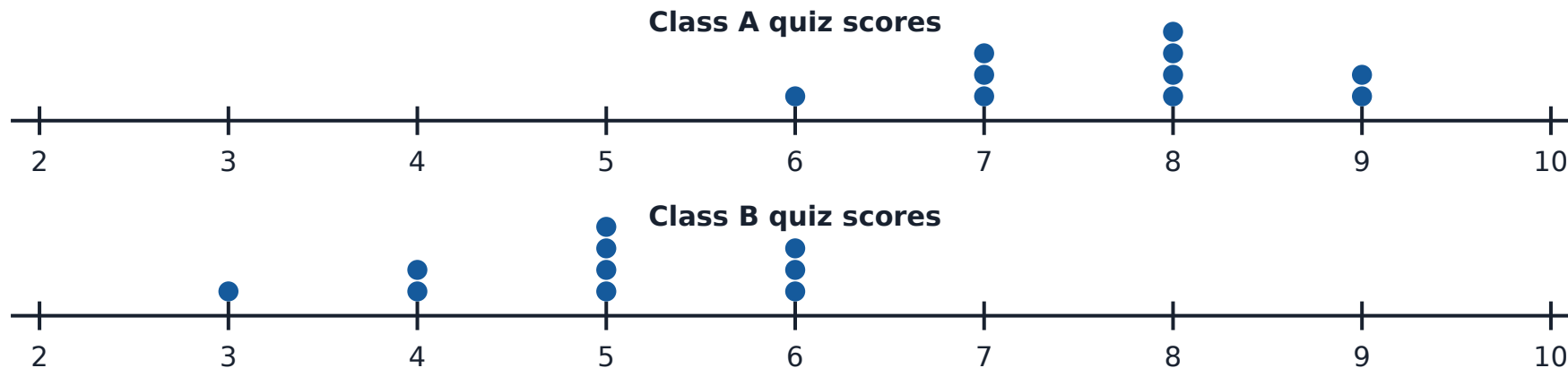
D. 5

SET 72 • 6.SP.B.4

FORWARD 1

Compare two dot plots • Grade 7 • Statistics & Probability • 7.SP.B.3

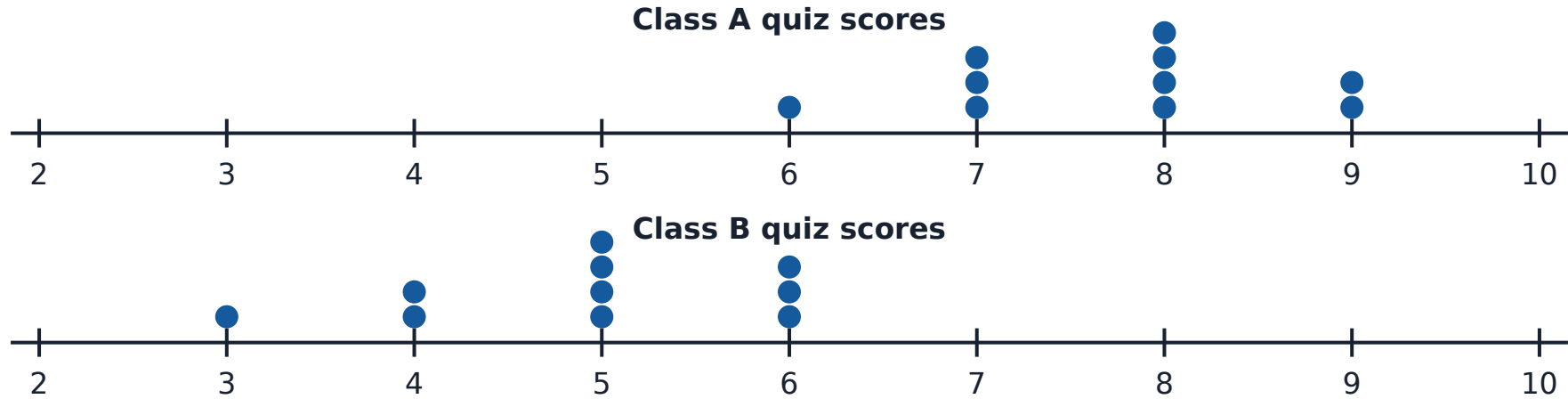
About how many points greater is the center of Class A's scores than Class B's?



Points: _____

True or false?

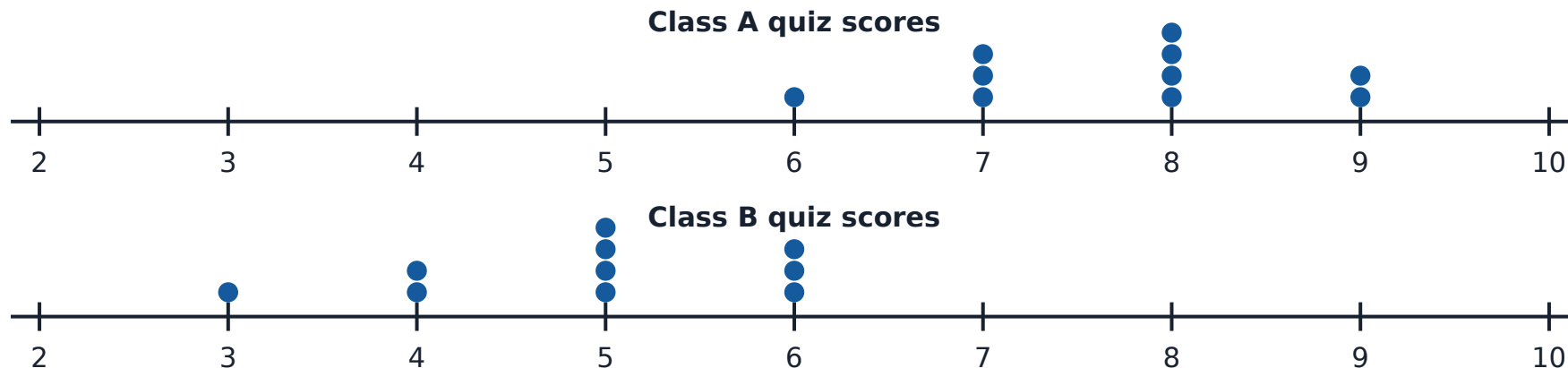
The two dot plots overlap.



A. True

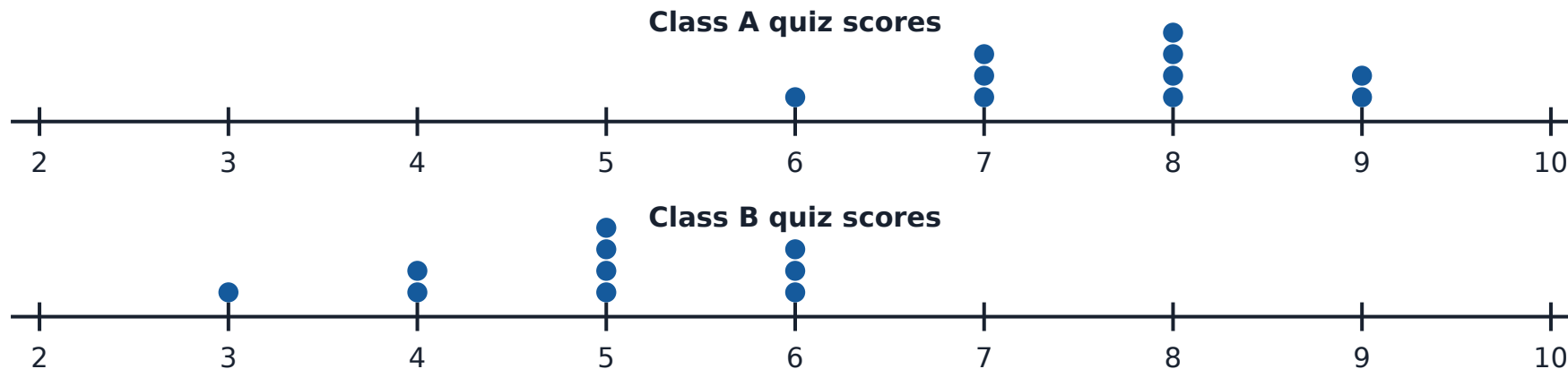
B. False

Which statement is true?



- A.** Class B has more variability.
- B.** The medians are equal.
- C.** Class B's scores are typically higher.
- D.** Class A's scores are typically higher.

What is the range of each class's scores?

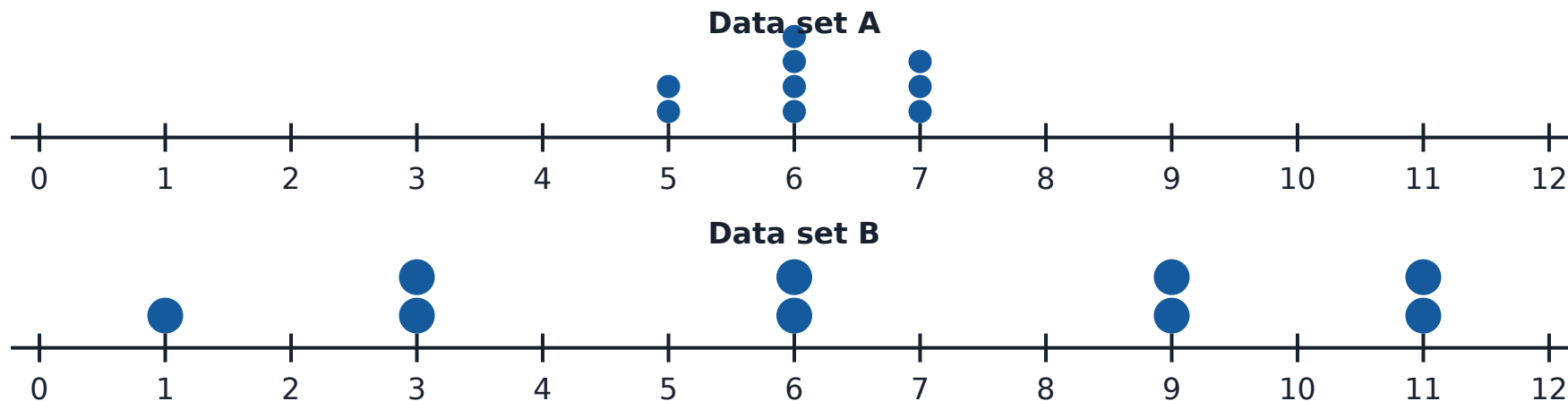


Class A: _____

Class B: _____

True or false?

Data set A and data set B have the same median.



A. True

B. False

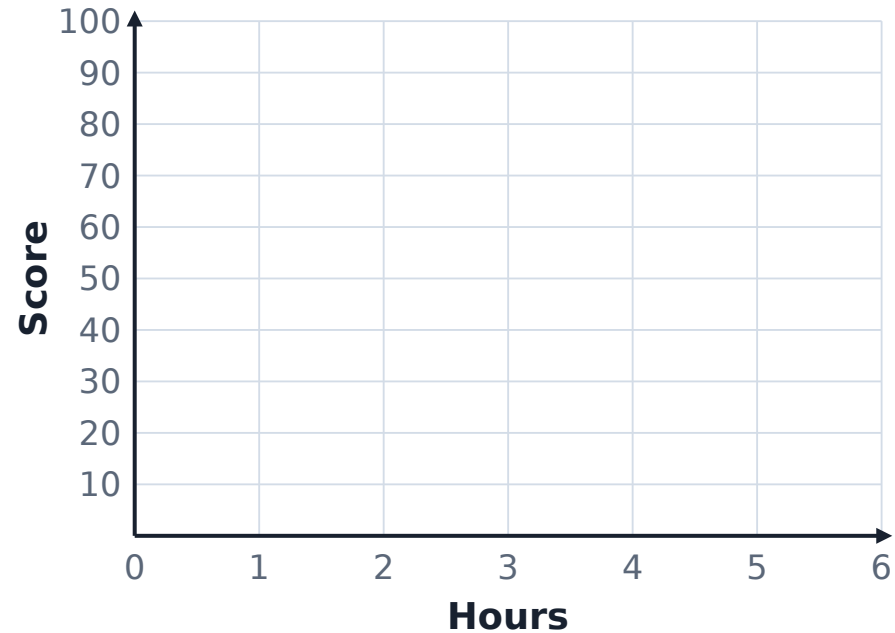
SET 72 • 6.SP.B.4

FORWARD 2

Construct a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

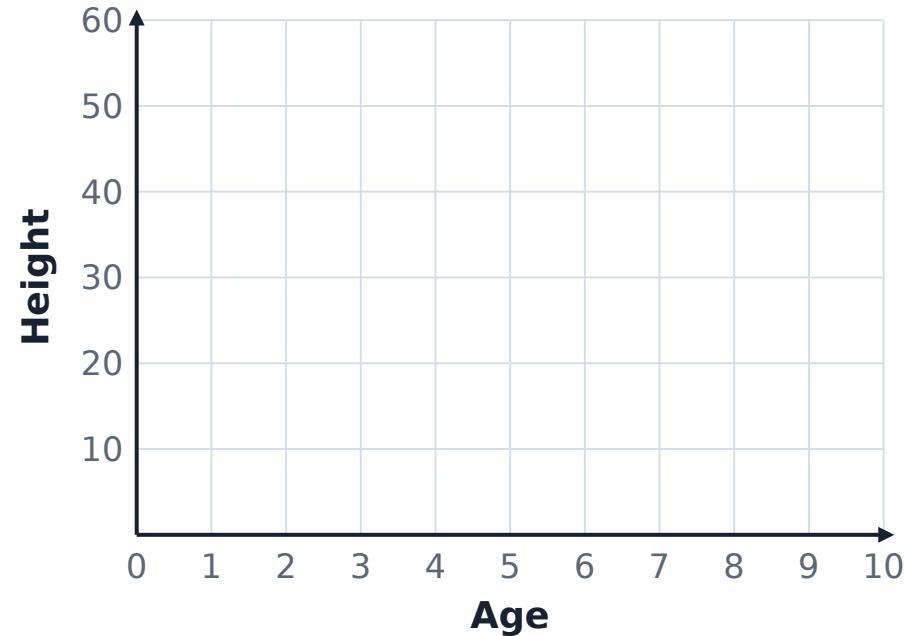
Make a scatter plot of the data.

Hours	1	2	3	4	5
Score	60	65	75	80	90



Make a scatter plot of the data.

Age	2	4	6	8
Height (in.)	34	40	46	50



A scatter plot shows age on the x-axis and height on the y-axis.

A 10-year-old is 54 inches tall. Which point shows this person?

A. (54, 10)

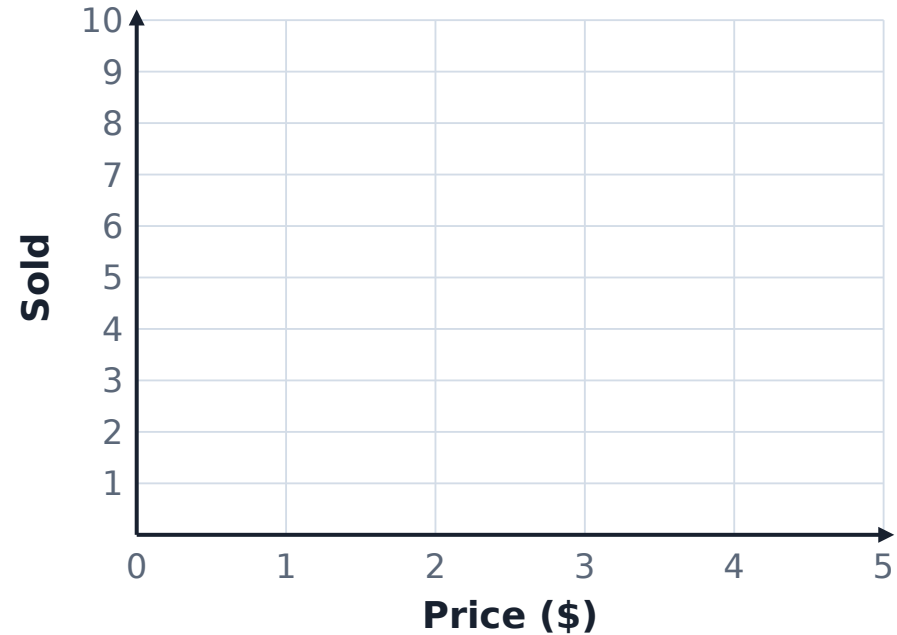
B. (54, 54)

C. (10, 10)

D. (10, 54)

Make a scatter plot of the data.

Price (\$)	1	2	3	4
Sold	9	7	4	2



True or false?

To make a scatter plot of height and arm span, each person is shown as one point.

- A.** True
- B.** False

Display numerical data in a histogram

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Drawing a scaled bar graph	3.MD.B.3
BACKWARD 2	Placing numbers in intervals	2.NBT.A.4
BACKWARD 3	Organizing data into categories	1.MD.C.4
FORWARD 1	Compare two populations shown in histograms	7.SP.B.4
FORWARD 2	Display numerical data for two variables in a scatter plot — neares...	8.SP.A.1

SET 73 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

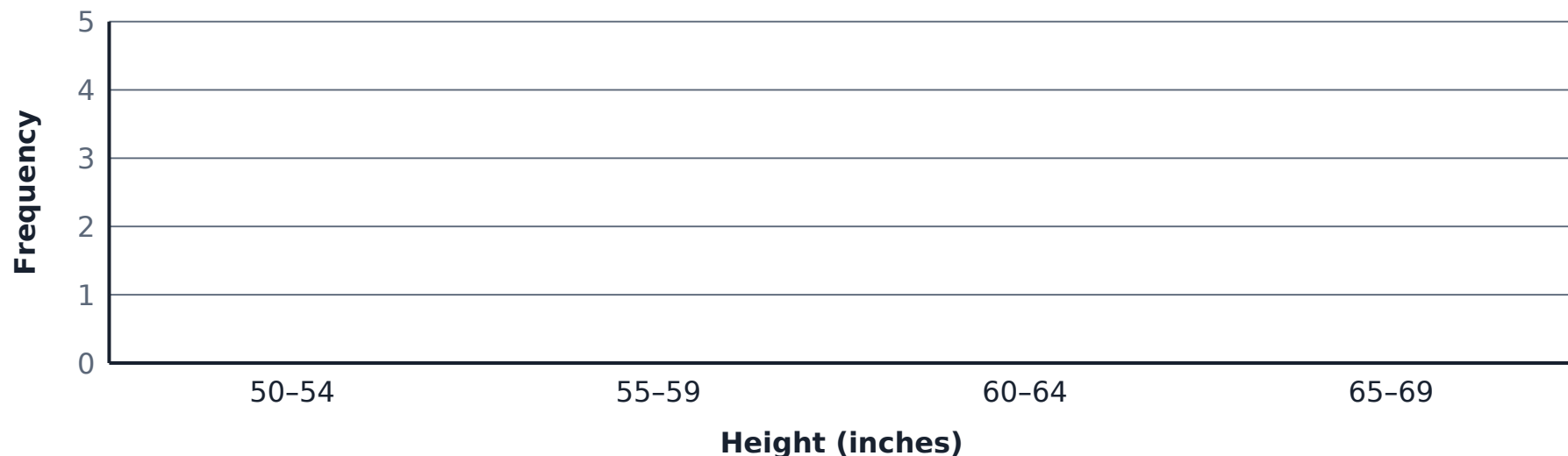
Books read: 3, 12, 15, 8, 22, 17, 25, 11, 31, 19

Make a histogram of the data.



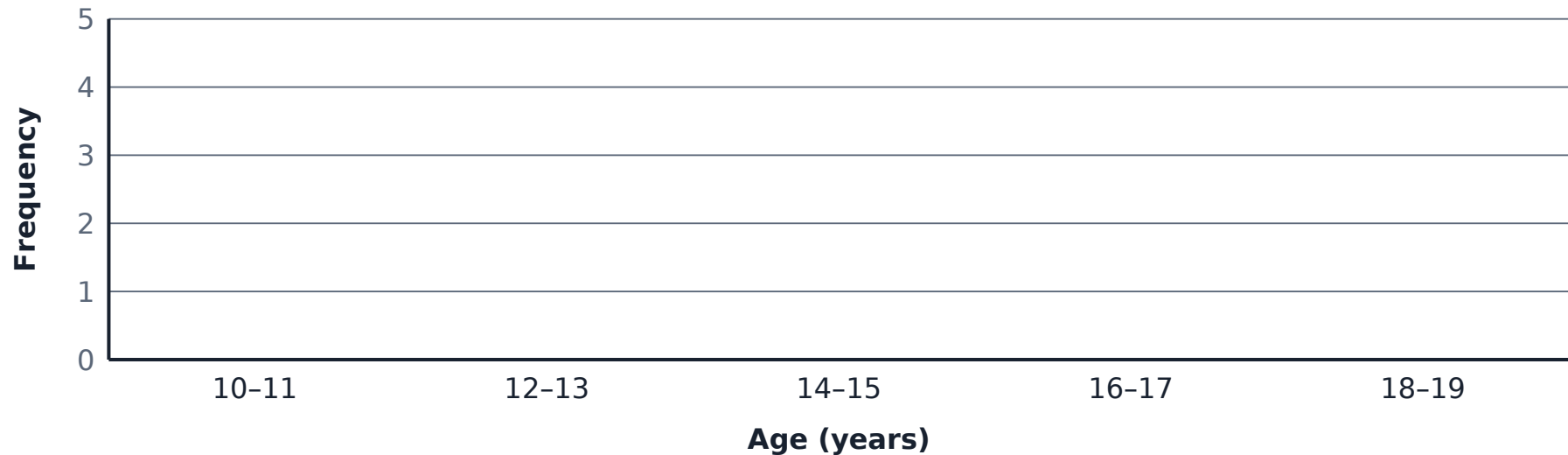
Heights (in.): 52, 58, 61, 55, 63, 57, 66, 60, 54, 59

Make a histogram of the data.



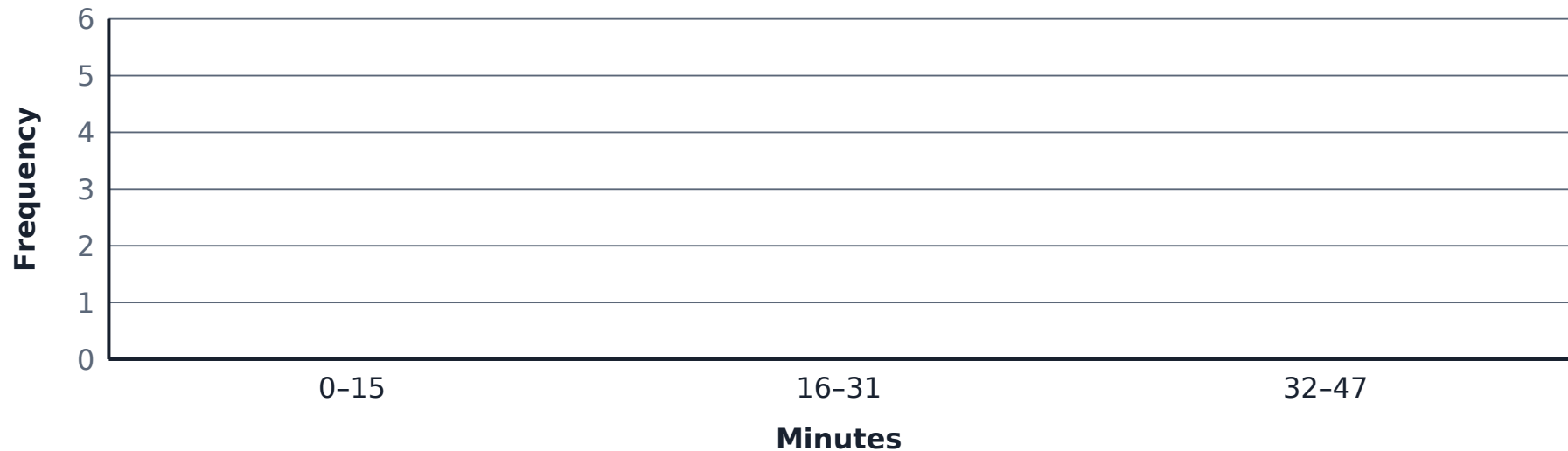
Ages: 11, 12, 12, 13, 14, 14, 14, 15, 16, 17, 19

Make a histogram of the data.



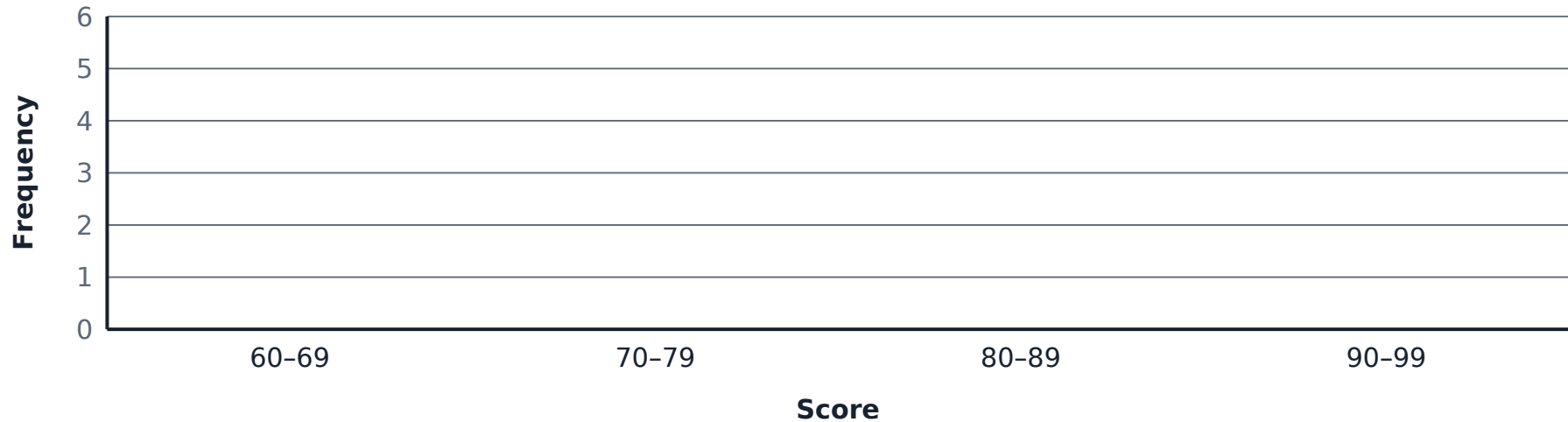
Minutes of exercise: 5, 18, 22, 25, 30, 34, 41, 45, 12, 28

Make a histogram of the data.



Test scores: 62, 75, 78, 81, 84, 88, 90, 93, 95, 71, 86, 79

Make a histogram of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

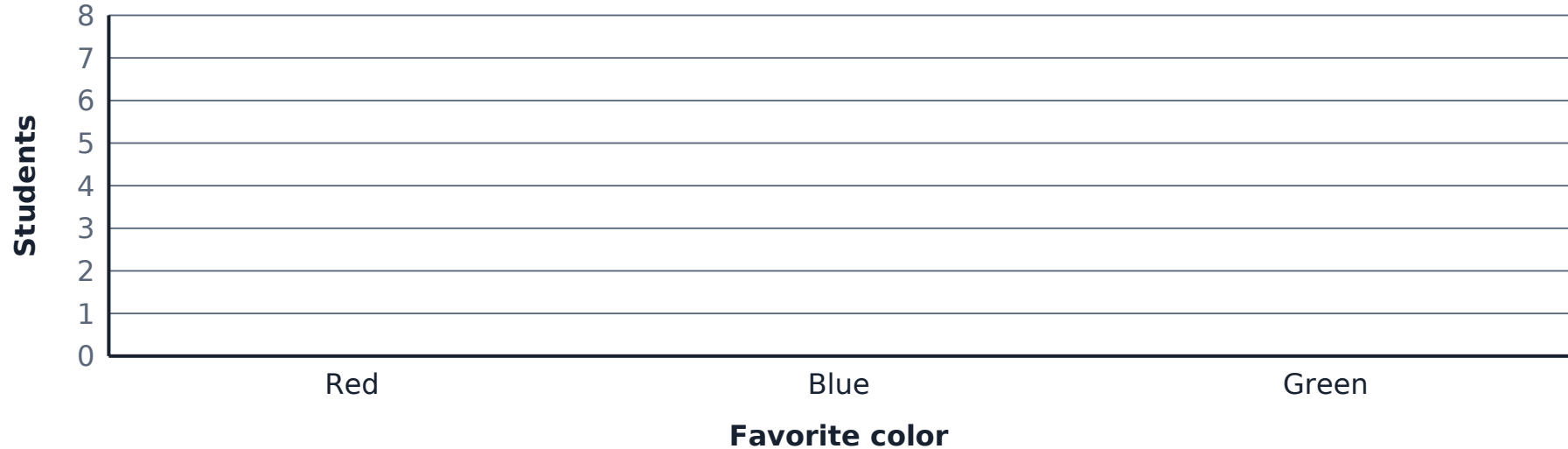
Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q5/5

SET 73 • 6.SP.B.4

BACKWARD 1

Drawing a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

Draw a bar graph: Red 4, Blue 6, Green 2.



On a bar graph, the bar for "Cats" ends at 5 on a scale that counts by 1s.

How many cats does the bar show?

Cats: _____

True or false?

On a bar graph with a scale of 2, a bar that stops halfway between 6 and 8 shows 7.

- A.** True
- B.** False

A bar graph has a scale that counts by 5s. A bar shows 20.

How many grid lines tall is the bar?

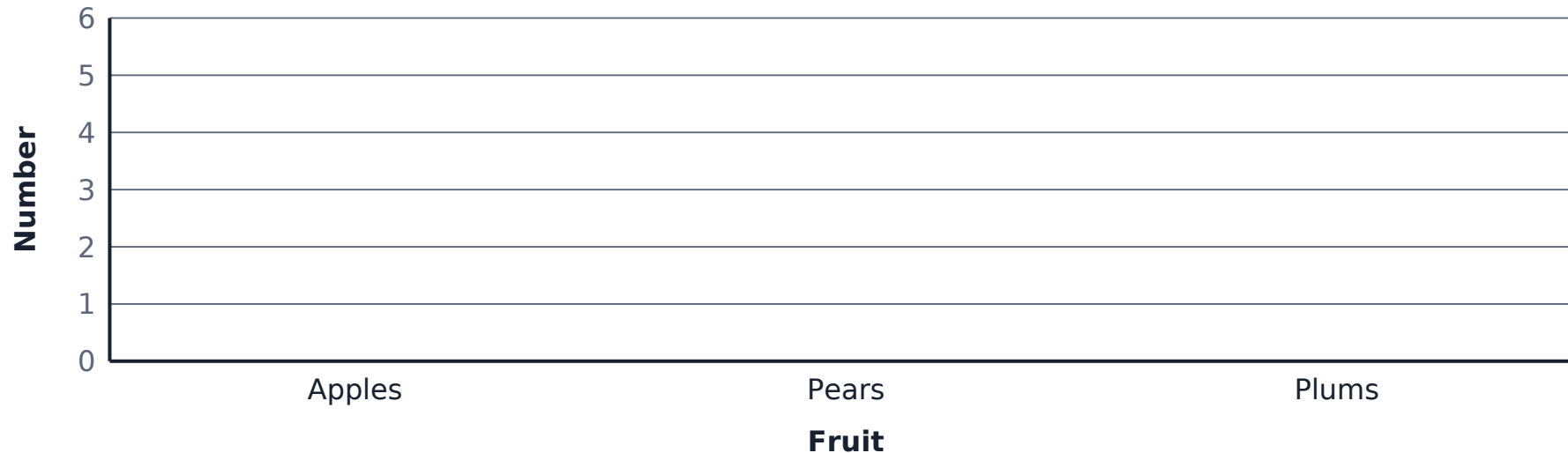
A. 4

B. 5

C. 15

D. 20

Draw a bar graph: Apples 3, Pears 1, Plums 5.



SET 73 • 6.SP.B.4

BACKWARD 2

Placing numbers in intervals • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Which interval contains 17?

- A.** 0–9
- B.** 10–19
- C.** 20–29
- D.** 30–39

True or false?

25 is in the interval 20–29.

A. True

B. False

Data: 3, 12, 15, 18, 22

How many of the numbers are in the interval 10–19?

Numbers: _____

Which number is in the interval 30–39?

- A.** 3
- B.** 29
- C.** 34
- D.** 40

True or false?

19 is in the interval 20–29.

A. True

B. False

SET 73 • 6.SP.B.4

BACKWARD 3

Organizing data into categories • Grade 1 • Measurement & Data • 1.MD.C.4

Data: cat, dog, cat, fish, dog, cat

How many answers are "cat"?

Cat: _____

True or false?

Data: red, blue, red, red. There are 3 answers of "red."

A. True

B. False

Data: A, B, A, C, A, B

Which category has the most?

A. A

B. B

C. C

Data: yes, no, yes, yes, no

How many more "yes" answers than "no" answers are there?

More: _____

Data: 3, 7, 12, 14, 18

How many numbers are less than 10?

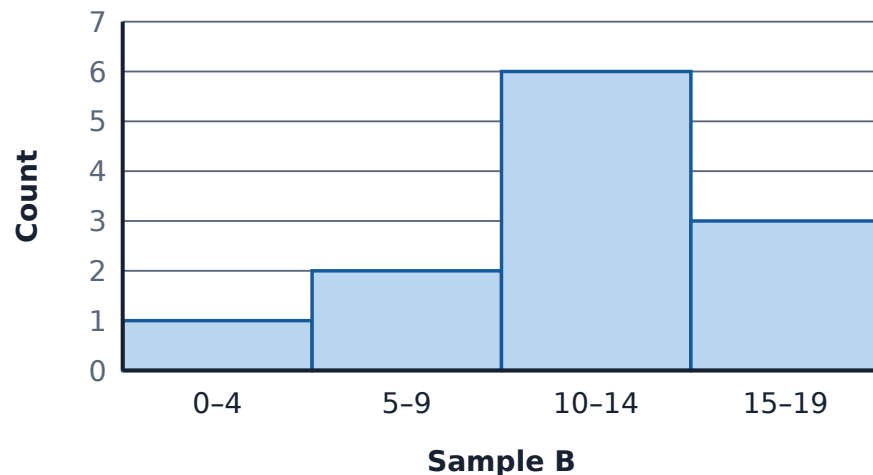
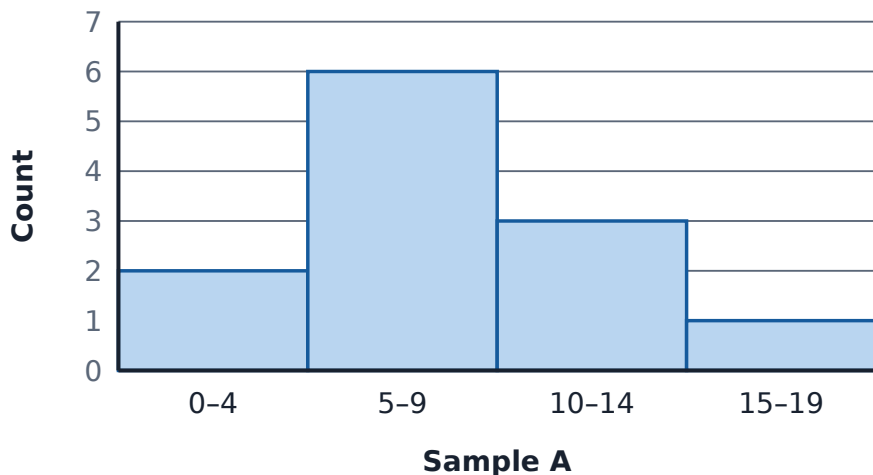
Numbers: _____

SET 73 • 6.SP.B.4

FORWARD 1

Compare two populations shown in histograms • Grade 7 • Statistics & Probability • 7.SP.B.4

Which sample tends to have greater values?

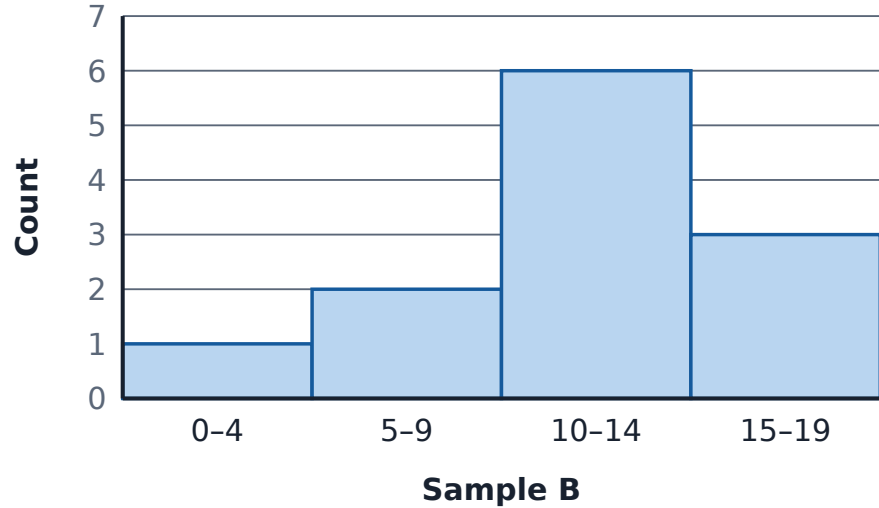
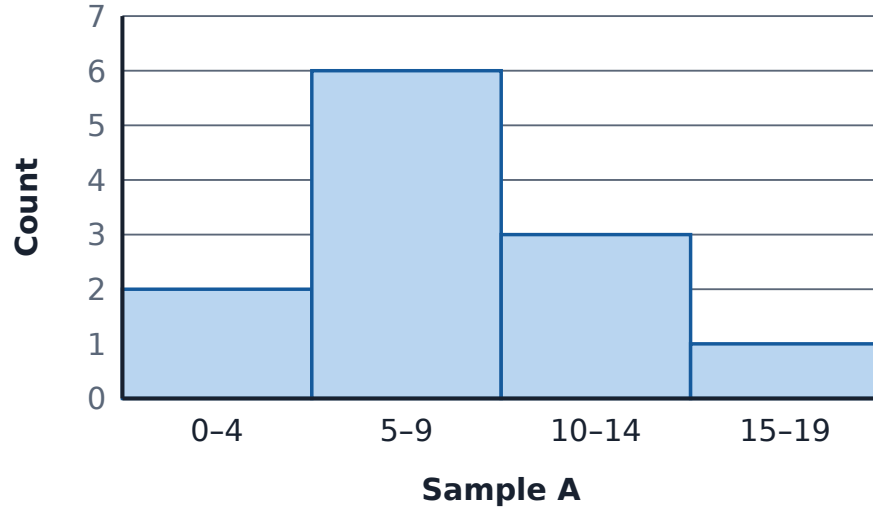


A. Sample A

B. Sample B

C. They are the same.

In which interval is the median of Sample A?



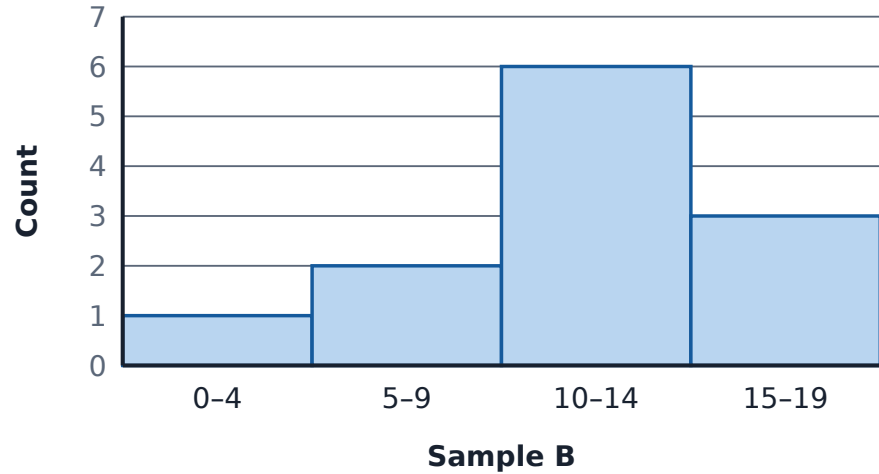
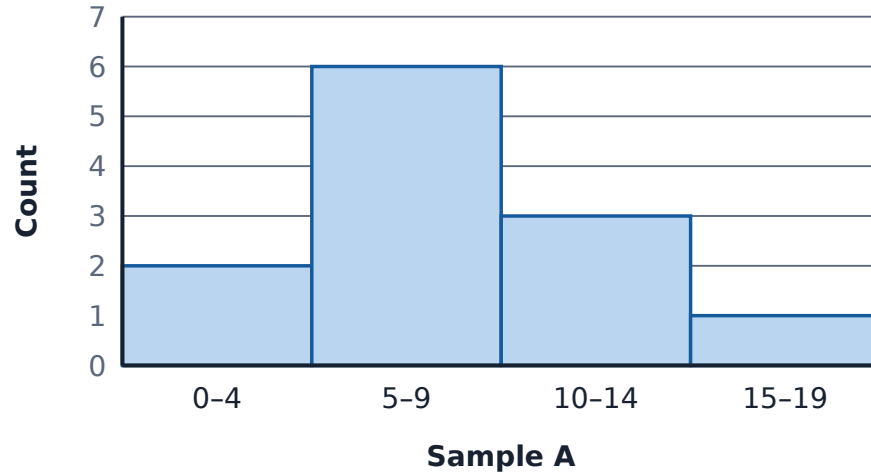
A. 0-4

B. 5-9

C. 10-14

D. 15-19

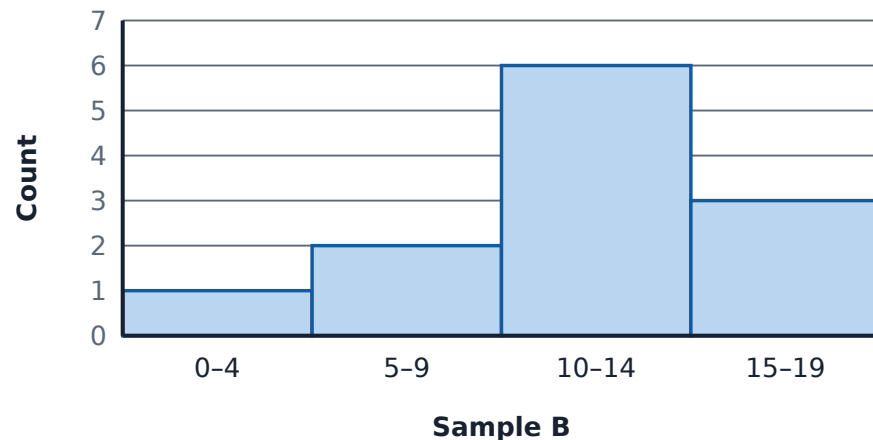
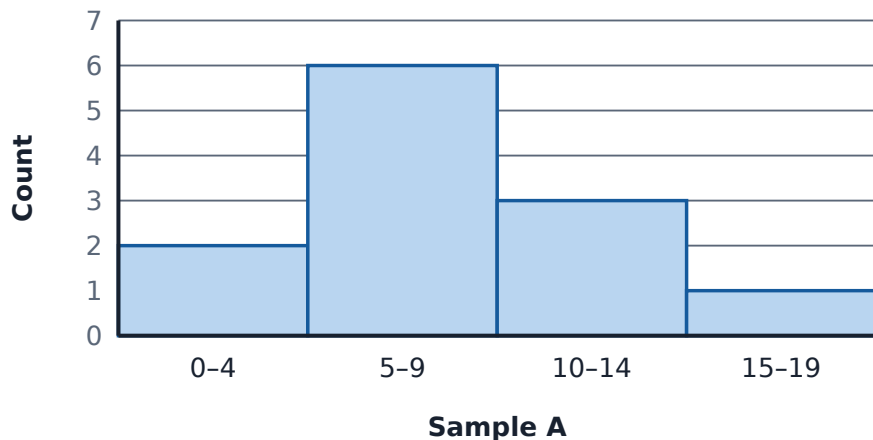
How many values are in each sample?



Values: _____

True or false?

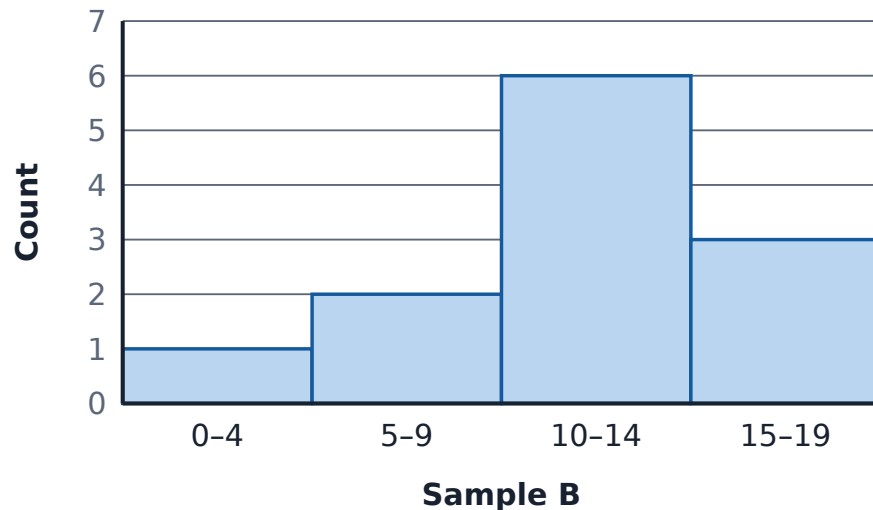
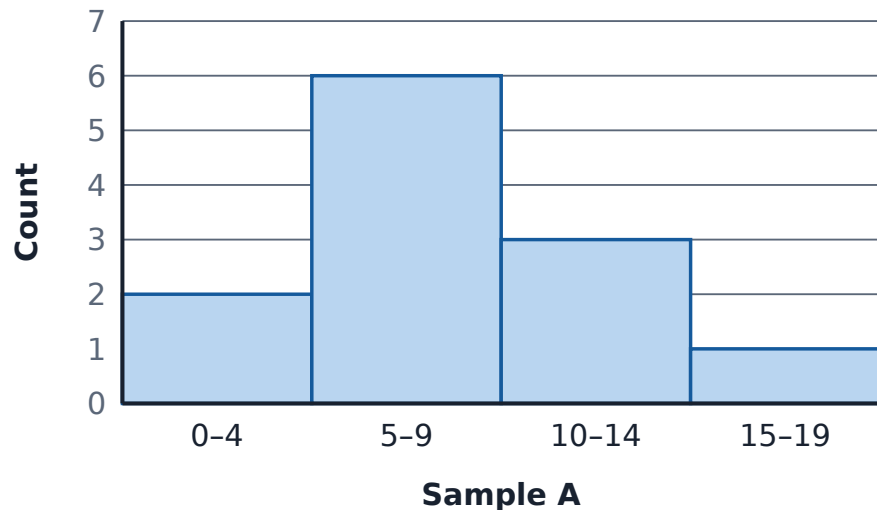
Sample B has more values from 15 to 19 than Sample A.



A. True

B. False

In which interval is the median of Sample B?



A. 0-4

B. 5-9

C. 10-14

D. 15-19

FORWARD 2

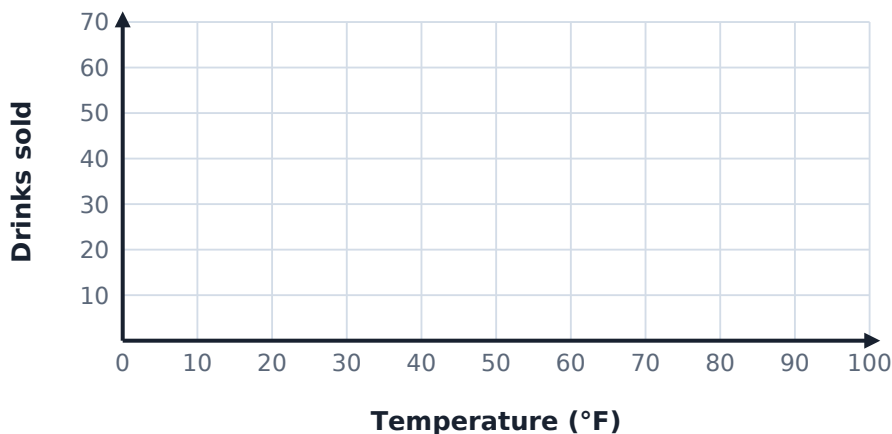
Display numerical data for two variables in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Make a scatter plot of the data.

Describe the association between temperature and drinks sold.

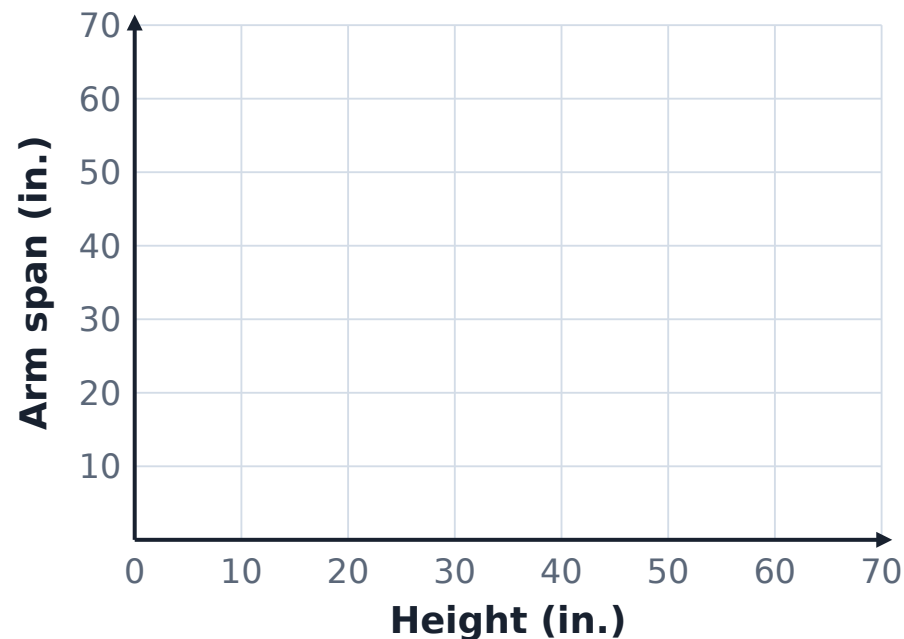
Temp (°F)	60	70	75	85	90
Drinks sold	20	35	40	55	65



Association: _____

Make a scatter plot of the data.

Height (in.)	50	54	58	62
Arm span (in.)	49	55	57	63



A data set gives the height and the shoe size of 20 students.

Which display shows how the two variables are related?

A. Dot plot

B. Scatter plot

C. Box plot

D. Histogram

True or false?

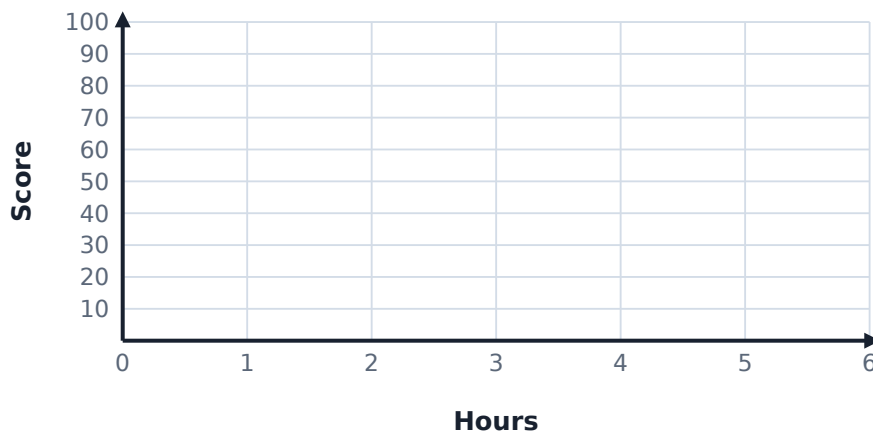
A histogram displays one numerical variable, while a scatter plot displays two numerical variables for each person or object.

- A.** True
- B.** False

Make a scatter plot of the data.

Which point does not fit the pattern?

Hours	1	2	3	4	5
Score	62	68	40	80	86



Point: _____

Display numerical data in a box plot

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	Halfway between two numbers	2.MD.B.6
BACKWARD 3	Reading a number line scale	4.MD.A.2
BACKWARD 4	Finding the median	6.SP.B.5.c
FORWARD 1	Compare populations with box plots	7.SP.B.4
FORWARD 2	Spread and outliers in scatter plots — nearest related (no direct G...	8.SP.A.1

SET 74 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

Data: 3, 5, 7, 8, 10, 12, 15

Make a box plot of the data on the number line.



Data: 12, 14, 15, 18, 20, 21, 25, 28

Make a box plot of the data on the number line.



Data: 40, 45, 45, 50, 55, 60, 70

Make a box plot of the data on the number line.



Data: 2, 4, 4, 5, 6, 8, 9, 11, 12

Make a box plot of the data on the number line.



Data: 21, 23, 24, 26, 28, 30, 31, 35

Make a box plot of the data on the number line.



SET 74 • 6.SP.B.4

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

15, 8, 23, 4, 16

Order: _____

Which list is in order from least to greatest?

A. 27, 21, 12, 7

B. 7, 21, 12, 27

C. 12, 7, 21, 27

D. 7, 12, 21, 27

True or false?

These numbers are in order from least to greatest: 30, 33, 31, 35

- A. True
- B. False

Order from least to greatest.

62, 26, 60, 20, 66

Order: _____

Which number is least?

A. 108

B. 118

C. 180

D. 801

SET 74 • 6.SP.B.4

BACKWARD 2

Halfway between two numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What number is halfway between 12 and 18?

Number: _____

What number is halfway between points A and B?



Number: _____

True or false?

The number halfway between 40 and 50 is 45.

- A. True
- B. False

What number is halfway between 6 and 10?

- A.** 4
- B.** 7
- C.** 8
- D.** 16

What number is halfway between 0 and 30?

Number: _____

SET 74 • 6.SP.B.4

BACKWARD 3

Reading a number line scale • Grade 4 • Measurement & Data • 4.MD.A.2

What number is at point A?



A = _____

What number is at point B?



B = _____

True or false?

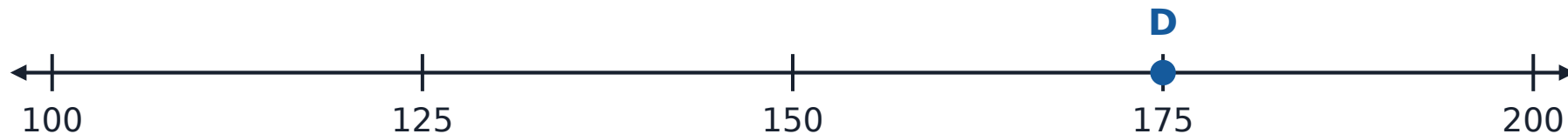
Point C is at 12.



A. True

B. False

What number is at point D?



A. 150

B. 170

C. 175

D. 180

What number is at point E?



E = _____

SET 74 • 6.SP.B.4

BACKWARD 4

Finding the median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

3, 5, 7, 8, 10, 12, 15

Median: _____

Find the median.

12, 14, 15, 18, 20, 21, 25, 28

Median: _____

True or false?

The median of 40, 45, 45, 50, 55, 60, 70 is 50.

A. True

B. False

What is the median of 2, 4, 4, 5, 6, 8, 9, 11, 12?

A. 5

B. 6

C. 7

D. 8

Find the median of the lower half.

3, 5, 7

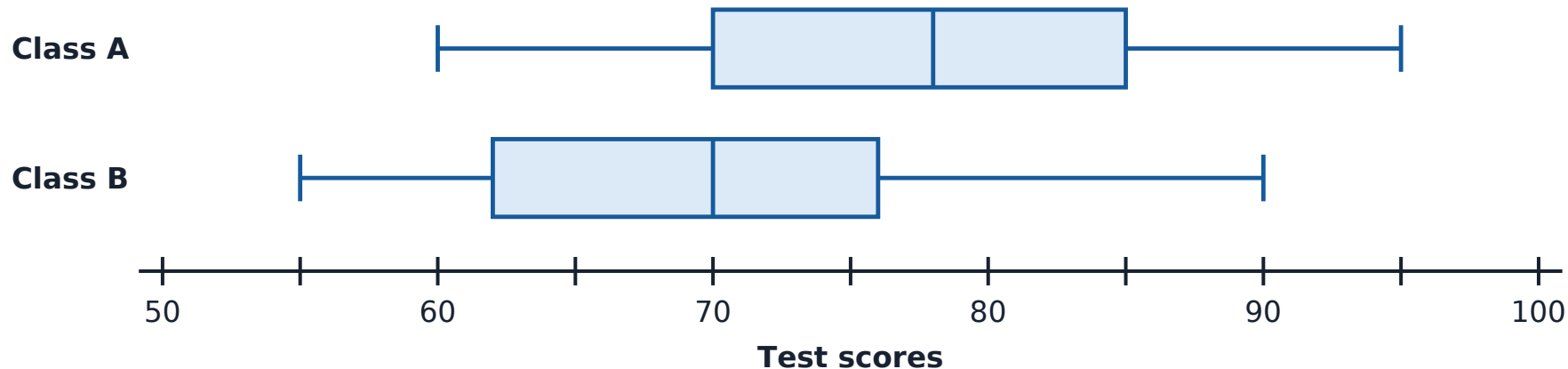
Median: _____

SET 74 • 6.SP.B.4

FORWARD 1

Compare populations with box plots • Grade 7 • Statistics & Probability •
7.SP.B.4

Which class has the greater median score?

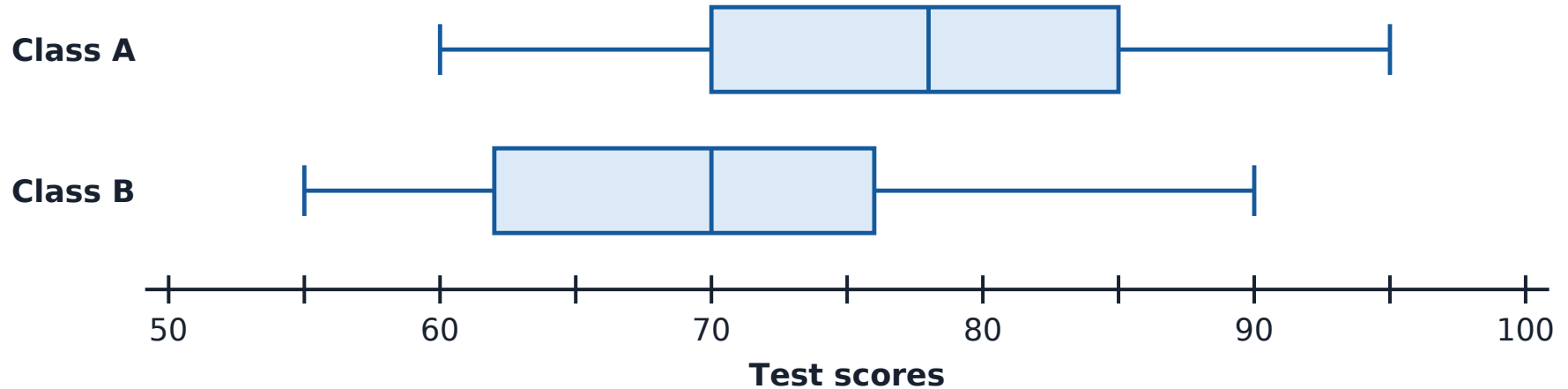


A. Class A

B. Class B

C. They are the same.

By how many points do the medians differ?



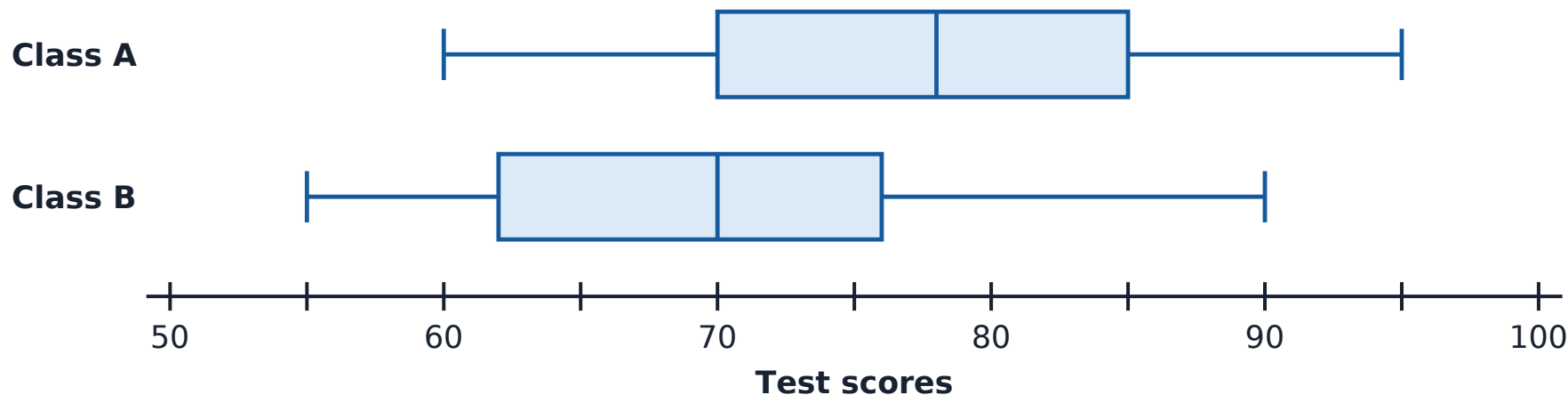
A. 3

B. 5

C. 8

D. 15

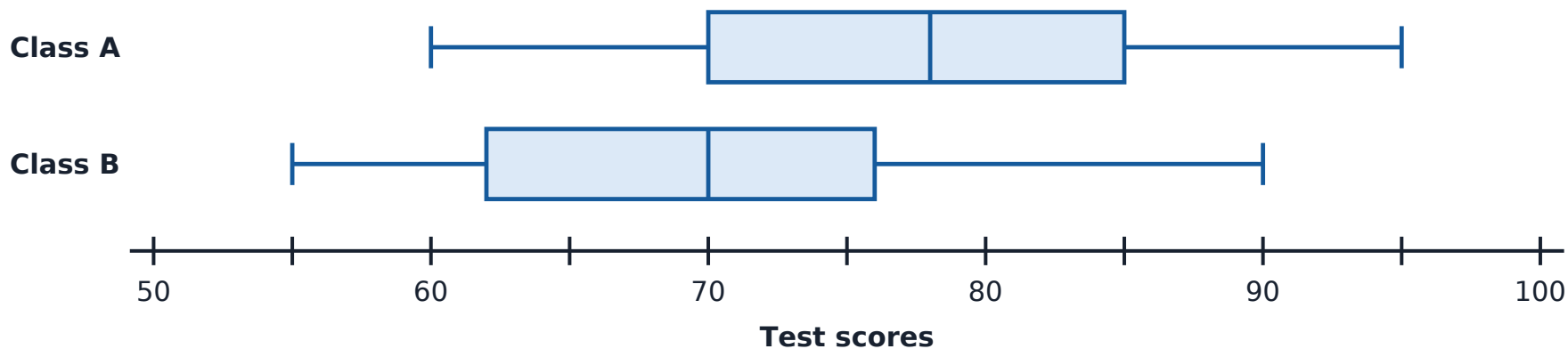
What is the interquartile range of Class A's scores?



IQR: _____

True or false?

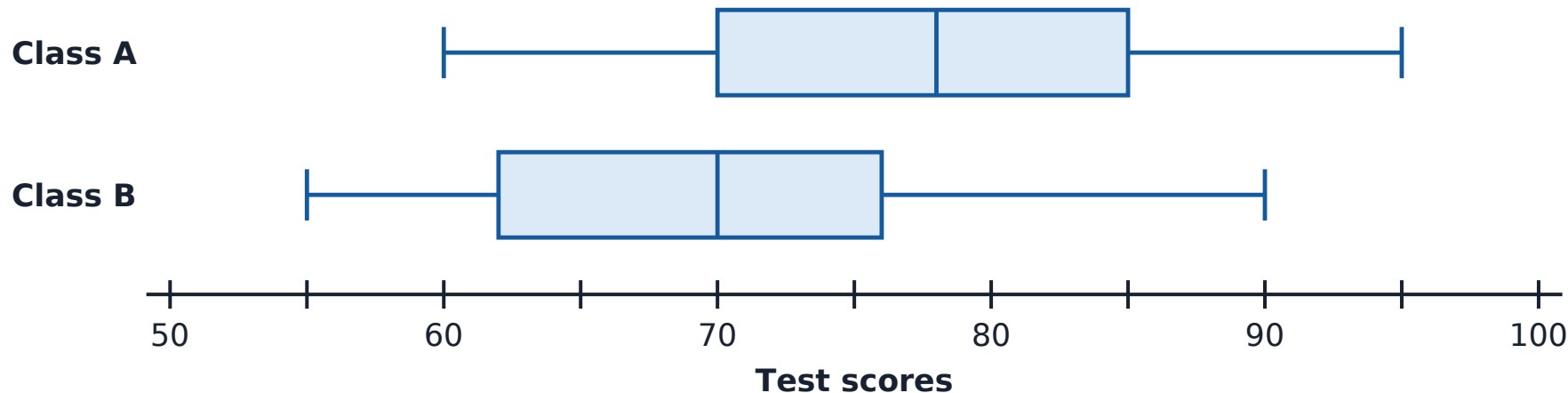
At least half of Class A scored 78 or higher.



A. True

B. False

Which class had the highest single score?



Class: _____

FORWARD 2

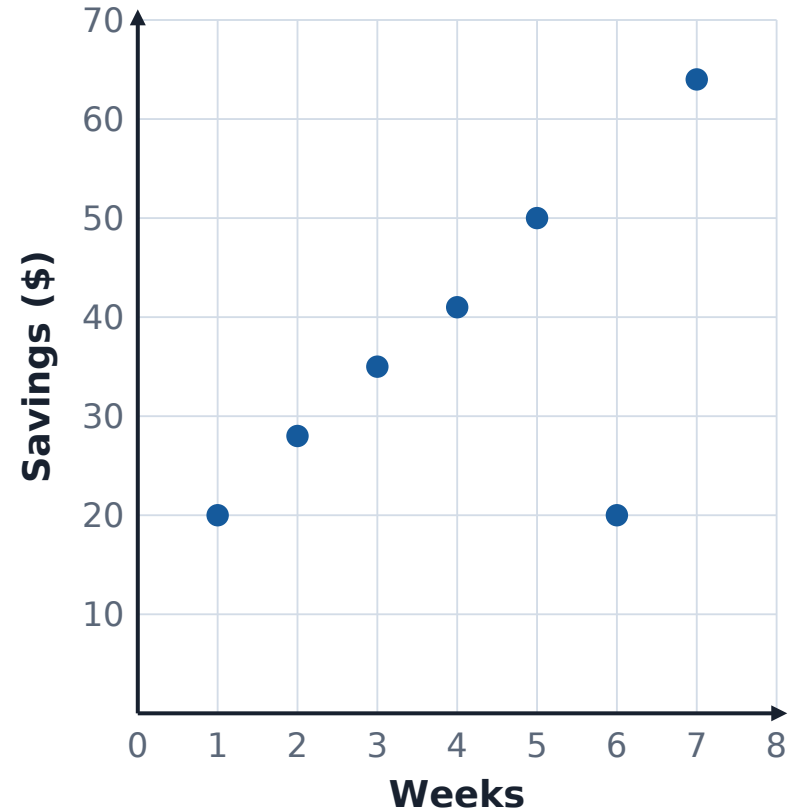
Spread and outliers in scatter plots • Grade 8 • Statistics & Probability •
8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Which point is an outlier?

Write its coordinates.

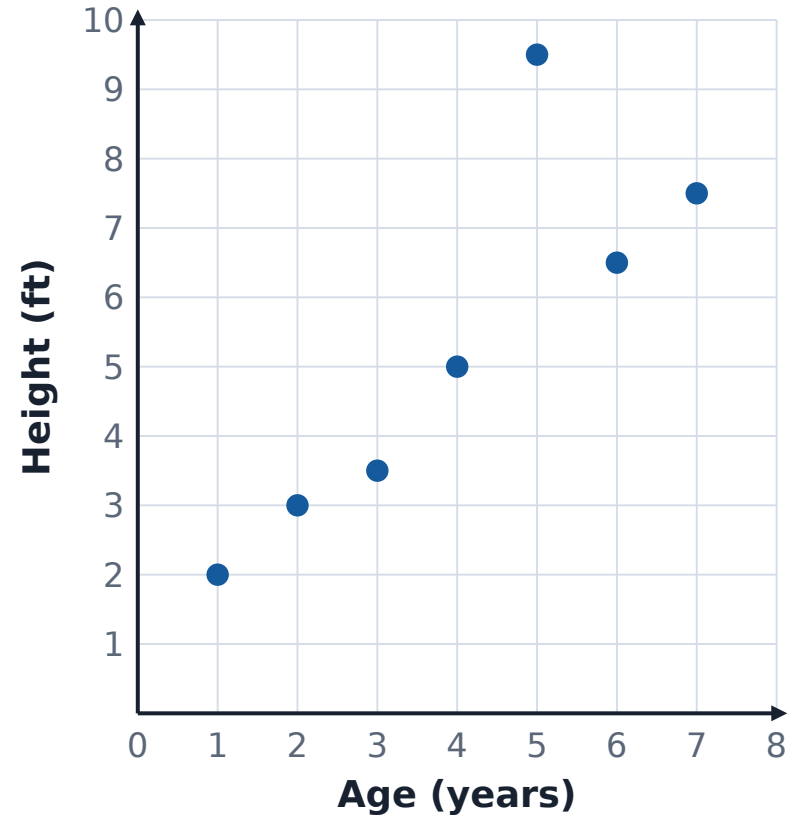
Outlier: _____



Which point is an outlier?

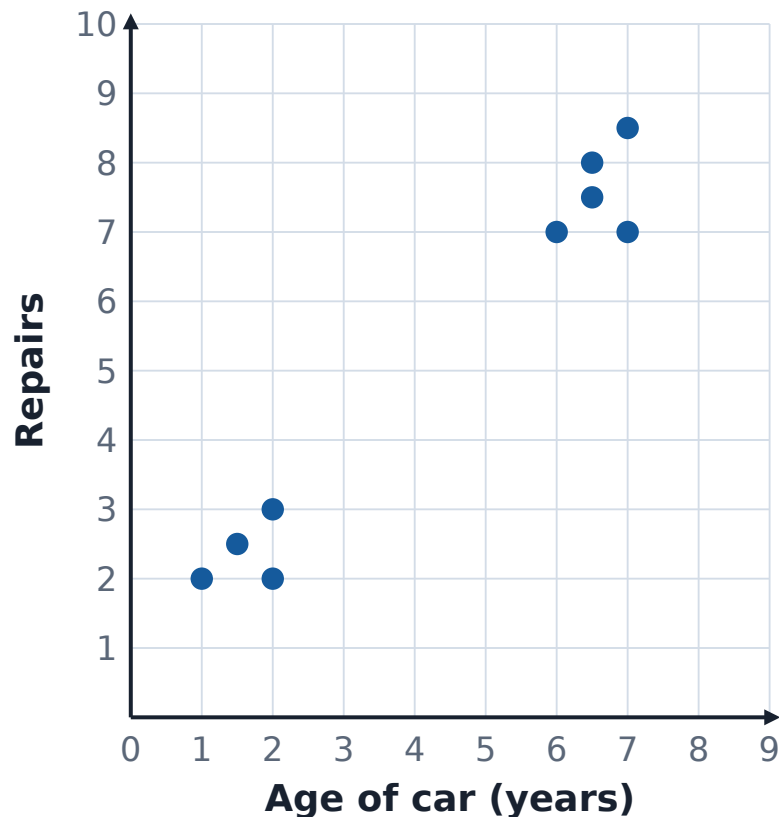
Write its coordinates.

Outlier: _____



Which describes the scatter plot?

- A.** A single outlier
- B.** A strong positive association
- C.** No pattern
- D.** Two clusters



True or false?

An outlier in a scatter plot is a point that is far from the overall pattern.

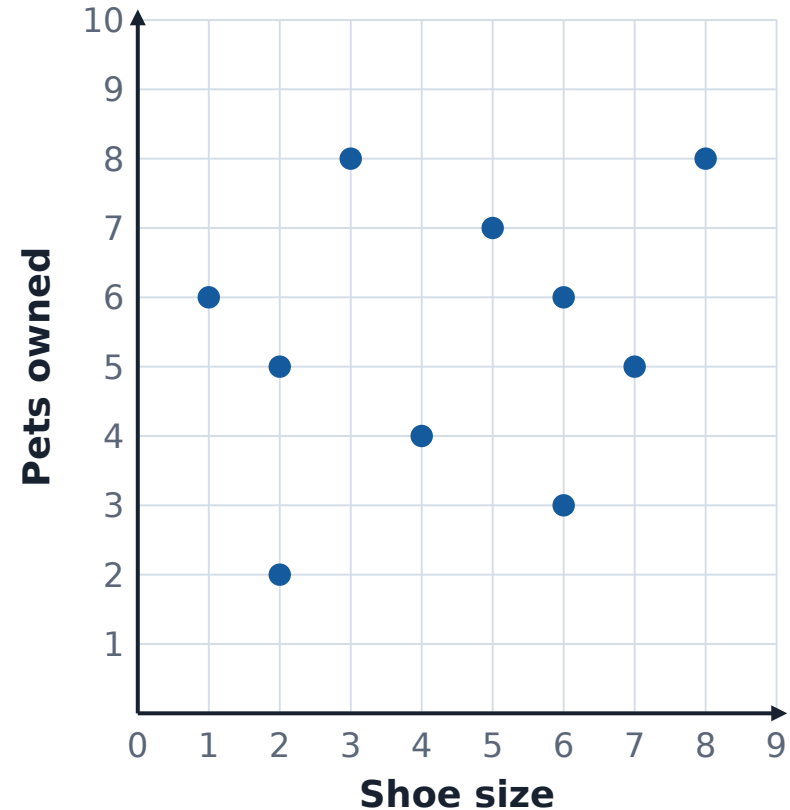
- A.** True
- B.** False

True or false?

The scatter plot shows no association.

A. True

B. False



Report the number of observations

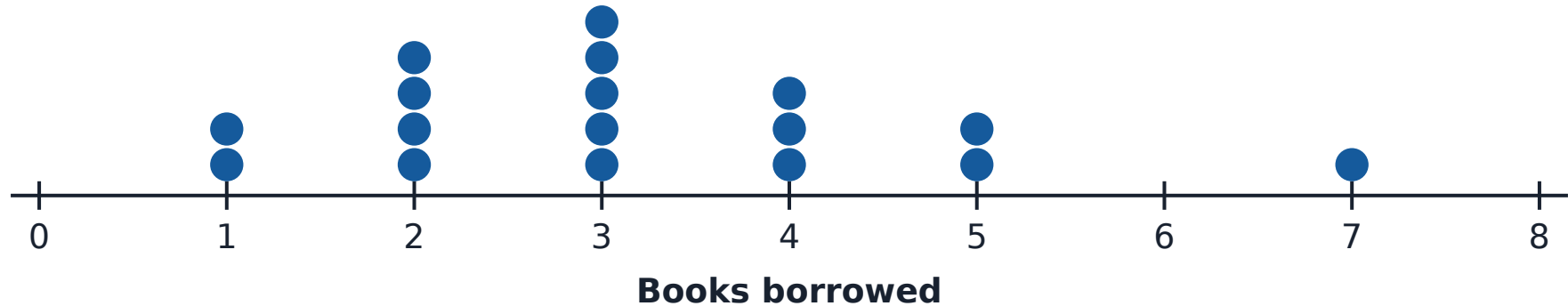
MAIN	Grade 6 main question	6.SP.B.5.a
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Totals from a bar graph	3.MD.B.3
BACKWARD 3	Adding several two-digit numbers	2.NBT.B.6
FORWARD 1	Use a random sample to make inferences	7.SP.A.2
FORWARD 2	Relative frequencies in two-way tables	8.SP.A.4

SET 75 • 6.SP.B.5.a

MAIN

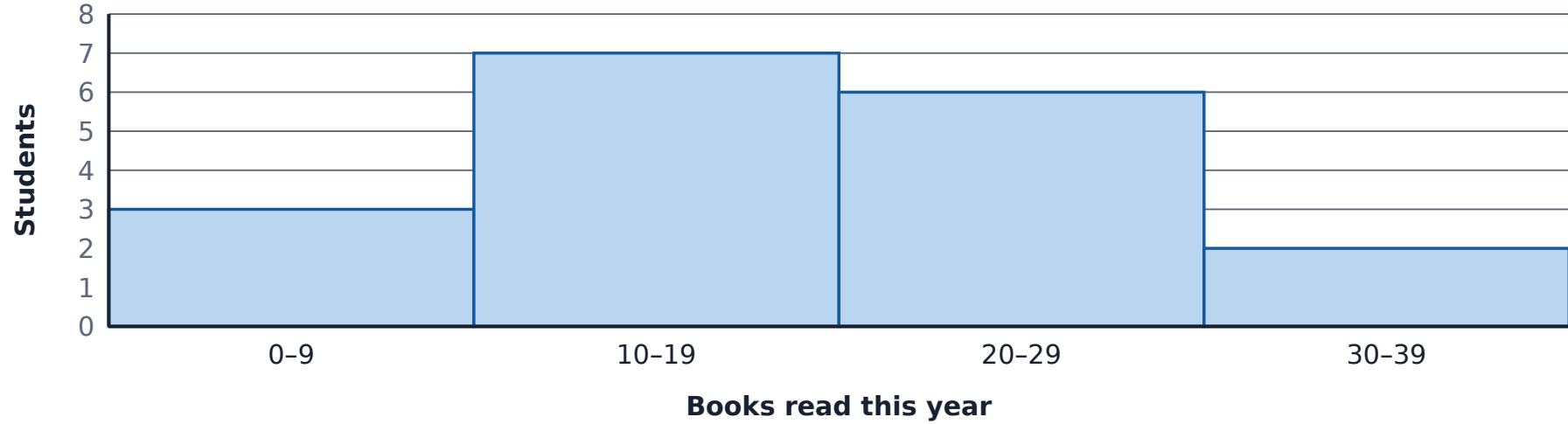
Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.a

The dot plot shows how many books students borrowed.
How many observations are in the data set?



Observations: _____

How many students are represented in the histogram?



Students: _____

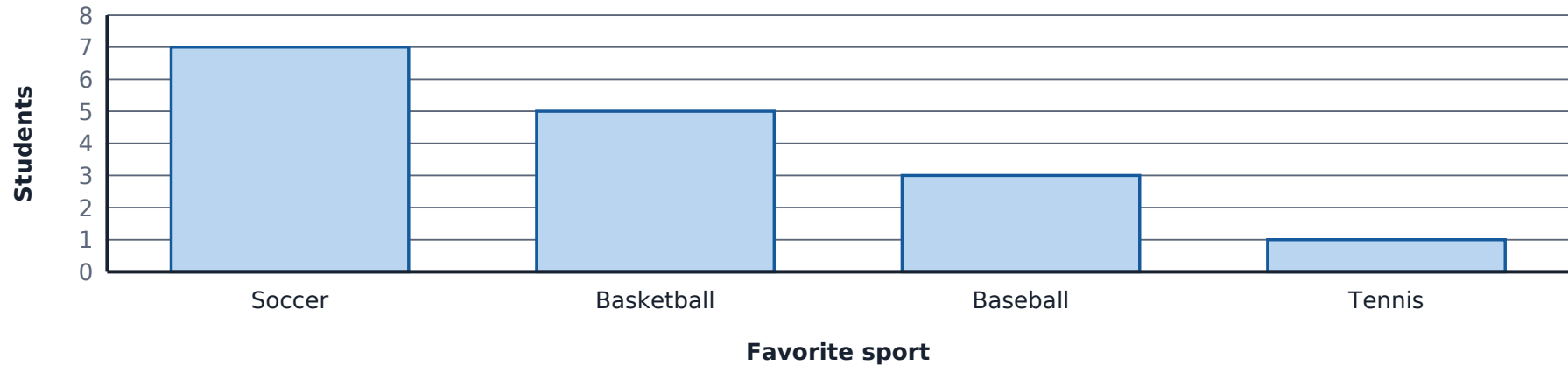
The table shows how many siblings students have.
How many observations are in the data set?

Siblings	0	1	2	3
Students	4	9	6	2

Observations: _____

The bar graph shows survey answers.

How many observations are in the data set?



Observations: _____

Data: 12, 15, 9, 15, 20, 11, 13, 18, 15, 10

How many observations are in the data set?

Observations: _____

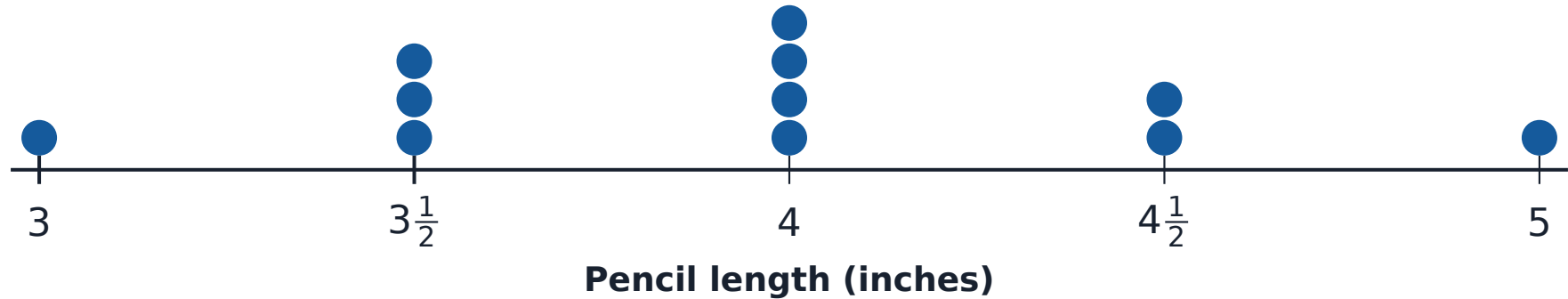
SET 75 • 6.SP.B.5.a

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

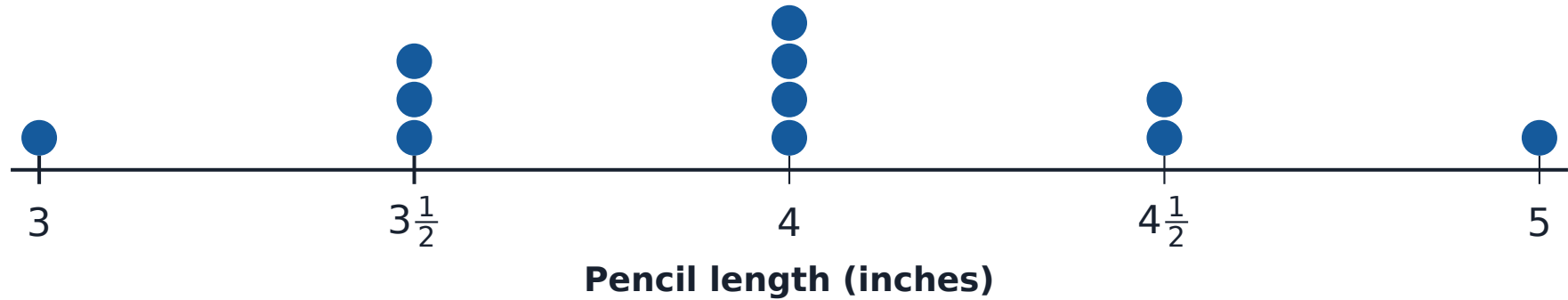
The line plot shows pencil lengths.

How many pencils are 4 inches long?



Pencils: _____

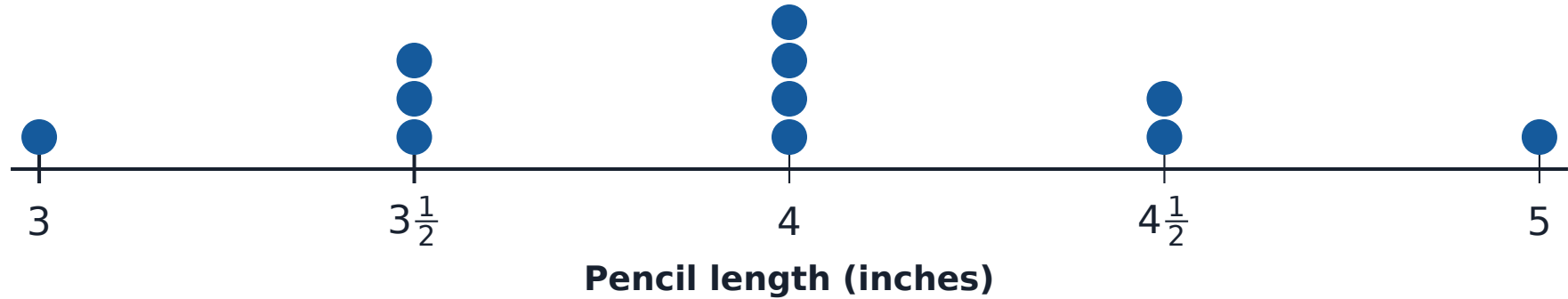
The line plot shows pencil lengths.
How many pencils were measured?



Pencils: _____

True or false?

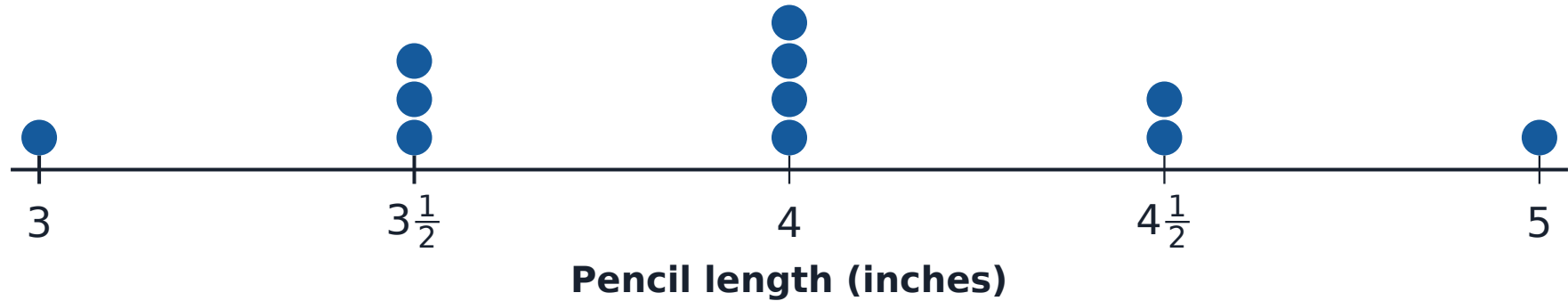
Two pencils are $4\frac{1}{2}$ inches long.



A. True

B. False

How many pencils are longer than 4 inches?



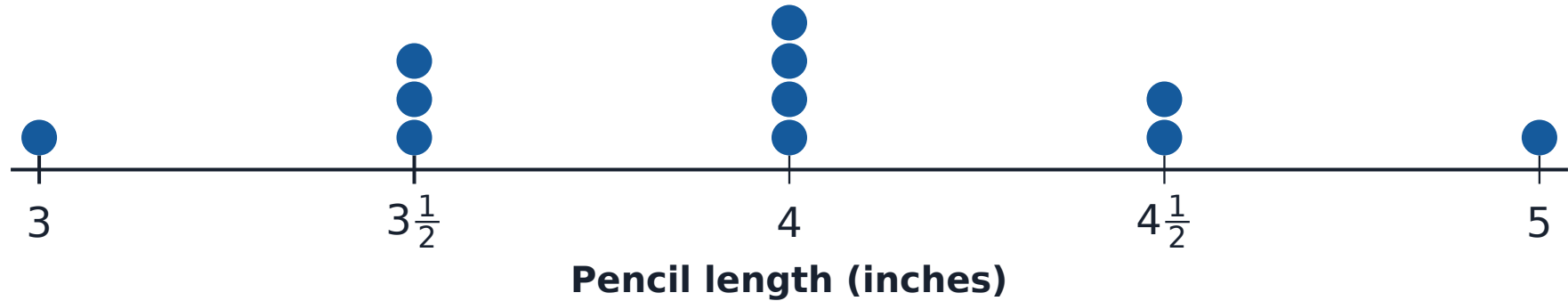
A. 2

B. 3

C. 4

D. 7

How many pencils are 5 inches long?



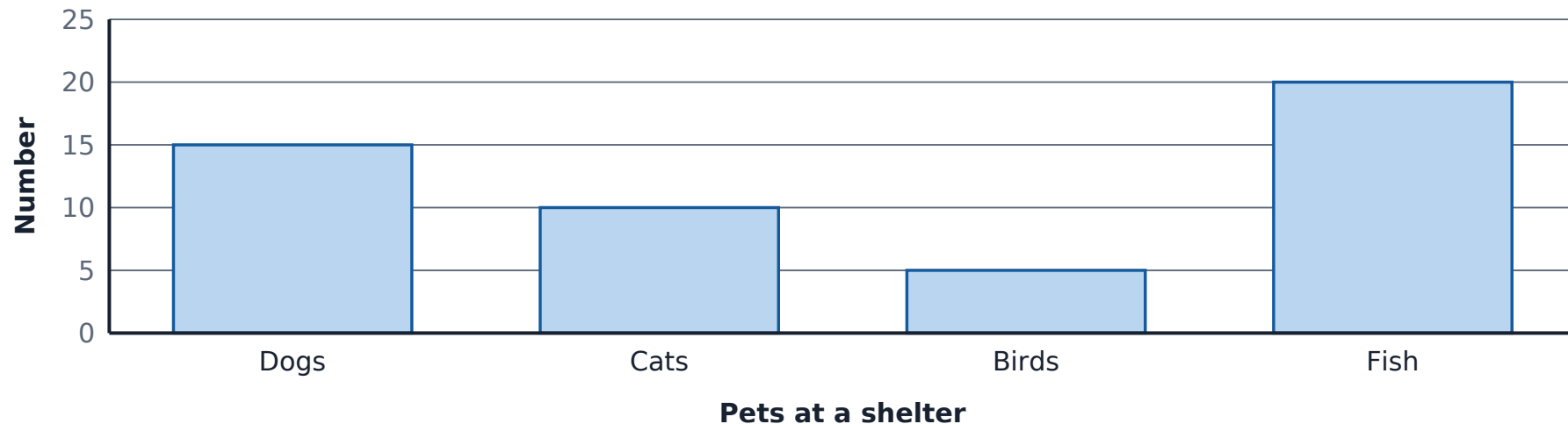
Pencils: _____

SET 75 • 6.SP.B.5.a

BACKWARD 2

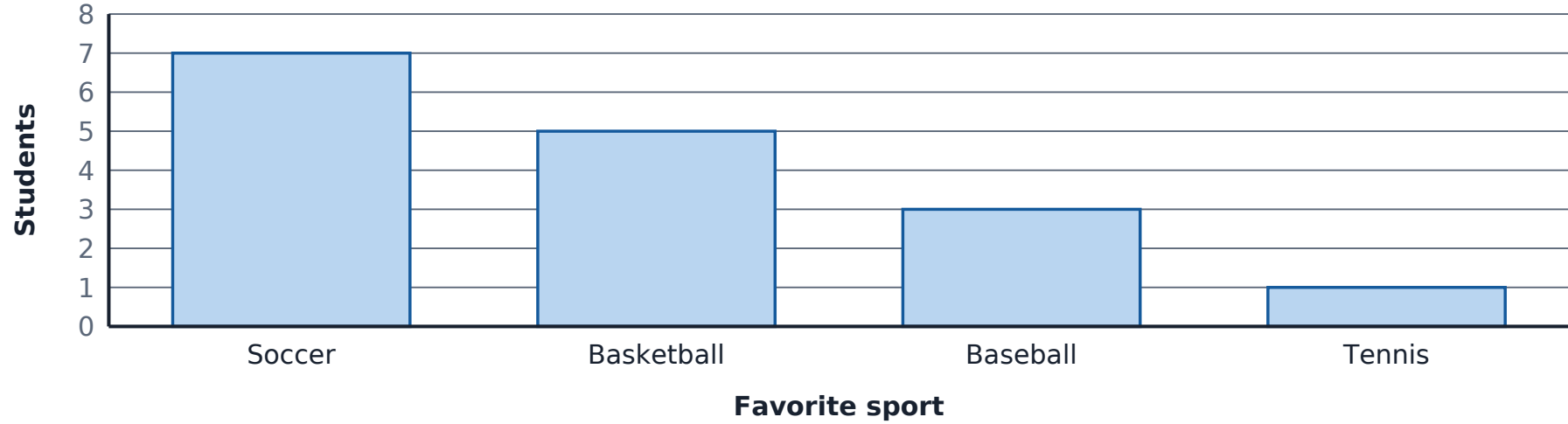
Totals from a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many pets are at the shelter in all?



Pets: _____

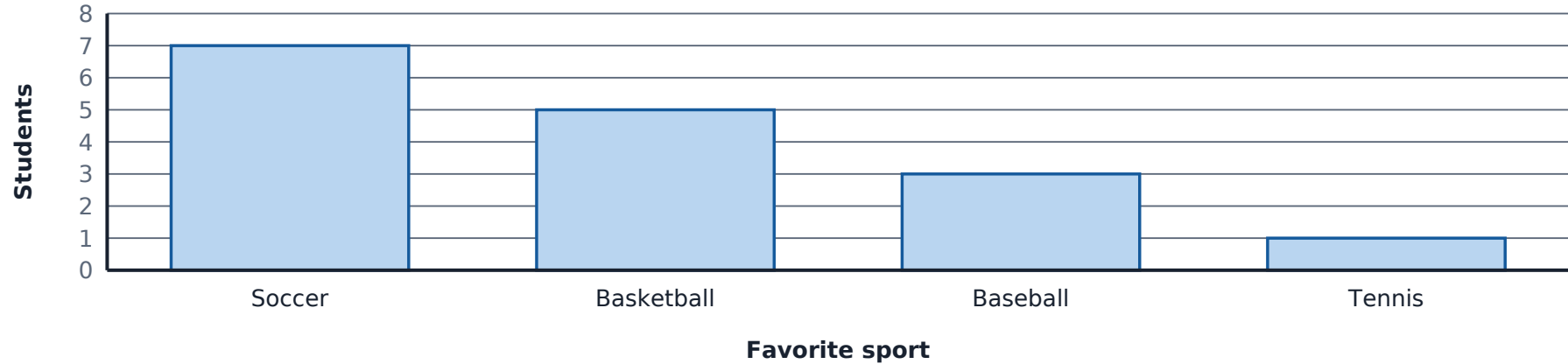
How many students chose soccer or tennis?



Students: _____

True or false?

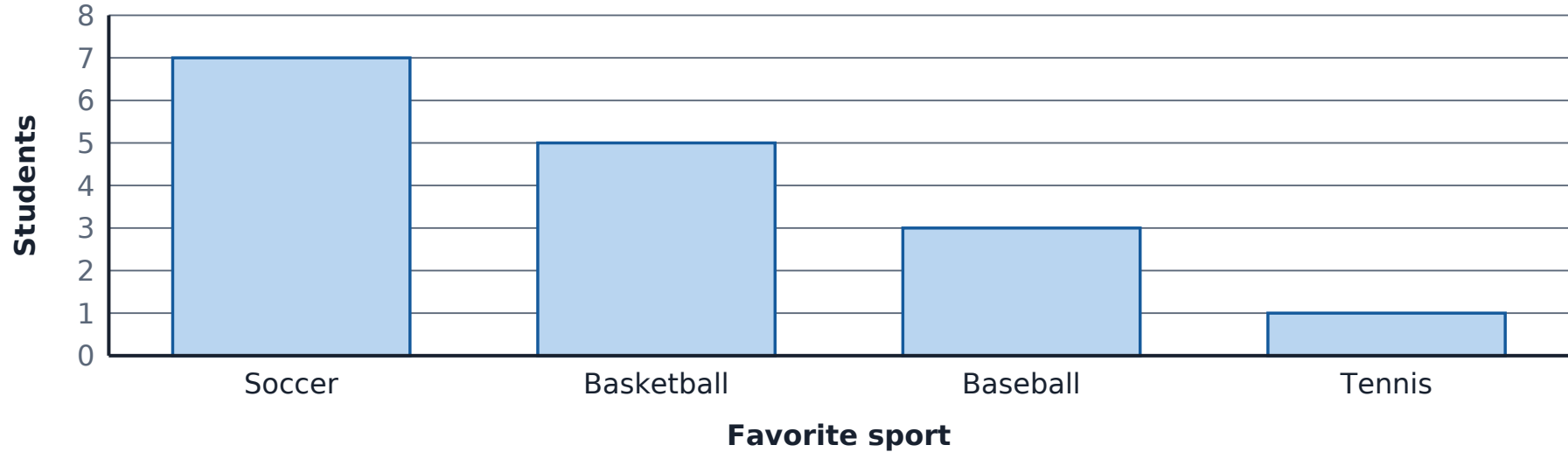
12 students chose soccer or basketball.



A. True

B. False

How many students did NOT choose soccer?



A. 5

B. 7

C. 9

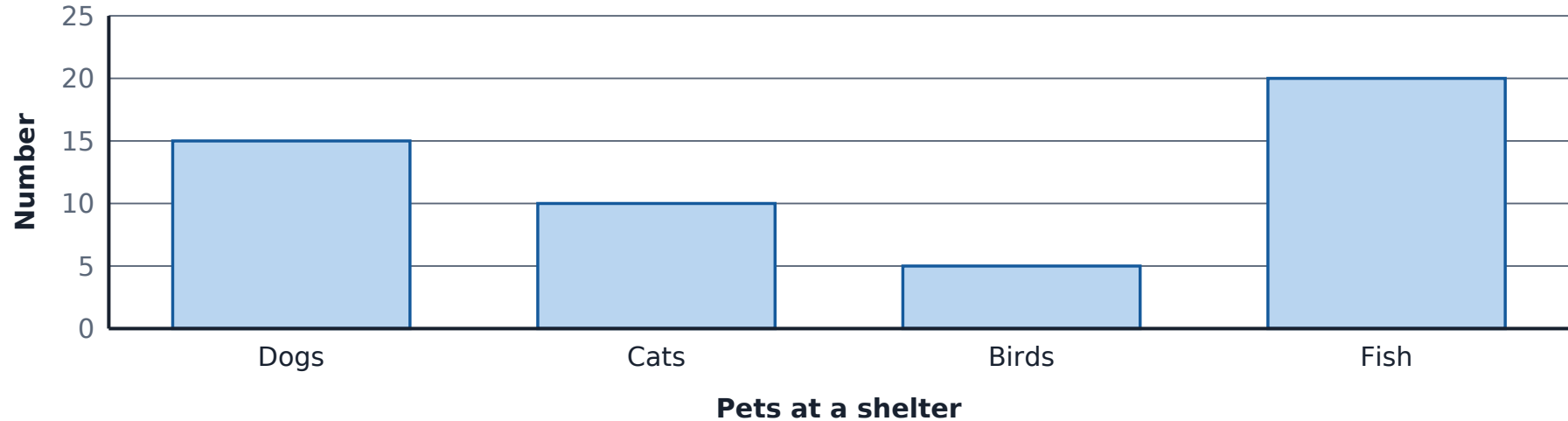
D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q4/5

How many dogs and cats are at the shelter together?



Number: _____

SET 75 • 6.SP.B.5.a

BACKWARD 3

Adding several two-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6

Add.

$$14 + 23 + 31 + 12$$

Sum: _____

Add.

$$25 + 18 + 30$$

Sum: _____

True or false?

$$16 + 16 + 20 + 8 = 60$$

A. True

B. False

$$41 + 27 + 13 + 9 = ?$$

- A.** 80
- B.** 90
- C.** 91
- D.** 100

Add.

$$35 + 22 + 18 + 15$$

Sum: _____

SET 75 • 6.SP.B.5.a

FORWARD 1

Use a random sample to make inferences • Grade 7 • Statistics & Probability •
7.SP.A.2

A researcher surveys 80 randomly chosen students out of 900.

What is the sample size?

A. 80

B. 820

C. 900

D. 980

True or false?

A larger random sample generally gives a better estimate about a population.

- A.** True
- B.** False

In a random sample of 25 students out of 500, 5 like jazz.

Estimate how many of the 500 students like jazz.

Estimate: _____

In a random sample of 50 voters, 30 support a new park. There are 2,000 voters.

About how many voters support the park?

A. 30

B. 600

C. 1,200

D. 1,500

Two random samples of 20 students each found 12 and 14 students who like pizza best.

What is the mean of the two sample results?

Mean: _____

SET 75 • 6.SP.B.5.a

FORWARD 2

Relative frequencies in two-way tables • Grade 8 • Statistics & Probability •
8.SP.A.4

What fraction of 7th graders have a pet? What fraction of 8th graders have a pet?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

Grade 7: _____

Grade 8: _____

Compare the relative frequencies. Which grade has a greater fraction of pet owners?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

A. Grade 7

B. Grade 8

C. They are the same.

What is the relative frequency of instrument players who also play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

A. 0.35

B. 0.47

C. 0.7

D. 14

Among students with no instrument, what fraction play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

Fraction: _____

True or false?

Students with no instrument are more likely to play sports than students who play an instrument.

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

A. True

B. False

Describe the attribute measured and its units

MAIN	Grade 6 main question	6.SP.B.5.b
BACKWARD 1	Choosing measurement units	4.MD.A.1
BACKWARD 2	What a bar graph shows	3.MD.B.3
BACKWARD 3	Choosing a measuring tool	2.MD.A.1
FORWARD 1	How data are collected: sampling methods	7.SP.A.1
FORWARD 2	Association between two categorical variables	8.SP.A.4

SET 76 • 6.SP.B.5.b

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.b

A data set lists the heights of 25 plants in centimeters.
What attribute is being measured, and in what unit?

Attribute: _____

Unit: _____

A data set records the time each student spends on homework.

Which unit is most likely?

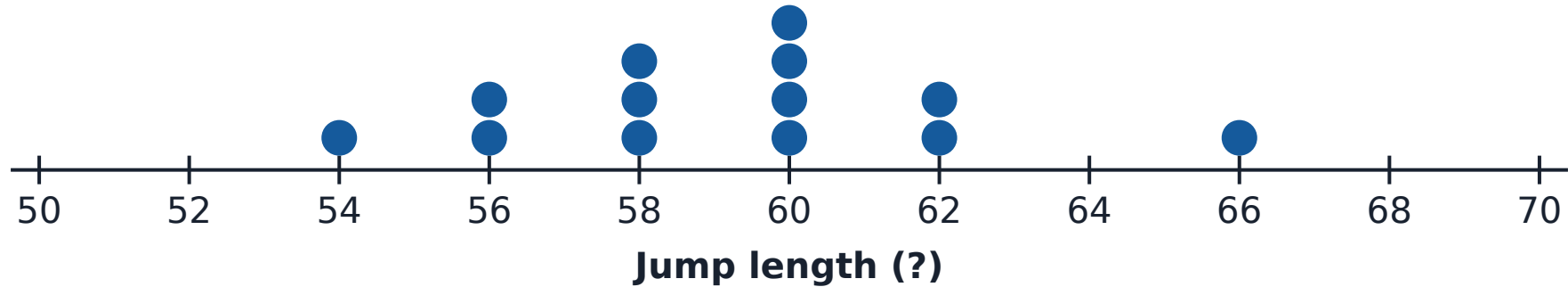
A. Degrees

B. Pounds

C. Inches

D. Minutes

The dot plot shows how far students jumped.
Which unit is most likely?



- A.** Inches **B.** Seconds **C.** Pounds **D.** Miles

A data set lists the mass of each of 30 apples.

What attribute is being measured, and what is a reasonable unit?

Attribute: _____

Unit: _____

A data set records how long each student's trip to school takes.

Which describes the attribute and its unit?

- A.** Distance; minutes
- B.** Time; minutes
- C.** Speed; pounds
- D.** Time; miles

SET 76 • 6.SP.B.5.b

BACKWARD 1

Choosing measurement units • Grade 4 • Measurement & Data • 4.MD.A.1

Which unit is best for measuring the length of a pencil?

- A.** Centimeters
- B.** Kilograms
- C.** Meters
- D.** Kilometers

True or false?

A kilogram is heavier than a gram.

A. True

B. False

Which unit measures liquid volume?

- A.** Meters
- B.** Liters
- C.** Grams
- D.** Seconds

Name one unit used to measure time.

Unit: _____

True or false?

An inch is longer than a foot.

A. True

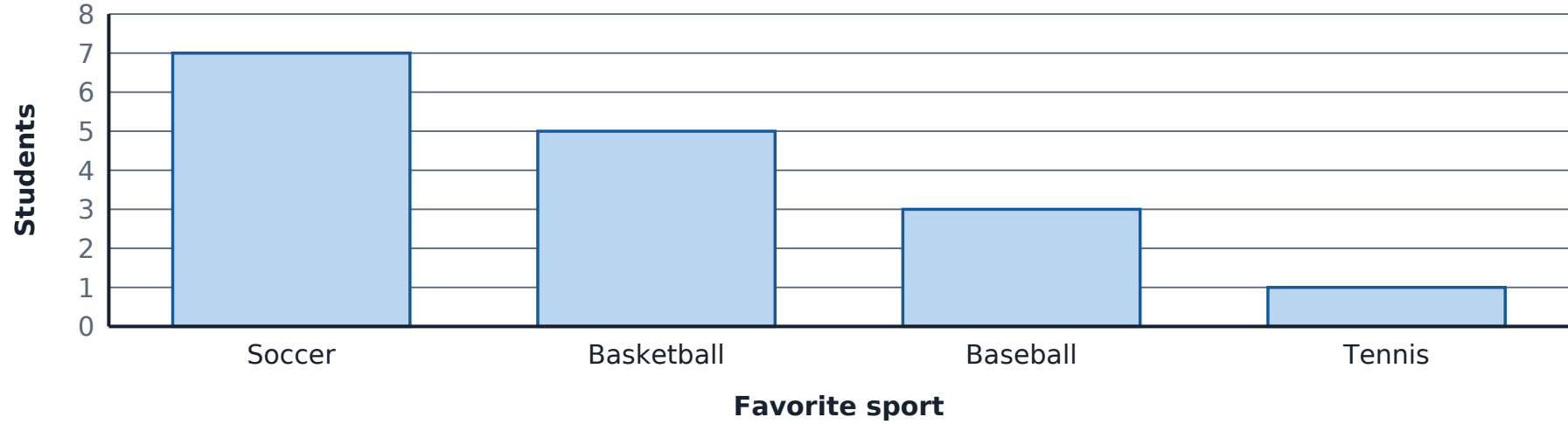
B. False

SET 76 • 6.SP.B.5.b

BACKWARD 2

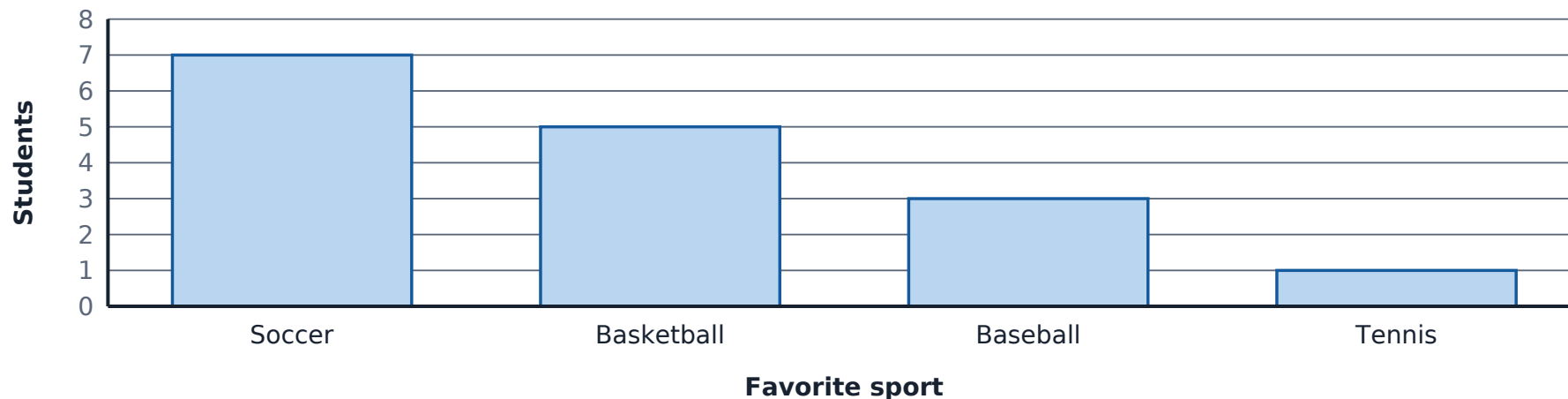
What a bar graph shows • Grade 3 • Measurement & Data • 3.MD.B.3

How many categories does the graph show?



Categories: _____

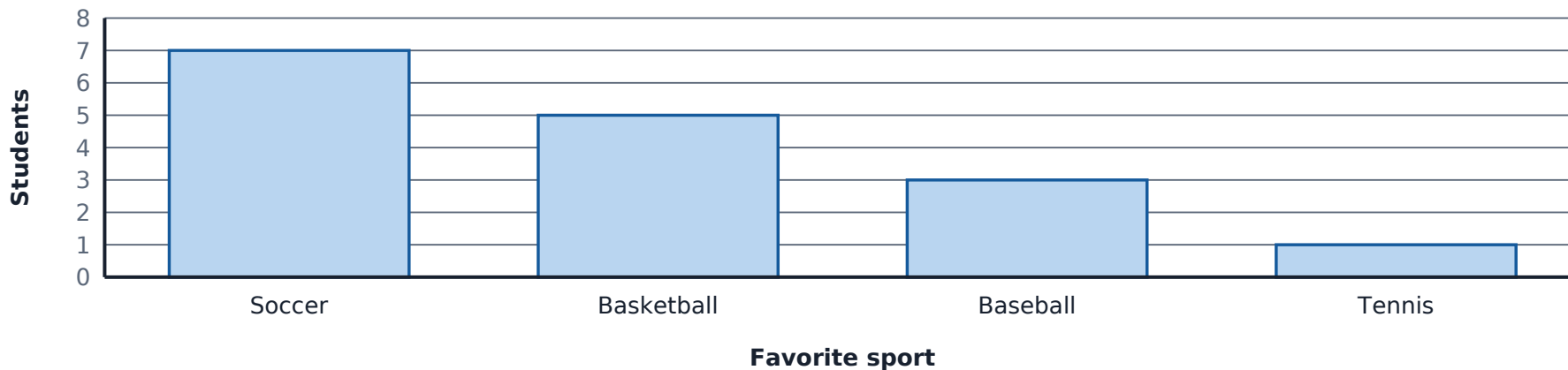
What does the height of each bar show?



- A.** The number of sports
- B.** The score of a game
- C.** The length of a game
- D.** The number of students

True or false?

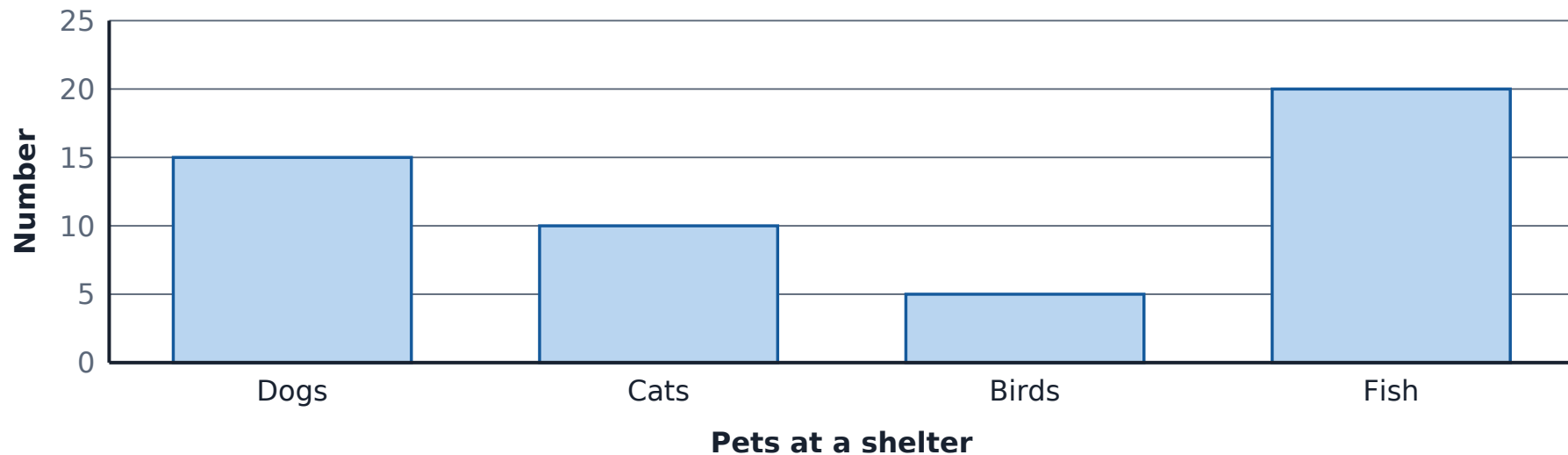
"Tennis" is one of the categories in the graph.



A. True

B. False

Which is a category in the graph?



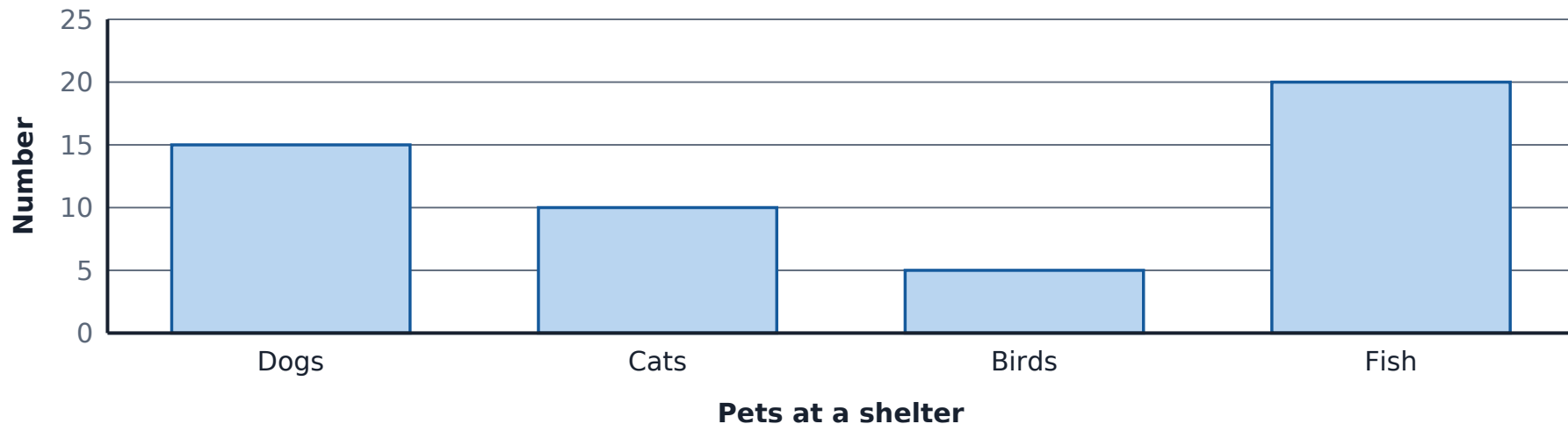
A. 25

B. Birds

C. Horses

D. Soccer

What is being counted in the graph?



Answer: _____

SET 76 • 6.SP.B.5.b

BACKWARD 3

Choosing a measuring tool • Grade 2 • Measurement & Data • 2.MD.A.1

Which tool measures the length of a table?

- A.** Tape measure
- B.** Thermometer
- C.** Scale
- D.** Clock

Which tool measures how heavy an apple is?

- A.** Ruler
- B.** Clock
- C.** Measuring cup
- D.** Scale

True or false?

A clock measures time.

A. True

B. False

Which tool measures temperature?

- A.** Thermometer
- B.** Scale
- C.** Ruler
- D.** Yardstick

Name a tool that measures the length of a pencil.

Tool: _____

SET 76 • 6.SP.B.5.b

FORWARD 1

How data are collected: sampling methods • Grade 7 • Statistics & Probability •
7.SP.A.1

Which method gives a random sample of the students in a school?

- A.** Ask the students in the library
- B.** Ask the first 20 students to arrive
- C.** Ask your friends
- D.** Draw names from a hat that holds every student's name

True or false?

Surveying people who walk out of a gym is a good way to learn how often all people in a town exercise.

A. True

B. False

A town wants to know how residents feel about a new library.

Describe a way to choose a random sample.

Method: _____

Which sample is most likely to be biased?

- A.** Asking only people at a dog park whether they like dogs
- B.** Picking every 10th name from a school list
- C.** Choosing 50 residents at random from a town list
- D.** Drawing 30 names from a hat of all members

True or false?

In a random sample, every member of the population has an equal chance of being chosen.

- A. True
- B. False

SET 76 • 6.SP.B.5.b

FORWARD 2

Association between two categorical variables • Grade 8 • Statistics & Probability • 8.SP.A.4

Which pair of variables is categorical and could be shown in a two-way table?

- A.** Hours slept and test score
- B.** Height and weight
- C.** Age and arm span
- D.** Grade level and way of getting to school

What fraction of 6th graders walk to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

Fraction: _____

What fraction of 8th graders walk to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

Fraction: _____

What is the relative frequency of 8th graders who ride to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

A. $\frac{1}{10}$

B. $\frac{5}{23}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

True or false?

8th graders are more likely to walk to school than 6th graders.

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

A. True

B. False

Find the mean of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Adding multi-digit numbers	4.NBT.B.4
BACKWARD 2	Dividing a total into equal groups	4.NBT.B.6
BACKWARD 3	Sharing a total equally	5.MD.B.2
FORWARD 1	Compare populations using means	7.SP.B.4
FORWARD 2	Use a linear model to make predictions — nearest related (no dire...	8.SP.A.3

SET 77 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 7, 9, 10, 15

Mean: _____

Five test scores are 82, 90, 75, 88, and 95.

What is the mean score?

Mean: _____

Find the mean.

12, 15, 18, 19

A. 15

B. 16

C. 16.5

D. 64

The dot plot shows goals scored in 8 games.
What is the mean number of goals?



Mean: _____

Find the mean.

3.5, 4.2, 5.0, 6.1, 6.2

Mean: _____

SET 77 • 6.SP.B.5.c

BACKWARD 1

Adding multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Add.

$$82 + 90 + 75 + 88 + 95$$

Sum: _____

Add.

$$145 + 238 + 97$$

Sum: _____

True or false?

$$56 + 64 + 70 = 190$$

A. True

B. False

$$125 + 175 + 200 = ?$$

A. 400

B. 450

C. 500

D. 550

Add.

$$1,250 + 875$$

Sum: _____

SET 77 • 6.SP.B.5.c

BACKWARD 2

Dividing a total into equal groups • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Divide.

$$430 \div 5$$

Quotient: _____

Divide.

$$64 \div 4$$

Quotient: _____

True or false?

$$108 \div 4 = 27$$

A. True

B. False

$$252 \div 6 = ?$$

A. 36

B. 41

C. 42

D. 48

Divide.

$$315 \div 7$$

Quotient: _____

SET 77 • 6.SP.B.5.c

BACKWARD 3

Sharing a total equally • Grade 5 • Measurement & Data • 5.MD.B.2

Two friends have 5 and 9 shells. They share the shells equally.

How many shells does each friend get?

Shells: _____

Three boxes hold 2, 7, and 9 books. The books are shared equally among the boxes.

How many books are in each box?

Books: _____

True or false?

Jars holding 4, 4, and 10 cups are shared equally.
Each jar then holds 6 cups.

- A.** True
- B.** False

Four children have 1, 2, 6, and 7 cards. They share the cards equally.

How many cards does each child get?

A. 3

B. 4

C. 5

D. 16

Two cups hold $\frac{1}{3}$ cup and $\frac{2}{3}$ cup of milk. The milk is shared equally.

How much is in each cup?

Cups: _____

SET 77 • 6.SP.B.5.c

FORWARD 1

Compare populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

A random sample of 6th graders has a mean height of 58 in. A random sample of 8th graders has a mean height of 64 in.

How much greater is the 8th graders' mean?

Difference: _____

Two random samples of plant heights have means of 24 cm and 30 cm. Each has a MAD of 3 cm.

How many MADs apart are the means?

A. 1

B. 2

C. 3

D. 6

True or false?

To compare two populations, you can compare the means of random samples from each population.

- A.** True
- B.** False

Sample A: mean 15.2 minutes. Sample B: mean 18.7 minutes.

Which population probably has greater values?

Population: _____

Ten random samples of 20 students gave these mean hours of sleep: 8.1, 8.3, 7.9, 8.0, 8.2, 8.1, 8.4, 7.8, 8.0, 8.2.

What is a good estimate of the population mean?

- A.** About 20 hours
- B.** About 7 hours
- C.** About 8.1 hours
- D.** About 9 hours

FORWARD 2

Use a linear model to make predictions • Grade 8 • Statistics & Probability •
8.SP.A.3

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The equation $y = 2.5x + 40$ models test score y after x hours of study.

Predict the score for 8 hours of study.

Score: _____

The model $y = 15x + 100$ gives the cost y of x tickets.
Predict the cost of 20 tickets.

Cost: _____

The model $y = 3x + 12$ gives a plant's height y (cm) after x weeks.

What height does the model predict after 5 weeks?

A. 15 cm

B. 27 cm

C. 36 cm

D. 60 cm

The model $y = -0.5x + 30$ gives the value y of a toy after x years.

Predict the value after 6 years.

Value: _____

True or false?

The model $y = 4x + 10$ predicts $y = 50$ when $x = 10$.

A. True

B. False

Find the median of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	The number halfway between two numbers	5.NBT.B.7
BACKWARD 3	Reading a line plot	4.MD.B.4
FORWARD 1	Compare populations using medians	7.SP.B.4
FORWARD 2	Draw and use a line of best fit — nearest related (no direct Grade...	8.SP.A.2

SET 78 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

12, 5, 9, 20, 7, 14

Median: _____

True or false?

The median of 3, 8, 8, 10, 21 is 8.

A. True

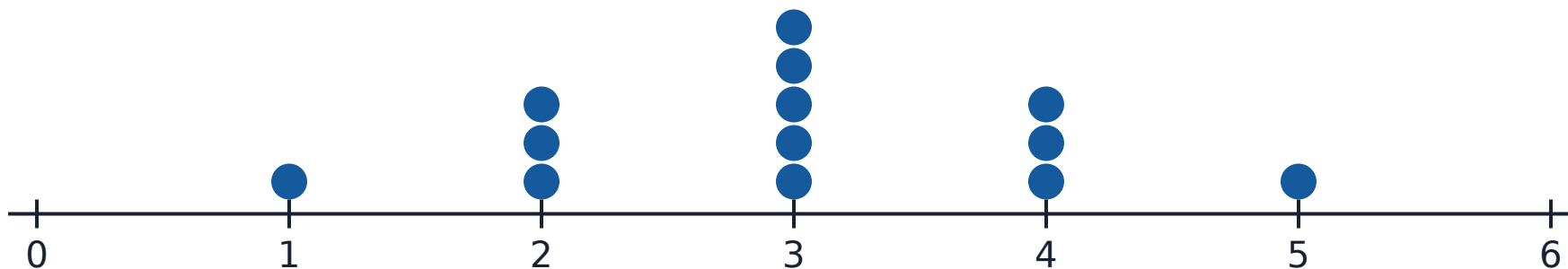
B. False

Heights (in inches): 58, 60, 61, 63, 63

What is the median height?

- A.** 60
- B.** 61
- C.** 61.5
- D.** 63

What is the median of the data in the dot plot?



Median: _____

Find the median.

4.5, 2.1, 3.8, 6.0

Median: _____

SET 78 • 6.SP.B.5.c

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

12, 5, 9, 20, 7, 14

Order: _____

Which list is in order from least to greatest?

A. 58, 60, 61, 63

B. 58, 61, 60, 63

C. 60, 58, 61, 63

D. 63, 61, 60, 58

True or false?

These numbers are in order from least to greatest: 75, 82, 88, 90, 95

- A. True
- B. False

Order from least to greatest.

47, 74, 44, 77

Order: _____

When 3, 9, and 6 are put in order, which number is in the middle?

- A.** 3
- B.** 6
- C.** 9
- D.** 18

SET 78 • 6.SP.B.5.c

BACKWARD 2

The number halfway between two numbers • Grade 5 • Number & Operations
in Base Ten • 5.NBT.B.7

Find the value.

$$(9 + 12) \div 2$$

Value: _____

Find the value.

$$(3.8 + 4.5) \div 2$$

Value: _____

True or false?

$$(6 + 9) \div 2 = 7.5$$

A. True

B. False

$$(20 + 25) \div 2 = ?$$

A. 22

B. 22.5

C. 23.5

D. 45

Find the value.

$$(3.5 + 4.5) \div 2$$

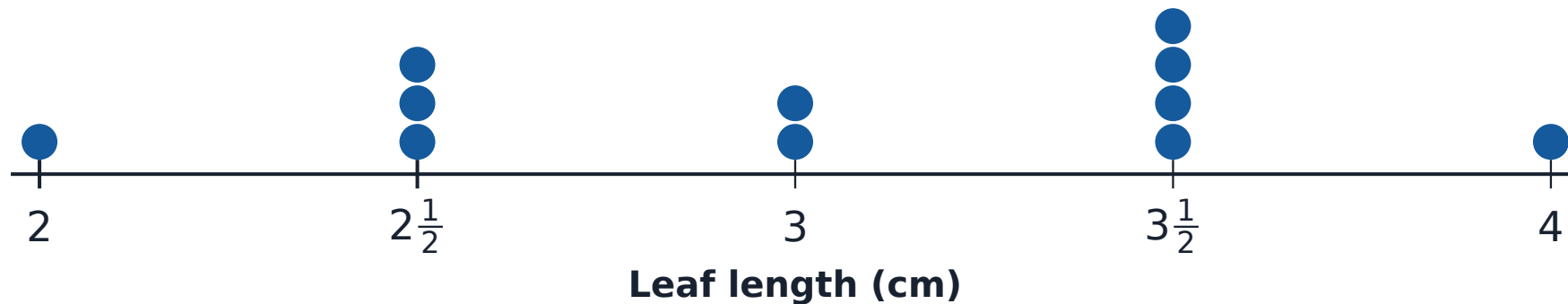
Value: _____

SET 78 • 6.SP.B.5.c

BACKWARD 3

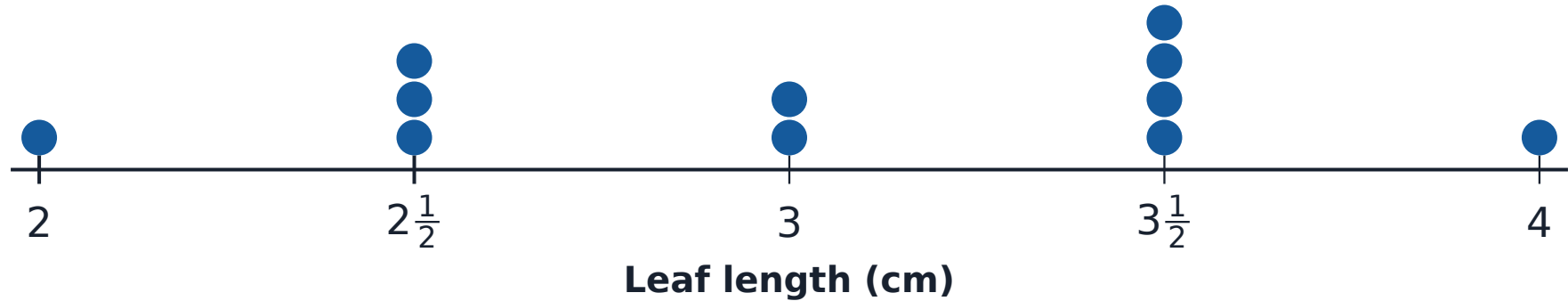
Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many leaves are $3\frac{1}{2}$ cm long?



Leaves: _____

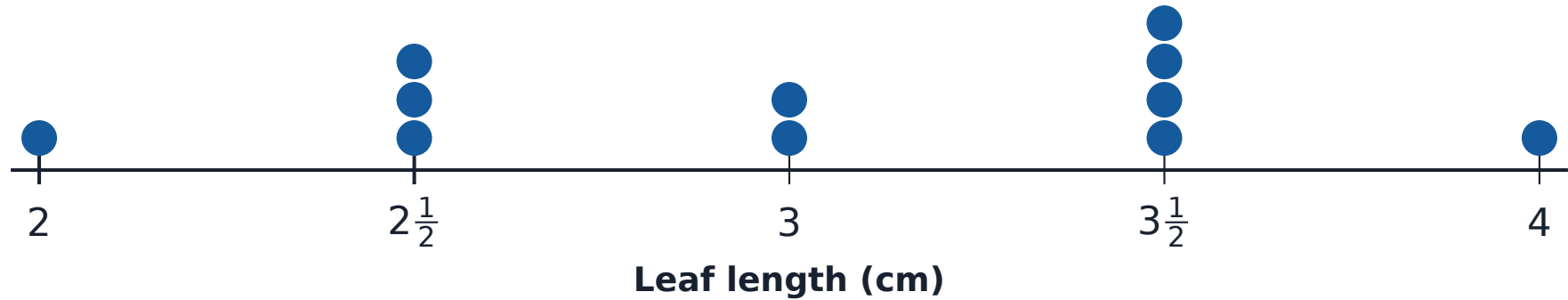
How many leaves were measured?



Leaves: _____

True or false?

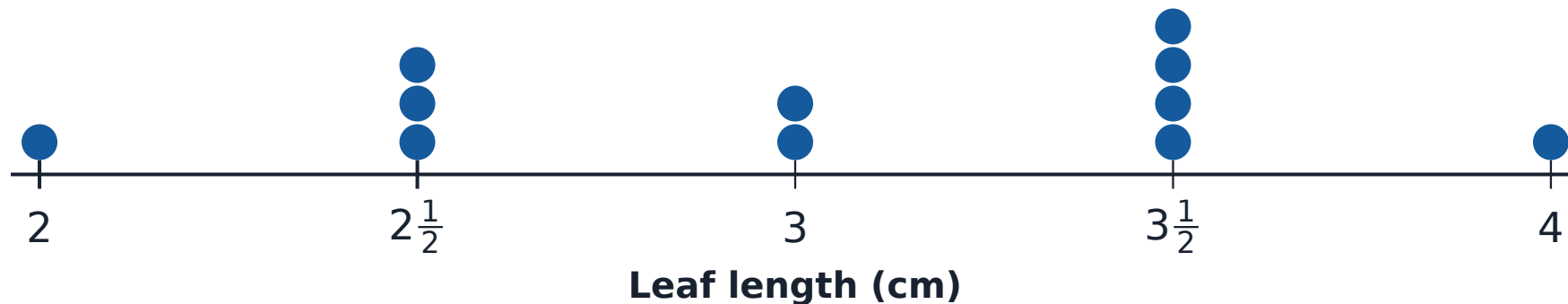
One leaf is 4 cm long.



A. True

B. False

Which leaf length is most common?



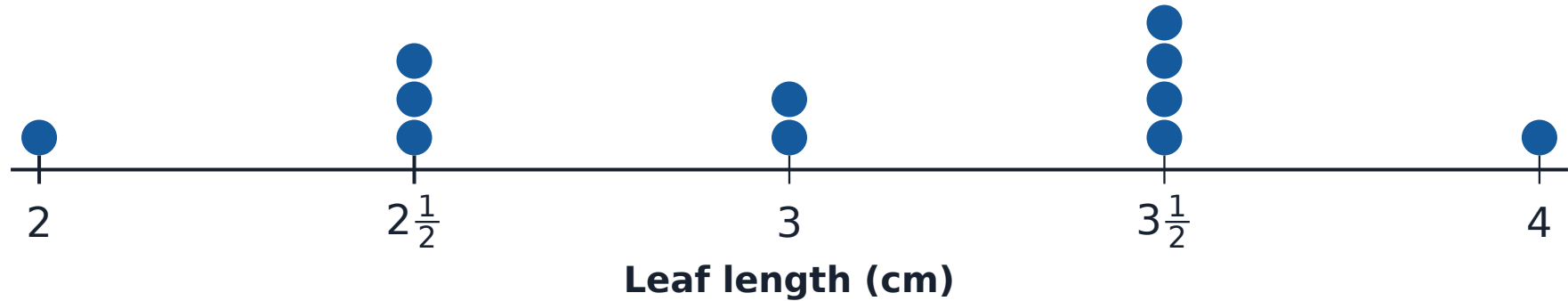
A. $2\frac{1}{2}$ cm

B. 3 cm

C. $3\frac{1}{2}$ cm

D. 4 cm

How many leaves are shorter than 3 cm?



Leaves: _____

SET 78 • 6.SP.B.5.c

FORWARD 1

Compare populations using medians • Grade 7 • Statistics & Probability •
7.SP.B.4

Sample A has a median of 12 and Sample B has a median of 15. The samples have similar variability.

Which conclusion is best?

- A.** Values in population A tend to be greater.
- B.** The populations are the same.
- C.** Values in population B tend to be greater.
- D.** No conclusion is possible.

In random samples, the median commute is 25 minutes in Town X and 18 minutes in Town Y. The IQRs are similar.

In which town are commutes typically longer?

Town: _____

Sample A: median 42, IQR 8. Sample B: median 50, IQR 8.

The difference in medians is how many times the IQR?

Times: _____

True or false?

When data are skewed, comparing medians is often better than comparing means.

A. True

B. False

Sample X has a median of 7.5 hours of sleep. Sample Y has a median of 8.5 hours. Both IQRs are 1 hour.

Which statement is best?

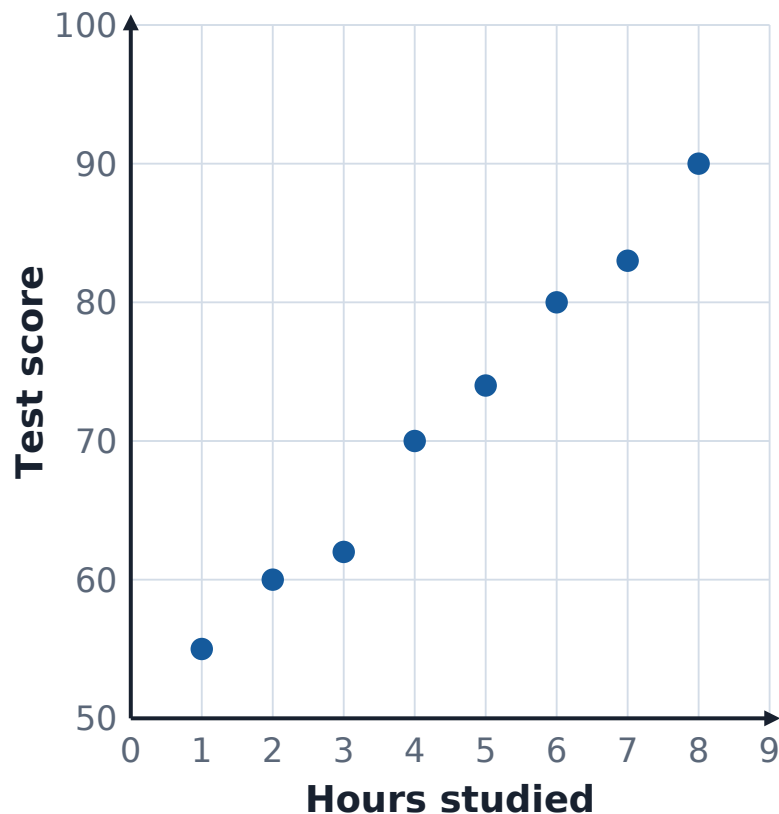
- A.** Population Y probably sleeps more.
- B.** Population X probably sleeps more.
- C.** Nothing can be said.
- D.** They sleep the same amount.

FORWARD 2

Draw and use a line of best fit • Grade 8 • Statistics & Probability • 8.SP.A.2

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Draw a line of best fit for the data.

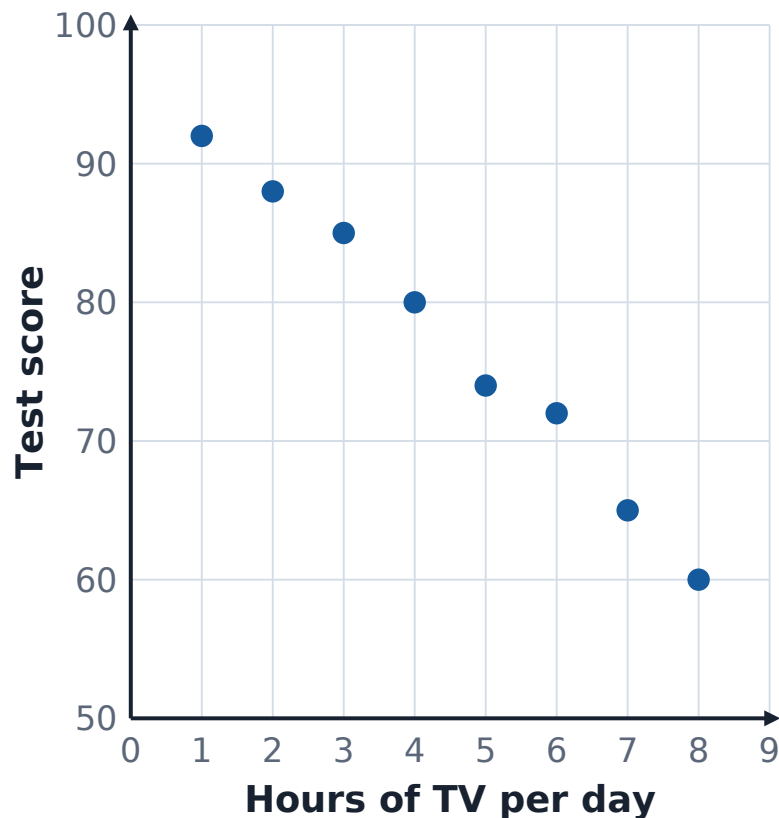


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q1/5 • NEAREST RELATED

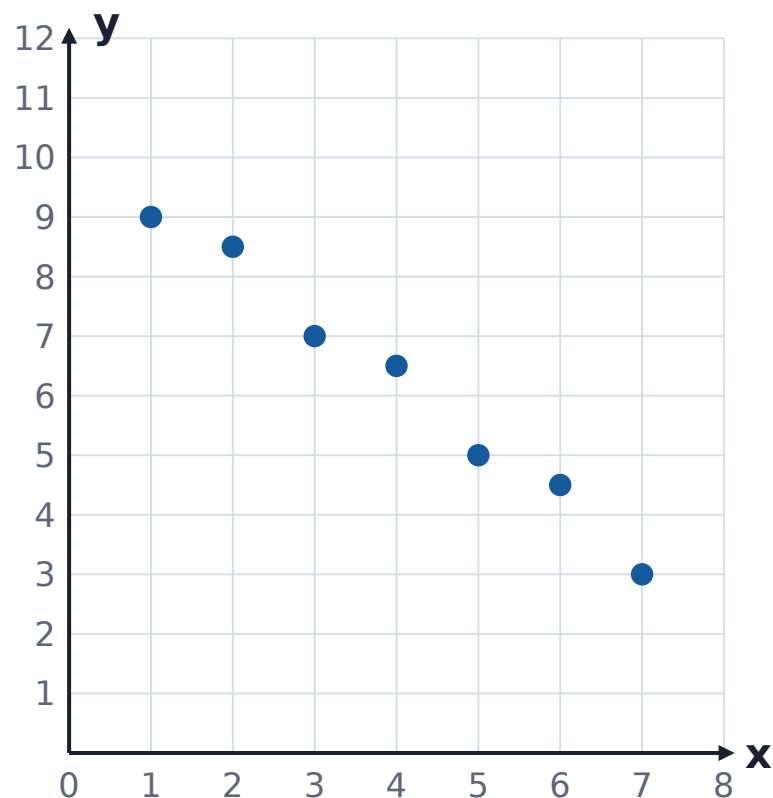
Draw a line of best fit for the data.



A line of best fit should have ____.

- A.** all points below it
- B.** no points near it
- C.** all points above it
- D.** about the same number of points above and below it

Draw a line of best fit for the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A line of best fit must pass through the first and last points.

A. True

B. False

Find the interquartile range of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	The number halfway between two numbers	5.NBT.B.7
BACKWARD 3	Finding a difference	2.NBT.B.5
FORWARD 1	Difference in centers as a multiple of the IQR	7.SP.B.3
FORWARD 2	Outliers and clusters in scatter plots — nearest related (no direct...	8.SP.A.1

SET 79 • 6.SP.B.5.c

MAIN

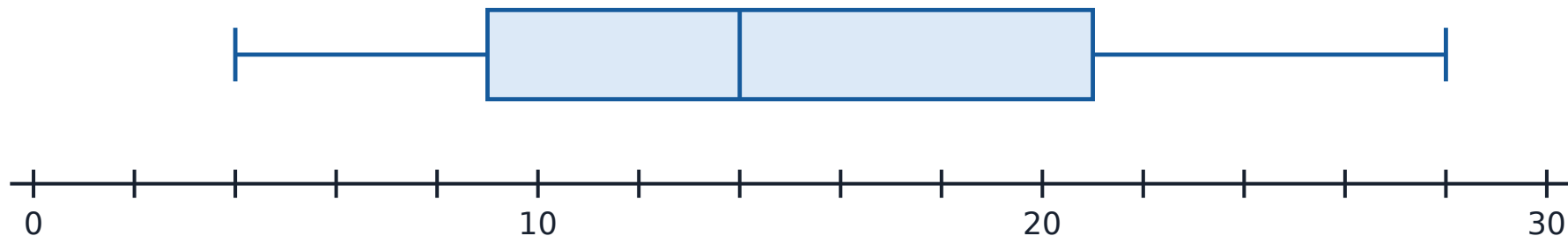
Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the interquartile range.

2, 4, 5, 7, 9, 11, 12, 15

IQR: _____

What is the interquartile range of the data in the box plot?



IQR: _____

What is the interquartile range?

10, 12, 15, 18, 20, 25, 30

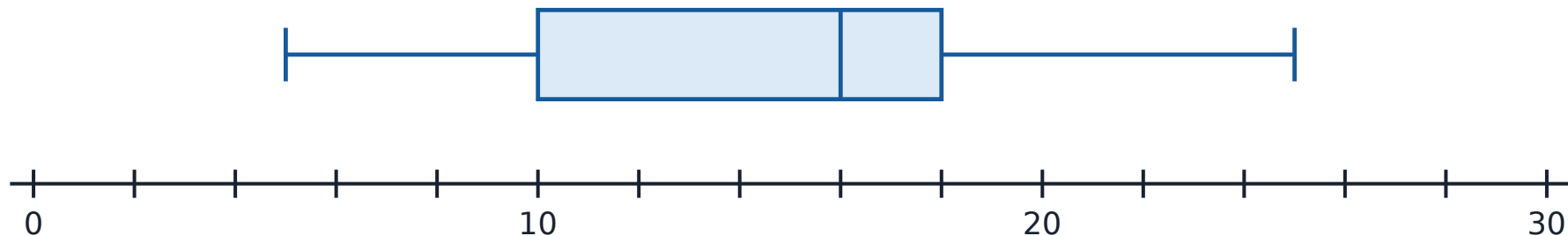
- A.** 5
- B.** 13
- C.** 18
- D.** 20

Find the interquartile range.

3, 6, 6, 8, 9, 12, 14, 15, 18

IQR: _____

What is the interquartile range of the data in the box plot?



IQR: _____

SET 79 • 6.SP.B.5.c

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

15, 9, 2, 12, 5

Order: _____

Which list is in order from least to greatest?

A. 18, 15, 12, 10

B. 10, 15, 12, 18

C. 12, 10, 15, 18

D. 10, 12, 15, 18

True or false?

These numbers are in order from least to greatest: 3, 6, 9, 8

- A. True
- B. False

Order from least to greatest.

25, 20, 30, 10

Order: _____

Which number is least?

A. 2

B. 5

C. 9

D. 12

SET 79 • 6.SP.B.5.c

BACKWARD 2

The number halfway between two numbers • Grade 5 • Number & Operations
in Base Ten • 5.NBT.B.7

Find the value.

$$(4 + 5) \div 2$$

Value: _____

Find the value.

$$(11 + 12) \div 2$$

Value: _____

True or false?

$$(7 + 10) \div 2 = 8.5$$

A. True

B. False

$$(15 + 18) \div 2 = ?$$

A. 16

B. 16.5

C. 17.5

D. 33

Find the value.

$$(2.5 + 3.5) \div 2$$

Value: _____

SET 79 • 6.SP.B.5.c

BACKWARD 3

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$25 - 12$$

Difference: _____

Subtract.

$$21 - 9$$

Difference: _____

True or false?

$$30 - 14 = 16$$

A. True

B. False

$$18 - 10 = ?$$

- A.** 7
- B.** 8
- C.** 9
- D.** 28

Subtract.

$$40 - 23$$

Difference: _____

SET 79 • 6.SP.B.5.c

FORWARD 1

Difference in centers as a multiple of the IQR • Grade 7 • Statistics & Probability • 7.SP.B.3

Two data sets have medians of 40 and 55. Each has an IQR of 5.

The difference in medians is how many times the IQR?

Times: _____

Two data sets have medians of 20 and 26. Each has an IQR of 3.

How many IQRs apart are the medians?

A. 2

B. 3

C. 6

D. 9

True or false?

Two data sets with medians that differ by less than one IQR probably overlap a lot.

- A. True
- B. False

Class A's median is 72 and Class B's median is 84. Each class has an IQR of 12.

The difference in medians is how many times the IQR?

Times: _____

Which pair of data sets has the LEAST overlap?

- A.** Medians 10 and 14, each IQR 8
- B.** Medians 10 and 30, each IQR 4
- C.** Medians 10 and 11, each IQR 3
- D.** Medians 10 and 12, each IQR 4

FORWARD 2

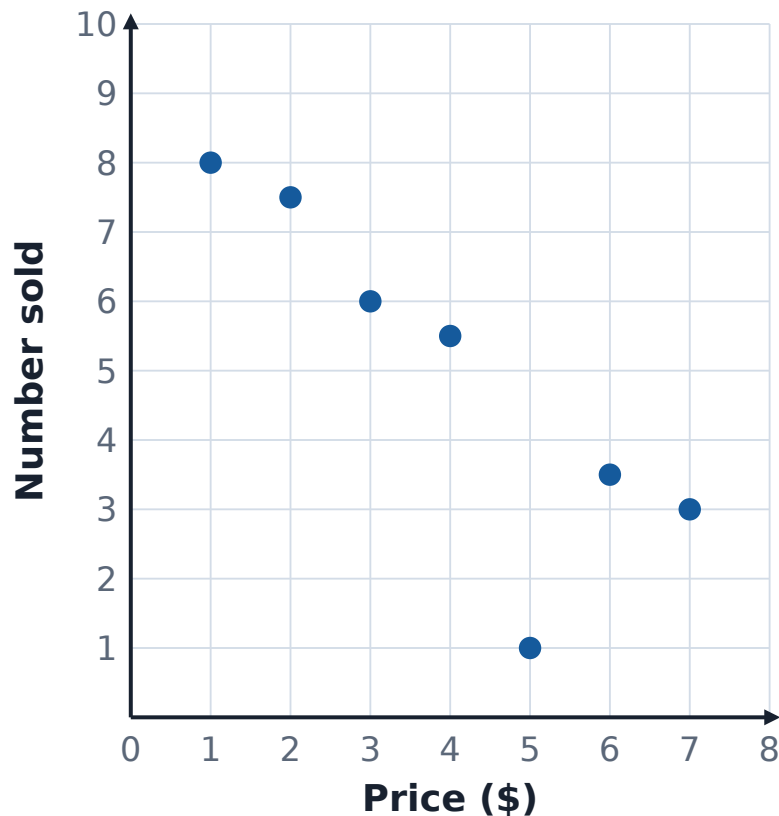
Outliers and clusters in scatter plots • Grade 8 • Statistics & Probability •
8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Which point is an outlier?

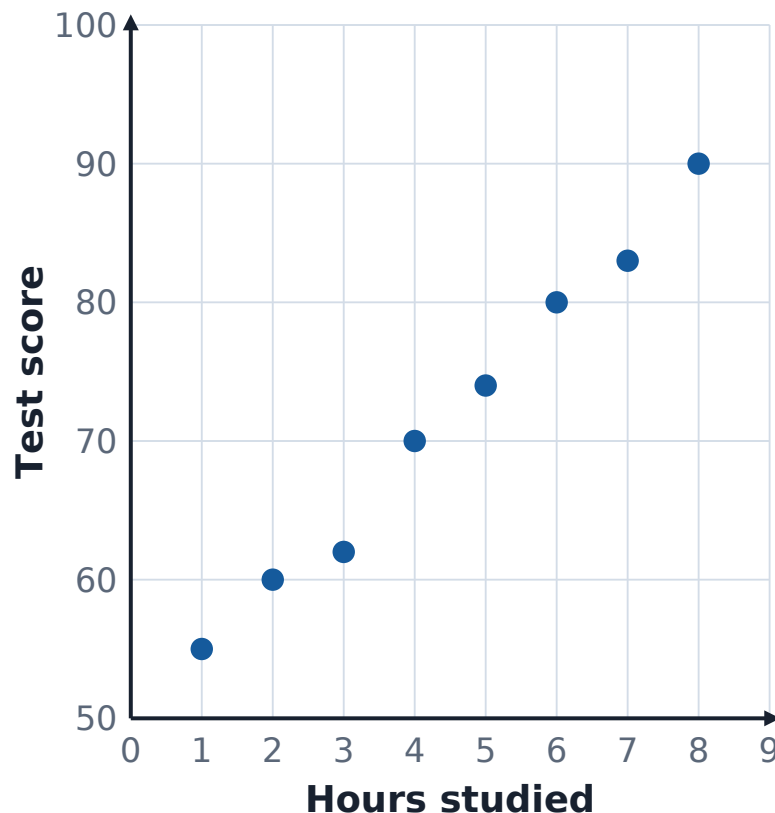
Write its coordinates.

Outlier: _____



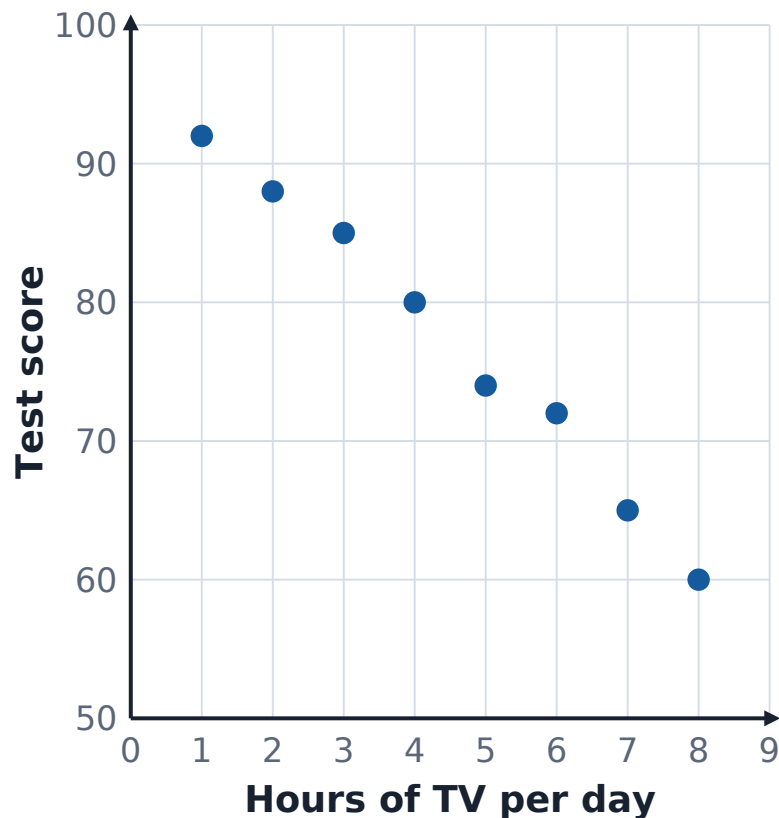
Which point would be an outlier if it were added to the scatter plot?

- A.** (5, 73)
- B.** (7, 50)
- C.** (7, 85)
- D.** (3, 63)



Which describes the association in the scatter plot?

- A.** No association
- B.** Negative linear
- C.** Nonlinear
- D.** Positive linear

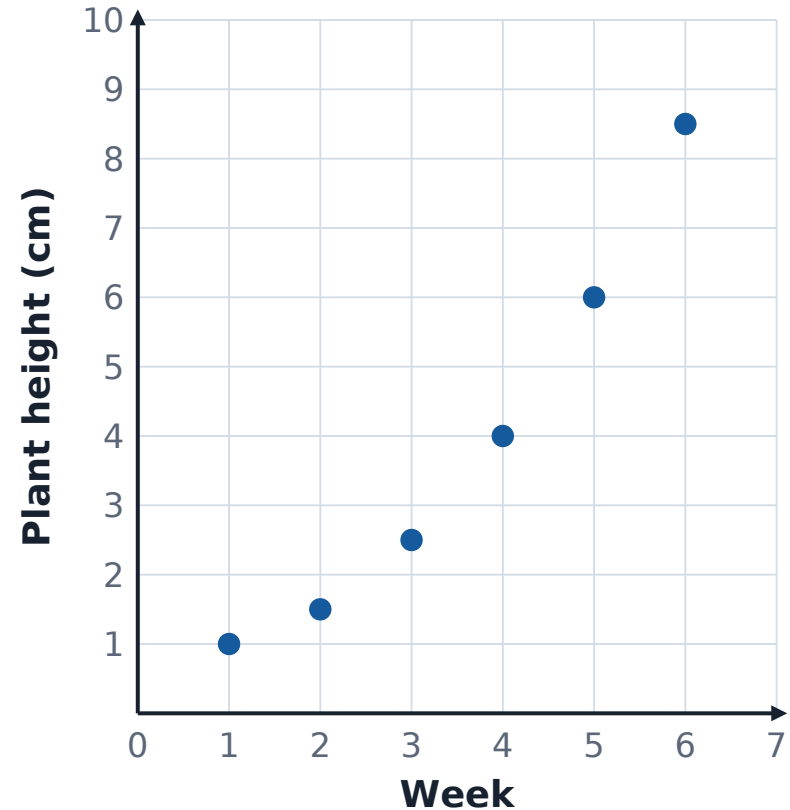


True or false?

The scatter plot has an outlier.

A. True

B. False



True or false?

Every scatter plot has at least one outlier.

- A.** True
- B.** False

Find the mean absolute deviation of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Sharing a total equally (the mean)	5.MD.B.2
BACKWARD 2	Distance as absolute value	6.NS.C.7.c
BACKWARD 3	Dividing to find an average distance	5.NBT.B.7
FORWARD 1	Difference in means as a multiple of the MAD	7.SP.B.3
FORWARD 2	Judge model fit by distances from the line	8.SP.A.2

SET 80 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean absolute deviation.

2, 4, 6, 8, 10

MAD: _____

What is the mean absolute deviation?

3, 3, 5, 7, 7

A. 0

B. 1.6

C. 2

D. 4

Find the mean absolute deviation.

5, 7, 8, 12

MAD: _____

Find the mean absolute deviation.

6, 8, 10, 12, 14, 16

MAD: _____

Find the mean absolute deviation.

20, 22, 25, 29

MAD: _____

SET 80 • 6.SP.B.5.c

BACKWARD 1

Sharing a total equally (the mean) • Grade 5 • Measurement & Data • 5.MD.B.2

Five jars hold 2, 4, 6, 8, and 10 cups of rice. The rice is shared equally among the jars.

How much is in each jar?

Cups: _____

Four friends have 3, 3, 7, and 7 marbles. They share the marbles equally.

How many marbles does each friend get?

Marbles: _____

True or false?

Bags with 1, 5, and 6 pounds of flour are shared equally. Each bag then has 4 pounds.

- A.** True
- B.** False

Six students have 2, 3, 3, 4, 5, and 7 pencils. They share equally.

How many pencils does each student get?

A. 3

B. 4

C. 5

D. 24

Three cups hold $\frac{1}{2}$, $\frac{1}{2}$, and 2 cups of water. The water is shared equally.

How much is in each cup?

Cups: _____

SET 80 • 6.SP.B.5.c

BACKWARD 2

Distance as absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|2 - 6|$$

Value: _____

How far is 9 from 6 on a number line?

Distance: _____

True or false?

$$|4 - 10| = 6$$

A. True

B. False

What is the distance between 3 and 8 on a number line?

A. -5

B. 5

C. 11

D. 24

Find the value.

$$|10 - 6|$$

Value: _____

SET 80 • 6.SP.B.5.c

BACKWARD 3

Dividing to find an average distance • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Divide.

$$12 \div 5$$

Quotient: _____

Divide.

$$8 \div 5$$

Quotient: _____

True or false?

$$18 \div 6 = 3$$

A. True

B. False

$$10 \div 4 = ?$$

A. 0.4

B. 2.4

C. 2.5

D. 40

Divide.

$$9 \div 4$$

Quotient: _____

SET 80 • 6.SP.B.5.c

FORWARD 1

Difference in means as a multiple of the MAD • Grade 7 • Statistics & Probability • 7.SP.B.3

Class A's mean is 72 and Class B's mean is 80. Each class has a MAD of 4.

The difference in means is how many times the MAD?

Times: _____

Two data sets have means of 15 and 21. Each has a MAD of 2.

How many MADs apart are the means?

A. 2

B. 3

C. 6

D. 12

True or false?

Two data sets have means that differ by 1 MAD.
The data sets probably overlap a lot.

- A. True
- B. False

Two teams have mean heights of 150 cm and 162 cm.
Each has a MAD of 4 cm.

The difference in means is how many times the MAD?

Times: _____

Which pair of data sets has the LEAST overlap?

- A.** Means 10 and 30, each MAD 2
- B.** Means 10 and 12, each MAD 2
- C.** Means 10 and 11, each MAD 3
- D.** Means 10 and 14, each MAD 4

SET 80 • 6.SP.B.5.c

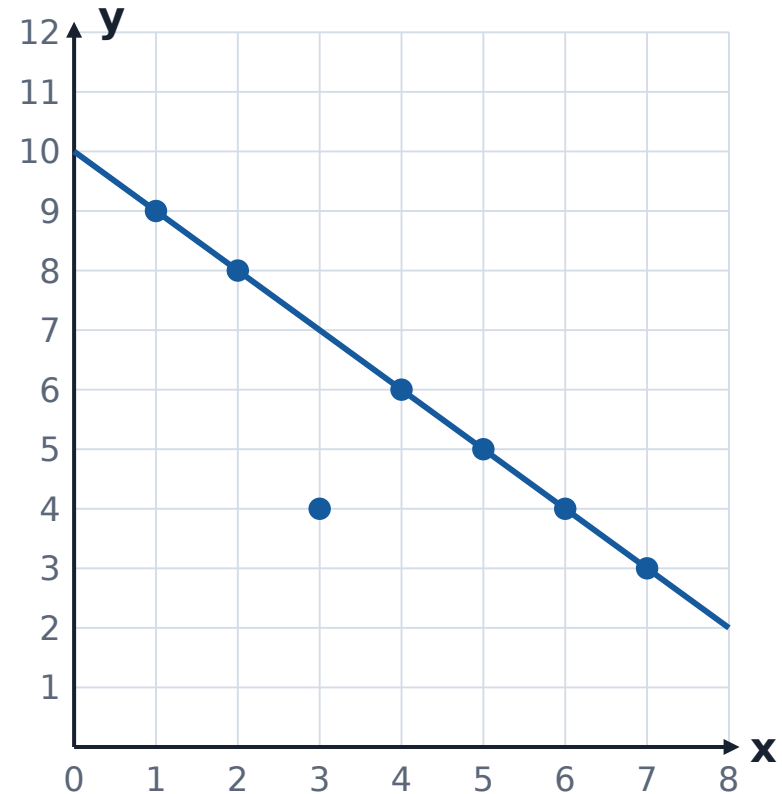
FORWARD 2

Judge model fit by distances from the line • Grade 8 • Statistics & Probability •
8.SP.A.2

Which point is farthest from the line?

Write its coordinates.

Point: _____

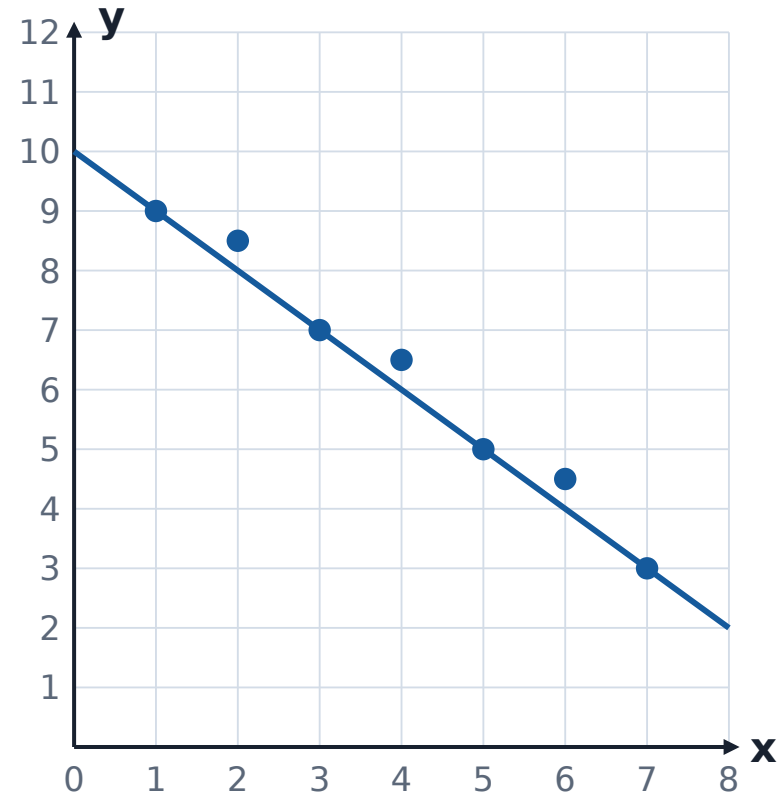


True or false?

The points are close to the line, so it is a good fit.

A. True

B. False

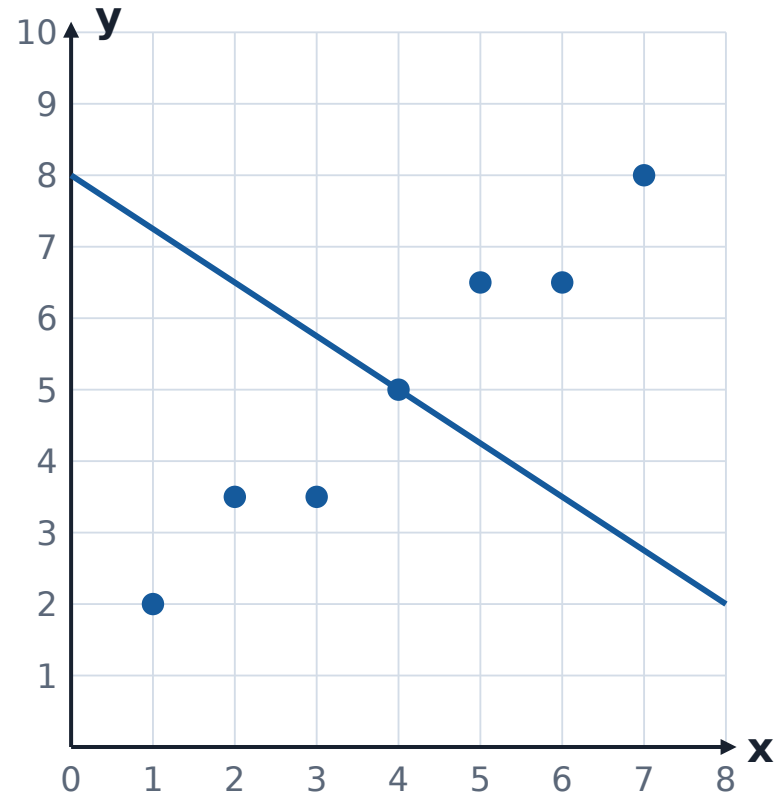


True or false?

The line is a good fit for the data.

A. True

B. False



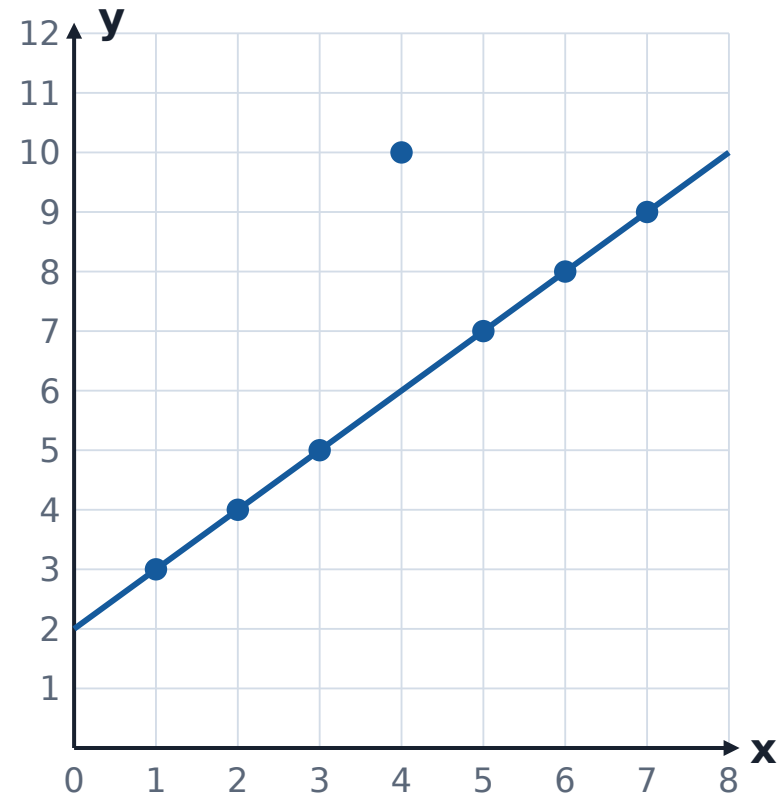
Which line fits data better?

- A.** A line that touches only one point
- B.** A line with all points far above it
- C.** A line with points close to it on both sides
- D.** A line with all points below it

For the point $(4, 10)$, the line gives $y = 6$.

How far is the point from the line, measured vertically?

Distance: _____



Describe the overall pattern and striking deviations in the context of the data

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Where data cluster and where there are gaps	6.SP.A.2
BACKWARD 2	Reading a dot plot in context	4.MD.B.4
BACKWARD 3	Greatest and least values	2.NBT.A.4
FORWARD 1	Compare the patterns of two distributions	7.SP.B.3
FORWARD 2	Describe patterns and deviations in scatter plots in context	8.SP.A.1

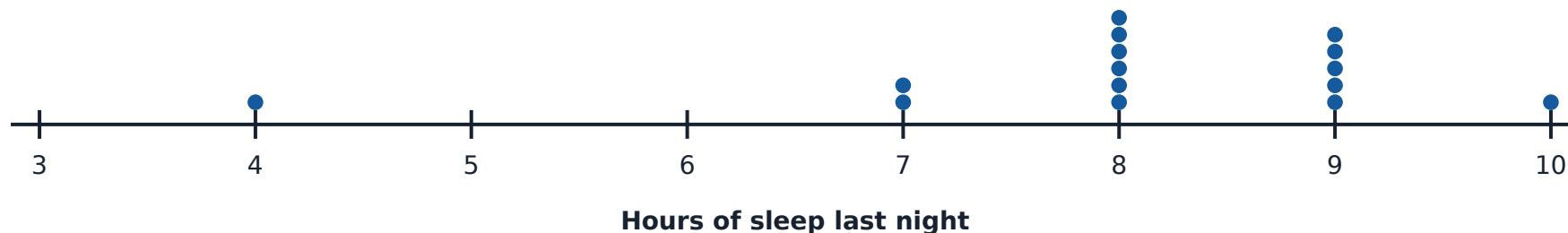
SET 81 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

The dot plot shows how many hours students slept last night.

Describe the overall pattern. Which value stands apart, and what might explain it?

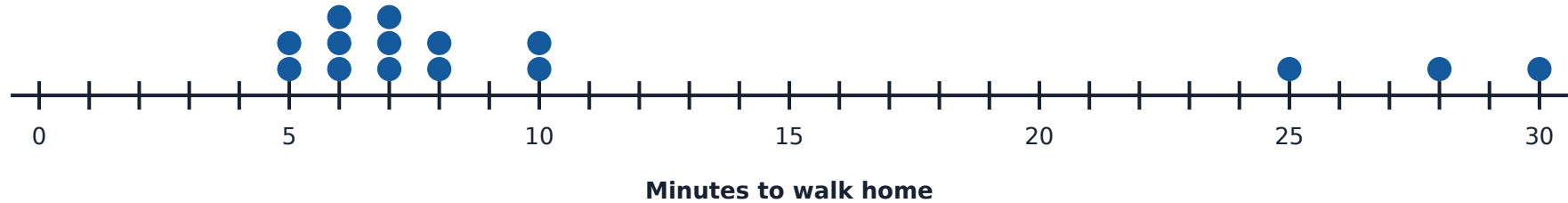


Overall pattern: _____

Value that stands apart: _____

The dot plot shows how long students take to walk home.

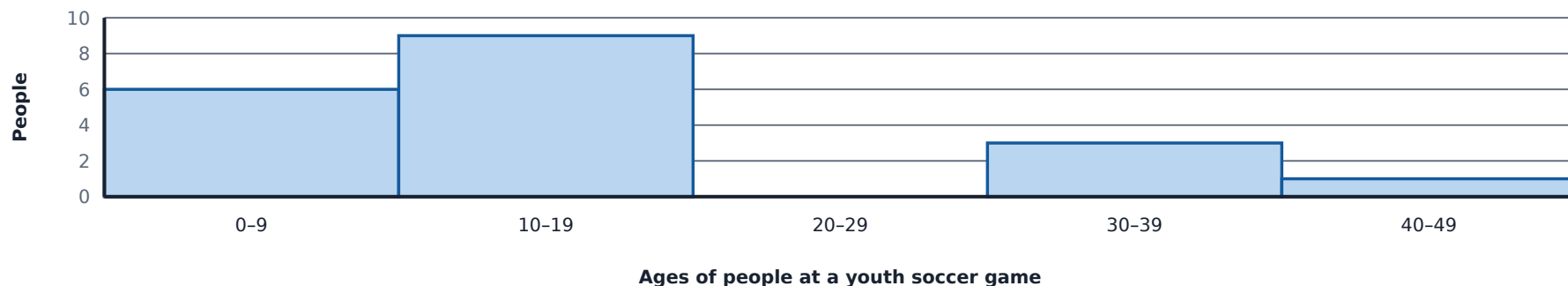
Which description best fits the data in context?



- A.** The times are spread evenly from 5 to 30 minutes.
- B.** All students take about the same amount of time.
- C.** Most students take 25 to 30 minutes.
- D.** Most students take 5 to 10 minutes; a few who live farther away take 25 to 30 minutes.

The histogram shows the ages of people at a youth soccer game.

Describe the overall pattern. Which ages stand apart, and who might they be?

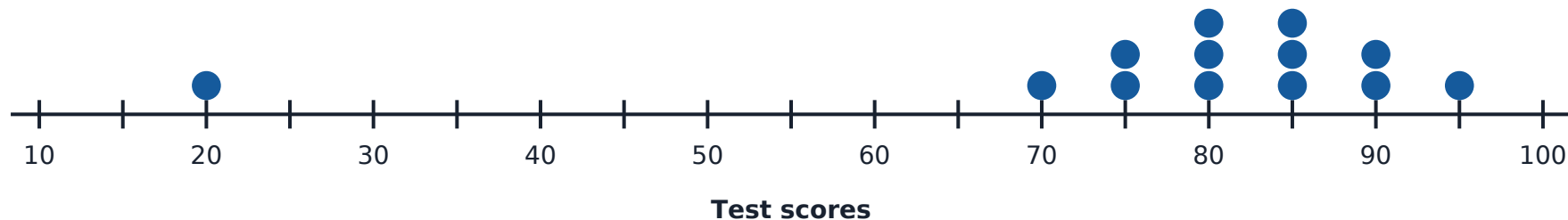


Overall pattern: _____

Ages that stand apart: _____

The dot plot shows the scores on a class test.

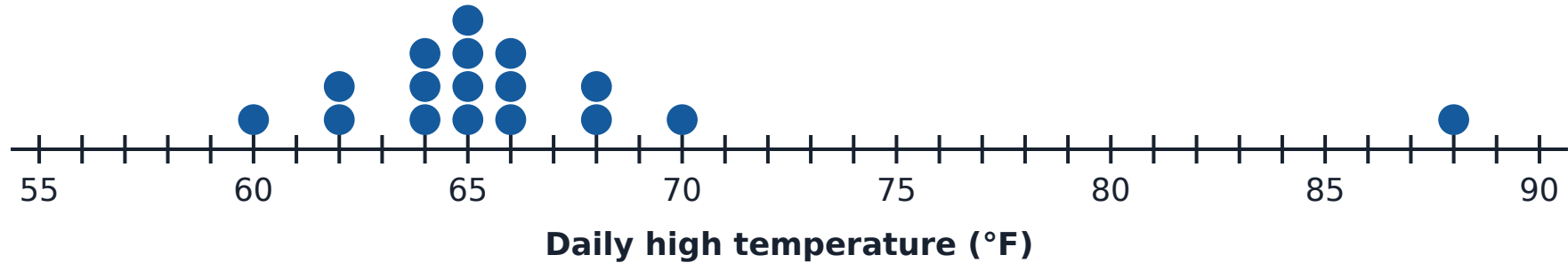
Which statement describes a striking deviation in context?



- A.** One student scored 20, far below the others, who scored from 70 to 95. **B.** Most students scored below 50.
- C.** The scores are spread evenly from 20 to 95. **D.** There are no unusual scores.

True or false?

The daily high temperatures cluster from 60°F to 70°F, with one unusually hot day at 88°F.



A. True

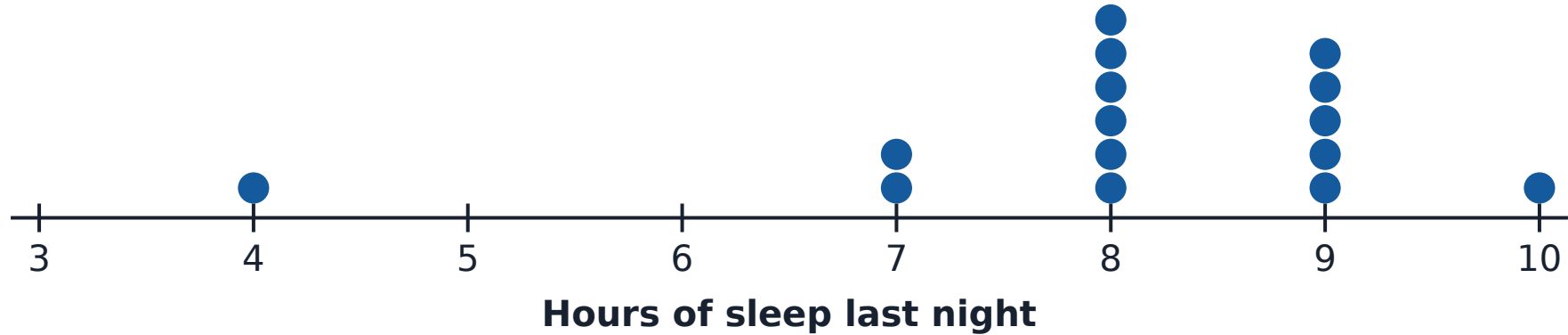
B. False

SET 81 • 6.SP.B.5.c

BACKWARD 1

Where data cluster and where there are gaps • Grade 6 • Statistics & Probability • 6.SP.A.2

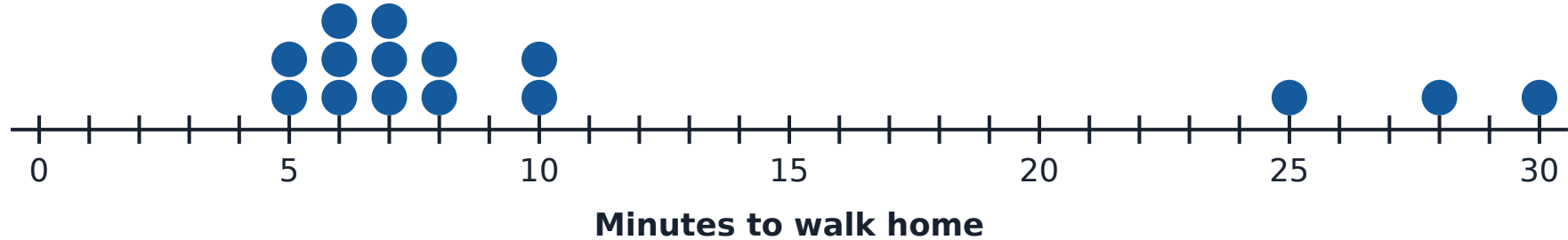
Between which two values do most of the data cluster?



Between: _____

True or false?

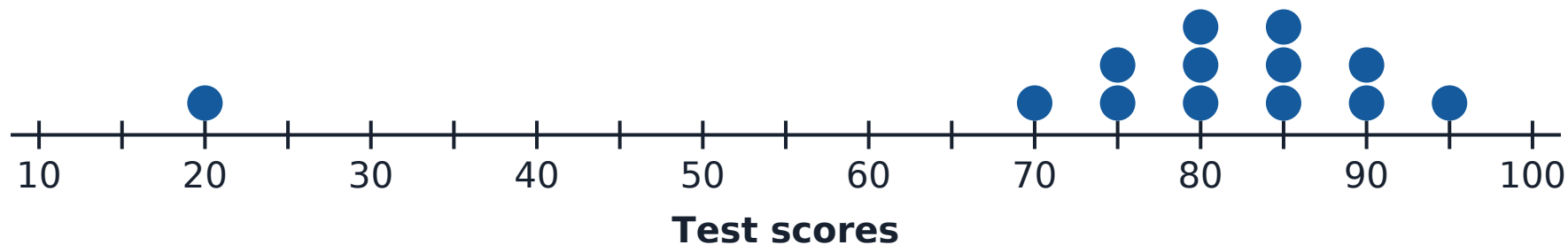
There is a gap in the data between 10 and 25 minutes.



A. True

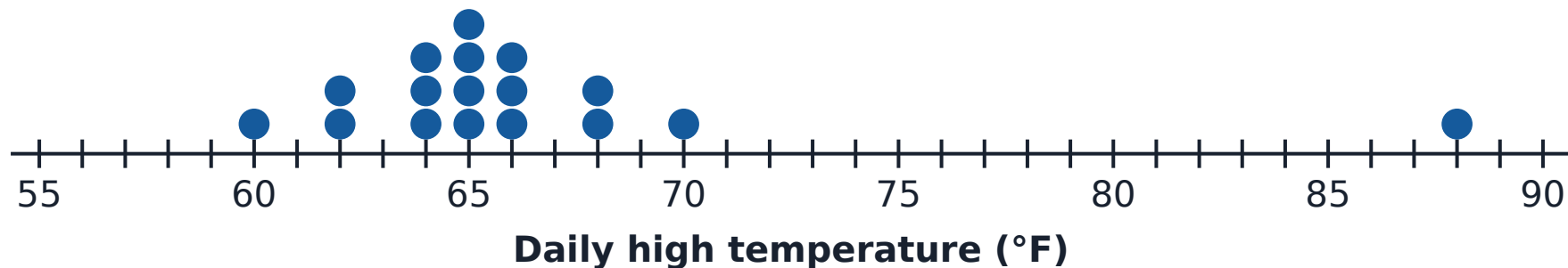
B. False

Where is there a gap in the data?



- A.** Between 70 and 80 **B.** Between 20 and 70
- C.** Between 85 and 95 **D.** There is no gap.

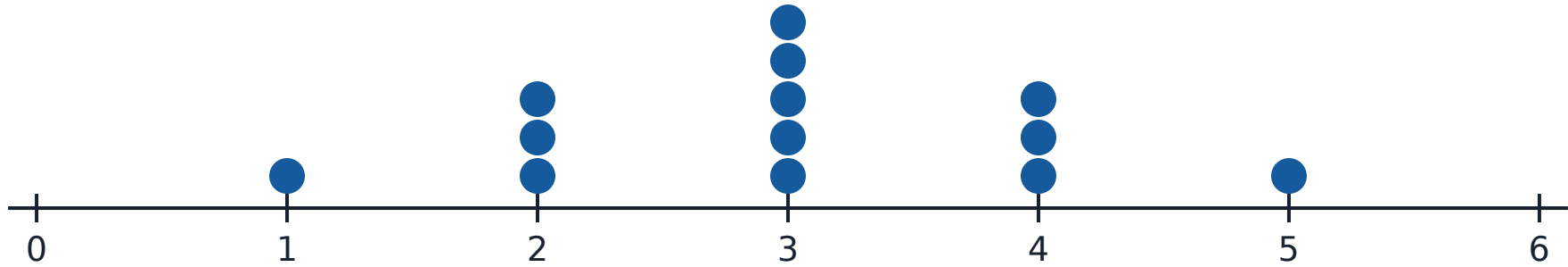
Between which two temperatures do most of the data cluster?



Between: _____

True or false?

The data have no gaps.



A. True

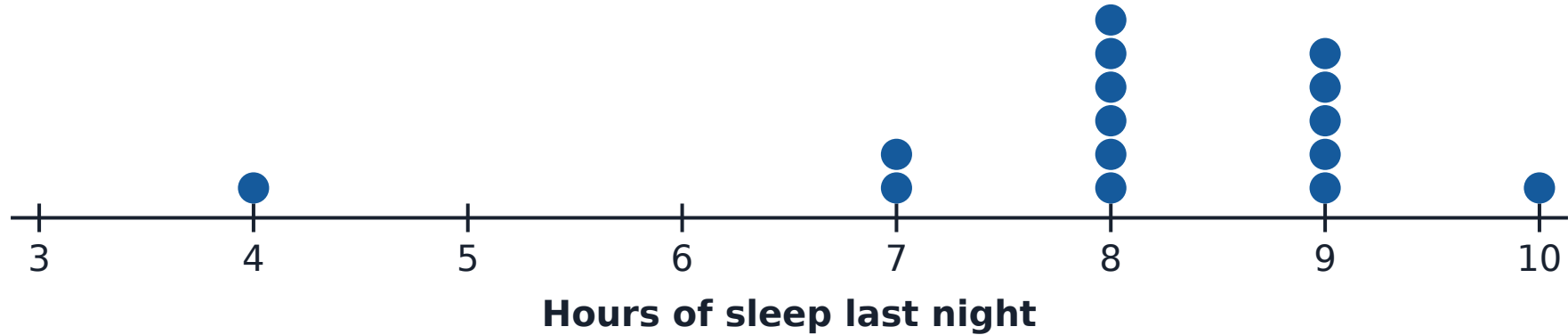
B. False

SET 81 • 6.SP.B.5.c

BACKWARD 2

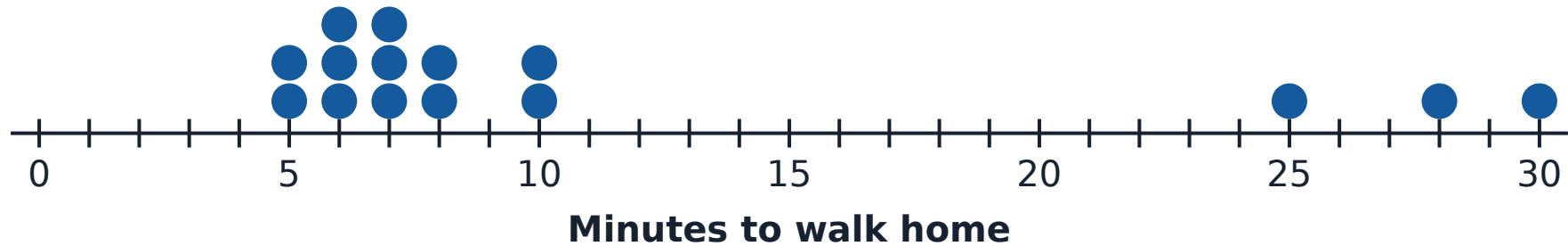
Reading a dot plot in context • Grade 4 • Measurement & Data • 4.MD.B.4

How many students slept 8 hours last night?



Students: _____

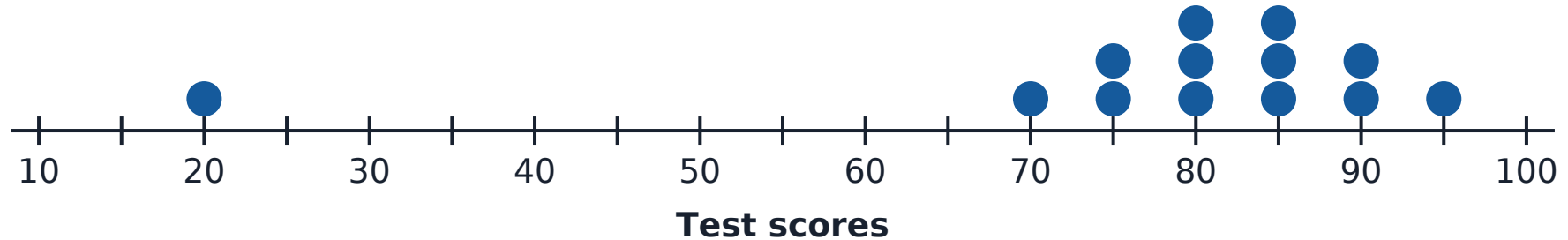
How many students take 7 minutes to walk home?



Students: _____

True or false?

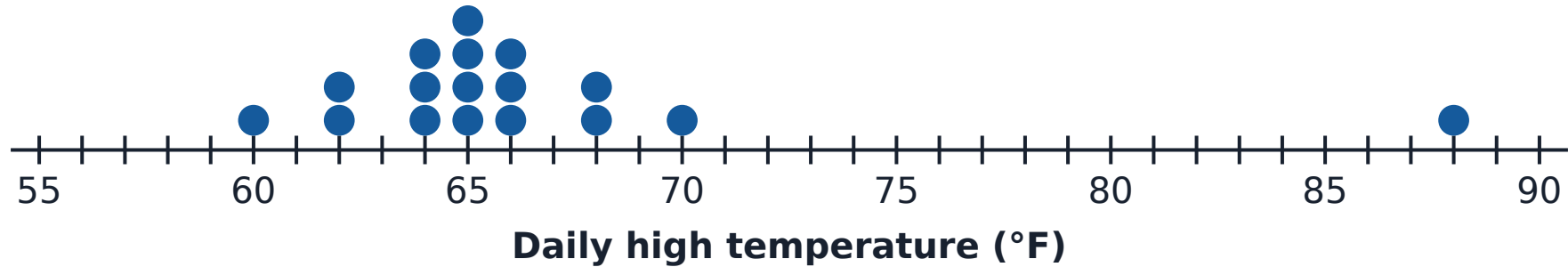
Three students scored 85 on the test.



A. True

B. False

How many days had a high temperature of 65°F?



A. 3

B. 4

C. 5

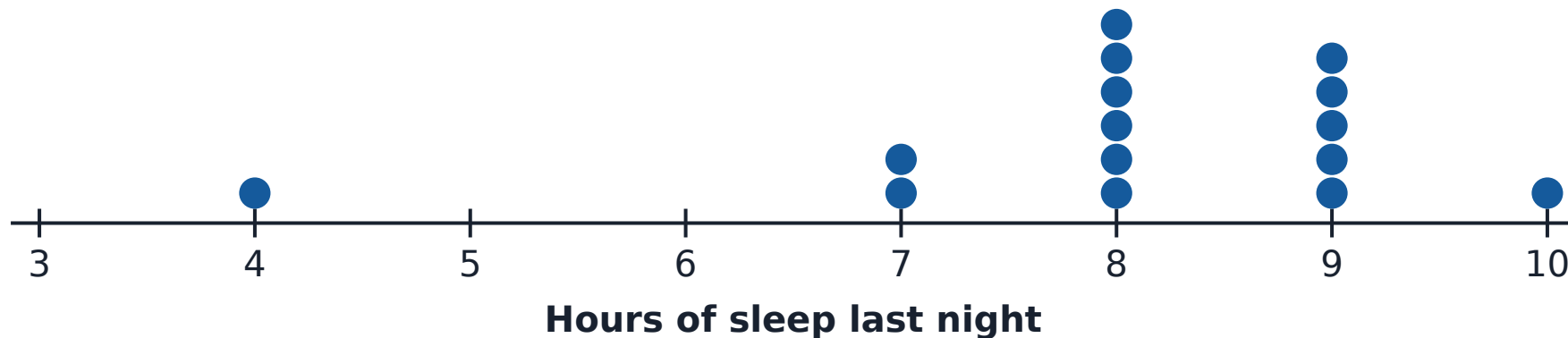
D. 65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q4/5

What does each dot in the dot plot represent?



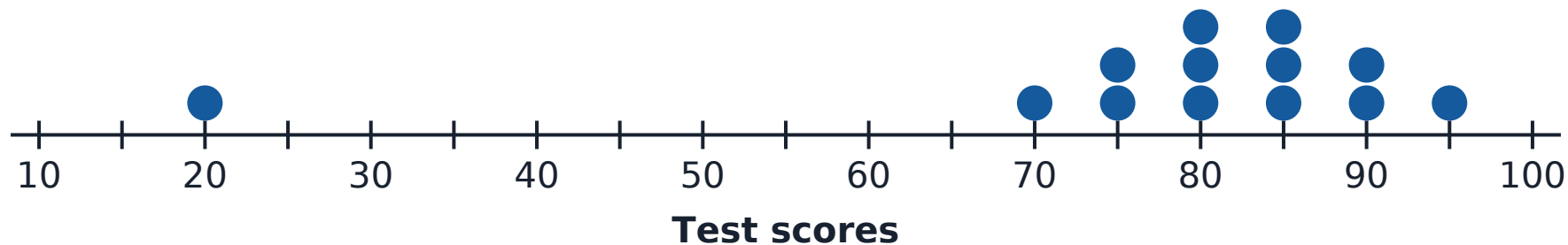
Each dot: _____

SET 81 • 6.SP.B.5.c

BACKWARD 3

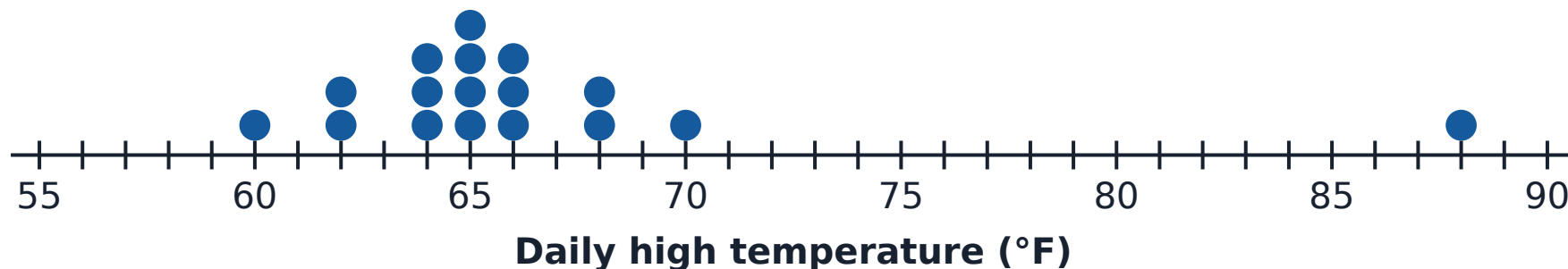
Greatest and least values • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

What is the least value in the data?



Least: _____

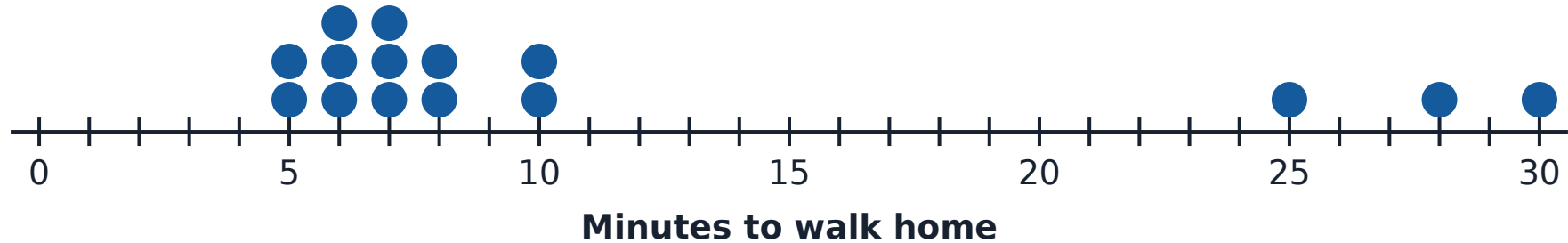
What is the greatest value in the data?



Greatest: _____

True or false?

The greatest walking time is 30 minutes.



A. True

B. False

Which number is least?

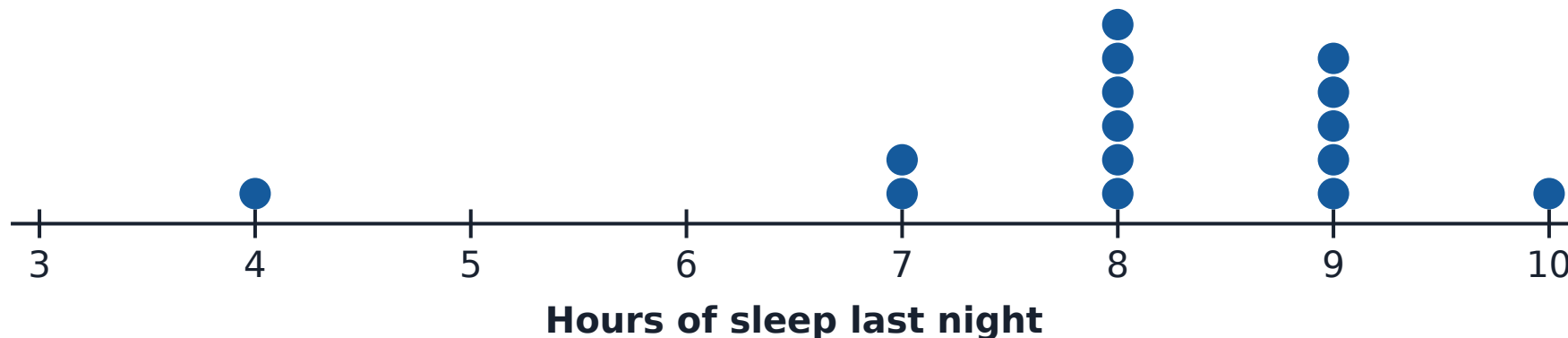
A. 20

B. 70

C. 75

D. 95

What is the least number of hours of sleep shown?



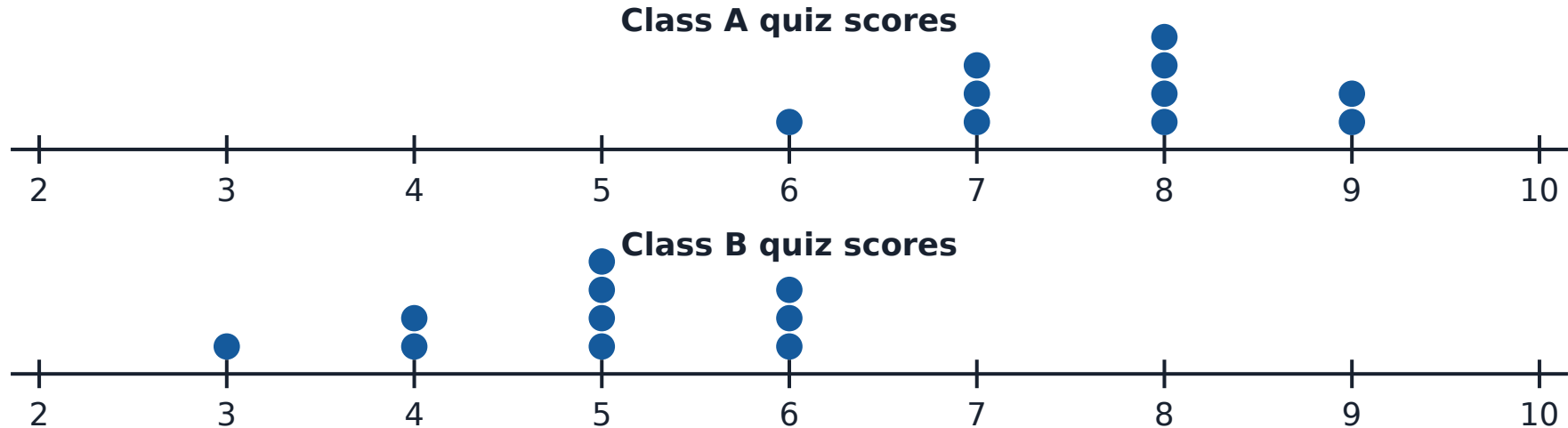
Least: _____

SET 81 • 6.SP.B.5.c

FORWARD 1

Compare the patterns of two distributions • Grade 7 • Statistics & Probability •
7.SP.B.3

Describe how the two distributions differ in context.



Comparison: _____

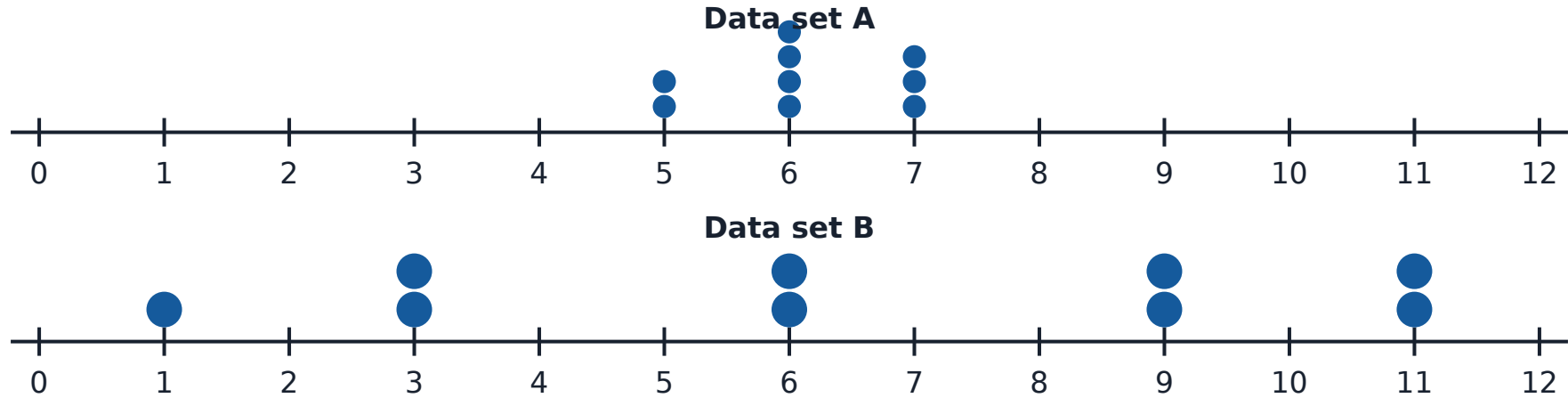
Which statement describes both gardens?



- A.** The gardens have the same center.
- B.** Garden A's plants are taller.
- C.** Garden B is much more spread out.
- D.** Both cluster tightly, and Garden B's plants are about 6 cm taller.

True or false?

Data set B is more spread out than data set A.



A. True

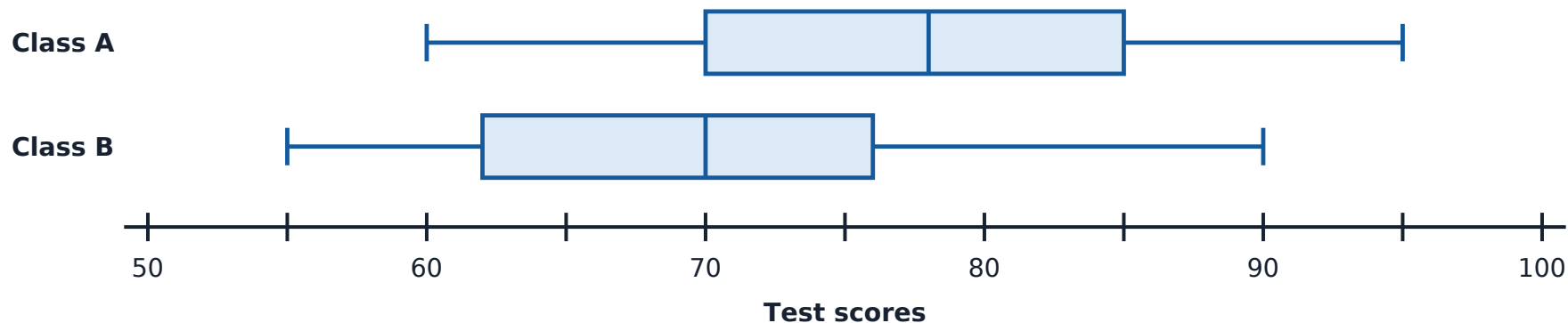
B. False

Class A's quiz scores cluster from 7 to 9, with one score of 2. Class B's scores cluster from 4 to 6 with no unusual values.

Which class has a striking deviation?

Class: _____

Which statement about the middle halves of the two box plots is true?



- A.** They overlap from 70 to 76.
- B.** They do not overlap.
- C.** They overlap from 55 to 95.
- D.** They are the same.

SET 81 • 6.SP.B.5.c

FORWARD 2

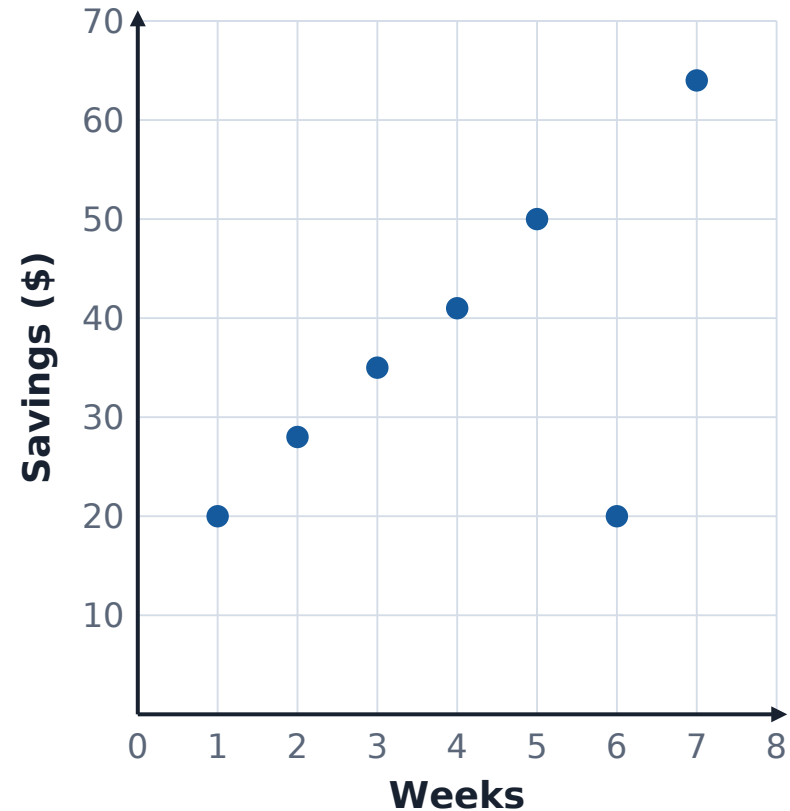
Describe patterns and deviations in scatter plots in context • Grade 8 •
Statistics & Probability • 8.SP.A.1

The scatter plot shows savings over several weeks.

Describe the overall pattern and the point that does not fit.

Pattern: _____

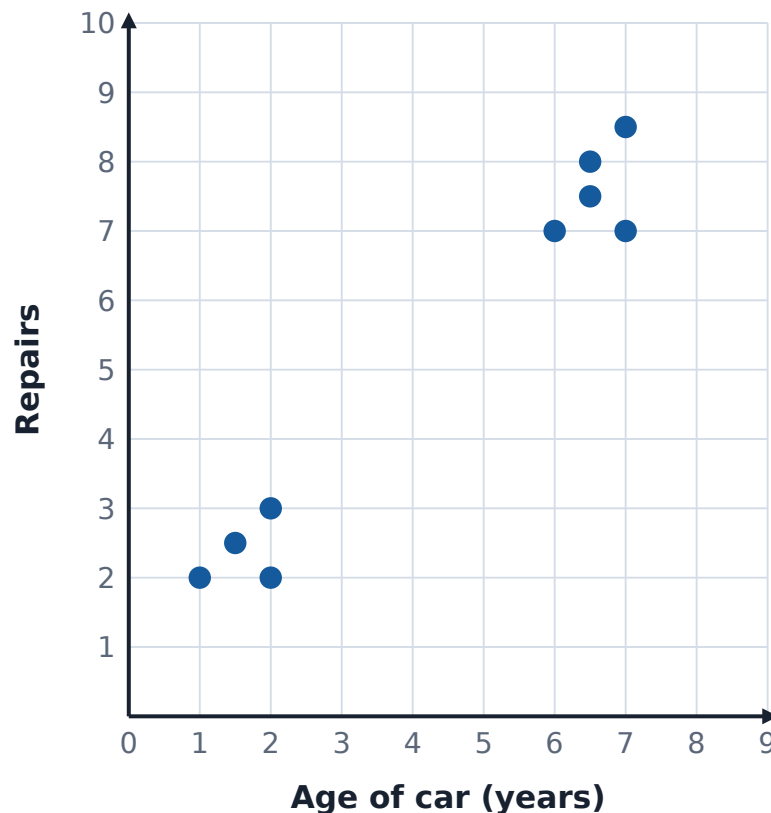
Point that does not fit: _____



The scatter plot shows the age of cars and the number of repairs.

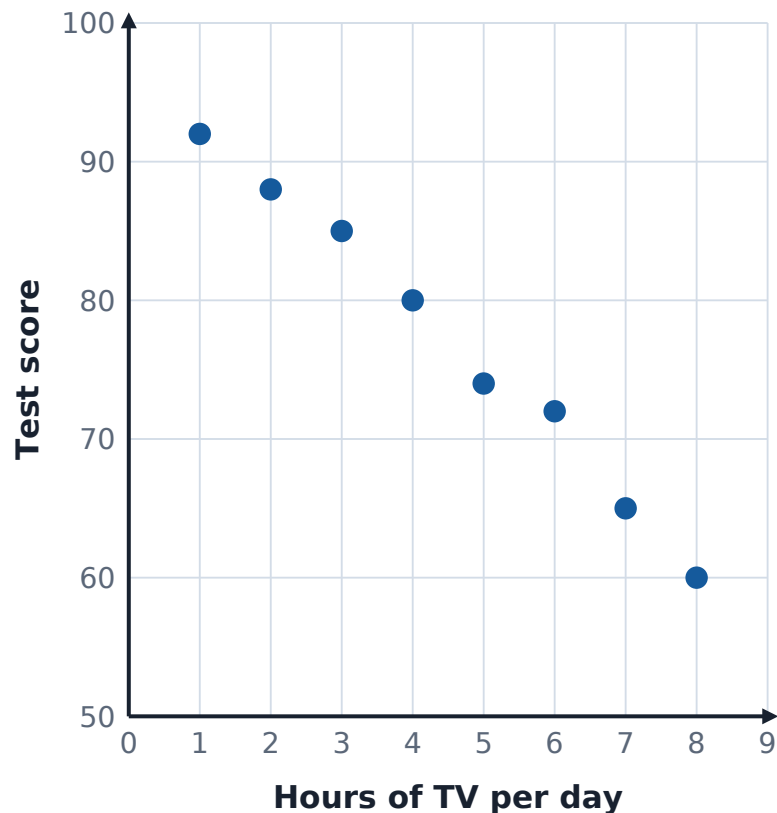
Which description fits the data in context?

- A.** Older cars have fewer repairs.
- B.** Newer cars have few repairs and older cars have many repairs, forming two groups.
- C.** All cars have the same number of repairs.
- D.** There is no pattern.



Which statement describes the association in context?

- A.** As hours of TV increase, test scores tend to increase.
- B.** Hours of TV and test scores are not related.
- C.** Every student scored the same.
- D.** As hours of TV increase, test scores tend to decrease.

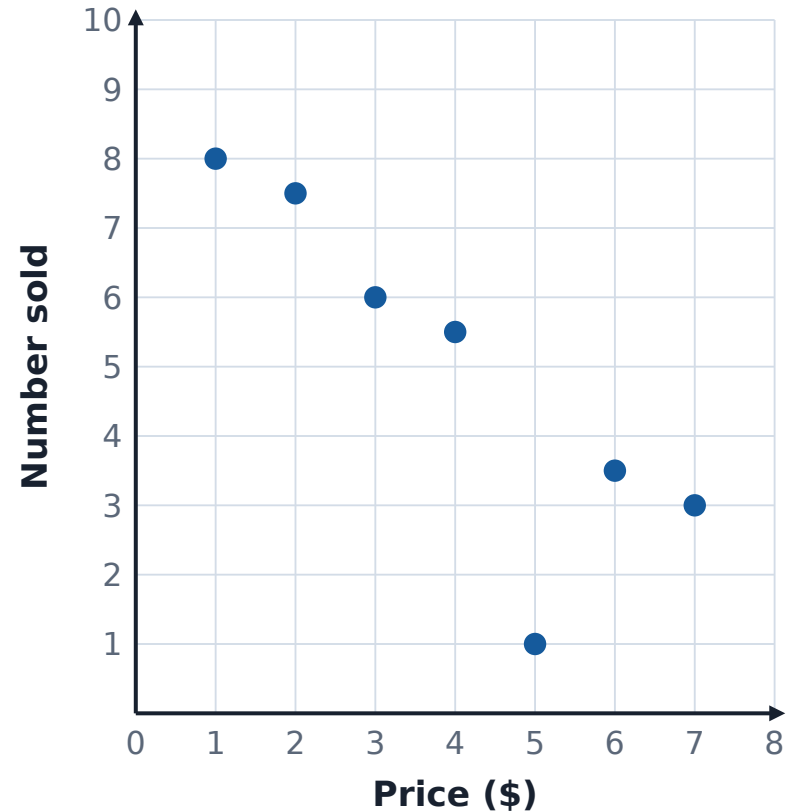


The scatter plot shows the price of an item and the number sold.

Which point does not fit the pattern? What might explain it?

Point: _____

Possible reason: _____

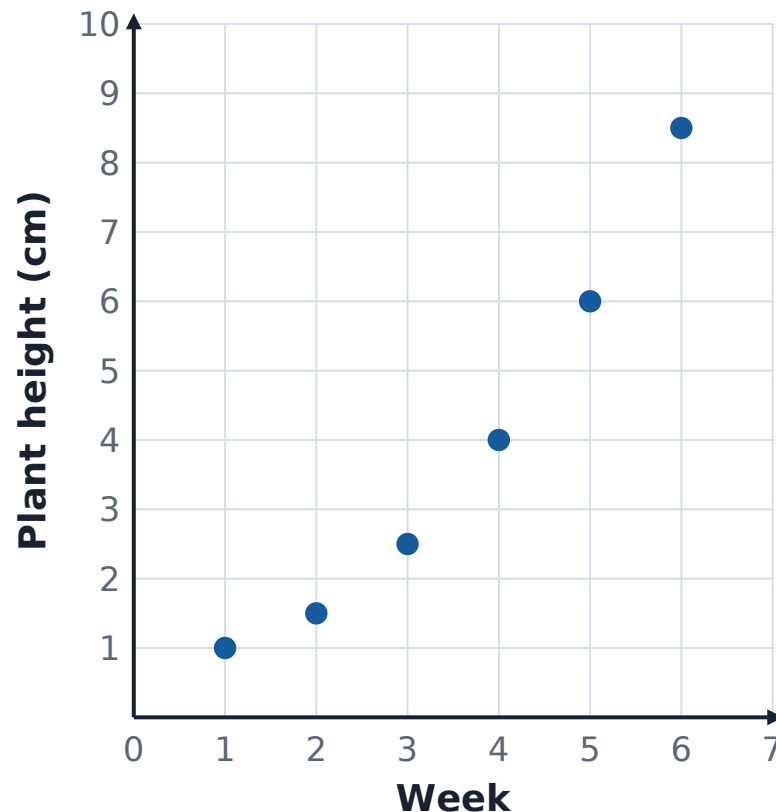


True or false?

The plant grows faster each week, so the pattern in the scatter plot is nonlinear.

A. True

B. False



Choose measures of center and variability based on the shape of the data

MAIN	Grade 6 main question	6.SP.B.5.d
BACKWARD 1	Shape of a distribution	6.SP.A.2
BACKWARD 2	Computing the mean and median	6.SP.B.5.c
BACKWARD 3	Reading a dot plot	4.MD.B.4
FORWARD 1	Compare populations using appropriate measures	7.SP.B.4
FORWARD 2	Judge how well a line fits the data — nearest related (no direct Gra...	8.SP.A.2

SET 82 • 6.SP.B.5.d

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.d

Data: 20, 22, 23, 25, 90

Which measure of center better describes a typical value?

A. Mean

B. Median

A data set is skewed right and has an outlier.

Which pair of measures best describes its center and spread?

A. Mode and range

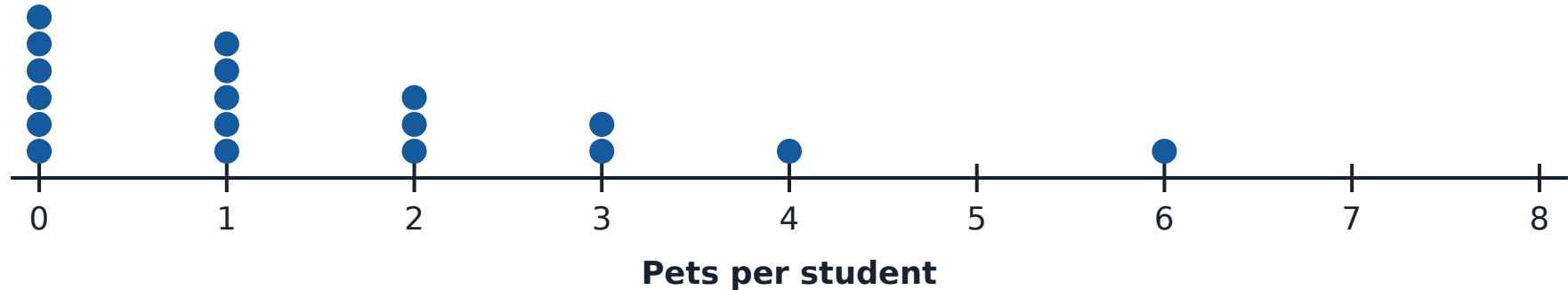
B. Median and IQR

C. Mean and range

D. Mean and MAD

The dot plot is skewed right.

Which measure of center better describes the typical number of pets?



A. Mean

B. Median

Salaries: \$30,000; \$32,000; \$35,000; \$38,000;
\$250,000

Which measure of center better describes a typical salary?

A. Mean

B. Median

True or false?

Heights of 20 sixth graders are symmetric with no outliers, so the mean is a good measure of center.

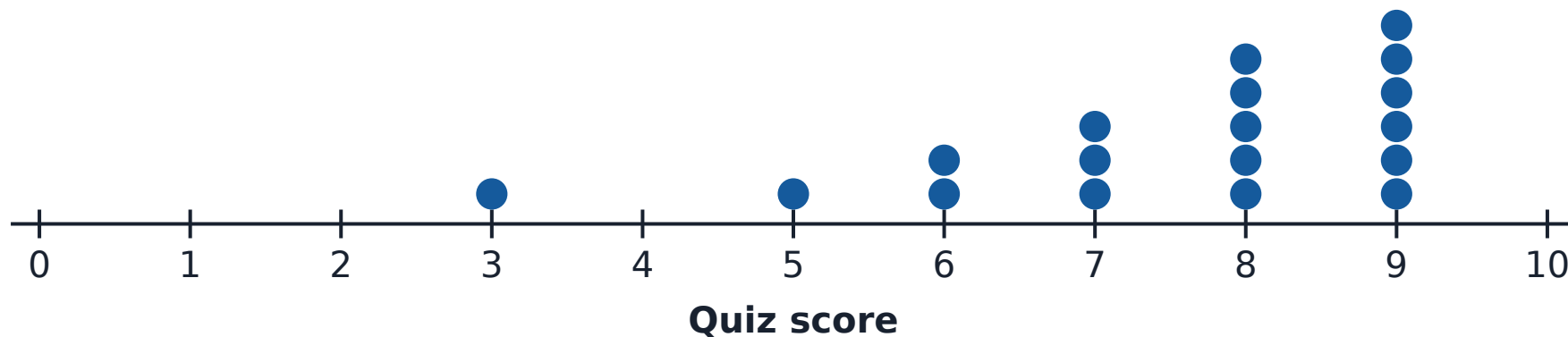
- A. True
- B. False

SET 82 • 6.SP.B.5.d

BACKWARD 1

Shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

Which word describes the shape of the data?



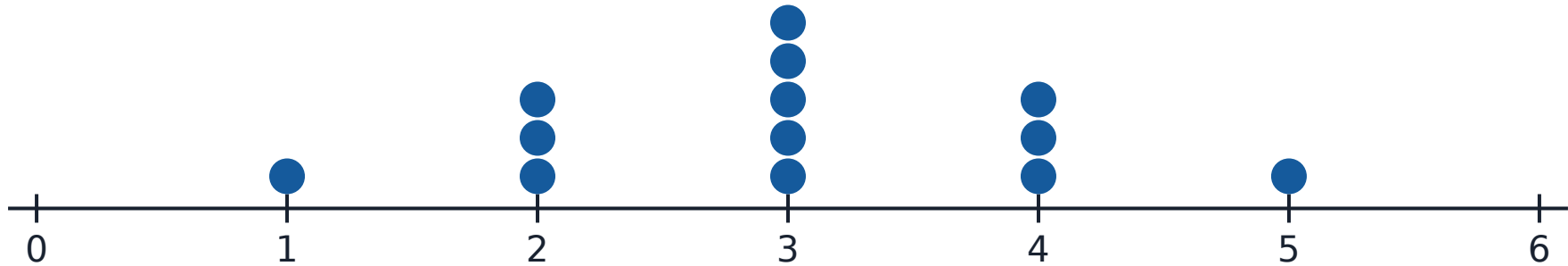
A. Skewed right

B. Symmetric

C. Skewed left

D. Uniform

Which word describes the shape of the data?



A. Skewed left

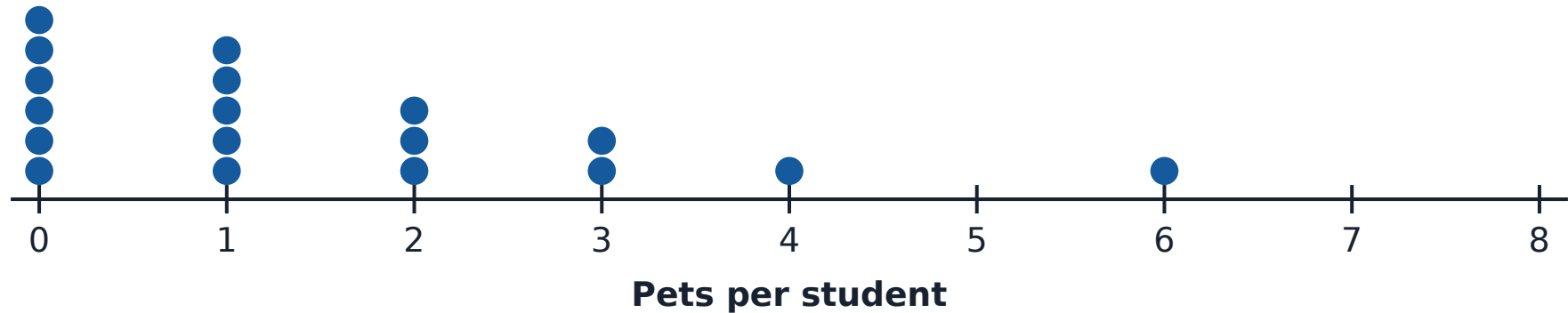
B. Symmetric

C. Uniform

D. Skewed right

True or false?

The data are skewed right.



A. True

B. False

True or false?

A data set with most values on the left and a long tail to the right is skewed right.

- A.** True
- B.** False

Which value in the data set is an outlier?

5, 6, 6, 7, 8, 30

- A.** 5
- B.** 6
- C.** 8
- D.** 30

SET 82 • 6.SP.B.5.d

BACKWARD 2

Computing the mean and median • Grade 6 • Statistics & Probability •
6.SP.B.5.c

Find the mean.

2, 3, 3, 4, 8

Mean: _____

Find the median.

2, 3, 3, 4, 8

Median: _____

True or false?

The mean of 10, 20, 30 is 20.

A. True

B. False

What is the median of 1, 4, 6, 9?

- A.** 4
- B.** 5
- C.** 6
- D.** 20

Find the mean.

5, 5, 6, 8, 11

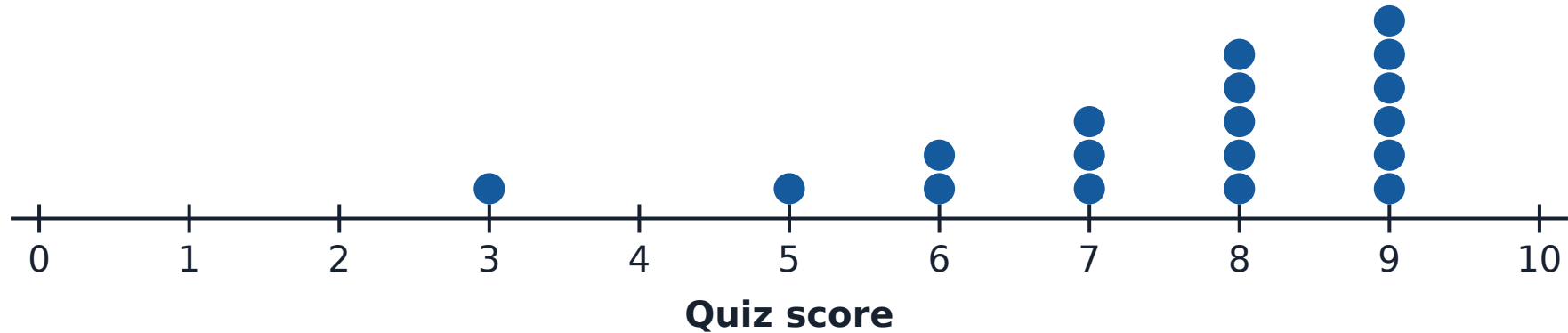
Mean: _____

SET 82 • 6.SP.B.5.d

BACKWARD 3

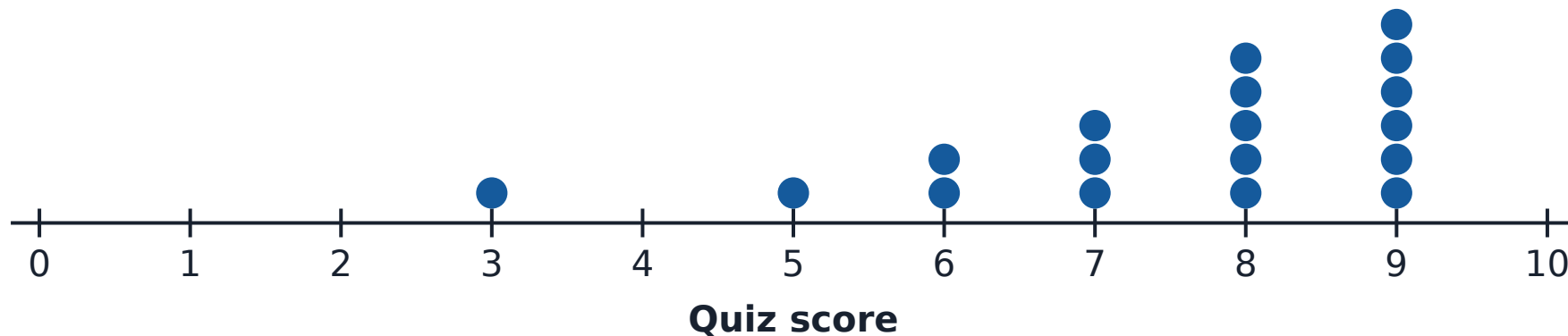
Reading a dot plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many students scored 9 on the quiz?



Students: _____

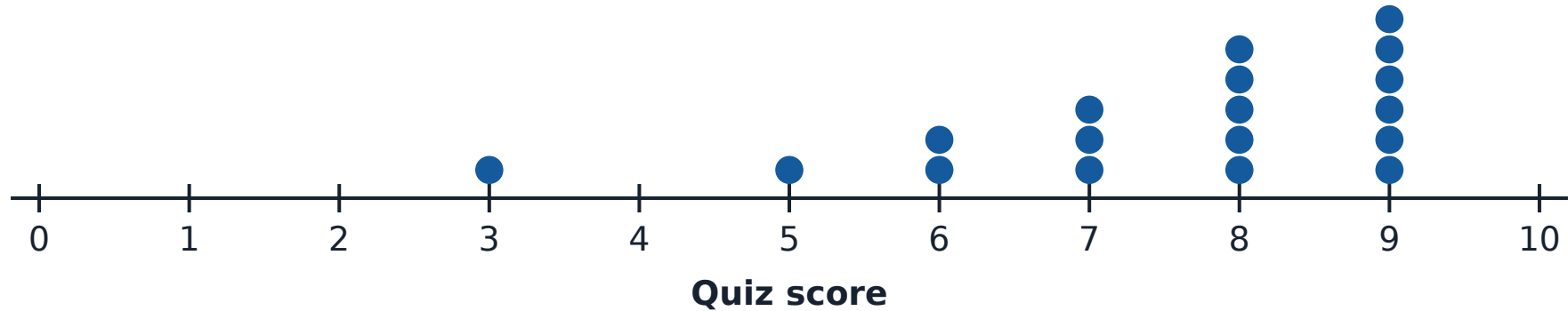
What is the lowest quiz score shown?



Score: _____

True or false?

Five students scored 8.



A. True

B. False

How many students took the quiz?



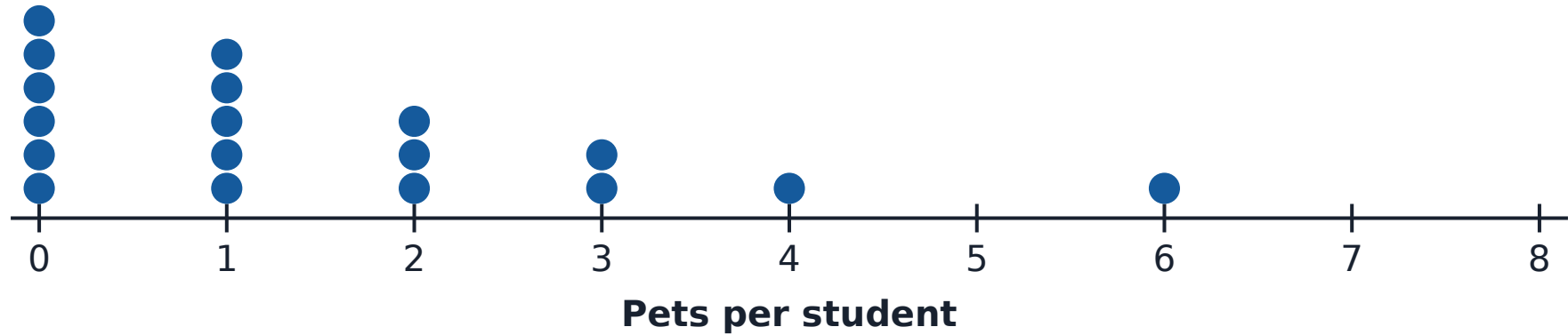
A. 6

B. 9

C. 15

D. 18

How many students had 0 pets?



Students: _____

SET 82 • 6.SP.B.5.d

FORWARD 1

Compare populations using appropriate measures • Grade 7 • Statistics & Probability • 7.SP.B.4

Two random samples of house prices are both skewed right with outliers.

Which measures are best for comparing them?

- A.** Medians and IQRs
- B.** Minimums
- C.** Means and MADs
- D.** Maximums

True or false?

When two samples are symmetric with no outliers, comparing their means is appropriate.

- A. True
- B. False

Sample A: median 30, IQR 6. Sample B: median 45, IQR 6.

The difference in medians is how many times the IQR?

Times: _____

Class A test scores have mean 78 and MAD 5. Class B test scores have mean 78 and MAD 12.

Which statement is true?

- A.** Class A scored higher.
- B.** Class B scored higher.
- C.** Class B's scores vary more.
- D.** Class A's scores vary more.

Random samples of puzzle times are both skewed right. Group X has a median of 14 minutes, and Group Y has a median of 20 minutes. Their IQRs are similar.

Which group typically takes longer?

Group: _____

FORWARD 2

Judge how well a line fits the data • Grade 8 • Statistics & Probability • 8.SP.A.2

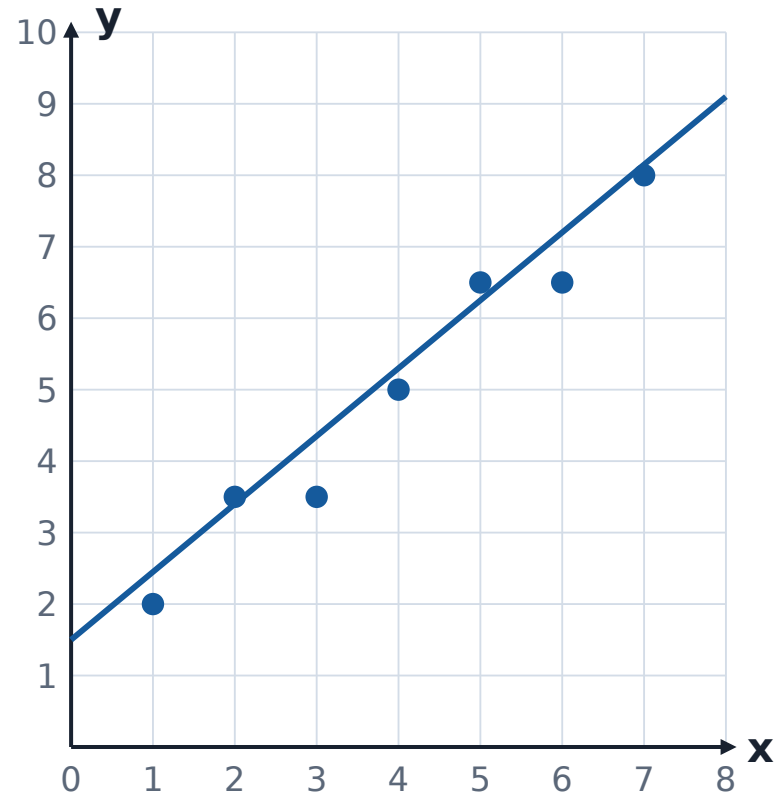
► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

True or false?

The line is a good fit for the data.

A. True

B. False

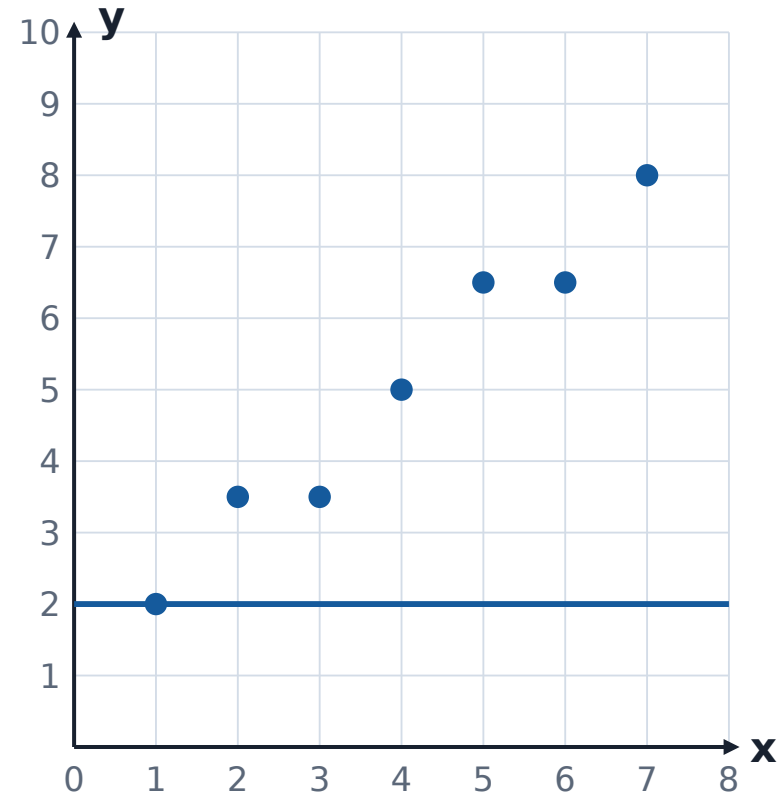


True or false?

The line is a good fit for the data.

A. True

B. False

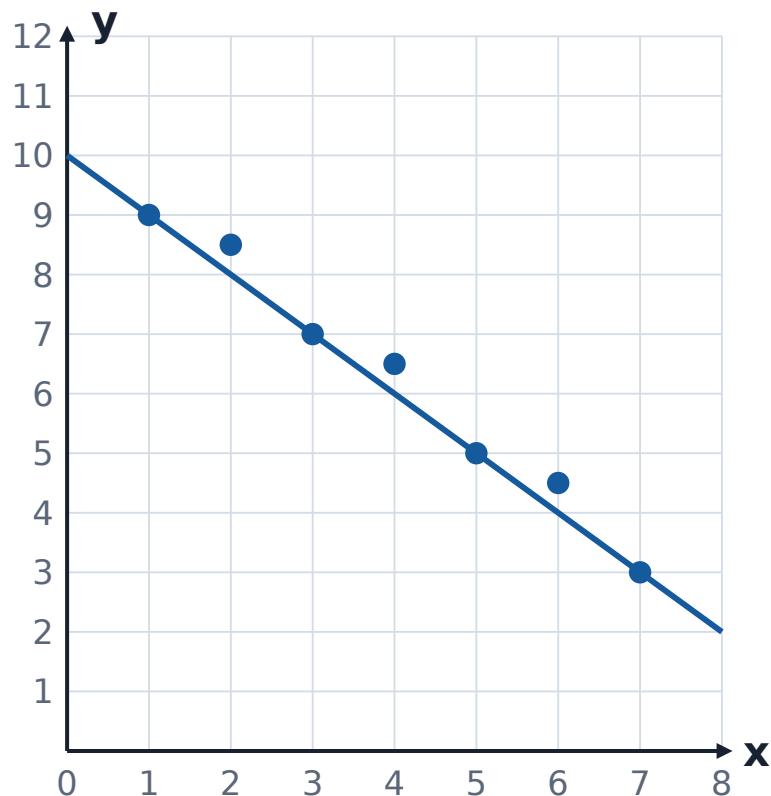


A line of best fit should show ____.

- A.** only the largest value
- B.** only the smallest value
- C.** only the first point
- D.** the overall trend of the data

Use the line of best fit to
predict y when $x = 2$.

$y =$ _____



Why should a scatter plot with no association NOT be modeled with a straight line?

- A.** There are too many points.
- B.** Lines must pass through the origin.
- C.** The axes are labeled.
- D.** There is no linear trend to model.

